MCQ 1:

A 40-year-old woman presents to the Emergency Department with vomiting and generalized abdominal pain. She has noticed a slight yellow tinged in her vision recently. On examination, her blood pressure is 90/50 mmHg, heart rate 60/min, respiratory rate 16/min, and temperature 37°C. What is the most likely diagnosis?

A. Anterolateral MI
B. Digoxin toxicity
C. Hypokalaemia
D. Hypothermia

Explanation:
The patient's symptoms of vomiting, abdominal pain, and yellow-tinged vision, along with her vital signs, are highly suggestive of **digoxin toxicity**. Digoxin toxicity can cause nausea, vomiting, abdominal pain, bradycardia, and visual disturbances (such as yellow vision). It is a common complication in patients on digoxin therapy, particularly with renal impairment or electrolyte disturbances.

Digoxin Toxicity

Definition: Digoxin toxicity occurs due to excessive levels of digoxin in the bloodstream, resulting in toxic effects on the heart and other organs.

Key Clinical Presentation: Nausea, vomiting, abdominal pain, visual disturbances (yellow vision), bradycardia, arrhythmias.

Investigations:

Best initial: Serum digoxin level, ECG

Confirmatory: Serum electrolytes (especially potassium), renal function tests.

Treatment Outline:

First-line: Discontinuation of digoxin, correcting electrolyte imbalances (especially potassium), and using digoxin-specific antibody (Digibind) in severe cases.

Second-line: Use of antiarrhythmic drugs (e.g., lidocaine or phenytoin) for arrhythmias.

Definitive: Digoxin-specific antibody (Digibind) for life-threatening toxicity.

Correct Answer: B. Digoxin toxicity

2. A 70-year-old man with a history of congestive heart failure presents to the clinic for follow-

up. He is currently on medical therapy but is concerned about improving his long-term survival. Which of the following agents has been shown to reduce mortality in patients with congestive heart failure?

A. Digitalis
B. Enalapril
C. Furosemide
D. Procainamide

Explanation:
Enalapril, an angiotensin-converting enzyme (ACE) inhibitor, has been shown to reduce mortality in patients with congestive heart failure. ACE inhibitors help by improving cardiac output, reducing afterload, and preventing harmful cardiac remodeling. They are considered a cornerstone of heart failure management.

Congestive Heart Failure (CHF)

Definition: A clinical syndrome where the heart is unable to pump blood effectively, leading to symptoms such as fluid retention, shortness of breath, and fatigue.

Key Clinical Presentation: Dyspnea, fatigue, edema, orthopnea, paroxysmal nocturnal dyspnea.

Investigations:

Best initial: Chest X-ray, echocardiogram, and serum B-type natriuretic peptide (BNP) levels.

Confirmatory: Echocardiogram showing reduced ejection fraction or diastolic dysfunction.

Treatment Outline:

First-line: ACE inhibitors (e.g., enalapril), beta-blockers (e.g., metoprolol), diuretics (e.g., furosemide).

Second-line: Aldosterone antagonists (e.g., spironolactone), angiotensin receptor blockers (ARBs), digoxin in selected patients.

Definitive: Heart transplantation or device implantation (e.g., ICD) for severe cases.

Correct Answer: B. Enalapril ✅

3. 72-year-old man with congestive heart failure has been prescribed furosemide to manage fluid retention. During follow-up, he develops symptoms of muscle weakness and fatigue. Which of the following is a recognized side effect of furosemide in congestive heart failure?

A. Hypokalaemia
B. Bronchospasm

C. Hypoglycaemia
D. Haemolytic anaemia

Explanation:
Hypokalaemia is a common side effect of **furosemide**, a loop diuretic, especially in patients with congestive heart failure. Furosemide increases the excretion of sodium, chloride, and potassium in the urine, which can lead to electrolyte imbalances, including low potassium levels. This is particularly concerning as hypokalaemia can lead to muscle weakness, fatigue, and arrhythmias.

Furosemide in Congestive Heart Failure

Definition: Furosemide is a loop diuretic used in the management of congestive heart failure to reduce fluid overload by promoting diuresis.

Key Clinical Presentation: Edema, pulmonary congestion, weight gain due to fluid retention.

Investigations:

Best initial: Electrolyte levels (especially potassium), renal function tests.

Confirmatory: None specific; diagnosis based on clinical presentation and patient history.

Treatment Outline:

First-line: Furosemide for fluid management, ACE inhibitors (e.g., enalapril), beta-blockers, aldosterone antagonists.

Second-line: Combination therapy with other diuretics (e.g., thiazides) in resistant cases.

Definitive: Heart transplantation for severe cases, or implantation of devices like an ICD.

Correct Answer: A. Hypokalaemia ✅

4. A 60-year-old patient with a history of ischaemic stroke presents with a regular narrow complex tachycardia. The patient is hemodynamically stable, and a decision is made to administer intravenous adenosine. In the presence of which of the following should the dose of adenosine be reduced?

A. Theophylline
B. Dipyridamole
C. Chronic renal impairment
D. Significant mitral regurgitation

Explanation:
In the presence of **chronic renal impairment**, the dose of adenosine should be reduced. This is because renal dysfunction can lead to delayed clearance of adenosine, increasing the risk of

excessive effects, such as severe bradycardia or hypotension. In contrast, **dipyridamole** enhances the effects of adenosine, and the dose should be increased, not reduced.

Adenosine Administration in Tachycardia

Definition: Adenosine is an antiarrhythmic used to treat certain types of supraventricular tachycardia by temporarily blocking the AV node and restoring normal rhythm.

Key Clinical Presentation: Regular narrow complex tachycardia, hemodynamically stable.

Investigations:

Best initial: Electrocardiogram (ECG) to confirm the rhythm.

Confirmatory: Diagnosis is typically clinical based on ECG findings and the response to adenosine.

Treatment Outline:

First-line: IV adenosine for narrow complex tachycardias.

Second-line: Additional antiarrhythmic agents or cardioversion if adenosine is ineffective.

Definitive: Treatment of underlying causes (e.g., catheter ablation for re-entrant arrhythmias).

Correct Answer: C. Chronic renal impairment ✅

5. A 65-year-old patient with chronic heart failure presents to the clinic for follow-up. The physician is considering medications to improve the patient's long-term survival. Which of the following drugs used in the treatment of chronic heart failure has been shown to increase survival?

A. Digoxin
B. Enalapril
C. Furosemide
D. Hydralazine

Explanation:
Enalapril, an angiotensin-converting enzyme (ACE) inhibitor, has been shown to increase survival in patients with chronic heart failure. ACE inhibitors improve heart function by reducing afterload, preventing harmful cardiac remodeling, and improving blood flow, which ultimately leads to improved survival rates in heart failure patients.

Chronic Heart Failure (CHF)

Definition: A condition where the heart is unable to pump sufficient blood to meet the body's needs, leading to symptoms such as fatigue, shortness of breath, and fluid retention.

Key Clinical Presentation: Dyspnea, orthopnea, fatigue, edema, paroxysmal nocturnal dyspnea.

Investigations:

Best initial: Chest X-ray, echocardiogram, and serum BNP levels.

Confirmatory: Echocardiogram showing reduced ejection fraction or diastolic dysfunction.

Treatment Outline:

First-line: ACE inhibitors (e.g., enalapril), beta-blockers (e.g., metoprolol), diuretics (e.g., furosemide).

Second-line: Aldosterone antagonists (e.g., spironolactone), angiotensin receptor blockers (ARBs), digoxin in selected cases.

Definitive: Heart transplantation or device implantation (e.g., ICD) for severe cases.

Correct Answer: B. Enalapril ✅

6. A 39-year-old man presents to the clinic with unexplained fever 19 days after an attack of acute rheumatic fever. Several of his teeth were extracted prior to his hospital admission. Clinical examination reveals a systolic murmur and splenomegaly. Laboratory results show elevated WBC (14.0 x 10^9/L), neutrophils at 72%, and urinalysis showing 25 erythrocytes per HPF. What is the most likely diagnosis?

A. Floppy mitral valve
B. Bacterial endocarditis
C. Acute myocardial infarction
D. Recurrent rheumatic heart disease

Explanation:
The patient's presentation of fever, systolic murmur, splenomegaly, and elevated WBC count following a recent attack of acute rheumatic fever is most consistent with **bacterial endocarditis**. The recent dental extractions increase the risk of bacteremia, which can lead to endocarditis, especially in patients with preexisting valvular heart disease from rheumatic fever.

Bacterial Endocarditis

Definition: Infection of the endocardial surface of the heart, often affecting the heart valves.

Key Clinical Presentation: Fever, heart murmur, splenomegaly, positive blood cultures, possible embolic phenomena.

Investigations:

- Best initial: Blood cultures
- Confirmatory: Echocardiography (to identify vegetations on heart valves)

Treatment Outline:

- First-line: IV antibiotics (e.g., ceftriaxone, gentamicin) based on blood culture sensitivity.
- Second-line: Surgery may be required for valve replacement in severe cases.

Correct Answer: B. Bacterial endocarditis ✅

7 . A 62-year-old man presents with recurrent retrosternal chest pain, which has become much worse over the last 3 weeks. He describes the pain as lasting longer than before and occurring even while sitting in a chair or lying in bed at night. Which of the following is the most likely type of angina?

A. Unstable
B. Heberden
C. Exertional
D. Prinzmetal

Explanation:
The patient's description of chest pain that occurs at rest, lasts longer, and is more frequent is characteristic of **unstable angina**. This form of angina occurs unpredictably and is associated with a higher risk of myocardial infarction (MI). It requires urgent evaluation and management.

Unstable Angina

Definition: A type of angina pectoris that occurs at rest or with minimal exertion and is more intense and prolonged than stable angina.

Key Clinical Presentation: Chest pain at rest, increasing frequency and duration, may be associated with diaphoresis or nausea.

Investigations:

- Best initial: ECG (may show ST-segment depression or T-wave inversion)
- Confirmatory: Cardiac biomarkers (e.g., troponins)

Treatment Outline:

- First-line: Antiplatelet therapy (e.g., aspirin), nitrates, beta-blockers

- Second-line: Coronary angiography, revascularization if indicated

Correct Answer: A. Unstable ✅

8.
MCQ:
A 52-year-old man with a 5-year history of hypertension is being treated with hydrochlorothiazide 25 mg/day. His current blood pressure is 155/101 mmHg, and his lab results show: sodium 141 mmol/L, potassium 4.1 mmol/L, creatinine 105 µmol/L, and fasting glucose 7.3 mmol/L. What is the most appropriate management modification?

A. Add a beta blocker
B. Add a calcium channel blocker
C. Double the hydrochlorothiazide dose
D. Add an angiotensin-converting enzyme inhibitor

Explanation:
The patient's blood pressure remains poorly controlled despite being on hydrochlorothiazide, and his fasting glucose is elevated, suggesting potential early signs of diabetes. **Adding an ACE inhibitor** is appropriate because ACE inhibitors not only lower blood pressure but also provide renal protection, which is beneficial in hypertensive patients, particularly those with early signs of diabetes.

Hypertension

Definition: A condition where the force of the blood against the artery walls is too high, potentially leading to complications like stroke, heart failure, or kidney damage.

Key Clinical Presentation: Elevated blood pressure readings, often asymptomatic until complications develop.

Investigations:

- Best initial: Blood pressure measurement
- Confirmatory: Ambulatory blood pressure monitoring (ABPM)

Treatment Outline:

- First-line: Lifestyle modifications (diet, exercise), ACE inhibitors (e.g., enalapril), calcium channel blockers.
- Second-line: Diuretics, beta-blockers, or ARBs if needed.

Correct Answer: D. Add an angiotensin-converting enzyme inhibitor ✅

9.
MCQ:
A 30-year-old intravenous drug user presents with fever, night sweats, chest pain, and arthralgia. On examination, painless erythematous lesions are noted on the palms of the hands, along with round erythematous lesions and central clearing noted in the retina and splinter hemorrhages under the fingernails. What is the most likely diagnosis?

A. Syphilis
B. HIV infection
C. Infectious hepatitis
D. Bacterial endocarditis

Explanation:
The combination of fever, night sweats, chest pain, splenomegaly, and erythematous lesions on the palms with retinal lesions and splinter hemorrhages under the fingernails is most consistent with **bacterial endocarditis**, specifically infective endocarditis. These findings, including retinal lesions (Roth spots) and splinter hemorrhages, are characteristic of this condition, especially in a patient with intravenous drug use, which is a significant risk factor for endocarditis.

Bacterial Endocarditis

Definition: Infection of the endocardial surface of the heart, most often involving heart valves, caused by bacteria entering the bloodstream.

Key Clinical Presentation: Fever, night sweats, heart murmur, splenomegaly, Roth spots, splinter hemorrhages, and positive blood cultures.

Investigations:

- Best initial: Blood cultures, echocardiogram
- Confirmatory: Transesophageal echocardiography for vegetations

Treatment Outline:

- First-line: IV antibiotics (e.g., vancomycin, ceftriaxone)
- Second-line: Surgery in severe cases, such as for valve replacement

Correct Answer: D. Bacterial endocarditis ✅

10.
MCQ:

A 77-year-old woman with poorly controlled long-standing type 2 diabetes and hypertension presents with dizziness upon standing rapidly. Current medications include aspirin, atenolol, insulin, metformin, and multivitamins. Sitting blood pressure is 140/85 mmHg, and standing blood pressure is 115/80 mmHg. What is the most likely explanation for this finding?

A. Hypoglycemia
B. Medication side effect
C. Autonomic neuropathy
D. Left ventricular dysfunction

Explanation:
The patient's dizziness upon standing, along with the drop in blood pressure from sitting to standing, is most likely due to **autonomic neuropathy**, a common complication of diabetes. This condition impairs the body's ability to regulate blood pressure upon standing, leading to orthostatic hypotension.

Autonomic Neuropathy

Definition: A form of diabetic neuropathy affecting the autonomic nervous system, leading to dysfunction in regulating blood pressure, heart rate, and other autonomic functions.

Key Clinical Presentation: Orthostatic hypotension, dizziness, syncope, and gastrointestinal symptoms.

Investigations:

- Best initial: Blood pressure measurements (sitting and standing)
- Confirmatory: Heart rate variability tests, tilt-table testing

Treatment Outline:

- First-line: Volume expansion (e.g., increased salt intake, fluid intake), compression stockings
- Second-line: Medications like fludrocortisone or midodrine

Correct Answer: C. Autonomic neuropathy ✅

11.
MCQ:
A 67-year-old man is brought to the Emergency Department by his family with confusion and a decreased level of consciousness. He has fever, diarrhea, and poor oral intake for the past 2 days. On examination, the JVP is not visible, heart sounds are normal, and the lungs are clear with no rhonchi or crepitations. The patient's extremities are cold, and peripheral pulses are

feeble. His blood pressure is 83/45 mmHg, heart rate 120/min, and respiratory rate 20/min. Which of the following is the most appropriate next step in management?

A. IV boluses of normal saline
B. Intravenous beta blockers
C. Nasogastric tube feeding
D. Dopamine infusion

Explanation:
The patient's presentation of confusion, hypotension, tachycardia, and cold extremities is most consistent with **hypovolemic shock** due to dehydration from diarrhea and poor oral intake. The most appropriate next step is to administer **IV boluses of normal saline** to restore circulating volume and stabilize blood pressure.

Hypovolemic Shock

Definition: A condition where the body does not have enough blood volume to maintain adequate tissue perfusion, often due to fluid loss or bleeding.

Key Clinical Presentation: Hypotension, tachycardia, cold extremities, altered mental status, low urine output.

Investigations:

- Best initial: Blood pressure measurement, blood tests for electrolytes, and renal function
- Confirmatory: Hemodynamic monitoring or central venous pressure measurement in severe cases

Treatment Outline:

- First-line: Fluid resuscitation with normal saline or Ringer's lactate
- Second-line: Vasopressors if fluid resuscitation fails to normalize blood pressure

Correct Answer: A. IV boluses of normal saline ✅

12.
MCQ:
A 40-year-old man is brought to the Emergency Department with progressive difficulty in breathing. He is being treated for bronchogenic carcinoma. On examination, the JVP is elevated, lungs are clear, and heart sounds are very quiet. His blood pressure is 88/50 mmHg, heart rate is 125/min, and respiratory rate is 24/min. Which of the following is the most appropriate test to confirm the diagnosis?

A. Chest X-ray
B. Echocardiogram
C. Electrocardiogram
D. Arterial blood gases

Explanation:
The combination of elevated JVP, hypotension, tachycardia, and a quiet heart is highly suggestive of **cardiac tamponade** secondary to malignancy, such as bronchogenic carcinoma. The most appropriate test to confirm this diagnosis is an **echocardiogram**, which will show the presence of fluid in the pericardial sac.

Cardiac Tamponade

Definition: A condition where fluid accumulates in the pericardium, leading to restricted filling of the heart and decreased cardiac output.

Key Clinical Presentation: Elevated JVP, hypotension, tachycardia, quiet heart sounds, pulsus paradoxus.

Investigations:

- Best initial: Echocardiogram (to assess pericardial effusion)
- Confirmatory: Pericardiocentesis if necessary

Treatment Outline:

- First-line: Pericardiocentesis to remove excess fluid
- Second-line: Pericardial window surgery if recurrent effusions

Correct Answer: B. Echocardiogram ✅

13.
MCQ:
A 52-year-old man was diagnosed with an acute inferior wall myocardial infarction. After intravenous morphine and IV nitroglycerin, his condition worsened. His JVP is now elevated, and his lungs are clear. His blood pressure dropped from 120/70 mmHg to 75/40 mmHg, and heart rate increased to 105/min. Which of the following is the most likely cause of the hypotension?

A. Cardiac tamponade
B. Complete heart block

C. Papillary muscle rupture
D. Right ventricular infarction

Explanation:
The patient’s worsening hypotension after an acute inferior wall myocardial infarction is most likely due to **right ventricular infarction**. This complication leads to reduced preload and hypotension, especially in patients with inferior wall MI, as the right ventricle is responsible for filling the left ventricle.

Right Ventricular Infarction

Definition: Myocardial infarction involving the right ventricle, often complicating inferior wall myocardial infarctions.

Key Clinical Presentation: Hypotension, elevated JVP, clear lungs, tachycardia.

Investigations:

- Best initial: ECG (right ventricular lead V4R)
- Confirmatory: Echocardiogram

Treatment Outline:

- First-line: IV fluids to increase preload, vasopressors if necessary
- Second-line: Inotropes if there is persistent hypotension

Correct Answer: D. Right ventricular infarction ✅

14.
MCQ:
A 70-year-old man presents with an acute confusional state. On examination, he has postural hypotension and dry mucous membranes. Laboratory results show a calcium level of 3.41 mmol/L (normal 2.15-2.62 mmol/L). Which of the following is the most appropriate initial treatment?

A. IV 0.9% sodium chloride
B. IV sodium pamidronate
C. Oral prednisolone
D. IV furosemide

Explanation:
The patient's elevated calcium level (hypercalcemia) combined with confusion and dehydration suggests **hypercalcemia** as the likely diagnosis. The most appropriate initial treatment for

hypercalcemia is **IV 0.9% sodium chloride** to promote diuresis and help lower calcium levels by increasing renal calcium excretion.

Hypercalcemia

Definition: An elevated serum calcium level, which can cause confusion, weakness, and renal impairment.

Key Clinical Presentation: Confusion, polyuria, dehydration, dry mucous membranes, and elevated calcium levels.

Investigations:

- Best initial: Serum calcium, renal function tests
- Confirmatory: Parathyroid hormone (PTH) and PTH-related peptide (PTHrP) levels to determine the cause

Treatment Outline:

- First-line: IV fluids (0.9% sodium chloride), loop diuretics if necessary
- Second-line: Bisphosphonates (e.g., pamidronate) or calcitonin for severe cases

Correct Answer: A. IV 0.9% sodium chloride ✅

15.
MCQ:
A 61-year-old woman with a 12-year history of type 2 diabetes presents with hypertension. She has been treated with insulin and was recently started on enalapril 10 mg daily for hypertension but discontinued it due to a cough. Laboratory results show: creatinine 142 µmol/L (normal 44-115 µmol/L), creatinine clearance 66 mL/min (normal 100-140 mL/min), and 24-hour urine protein 1500 mg/24 hr (normal 0-150 mg/24 hr). Which of the following is the most appropriate therapy?

A. Beta blocker
B. Thiazide diuretic
C. Calcium channel blocker
D. Angiotensin receptor blocker

Explanation:
This patient with hypertension, diabetic nephropathy (evidenced by proteinuria), and chronic kidney disease would benefit from an **angiotensin receptor blocker (ARB)**. ARBs, like losartan, can reduce proteinuria and help slow the progression of kidney disease, making them an appropriate choice for patients with diabetic nephropathy.

Diabetic Nephropathy

Definition: Kidney damage resulting from long-standing diabetes, characterized by albuminuria, reduced glomerular filtration rate, and progression to end-stage renal disease.

Key Clinical Presentation: Hypertension, proteinuria, and worsening renal function in a patient with diabetes.

Investigations:

- Best initial: Urinary albumin-to-creatinine ratio, serum creatinine
- Confirmatory: Kidney biopsy if necessary

Treatment Outline:

- First-line: Angiotensin receptor blockers (ARBs) or ACE inhibitors
- Second-line: Diuretics (e.g., furosemide) for fluid management

Correct Answer: D. Angiotensin receptor blocker ✅

16.
MCQ:
A 70-year-old patient presents with sudden onset of pain in the left lower limb. The pain is severe and associated with numbness. The patient had an acute myocardial infarction 2 weeks ago and was discharged 24 hours before this presentation. The left leg is cold and pale, and the right leg is normal. Which of the following is the most likely diagnosis?

A. Acute arterial thrombosis
B. Acute arterial embolus
C. Deep vein thrombosis
D. Dissecting aneurysm

Explanation:
The patient's sudden onset of severe pain, numbness, and coldness in the left lower limb following a recent myocardial infarction suggests an **acute arterial embolus**. This is commonly caused by a thrombus from the heart (especially after an MI) that travels to the peripheral arteries, resulting in limb ischemia.

Acute Arterial Embolus

Definition: An embolus that lodges in a peripheral artery, often following a recent cardiac event, leading to ischemia and pain in the affected limb.

Key Clinical Presentation: Sudden, severe limb pain, coldness, pallor, and numbness in the affected limb.

Investigations:

- Best initial: Doppler ultrasound of the affected limb
- Confirmatory: Angiography to visualize the embolus

Treatment Outline:

- First-line: Anticoagulation and thrombolysis
- Second-line: Surgery (embolectomy) if necessary

Correct Answer: B. Acute arterial embolus ✅

17.
MCQ:
A patient with atrial fibrillation is on warfarin 5 mg daily. His latest INR is 7.0, and there is no evidence of abnormal bleeding. Which of the following is the most appropriate action?

A. Increase warfarin to 7.5 mg daily
B. Decrease warfarin to 2.5 mg daily
C. Continue with the same dose of warfarin
D. Hold warfarin and repeat INR next day

Explanation:
An INR of 7.0 is significantly elevated, indicating a high risk of bleeding. The most appropriate course of action is to **hold warfarin and repeat INR the next day** to assess whether the INR decreases on its own. Adjustments in warfarin dosage should be made based on further INR readings.

Warfarin Management

Definition: Warfarin is an oral anticoagulant used to prevent thromboembolic events in atrial fibrillation and other conditions.

Key Clinical Presentation: Elevated INR, risk of bleeding.

Investigations:

- Best initial: INR measurement
- Confirmatory: None needed

Treatment Outline:

- First-line: Adjust warfarin dose based on INR
- Second-line: Vitamin K or fresh frozen plasma if bleeding or very high INR

Correct Answer: D. Hold warfarin and repeat INR next day ✅

18.
MCQ:
A 73-year-old man presents to the Primary Health Care Clinic with severe shortness of breath at night and swelling of the ankles. He has reduced his activity over the past year due to chest pain on exertion. Physical examination reveals crackles over both lungs, enlargement of the liver, and pitting edema of the ankles. His blood pressure is 140/90 mmHg, heart rate 95/min, and respiratory rate 24/min. Which of the following is the most appropriate immediate management?

A. Digoxin
B. Atenolol
C. Furosemide
D. Hydrochlorothiazide

Explanation:
The patient's symptoms of shortness of breath, edema, and crackles in the lungs suggest **heart failure**, likely due to left-sided heart failure with pulmonary congestion. **Furosemide**, a loop diuretic, is the first-line treatment to reduce fluid overload and alleviate symptoms of heart failure.

Heart Failure

Definition: A condition where the heart is unable to pump blood effectively, leading to fluid retention, shortness of breath, and fatigue.

Key Clinical Presentation: Shortness of breath, orthopnea, edema, and fatigue.

Investigations:

- Best initial: Chest X-ray, echocardiogram, BNP levels
- Confirmatory: Echocardiogram showing reduced ejection fraction

Treatment Outline:

- First-line: Diuretics (e.g., furosemide), ACE inhibitors, beta-blockers
- Second-line: Aldosterone antagonists (e.g., spironolactone)

Correct Answer: C. Furosemide ✅

19.
MCQ:
A 60-year-old diabetic patient presents for follow-up. The patient takes metformin and has fasting blood glucose levels of 5.5 mmol/L and an HbA1c of 5.8%. Physical examination is normal, including a neurologic exam of the lower extremities, and there are no signs of retinopathy. His blood pressure is 149/90 mmHg. Which of the following is the most appropriate management?

A. No change in therapy
B. Add ACE inhibitor
C. Add beta-blocker
D. Add sulfonylurea

Explanation:
This patient with well-controlled diabetes and hypertension (but elevated blood pressure) should benefit from adding an **ACE inhibitor**. ACE inhibitors are shown to reduce blood pressure and protect the kidneys in patients with diabetes, particularly those with elevated proteinuria or high blood pressure.

Hypertension and Diabetes Management

Definition: The management of hypertension in diabetic patients to reduce cardiovascular risk and prevent diabetic nephropathy.

Key Clinical Presentation: Elevated blood pressure in a diabetic patient.

Investigations:

- Best initial: Blood pressure measurement
- Confirmatory: None required

Treatment Outline:

- First-line: ACE inhibitors, lifestyle modifications (e.g., diet, exercise)
- Second-line: Diuretics, calcium channel blockers, or beta-blockers

Correct Answer: B. Add ACE inhibitor ✅

20.
MCQ:
A 25-year-old patient is diagnosed with diabetic ketoacidosis. His blood pressure is 80/50

mmHg, heart rate is 140/min, respiratory rate is 26/min, and temperature is 36.6°C. Laboratory results show: random glucose 52.7 mmol/L, potassium 6 mmol/L, and bicarbonate 5 mmol/L. Which of the following is the best initial management?

A. Dopamine drip to raise blood pressure above 90 mm/Hg
B. 2L of normal saline with KCl 20 mEq/L intravenously
C. 20-unit regular insulin intravenously, followed by 5% dextrose in water at 200 mL/h
D. Normal saline 2L as an intravenous bolus and insulin intravenous infusion at 0.1 U/kg/hr

Explanation:
The patient is in diabetic ketoacidosis (DKA), with severe dehydration and electrolyte abnormalities. The best initial treatment involves **normal saline** for volume resuscitation and an **insulin infusion** to correct hyperglycemia and ketosis. This approach is essential to manage DKA effectively and prevent complications such as cerebral edema.

Diabetic Ketoacidosis (DKA)

Definition: A severe complication of diabetes characterized by hyperglycemia, metabolic acidosis, and ketonemia.

Key Clinical Presentation: Polyuria, polydipsia, vomiting, dehydration, confusion, and a fruity odor on the breath.

Investigations:

- Best initial: Blood glucose, serum ketones, arterial blood gas (ABG)
- Confirmatory: Urine ketones, serum bicarbonate

Treatment Outline:

- First-line: IV fluids (normal saline), insulin infusion (0.1 U/kg/hr)
- Second-line: Potassium supplementation as required

Correct Answer: D. Normal saline 2L as an intravenous bolus and insulin intravenous infusion at 0.1 U/kg/hr ✅

21.
MCQ:
A 55-year-old man with a family history of type 2 diabetes undergoes screening. His blood pressure is 130/85 mmHg, heart rate 80/min, height 185 cm, weight 91 kg. Laboratory results

show: fasting glucose 7.4 mmol/L, HbA1c 6.3%. Which of the following should be recommended?

A. Random plasma glucose
BHbA1c after 6 weeks
C. Fasting plasma glucose after 3 months
D. 2-hour 75g oral glucose tolerance test

Explanation:
The patient has an elevated fasting glucose and HbA1c, which suggests he may be at risk for **type 2 diabetes**. Given his elevated HbA1c, the most appropriate next step would be to **repeat the fasting plasma glucose** after 3 months to confirm the diagnosis of prediabetes or diabetes.

Type 2 Diabetes Screening

Definition: Screening for type 2 diabetes in high-risk individuals, such as those with a family history or obesity.

Key Clinical Presentation: Elevated blood glucose levels, family history of diabetes, and obesity.

Investigations:

- Best initial: Fasting plasma glucose, HbA1c, and possibly an oral glucose tolerance test (OGTT)

Treatment Outline:

- First-line: Lifestyle modifications (e.g., weight loss, exercise)
- Second-line: Metformin for blood sugar control in diagnosed patients

Correct Answer: C. Fasting plasma glucose after 3 months ✅

22.
MCQ:
A 55-year-old man with a family history of type 2 diabetes presents with polyuria. His blood pressure is 130/85 mmHg, heart rate 80/min, height 185 cm, weight 91 kg. Laboratory results show: fasting glucose 7.0 mmol/L, HbA1c 7.1%. Which of the following is the most likely diagnosis?

A. Prediabetes
B. Type 2 diabetes

C. Impaired glucose tolerance
D. Mature onset diabetes of youth

Explanation:
This patient's elevated fasting glucose and HbA1c levels are consistent with **type 2 diabetes**. An HbA1c above 6.5% is diagnostic of diabetes, and his symptoms of polyuria further support this diagnosis.

Type 2 Diabetes

Definition: A chronic condition where the body becomes resistant to insulin or the pancreas does not produce sufficient insulin.

Key Clinical Presentation: Polyuria, polydipsia, increased hunger, and blurred vision.

Investigations:

- Best initial: Fasting plasma glucose, HbA1c, and oral glucose tolerance test (OGTT)
- Confirmatory: HbA1c ≥6.5%

Treatment Outline:

- First-line: Lifestyle modifications (e.g., weight loss, exercise), metformin
- Second-line: Insulin or other oral hypoglycemics

Correct Answer: B. Type 2 diabetes ✅

23.
MCQ:
A 45-year-old man with type 2 diabetes for 3 months is on metformin 500 mg TDS. He comes for his first follow-up, and laboratory results show: fasting glucose 7.4 mmol/L, creatinine 125 μmol/L, HbA1c 6.8%. Which of the following investigations is recommended to assess his renal function annually?

A. Serum creatinine
B. Urine microalbumin
C. Urine albumin/creatinine ratio
D. Estimated glomerular filtration rate

Explanation:
For patients with diabetes, **urine microalbumin** or the **urine albumin/creatinine ratio** is

recommended annually to assess for early signs of diabetic nephropathy. This is an important step in preventing progression to end-stage renal disease.

Diabetic Nephropathy

Definition: Kidney damage due to long-standing diabetes, often starting with albuminuria.

Key Clinical Presentation: Proteinuria, reduced renal function, hypertension.

Investigations:

- Best initial: Urine albumin/creatinine ratio
- Confirmatory: Kidney biopsy if necessary

Treatment Outline:

- First-line: ACE inhibitors or ARBs
- Second-line: SGLT-2 inhibitors in patients with significant albuminuria

Correct Answer: C. Urine albumin/creatinine ratio ✅

24.
MCQ:
A 55-year-old woman with type 2 diabetes for 20 years is on an 1800 kcal/day diet and takes metformin 850 mg three times daily. Her blood pressure is 130/85 mmHg, heart rate 80/min, and laboratory results show: fasting glucose 7.1 mmol/L, HbA1c 7.8%. What should be the target HbA1c percentage?

A. <5.0
B. <5.5
C. <6.0
D. <7.0

Explanation:
For most diabetic patients, the **target HbA1c** is typically **less than 7.0%**, as this level has been shown to reduce the risk of complications without increasing the risk of hypoglycemia.

Type 2 Diabetes Management

Definition: Management of blood glucose in diabetic patients to reduce complications.

Key Clinical Presentation: Poorly controlled blood glucose, elevated HbA1c.

Investigations:

- Best initial: Fasting glucose, HbA1c, and postprandial glucose testing

Treatment Outline:

- First-line: Metformin, lifestyle modifications
- Second-line: Insulin or other oral hypoglycemics if HbA1c remains high

Correct Answer: D. <7.0 ✅

25.
MCQ:
A 43-year-old man presents with headache and neck stiffness. CSF analysis reveals cloudy appearance, low glucose, high protein, and 100 white cells/mm^3 (70% polymorphs). CT brain is normal. Which of the following is the most likely cause of the infection?

A. Cryptococcal
B. Tuberculous
C. Bacterial
D. Viral

Explanation:
The patient's symptoms of headache, neck stiffness, and CSF findings (low glucose, high protein, and a high white cell count with polymorphs) are most consistent with **bacterial meningitis**. The cloudy CSF, elevated white cells, and low glucose point towards a bacterial etiology, rather than viral or fungal.

Bacterial Meningitis

Definition: Infection of the meninges leading to inflammation of the brain and spinal cord coverings, typically caused by bacteria.

Key Clinical Presentation: Fever, headache, neck stiffness, altered mental status.

Investigations:

- Best initial: CSF analysis (low glucose, high protein, high white cells)
- Confirmatory: Blood cultures, CSF cultures for bacteria

Treatment Outline:

- First-line: IV antibiotics (e.g., ceftriaxone, vancomycin)

- Second-line: Dexamethasone for inflammation

Correct Answer: C. Bacterial ✅

26.
MCQ:
A 22-year-old man presents with altered sensorium for the last 2 days and headaches for 5 days, with a history of fever for the preceding month. On examination, there is nucal rigidity. CSF analysis shows: 240 cells/µL (73% lymphocytes), glucose 2.7 mmol/L, protein 3.6 g/L. What is the most likely diagnosis?

A. Septicaemia
B. Tubercular meningitis
C. Pyogenic meningitis
D. Viral meningoencephalitis

Explanation:
The patient's history of fever and headache, along with CSF findings of elevated protein, low glucose, and a lymphocytic predominance, is suggestive of **tubercular meningitis**. This condition typically presents with chronic symptoms and the CSF analysis shows a high protein level and low glucose with lymphocytic predominance.

Tubercular Meningitis

Definition: A form of meningitis caused by Mycobacterium tuberculosis, often associated with prolonged fever and neurological symptoms.

Key Clinical Presentation: Fever, headache, neck stiffness, altered mental status, with a prolonged course.

Investigations:

- Best initial: CSF analysis (low glucose, high protein, lymphocytic predominance)
- Confirmatory: Mycobacterial culture or PCR from CSF

Treatment Outline:

- First-line: Anti-tuberculosis therapy (e.g., rifampin, isoniazid, pyrazinamide, ethambutol)
- Second-line: Corticosteroids to reduce inflammation

Correct Answer: B. Tubercular meningitis ✅

26.
MCQ:
A 29-year-old woman comes to the Emergency Department with vomiting, profuse watery diarrhea, and abdominal cramps. The symptoms started within 4 hours of eating steak at a local restaurant. What is the most likely cause?

A. Escherichia coli
B. Salmonella spp.
C. Campylobacter jejuni
D. Staphylococcus aureus

Explanation:
The rapid onset of vomiting and diarrhea after eating food contaminated with **Staphylococcus aureus** is a classic presentation of food poisoning caused by preformed toxins in the food. The rapid onset (within 4 hours) of symptoms points towards this pathogen.

Staphylococcal Food Poisoning

Definition: Gastroenteritis caused by ingestion of food contaminated with enterotoxins produced by Staphylococcus aureus.

Key Clinical Presentation: Vomiting, diarrhea, abdominal cramps, usually within 2-6 hours of eating contaminated food.

Investigations:

- Best initial: Clinical diagnosis
- Confirmatory: Stool culture or toxin detection

Treatment Outline:

- First-line: Supportive care (hydration, antiemetics)
- Second-line: Antibiotics if secondary infection is suspected

Correct Answer: D. Staphylococcus aureus ✅

27.
MCQ:
A 47-year-old patient presents with 12 hours of severe pain and swelling of the right first metatarsophalangeal joint. On examination, the affected joint is erythematous and tender.

Blood pressure 110/65 mmHg, heart rate 105/min, respiratory rate 12/min, temperature 38°C. Which of the following is the most likely cause?

A. Sodium urate crystals
B. Staphylococcus aureus
C. Disseminated gonorrhea
D. Calcium pyrophosphate crystals

Explanation:
The patient's acute onset of pain, erythema, and tenderness in the first metatarsophalangeal joint, with the characteristic rapid onset, is typical of **gout**, which is caused by the deposition of sodium urate crystals.

Gout

Definition: A form of inflammatory arthritis caused by the deposition of uric acid crystals in the joints, leading to sudden and severe joint pain.

Key Clinical Presentation: Acute pain, erythema, and swelling in the big toe or other joints, often with a history of hyperuricemia.

Investigations:

- Best initial: Joint aspiration and polarized light microscopy showing negatively birefringent urate crystals
- Confirmatory: Serum uric acid levels (although not always diagnostic)

Treatment Outline:

- First-line: NSAIDs, colchicine
- Second-line: Corticosteroids for refractory cases

Correct Answer: A. Sodium urate crystals ✅

28.
MCQ:
A 32-year-old woman complains of severe joint pain in the fingers of both hands, especially in the morning during the winter months. Examination confirms deformity of her fingers. What is the most likely diagnosis?

A. Gout
B. Hemophilia

C. Rheumatoid arthritis
D. Primary degenerative osteoarthritis

Explanation:
The patient's presentation of joint pain, especially in the morning with deformities in the fingers, is consistent with **rheumatoid arthritis**. This condition often presents with symmetrical joint involvement, particularly in the hands, and is worse in the morning.

Rheumatoid Arthritis

Definition: A chronic autoimmune disease characterized by inflammation of the joints, especially in the hands and feet.

Key Clinical Presentation: Morning stiffness, symmetrical joint pain, deformities, and swelling.

Investigations:

- Best initial: Rheumatoid factor, anti-CCP antibodies
- Confirmatory: X-rays showing joint erosions, synovial fluid analysis

Treatment Outline:

- First-line: Methotrexate, NSAIDs
- Second-line: Biologic agents (e.g., TNF inhibitors) for refractory disease

Correct Answer: C. Rheumatoid arthritis ✅

29.
MCQ:
A 74-year-old woman complains of deep, aching pain in her hip, aggravated by walking. The pain keeps her awake at night and is stiff for several hours in the morning. What is the most likely cause of the pain?

A. Depression
B. Osteomyelitis
C. Osteoporosis
D. Osteoarthritis

Explanation:
The patient's presentation of hip pain aggravated by walking, particularly at night, with morning stiffness, is most consistent with **osteoarthritis**, which commonly affects the hip joint in older adults.

Osteoarthritis

Definition: A degenerative joint disease characterized by the breakdown of articular cartilage, leading to pain and stiffness.

Key Clinical Presentation: Joint pain, stiffness, and decreased range of motion, particularly in weight-bearing joints like the hips and knees.

Investigations:

- Best initial: X-ray showing joint space narrowing and osteophytes
- Confirmatory: MRI if needed for soft tissue assessment

Treatment Outline:

- First-line: NSAIDs for pain management, physical therapy
- Second-line: Joint replacement in advanced cases

Correct Answer: D. Osteoarthritis ✅

30.
Question:
A 60-year-old woman complains of pain in her right knee for the last 3 years. The pain is worse with activity and relieved by rest and analgesics. Physical examination reveals a varus knee deformity and tenderness over the medial joint line. What is the most likely diagnosis?

Explanation:
The patient's chronic knee pain, relieved by rest and analgesics, along with varus deformity and tenderness over the medial joint line, is most consistent with **osteoarthritis**. This condition is common in older adults and presents with joint pain that worsens with activity and improves with rest.

Osteoarthritis

Definition: A degenerative joint disease characterized by the breakdown of articular cartilage.

Key Clinical Presentation: Joint pain, stiffness, and decreased range of motion, particularly in weight-bearing joints like the knees.

Investigations:

- Best initial: X-ray showing joint space narrowing, osteophytes, and varus deformity.
- Confirmatory: MRI if needed.

Treatment Outline:

- First-line: NSAIDs for pain relief, physical therapy
- Second-line: Joint replacement in advanced cases

Correct Answer: D. Osteoarthritis ✅

31.
Question:
A 58-year-old man presents to the clinic with severe pain, swelling, and redness of the first metatarsophalangeal joint. Joint fluid is withdrawn and shows yellow, opaque color with 100,000 WBCs per mm^3 (75% PMNs). Polarized light microscopy reveals negatively birefringent urate crystals. Which treatment should be avoided?

Explanation:
This is a typical case of **gout** characterized by the presence of urate crystals in the joint fluid. **Allopurinol** should not be used during an acute gout attack as it can precipitate further flare-ups. The preferred treatment for acute gout involves NSAIDs or colchicine.

Gout

Definition: A form of inflammatory arthritis caused by uric acid crystal deposition in joints.

Key Clinical Presentation: Acute, severe pain, swelling, and erythema in the big toe or other joints.

Investigations:

- Best initial: Joint aspiration showing urate crystals under polarized light.
- Confirmatory: Serum uric acid levels (although not always diagnostic).

Treatment Outline:

- First-line: NSAIDs, colchicine
- Second-line: Corticosteroids for refractory cases

Correct Answer: C. Allopurinol ✅

32.
Question:
Which of the following crystals is responsible for the arthritic symptoms of pseudogout?

Explanation:
The cause of pseudogout is the deposition of **calcium pyrophosphate dihydrate (CPPD)** crystals in the joints, which lead to symptoms similar to gout.

Pseudogout

Definition: A form of arthritis caused by the deposition of calcium pyrophosphate crystals in the joints.

Key Clinical Presentation: Acute pain, swelling, and erythema, commonly affecting the knee or wrist.

Investigations:

- Best initial: Joint aspiration showing CPPD crystals under polarized light.
- Confirmatory: X-rays showing chondrocalcinosis.

Treatment Outline:

- First-line: NSAIDs, colchicine
- Second-line: Corticosteroids for severe cases

Correct Answer: D. Calcium pyrophosphate ✅

33.
Question:
A young man presents with intermittent hemoptysis for 2 months, blood in his urine, weight gain, and "puffy eyes." Serum creatinine is elevated. Lung biopsy shows serum immunoglobulin on basement membranes. What is the most likely diagnosis?

Explanation:
The combination of hemoptysis, blood in the urine, weight gain, "puffy eyes," and elevated creatinine suggests **Goodpasture syndrome**, a rare autoimmune disorder where antibodies attack the basement membranes in the lungs and kidneys, leading to pulmonary hemorrhage and glomerulonephritis.

Goodpasture Syndrome

Definition: A rare autoimmune disease characterized by the presence of anti-glomerular basement membrane antibodies, affecting the lungs and kidneys.

Key Clinical Presentation: Hemoptysis, hematuria, proteinuria, kidney failure, and pulmonary symptoms.

Investigations:

- Best initial: Anti-glomerular basement membrane antibody test
- Confirmatory: Kidney biopsy showing linear IgG deposition along the basement membrane.

Treatment Outline:

- First-line: Plasmapheresis, corticosteroids, immunosuppressive drugs
- Second-line: Cyclophosphamide in severe cases

Correct Answer: B. Goodpasture syndrome ✅

33.
Question:
A 45-year-old man presents with difficulty breathing and nocturnal cough for the past 3 weeks. Spirometry shows a lower-than-normal FEV/FVC ratio. What is consistent with these findings?

Explanation:
A reduced FEV/FVC ratio is characteristic of **obstructive lung disease**, such as chronic obstructive pulmonary disease (COPD) or asthma. The patient's nocturnal cough and difficulty breathing suggest **asthma** or **COPD**.

Chronic Obstructive Pulmonary Disease (COPD)

Definition: A progressive lung disease characterized by airflow limitation, commonly due to smoking.

Key Clinical Presentation: Chronic cough, shortness of breath, and wheezing, especially on exertion.

Investigations:

- Best initial: Spirometry showing a reduced FEV/FVC ratio
- Confirmatory: Chest X-ray or CT scan to assess lung damage.

Treatment Outline:

- First-line: Bronchodilators (beta-agonists, anticholinergics)
- Second-line: Inhaled corticosteroids for exacerbations

Correct Answer: D. Increased airway resistance ✅

34.
Question:
What is the optimal duration of anticoagulation following an initial episode of pulmonary embolism?

Explanation:
The optimal duration of anticoagulation for **pulmonary embolism** (PE) is typically **3 months** for a first episode, particularly if there is no underlying long-term risk factor.

Pulmonary Embolism

Definition: A blockage of a pulmonary artery or its branches by a clot, leading to impaired blood flow to the lungs.

Key Clinical Presentation: Sudden onset of shortness of breath, chest pain, and sometimes hemoptysis.

Investigations:

- Best initial: CT pulmonary angiography or V/Q scan
- Confirmatory: Pulmonary angiography (rarely performed now)

Treatment Outline:

- First-line: Anticoagulation with heparin or warfarin
- Second-line: Thrombolytic therapy for massive PE

Correct Answer: B. 3 months ✅

35.
Question:
A 38-year-old woman with pleurisy requires a thoracentesis. The needle should be placed on the midaxillary line of the affected side. Which rib level should the needle penetrate through?

Explanation:
The needle for thoracentesis should be inserted between the 6th and 7th ribs at the midaxillary line to avoid injury to vital structures while ensuring adequate fluid drainage.

Key Exam Points:

- Best initial test: Chest X-ray to locate the fluid accumulation.
- Diagnostic test: Thoracentesis to analyze pleural fluid.

Correct Answer: C. Between the 6th and 7th rib ✅

36.
Question:
A 72-year-old man presents with fever, night sweats, and cough for 2 weeks. Chest X-ray shows right upper lobe consolidation. What is the best next step in management?

Explanation:
The patient's symptoms and chest X-ray findings suggest a possible infectious cause, most likely **tuberculosis (TB)**. The next step is to **isolate the patient** in a single, negative pressure room to prevent transmission and to proceed with diagnostic tests.

Key Exam Points:

- Best initial test: Sputum smear for acid-fast bacilli (AFB).
- Confirmatory test: Sputum culture for Mycobacterium tuberculosis.

Correct Answer: D. Isolate the patient in a single, negative pressure room ✅

37.
Question:
A 55-year-old patient with chronic obstructive pulmonary disease presents with severe respiratory distress and hypoxemia. The pH is 7.3 with no improvement after bronchodilators. What is the best next step in management?

Explanation:
This patient with COPD in severe respiratory distress and a low pH (indicating respiratory acidosis) should be treated with **non-invasive positive pressure ventilation (NIPPV)** to improve oxygenation and reduce the work of breathing.

Key Exam Points:

- First-line treatment: NIPPV for acute exacerbations of COPD.
- Diagnostic test: Arterial blood gases to monitor progress.

Correct Answer: D. Non-invasive positive pressure ventilation ✅

38.
Question:
A 28-year-old patient with mild intermittent asthma has daytime symptoms 1-2 times a week. Which of the following should be given when necessary?

Explanation:
For mild intermittent asthma, **short-acting beta-2 agonists (SABA)** like salbutamol are the first-line treatment for symptom relief during episodes.

Key Exam Points:

- First-line treatment for intermittent asthma: Inhaled short-acting beta-2 agonist.
- Follow-up treatment: Consider inhaled corticosteroids if symptoms increase.

Correct Answer: A. Inhaled short-acting beta-2 agonist ✅

39.
Question:
A 55-year-old man with a history of smoking presents with shortness of breath and a productive cough. Chest X-ray is normal. Which of the following is the most likely diagnosis?

Explanation:
Given the patient's smoking history and chronic symptoms, **chronic obstructive pulmonary disease (COPD)** is the most likely diagnosis, despite the normal chest X-ray. It is a common cause of chronic cough and dyspnea.

Key Exam Points:

- First-line treatment: Inhaled bronchodilators.
- Diagnostic test: Spirometry to assess airflow obstruction.

Correct Answer: D. Chronic obstructive pulmonary disease ✅

40.

Question:
A 25-year-old patient presents with chest pain and dyspnea. Chest X-ray shows absent lung markings on the left side. What is the most likely diagnosis?

Explanation:
The findings of absent lung markings and a clinical presentation of chest pain and dyspnea are consistent with a **spontaneous pneumothorax**, a condition where air enters the pleural space, causing lung collapse.

Key Exam Points:

- Best initial test: Chest X-ray.
- Confirmatory test: CT scan for detailed assessment if needed.

Correct Answer: A. Spontaneous pneumothorax ✅

41.

Question:
A 45-year-old man with a history of a chest infection presents with a pleural effusion. Chest X-ray shows right-sided pleural effusion. What is the most likely diagnosis?

Explanation:
A **para-pneumonic effusion** is the most likely diagnosis, which occurs as a complication of pneumonia, typically presenting with pleuritic chest pain, fever, and dyspnea.

Key Exam Points:

- First-line treatment: Antibiotics for pneumonia.
- Diagnostic test: Thoracentesis to analyze the pleural fluid.

Correct Answer: D. Para-pneumonic effusion ✅

42.

Question:
A 55-year-old man presents with a productive cough, yellow sputum, and blood-stained sputum. Chest X-ray shows a dense infiltrate in the right lower lobe. What is the most likely diagnosis?

Explanation:
This patient's symptoms and chest X-ray findings are consistent with **community-acquired pneumonia** (CAP), a common infection, particularly in smokers, with consolidation seen on chest X-ray.

Key Exam Points:

- First-line treatment: Empiric antibiotics based on local resistance patterns.
- Diagnostic test: Sputum culture to identify the pathogen.

Correct Answer: A. Community-acquired pneumonia ✅

43.
Question:
A 27-year-old woman with asthma uses a rescue inhaler at least once a day. What should be added to her treatment?

Explanation:
For patients with moderate persistent asthma who require daily use of a rescue inhaler, a **long-acting beta-2 agonist (LABA)** should be added to the regimen to provide better symptom control.

Key Exam Points:

- First-line treatment: Inhaled corticosteroids (ICS).
- Add-on therapy: LABAs or leukotriene inhibitors for better control.

Correct Answer: C. Long-acting beta-2 agonist to steroid regimen ✅

44.
Question:
A 42-year-old man with asthma presents with daily symptoms despite using a long-acting β-agonist and high-dose corticosteroid. What is the most appropriate next step?

Explanation:
For poorly controlled asthma, despite inhaled corticosteroids and LABAs, the addition of a **leukotriene inhibitor** is appropriate to improve control and reduce inflammation.

Key Exam Points:

- First-line treatment: Inhaled corticosteroids.
- Add-on therapy: Leukotriene inhibitors or biologics for severe asthma.

Correct Answer: B. Add a leukotriene inhibitor ✅

45.
Question:
A 67-year-old patient presents with chest pain, shortness of breath, and hemoptysis following a long car trip. What is the most likely diagnosis?

Explanation:
The most likely diagnosis is **pulmonary embolism (PE)**, given the patient's history of immobility, chest pain, dyspnea, and hemoptysis. PE is a common complication of prolonged immobility.

Key Exam Points:

- First-line treatment: Anticoagulation with heparin or warfarin.
- Diagnostic test: CT pulmonary angiography.

Correct Answer: C. Pulmonary embolism ✅

46.

MCQ

45-year-old nurse working in the pulmonary clinic complains of fever and a productive cough for the last 3 months. She has received 2 courses of oral antibiotics, but showed no improvement. Which of the following is the most likely diagnosis?

A. Pulmonary tuberculosis
B. H. Influenza pneumonia
C. Infectious mononucleosis
D. Streptococcal pneumonia

Explanation:

The patient's symptoms of prolonged fever and a productive cough lasting for 3 months with no improvement despite antibiotic therapy are strongly suggestive of **pulmonary tuberculosis (TB)**. TB is a chronic infection that often presents with these nonspecific symptoms, particularly in healthcare workers who are at higher risk for exposure. A prolonged, unresponsive cough, especially in the context of a healthcare setting, is characteristic of TB.

Pulmonary Tuberculosis (TB)

Definition: A bacterial infection caused by *Mycobacterium tuberculosis*, primarily affecting the lungs but can spread to other organs.

Key Clinical Presentation: Persistent cough, hemoptysis, fever, night sweats, weight loss, and fatigue. Risk factors include healthcare workers, immunocompromised individuals, and exposure to TB patients.

Investigations:

- **Best initial:** Chest X-ray (can show infiltrates, cavitations, or fibrosis).
- **Confirmatory:** Sputum culture for *Mycobacterium tuberculosis*, or sputum acid-fast bacillus (AFB) smear.

Treatment Outline:

- **First-line:** Rifampin, Isoniazid, Pyrazinamide, and Ethambutol for 6-9 months (RIPE regimen)
- **Second-line:** Second-line drugs for drug-resistant TB (if necessary)
- **Definitive:** Long-term antimicrobial therapy and follow-up with sputum cultures.

Correct Answer: A. Pulmonary tuberculosis ✅

47.

A 55-year-old patient presents to the clinic with shortness of breath and a cough productive of yellow sputum for the past 2 days. There has also been blood-stained sputum for the last 3 hours. The patient has smoked 2 packs of cigarettes a day for the past 35 years. On examination, there are rhonchi, wheezing, and crepitations heard over the right hemothorax.

Blood pressure: 110/70 mmHg
Heart rate: 84 /min
Respiratory rate: 22 /min
Temperature: 38.8°C
Oxygen saturation: 95% on room air
Chest X-ray: It shows a dense infiltrate in the right lower lobe.

Which of the following is the most likely diagnosis?

A. Community acquired pneumonia
B. Acute bronchitis
C. Tuberculosis
D. COPD

Explanation:

The patient's presentation of shortness of breath, cough with yellow and blood-stained sputum, and the chest X-ray showing a dense infiltrate in the right lower lobe is most consistent with **community acquired pneumonia (CAP)**. CAP is a bacterial infection of the lung parenchyma, typically presenting with fever, productive cough, and infiltrates on chest X-ray. The patient's significant smoking history increases the risk of respiratory infections like pneumonia.

Community Acquired Pneumonia (CAP)

Definition: Acute infection of the lung parenchyma, typically caused by bacteria, in individuals who acquire the infection outside of a healthcare setting.

Key Clinical Presentation: Productive cough, fever, pleuritic chest pain, dyspnea, blood-stained sputum, and abnormal breath sounds (rhonchi, crepitations).

Investigations:

- **Best initial:** Chest X-ray (shows consolidation or infiltrates).
- **Confirmatory:** Sputum culture or blood culture to identify the causative organism.

Treatment Outline:

- **First-line:** Empiric antibiotic therapy, including a macrolide (e.g., azithromycin) or fluoroquinolone (e.g., levofloxacin) for typical pathogens like *Streptococcus pneumoniae*.
- **Second-line:** Consider broader spectrum antibiotics if hospitalized or suspected resistant organisms.
- **Definitive:** Tailored therapy based on culture results.

Correct Answer: A. Community acquired pneumonia ✅

48.

MCQ:

A 27-year-old woman with asthma has not had a severe attack for the last 3 months. However, she has a history of cough and wheezing 2-3 times each week.

Which of the following is the most appropriate management?

A. Ipratropium bromide
B. Inhaled corticosteroid
C. Salbutamol
D. Salmeterol

Explanation:

For a patient with asthma who experiences intermittent symptoms such as cough and wheezing 2-3 times per week but does not have severe attacks, the **most appropriate management** is an **inhaled corticosteroid** (option B). Inhaled corticosteroids (ICS) are the cornerstone of asthma management for patients with persistent symptoms, as they reduce inflammation and prevent symptom exacerbation. ICS are used for daily control of asthma symptoms and help reduce the frequency of attacks.

Asthma Management

Definition: Chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Wheezing, cough, shortness of breath, and chest tightness, often exacerbated by triggers like allergens, exercise, or respiratory infections.

Investigations:

- **Best initial:** Spirometry or peak flow measurement to assess lung function.
- **Confirmatory:** Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge test).

Treatment Outline:

- **First-line:** Inhaled corticosteroids for daily control of inflammation (e.g., fluticasone, budesonide).
- **Second-line:** Long-acting beta-agonists (LABAs) added if needed for additional symptom control.
- **Rescue (for acute symptoms):** Short-acting beta-agonists (e.g., salbutamol) for quick relief during exacerbations.

Correct Answer: B. Inhaled corticosteroid ✅

49.

A 21-year-old woman comes to the clinic with episodes of shortness of breath on exertion for one year. She has to stop after 15-20 minutes of exercise, because of dyspnoea and coughing, which get better after about one hour of rest. There are no symptoms at rest. Physical examination and spirometry are normal.

Blood pressure 125/75 mmHg
Heart rate 80 /min
Respiratory rate 12 /min
Temperature 37.0° C

Which of the following is the most appropriate inhaled treatment before exercise?

A. Salmeterol
B. Salbutamol
C. Budesonide
D. Ipratropium

Explanation:

This patient presents with **exercise-induced asthma** (EIA), characterized by shortness of breath and cough triggered by exercise, which improves with rest. The most appropriate inhaled treatment to prevent symptoms before exercise is a **short-acting beta-agonist (SABA)** like **salbutamol** (option B). Salbutamol is commonly used as a **rescue inhaler** and can be used 15-30 minutes before exercise to prevent bronchoconstriction.

Exercise-Induced Asthma

Definition: A form of asthma triggered by physical activity, leading to reversible airway obstruction, wheezing, cough, and shortness of breath during or after exercise.

Key Clinical Presentation: Dyspnoea and coughing during exercise, relieved by rest, with normal symptoms at rest.

Investigations:

- **Best initial:** Spirometry or peak flow measurement during or after exercise.
- **Confirmatory:** Methacholine challenge if needed.

Treatment Outline:

- **First-line:** Short-acting beta-agonists (e.g., salbutamol) used before exercise to prevent bronchoconstriction.
- **Second-line:** Long-acting beta-agonists (e.g., salmeterol) or leukotriene inhibitors for persistent symptoms.

Correct Answer: B. Salbutamol ✅

50.

A 42-year-old man with asthma presents with daily symptoms despite regularly using a long-acting β-agonist inhaler and an inhaled high-dose corticosteroid. He also uses a short-acting β-agonist inhaler as "rescue" medication. These episodes also wake him from sleep 2-3 times a week. Physical examination shows bilateral wheezing.

Blood pressure 140/85 mmHg
Heart rate 90 /min
Respiratory rate 18 /min
Temperature 36.7° C
Forced Expiratory Volume in 1 Second:
Before bronchodilator: 65% (25-75%)
After bronchodilators: 91%

Which of the following is the most appropriate next step in management?

A. Add theophylline
B. Add a leukotriene inhibitor
C. Observe the patient using his inhaler
D. Treat for gastroesophageal reflux disease

Explanation:

This patient has poorly controlled asthma, as evidenced by daily symptoms, nighttime awakenings, and an FEV1 of 65% (indicating moderate asthma). The most appropriate next step is to **add a leukotriene inhibitor** (option B). Leukotriene inhibitors (e.g., montelukast) are effective for patients with asthma that is not adequately controlled by inhaled corticosteroids and long-acting beta-agonists. These medications can help reduce inflammation and improve asthma control, particularly for patients with nighttime symptoms.

Asthma Management

Definition: Chronic inflammatory disorder of the airways, leading to episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Daily symptoms, nocturnal awakenings, use of short-acting beta-agonists for relief, and spirometry showing reduced FEV1.

Investigations:

- **Best initial:** Spirometry for lung function.

- **Confirmatory:** Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge).

Treatment Outline:

- **First-line:** Inhaled corticosteroids (ICS) and long-acting beta-agonists (LABAs).
- **Second-line:** Leukotriene inhibitors, theophylline, or biologic agents for more severe asthma.
- **Rescue:** Short-acting beta-agonists (SABA) for acute symptom relief.

Correct Answer: B. Add a leukotriene inhibitor ✅

51:

A 67-year-old patient presents to the Emergency Department with chest pain, shortness of breath, and haemoptysis following a car trip for more than 5 hours (see report). ECG: Right axis deviation, an S1-Q3-T3 pattern, and a right bundle branch block.

Which of the following is the most likely diagnosis?

A. Acute myocardial infarction
B. Bronchogenic carcinoma
C. Pulmonary embolism
D. Pericarditis

Explanation:

The patient's clinical presentation with chest pain, shortness of breath, haemoptysis, and prolonged immobility (after a long car trip) raises suspicion for **pulmonary embolism (PE)** (option C). The ECG showing **S1-Q3-T3 pattern** and **right axis deviation** are classic signs of PE. A right bundle branch block may also be seen in PE due to strain on the right side of the heart from obstructed pulmonary circulation.

Pulmonary Embolism (PE)

Definition: Blockage of the pulmonary arteries by a thrombus, often originating from deep vein thrombosis (DVT).

Key Clinical Presentation: Sudden onset of chest pain, shortness of breath, haemoptysis, tachycardia, hypoxia, and signs of right heart strain (e.g., S1-Q3-T3 pattern on ECG).

Investigations:

- **Best initial:** CT pulmonary angiography (CTPA)

- **Confirmatory:** Pulmonary angiography (gold standard), D-dimer

Treatment Outline:

- **First-line:** Anticoagulation with heparin or low molecular weight heparin (LMWH)
- **Second-line:** Thrombolysis or surgical embolectomy for massive PE
- **Definitive:** Ongoing anticoagulation for 3–6 months, depending on risk factors.

Correct Answer: C. Pulmonary embolism ✅

MCQ 52:

A 55-year-old smoker presents to a clinic with progressive shortness of breath (see report). Chest X-ray: It reveals a massive right pleural effusion.

Which of the following is the best next step in management?

A. Thoracentesis
B. High resolution CT scan
C. Intercostal tube under water seal
D. Repeated bronchodilator inhalations

Explanation:

The presence of a **massive pleural effusion** in a smoker raises concern for potential **malignant pleural effusion**, which could be caused by lung cancer or other malignancies. The best next step in management is to perform **thoracentesis** (option A), which allows for both diagnostic and therapeutic relief. Thoracentesis will help determine the etiology of the effusion by analyzing the pleural fluid (e.g., cytology for malignancy, cell count, pH, and other markers).

Pleural Effusion

Definition: Accumulation of fluid in the pleural space, which can be transudative or exudative, depending on the underlying cause.

Key Clinical Presentation: Progressive shortness of breath, pleuritic chest pain, decreased breath sounds, dullness to percussion, and egophony on examination.

Investigations:

- **Best initial:** Chest X-ray to identify the effusion.

- **Confirmatory:** Thoracentesis to obtain pleural fluid for analysis (cell count, protein, LDH, pH, and cytology).

Treatment Outline:

- **First-line:** Thoracentesis for diagnostic and therapeutic purposes
- **Second-line:** Chest tube drainage or pleurodesis if the effusion is recurrent or malignant
- **Definitive:** Management of the underlying cause (e.g., chemotherapy for malignancy, diuretics for heart failure).

Correct Answer: A. Thoracentesis ✅

MCQ 53:

A 28-year-old patient develops the sudden onset of dyspnoea and tachycardia 5 days post knee surgery.

Which of the following is the most likely diagnosis?

A. Pulmonary embolism
B. Myocardial infarction
C. Ventricular tachycardia
D. Pulmonary emphysema

Explanation:

The patient's **sudden onset of dyspnoea** and **tachycardia** 5 days post-surgery, especially knee surgery, raises strong suspicion for **pulmonary embolism (PE)** (option A). PE commonly occurs in post-surgical patients, particularly after orthopedic surgeries like knee replacements, due to deep vein thrombosis (DVT) embolizing to the lungs. This condition can present with shortness of breath, tachycardia, and other signs of right heart strain.

Pulmonary Embolism (PE)

Definition: Blockage of a pulmonary artery or one of its branches by a clot, typically arising from a deep vein thrombosis.

Key Clinical Presentation: Sudden onset of dyspnoea, tachycardia, pleuritic chest pain, hypoxia, and, if severe, signs of right heart failure.

Investigations:

- **Best initial:** D-dimer, followed by CT pulmonary angiography (CTPA) for diagnosis.
- **Confirmatory:** Pulmonary angiography (gold standard).

Treatment Outline:

- **First-line:** Anticoagulation therapy with heparin or low molecular weight heparin (LMWH).
- **Second-line:** Thrombolytic therapy or surgical embolectomy in cases of massive PE.
- **Definitive:** Long-term anticoagulation therapy depending on the cause.

Correct Answer: A. Pulmonary embolism ✅

MCQ 54:

A patient with stage IIIB small cell lung cancer complains of new onset of back pain for the past 12 hours. Initial neurological examination is normal.

Which of the following is the most appropriate immediate action to take?

A. MRI
B. Radiation therapy
C. Take no immediate action
D. High dose IV corticosteroids and MRI

Explanation:

A patient with **small cell lung cancer** (SCLC) presenting with new **back pain** raises suspicion for **spinal cord compression** due to metastasis (which is common in lung cancer). The most appropriate immediate action is to administer **high dose IV corticosteroids** to reduce inflammation and prevent further damage to the spinal cord, followed by **MRI** to confirm the diagnosis. Early intervention is critical to prevent permanent neurological damage.

Spinal Cord Compression in Cancer

Definition: Compression of the spinal cord due to tumor growth, typically seen in patients with metastatic cancer.

Key Clinical Presentation: Back pain, sensory loss, motor deficits, bladder/bowel dysfunction, and worsening neurological symptoms.

Investigations:

- **Best initial:** MRI of the spine
- **Confirmatory:** MRI with gadolinium to assess the extent of compression.

Treatment Outline:

- **First-line:** High dose corticosteroids (e.g., dexamethasone) to reduce swelling and prevent further neurological damage.
- **Second-line:** Radiation therapy for metastatic tumors compressing the spinal cord.
- **Definitive:** Surgery if needed for decompressing the spine in certain cases.

Correct Answer: D. High dose IV corticosteroids and MRI ✅

MCQ 55:

On a follow-up consultation of a patient with squamous cell carcinoma, some abnormalities are noted in blood analysis (see lab results).

Test Result:

- **Parathyroid hormone**: 1.0 (Normal: 1.1-5.3 pmol/L)
- **Calcium**: 3.30 (Normal: 2.15-2.62 mmol/L)
- **Calcium ionised**: 1.6 (Normal: 1.1-1.3 mmol/L)
- **Sodium**: 140 (Normal: 134-146 mmol/L)
- **Potassium**: 4.3 (Normal: 3.5-5.1 mmol/L)

Which of the following is the most likely cause of the metabolic abnormality?

A. Renal failure
B. Primary hyperparathyroidism
C. Secondary hyperparathyroidism
D. Secretion of PTH-related peptide by the tumour

Explanation:

The patient's **elevated calcium levels** (both total and ionized) in the context of **low parathyroid hormone (PTH)** levels strongly suggest **secretion of PTH-related peptide (PTHrP) by the tumour** (option D). This is commonly seen in **squamous cell carcinoma**, particularly in the lungs, where PTHrP secretion leads to **hypercalcemia of malignancy**. PTHrP mimics the action of PTH, causing increased calcium reabsorption from bones and the kidneys, but it does not lead to increased PTH production.

Hypercalcemia of Malignancy due to PTHrP

Definition: Hypercalcemia resulting from the secretion of PTH-related peptide by tumors, leading to elevated calcium levels without an increase in PTH levels.

Key Clinical Presentation: Weight loss, fatigue, confusion, dehydration, polyuria, and polydipsia in patients with cancer.

Investigations:

- **Best initial:** Serum calcium and PTH levels
- **Confirmatory:** Measurement of PTH-related peptide (PTHrP)

Treatment Outline:

- **First-line:** IV fluids, bisphosphonates (e.g., pamidronate), or denosumab
- **Second-line:** Corticosteroids for specific types of malignancy-related hypercalcemia.
- **Definitive:** Treatment of the underlying malignancy.

Correct Answer: D. Secretion of PTH-related peptide by the tumour ✅

MCQ 56:

A 78-year-old man, a current smoker with a 50 pack-year smoking history, complains of shortness of breath, dry cough, and fatigue progressing over the past 8 weeks. The patient also reports a significant weight loss, corresponding to approximately 15% of body weight over the last 12 weeks. Physical examination is normal.

Which of the following is the most likely diagnosis?

A. COPD
B. Pneumonia
C. Lung cancer
D. Interstitial lung disease

Explanation:

The patient's **weight loss** (15% of body weight) alongside **progressive shortness of breath**, **dry cough**, and his significant smoking history points towards **lung cancer** (option C). Smoking is the primary risk factor for **lung cancer**, and the symptoms described are common in patients with **non-small cell lung cancer** (NSCLC), especially **squamous cell carcinoma**, which often presents with cough, weight loss, and fatigue. The lack of acute symptoms (such as fever or purulent sputum) makes **pneumonia** less likely.

Lung Cancer

Definition: A malignancy arising from the epithelial cells of the lung, commonly associated with smoking.

Key Clinical Presentation: Chronic cough, weight loss, hemoptysis, dyspnoea, chest pain, and history of smoking.

Investigations:

- **Best initial:** Chest X-ray, followed by CT scan of the chest
- **Confirmatory:** Biopsy of the lesion (either via bronchoscopy or percutaneous biopsy)

Treatment Outline:

- **First-line:** Surgical resection if localized; chemotherapy and/or radiation for advanced disease.
- **Second-line:** Targeted therapy (e.g., EGFR inhibitors, ALK inhibitors) for specific mutations.
- **Definitive:** Multidisciplinary approach including oncology, surgery, and radiation.

Correct Answer: C. Lung cancer ✅

MCQ 56:

A 20-year-old man with sickle cell anemia presents with right-sided chest pain and shortness of breath for 2 days. It was associated with pain in both legs and back. On examination, fine crackles were heard in the right middle and lower zones.

Blood pressure: 125/75 mmHg
Heart rate: 92/min
Respiratory rate: 25/min
Temperature: 37.9°C
Oxygen saturation: 91% on room air

Which of the following is the most likely diagnosis?

A. Viral pericarditis
B. Pulmonary embolism
C. Acute chest syndrome
D. Pneumocystis carinii pneumonia

Explanation:

The most likely diagnosis is **acute chest syndrome (ACS)** (option C). ACS is a common and life-threatening complication of **sickle cell disease** characterized by chest pain, hypoxia, fever, and a new infiltrate on chest imaging. It is often triggered by infection or pulmonary infarction. This patient's symptoms, including chest pain, shortness of breath, fine crackles, and the sickle cell disease background, are classic for ACS. The presence of pain in both legs and back suggests a possible vaso-occlusive crisis, which can exacerbate ACS.

Acute Chest Syndrome (ACS)

Definition: A common and serious complication of sickle cell disease, characterized by chest pain, fever, hypoxia, and new pulmonary infiltrates on imaging.

Key Clinical Presentation: Chest pain, shortness of breath, fever, hypoxia, new lung infiltrates on chest X-ray, pain in extremities due to vaso-occlusion.

Investigations:

- **Best initial:** Chest X-ray (to identify infiltrates or signs of pulmonary disease)
- **Confirmatory:** Arterial blood gas to assess oxygenation, and blood cultures if infection is suspected.

Treatment Outline:

- **First-line:** Oxygen therapy, pain management, antibiotics (if infection is suspected), blood transfusion to reduce sickling.
- **Second-line:** Exchange transfusion for severe cases.
- **Definitive:** Address the underlying sickle cell disease and prevent recurrence by hydroxyurea.

Correct Answer: C. Acute chest syndrome ✅

MCQ 57:

A 20-year-old man with sickle cell anemia presents with right-sided chest pain and shortness of breath for 2 days. It is associated with pain in both legs and back. On examination, fine crackles are heard in the right middle and lower zones.

Blood pressure: 125/75 mmHg
Heart rate: 92/min
Respiratory rate: 25/min
Temperature: 37.9°C
Oxygen saturation: 91% on room air

What is the most effective drug to reduce this incidence?

A. Folic acid
B. Hydroxyurea
C. 5-azacytidine
D. Erythropoietin

Explanation:

The most effective drug to reduce the incidence of **acute chest syndrome** and vaso-occlusive crises in sickle cell disease is **hydroxyurea (option B)**. Hydroxyurea works by increasing fetal hemoglobin (HbF) production, which reduces the sickling of red blood cells, thereby decreasing the frequency of complications like acute chest syndrome. It has been shown to reduce both the frequency and severity of sickle cell crises, including acute chest syndrome.

Hydroxyurea for Sickle Cell Disease

Definition: Hydroxyurea is a chemotherapeutic agent that increases the production of fetal hemoglobin, reducing the sickling of red blood cells in patients with sickle cell disease.

Key Clinical Presentation: Chronic pain episodes, acute chest syndrome, frequent hospitalizations, and stroke.

Investigations:

- **Best initial:** Blood counts to monitor for leukopenia and other side effects of hydroxyurea.
- **Confirmatory:** Clinical improvement with reduced episodes of crises and acute chest syndrome.

Treatment Outline:

- **First-line:** Hydroxyurea for long-term management to reduce the frequency of sickle cell crises.
- **Second-line:** Blood transfusions and other supportive measures during acute crises.

Correct Answer: B. Hydroxyurea ✅

MCQ 58:

A 78-year-old man presents with a 2-year history of slow and progressive deterioration in his memory. The patient is able to perform some activities of daily living, including dressing and bathing. His personality has changed from that of a kind and caring person into someone who has periods of agitation and aggression. Neurologic and musculoskeletal examination cannot be carried out.

Which of the following is the most likely diagnosis?

A. Mixed dementia
B. Alzheimer's disease
C. Multi-infarct dementia
D. Major depressive disorder

Explanation:

The most likely diagnosis is **Alzheimer's disease (option B)**. The gradual decline in memory, along with personality changes and agitation, are classic signs of Alzheimer's disease. Alzheimer's is the most common cause of dementia in older adults and is characterized by a slow and progressive decline in cognitive functions such as memory, thinking, and reasoning.

Alzheimer's Disease

Definition: A neurodegenerative disorder characterized by progressive cognitive decline, particularly in memory, and changes in behavior.

Key Clinical Presentation: Memory loss, personality changes (e.g., agitation, aggression), and difficulty with activities of daily living.

Investigations:

- **Best initial:** Clinical assessment, cognitive testing (e.g., MMSE), and neuroimaging (MRI or CT) to rule out other causes.
- **Confirmatory:** Post-mortem examination showing amyloid plaques and neurofibrillary tangles.

Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (donepezil, rivastigmine) to improve cognitive function temporarily.
- **Second-line:** NMDA receptor antagonists (memantine) for moderate to severe disease.
- **Definitive:** Supportive care and management of symptoms.

Correct Answer: B. Alzheimer's disease ✅

MCQ 59:

A 46-year-old woman with schizophrenia presents to the clinic with a 1-month history of abnormal orofacial movements. She has been prescribed amisulpride 600mg for many years and procyclidine 5 mg daily.

Which of the following is the most likely cause?

A. Tardive dyskinesia
B. Catatonic phenomenon
C. Cerebrovascular accident

Explanation:

The most likely cause is **tardive dyskinesia (option A)**. Tardive dyskinesia is a common side effect of long-term antipsychotic use, especially with dopamine receptor antagonists like amisulpride. It presents with involuntary, repetitive movements, particularly of the face, tongue, and jaw. The patient's history of taking amisulpride, a typical antipsychotic, increases the likelihood of this diagnosis.

Tardive Dyskinesia

Definition: A movement disorder characterized by involuntary, repetitive movements, particularly in the orofacial region, due to prolonged use of antipsychotic medications.

Key Clinical Presentation: Involuntary movements of the face, tongue, and limbs, usually occurring after prolonged use of antipsychotic medications.

Investigations:

- **Best initial:** Clinical assessment to identify characteristic movements.
- **Confirmatory:** Diagnosis is largely clinical; no specific confirmatory test is available.

Treatment Outline:

- **First-line:** Discontinuation or reduction of the offending antipsychotic medication, and switching to a lower-potency agent.
- **Second-line:** Medications like tetrabenazine or valbenazine, and adjunctive use of vitamin E.
- **Definitive:** Prevention is key by minimizing antipsychotic use and monitoring for early signs.

Correct Answer: A. Tardive dyskinesia ✅

60:

A 67-year-old patient presents to the Emergency Department with chest pain, shortness of breath, and haemoptysis following a car trip for more than 5 hours (see report). ECG: Right axis deviation, an S1-Q3-T3 pattern, and a right bundle branch block.

Which of the following is the most likely diagnosis?

A. Acute myocardial infarction
B. Bronchogenic carcinoma
C. Pulmonary embolism
D. Pericarditis

Explanation:

The patient's clinical presentation with chest pain, shortness of breath, haemoptysis, and prolonged immobility (after a long car trip) raises suspicion for **pulmonary embolism (PE)** (option C). The ECG showing **S1-Q3-T3 pattern** and **right axis deviation** are classic signs of PE. A right bundle branch block may also be seen in PE due to strain on the right side of the heart from obstructed pulmonary circulation.

Pulmonary Embolism (PE)

Definition: Blockage of the pulmonary arteries by a thrombus, often originating from deep vein thrombosis (DVT).

Key Clinical Presentation: Sudden onset of chest pain, shortness of breath, haemoptysis, tachycardia, hypoxia, and signs of right heart strain (e.g., S1-Q3-T3 pattern on ECG).

Investigations:

- **Best initial:** CT pulmonary angiography (CTPA)
- **Confirmatory:** Pulmonary angiography (gold standard), D-dimer

Treatment Outline:

- **First-line:** Anticoagulation with heparin or low molecular weight heparin (LMWH)
- **Second-line:** Thrombolysis or surgical embolectomy for massive PE
- **Definitive:** Ongoing anticoagulation for 3–6 months, depending on risk factors.

Correct Answer: C. Pulmonary embolism ✅

MCQ 61:

A 55-year-old smoker presents to a clinic with progressive shortness of breath (see report). Chest X-ray: It reveals a massive right pleural effusion.

Which of the following is the best next step in management?

A. Thoracentesis
B. High resolution CT scan
C. Intercostal tube under water seal
D. Repeated bronchodilator inhalations

Explanation:

The presence of a **massive pleural effusion** in a smoker raises concern for potential **malignant pleural effusion**, which could be caused by lung cancer or other malignancies. The best next step in management is to perform **thoracentesis** (option A), which allows for both diagnostic and therapeutic relief. Thoracentesis will help determine the etiology of the effusion by analyzing the pleural fluid (e.g., cytology for malignancy, cell count, pH, and other markers).

Pleural Effusion

Definition: Accumulation of fluid in the pleural space, which can be transudative or exudative, depending on the underlying cause.

Key Clinical Presentation: Progressive shortness of breath, pleuritic chest pain, decreased breath sounds, dullness to percussion, and egophony on examination.

Investigations:

- **Best initial:** Chest X-ray to identify the effusion.
- **Confirmatory:** Thoracentesis to obtain pleural fluid for analysis (cell count, protein, LDH, pH, and cytology).

Treatment Outline:

- **First-line:** Thoracentesis for diagnostic and therapeutic purposes
- **Second-line:** Chest tube drainage or pleurodesis if the effusion is recurrent or malignant
- **Definitive:** Management of the underlying cause (e.g., chemotherapy for malignancy, diuretics for heart failure).

Correct Answer: A. Thoracentesis ✅

MCQ 62:

A 28-year-old patient develops the sudden onset of dyspnoea and tachycardia 5 days post knee surgery.

Which of the following is the most likely diagnosis?

A. Pulmonary embolism
B. Myocardial infarction
C. Ventricular tachycardia
D. Pulmonary emphysema

Explanation:

The patient's **sudden onset of dyspnoea** and **tachycardia** 5 days post-surgery, especially knee surgery, raises strong suspicion for **pulmonary embolism (PE)** (option A). PE commonly occurs in post-surgical patients, particularly after orthopedic surgeries like knee replacements, due to deep vein thrombosis (DVT) embolizing to the lungs. This condition can present with shortness of breath, tachycardia, and other signs of right heart strain.

Pulmonary Embolism (PE)

Definition: Blockage of a pulmonary artery or one of its branches by a clot, typically arising from a deep vein thrombosis.

Key Clinical Presentation: Sudden onset of dyspnoea, tachycardia, pleuritic chest pain, hypoxia, and, if severe, signs of right heart failure.

Investigations:

- **Best initial:** D-dimer, followed by CT pulmonary angiography (CTPA) for diagnosis.
- **Confirmatory:** Pulmonary angiography (gold standard).

Treatment Outline:

- **First-line:** Anticoagulation therapy with heparin or low molecular weight heparin (LMWH).
- **Second-line:** Thrombolytic therapy or surgical embolectomy in cases of massive PE.
- **Definitive:** Long-term anticoagulation therapy depending on the cause.

Correct Answer: A. Pulmonary embolism ✅

MCQ 63:

A patient with stage IIIB small cell lung cancer complains of new onset of back pain for the past 12 hours. Initial neurological examination is normal.

Which of the following is the most appropriate immediate action to take?

A. MRI
B. Radiation therapy
C. Take no immediate action
D. High dose IV corticosteroids and MRI

Explanation:

A patient with **small cell lung cancer** (SCLC) presenting with new **back pain** raises suspicion for **spinal cord compression** due to metastasis (which is common in lung cancer). The most appropriate immediate action is to administer **high dose IV corticosteroids** to reduce inflammation and prevent further damage to the spinal cord, followed by **MRI** to confirm the diagnosis. Early intervention is critical to prevent permanent neurological damage.

Spinal Cord Compression in Cancer

Definition: Compression of the spinal cord due to tumor growth, typically seen in patients with metastatic cancer.

Key Clinical Presentation: Back pain, sensory loss, motor deficits, bladder/bowel dysfunction, and worsening neurological symptoms.

Investigations:

- **Best initial:** MRI of the spine
- **Confirmatory:** MRI with gadolinium to assess the extent of compression.

Treatment Outline:

- **First-line:** High dose corticosteroids (e.g., dexamethasone) to reduce swelling and prevent further neurological damage.
- **Second-line:** Radiation therapy for metastatic tumors compressing the spinal cord.
- **Definitive:** Surgery if needed for decompressing the spine in certain cases.

Correct Answer: D. High dose IV corticosteroids and MRI ✅

MCQ 64:

On a follow-up consultation of a patient with squamous cell carcinoma, some abnormalities are noted in blood analysis (see lab results).

Test Result:

- **Parathyroid hormone**: 1.0 (Normal: 1.1-5.3 pmol/L)
- **Calcium**: 3.30 (Normal: 2.15-2.62 mmol/L)
- **Calcium ionised**: 1.6 (Normal: 1.1-1.3 mmol/L)
- **Sodium**: 140 (Normal: 134-146 mmol/L)
- **Potassium**: 4.3 (Normal: 3.5-5.1 mmol/L)

Which of the following is the most likely cause of the metabolic abnormality?

A. Renal failure
B. Primary hyperparathyroidism
C. Secondary hyperparathyroidism
D. Secretion of PTH-related peptide by the tumour

Explanation:

The patient's **elevated calcium levels** (both total and ionized) in the context of **low parathyroid hormone (PTH)** levels strongly suggest **secretion of PTH-related peptide (PTHrP) by the tumour** (option D). This is commonly seen in **squamous cell carcinoma**, particularly in the lungs, where PTHrP secretion leads to **hypercalcemia of malignancy**. PTHrP mimics the action of PTH, causing increased calcium reabsorption from bones and the kidneys, but it does not lead to increased PTH production.

Hypercalcemia of Malignancy due to PTHrP

Definition: Hypercalcemia resulting from the secretion of PTH-related peptide by tumors, leading to elevated calcium levels without an increase in PTH levels.

Key Clinical Presentation: Weight loss, fatigue, confusion, dehydration, polyuria, and polydipsia in patients with cancer.

Investigations:

- **Best initial:** Serum calcium and PTH levels
- **Confirmatory:** Measurement of PTH-related peptide (PTHrP)

Treatment Outline:

- **First-line:** IV fluids, bisphosphonates (e.g., pamidronate), or denosumab
- **Second-line:** Corticosteroids for specific types of malignancy-related hypercalcemia.

- **Definitive:** Treatment of the underlying malignancy.

Correct Answer: D. Secretion of PTH-related peptide by the tumour ✅

MCQ 65:

A 78-year-old man, a current smoker with a 50 pack-year smoking history, complains of shortness of breath, dry cough, and fatigue progressing over the past 8 weeks. The patient also reports a significant weight loss, corresponding to approximately 15% of body weight over the last 12 weeks. Physical examination is normal.

Which of the following is the most likely diagnosis?

A. COPD
B. Pneumonia
C. Lung cancer
D. Interstitial lung disease

Explanation:

The patient's **weight loss** (15% of body weight) alongside **progressive shortness of breath**, **dry cough**, and his significant smoking history points towards **lung cancer** (option C). Smoking is the primary risk factor for **lung cancer**, and the symptoms described are common in patients with **non-small cell lung cancer** (NSCLC), especially **squamous cell carcinoma**, which often presents with cough, weight loss, and fatigue. The lack of acute symptoms (such as fever or purulent sputum) makes **pneumonia** less likely.

Lung Cancer

Definition: A malignancy arising from the epithelial cells of the lung, commonly associated with smoking.

Key Clinical Presentation: Chronic cough, weight loss, hemoptysis, dyspnoea, chest pain, and history of smoking.

Investigations:

- **Best initial:** Chest X-ray, followed by CT scan of the chest
- **Confirmatory:** Biopsy of the lesion (either via bronchoscopy or percutaneous biopsy)

Treatment Outline:

- **First-line:** Surgical resection if localized; chemotherapy and/or radiation for advanced disease.
- **Second-line:** Targeted therapy (e.g., EGFR inhibitors, ALK inhibitors) for specific mutations.
- **Definitive:** Multidisciplinary approach including oncology, surgery, and radiation.

Correct Answer: C. Lung cancer ✅

MCQ 66:

A 20-year-old man with sickle cell anemia presents with right-sided chest pain and shortness of breath for 2 days. It is associated with pain in both legs and back. On examination, fine crackles are heard in the right middle and lower zones.

Blood pressure: 125/75 mmHg
Heart rate: 92/min
Respiratory rate: 25/min
Temperature: 37.9°C
Oxygen saturation: 91% on room air

What is the most effective drug to reduce this incidence?

A. Folic acid **Which of the following is the peak of nicotine withdrawal symptoms after smoking cessation?**

A. 1 day
B. 2 to 4 days
C. 5 to 7 days
D. 8 to 10 days

Explanation:
The peak of nicotine withdrawal symptoms typically occurs **2 to 4 days** after smoking cessation. This is when symptoms such as irritability, anxiety, and cravings for nicotine are at their most intense. Symptoms usually begin to subside after this period.

Nicotine Withdrawal
Definition: A group of symptoms that occur after cessation of nicotine use, typically within hours to days.
Key Clinical Presentation: Irritability, difficulty concentrating, increased appetite, and cravings for nicotine.
Investigations: Diagnosis is clinical based on symptoms after smoking cessation.
Treatment Outline:

- **First-line:** Nicotine replacement therapy (gum, patch, lozenge), bupropion, or varenicline.
- **Second-line:** Behavioral therapy and support groups.

Correct Answer: B. 2 to 4 days ✅

MCQ 67:

Which of the following is an alternative treatment for severe depression?

A. Psychotherapy
B. Tricyclic antidepressants
C. Electroconvulsive therapy
D. Electroencephalogram therapy

Explanation:
Electroconvulsive therapy (ECT) (option C) is an effective alternative treatment for severe depression, especially when patients do not respond to medications or are severely ill. ECT involves the use of electrical currents to stimulate the brain and is typically used in treatment-resistant cases or when rapid response is required.

Severe Depression
Definition: A mood disorder characterized by persistent sadness, loss of interest, and other symptoms affecting daily functioning.
Key Clinical Presentation: Depressed mood, anhedonia, changes in sleep and appetite, cognitive impairment.
Investigations: Clinical diagnosis based on DSM-5 criteria.
Treatment Outline:

- **First-line:** Antidepressants (SSRIs, SNRIs).
- **Second-line:** Psychotherapy (Cognitive Behavioral Therapy, Interpersonal Therapy).
- **Definitive:** ECT for treatment-resistant or severe cases.

Correct Answer: C. Electroconvulsive therapy ✅

MCQ 68:

Which of the following is the antidote for an opiate overdose?

A. Naloxone
B. Flumazenil
C. Lorazepam
D. Buprenorphine

Explanation:
Naloxone (option A) is the antidote for opioid overdose. It is an opioid antagonist that rapidly reverses the effects of opioid toxicity, including respiratory depression and sedation. Naloxone can be administered intranasally or intramuscularly to quickly reverse life-threatening overdose symptoms.

Opiate Overdose
Definition: Life-threatening toxicity caused by excessive use of opioids, leading to respiratory depression, altered mental status, and potential death.
Key Clinical Presentation: Respiratory depression, pinpoint pupils, unconsciousness, and hypoxia.
Investigations: Diagnosis is clinical, based on presentation and history of opioid use.
Treatment Outline:

- **First-line:** Naloxone administration.
- **Second-line:** Supportive care (oxygen, mechanical ventilation).

Correct Answer: A. Naloxone ✅

MCQ 69:

Which of the following is the most likely side effect of bupropion when prescribed for smoking cessation?

A. Arrhythmia
B. Xerostomia
C. Headache
D. Seizure

Explanation:
Seizure (option D) is a known serious side effect of **bupropion**, particularly at higher doses. It lowers the seizure threshold, which can lead to seizures in susceptible individuals, especially those with a history of seizures or eating disorders. Patients using bupropion should be monitored closely for this risk.

Bupropion for Smoking Cessation
Definition: A norepinephrine-dopamine reuptake inhibitor used as an aid in smoking cessation.
Key Clinical Presentation: Used in patients who wish to quit smoking to reduce withdrawal symptoms and cravings.
Investigations: Monitor for adverse effects such as seizures.
Treatment Outline:

- **First-line:** Bupropion, nicotine replacement therapy (NRT), or varenicline.
- **Second-line:** Behavioral therapy for support.

Correct Answer: D. Seizure ✅

MCQ 70:

A 24-year-old inactive woman presents to the clinic with a 5-month history of amenorrhea. On examination, she has a body mass index of 20, stable for the last 5 years (see lab results). Pregnancy test: Negative. Which of the following is the most likely diagnosis?

A. Eating disorder
B. Illegal drug use
C. Pituitary neoplasm
D. Chronic depression

Explanation:
Eating disorder (option A), particularly **anorexia nervosa**, is a likely cause of amenorrhea in an inactive young woman with a normal BMI but possibly restrictive eating habits. This condition often leads to hormonal imbalances, which result in the cessation of menstruation. A detailed history of eating habits and psychological factors should be explored to confirm the diagnosis.

Eating Disorders
Definition: A range of psychological disorders characterized by abnormal or disturbed eating habits.
Key Clinical Presentation: Amenorrhea, weight loss, body image issues, restrictive eating behaviors.
Investigations: Clinical assessment, lab tests (hormonal levels), and screening for eating disorders.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT), nutritional rehabilitation.
- **Second-line:** Pharmacotherapy for associated depression or anxiety.

Correct Answer: A. Eating disorder ✅

MCQ 71:

Which of the following factors is the most important in determining the success of a smoking cessation program?

A. Patient's desire to quit smoking
B. Behaviour-modification
C. Pharmacologic agents
D. Medical counseling

Explanation:
Patient's desire to quit smoking (option A) is the most important factor in the success of a smoking cessation program. Motivation and commitment to quitting are fundamental to success, as they determine the patient's willingness to adhere to treatment strategies, including pharmacologic agents, behavior-modification techniques, and medical counseling.

Smoking Cessation
Definition: The process of discontinuing tobacco smoking.
Key Clinical Presentation: Cravings, irritability, difficulty concentrating, and withdrawal symptoms.
Investigations: Clinical assessment of nicotine dependence and readiness to quit.
Treatment Outline:

- **First-line:** Behavioral therapy and pharmacotherapy (e.g., nicotine replacement therapy, bupropion, varenicline).
- **Second-line:** Group therapy or support groups for motivation.

Correct Answer: A. Patient's desire to quit smoking ✅

MCQ 72:

A 45-year-old man newly diagnosed with type 2 diabetes presents to the clinic for further management. He is on time for the appointment, but because of an emergency case, the physician arrives late to the clinic. When the physician walks in, the patient appears extremely angry. Which of the following is the most appropriate response by the physician?

A. Acknowledge the patient's anger
B. Explore the reasons for the anger
C. Help the patient understand their anger
D. Empathize with the newly diagnosed diabetic patient

Explanation:
The most appropriate response is **A. Acknowledge the patient's anger**. Acknowledging the patient's feelings is a vital first step in de-escalating the situation. It helps to validate their emotions and creates a space for the physician to explore the underlying issues more effectively. Immediate exploration or explaining the patient's anger could escalate the situation.

Patient Communication
Definition: The process of addressing patients' emotions and concerns in a respectful and supportive manner.
Key Clinical Presentation: Angry or frustrated patients, especially in stressful medical situations.
Investigations: Clinical evaluation of emotional response and behavioral cues.
Treatment Outline:

- **First-line:** Active listening and acknowledgment of emotions.
- **Second-line:** Exploring underlying issues with empathy and patience.

Correct Answer: A. Acknowledge the patient's anger ✅

73:

A 41-year-old man presents to the clinic requesting a complete examination to exclude colon cancer. He recently had some investigations done and provided a list of test results. He complains that this fear is interfering severely with his work and social activities. Which of the following is the most likely diagnosis?

A. Hypochondriasis
B. Factitious disorder
C. Somatization disorder
D. Obsessive-compulsive disorder

Explanation:
The most likely diagnosis is **Hypochondriasis** (option A), also known as illness anxiety disorder. This condition is characterized by excessive preoccupation with having a serious illness despite

having no or minimal physical symptoms. The patient's fear of colon cancer and its impact on his life, despite negative test results, supports this diagnosis.

Hypochondriasis (Illness Anxiety Disorder)
Definition: A mental disorder marked by excessive worry about having or developing a serious illness, even without significant symptoms.
Key Clinical Presentation: Preoccupation with health concerns, frequent seeking of medical tests, and fear of serious illness despite reassurance.
Investigations: Clinical evaluation and screening for other psychological disorders.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT)
- **Second-line:** Antidepressants (SSRIs) for associated anxiety or depression.

Correct Answer: A. Hypochondriasis ✅

74:

A 16-year-old boy learns about the death of his father and within hours, his speech becomes disorganized, he is walking outside of his house with no clothes on, and he claims that his father is inside the house and telling him to do these things in his honor. The symptoms resolved after 3 days, and the son returned to full pre-morbid level of function but is obviously confused over the events of the past few days. Which of the following is the most likely diagnosis?

A. Schizophreniform disorder
B. Schizoaffective disorder
C. Brief psychotic disorder
D. Major depressive disorder

Explanation:
The most likely diagnosis is **Brief psychotic disorder** (option C), which involves the sudden onset of psychotic symptoms, including disorganized speech, delusions, and sometimes catatonia, triggered by a stressful event (in this case, the death of the father). The symptoms resolve within a month, which is characteristic of brief psychotic disorder.

Brief Psychotic Disorder
Definition: A psychotic disorder characterized by the sudden onset of delusions, hallucinations, disorganized speech, or behavior, lasting at least one day but less than one month.
Key Clinical Presentation: Acute psychosis after a significant stressor, with full recovery within a month.

Investigations: Clinical diagnosis based on history and symptom duration.
Treatment Outline:

- **First-line:** Antipsychotics (e.g., risperidone or olanzapine) for short-term management.
- **Second-line:** Psychotherapy after symptom resolution to address the trauma.

Correct Answer: C. Brief psychotic disorder ✅

75:

A 35-year-old healthy man presents to the clinic with a 3-month history of feeling depressed. He is distressed at work and burnt-out. 4 months earlier, he was promoted to a managerial position and was having an interpersonal conflict with 2 of the employees. Which of the following is the most appropriate next step in management?

A. Selective serotonin reuptake inhibitors
B. TCA tricyclic antidepressant
C. Supportive psychotherapy
D. Phenothiazine

Explanation:
The most appropriate next step in this patient's management is **supportive psychotherapy** (option C). This patient appears to be experiencing work-related stress and burnout, which can be effectively managed with psychotherapy. Medication may be considered if his symptoms do not improve with psychotherapy or if they are significantly impairing his daily functioning.

Supportive Psychotherapy
Definition: A form of therapy that provides emotional support, helps patients cope with stressors, and improves problem-solving and coping mechanisms.
Key Clinical Presentation: Symptoms of burnout, anxiety, or mild depressive symptoms due to stress.
Investigations: Clinical interview to assess the severity of symptoms.
Treatment Outline:

- **First-line:** Psychotherapy (supportive therapy, stress management techniques).
- **Second-line:** Medication if necessary, such as SSRIs, if depression or anxiety becomes more prominent.

Correct Answer: C. Supportive psychotherapy ✅

MCQ 76:

An 18-year-old man presents to the Emergency Department with a 20-minute history of light-headedness, headaches, and nausea. He complains of tingling around the mouth and in the fingertips. On examination, he is anxious, tremulous, sweating, and breathing heavily. Which of the following is the most appropriate first action?

A. Patient rebreathe into a paper bag
B. Conduct an amobarbital interview
C. Blood alcohol concentration
D. CT Brain scan

Explanation:
The most appropriate first action is to have the patient **rebreathe into a paper bag** (option A). This patient is presenting with symptoms consistent with hyperventilation syndrome, often triggered by anxiety. Rebreathing into a paper bag can help restore carbon dioxide levels and alleviate symptoms.

Hyperventilation Syndrome
Definition: A condition caused by rapid or deep breathing that leads to a decrease in carbon dioxide levels, resulting in symptoms like dizziness, tingling, and shortness of breath.
Key Clinical Presentation: Anxiety, hyperventilation, tingling, dizziness.
Investigations: Clinical evaluation; rule out other causes like metabolic or neurological conditions.
Treatment Outline:

- **First-line:** Rebreathing into a paper bag to help with symptoms.
- **Second-line:** Anxiolytics if needed, or further psychological support.

Correct Answer: A. Patient rebreathe into a paper bag ✅

MCQ 77:

Which of the following atypical antipsychotic agents is associated with the most weight gain?

A. Ziprasidone
B. Aripiprazole
C. Olanzapine
D. Quetiapine

Explanation:
The atypical antipsychotic associated with the most weight gain is **Olanzapine** (option C). Olanzapine is known to have a significant side effect of weight gain and metabolic disturbances compared to other antipsychotics.

Atypical Antipsychotics
Definition: A class of antipsychotic drugs used primarily for the treatment of schizophrenia and bipolar disorder.
Key Clinical Presentation: Schizophrenia, bipolar disorder, and severe anxiety.
Investigations: Clinical monitoring of weight, glucose levels, and lipid profiles.
Treatment Outline:

- **First-line:** Olanzapine for patients requiring atypical antipsychotic therapy.
- **Second-line:** Other antipsychotics like aripiprazole if weight gain is a significant concern.

Correct Answer: C. Olanzapine ✅

MCQ 78:

Which of the following is the best treatment for generalized anxiety disorder?

A. Buspirone
B. Bupropion
C. Alprazolam
D. Escilopram

Explanation:
The most appropriate treatment for generalized anxiety disorder is **Escitalopram** (option D), an SSRI. SSRIs are considered first-line therapy for GAD due to their efficacy and safety profile. Buspirone (option A) is another option, but SSRIs like escitalopram are typically preferred.

Generalized Anxiety Disorder (GAD)
Definition: A condition characterized by excessive, uncontrollable worry about a variety of topics, often accompanied by physical symptoms like restlessness and fatigue.
Key Clinical Presentation: Excessive worry, restlessness, muscle tension, and sleep disturbances.
Investigations: Clinical evaluation and anxiety scales (e.g., GAD-7).
Treatment Outline:

- **First-line:** SSRIs (e.g., escitalopram, sertraline).

- **Second-line:** Benzodiazepines (for acute episodes) or buspirone for long-term management.

Correct Answer: D. Escitalopram ✅

79:

For which of the following childhood psychiatric disorders, clozapine is the most appropriate medication?

A. Bipolar illness
B. Schizophrenia
C. Substance abuse
D. Major depression

Explanation:
Clozapine (option B) is the most appropriate medication for **Schizophrenia** in children, particularly for patients who do not respond to other antipsychotic medications. It is often used in refractory cases of schizophrenia due to its effectiveness and relatively unique side effect profile.

Schizophrenia in Children
Definition: A severe mental disorder involving distorted thinking, hallucinations, and severe emotional withdrawal, typically diagnosed in late adolescence or early adulthood.
Key Clinical Presentation: Hallucinations, delusions, disorganized speech, and behavior.
Investigations: Diagnosis is clinical, supported by psychiatric evaluation and imaging (e.g., MRI).
Treatment Outline:

- **First-line:** Antipsychotic medications like clozapine in refractory cases.
- **Second-line:** Other atypical antipsychotics if clozapine is ineffective or not well tolerated.

Correct Answer: B. Schizophrenia ✅

80:

Which of the following antidepressants has a high efficacy in childhood and adolescent depression?

A. Lithium
B. Clozapine

C. Fluoxetine
D. Imipramine

Explanation:
Fluoxetine (option C) is the most appropriate and effective antidepressant for **childhood and adolescent depression**. It is an SSRI and has been proven to be safe and effective for this age group in treating major depressive disorder.

Childhood and Adolescent Depression
Definition: A mood disorder characterized by persistent feelings of sadness and loss of interest in activities, affecting children and adolescents.
Key Clinical Presentation: Depressed mood, irritability, loss of interest in activities, difficulty concentrating.
Investigations: Diagnosis is clinical, based on DSM-5 criteria.
Treatment Outline:

- **First-line:** SSRIs like fluoxetine or cognitive behavioral therapy (CBT).
- **Second-line:** Tricyclic antidepressants (e.g., imipramine) if SSRIs are ineffective.

Correct Answer: C. Fluoxetine ✅

81:

A 30-year-old man presents to the clinic with a 3-month history of aliens taking to him and putting thoughts into his mind. Which of the following is the most appropriate management?

A. Behavioural therapy
B. Anti-psychotic therapy
C. Bereavement counselling
D. Anti-depressive medications

Explanation:
The most appropriate management for this patient is **anti-psychotic therapy** (option B). This patient is presenting with **delusions**, a hallmark symptom of psychosis. Antipsychotic medications are the first-line treatment for conditions like schizophrenia or delusional disorder.

Psychosis
Definition: A mental disorder characterized by impaired reality testing, including symptoms like delusions, hallucinations, or disorganized thinking.

Key Clinical Presentation: Delusions, hallucinations, disorganized speech and behavior.
Investigations: Diagnosis is clinical, confirmed by psychiatric evaluation.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, olanzapine).
- **Second-line:** Cognitive-behavioral therapy and psychosocial support for long-term management.

Correct Answer: B. Anti-psychotic therapy ✅

82:

A 55-year-old man presents to the clinic with history of difficulty maintaining erections and impatience that occurs during stressful periods of life. He is currently under stress at work. On examination, he has normal physical findings. Which of the following is the first step in management?

A. Prescribe sildenafil
B. Refer to an endocrinologist
C. Refer to a urologist for evaluation
D. Encourage use of relaxation strategies

Explanation:
The first step in management is to **encourage use of relaxation strategies** (option D), particularly if the patient's erectile dysfunction is related to stress or anxiety. Sildenafil (option A) may be prescribed later if there is no improvement, but the first step is to address the underlying psychological cause.

Erectile Dysfunction
Definition: The inability to achieve or maintain an erection for satisfactory sexual performance.
Key Clinical Presentation: Difficulty with erections, particularly during times of stress or anxiety.
Investigations: Clinical evaluation, including a full medical and psychological history.
Treatment Outline:

- **First-line:** Psychological interventions like stress reduction techniques.
- **Second-line:** Pharmacologic treatments (e.g., sildenafil) if psychological therapies are ineffective.

Correct Answer: D. Encourage use of relaxation strategies ✅

83:

A 38-year-old woman presents with a 3-month history of recurrent intense fear accompanied by irritability, excessive worry, palpitations, and chest tightness. She is avoiding social gatherings fearing of having same symptoms. Physical examination findings are normal. Which of the following medications is the most appropriate first line of treatment?

A. Selective serotonin reuptake inhibitor
B. Monoamine oxidase inhibitor
C. Tricyclic antidepressants
D. Benzodiazepine

Explanation:
The most appropriate first-line treatment for **panic disorder** is a **Selective serotonin reuptake inhibitor (SSRI)** (option A). SSRIs are effective for both short-term and long-term management of panic disorder and are preferred due to their safety profile.

Panic Disorder
Definition: A disorder characterized by recurrent and unexpected panic attacks, along with persistent fear of future attacks.
Key Clinical Presentation: Recurrent panic attacks, anxiety, physical symptoms (e.g., palpitations, sweating).
Investigations: Clinical evaluation, with diagnostic criteria based on DSM-5.
Treatment Outline:

- **First-line:** SSRIs (e.g., escitalopram) and Cognitive Behavioral Therapy (CBT).
- **Second-line:** Benzodiazepines (e.g., lorazepam) for short-term management during acute attacks.

Correct Answer: A. Selective serotonin reuptake inhibitor ✅

84:

A 5-year-old boy is brought to the clinic by his mother with history of impaired language development, impaired intelligence, compulsive repetitive behaviour, and a preoccupation with inanimate objects. Which of the following is the most likely diagnosis?

A. Autism
B. Dyslexia

C. Conductive hearing loss
D. Manic depressive disorder

Explanation:
The most likely diagnosis is **Autism** (option A). This child presents with impaired language, intellectual delays, and repetitive behavior, which are hallmark signs of autism spectrum disorder (ASD).

Autism Spectrum Disorder (ASD)
Definition: A neurodevelopmental disorder characterized by impairments in social interaction, communication, and the presence of restricted and repetitive behaviors.
Key Clinical Presentation: Impaired language development, repetitive behaviors, preoccupation with inanimate objects, and intellectual impairment.
Investigations: Diagnosis is clinical, based on developmental history and behavioral observations.
Treatment Outline:

- **First-line:** Behavioral therapies, including Applied Behavioral Analysis (ABA), and social skills training.
- **Second-line:** Medication for managing symptoms (e.g., antipsychotics for severe behavioral issues).

Correct Answer: A. Autism ✅

85:

A mother of a 3-month-old baby presents to the clinic and tells the physician that she is "going crazy." She states that since the baby has been born, she has continuously visualized a snake crawling into the baby's crib at night. She states that she knows it is not likely to happen but cannot get the thought out of her mind. She checks the baby's crib up to 50 times per night to ensure that there is no snake. Which of the following is the most likely diagnosis?

A. Delusions
B. Obsessions
C. Hallucinations
D. Postpartum psychosis

Explanation:
The most likely diagnosis is **Obsessions** (option B). The mother is experiencing repetitive,

intrusive thoughts about the snake, which she knows are irrational, but cannot control, a hallmark of obsessive-compulsive disorder (OCD).

Obsessive-Compulsive Disorder (OCD)
Definition: A mental health disorder marked by recurrent, intrusive thoughts (obsessions) and repetitive behaviors or mental acts (compulsions) performed to alleviate anxiety.
Key Clinical Presentation: Repetitive, intrusive thoughts (obsessions) and compulsive behaviors to reduce anxiety (e.g., checking, washing).
Investigations: Diagnosis is clinical, often using the Yale-Brown Obsessive Compulsive Scale.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with exposure and response prevention (ERP), and SSRIs (e.g., fluoxetine).
- **Second-line:** Other antidepressants or medication for refractory cases.

Correct Answer: B. Obsessions ✅

86:

A 50-year-old woman presents to the clinic with a history of anxiety. 2 months ago, during a meeting at her job, she became very anxious, sweaty, and short of breath. She had to leave the meeting in order to calm down. Since then, she has avoided meeting rooms out of fear that she will have a recurrence of her symptoms. Which of the following is the most likely diagnosis?

A. Panic disorder
B. Specific anxiety disorder
C. Generalized anxiety disorder
D. Post-traumatic stress disorder

Explanation:
The most likely diagnosis is **Panic disorder** (option A). The woman experiences sudden episodes of intense anxiety, including physical symptoms like sweating and shortness of breath, and has started avoiding situations that could trigger a recurrence, characteristic of panic disorder.

Panic Disorder
Definition: A mental health disorder characterized by recurrent, unexpected panic attacks, followed by persistent concern about future attacks and avoidance behaviors.
Key Clinical Presentation: Sudden onset of intense anxiety or fear, with physical symptoms such as sweating, palpitations, or shortness of breath.

Investigations: Diagnosis is clinical, based on the presence of recurrent panic attacks and avoidance behavior.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) and SSRIs.
- **Second-line:** Benzodiazepines for acute symptom relief.

Correct Answer: A. Panic disorder ✅

87:

A 24-year-old man presents to the clinic with a history of gradual fear of automobiles. He has always been anxious about cars, but now can no longer look at them without experiencing shortness of breath, sweating, and fear. Which of the following disorders is the most likely diagnosis?

A. Panic
B. Specific anxiety
C. Generalized anxiety
D. Post-traumatic stress

Explanation:
The most likely diagnosis is **Specific anxiety disorder** (option B). This patient has a clear, specific fear of automobiles that has caused significant distress and avoidance, which is characteristic of a specific phobia.

Specific Phobia (Automobile-related)
Definition: A marked fear or anxiety about a specific object or situation (in this case, automobiles), leading to avoidance behavior.
Key Clinical Presentation: Fear of a specific object or situation, resulting in distress or avoidance.
Investigations: Diagnosis is clinical, often based on the specific phobia type (e.g., cars).
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with exposure therapy.
- **Second-line:** Medications like SSRIs for co-occurring anxiety.

Correct Answer: B. Specific anxiety ✅

88:

A 21-year-old woman presents to the clinic with a history of red dry skin on her hands. She washes her hands up to 70 times per day because she always feels dirty. Washing her hands relieves her anxiety, but as she touches anything, she wants to wash again. Which of the following disorders is the most likely diagnosis?

A. Obsessive compulsive disorder
B. Dysmorphic disorder
C. Phobic disorder
D. Panic disorder

Explanation:
The most likely diagnosis is **Obsessive-compulsive disorder (OCD)** (option A). The patient's compulsive hand-washing behavior is a response to intrusive thoughts about cleanliness, which is characteristic of OCD.

Obsessive-Compulsive Disorder (OCD)
Definition: A disorder characterized by obsessive thoughts and compulsive behaviors, such as repetitive hand-washing, that are performed to reduce anxiety.
Key Clinical Presentation: Intrusive thoughts (obsessions) leading to repetitive behaviors (compulsions).
Investigations: Diagnosis is clinical, supported by tools like the Yale-Brown Obsessive-Compulsive Scale.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with exposure and response prevention (ERP) and SSRIs.
- **Second-line:** Antipsychotics or other SSRIs if needed.

Correct Answer: A. Obsessive compulsive disorder ✅

89: A 60-year-old woman presents with pain in her right knee for the last 3 years. The pain is worse with activity and relieved by rest and analgesics. Physical examination reveals a varus knee deformity and tenderness over the medial joint line. What is the most likely diagnosis?

A. Inflammatory arthritis
B. Medial meniscus tear
C. Septic arthritis
D. Osteoarthritis

Explanation:

The patient presents with chronic knee pain that is worse with activity and relieved by rest, which is characteristic of **osteoarthritis (OA)**. The varus deformity and tenderness over the medial joint line are typical findings in **medial compartment OA**.

Osteoarthritis (OA)

Definition: A degenerative joint disease characterized by the breakdown of articular cartilage and underlying bone. Commonly affects weight-bearing joints like the knees.

Key Clinical Presentation: Chronic, insidious onset of joint pain, worse with activity and relieved by rest, accompanied by joint deformity (e.g., varus/valgus deformities) and tenderness at the joint line.

Investigations:

- **Best initial:** X-ray (joint space narrowing, osteophytes, subchondral sclerosis)
- **Confirmatory:** MRI for detailed cartilage and soft tissue assessment (if needed for surgical planning)

Treatment Outline:

- **First-line:** Weight management, physical therapy, NSAIDs
- **Second-line:** Intra-articular corticosteroid injections, hyaluronic acid injections
- **Definitive:** Joint replacement surgery (in severe cases)

Correct Answer: D. Osteoarthritis ✅

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C. Septic arthritis
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- **Definitive:** Joint replacement surgery (in severe cases)

Correct Answer: D. Osteoarthritis ✅

91:

A 45-year-old nurse working in the pulmonary clinic complains of fever and a productive cough for the last 3 months. She has received 2 courses of oral antibiotics, but showed no improvement. Which of the following is the most likely diagnosis?

A. Pulmonary tuberculosis
B. H. Influenza pneumonia
C. Infectious mononucleosis
D. Streptococcal pneumonia

Explanation:

The patient's symptoms of prolonged fever and a productive cough lasting for 3 months with no improvement despite antibiotic therapy are strongly suggestive of **pulmonary tuberculosis (TB)**.

TB is a chronic infection that often presents with these nonspecific symptoms, particularly in healthcare workers who are at higher risk for exposure. A prolonged, unresponsive cough, especially in the context of a healthcare setting, is characteristic of TB.

Pulmonary Tuberculosis (TB)

Definition: A bacterial infection caused by *Mycobacterium tuberculosis*, primarily affecting the lungs but can spread to other organs.

Key Clinical Presentation: Persistent cough, hemoptysis, fever, night sweats, weight loss, and fatigue. Risk factors include healthcare workers, immunocompromised individuals, and exposure to TB patients.

Investigations:

- **Best initial:** Chest X-ray (can show infiltrates, cavitations, or fibrosis).
- **Confirmatory:** Sputum culture for *Mycobacterium tuberculosis*, or sputum acid-fast bacillus (AFB) smear.

Treatment Outline:

- **First-line:** Rifampin, Isoniazid, Pyrazinamide, and Ethambutol for 6-9 months (RIPE regimen)
- **Second-line:** Second-line drugs for drug-resistant TB (if necessary)
- **Definitive:** Long-term antimicrobial therapy and follow-up with sputum cultures.

Correct Answer: A. Pulmonary tuberculosis ✅

MCQ 92:

A 45-year-old nurse working in the pulmonary clinic complains of fever and a productive cough for the last 3 months. She has received 2 courses of oral antibiotics, but showed no improvement. Which of the following is the most likely diagnosis?

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D. Streptococcal pneumonia

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Treatment Outline:

- **First-line:** Rifampin, Isoniazid, Pyrazinamide, and Ethambutol for 6-9 months (RIPE regimen)
- **Second-line:** Second-line drugs for drug-resistant TB (if necessary)
- **Definitive:** Long-term antimicrobial therapy and follow-up with sputum cultures.

Correct Answer: A. Pulmonary tuberculosis ✅

93:

A 55-year-old patient presents to the clinic with shortness of breath and a cough productive of yellow sputum for the past 2 days. There has also been blood-stained sputum for the last 3 hours. The patient has smoked 2 packs of cigarettes a day for the past 35 years. On examination, there are rhonchi, wheezing, and crepitations heard over the right hemothorax.

Blood pressure: 110/70 mmHg
Heart rate: 84 /min
Respiratory rate: 22 /min
Temperature: 38.8°C
Oxygen saturation: 95% on room air
Chest X-ray: It shows a dense infiltrate in the right lower lobe.

Which of the following is the most likely diagnosis?

A. Community acquired pneumonia
B. Acute bronchitis
C. Tuberculosis
D. COPD

Explanation:

The patient's presentation of shortness of breath, cough with yellow and blood-stained sputum, and the chest X-ray showing a dense infiltrate in the right lower lobe is most consistent with **community acquired pneumonia (CAP)**. CAP is a bacterial infection of the lung parenchyma, typically presenting with fever, productive cough, and infiltrates on chest X-ray. The patient's significant smoking history increases the risk of respiratory infections like pneumonia.

Community Acquired Pneumonia (CAP)

Definition: Acute infection of the lung parenchyma, typically caused by bacteria, in individuals who acquire the infection outside of a healthcare setting.

Key Clinical Presentation: Productive cough, fever, pleuritic chest pain, dyspnea, blood-stained sputum, and abnormal breath sounds (rhonchi, crepitations).

Investigations:

- **Best initial:** Chest X-ray (shows consolidation or infiltrates).
- **Confirmatory:** Sputum culture or blood culture to identify the causative organism.

Treatment Outline:

- **First-line:** Empiric antibiotic therapy, including a macrolide (e.g., azithromycin) or fluoroquinolone (e.g., levofloxacin) for typical pathogens like *Streptococcus pneumoniae*.
- **Second-line:** Consider broader spectrum antibiotics if hospitalized or suspected resistant organisms.
- **Definitive:** Tailored therapy based on culture results.

Correct Answer: A. Community acquired pneumonia ✅

MCQ 94:

A 27-year-old woman with asthma has not had a severe attack for the last 3 months. However, she has a history of cough and wheezing 2-3 times each week.

Which of the following is the most appropriate management?

A. Ipratropium bromide
B. Inhaled corticosteroid
C. Salbutamol
D. Salmeterol

Explanation:

For a patient with asthma who experiences intermittent symptoms such as cough and wheezing 2-3 times per week but does not have severe attacks, the **most appropriate management** is an **inhaled corticosteroid** (option B). Inhaled corticosteroids (ICS) are the cornerstone of asthma management for patients with persistent symptoms, as they reduce inflammation and prevent symptom exacerbation. ICS are used for daily control of asthma symptoms and help reduce the frequency of attacks.

Asthma Management

Definition: Chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Wheezing, cough, shortness of breath, and chest tightness, often exacerbated by triggers like allergens, exercise, or respiratory infections.

Investigations:

- **Best initial:** Spirometry or peak flow measurement to assess lung function.
- **Confirmatory:** Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge test).

Treatment Outline:

- **First-line:** Inhaled corticosteroids for daily control of inflammation (e.g., fluticasone, budesonide).
- **Second-line:** Long-acting beta-agonists (LABAs) added if needed for additional symptom control.
- **Rescue (for acute symptoms):** Short-acting beta-agonists (e.g., salbutamol) for quick relief during exacerbations.

Correct Answer: B. Inhaled corticosteroid ✅

MCQ 95:

A 21-year-old woman comes to the clinic with episodes of shortness of breath on exertion for one year. She has to stop after 15-20 minutes of exercise, because of dyspnoea and coughing, which get better after about one hour of rest. There are no symptoms at rest. Physical examination and spirometry are normal.

Blood pressure 125/75 mmHg
Heart rate 80 /min
Respiratory rate 12 /min
Temperature 37.0° C

Which of the following is the most appropriate inhaled treatment before exercise?

A. Salmeterol
B. Salbutamol
C. Budesonide
D. Ipratropium

Explanation:

This patient presents with **exercise-induced asthma** (EIA), characterized by shortness of breath and cough triggered by exercise, which improves with rest. The most appropriate inhaled treatment to prevent symptoms before exercise is a **short-acting beta-agonist (SABA)** like **salbutamol** (option B). Salbutamol is commonly used as a **rescue inhaler** and can be used 15-30 minutes before exercise to prevent bronchoconstriction.

Exercise-Induced Asthma

Definition: A form of asthma triggered by physical activity, leading to reversible airway obstruction, wheezing, cough, and shortness of breath during or after exercise.

Key Clinical Presentation: Dyspnoea and coughing during exercise, relieved by rest, with normal symptoms at rest.

Investigations:

- **Best initial:** Spirometry or peak flow measurement during or after exercise.
- **Confirmatory:** Methacholine challenge if needed.

Treatment Outline:

- **First-line:** Short-acting beta-agonists (e.g., salbutamol) used before exercise to prevent bronchoconstriction.

- **Second-line:** Long-acting beta-agonists (e.g., salmeterol) or leukotriene inhibitors for persistent symptoms.

Correct Answer: B. Salbutamol ✅

MCQ 96:

A 42-year-old man with asthma presents with daily symptoms despite regularly using a long-acting β-agonist inhaler and an inhaled high-dose corticosteroid. He also uses a short-acting β-agonist inhaler as "rescue" medication. These episodes also wake him from sleep 2-3 times a week. Physical examination shows bilateral wheezing.

Blood pressure 140/85 mmHg
Heart rate 90 /min
Respiratory rate 18 /min
Temperature 36.7° C
Forced Expiratory Volume in 1 Second:
Before bronchodilator: 65% (25-75%)
After bronchodilators: 91%

Which of the following is the most appropriate next step in management?

A. Add theophylline
B. Add a leukotriene inhibitor
C. Observe the patient using his inhaler
D. Treat for gastroesophageal reflux disease

Explanation:

This patient has poorly controlled asthma, as evidenced by daily symptoms, nighttime awakenings, and an FEV1 of 65% (indicating moderate asthma). The most appropriate next step is to **add a leukotriene inhibitor** (option B). Leukotriene inhibitors (e.g., montelukast) are effective for patients with asthma that is not adequately controlled by inhaled corticosteroids and long-acting beta-agonists. These medications can help reduce inflammation and improve asthma control, particularly for patients with nighttime symptoms.

Asthma Management

Definition: Chronic inflammatory disorder of the airways, leading to episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Daily symptoms, nocturnal awakenings, use of short-acting beta-agonists for relief, and spirometry showing reduced FEV1.

Investigations:

- **Best initial:** Spirometry for lung function.
- **Confirmatory:** Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge).

Treatment Outline:

- **First-line:** Inhaled corticosteroids (ICS) and long-acting beta-agonists (LABAs).
- **Second-line:** Leukotriene inhibitors, theophylline, or biologic agents for more severe asthma.
- **Rescue:** Short-acting beta-agonists (SABA) for acute symptom relief.

Correct Answer: B. Add a leukotriene inhibitor ✅

MCQ 97:

A 67-year-old patient presents to the Emergency Department with chest pain, shortness of breath, and haemoptysis following a car trip for more than 5 hours (see report). ECG: Right axis deviation, an S1-Q3-T3 pattern, and a right bundle branch block.

Which of the following is the most likely diagnosis?

A. Acute myocardial infarction
B. Bronchogenic carcinoma
C. Pulmonary embolism
D. Pericarditis

Explanation:

The patient's clinical presentation with chest pain, shortness of breath, haemoptysis, and prolonged immobility (after a long car trip) raises suspicion for **pulmonary embolism (PE)** (option C). The ECG showing **S1-Q3-T3 pattern** and **right axis deviation** are classic signs of PE. A right bundle branch block may also be seen in PE due to strain on the right side of the heart from obstructed pulmonary circulation.

Pulmonary Embolism (PE)

Definition: Blockage of the pulmonary arteries by a thrombus, often originating from deep vein thrombosis (DVT).

Key Clinical Presentation: Sudden onset of chest pain, shortness of breath, haemoptysis, tachycardia, hypoxia, and signs of right heart strain (e.g., S1-Q3-T3 pattern on ECG).

Investigations:

- **Best initial:** CT pulmonary angiography (CTPA)
- **Confirmatory:** Pulmonary angiography (gold standard), D-dimer

Treatment Outline:

- **First-line:** Anticoagulation with heparin or low molecular weight heparin (LMWH)
- **Second-line:** Thrombolysis or surgical embolectomy for massive PE
- **Definitive:** Ongoing anticoagulation for 3–6 months, depending on risk factors.

Correct Answer: C. Pulmonary embolism ✅

MCQ 98:

A 55-year-old smoker presents to a clinic with progressive shortness of breath (see report). Chest X-ray: It reveals a massive right pleural effusion.

Which of the following is the best next step in management?

A. Thoracentesis
B. High resolution CT scan
C. Intercostal tube under water seal
D. Repeated bronchodilator inhalations

Explanation:

The presence of a **massive pleural effusion** in a smoker raises concern for potential **malignant pleural effusion**, which could be caused by lung cancer or other malignancies. The best next step in management is to perform **thoracentesis** (option A), which allows for both diagnostic and therapeutic relief. Thoracentesis will help determine the etiology of the effusion by analyzing the pleural fluid (e.g., cytology for malignancy, cell count, pH, and other markers).

Pleural Effusion

Definition: Accumulation of fluid in the pleural space, which can be transudative or exudative, depending on the underlying cause.

Key Clinical Presentation: Progressive shortness of breath, pleuritic chest pain, decreased breath sounds, dullness to percussion, and egophony on examination.

Investigations:

- **Best initial:** Chest X-ray to identify the effusion.
- **Confirmatory:** Thoracentesis to obtain pleural fluid for analysis (cell count, protein, LDH, pH, and cytology).

Treatment Outline:

- **First-line:** Thoracentesis for diagnostic and therapeutic purposes
- **Second-line:** Chest tube drainage or pleurodesis if the effusion is recurrent or malignant
- **Definitive:** Management of the underlying cause (e.g., chemotherapy for malignancy, diuretics for heart failure).

Correct Answer: A. Thoracentesis ✅

MCQ 99:

A 20-year-old woman presents to the clinic with a history of abdominal pain, nausea, vomiting, localized weakness to the right leg, and has a history of irregular menses. In the past, she has sought treatment for head, back, joint, and chest pain. Factitious disorder and malingering have been excluded as diagnoses. Which of the following is the most likely diagnosis?

A. Hypochondriasis
B. Conversion disorder
C. Somatization disorder
D. Somatoform pain disorder

Explanation:
The most likely diagnosis is **Conversion disorder** (option B), which involves neurological symptoms such as weakness or paralysis that cannot be explained by medical conditions. The symptoms often arise after a stressful event and may affect motor or sensory function, with no clear medical cause.

Conversion Disorder
Definition: A psychiatric condition characterized by neurological symptoms, such as weakness, numbness, or paralysis, that are inconsistent with or cannot be explained by medical or neurological conditions.
Key Clinical Presentation: Motor or sensory dysfunction (e.g., weakness, paralysis, blindness)

without a medical cause, often following a stressor.
Investigations: Clinical evaluation to rule out neurological or medical conditions.
Treatment Outline:

- **First-line:** Psychotherapy (particularly CBT)
- **Second-line:** Stress management techniques and support groups.

Correct Answer: B. Conversion disorder ✅

MCQ 100:

Which of the following investigations should be recommended for an elderly patient with cognitive deficit?

A. Brief IQ test
B. Hearing loss evaluation
C. Wechsler memory scale
D. Involuntary movement test

Explanation:
The most appropriate investigation is **Hearing loss evaluation** (option B). Cognitive deficits in elderly patients can often be exacerbated by sensory impairments like hearing loss, which can make it difficult for patients to engage with their environment or participate in cognitive assessments. Addressing hearing impairment can improve cognitive function and communication.

Hearing Loss Evaluation
Definition: A diagnostic assessment to determine the presence and severity of hearing loss, often conducted with audiometry or other auditory tests.
Key Clinical Presentation: Difficulty hearing, misunderstanding speech, lack of response to auditory stimuli.
Investigations: Audiological assessment, including pure tone audiometry.
Treatment Outline:

- **First-line:** Hearing aids if hearing loss is confirmed.
- **Second-line:** Referral to an audiologist or ENT specialist if needed.

Correct Answer: B. Hearing loss evaluation ✅

MCQ 101:

An 8-year-old boy recently moved to another city. Soon after moving, school started, but the boy hesitated to make friends with anyone and had a hard time paying attention in class. Which of the following is the most likely diagnosis?

A. Dysthymia
B. Hypomania
C. Depression
D. Adjustment disorder

Explanation:
The most likely diagnosis is **Adjustment disorder** (option D). This condition is characterized by emotional or behavioral symptoms in response to a specific stressor, such as moving to a new city and starting school. Symptoms may include difficulties with attention and making friends, but they typically improve as the individual adjusts to the new environment.

Adjustment Disorder
Definition: A psychological response to identifiable stressors resulting in emotional or behavioral symptoms that are disproportionate or excessive.
Key Clinical Presentation: Symptoms include anxiety, depression, difficulty adjusting to a new environment or life event.
Investigations: Clinical evaluation to rule out other psychiatric disorders.
Treatment Outline:

- **First-line:** Psychotherapy (CBT, family therapy).
- **Second-line:** Medications (SSRIs) if needed for symptoms like anxiety or depression.

Correct Answer: D. Adjustment disorder ✅

MCQ 102:

A 48-year-old man presents for a routine visit, while reporting his medical history, he reveals a significant alcohol intake. The physician wants to assess the drinking further using the cut-annoyed-guilty-eye (CAGE) screening questionnaire. Which of the following is a question in the CAGE screening questionnaire?

A. "Do you eat more after you drink?"
B. "Do you think you are an alcoholic?"
C. "Have you ever felt guilty about your drinking?"
D. "How many drinks do you consume your daily bags?"

Explanation:
The correct question in the CAGE screening questionnaire is **"Have you ever felt guilty about your drinking?"** (option C). This is one of the key components of the CAGE questionnaire, which helps assess alcohol use and dependence by asking about **Cutting down**, **Annoyance**, **Guilt**, and **Eye-opener**.

CAGE Screening Questionnaire
Definition: A simple and effective screening tool for alcohol use disorder.
Key Clinical Presentation: Patients with alcohol use disorder may feel guilty about their drinking, try to cut down, or experience irritability when questioned about their drinking.
Investigations: Use of the CAGE questionnaire as a screening tool.
Treatment Outline:

- **First-line:** Behavioral interventions, counseling, or alcohol support groups.
- **Second-line:** Pharmacotherapy (e.g., disulfiram, acamprosate) for alcohol use disorder.

Correct Answer: C. "Have you ever felt guilty about your drinking?" ✅

MCQ 103:

A 46-year-old woman presents to the clinic with an 8-month history of irritability, excessive worry, and difficulty falling asleep and concentrating. Which of the following is the most likely diagnosis?

A. Mood disorder not otherwise specified
B. Major depressive disorder with anxiety
C. Anxiety disorder with depression
D. Generalized anxiety disorder

Explanation:
The most likely diagnosis is **Generalized anxiety disorder (GAD)** (option D). GAD is characterized by chronic, excessive worry and associated physical symptoms, including irritability, sleep disturbances, and concentration problems. In this case, the anxiety is long-standing and pervasive, suggesting GAD rather than a mood disorder or comorbid anxiety and depression.

Generalized Anxiety Disorder (GAD)
Definition: A disorder characterized by excessive, uncontrollable worry about various aspects of life, often accompanied by physical symptoms.
Key Clinical Presentation: Excessive worry, irritability, difficulty concentrating, sleep disturbances.

Investigations: Clinical evaluation, anxiety scales (e.g., GAD-7).
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT), SSRIs (Selective Serotonin Reuptake Inhibitors).
- **Second-line:** Benzodiazepines for short-term management during acute exacerbations.

Correct Answer: D. Generalized anxiety disorder ✅

MCQ 104:

A 42-year-old woman is concerned about losing her job because of tiredness and would like to take antidepressants because of having a difficult time falling asleep and getting up in the morning. Her sleep disturbance is related to the inability to relax after getting into bed. She is repeatedly checking to make sure that the door is locked, the oven is off, and the children's school supplies are organized. A similar routine occurs in the morning. She states that her behavior is out of control, but if it is not done, she will not be able to sleep or leave the house. Which of the following disorders is the most likely diagnosis?

A. Obsessive-compulsive disorder
B. Generalized anxiety disorder
C. Somatization disorder
D. Delusional disorder

Explanation:
The most likely diagnosis is **Obsessive-compulsive disorder (OCD)** (option A). OCD is characterized by persistent, intrusive thoughts (obsessions) and repetitive behaviors (compulsions) performed to reduce anxiety. In this case, the patient's compulsive checking behaviors and anxiety related to not performing these actions are hallmark features of OCD.

Obsessive-Compulsive Disorder (OCD)
Definition: A disorder characterized by recurrent, intrusive thoughts (obsessions) and repetitive behaviors (compulsions) performed to alleviate the anxiety caused by the obsessions.
Key Clinical Presentation: Recurrent obsessions and compulsions that interfere with daily life.
Investigations: Clinical evaluation using OCD diagnostic criteria.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT) (Exposure and Response Prevention), SSRIs (Selective Serotonin Reuptake Inhibitors).
- **Second-line:** Clomipramine (TCA) if SSRIs are ineffective.

Correct Answer: A. Obsessive-compulsive disorder ✅

MCQ 105:

A 35-year-old healthy man presents to the clinic with a 3-month history of feeling depressed. He is distressed at work and burnt-out. 4 months earlier, he was promoted to a managerial position and was having an interpersonal conflict with 2 of the employees. Which of the following is the most appropriate next step in management?

A. Selective serotonin reuptake inhibitors
B. TCA tricyclic antidepressant
C. Supportive psychotherapy
D. Phenothiazine

Explanation:
The most appropriate next step in this patient's management is **supportive psychotherapy** (option C). This patient appears to be experiencing work-related stress and burnout, which can be effectively managed with psychotherapy. Medication may be considered if his symptoms do not improve with psychotherapy or if they are significantly impairing his daily functioning.

Supportive Psychotherapy
Definition: A form of therapy that provides emotional support, helps patients cope with stressors, and improves problem-solving and coping mechanisms.
Key Clinical Presentation: Symptoms of burnout, anxiety, or mild depressive symptoms due to stress.
Investigations: Clinical interview to assess the severity of symptoms.
Treatment Outline:

- **First-line:** Psychotherapy (supportive therapy, stress management techniques).
- **Second-line:** Medication if necessary, such as SSRIs, if depression or anxiety becomes more prominent.

Correct Answer: C. Supportive psychotherapy ✅

MCQ 106:

A 30-year-old man with a history of drug abuse presents to the clinic weekly with a variety of health complaints, which vary each time, and are greatly exaggerated. When he is unaware of being observed he appears asymptomatic. Which of the following is the most likely diagnosis?

A. Depression
B. Malingering
C. Somatic delusion
D. Conversion reaction

Explanation:
The most likely diagnosis is **Malingering** (option B). The patient's exaggerated and inconsistent symptoms that resolve when not being observed suggest a purposeful intention to gain attention, benefits, or avoid responsibilities, which is characteristic of malingering.

Malingering
Definition: Intentional production of false or grossly exaggerated symptoms motivated by external incentives (e.g., financial compensation, avoiding work).
Key Clinical Presentation: Inconsistent or exaggerated symptoms that are voluntarily produced and may resolve when the patient is not being observed.
Investigations: Diagnosis is clinical, with a focus on identifying inconsistencies in the patient's story and behavior.
Treatment Outline:

- **First-line:** Address underlying reasons for malingering (e.g., social, economic).
- **Second-line:** Referral for psychological evaluation if necessary.

Correct Answer: B. Malingering ✅

MCQ 107:

A 19-year-old woman presents to the clinic for a routine physical check-up. On examination, she has fine hair all over the body. She feels fat and subsequently diets. BMI 16.8 kg/m². Which of the following is the most likely diagnosis?

A. Bulimia
B. Anorexia nervosa
C. Body dysmorphic disorder
D. Generalized anxiety disorder

Explanation:
The most likely diagnosis is **Anorexia nervosa** (option B). The patient has a significantly low BMI, preoccupation with body image, and abnormal eating behaviors, which are key features of anorexia nervosa.

Anorexia Nervosa
Definition: A psychiatric disorder characterized by an intense fear of gaining weight, refusal to maintain a normal body weight, and a distorted body image.
Key Clinical Presentation: Significant weight loss, intense fear of weight gain, and preoccupation with food and body shape.
Investigations: Diagnosis is clinical, with assessments of weight, eating patterns, and psychological status.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) and nutritional rehabilitation.
- **Second-line:** Pharmacotherapy (e.g., SSRIs) for co-occurring depression or anxiety.

Correct Answer: B. Anorexia nervosa ✅

MCQ 108:

A 48-year-old woman calls the clinic expressing fear of leaving the house, and has been inside for 2 months. She becomes anxious at the thought of walking out the front door and is now unable to enter the room containing the front door. Which of the following is the most likely diagnosis?

A. Depression
B. Malingering
C. Agoraphobia
D. Substance abuse

Explanation:
The most likely diagnosis is **Agoraphobia** (option C). This patient is experiencing a marked fear of leaving her home and avoidance behavior, which are characteristic of agoraphobia.

Agoraphobia
Definition: A type of anxiety disorder where individuals experience fear of being in situations where escape may be difficult or help unavailable in the event of a panic attack.
Key Clinical Presentation: Avoidance of situations such as leaving the home, using public transportation, or being in crowded places.
Investigations: Diagnosis is clinical, based on the history of avoidance behaviors and anxiety in specific situations.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with exposure therapy.

- **Second-line:** SSRIs for anxiety management.

Correct Answer: C. Agoraphobia ✅

MCQ 108:

A 27-year-old man presents to the clinic with a history of sweating, palpitations, nausea, and a "panicky feeling" when required to deliver a speech at work. He has no other difficulties at work. Which of the following is the most likely diagnosis?

A. Malingering
B. Agoraphobia
C. Performance anxiety
D. Generalized anxiety disorder

Explanation:
The most likely diagnosis is **Performance anxiety** (option C). The patient experiences anxiety and physical symptoms in a specific situation—giving a speech—without generalized anxiety outside of that context.

Performance Anxiety
Definition: A form of situational anxiety characterized by fear and physical symptoms during performance-related events (e.g., speaking in public).
Key Clinical Presentation: Anxiety symptoms (sweating, palpitations) triggered by specific performance situations, without broader anxiety.
Investigations: Diagnosis is clinical, based on the situational nature of the anxiety.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with relaxation techniques.
- **Second-line:** Beta-blockers for acute anxiety management.

Correct Answer: C. Performance anxiety ✅

MCQ 109:

A 56-year-old woman presents to the clinic with a history of constant counting of floor tiles, cars, items on a dinner plate, and words in a conversation. She is upset by the counting but becomes anxious and agitated when she tries to stop herself from counting. Which of the following is the most likely diagnosis?

A. Delusion
B. Obsession
C. Compulsion
D. Hallucination

Explanation:
The most likely diagnosis is **Compulsion** (option C). The patient is performing repetitive behaviors (counting) to relieve anxiety caused by her obsessive thoughts.

Obsessive-Compulsive Disorder (OCD)
Definition: A disorder characterized by obsessive thoughts and compulsive behaviors performed to reduce anxiety.
Key Clinical Presentation: Intrusive thoughts (obsessions) and repetitive behaviors (compulsions) that are difficult to control and cause distress.
Investigations: Diagnosis is clinical, supported by the Yale-Brown Obsessive Compulsive Scale.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with exposure and response prevention (ERP) and SSRIs.
- **Second-line:** Antipsychotics if SSRIs are ineffective.

Correct Answer: C. Compulsion ✅

MCQ 110:

A 61-year-old man presents with history of constantly thinking about aliens landing in his backyard, especially when away from home. Despite knowing that aliens do not exist, he feels overwhelmed with the idea that they have landed. He fears that he is "going insane". Which of the following is the most likely diagnosis?

A. Delusion
B. Obsession
C. Compulsion
D. Hallucination

Explanation:
The most likely diagnosis is **Delusion** (option A). The patient has a false belief that aliens have landed, despite evidence to the contrary, and he feels overwhelmed by this belief, which is a characteristic of a delusion.

Delusion
Definition: A fixed, false belief that is not based on reality.

Key Clinical Presentation: The belief is unshakable even in the face of contrary evidence, and it often leads to distress.
Investigations: Diagnosis is clinical, based on the patient's history and presentation.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone).
- **Second-line:** Cognitive behavioral therapy (CBT) for management.

Correct Answer: A. Delusion ✅

MCQ 111:

A 38-year-old man presents to the clinic with a history of hearing voices coming from the toaster and refrigerator, warning that all the food is poisoned. Which of the following is the most likely diagnosis?

A. Delusional thinking
B. Visual hallucination
C. Disorganized thinking
D. Auditory hallucination

Explanation:
The most likely diagnosis is **Auditory hallucination** (option D). The patient is hearing voices that others cannot hear, which is a classic example of auditory hallucinations.

Auditory Hallucination
Definition: The perception of sounds, often voices, that are not present.
Key Clinical Presentation: The patient hears voices or sounds that no one else can hear, often with distressing content.
Investigations: Diagnosis is clinical, typically supported by psychiatric assessment.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., olanzapine, risperidone).
- **Second-line:** CBT or supportive psychotherapy.

Correct Answer: D. Auditory hallucination ✅

MCQ 112:

A 27-year-old man presents to the clinic for a routine check-up. On examination, he keeps looking to the right frequently. When questioned, he said, "My mother is there, but no one can see her except me". He claims that she comes and goes, even though the family indicates she died when the patient was a child. Which of the following is the most likely diagnosis?

A. Delusion
B. Visual hallucination
C. Auditory hallucination
D. Episode of disorganized thinking

Explanation:
The most likely diagnosis is **Visual hallucination** (option B). The patient is perceiving his deceased mother, which is a visual hallucination, as the patient reports seeing someone who is not physically present.

Visual Hallucination
Definition: Seeing objects, people, or images that are not actually present.
Key Clinical Presentation: The patient reports seeing things that are not observable by others, which can cause distress or confusion.
Investigations: Diagnosis is clinical, supported by psychiatric evaluation.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., quetiapine, risperidone).
- **Second-line:** Psychotherapy or CBT for managing delusions.

Correct Answer: B. Visual hallucination ✅

MCQ 113:

A 17-year-old high school athlete presents to the clinic for an annual physical check-up. On examination, he has gained 20 kg in the past 4 months and has very well-developed musculature of the upper body, acne over the face and back, and foul breath. Heart rate 94 /min, temperature 37.6° C, blood pressure 138/89 mmHg, respiratory rate 14 /min. Which of the following should the patient be screened for?

A. Depression
B. Cocaine use
C. Thyroid tumor
D. Anabolic steroid use

Explanation:
The most likely diagnosis is **Anabolic steroid use** (option D). The rapid weight gain, muscular development, acne, and foul breath are all signs of anabolic steroid abuse.

Anabolic Steroid Use
Definition: The use of synthetic substances that mimic the effects of testosterone to increase muscle mass and improve athletic performance.
Key Clinical Presentation: Rapid muscle growth, acne, increased aggression, foul body odor, and in some cases, altered mood and behavior.
Investigations: Diagnosis is clinical, supported by screening for steroids in urine or blood.
Treatment Outline:

- **First-line:** Discontinuation of steroid use and counseling.
- **Second-line:** Supportive therapy and management of side effects (e.g., acne).

Correct Answer: D. Anabolic steroid use ✅

MCQ 114:

A 52-year-old man presents to the clinic for a routine physical check-up. He has a history of pulling out hair when under stress. On examination, he has diffuse hair thinning along with the absence of eyelashes. Which of the following is the most likely diagnosis?

A. Post-traumatic stress disorder
B. Cicatricial syndrome
C. Trichotillomania
D. Alopecia areata

Explanation:
The most likely diagnosis is **Trichotillomania** (option C). The patient's history of pulling out hair and evidence of hair loss is characteristic of trichotillomania, a condition where individuals have a compulsion to pull out their hair.

Trichotillomania
Definition: A mental health condition characterized by the irresistible urge to pull out one's hair, leading to hair loss.
Key Clinical Presentation: Hair loss in areas where the patient habitually pulls; absence of eyelashes and other body hair.
Investigations: Diagnosis is clinical, often made through a detailed history and observation of

hair-pulling behavior.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with habit reversal training.
- **Second-line:** Antidepressants (e.g., SSRIs) if comorbid anxiety or depression is present.

Correct Answer: C. Trichotillomania ✅

MCQ 115:

A 15-year-old girl presents to the clinic with history of tetany, swollen glands, and eroded dental enamel (see lab result). Electrolytes test Result: Potassium 2.8 (Normal: 3.5-5.1 mmol/L). Which of the following is the most likely diagnosis?

A. Bulimia nervosa
B. Substance abuse
C. Anorexia nervosa
D. Alcohol withdrawal

Explanation:
The most likely diagnosis is **Anorexia nervosa** (option C). The electrolyte imbalance with low potassium, along with symptoms of tetany, swollen glands, and eroded dental enamel, is consistent with anorexia nervosa, where purging behavior often leads to electrolyte disturbances and dental erosion.

Anorexia Nervosa
Definition: A psychiatric disorder characterized by restrictive eating, excessive exercise, and an intense fear of gaining weight, leading to malnutrition and physical complications.
Key Clinical Presentation: Low body weight, preoccupation with food, compensatory behaviors such as vomiting, use of laxatives, and electrolyte disturbances.
Investigations: Electrolyte disturbances (hypokalemia), serum calcium, and magnesium levels.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT).
- **Second-line:** Nutritional rehabilitation and possible inpatient care.

Correct Answer: C. Anorexia nervosa ✅

MCQ 116:

A 40-year-old drug addict man presents to the Emergency Department with 1-hour history of unconsciousness. On examination, he has pinpoint pupils and a respiratory rate of 8/minute. Which of the following is the most appropriate treatment?

A. Atropine
B. Naloxone
C. Furosemide
D. N-acetylcysteine

Explanation:
The most appropriate treatment is **Naloxone** (option B). The patient's pinpoint pupils and respiratory depression are indicative of opioid overdose, and naloxone, an opioid antagonist, is the treatment of choice to reverse the effects of opioid toxicity.

Opioid Overdose
Definition: Poisoning caused by opioids, leading to CNS depression, respiratory depression, and pinpoint pupils.
Key Clinical Presentation: Pinpoint pupils, respiratory depression, altered mental status.
Investigations: Clinical diagnosis; no need for lab testing if clinical features are suggestive.
Treatment Outline:

- **First-line:** Naloxone administration (IV or IM).
- **Second-line:** Supportive care, including mechanical ventilation if necessary.

Correct Answer: B. Naloxone ✅

MCQ 117:

A 53-year-old woman with a history of depression presents to the Emergency Department with history of nausea, vomiting, epigastric pain, tinnitus, and shortness of breath 6 hours after consuming 50 tablets of aspirin (see lab result). Test Result: Plasma salicylate concentration 54 (Normal: 15-30 mg/dL). Which of the following is the most appropriate initial management?

A. Hemodialysis
B. Gastric lavage
C. Activated charcoal
D. Urine alkalinization

Explanation:
The most appropriate initial management is **Activated charcoal** (option C). In aspirin overdose,

activated charcoal is the first-line treatment within 1-2 hours of ingestion to reduce further absorption. Urine alkalinization (option D) is also important, but activated charcoal is typically the first intervention.

Salicylate Toxicity
Definition: Toxicity caused by an overdose of aspirin (acetylsalicylic acid), leading to respiratory alkalosis, metabolic acidosis, and multi-organ dysfunction.
Key Clinical Presentation: Nausea, vomiting, tinnitus, tachypnea, altered mental status, and elevated salicylate levels.
Investigations: Plasma salicylate concentration, arterial blood gases.
Treatment Outline:

- **First-line:** Activated charcoal within 1-2 hours of ingestion.
- **Second-line:** Urine alkalinization and possible hemodialysis for severe toxicity.

Correct Answer: C. Activated charcoal ✅

MCQ 118:

A 34-year-old healthy man presents to the clinic with a 2-year history of worsening sad mood, poor appetite, insomnia, fatigue, low self-esteem, and feeling of hopelessness. Which of the following is the most likely diagnosis?

A. Organic mood disorder
B. Depressive disorder
C. Minor depression
D. Dysthymia

Explanation:
The most likely diagnosis is **Dysthymia** (option D). Dysthymia (now called Persistent Depressive Disorder in DSM-5) is characterized by a chronically low mood lasting for at least 2 years with symptoms such as poor appetite, fatigue, and low self-esteem.

Persistent Depressive Disorder (Dysthymia)
Definition: A chronic form of depression characterized by a low mood lasting for at least 2 years, with other depressive symptoms.
Key Clinical Presentation: Low mood, fatigue, poor appetite, insomnia, and low self-esteem.
Investigations: Clinical diagnosis based on symptoms lasting at least 2 years.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT), selective serotonin reuptake inhibitors (SSRIs).
- **Second-line:** Tricyclic antidepressants (TCAs) if SSRIs are ineffective.

Correct Answer: D. Dysthymia ✅

MCQ 119:

A 9-year-old girl is brought to the clinic by her mother with a bald patch on her scalp. Her mother noted that she plucks her hair while studying and under stress. Which of the following is the most appropriate treatment?

A. Olanzapine
B. Bupropion
C. Fluoxetine
D. Lithium

Explanation:
The most appropriate treatment is **Fluoxetine** (option C). The girl's behavior is consistent with **Trichotillomania**, a condition where individuals pull out their hair in response to stress or anxiety. SSRIs, such as fluoxetine, are commonly used in the treatment of trichotillomania.

Trichotillomania
Definition: A psychiatric disorder characterized by the compulsive urge to pull out one's hair, often leading to bald patches.
Key Clinical Presentation: Repetitive hair-pulling behavior, bald patches, and increased stress or anxiety.
Investigations: Diagnosis is clinical based on the patient's history and behavior.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) with habit reversal training, SSRIs (e.g., fluoxetine).
- **Second-line:** Clomipramine or other medications if CBT and SSRIs are ineffective.

Correct Answer: C. Fluoxetine ✅

MCQ 120:

A 20-year-old man presents to the clinic with poor personal hygiene, echolalia, echopraxia, muttering to self, sitting in an odd posture for hours, and insomnia for the past 2 months. On examination, his orientation and memory are normal. Which of the following is the most appropriate treatment?

A. Oxcarbazepine
B. Amisulpride
C. Venlafaxine
D. Lithium

Explanation:
The most appropriate treatment is **Amisulpride** (option B). The patient's presentation suggests a psychotic disorder, and amisulpride, an atypical antipsychotic, is effective in managing symptoms like echolalia, echopraxia, and disorganized behavior.

Schizophrenia
Definition: A chronic psychiatric disorder characterized by psychotic symptoms such as hallucinations, delusions, disorganized thinking, and abnormal behaviors.
Key Clinical Presentation: Disorganized speech and behavior, hallucinations, and impaired reality testing.
Investigations: Clinical diagnosis with psychiatric assessment.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., amisulpride, risperidone).
- **Second-line:** Clozapine for treatment-resistant cases.

Correct Answer: B. Amisulpride ✅

MCQ 121:

A 45-year-old woman presents to the Primary Health Care Clinic with symptoms of anxiety due to difficulties in her relationship with her co-workers. Her examination was normal and the doctor has decided that there is no need for any investigation. Which of the following would be the most effective approach for counselling?

A. Instructing how to deal with others
B. Empathy during consultation
C. Asking humorous questions
D. Standing up for her rights

Explanation:
The most effective approach in this case is **Empathy during consultation** (option B). Empathy is key in building a therapeutic alliance and allows the patient to feel understood and supported, which is critical for anxiety management in a non-judgmental, supportive setting.

Generalized Anxiety Disorder
Definition: A mental health condition characterized by excessive and uncontrollable worry, often leading to physical symptoms like restlessness, fatigue, and difficulty concentrating.
Key Clinical Presentation: Chronic worry about multiple life domains, difficulty controlling worry, physical symptoms (e.g., tension, sleep disturbance).
Investigations: Clinical diagnosis based on DSM-5 criteria.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT), SSRI/SNRI medications.
- **Second-line:** Benzodiazepines for short-term relief.

Correct Answer: B. Empathy during consultation ✅

MCQ 122:

A 55-year-old smoker presents to the Primary Health Care Clinic for the result of the biopsy of a lesion in his right lung. When informed that, he has cancer in his right lung, he asked with anxiety, "Is it a mistake?". Which of the following is the most appropriate description for the patient's reaction?

A. Anger
B. Denial
C. Bargaining
D. Depression

Explanation:
The most appropriate description of the patient's reaction is **Denial** (option B). Denial is a common psychological defense mechanism where individuals refuse to accept reality or facts. The patient's disbelief in the diagnosis reflects this defensive response.

Cancer Diagnosis
Definition: Malignant growth in tissues that can invade surrounding areas and spread to other parts of the body.
Key Clinical Presentation: Unexplained weight loss, cough, hemoptysis, and dyspnea.

Investigations: Biopsy, CT scans, and sputum cytology for confirmation.
Treatment Outline:

- **First-line:** Chemotherapy, radiotherapy, and/or surgery depending on cancer stage.
- **Second-line:** Palliative care for advanced stages.

Correct Answer: B. Denial ✅

MCQ 123:

A 30-year-old man presents to the clinic with a 10-month history of restlessness, difficulty concentrating, and disturbance in his sleep pattern. He is diagnosed with generalized anxiety disorder. Which of the following is the most appropriate treatment?

A. Alprazolam
B. Bupropion
C. Haloperidol
D. Naltrexone

Explanation:
The most appropriate treatment is **Alprazolam** (option A). Alprazolam, a benzodiazepine, is useful for short-term management of acute anxiety symptoms. However, long-term treatment should focus on SSRIs (Selective Serotonin Reuptake Inhibitors), which are preferred for chronic management.

Generalized Anxiety Disorder
Definition: A psychiatric disorder characterized by excessive worry and anxiety lasting for at least 6 months.
Key Clinical Presentation: Restlessness, fatigue, difficulty concentrating, irritability, muscle tension, and sleep disturbances.
Investigations: Clinical diagnosis based on DSM-5 criteria.
Treatment Outline:

- **First-line:** SSRIs (e.g., sertraline), CBT.
- **Second-line:** Benzodiazepines like alprazolam for short-term relief, though long-term use is not recommended.

Correct Answer: A. Alprazolam ✅

MCQ 124:

An 18-year-old man presents to the Emergency Department with 1-week history of euphoria, increased blood pressure, tachycardia, nausea, weight loss, psychomotor agitation, chills, sweating, and visual hallucinations. Which of the following is the most likely diagnosis?

A. Amphetamine intoxication
B. Cannabis intoxication
C. Acute schizophrenia
D. Cocaine withdrawal

Explanation:
The most likely diagnosis is **Amphetamine intoxication** (option A). The symptoms of euphoria, increased blood pressure, tachycardia, and psychomotor agitation are consistent with stimulant use, specifically amphetamines.

Substance Use and Intoxication
Definition: The use of substances leading to acute psychological and physical symptoms.
Key Clinical Presentation: Euphoria, tachycardia, hypertension, psychomotor agitation, and possible hallucinations.
Investigations: Urine toxicology screen for amphetamines.
Treatment Outline:

- **First-line:** Supportive care, hydration, and monitoring.
- **Second-line:** Benzodiazepines for agitation, antipsychotics for hallucinations.

Correct Answer: A. Amphetamine intoxication ✅

MCQ 125:

A 3-year-old child is brought to the clinic for his yearly physical examination. His history revealed various behavioural issues, including poor interaction with peers and family, delayed language development, repetitive movements, and difficulty of accepting change. The child stiffens physically when touched, does not respond to the examiner, and does not make eye contact. Which of the following is the most likely diagnosis?

A. Attention deficit hyperactivity disorder
B. Asperger's syndrome
C. Autistic disorder
D. Down syndrome

Explanation:
The most likely diagnosis is **Autistic disorder** (option C). The child's difficulty with social interactions, repetitive movements, and language delay points to autism spectrum disorder (ASD), a developmental disorder.

Autistic Disorder (Autism Spectrum Disorder)
Definition: A developmental disorder characterized by challenges in communication, social interaction, and repetitive behaviors.
Key Clinical Presentation: Delayed speech, difficulty interacting with others, repetitive movements, and resistance to change.
Investigations: Developmental screening and observation.
Treatment Outline:

- **First-line:** Behavioral therapy, such as Applied Behavior Analysis (ABA).
- **Second-line:** Medications to address specific symptoms (e.g., stimulants for hyperactivity).

Correct Answer: C. Autistic disorder ✅

MCQ 126:

A 23-year-old woman is admitted to the Psychiatric Unit after causing superficial lacerations to her wrists. The reason is that she felt her therapist had abandoned her. She has a depressed mood that varies by the hour. On admission, she claimed that she heard a voice speaking to her, although she currently denies it. She caused a disagreement between the psychiatric resident and the nursing staff, who have conflicting views of her behavior in the unit. Which of the following personality disorders is the most likely diagnosis?

A. Antisocial
B. Borderline
C. Obsessive
D. Schizoid

Explanation:
The most likely diagnosis is **Borderline Personality Disorder (BPD)** (option B). BPD is characterized by intense, unstable relationships, mood swings, impulsivity, self-harm, and feelings of abandonment. The patient's behavior and symptoms, including self-harming due to perceived abandonment and mood variability, are indicative of BPD.

Borderline Personality Disorder (BPD)
Definition: A personality disorder marked by instability in mood, relationships, self-image, and behavior.
Key Clinical Presentation: Impulsive behaviors, fear of abandonment, self-harm, mood instability, and distorted self-image.
Investigations: Diagnosis is clinical, based on the DSM-5 criteria.
Treatment Outline:

- **First-line:** Dialectical Behavior Therapy (DBT), psychotherapy.
- **Second-line:** Medications like mood stabilizers or antidepressants for managing symptoms.

Correct Answer: B. Borderline ✅

MCQ 127:

A 40-year-old man presents to the Emergency Department with irritability, suspiciousness, overactivity, and poor hygiene. Emergency Department doctors regarded his behavior as abnormal and called for a psychiatrist. Which of the following is the most likely diagnosis?

A. Acute confusional state
B. Alzheimer's Disease
C. Schizophrenia
D. Psychosis

Explanation:
The most likely diagnosis is **Schizophrenia** (option C). Schizophrenia is characterized by symptoms like irritability, suspiciousness (paranoia), overactivity, and disorganized behavior, including poor hygiene. The patient's behavior is consistent with a psychotic episode, typical of schizophrenia.

Schizophrenia
Definition: A chronic, severe mental health disorder characterized by disturbances in thought, perception, and behavior.
Key Clinical Presentation: Hallucinations, delusions, disorganized thinking, and behavioral dysfunction.
Investigations: Clinical diagnosis, supported by imaging (e.g., MRI) and psychological evaluation.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, olanzapine).

- **Second-line:** Cognitive-behavioral therapy (CBT), psychosocial support.

Correct Answer: C. Schizophrenia ✅

MCQ 128:

A 58-year-old man presents to the clinic with a history of disturbed recent memory as a cardinal feature of his disturbed state of mind along with behavioral changes. Which of the following is the most likely diagnosis?

A. Multiple Sclerosis
B. Alzheimer's Disease
C. Huntington's chorea
D. Multi-infarct dementia

Explanation:
The most likely diagnosis is **Alzheimer's Disease** (option B). Alzheimer's is a progressive neurodegenerative disorder characterized primarily by memory impairment, with later behavioral changes. It often presents with early memory loss and cognitive decline.

Alzheimer's Disease
Definition: A neurodegenerative disease that causes progressive memory loss, confusion, and changes in behavior.
Key Clinical Presentation: Early memory impairment, difficulty with daily tasks, confusion, and personality changes.
Investigations: Clinical diagnosis with imaging (CT/MRI), neuropsychological testing.
Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (donepezil).
- **Second-line:** NMDA receptor antagonists (memantine) for moderate to severe stages.

Correct Answer: B. Alzheimer's Disease ✅

MCQ 129:

A 30-year-old man presents to the clinic complaining of reduced energy, lack of enjoyment, reduced sleep, reduced appetite, suicidal ideas, and lack of self-confidence. Examination and lab results are normal. Which of the following is the most likely diagnosis?

A. Anxiety disorders
B. Depressive disorders
C. Drug-induced psychoses
D. Dissociative conversion disorder

Explanation:
The most likely diagnosis is **Depressive disorders** (option B). The patient's symptoms, including lack of energy, reduced sleep, appetite loss, suicidal thoughts, and low self-esteem, are hallmark signs of a depressive disorder.

Depressive Disorder
Definition: A mood disorder characterized by persistent feelings of sadness, lack of interest in activities, and other emotional and physical symptoms.
Key Clinical Presentation: Depressed mood, loss of interest, fatigue, sleep disturbances, appetite changes, and suicidal ideation.
Investigations: Diagnosis is clinical, based on DSM-5 criteria.
Treatment Outline:

- **First-line:** SSRIs (e.g., sertraline), CBT.
- **Second-line:** SNRIs, tricyclic antidepressants (TCAs) if SSRIs are ineffective.

Correct Answer: B. Depressive disorders ✅

MCQ 130:

A 28-year-old man was brought to the Emergency Department complaining that his neighbors are trying to harm him and are plotting to kill him. On questioning, he was very sure about this and said that he is still hearing their voices talking with each other, though none of them were present. Which of the following is the most likely diagnosis?

A. Cognitive and perceptive disturbance
B. Depersonalization and derealization
C. Obsessive-compulsive disorder
D. Hallucinations and delusions

Explanation:
The most likely diagnosis is **Hallucinations and delusions** (option D). The patient exhibits symptoms of paranoia and auditory hallucinations, both of which are indicative of delusional thinking and psychosis.

Psychosis
Definition: A mental health condition characterized by a disconnection from reality, often involving delusions and hallucinations.
Key Clinical Presentation: Delusions (false beliefs) and hallucinations (false perceptions), often in a psychotic state.
Investigations: Clinical diagnosis, possibly supported by imaging (e.g., CT/MRI), and psychological assessment.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, haloperidol).
- **Second-line:** Psychotherapy, social support.

Correct Answer: D. Hallucinations and delusions ✅

MCQ 131:

A 26-year-old woman presents to the Psychiatric Outpatient Clinic with a history of psychomotor excitation and irritability. She has grandiose delusions and auditory hallucinations. On a mental state examination, she exhibited a lack of insight and flight of ideas. Which of the following disorders is the most likely diagnosis?

A. Neurosis
B. Psychoses
C. Personality
D. Dissociative conversion

Explanation:
The most likely diagnosis is **Psychoses** (option B). The patient exhibits symptoms of psychosis, including grandiose delusions, auditory hallucinations, psychomotor agitation, and flight of ideas, which are typical signs of a psychotic disorder such as **Schizophrenia** or **Bipolar Disorder with psychotic features**.

Psychoses
Definition: A severe mental disorder characterized by a disconnection from reality, involving hallucinations and delusions.
Key Clinical Presentation: Delusions, hallucinations, disorganized thinking, and impaired insight.
Investigations: Clinical diagnosis based on psychiatric evaluation and observation.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., olanzapine, risperidone).

- **Second-line:** Cognitive Behavioral Therapy (CBT), social support.

Correct Answer: B. Psychoses ✅

MCQ 132:

A 32-year-old man presents to the clinic with a 2-year history of getting orgasm too readily during sexual intercourse. He complains that during sexual intercourse, he has palpitations and worrying thoughts in his mind all the time. On examination, he has normal findings. Which ofthe following is the most likely diagnosis?

A. Anxiety disorder
B. Depressive disorder
C. Bipolar affective disorder
D. Primary erectile dysfunction

Explanation:
The most likely diagnosis is **Anxiety disorder** (option A). The patient’s symptoms, including palpitations, worrying thoughts, and premature orgasm, point toward an anxiety-related disorder, possibly **performance anxiety**. His focus on worrying during sexual activity indicates **psychological factors** contributing to his symptoms.

Anxiety Disorder
Definition: A mental health disorder characterized by excessive worry, nervousness, and physical symptoms such as palpitations.
Key Clinical Presentation: Excessive worry, physical manifestations like palpitations, shortness of breath, and irritability.
Investigations: Diagnosis is clinical, based on the patient's history and symptoms.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT), selective serotonin reuptake inhibitors (SSRIs).
- **Second-line:** Benzodiazepines for short-term relief.

Correct Answer: A. Anxiety disorder ✅

MCQ 133:

A 16-year-old girl presents for a follow-up visit complaining that she is unable to eat because of an intense desire to reduce her weight. Sometimes she develops an impulse to eat excessively and she then follows the impulse. She intermittently self-induces vomiting. On examination, she looks weak. Which of the following is the most likely diagnosis?

A. Bulimia nervosa
B. Anorexia nervosa
C. Personality disorder
D. Obsessive compulsive disorder

Explanation:
The most likely diagnosis is **Bulimia nervosa** (option A). The patient displays characteristic features of bulimia nervosa, including cycles of binge eating followed by compensatory behaviors like self-induced vomiting to prevent weight gain.

Bulimia Nervosa
Definition: An eating disorder characterized by episodes of binge eating followed by inappropriate behaviors to prevent weight gain (e.g., vomiting, excessive exercise).
Key Clinical Presentation: Recurrent episodes of binge eating, inappropriate compensatory behaviors (e.g., vomiting), and fear of weight gain.
Investigations: Diagnosis is clinical, supported by lab tests (e.g., electrolyte imbalances) and physical exam.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT), antidepressants (SSRIs).
- **Second-line:** Nutritional counseling and family therapy.

Correct Answer: A. Bulimia nervosa ✅

MCQ 134:

A 20-year-old student presents to the Psychiatric Outpatient Clinic complaining that she feels embarrassed, dizzy, and tremulous whenever she tries to answer verbally in the classroom. Which of the following is the best treatment?

A. Small doses of alprazolam
B. Behavioral therapy
C. Relaxation training
D. Bupropion

Explanation:
The best treatment is **Behavioral therapy** (option B), specifically **Cognitive Behavioral Therapy (CBT)**. This approach is highly effective for addressing performance anxiety and improving social confidence.

Social Anxiety Disorder
Definition: A psychological condition characterized by intense fear of social situations due to a fear of being judged or humiliated.
Key Clinical Presentation: Avoidance of social situations, embarrassment, physical symptoms like dizziness and sweating in social or performance settings.
Investigations: Diagnosis is clinical, based on the history of avoidance and anxiety in social situations.
Treatment Outline:

- **First-line:** Cognitive Behavioral Therapy (CBT).
- **Second-line:** SSRIs, benzodiazepines for short-term anxiety relief.

Correct Answer: B. Behavioral therapy ✅

MCQ 135

A 21-year-old man with depression presents for a follow-up visit. During the consultation, the psychiatrist asked, "Do you think that you are mentally ill?" Which of the following findings is the psychiatrist trying to assess?

A. Insight
B. Judgment
C. Compliance
D. Guilt feeling

Explanation:
The psychiatrist is assessing **Insight** (option A), which refers to the patient's understanding of their illness and its impact on their life. Insight into one's mental health is crucial in determining the course of treatment and its effectiveness.

Insight
Definition: The ability of an individual to understand and acknowledge their mental health condition and its effects on their behavior.
Key Clinical Presentation: Ability to recognize symptoms and their connection to the disorder.

Investigations: Assessed through clinical interviews and questioning.
Treatment Outline:

- **First-line:** Supportive psychotherapy to improve insight.
- **Second-line:** Psychoeducation and medications for managing symptoms.

Correct Answer: A. Insight ✅

MCQ 136:

A 23-year-old college student was brought to the Psychiatric Outpatient Clinic with a 3-month history of persecutory delusions, hearing voices commenting on his actions, and disorganized behavior without disturbance of mood, then returning to normal with no medication. Which of the following is the most likely diagnosis?

A. Schizophreniform disorder
B. Brief psychotic disorder
C. Delusional disorder
D. Schizophrenia

Explanation:
The most likely diagnosis is **Brief psychotic disorder** (option B). The patient's symptoms, including persecutory delusions, auditory hallucinations, and disorganized behavior, which resolve within a month and return to normal without medication, are characteristic of brief psychotic disorder. The key feature is that the episode lasts for at least one day but less than a month and returns to baseline.

Brief Psychotic Disorder
Definition: A sudden onset of psychotic symptoms lasting for at least 1 day but less than 1 month, often triggered by a stressful event.
Key Clinical Presentation: Delusions, hallucinations, disorganized speech or behavior.
Investigations: Clinical diagnosis; no laboratory test required.
Treatment Outline:

- **First-line:** Antipsychotic medications, psychotherapy.
- **Second-line:** Supportive care and stress management.

Correct Answer: B. Brief psychotic disorder ✅

MCQ 137:

A 34-year-old man presents to the clinic for a routine visit. During the consultation, he speaks loudly and rapidly without completely expressing one thought before beginning another. Speech patterns are rhyming and the patient is easily distracted by cues in the immediate surroundings. Which of the following is the most likely description for these findings?

A. Flight of ideas
B. Perseveration
C. Thought insertion
D. Loosening of associations

Explanation:
The most likely description is **Flight of ideas** (option A). Flight of ideas is characterized by rapid speech with frequent shifts from one topic to another, often accompanied by a loose connection between thoughts. It is commonly seen in manic episodes and some psychotic disorders.

Flight of Ideas
Definition: A rapid flow of thoughts with shifting from one topic to another, often making it difficult to follow the speaker.
Key Clinical Presentation: Rapid, pressured speech, fragmented thoughts, distractibility.
Investigations: Clinical diagnosis; assessment through mental status examination.
Treatment Outline:

- **First-line:** Mood stabilizers (e.g., lithium), antipsychotics for psychotic symptoms.
- **Second-line:** Psychotherapy, especially cognitive behavioral therapy.

Correct Answer: A. Flight of ideas ✅

MCQ 138

A 22-year-old man presents to the clinic with a history of approaching strangers and asking them the same inappropriate question without stopping when instructed to do so. The patient seemed unaware of the fact that this was an inappropriate behavior. Which of the following is the most likely description for this behavior?

A. Flight Of ideas
B. Perseveration
C. Thought insertion
D. Loosening of associations

Explanation:
The most likely description is **Perseveration** (option B). Perseveration refers to the repetitive and involuntary continuation of a thought or action, even when it is no longer appropriate or relevant. This can be seen in certain neurological conditions, including schizophrenia and brain injury.

Perseveration
Definition: The repetition of a particular response (such as a word, phrase, or action) despite the cessation of the stimulus.
Key Clinical Presentation: Inability to stop a particular thought or action despite being instructed to do so.
Investigations: Clinical diagnosis, with possible neuroimaging or neurological examination.
Treatment Outline:

- **First-line:** Cognitive behavioral therapy (CBT) to address compulsive behaviors.
- **Second-line:** Medications like antipsychotics in psychotic disorders.

Correct Answer: B. Perseveration ✅

MCQ 139;

A mother brings her 2-week-old infant to the Well-Baby Clinic for a check-up. During the visit, the physician found the child to be normal but the mother gives inappropriate answers to the questions. She appeared disoriented in time and date. She states that the child is deformed and unlikely to live long. Which of the following is the most likely diagnosis?

A. Postnatal non-bonding
B. Postpartum psychosis
C. Neurosis
D. Fatigue

Explanation:
The most likely diagnosis is **Postpartum psychosis** (option B). The mother exhibits disorientation, inappropriate responses, and delusional thinking about the child, which are characteristic features of postpartum psychosis. This condition involves severe mood changes, hallucinations, and delusions following childbirth.

Postpartum Psychosis
Definition: A severe mental illness that can occur after childbirth, characterized by delusions, hallucinations, and severe mood swings.

Key Clinical Presentation: Disorientation, delusions, hallucinations, manic or depressive symptoms.
Investigations: Clinical diagnosis, psychiatric evaluation.
Treatment Outline:

- **First-line:** Antipsychotics (e.g., risperidone), mood stabilizers (e.g., lithium).
- **Second-line:** Electroconvulsive therapy (ECT) in severe cases.

Correct Answer: B. Postpartum psychosis ✅

MCQ 140:

A 34-year-old woman presents to the clinic for postpartum care. She has experienced tearfulness and irritability during the 1 week following the delivery of her child. This episode resolved spontaneously and her mood has been normal since then. Which of the following is the most likely diagnosis?

A. Maternity blues
B. Postnatal depression
C. Postpartum psychosis
D. Postnatal non-bonding

Explanation:
The most likely diagnosis is **Maternity blues** (option A). The symptoms of tearfulness and irritability occurring within the first week postpartum, which resolve spontaneously, are typical of maternity blues. This condition affects many women following childbirth and usually resolves within a few days.

Maternity Blues
Definition: A mild mood disturbance characterized by tearfulness, irritability, and emotional instability in the first few days after childbirth.
Key Clinical Presentation: Tearfulness, irritability, mood swings, usually resolving within 2 weeks postpartum.
Investigations: Clinical diagnosis based on symptoms.
Treatment Outline:

- **First-line:** Supportive care, reassurance.
- **Second-line:** In severe cases, short-term use of antidepressants.

Correct Answer: A. Maternity blues ✅

MCQ 140:

Which of the following conditions is most often associated with manic behaviour?

A. Paranoid schizophrenia
B. Bipolar affective disorder
C. Grandiose mood disorder
D. Generalized anxiety disorder

Explanation:
The most likely diagnosis is **Bipolar affective disorder** (option B). Manic behavior, characterized by elevated mood, increased activity, and impulsivity, is a hallmark of bipolar disorder, especially during manic episodes.

Bipolar Affective Disorder
Definition: A mood disorder characterized by alternating episodes of mania (elevated mood, excessive activity) and depression.
Key Clinical Presentation: Episodes of mania, depressive episodes, irritability, and impaired judgment.
Investigations: Clinical diagnosis based on history and mental status examination.
Treatment Outline:

- **First-line:** Lithium, anticonvulsants, or antipsychotics.
- **Second-line:** Antidepressants (with caution during manic episodes).

Correct Answer: B. Bipolar affective disorder ✅

MCQ 141

A 21-year-old woman presents to the clinic with history of changes in the voice that she hears. Until now, it had been merely making critical comments about everything she did. For the past 2 weeks, however, she thinks the voices seemed to have been able to "suck thoughts out" of her and send them to others. Which of the following is the most likely diagnosis?

A. Mania
B. Agoraphobia
C. Schizophrenia
D. Mood disorder

Explanation:
The most likely diagnosis is **Schizophrenia** (option C). The patient's auditory hallucinations, where she believes the voices are able to manipulate her thoughts, are consistent with the psychotic features of schizophrenia. Schizophrenia often presents with hallucinations, delusions, and disorganized thinking.

Schizophrenia
Definition: A chronic mental health disorder characterized by symptoms such as hallucinations, delusions, disorganized thinking, and cognitive dysfunction.
Key Clinical Presentation: Auditory hallucinations, delusions, impaired reality testing.
Investigations: Clinical diagnosis, often supported by psychiatric evaluation.
Treatment Outline:

- **First-line:** Antipsychotics (e.g., risperidone, olanzapine).
- **Second-line:** Psychosocial interventions and cognitive behavioral therapy (CBT).

Correct Answer: C. Schizophrenia ✅

MCQ 142:

A 26-year-old man presents to the clinic with history of unusual behavior. He keeps a blanket over the television believing that the government uses the actors to spy through the TV. He believes that the instructions to cover the TV came from God who spoke through the light bulb. Which of the following is the most likely diagnosis?

A. Schizophrenia
B. Agoraphobia
C. Dementia
D. Mania

Explanation:
The most likely diagnosis is **Schizophrenia** (option A). The patient is experiencing delusions (the belief that the government is spying through the TV and receiving instructions from God through a light bulb), which are a hallmark of schizophrenia.

Schizophrenia
Definition: A severe mental illness that affects how a person thinks, feels, and behaves, often characterized by delusions and hallucinations.
Key Clinical Presentation: Paranoia, delusions, hallucinations, disorganized speech.

Investigations: Diagnosis based on clinical presentation, mental status examination.
Treatment Outline:

- **First-line:** Antipsychotics (e.g., haloperidol, clozapine).
- **Second-line:** Supportive therapy, rehabilitation.

Correct Answer: A. Schizophrenia ✅

MCQ 143:

A 25-year-old healthy man complains of episodes of chest tightness, palpitations, dizziness and breathlessness. These episodes resolved after about 5 minutes. He remains normal between episodes and the interval may extend for several days before a new episode occurs. He has had constant anxiety about having another attack and feels sad about the situation. The patient cannot identify a triggering mechanism. Which of the following is the most likely diagnosis?

A. Generalized anxiety disorder
B. Major depressive disorder
C. Phobic reaction
D. Panic disorder

Explanation:
The most likely diagnosis is **Panic disorder** (option D). Panic disorder is characterized by recurrent, unexpected panic attacks, which include symptoms like chest tightness, palpitations, dizziness, and breathlessness. The patient also exhibits anxiety about having future attacks.

Panic Disorder
Definition: Recurrent and unexpected panic attacks, often accompanied by intense fear and physical symptoms such as palpitations and shortness of breath.
Key Clinical Presentation: Sudden onset of chest pain, palpitations, dizziness, and anxiety.
Investigations: Diagnosis is based on clinical evaluation and assessment of symptoms.
Treatment Outline:

- **First-line:** Cognitive behavioral therapy (CBT), selective serotonin reuptake inhibitors (SSRIs).
- **Second-line:** Benzodiazepines for acute relief.

Correct Answer: D. Panic disorder ✅

MCQ 144

A 30-year-old man gives a history of pancreatic carcinoma and reports having lost all his prescriptions. He says that despite having recently completed chemotherapy he has developed new symptoms and is afraid of a recurrence. There is no medical records for him in the hospitals. When confronted with this information, the patient ran away from the clinic. Which of the following is the most likely diagnosis?

A. Conversion disorder
B. Factitious disorder
C. Hypochondriasis
D. Malingering

Explanation:
The most likely diagnosis is **Factitious disorder** (option B). This disorder involves the intentional production or feigning of symptoms for the purpose of assuming the sick role. The patient's behavior, including running away from the clinic when confronted with the lack of medical records, suggests factitious disorder.

Factitious Disorder
Definition: A condition in which a person intentionally produces or feigns physical or psychological symptoms to gain attention or sympathy.
Key Clinical Presentation: Falsification of medical history or symptoms, often for secondary gain such as attention.
Investigations: Diagnosis is primarily clinical, with a detailed medical history and evaluation.
Treatment Outline:

- **First-line:** Psychological therapy, including cognitive behavioral therapy (CBT).
- **Second-line:** Medication management for co-occurring psychiatric disorders.

Correct Answer: B. Factitious disorder ✅

MCQ 145:

A 40-year-old woman presents to the clinic with 2-month history of a sleep disturbance, loss of interest, low energy, poor memory, and suicidal ideation. On examination, she looks weak

and inactive. Laboratory investigations are normal. Which of the following is the most likely diagnosis?

A. Dysthymia
B. Bipolar disorder
C. Major depression
D. Minor depression

Explanation:
The most likely diagnosis is **Major depression** (option C). The patient's symptoms, including sleep disturbance, loss of interest, low energy, poor memory, and suicidal ideation, are characteristic of a major depressive episode. These symptoms have persisted for more than two weeks, which fits the diagnostic criteria for major depressive disorder.

Major Depression
Definition: A mood disorder characterized by persistent feelings of sadness, hopelessness, and loss of interest in activities that were previously enjoyed.
Key Clinical Presentation: Depressed mood, fatigue, sleep disturbances, poor concentration, and suicidal thoughts.
Investigations: Diagnosis is clinical, but lab investigations may be done to rule out other conditions.
Treatment Outline:

- **First-line:** Antidepressants (SSRIs, SNRIs) + psychotherapy (CBT).
- **Second-line:** TCA (tricyclic antidepressants) or other classes of antidepressants if first-line medications fail.
- **Definitive:** Long-term maintenance therapy to prevent relapse.

Correct Answer: C. Major depression ✅

MCQ 146:

A 19-year-old healthy man is brought to the clinic for evaluation of progressive erratic behaviour. Over the past 4 months, he has believed that people on television are reading his thoughts. He expressed the presence of supernatural powers and that he has begun to act out in response to these ideations. He has not caused injuries, but the trends are becoming more violent. On examination, he is normal. Which of the following is the most likely diagnosis?

A. Schizophrenia
B. Conversion disorder

C. Rapid cycling depression
D. Manic phase of bipolar disorder

Explanation:
The most likely diagnosis is **Schizophrenia** (option A). The patient's delusions (belief that people on television are reading his thoughts and the presence of supernatural powers), along with the onset of erratic and violent behavior, are indicative of schizophrenia. Schizophrenia is characterized by positive symptoms (delusions, hallucinations) and a decline in functioning.

Schizophrenia
Definition: A severe, chronic mental disorder characterized by disturbances in thought processes, perception, emotional regulation, and behavior.
Key Clinical Presentation: Delusions, hallucinations, disorganized thinking and behavior, social withdrawal.
Investigations: Diagnosis is clinical based on symptoms and history.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, clozapine).
- **Second-line:** Cognitive-behavioral therapy (CBT), family therapy.

Correct Answer: A. Schizophrenia ✅

MCQ 147:

Which of the following is the most appropriate management plan for major depression?

A. Select any drug to start as all have equal efficacy
B. Initiate therapy with one medication for 2 weeks
C. Stop medication with symptomatic improvement
D. Change medications if the patient doesn’t respond

Explanation:
The correct answer is **D. Change medications if the patient doesn’t respond**. In the management of major depression, if there is no response to the initial antidepressant after an adequate trial (usually 6-8 weeks), it is appropriate to change medications. Antidepressants often take several weeks to show their full effect, so therapy should not be stopped prematurely.

Major Depression
Definition: A mood disorder characterized by persistent feelings of sadness, hopelessness, and loss of interest in activities.

Key Clinical Presentation: Depressed mood, fatigue, sleep disturbances, poor concentration, suicidal thoughts.
Investigations: Diagnosis is clinical.
Treatment Outline:

- **First-line:** SSRIs or SNRIs.
- **Second-line:** TCAs or other antidepressants if first-line treatments fail.
- **Definitive:** Long-term treatment and monitoring to prevent relapse.

Correct Answer: D. Change medications if the patient doesn't respond ✅

MCQ 148:

Which of the following systems is stimulated by stressful situations?

A. Urinary
B. Respiratory
C. Sympathetic
D. Parasympathetic

Explanation:
The correct answer is **C. Sympathetic**. In response to stress, the sympathetic nervous system is activated, which is responsible for the "fight or flight" response. This leads to increased heart rate, blood pressure, and energy availability.

Sympathetic Nervous System
Definition: Part of the autonomic nervous system that mediates the body's response to stress, preparing the body for rapid action.
Key Clinical Presentation: Increased heart rate, dilated pupils, sweating, and elevated blood pressure.
Investigations: No specific investigations; observed in response to stress.
Treatment Outline:

- **First-line:** Stress management techniques, relaxation, and physical exercise.
- **Second-line:** Medication for chronic stress or anxiety (e.g., beta-blockers, anxiolytics).

Correct Answer: C. Sympathetic ✅

MCQ 149:

Stress leads to an increase in cortisol and catecholamine secretion that can alter the levels of lymphocyte production that fight disease. Which of the following is the most likely explanation for these effects?

A. Interpretation of the sitting
B. Psychoneuroimmunology
C. Vaccination effect
D. Personal effects

Explanation:
The correct answer is **B. Psychoneuroimmunology**. This field studies the interaction between psychological processes, the nervous system, and immune function. Stress activates the sympathetic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis, which in turn influences immune response.

Psychoneuroimmunology
Definition: The study of the interaction between the mind (psychology), the nervous system, and the immune system.
Key Clinical Presentation: Stress-induced changes in immune function, which can impact susceptibility to disease.
Investigations: Stress tests, cortisol levels, and immune function tests.
Treatment Outline:

- **First-line:** Stress reduction, relaxation techniques, mindfulness.
- **Second-line:** Cognitive behavioral therapy (CBT) for managing chronic stress.

Correct Answer: B. Psychoneuroimmunology ✅

MCQ 150:

Which of the following is the main difference between delirium and dementia?

A. Disturbance of consciousness
B. Multiple cognitive deficits
C. Memory impairment
D. Aphasia

Explanation:
The correct answer is **A. Disturbance of consciousness**.
Delirium is characterized by an acute disturbance in attention and awareness, with a fluctuating

course, often triggered by a medical illness or drug use. Dementia, on the other hand, involves chronic cognitive decline without the acute disturbance of consciousness.

Delirium vs. Dementia
Definition:

- **Delirium:** Acute confusion characterized by impaired attention, fluctuating consciousness, and cognitive dysfunction.
- **Dementia:** Chronic, progressive decline in cognitive function, typically without an acute disturbance in consciousness.

Key Clinical Presentation:

- **Delirium:** Acute onset, fluctuating course, disorientation, altered consciousness.
- **Dementia:** Progressive memory loss, poor judgment, impaired cognitive function, no acute confusion.

Investigations:

- **Delirium:** Clinical diagnosis, laboratory tests to rule out metabolic, infectious causes.
- **Dementia:** Neuroimaging, neuropsychological testing.

Treatment Outline:

- **Delirium:** Treat underlying cause (infection, metabolic imbalance).
- **Dementia:** Symptom management (cholinesterase inhibitors, environmental modifications).

Correct Answer: A. Disturbance of consciousness ✅

MCQ 151:

A 72-year-old man experiences inability to sustain attention, disorganized thinking, hallucinations, and disorientation commencing 12 hours after an aortic femoral popliteal bypass. The symptoms fluctuated during the next 2 days. Which of the following is the most likely diagnosis?

A. Multi-infarct dementia
B. Depression
C. Delirium
D. Mania

Explanation:
The correct answer is **C. Delirium**.
Delirium is an acute confusional state that can occur after surgery, especially in elderly patients. It is characterized by fluctuating consciousness, impaired attention, and disorganized thinking. The fact that symptoms began post-operatively and fluctuate in nature supports the diagnosis of delirium.

Delirium
Definition: Acute onset of confusion, attention deficit, disorganized thinking, and altered consciousness.
Key Clinical Presentation: Sudden onset, fluctuating mental state, hallucinations, disorientation.
Investigations: Clinical diagnosis, rule out infections, metabolic disturbances, and medications.
Treatment Outline:

- **First-line:** Treat underlying causes (e.g., infections, metabolic abnormalities).
- **Second-line:** Supportive care, antipsychotic medications for agitation if necessary.

Correct Answer: C. Delirium ✅

MCQ 152:

A 70-year-old man with Alzheimer's dementia becomes agitated and aggressive after developing hallucinations and delusions. A complete evaluation does not reveal any environmental, medical, or psychiatric cause and behavioural interventions failed to improve the agitation. Which of the following is the most appropriate next step in management?

A. Thioridazine
B. Haloperidol
C. Olanzapine
D. Buspirone

Explanation:
The correct answer is **C. Olanzapine**.
In Alzheimer's dementia, if agitation, aggression, or psychotic symptoms (like delusions and hallucinations) are present, atypical antipsychotics such as **Olanzapine** are often used. These medications are effective in managing these symptoms and have a better safety profile than typical antipsychotics like haloperidol, which are associated with more side effects in elderly patients.

Alzheimer's Dementia Management
Definition: A neurodegenerative disorder marked by cognitive decline, memory impairment, and behavioral changes.
Key Clinical Presentation: Memory loss, confusion, and changes in behavior, including aggression and agitation in advanced stages.
Investigations: Clinical diagnosis, neuroimaging, neuropsychological testing.
Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (Donepezil) for cognitive symptoms.
- **Second-line:** Atypical antipsychotics (Olanzapine, Quetiapine) for agitation and psychotic symptoms.
- **Definitive:** Symptom management, including non-pharmacological approaches.

Correct Answer: C. Olanzapine ✅

MCQ 153

A 78-year-old man is brought to the Outpatient Clinic with a 2-year history of slow and progressive deterioration in his memory. He is able to perform some activities of daily living, including dressing and bathing as well as cooking. He has personality changes, once a kind and caring parent he now displays periods of both agitation and aggression. Neurologic and musculoskeletal examination cannot be carried out. Which of the following is the best management?

A. Cost-effective laboratory investigation
B. Prescribe Risperidone for the patient
C. Admission to a chronic care facility
D. Refer to a geriatrician

Explanation:
The correct answer is **D. Refer to a geriatrician**.
Given the patient's slow progressive memory loss, personality changes, and agitation, it is likely that he has dementia, possibly Alzheimer's disease or another form of neurodegenerative disorder. A referral to a geriatrician is appropriate for further assessment, management, and to consider pharmacological and non-pharmacological interventions.

Dementia Management
Definition: A group of symptoms affecting memory, thinking, and social abilities, which interfere with daily functioning.

Key Clinical Presentation: Gradual memory loss, personality changes, and cognitive decline.
Investigations: Neuroimaging (CT, MRI), neuropsychological testing.
Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (Donepezil, Rivastigmine).
- **Second-line:** Antipsychotics for agitation or hallucinations, under careful supervision.
- **Definitive:** Long-term care planning, including geriatric support.

Correct Answer: D. Refer to a geriatrician ✅

MCQ 154:

A 30-year-old man presents to the Outpatient Clinic with manifestations of opioid withdrawal. After full investigations, he was diagnosed as a heroin addict. As part of the rehabilitation process, he was prescribed a substitute for the addictive drug. Which of the following drugs should be prescribed?

A. Naloxone
B. Papaverine
C. Methadone
D. Dextromethorphane

Explanation:
The correct answer is **C. Methadone**.
Methadone is a long-acting opioid agonist used in the management of opioid dependence. It helps reduce cravings and withdrawal symptoms in individuals who are addicted to opioids like heroin. It is used in maintenance therapy and in opioid substitution programs.

Opioid Use Disorder Management
Definition: A chronic disease characterized by the compulsive use of opioids despite harmful consequences.
Key Clinical Presentation: Cravings, withdrawal symptoms, and continued opioid use.
Investigations: Clinical evaluation, urine drug screen.
Treatment Outline:

- **First-line:** Methadone, buprenorphine for opioid substitution therapy.
- **Second-line:** Cognitive-behavioral therapy (CBT), contingency management.
- **Definitive:** Long-term rehabilitation programs, harm reduction strategies.

Correct Answer: C. Methadone ✅

MCQ 155:

Which of the following is described as "False belief not in accordance with a person's intelligence or culture"?

A. Ellusion
B. Delusion
C. Hallucination
D. Somatisation

Explanation:
The correct answer is **B. Delusion**.
A delusion is a false belief that is strongly held despite evidence to the contrary and is not aligned with the individual's intelligence or cultural background. Delusions are commonly seen in psychiatric disorders such as schizophrenia.

Delusion
Definition: A false belief that is not based on reality and is inconsistent with the individual's cultural background or education.
Key Clinical Presentation: False beliefs, often paranoid, grandiose, or persecutory.
Investigations: Clinical evaluation, psychiatric assessment.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., Olanzapine, Risperidone).
- **Second-line:** Cognitive behavioral therapy for managing symptoms.

Correct Answer: B. Delusion ✅

MCQ 156:

A 65-year-old man with congestive heart failure and osteoarthritis moved into a nursing home 3 months ago. During the past 4 weeks, the patient has lost 3.6 kg, lost interest in all social activities, and has been crying almost every day. The mood is worse in the morning. At times, especially when the mood is impaired, the patient's short-term memory appears to be lost. Which of the following is the most likely diagnosis?

A. Multi-infarct dementia
B. Alzheimer’s disease
C. Hypothyroidism
D. Depression

Explanation:
The correct answer is **D. Depression**.
This patient exhibits classic symptoms of depression, including weight loss, loss of interest in activities, crying, and morning mood worsening. Depression can also cause cognitive symptoms like impaired memory, which may mimic dementia, but this patient's symptoms are more consistent with depression.

Depression
Definition: A mood disorder characterized by persistent sadness, loss of interest in activities, and impaired functioning.
Key Clinical Presentation: Depressed mood, weight loss, insomnia or hypersomnia, cognitive impairment.
Investigations: Clinical diagnosis, screening tools like the PHQ-9.
Treatment Outline:

- **First-line:** Antidepressants (SSRIs like Sertraline, or SNRIs).
- **Second-line:** Cognitive behavioral therapy (CBT), psychotherapy.

Correct Answer: D. Depression ✅

MCQ 157:

A 19-year-old girl presents with a 12 kg weight loss over the last 6 months. She feels fat with an intense fear of gaining weight. Over the last 4 months, the patient has had amenorrhea. On examination, she has lanugo hair distribution. Height 167 cm weight 42 kg. Which of the following is the most likely diagnosis?

A. Bulimia
B. Anorexia nervosa
C. Masked depression
D. Generalized anxiety disorder

Explanation:
The correct answer is **B. Anorexia nervosa**.
Anorexia nervosa is characterized by intense fear of gaining weight, significant weight loss,

amenorrhea, and body image disturbance. The presence of lanugo hair is a typical finding in anorexia nervosa due to the body's attempt to keep warm in the face of malnutrition.

Anorexia Nervosa
Definition: A psychiatric eating disorder marked by severe restriction of food intake, an intense fear of gaining weight, and body image distortion.
Key Clinical Presentation: Significant weight loss, intense fear of gaining weight, amenorrhea, lanugo.
Investigations: Clinical evaluation, BMI calculation, laboratory tests for nutritional deficiencies.
Treatment Outline:

- **First-line:** Cognitive behavioral therapy (CBT), nutritional rehabilitation.
- **Second-line:** Antidepressants (SSRIs) for comorbid anxiety/depression.

Correct Answer: B. Anorexia nervosa ✅

MCQ 158:

A 29-year-old woman presents to the clinic with recurrent attacks of palpitations, chest pain, difficulty in breathing, and a fear of dying. She also trembles, experiences numbness and tingling sensations in the hands and feet. The past medical history is not significant. On examination, she looks tense, and the hands are cold with fine tremors. Heart rate 108/min, regular Temperature 37.2°C Blood pressure 120/82 mmHg Respiratory rate 14/min. Which of the following is the most likely diagnosis?

A. Pheochromocytoma
B. Hyperthyroidism
C. Panic disorder
D. Agoraphobia

Explanation:
The correct answer is **C. Panic disorder**.
The patient presents with recurrent episodes of intense fear or discomfort, physical symptoms like palpitations, chest pain, breathlessness, and tingling, all of which are characteristic of panic attacks, a hallmark of panic disorder.

Panic Disorder
Definition: A psychiatric disorder characterized by recurrent, unexpected panic attacks and persistent concern about having additional attacks.
Key Clinical Presentation: Rapid onset of symptoms, physical signs like tachycardia, sweating,

breathlessness, and fear of dying during the attacks.
Investigations: Clinical diagnosis based on symptom criteria (DSM-5).
Treatment Outline:

- **First-line:** SSRIs (Sertraline, Fluoxetine) and cognitive behavioral therapy (CBT).
- **Second-line:** Benzodiazepines for acute management of attacks.

Correct Answer: C. Panic disorder ✅

MCQ 159

A 29-year-old man presents with a 2-year history of intense fear before giving classes as a teacher in a secondary school. He claims that it is only a matter of time before a major teaching mistake is made. Which of the following phobias is the most likely diagnosis?

A. Social
B. Mixed
C. Specific
D. Agoraphobia

Explanation:
The correct answer is **A. Social phobia**.
Social phobia (or social anxiety disorder) involves a fear of being judged or scrutinized in social situations, such as public speaking or performing tasks in front of others. The patient's fear of making mistakes while teaching is typical of social anxiety.

Social Phobia
Definition: A psychiatric condition characterized by an intense fear of social situations and potential embarrassment or negative evaluation.
Key Clinical Presentation: Fear of public speaking, eating in front of others, social gatherings, or performing in front of others.
Investigations: Clinical evaluation, often using the Social Phobia Inventory (SPIN).
Treatment Outline:

- **First-line:** Cognitive behavioral therapy (CBT), SSRIs (Sertraline).
- **Second-line:** Beta-blockers for performance anxiety, benzodiazepines for acute anxiety.

Correct Answer: A. Social ✅

MCQ 160:

A 52-year-old woman presents with a 5-day history of insomnia and crying because of the death of a sibling. The patient has not slept for the last 2 days. Which of the following is the most appropriate short course of therapy?

A. Fluoxetine
B. Imipramine
C. Lorazepam
D. Chlorpromazine

Explanation:
The correct answer is **C. Lorazepam**.
Given the short duration of symptoms following the death of a sibling, the patient may be experiencing acute grief. The most appropriate short-term intervention for insomnia and distress caused by grief would be a short-acting benzodiazepine like lorazepam, which provides immediate relief for anxiety and sleep disturbances. Antidepressants like fluoxetine are not typically indicated immediately after a bereavement unless the symptoms evolve into a major depressive episode.

Acute Grief Reaction
Definition: The emotional response to a significant loss, which can lead to symptoms such as sadness, insomnia, and crying.
Key Clinical Presentation: Loss of interest, crying, insomnia, sadness, and distress post-loss.
Investigations: Clinical evaluation.
Treatment Outline:

- **First-line:** Supportive therapy and grief counseling.
- **Second-line:** Short-term use of benzodiazepines (e.g., Lorazepam) for sleep or anxiety.

Correct Answer: C. Lorazepam ✅

MCQ 161:

Which of the following factors indicates a poor prognosis for schizophrenia?

A. Acute onset of psychosis
B. Onset during adolescence

C. Family history of schizophrenia
D. Anxiety during psychotic episodes

Explanation:
The correct answer is **B. Onset during adolescence**.
Schizophrenia that presents during adolescence is generally associated with a poorer prognosis, as it is typically more severe and is linked with more significant cognitive and functional impairments. An acute onset of psychosis (Option A) generally has a better prognosis compared to a gradual onset, while a family history (Option C) does not directly correlate with prognosis. Anxiety during episodes (Option D) is not a key indicator of prognosis.

Schizophrenia
Definition: A chronic, severe mental disorder that affects thinking, emotional regulation, and behavior.
Key Clinical Presentation: Delusions, hallucinations, disorganized speech, and negative symptoms such as flat affect.
Investigations: Clinical diagnosis based on DSM-5 criteria, psychiatric evaluation.
Treatment Outline:

- **First-line:** Antipsychotics (Olanzapine, Risperidone).
- **Second-line:** Cognitive behavioral therapy (CBT) for managing symptoms.

Correct Answer: B. Onset during adolescence ✅

MCQ 162:

A 72-year-old man who lost his spouse 2 months ago after a prolonged illness presents with a sad mood, decreased appetite, and sleep disturbance. Which of the following is the most likely diagnosis?

A. Dysthymia
B. Bereavement
C. Major depression
D. Minor depression

Explanation:
The correct answer is **B. Bereavement**.
Given the patient's recent loss of a spouse, the most likely diagnosis is an appropriate grief response, or bereavement. While the symptoms may overlap with depression, bereavement is a

normal reaction to loss and does not typically require antidepressant treatment unless the symptoms evolve into major depression.

Bereavement
Definition: The period of mourning and adjustment after the death of a loved one, which may cause sadness, sleep disturbance, and changes in appetite.
Key Clinical Presentation: Sadness, mourning, loss of appetite, sleep disturbances following a significant loss.
Investigations: Clinical evaluation.
Treatment Outline:

- **First-line:** Supportive therapy, grief counseling.
- **Second-line:** Antidepressants if symptoms progress to major depressive disorder.

Correct Answer: B. Bereavement ✅

MCQ 163:

Which of the following drug mechanisms of action is most appropriate to treat obsessional behavioural disorder?

A. Blocks serotonin production
B. Increases the availability of serotonin
C. Increases the production of serotonin
D. Decreases the availability of serotonin

Explanation:
The correct answer is **B. Increases the availability of serotonin**.
Obsessive-compulsive disorder (OCD) is most commonly treated with selective serotonin reuptake inhibitors (SSRIs), which increase the availability of serotonin in the synaptic cleft. These medications help reduce the severity of obsessive thoughts and compulsive behaviors.

Obsessive-Compulsive Disorder (OCD)
Definition: A chronic psychiatric disorder characterized by unwanted repetitive thoughts (obsessions) and behaviors (compulsions).
Key Clinical Presentation: Repetitive behaviors or mental acts performed to reduce anxiety caused by obsessions.
Investigations: Clinical evaluation, often using the Yale-Brown Obsessive Compulsive Scale (Y-BOCS).
Treatment Outline:

- **First-line:** SSRIs (Fluoxetine, Sertraline).
- **Second-line:** Cognitive-behavioral therapy (CBT) focusing on exposure and response prevention (ERP).

Correct Answer: B. Increases the availability of serotonin ✅

MCQ 164:

Which of the following medications causes insomnia, anxiety, and restlessness?

A. Tricyclic antidepressant
B. Tetracyclic Antidepressant
C. Monoamine oxidase inhibitor
D. Selective serotonin-reuptake inhibitor

Explanation:
The correct answer is **D. Selective serotonin-reuptake inhibitor**.
Selective serotonin-reuptake inhibitors (SSRIs), such as fluoxetine and sertraline, are commonly associated with side effects such as insomnia, anxiety, and restlessness, especially early in treatment. SSRIs increase serotonin levels in the brain, which can lead to overstimulation, resulting in these symptoms. Other classes of antidepressants like tricyclic antidepressants (TCAs) or monoamine oxidase inhibitors (MAOIs) are more commonly linked with sedative effects or other side effects rather than insomnia or restlessness.

Selective Serotonin-Reuptake Inhibitors (SSRIs)
Definition: A class of medications commonly used to treat depression and anxiety disorders by increasing serotonin availability in the brain.
Key Clinical Presentation: Insomnia, anxiety, and restlessness during the initial period of treatment.
Investigations: Clinical evaluation, patient history.
Treatment Outline:

- **First-line:** SSRIs (Fluoxetine, Sertraline).
- **Second-line:** Cognitive-behavioral therapy (CBT), SNRIs if SSRIs are ineffective.

Correct Answer: D. Selective serotonin-reuptake inhibitor ✅

MCQ 165:

Which of the following is a potential side effect of amitriptyline?

A. Dystonia
B. Weight gain
C. Hypersalivation
D. Hyperpigmentation

Explanation:
The correct answer is **B. Weight gain**.
Amitriptyline, a tricyclic antidepressant, is associated with weight gain, which is a common side effect. This is due to its antihistaminic properties, which can increase appetite and slow metabolism. Dystonia (A) and hypersalivation (C) are not common side effects of amitriptyline, while hyperpigmentation (D) is more commonly associated with other medications.

Amitriptyline (Tricyclic Antidepressant)
Definition: A TCA used for the treatment of depression, anxiety, and chronic pain.
Key Clinical Presentation: Weight gain, sedation, dry mouth, constipation.
Investigations: Clinical evaluation of side effects, blood tests for metabolic changes.
Treatment Outline:

- **First-line:** SSRIs for elderly or those with risk of side effects.
- **Second-line:** Amitriptyline for specific conditions like chronic pain.

Correct Answer: B. Weight gain ✅

MCQ 166:

A 12-year-old obese boy presents to the Primary Care Clinic with a history of being teased by children at school and, as a result, has considered "taking a lot of pills to just go to sleep and not wake up." Which of the following is the most appropriate action?

A. Weight loss advice
B. Reassuring the patient
C. Immediate psychiatry referral
D. Prescription of a fast-acting antidepressant

Explanation:
The correct answer is **C. Immediate psychiatry referral**.
The child’s mention of suicidal ideation, especially in the context of weight-related bullying, is

concerning for potential self-harm and warrants immediate psychiatric evaluation. While weight loss advice (A) is important, the priority should be addressing the mental health risk. Reassurance (B) is not sufficient, and prescribing an antidepressant (D) without proper evaluation is inappropriate.

Suicide Prevention and Child Mental Health
Definition: Immediate mental health intervention is necessary for any child or adolescent with suicidal ideation.
Key Clinical Presentation: Suicidal thoughts, social isolation, weight-related distress.
Investigations: Mental health assessment, suicide risk evaluation.
Treatment Outline:

- **First-line:** Referral for psychiatric evaluation, cognitive-behavioral therapy (CBT).
- **Second-line:** Medication if necessary after evaluation.

Correct Answer: C. Immediate psychiatry referral ✅

MCQ 167:

A 76-year-old man presents to the Psychiatry Clinic for follow-up accompanied by his son. He has progressive memory loss, difficulty in finding words, wandering, and agitation, as well as urinary and fecal incontinence for the last 3 years. On examination, he was conscious but not oriented in time and place; his short memory was severely impaired with an intact long memory (see report). Brain CT scan: Atrophied hippocampal and temporal lobes. Which of the following chromosomes harbors the mutated gene responsible for this presentation?

A. 1
B. 15
C. 21
D. X

Explanation:
The correct answer is **C. 21**.
The patient's presentation is consistent with Alzheimer's disease, which is commonly associated with genetic mutations on chromosome 21. Alzheimer’s disease often presents with memory loss, difficulty finding words, and behavioral changes, as well as atrophy of the hippocampus and temporal lobes on imaging. Chromosome 21 is linked to early-onset familial Alzheimer's disease, particularly in the case of genetic mutations related to amyloid precursor protein (APP).

Alzheimer's Disease
Definition: A neurodegenerative disorder characterized by progressive memory loss and cognitive decline.
Key Clinical Presentation: Memory loss, word-finding difficulty, agitation, incontinence.
Investigations: Brain imaging (CT, MRI), genetic testing (for familial cases).
Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (Donepezil, Rivastigmine).
- **Second-line:** NMDA receptor antagonist (Memantine) for moderate to severe stages.

Correct Answer: C. 21 ✅

MCQ 168:

Which of the following clinical signs is caused by a lesion in Wernicke's area?

A. Can understand but unable to say words easily
B. Cannot understand spoken or written words
C. Loss of ability to recognize faces
D. Loss of short-term memory

Explanation:
The correct answer is **B. Cannot understand spoken or written words**.
A lesion in Wernicke's area (located in the left temporal lobe) causes **Wernicke's aphasia**, which is characterized by **impaired comprehension** of both spoken and written language, while speech production remains fluent but nonsensical (without proper meaning). This contrasts with Broca's aphasia, where speech production is impaired but comprehension remains intact.

Wernicke's Aphasia
Definition: A language disorder caused by damage to Wernicke's area, resulting in fluent but meaningless speech and poor comprehension.
Key Clinical Presentation: Difficulty understanding spoken and written language, fluent but nonsensical speech.
Investigations: Neuroimaging (CT/MRI), clinical language assessment.
Treatment Outline:

- **First-line:** Speech and language therapy, supportive care.
- **Second-line:** Antidepressants or antipsychotics if associated psychiatric symptoms occur.

Correct Answer: B. Cannot understand spoken or written words ✅

MCQ 169:

A 55-year-old woman presents to the Primary Health Care Centre complaining of loss of interest and pleasure in daily activities for 3 weeks. She also complains that she has difficulty concentrating and sleeping. Further, she has also expressed that she feels that she is worthless in the world and therefore wants to end her life. Which of the following biochemical substance deficiency is the most likely cause?

A. Aspartate
B. Serotonin
C. Glutamate
D. Acetylcholine

Explanation:
The correct answer is **B. Serotonin**.
The patient's symptoms — including **loss of interest**, **difficulty concentrating**, **sleep disturbances**, and **suicidal ideation** — are indicative of **major depressive disorder (MDD)**, which is most commonly associated with a **serotonin deficiency** in the brain. Serotonin plays a crucial role in mood regulation, and its deficiency is linked to mood disorders like depression.

Major Depressive Disorder (MDD)
Definition: A mood disorder characterized by persistent sadness, loss of interest, sleep disturbances, and suicidal thoughts.
Key Clinical Presentation: Depressed mood, anhedonia, concentration difficulty, suicidal ideation.
Investigations: Clinical evaluation, screening tools (e.g., PHQ-9), serotonin level assessment (if available).
Treatment Outline:

- **First-line:** SSRIs (Fluoxetine, Sertraline).
- **Second-line:** SNRIs (Duloxetine), psychotherapy.

Correct Answer: B. Serotonin ✅

MCQ 170:

Which of the following reuptake inhibitors is the best treatment for depression?

A. DOPA
B. GABA
C. Serotonin
D. Acetylcholine

Explanation:
The correct answer is **C. Serotonin**.
Selective serotonin reuptake inhibitors (SSRIs), which work by increasing serotonin levels in the brain, are the **first-line treatment for depression**. Serotonin reuptake inhibitors block the reabsorption (reuptake) of serotonin in the brain, making more serotonin available to improve mood and emotional well-being.

Antidepressant Treatment
Definition: Medications that treat depression by regulating neurotransmitters like serotonin.
Key Clinical Presentation: Low mood, lack of interest, sleep disturbances, appetite changes.
Investigations: Clinical evaluation for depression, response to antidepressant therapy.
Treatment Outline:

- **First-line:** SSRIs (Fluoxetine, Sertraline).
- **Second-line:** SNRIs (Duloxetine) or other classes like TCAs.

Correct Answer: C. Serotonin ✅

MCQ 171:

Which of the following drugs used in Alzheimer's disease is most likely to cause hepatotoxicity?

A. Tacrine
B. Donepezil
C. Rivastigmine
D. Galantamine

Explanation:
The correct answer is **A. Tacrine**.
Tacrine, an acetylcholinesterase inhibitor, was used in the past for treating Alzheimer's disease but was discontinued due to its **hepatotoxicity**. It can cause significant liver damage, which is not seen with other acetylcholinesterase inhibitors like donepezil, rivastigmine, and galantamine.

Alzheimer's Disease Treatment
Definition: A neurodegenerative condition characterized by cognitive decline, memory loss, and behavioral changes.
Key Clinical Presentation: Memory loss, confusion, difficulty with communication.
Investigations: Clinical assessment, brain imaging, cognitive testing.
Treatment Outline:

- **First-line:** Acetylcholinesterase inhibitors (Donepezil, Rivastigmine).
- **Second-line:** NMDA receptor antagonists (Memantine).

Correct Answer: A. Tacrine ✅

MCQ 172:

Which of the following antipsychotic medications is most likely to cause weight gain?

A. Clozapine
B. Olanzapine
C. Risperidone
D. Ziprasidone

Explanation:
The correct answer is **B. Olanzapine**.
Olanzapine, an atypical antipsychotic, is known to cause **significant weight gain**, which is a common side effect. Other atypical antipsychotics, like risperidone, clozapine, and ziprasidone, can also cause weight gain, but olanzapine is most strongly associated with this effect.

Atypical Antipsychotics
Definition: A class of antipsychotic drugs that are used to treat conditions like schizophrenia and bipolar disorder, known for causing fewer movement-related side effects than typical antipsychotics.
Key Clinical Presentation: Sedation, weight gain, metabolic changes.
Investigations: Monitoring for metabolic side effects (weight, glucose, lipids).
Treatment Outline:

- **First-line:** Atypical antipsychotics (Olanzapine, Risperidone).
- **Second-line:** Typical antipsychotics if needed.

Correct Answer: B. Olanzapine ✅

MCQ 173:

A 35-year-old depressed housewife was started with antidepressants. After 2 weeks of taking the medication, the patient started complaining of constipation. Which of the following groups of antidepressants is the most likely cause?

A. Tricyclic antidepressants
B. Monoamine oxidase inhibitors
C. Selective serotonin reuptake inhibitors
D. Selective nor-adrenaline and serotonin inhibitors

Explanation:
The correct answer is **A. Tricyclic antidepressants**.
Tricyclic antidepressants (TCAs) such as **amitriptyline** are known to cause **anticholinergic side effects**, including **constipation**, dry mouth, and urinary retention. These side effects are related to their **muscarinic receptor antagonism**. Other antidepressant classes (SSRIs, SNRIs, and MAOIs) are less likely to cause such significant gastrointestinal symptoms.

Antidepressant Treatment Side Effects
Definition: Medications used to treat depression, each with potential side effects.
Key Clinical Presentation: Constipation, dry mouth, blurred vision (due to anticholinergic effects).
Investigations: Clinical evaluation of side effects, patient history.
Treatment Outline:

- **First-line:** For depressive treatment, SSRIs or SNRIs.
- **Second-line:** If TCA-related constipation occurs, consider switching to an SSRI.

Correct Answer: A. Tricyclic antidepressants ✅

MCQ 174:

Which of the following conditions is the most likely cause of dementia?

A. Spinocerebellar ataxia
B. Vitamin B12 deficiency
C. Idiopathic parkinsonism
D. Amyotrophic lateral sclerosis

Explanation:
The correct answer is **B. Vitamin B12 deficiency**.
Vitamin B12 deficiency can cause **subacute combined degeneration** of the spinal cord, leading to cognitive decline and dementia. It can also cause **peripheral neuropathy** and **psychiatric symptoms** like depression and memory impairment. This is a reversible cause of dementia when treated appropriately.

Dementia
Definition: A cognitive disorder characterized by memory loss, language difficulties, and impaired executive function.
Key Clinical Presentation: Memory impairment, confusion, mood changes, difficulty performing daily tasks.
Investigations: B12 levels, MRI of the brain, cognitive testing.
Treatment Outline:

- **First-line:** Vitamin B12 supplementation (oral or injections).
- **Second-line:** Cognitive rehabilitation if needed.

Correct Answer: B. Vitamin B12 deficiency ✅

MCQ 175:

A 15-year-old girl who had a serious disagreement with her friend. She did not want to meet her friend or talk to her anymore. If the girl saw her friend coming towards her or sitting in a room, she would move away and not go in that direction. Which of the following defense mechanisms is the most likely explanation for this behaviour?

A. Denial
B. Acting out
C. Avoidance
D. Repression

Explanation:
The correct answer is **C. Avoidance**.
Avoidance is a defense mechanism where an individual actively **avoids situations, people, or thoughts** that are emotionally distressing or anxiety-provoking. In this case, the girl avoids her friend because of the emotional discomfort caused by the disagreement. **Denial**, **acting out**, and **repression** are other defense mechanisms, but they do not directly fit this scenario.

Defense Mechanisms
Definition: Psychological strategies used by individuals to cope with reality and maintain self-image.
Key Clinical Presentation: Active avoidance of distressing stimuli.
Investigations: Clinical evaluation of the patient's emotional response.
Treatment Outline:

- **First-line:** Therapy (Cognitive Behavioral Therapy) to address avoidance behavior.
- **Second-line:** Psychodynamic therapy if avoidance is a recurring problem.

Correct Answer: C. Avoidance ✅

MCQ 176

A 65-year-old man with depression presents to the clinic for a follow-up visit. The physician did his consultation using a bio-psycho-social model of treatment. Which of the following is a characteristic of this treatment model?

A. Evidence based medicine
B. Traditional allopathy
C. Geriatric medicine
D. Holistic medicine

Explanation:
The correct answer is **D. Holistic medicine**.
The **bio-psycho-social model** takes a **holistic approach** to treatment, considering not just the biological aspects (e.g., neurochemistry) but also the psychological and social factors (e.g., family dynamics, lifestyle). This model is comprehensive and focuses on treating the whole person, not just the disease.

Bio-psycho-social Model
Definition: A model that integrates biological, psychological, and social factors in understanding and treating illness.
Key Clinical Presentation: Focus on the whole person and all contributing factors.
Investigations: Comprehensive evaluation of mental health, lifestyle, and social context.
Treatment Outline:

- **First-line:** Multidisciplinary treatment involving medication, therapy, and social support.
- **Second-line:** Alternative therapies, including complementary medicine if appropriate.

Correct Answer: D. Holistic medicine ✅

MCQ 177

A 36-year-old diabetic woman presents to the clinic for follow-up visit, and after an initial assessment, the physician wished to perform a physical examination. He first asked her permission, which was granted. Which of the following ethical measures was the physician adopting?

A. Obtain informed consent
B. Respect for the patient
C. Fulfil his responsibility
D. Show his efficiency

Explanation:
The correct answer is **A. Obtain informed consent**.
Obtaining **informed consent** is an essential ethical practice where the patient is provided with all the relevant information regarding the procedure or treatment, and their agreement is sought before proceeding. This is an important ethical principle in healthcare, especially when touching or examining a patient.

Informed Consent
Definition: The process of ensuring that the patient understands and agrees to the treatment or procedure.
Key Clinical Presentation: Asking for permission before performing procedures or examinations.
Investigations: Assessing the patient's capacity to understand the information provided.
Treatment Outline:

- **First-line:** Clear communication and documentation of consent.
- **Second-line:** If patient is unable to consent, involve family or legal guardians.

Correct Answer: A. Obtain informed consent ✅

MCQ 177:

A 40-year-old diabetic man was treated appropriately with small doses of insulin. On next follow-up visit, he bought a very expensive watch as a gesture of gratitude to his physician. Which of the following is the most appropriate response?

A. Accept and advise not to repeat
B. Refuse and give a warning
C. Politely refuse the gift
D. Accept from the rich

Explanation:
The correct answer is **C. Politely refuse the gift**.
It is ethically inappropriate for healthcare providers to accept gifts from patients as it may compromise professional boundaries and objectivity. The most appropriate response is to politely refuse the gift while acknowledging the patient's gratitude. Accepting gifts could lead to conflicts of interest, real or perceived.
Professional Ethics
Definition: The set of ethical standards and principles that govern the conduct of healthcare providers.
Key Clinical Presentation: Patient offers a gift or token of appreciation.
Investigations: Assess if the acceptance of the gift could compromise professional objectivity.
Treatment Outline:

- **First-line:** Politely refuse the gift to maintain professional boundaries.
- **Second-line:** Educate the patient on alternative ways of showing appreciation.

Correct Answer: C. Politely refuse the gift ✅

MCQ 178

A 36-year-old newly diagnosed hypertensive man presents for follow-up visit. After a thorough assessment, the physician discussed with him his diagnosis, the etiology of his illness, the probable side effects of the prescribed medicine, and the prognosis of his illness and finally the restrictions the patient might experience due to the illness. Which of the following professional skills is the physician using?

A. Developing rapport
B. Improving communication
C. Providing informational care
D. Developing bonding with the patient

Explanation:
The correct answer is **C. Providing informational care**.
In this scenario, the physician is offering **informational care** by providing the patient with

detailed information about his condition, treatment options, and potential restrictions. This is an essential part of the physician's role in ensuring that patients understand their health status and treatment plan.
Professional Skills in Medical Practice
Definition: Providing patients with relevant, accurate, and timely information to aid in decision-making.
Key Clinical Presentation: Giving detailed explanations about diagnosis, treatment, and prognosis.
Investigations: Evaluate how well the patient comprehends the information provided.
Treatment Outline:

- **First-line:** Provide clear, empathetic, and evidence-based information.
- **Second-line:** Use additional educational resources if the patient requires more clarification.

Correct Answer: C. Providing informational care ✅

MCQ 178:

Which of the following is the most useful communication technique when taking a patient history?

A. Humour
B. Silent listening
C. Specific questions
D. Open-ended questions

Explanation:
The correct answer is **D. Open-ended questions**.
Open-ended questions encourage the patient to express themselves more freely and provide a more comprehensive history. These types of questions help in building rapport and allowing the patient to share important details that may not come out with more closed, specific questions.
Communication Techniques
Definition: Methods used by healthcare professionals to gather information from patients effectively.
Key Clinical Presentation: Eliciting a comprehensive patient history.
Investigations: Evaluate the patient's responses and use open-ended questions to explore further.
Treatment Outline:

- **First-line:** Use open-ended questions to facilitate communication and gather comprehensive information.
- **Second-line:** Ask specific questions if needed to clarify or narrow down the patient's responses.

Correct Answer: D. Open-ended questions ✅

MCQ 179:

Which of the following is an example of the open-ended questions in consultation?

A. "Tell me about the pain."
B. "Tell me about the pain in your chest."
C. "Point to the area of pain in your chest."
D. "Have you seen a doctor in the past 6 months?"

Explanation:
The correct answer is **A. "Tell me about the pain."**
This is an open-ended question because it invites the patient to provide a detailed response. Open-ended questions allow patients to describe their symptoms and experiences in their own words, providing richer information for the clinician.
Open-Ended Questions
Definition: Questions that allow the patient to give a full, descriptive answer.
Key Clinical Presentation: Eliciting detailed patient responses.
Investigations: Use open-ended questions in the early stages of the consultation to gain a comprehensive understanding of the patient's concerns.
Treatment Outline:

- **First-line:** Ask open-ended questions to start the consultation and gather essential information.
- **Second-line:** Use follow-up questions to clarify or expand upon the initial responses.

Correct Answer: A. "Tell me about the pain." ✅

MCQ 180:

A 6-year-old boy is brought to the clinic by his mother. She noticed that the child has been eating clay and paper for the last 4 months. Which of the following is the most appropriate initial treatment?

A. Fluoxetine
B. Imipramine
C. Clonazepam
D. Behavioural therapy

Explanation:
The correct answer is **D. Behavioural therapy**.
The behavior described, eating clay and paper, is a form of **pica**, which can be addressed through **behavioral therapy**. This therapy aims to modify the child's behavior by reinforcing positive actions and discouraging the ingestion of non-food items. Medications like **fluoxetine** or **imipramine** are not first-line treatments for pica.
Behavioral Therapy for Pica
Definition: A form of therapy focused on changing maladaptive behaviors.
Key Clinical Presentation: Consumption of non-food items such as clay or paper.
Investigations: Psychological evaluation to understand the underlying causes of pica.
Treatment Outline:

- **First-line:** Behavioral therapy to address pica.
- **Second-line:** Address any underlying psychological or nutritional deficiencies.

Correct Answer: D. Behavioural therapy ✅

MCQ 181:

A 34-year-old woman presents to the clinic with a 6-month history of extreme fear of going to the park, the zoo, and sporting events. This has worsened to the point where she has stopped going out of the house. Which of the following is the most likely diagnosis?

A. Conversion disorder
B. Schizophrenia
C. Social phobia
D. Agoraphobia

Explanation:
The correct answer is **D. Agoraphobia**.
Agoraphobia is characterized by the fear of being in situations where escape might be difficult

or help unavailable, such as in public spaces. The patient is avoiding specific places and has stopped leaving the house altogether, which is indicative of agoraphobia.
Anxiety Disorders
Definition: Agoraphobia involves intense fear of being in places where escape might be difficult or help unavailable.
Key Clinical Presentation: Fear of specific places, avoiding leaving the house.
Investigations: Clinical interview and symptom assessment.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT)
- **Second-line:** Medication (SSRI or benzodiazepine for acute anxiety).

Correct Answer: D. Agoraphobia ✅

MCQ 182

A 45-year-old man presents in an anxious, tearful state. He complains of chronic, upper abdominal pain and is certain that he has stomach cancer but no one believes him. He has seen 6 different physicians in the past year and has had a GI series and barium enema performed, an ultrasound of the abdomen twice, and a CT scan. The patient was thinking of next going to a Herbal Clinic. Which of the following is the most likely diagnosis?

A. Somatoform pain disorder
B. Somatization disorder
C. Conversion disorder
D. Hypochondriasis

Explanation:
The correct answer is **D. Hypochondriasis** (now referred to as **Illness Anxiety Disorder**).
Hypochondriasis is characterized by a preoccupation with having or acquiring a serious illness despite medical reassurance and negative test results. The patient has excessive concern about his health and seeks numerous medical opinions without accepting reassurance from doctors.
Somatoform Disorders
Definition: Hypochondriasis involves excessive worry about having a serious disease, even after negative tests.
Key Clinical Presentation: Preoccupation with illness, seeking multiple doctors' opinions.
Investigations: Multiple unnecessary medical tests.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT)
- **Second-line:** Medication (SSRIs or anxiolytics).

Correct Answer: D. Hypochondriasis ✅

MCQ 183:

A 43-year-old woman delivered a full-term infant. On examination, the infant has epicanthic folds, delayed milestones, a flat facial profile, simian palmar creases, and an atrial septal defect. Which of the following is the most likely diagnosis?

A. Trisomy 12
B. Trisomy 21
C. Di-George syndrome
D. Robertsonian translocation

Explanation:
The correct answer is **B. Trisomy 21** (Down syndrome).
Trisomy 21 is characterized by features such as epicanthic folds, a flat facial profile, simian palmar creases, developmental delay, and congenital heart defects like an atrial septal defect. These findings are consistent with Down syndrome.
Genetic Disorders
Definition: Trisomy 21 is a genetic condition caused by an extra copy of chromosome 21, leading to developmental and physical abnormalities.
Key Clinical Presentation: Epicanthic folds, simian crease, developmental delay, congenital heart defects.
Investigations: Karyotyping to confirm the trisomy 21 diagnosis.
Treatment Outline:

- **First-line:** Early intervention, including physical therapy and developmental support.
- **Second-line:** Medical management for associated conditions like congenital heart defects.

Correct Answer: B. Trisomy 21 ✅

MCQ 184

A 16-year-old girl is brought by her father to the Emergency Department with complaints of tachypnoea, light-headedness, paraesthesia, and palpitations. Her father gives the history that she has recently failed her math test. Which of the following is the most likely diagnosis?

A. Oppositional defiant disorder
B. Hyperventilation syndrome
C. Munchausen syndrome
D. Anxiety disorder

Explanation:
The correct answer is **B. Hyperventilation syndrome**.
Hyperventilation syndrome often occurs in response to anxiety or stress, which is evident in this patient who has experienced a stressful event (failing a test). Symptoms like tachypnoea, light-headedness, and paraesthesia are common in hyperventilation.
Anxiety Disorders
Definition: Hyperventilation syndrome is caused by rapid, shallow breathing, leading to respiratory alkalosis and symptoms such as light-headedness and paraesthesia.
Key Clinical Presentation: Tachypnoea, light-headedness, anxiety-related symptoms.
Investigations: Diagnosis is clinical, with improvement after controlled breathing.
Treatment Outline:

- **First-line:** Breathing exercises to reduce hyperventilation.
- **Second-line:** Stress management techniques.

Correct Answer: B. Hyperventilation syndrome ✅

MCQ 185

A 27-year-old man presents to the clinic with a 1-month history of low mood and lack of sleep. It was associated with weight gain, isolation, and concentration impairment. His wife mentioned that he bought a very expensive car and thought that he was a wealthy man 3 months ago. He is very talkative and difficult to interrupt. Which of the following medications is the most effective for long-term management?

A. Lithium
B. Bupropion
C. Olanzapine
D. Mirtazapine

Explanation:
The correct answer is **A. Lithium**.
This patient's symptoms, including low mood, sleep disturbance, weight gain, impulsivity, and grandiosity, suggest **bipolar disorder**. **Lithium** is considered the gold standard treatment for long-term management of bipolar disorder, particularly for mood stabilization.
Mood Disorders
Definition: Bipolar disorder is a mood disorder characterized by alternating episodes of mania/hypomania and depression.
Key Clinical Presentation: Low mood, sleep disturbances, grandiosity, impulsivity.
Investigations: Diagnosis is clinical, with history of manic episodes and depressive symptoms.
Treatment Outline:

- **First-line:** Lithium for mood stabilization.
- **Second-line:** Anticonvulsants or atypical antipsychotics if lithium is not effective.

Correct Answer: A. Lithium ✅

MCQ 186:

Which of the following is the diagnostic criteria for Attention Deficit Hyperactivity Disorder in children, according to the International Classification of Diseases-10?

A. 1 inattentive, 3 hyperactive and 1 impulsive symptoms
B. 1 inattentive, 3 hyperactive and 3 impulsive symptoms
C. 3 inattentive, 3 hyperactive and 1 impulsive symptoms
D. 6 inattentive, 3 hyperactive and 1 impulsive symptoms

Explanation:
The correct answer is **D. 6 inattentive, 3 hyperactive and 1 impulsive symptoms**.
According to the ICD-10 criteria for Attention Deficit Hyperactivity Disorder (ADHD) in children, the diagnosis is made when there are at least **6 symptoms** of inattention, **3 symptoms** of hyperactivity, and **1 symptom** of impulsivity that persist for at least **6 months**.

Attention Deficit Hyperactivity Disorder (ADHD)
Definition: ADHD is a neurodevelopmental disorder characterized by persistent inattention, hyperactivity, and impulsivity.
Key Clinical Presentation: Inattention, hyperactivity, impulsivity that impacts functioning in multiple settings.
Investigations: Clinical assessment and parent/teacher reports using ADHD-specific rating

scales.
Treatment Outline:

- **First-line:** Behavioral therapy and stimulant medications (e.g., methylphenidate).
- **Second-line:** Non-stimulant medications (e.g., atomoxetine).

Correct Answer: D. 6 inattentive, 3 hyperactive and 1 impulsive symptoms ✅

MCQ 187:

A 23-year-old woman with frequent panic attacks, presents to the clinic complaining that during her last attack 3 days ago, she felt that she was disconnected from the world, as if the environment is strange. Which of the following would best describe her feelings?

A. Illusion
B. Derealisation
C. Dull perception
D. Depersonalization

Explanation:
The correct answer is **B. Derealisation**.
Derealisation refers to the feeling that the external world is unreal or strange, which is described by the patient in this scenario. It is commonly associated with anxiety disorders and panic attacks.
Disorders of Perception
Definition: Derealisation is the alteration in perception where the surroundings seem strange or distant, while **depersonalization** refers to the feeling of being detached from oneself.
Key Clinical Presentation: A feeling that the world around the person is strange or unreal, often occurring during panic attacks.
Investigations: Diagnosis is clinical, based on patient's description of symptoms.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) and relaxation techniques for panic disorder.
- **Second-line:** Medications like SSRIs or benzodiazepines for acute relief.

Correct Answer: B. Derealisation ✅

MCQ 188:

A 30-year-old man has a personality disorder, which is characterized by being aggressive and violent. Which of the following is most likely associated with violence?

A. Low levels of serotonin
B. High levels of serotonin
C. Low levels of endorphins
D. High levels of endorphins

Explanation:
The correct answer is **A. Low levels of serotonin**.
Low serotonin levels have been linked to increased aggression and violent behavior. This is commonly seen in patients with personality disorders, particularly those with impulsive aggression.
Personality Disorders
Definition: Personality disorders often involve long-standing patterns of behavior that deviate from societal expectations, and these can include aggression and impulsivity.
Key Clinical Presentation: Aggression, impulsivity, violent outbursts.
Investigations: Psychological evaluation, possible hormone and neurotransmitter levels.
Treatment Outline:

- **First-line:** Psychotherapy (e.g., Dialectical behavior therapy for Borderline Personality Disorder).
- **Second-line:** Medications (e.g., SSRIs for impulsivity and aggression).

Correct Answer: A. Low levels of serotonin ✅

MCQ 189

Which of the following is associated with dyslexia?

A. Abnormal gait
B. Brainstem defects
C. Verbal language difficulties
D. Defect in visual or hearing acuity

Explanation:
The correct answer is **C. Verbal language difficulties**.
Dyslexia is a specific learning disability that primarily affects reading, spelling, and sometimes writing, due to difficulties with phonological processing. It is not associated with deficits in

visual or hearing acuity but rather with verbal language and processing difficulties.
Learning Disabilities
Definition: Dyslexia is a learning disorder marked by difficulties with word recognition, decoding, and spelling, despite normal intelligence.
Key Clinical Presentation: Difficulty reading, spelling, and writing, with normal intelligence.
Investigations: Diagnosis is clinical, often confirmed through reading and language assessments.
Treatment Outline:

- **First-line:** Special education services and reading interventions.
- **Second-line:** Behavioral therapy, phonics-based approaches.

Correct Answer: C. Verbal language difficulties ✅

MCQ 190

Which of the following is the most likely clinical presentation of delusional patients?

A. Disturbance in perception
B. Rare in manic disorders
C. Found only in schizophrenia
D. Disturbance of thought content

Explanation:
The correct answer is **D. Disturbance of thought content**.
Delusions are fixed, false beliefs that are not in line with reality, which can affect the content of a person's thinking. Patients with delusions typically experience disturbances in thought content rather than in perception (which would be seen in hallucinations).
Psychotic Disorders
Definition: Delusions are false beliefs that are resistant to reasoning or contrary evidence, often seen in psychotic disorders.
Key Clinical Presentation: Beliefs not aligned with reality, such as paranoia, grandiosity, or persecution.
Investigations: Psychiatric evaluation and mental status examination.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., haloperidol, risperidone).
- **Second-line:** Cognitive-behavioral therapy for delusions.

Correct Answer: D. Disturbance of thought content ✅

MCQ 191:

A 7-year-old boy is brought by his mother because he was unable to sit in school without interrupting the class, wandering around the classroom or poking his neighbours. On examination, he has normal growth parameters. Which of the following is the best management?

A. Bupropion
B. Paroxetine
C. Guanfacine
D. Dextroamphetamine

Explanation:
The correct answer is **D. Dextroamphetamine**.
The child exhibits symptoms consistent with **Attention Deficit Hyperactivity Disorder (ADHD)**, which includes difficulty staying seated, impulsivity, and inattentiveness. **Dextroamphetamine** is a stimulant medication commonly used to treat ADHD.
Attention Deficit Hyperactivity Disorder (ADHD)
Definition: A neurodevelopmental disorder characterized by inattention, hyperactivity, and impulsivity.
Key Clinical Presentation: Inability to sit still, distractibility, impulsive behavior.
Investigations: Diagnosis is clinical, based on behavior assessment and rating scales from parents and teachers.
Treatment Outline:

- **First-line:** Stimulant medications (e.g., dextroamphetamine, methylphenidate).
- **Second-line:** Non-stimulants (e.g., guanfacine).
  Correct Answer: D. Dextroamphetamine ✅

MCQ 192:

A 30-year-old woman presents with a complaint of lower back pain for 1 year. During the last 2 years, she had presented with recurrent abdominal pain, bloating, headache, and migratory joint pain. Her menstrual periods are irregular and heavy. She was terminated twice from the job due to frequent absence from work. She is upset that proper diagnosis or treatment was not given to her. She had multiple laboratory tests done during the last 2 years and all were

normal. The physical examination is unremarkable. Which of the following is the most likely diagnosis?

A. Malingering
B. Hypochondriasis
C. Factitious disorder
D. Somatization disorder

Explanation:
The correct answer is **D. Somatization disorder**.
Somatization disorder involves the presence of multiple physical complaints (e.g., abdominal pain, back pain, headaches) without an identifiable medical cause, which is characteristic of the patient's symptoms. This disorder is often marked by frequent healthcare visits and unexplained physical symptoms.
Somatization Disorder
Definition: A psychiatric condition characterized by multiple unexplained physical symptoms that are not explained by medical tests.
Key Clinical Presentation: Multiple physical symptoms with no medical cause, often leading to frequent healthcare visits.
Investigations: Diagnosis is clinical, often based on exclusion of medical causes and assessment of symptom patterns.
Treatment Outline:

- **First-line:** Cognitive-behavioral therapy (CBT) and psychotherapy.
- **Second-line:** Medications for symptom management (e.g., antidepressants).
  Correct Answer: D. Somatization disorder ✅

MCQ 193:

A 23-year-old man with bronchial asthma presents with a 5-year history of severe anxiety in social gatherings. He often has sweating and palpitations when he leads discussions at work. He prefers to sit alone at home. He notices that his presenting complaint is provoked by salbutamol. His physical examination is unremarkable. Which of the following is the most appropriate therapy?

A. Sertraline
B. Bupropion
C. Alprazolam
D. Propranolol

Explanation:
The correct answer is **A. Sertraline**.
This patient most likely has **social anxiety disorder (SAD)**, which is characterized by an intense fear of being judged negatively in social situations. **Sertraline**, a selective serotonin reuptake inhibitor (SSRI), is the most appropriate first-line treatment for this condition.
Social Anxiety Disorder
Definition: A marked fear or anxiety about one or more social situations where the individual is exposed to possible scrutiny.
Key Clinical Presentation: Fear of embarrassment in social situations, avoidance of public speaking, physical symptoms like sweating and palpitations.
Investigations: Diagnosis is clinical, often based on the patient's reported symptoms and clinical interview.
Treatment Outline:

- **First-line:** SSRIs (e.g., sertraline, fluoxetine) or CBT.
- **Second-line:** Benzodiazepines (e.g., alprazolam) for short-term use.
  Correct Answer: A. Sertraline ✅

MCQ 194:

A 28-year-old woman with bipolar disorder, treated for 6 months, presents for follow-up visit. On examination, she has 8 kg weight gain, generalized weakness, and hair loss. Which of the following is the most likely cause?

A. Hypothalamic disorder
B. Hypothyroidism
C. Verapamil
D. Lithium

Explanation:
The correct answer is **D. Lithium**.
Lithium is a common medication used to treat bipolar disorder, and weight gain, weakness, and hair loss are known side effects of lithium.
Bipolar Disorder
Definition: A mood disorder characterized by periods of mania and depression.
Key Clinical Presentation: Episodes of elevated mood (mania) and low mood (depression).
Investigations: Diagnosis is clinical, based on the pattern of mood episodes.
Treatment Outline:

- **First-line:** Lithium, anticonvulsants (e.g., valproate, lamotrigine).
- **Second-line:** Antipsychotics for acute mania.
  Correct Answer: D. Lithium ✅

MCQ 195:

A 56-year-old man with schizophrenia is started on neuroleptic medications to treat his psychotic symptoms. Which of the following side effects is the patient at greatest risk?

A. Seizures
B. Akathisia
C. Hyperthermia
D. Myocardial infarction

Explanation:
The correct answer is **B. Akathisia**.
Akathisia, which involves an intense feeling of restlessness and the need to move, is a common side effect of **neuroleptic medications** (antipsychotics). It is more frequent with first-generation antipsychotics (e.g., haloperidol).
Neuroleptic Side Effects
Definition: Neuroleptic medications, used to treat psychosis, are associated with several side effects, including movement disorders.
Key Clinical Presentation: Restlessness, inability to sit still, pacing.
Investigations: Clinical examination to assess the presence of abnormal movements.
Treatment Outline:

- **First-line:** Anticholinergic medications (e.g., benztropine) for akathisia.
- **Second-line:** Dose reduction or switching to a second-generation antipsychotic.
  Correct Answer: B. Akathisia ✅

MCQ 196:

Which of the following is the most likely benefit of using antipsychotic therapy in schizophrenic patients?

A. Improve apathy
B. Reduce delusions

C. Improve thinking ability
D. Increase social interaction

Explanation:
The correct answer is **B. Reduce delusions**.
Antipsychotic medications are primarily used to manage the **positive symptoms of schizophrenia**, including **delusions** and **hallucinations**. These medications have a significant effect on reducing delusions, which are false beliefs that are not in accordance with reality.
Schizophrenia
Definition: A chronic, severe mental health disorder characterized by psychosis, which includes delusions, hallucinations, disorganized thinking, and negative symptoms (e.g., social withdrawal, apathy).
Key Clinical Presentation: Delusions, auditory hallucinations, disorganized speech, and thought disorder.
Investigations: Diagnosis is clinical, based on symptom history and exclusion of other causes.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, olanzapine)
- **Second-line:** Cognitive-behavioral therapy, supportive therapy.
  Correct Answer: B. Reduce delusions ✅

MCQ 197:

A 23-year-old man is brought to the psychiatrist clinic because for the first time he started to hear some voices talking to him with unknown sources. On examination, he was apathetic and emotionally withdrawn. Which of the following is the best management?

A. Olanzapine
B. Fluoxetine
C. Cognitive behavioral therapy
D. Antipsychotic and cognitive behavioral therapy

Explanation:
The correct answer is **D. Antipsychotic and cognitive behavioral therapy**.
This patient is presenting with **first-episode psychosis**, likely indicating **schizophrenia** or another psychotic disorder. The best treatment involves **antipsychotic medication** (such as **olanzapine**) to manage psychotic symptoms like **auditory hallucinations**. **Cognitive behavioral therapy** (CBT) is also beneficial for addressing cognitive distortions and promoting recovery.
Schizophrenia

Definition: A mental health disorder characterized by persistent psychotic symptoms such as delusions, hallucinations, and disorganized behavior.
Key Clinical Presentation: Hallucinations, delusions, emotional withdrawal, apathy.
Investigations: Diagnosis is clinical and may involve imaging and lab work to rule out other causes.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., olanzapine, risperidone)
- **Second-line:** Cognitive behavioral therapy, family therapy.
  Correct Answer: D. Antipsychotic and cognitive behavioral therapy ✅

MCQ 198:

A 45-year-old man with recurrent episodes of schizophrenia is on maintenance therapy to maintain suppression of symptoms, prevent relapse, improve quality of life, and support engagement in psychosocial therapy. Which of the following is the best outcome following the maintenance therapy?

A. Complete resolution
B. 5% will achieve remission
C. 70% will have a satisfactory quality of life
D. 33% obtain long-lasting symptom reduction

Explanation:
The correct answer is **D. 33% obtain long-lasting symptom reduction**.
Schizophrenia is a chronic condition, and **maintenance therapy** aims to control symptoms, prevent relapse, and improve the patient's quality of life. A **significant percentage** of patients may achieve **long-lasting symptom reduction**, but complete resolution is rare.
Schizophrenia
Definition: A chronic psychiatric disorder that affects thinking, emotional regulation, and behavior.
Key Clinical Presentation: Positive symptoms (delusions, hallucinations) and negative symptoms (apathy, withdrawal).
Investigations: Clinical diagnosis based on symptoms and exclusion of other causes.
Treatment Outline:

- **First-line:** Antipsychotics (maintenance treatment).

- **Second-line:** Psychosocial therapy, cognitive-behavioral therapy.
  Correct Answer: D. 33% obtain long-lasting symptom reduction ✅

MCQ 199:

Which of the following selective serotonin reuptake inhibitors is the most toxic in overdose?

A. Sertraline
B. Citalopram
C. Paroxetine
D. Fluoxetine

Explanation:
The correct answer is **B. Citalopram**.
Citalopram has been associated with a higher risk of **QT prolongation** and **cardiotoxicity** compared to other SSRIs, making it **more toxic in overdose**.
Selective Serotonin Reuptake Inhibitors (SSRIs)
Definition: A class of drugs commonly used to treat depression and anxiety disorders by increasing serotonin levels in the brain.
Key Clinical Presentation: Used to treat depression, anxiety, OCD, and PTSD.
Investigations: Monitoring for adverse effects, particularly cardiac effects in overdose situations.
Treatment Outline:

- **First-line:** SSRIs (e.g., fluoxetine, sertraline, citalopram).
- **Second-line:** SNRIs, tricyclic antidepressants.
  Correct Answer: B. Citalopram ✅

MCQ 200:

A 23-year-old woman is newly diagnosed with paranoid schizophrenia based on presence of both negative and positive symptoms of schizophrenia. Which of the following is the most effective medication?

A. Clozapine
B. Trazodone
C. Fluphenazine
D. Chlorpromazine

Explanation:
The correct answer is **A. Clozapine**.
Clozapine is the most effective **atypical antipsychotic** for **treatment-resistant schizophrenia**, especially when the patient has not responded to other antipsychotics. It is also known to reduce **both positive and negative symptoms** of schizophrenia.
Schizophrenia
Definition: A chronic psychiatric disorder that leads to disturbances in thinking, perception, emotions, and behavior.
Key Clinical Presentation: Hallucinations, delusions, disorganized behavior.
Investigations: Diagnosis is based on clinical criteria, ruling out other medical causes.
Treatment Outline:

- **First-line:** Antipsychotic medications (e.g., risperidone, clozapine).
- **Second-line:** Cognitive-behavioral therapy, social skills training.
  Correct Answer: A. Clozapine ✅

MCQ 201:

A 32-year-old man was brought to the Emergency Department after he was involved in a road traffic accident. He was diagnosed with a massive intra-abdominal bleeding due to a severe splenic injury. He underwent exploratory laparotomy and splenectomy. Which of the following is most likely to be low in the early postoperative period?

A. Insulin
B. Glucagon
C. Prolactin
D. Vasopressin

Explanation:
The correct answer is **D. Vasopressin**.
After **splenectomy**, there is often **hypovolemia** and **sepsis**, which can lead to **disruption of the hypothalamic-pituitary axis**, specifically affecting **vasopressin** secretion. In the early postoperative period, **vasopressin levels** may be **low**, contributing to **diabetes insipidus** due to insufficient antidiuretic hormone (ADH) response.
Splenectomy
Definition: Surgical removal of the spleen, often due to trauma or disease.
Key Clinical Presentation: Postoperative recovery can be complicated by **infection** (increased

risk of sepsis) and **impaired immune function**.
Investigations: Monitoring for **fluid balance** and **electrolyte abnormalities** (including **sodium** and **potassium**).
Treatment Outline:

- **First-line:** Hydration, **vasopressin** supplementation if necessary.
- **Second-line:** Antibiotic prophylaxis for infection.
  Correct Answer: D. Vasopressin ✅

MCQ 202

A 27-year-old man admitted to the Intensive Care Unit 4 weeks ago with severe infected necrotizing pancreatitis. He underwent several interventions for sepsis control and drainage of collections. He is on mechanical ventilation and inotropes. Which of the following metabolic responses is most expected?

A. Reduced insulin resistance
B. Reduced gluconeogenesis
C. Increased lipolysis
D. Hypoglycaemia

Explanation:
The correct answer is **C. Increased lipolysis**.
In patients with severe **necrotizing pancreatitis** and **sepsis**, **increased lipolysis** is a common response due to **stress-induced hormonal changes** (e.g., increased **catecholamines**, **glucocorticoids**, and **growth hormone**). These hormones stimulate the breakdown of **fat stores** to provide energy for the body during **acute illness**.
Acute Pancreatitis and Sepsis
Definition: Acute inflammation of the pancreas due to enzyme autodigestion, often complicated by infection and systemic inflammatory response syndrome (SIRS).
Key Clinical Presentation: Abdominal pain, nausea, vomiting, and systemic signs of sepsis (e.g., fever, tachycardia).
Investigations: Serum amylase, **lipase**, imaging for complications, **blood cultures** for sepsis.
Treatment Outline:

- **First-line:** Fluid resuscitation, antibiotics, and nutritional support.
- **Second-line:** Insulin therapy for hyperglycemia, glucose monitoring.
  Correct Answer: C. Increased lipolysis ✅

MCQ 203

A 38-year-old man who is known to have Crohn's disease, presented to the Emergency Department with a history of 8-12 episodes of diarrhoea per day over the last 5 days. On examination, he was dehydrated; abdomen was distended with generalized muscles weakness (see report). ECG: Flattened T-waves. Which of the following is the most likely electrolyte abnormality?

A. Hypomagnesaemia
B. Hyponatremia
C. Hypocalcaemia
D. Hypokalaemia

Explanation:
The correct answer is **D. Hypokalaemia**.
Crohn's disease and severe **diarrhea** can lead to significant **potassium loss**, resulting in **hypokalemia**, which is commonly seen in patients with chronic diarrhea. **Flattened T-waves** on ECG are a classic finding of **hypokalemia**.
Crohn's Disease
Definition: A chronic inflammatory bowel disease that can affect any part of the gastrointestinal tract, leading to symptoms like diarrhea, abdominal pain, weight loss, and malabsorption.
Key Clinical Presentation: Diarrhea, abdominal pain, weight loss, and dehydration.
Investigations: Stool studies, **electrolyte levels**, **colonoscopy**.
Treatment Outline:

- **First-line:** Corticosteroids and immunosuppressive therapy for inflammation.
- **Second-line:** Potassium replacement, IV fluids for dehydration.
  Correct Answer: D. Hypokalaemia ✅

MCQ 204

A 33-year-old man is known to have renal impairment, underwent appendectomy for acute appendicitis. On the first day after surgery, he was complaining of palpitation (see lab result and report). Test Result Normal Value Potassium 6.5 3.5-5.1 mmol/L ECG: Tented T-waves. Which of the following is the most appropriate initial step in management?

A. Intravenous infusion of dextrose with insulin
B. Intravenous infusion of sodium bicarbonate

C. Intravenous calcium gluconate
D. Immediate haemodialysis

Explanation:
The correct answer is **C. Intravenous calcium gluconate**.
The **elevated potassium level** (**6.5 mmol/L**) is indicative of **hyperkalemia**, which can cause **cardiac arrhythmias** (e.g., **tented T-waves** on ECG). The initial management of **severe hyperkalemia** involves **calcium gluconate** to stabilize the myocardial membrane and prevent arrhythmias. Other treatments (e.g., insulin, sodium bicarbonate) can be used to shift potassium into cells, but **calcium gluconate** is the **first step** to prevent immediate cardiac complications.
Hyperkalemia
Definition: Elevated serum potassium levels, which can cause life-threatening arrhythmias.
Key Clinical Presentation: ECG changes, weakness, palpitations.
Investigations: Serum potassium, ECG, renal function tests.
Treatment Outline:

- **First-line:** Calcium gluconate IV, sodium bicarbonate, insulin with glucose.
- **Second-line:** Hemodialysis if the patient is refractory to medical treatment.
  Correct Answer: C. Intravenous calcium gluconate ✅

MCQ 205:

A 55-year-old woman known to have chronic duodenal ulcer, presented to the Emergency Department with history of frequent vomiting for 2 weeks. Examination revealed dehydration, with distended upper abdomen and normal bowel sounds (see report). Upper Gastrointestinal Endoscopy: Severely narrowed pylorus due to chronic peptic ulcer disease. Which of the following is the most expected?

A. High urinary potassium
B. Respiratory alkalosis
C. Metabolic acidosis
D. Hypokalaemia

Explanation:
The correct answer is **D. Hypokalaemia**.
Chronic vomiting leads to **loss of hydrochloric acid**, resulting in **metabolic alkalosis**. In addition, the **dehydration** and loss of potassium (due to vomiting and reduced oral intake) will cause **hypokalaemia**. Therefore, **hypokalaemia** is the most expected electrolyte disturbance in this

patient with chronic vomiting and pyloric stenosis due to peptic ulcer disease.
Chronic Duodenal Ulcer
Definition: A condition in which a peptic ulcer affects the duodenum, causing inflammation, ulceration, and possible stenosis.
Key Clinical Presentation: Abdominal pain, vomiting, weight loss, and signs of obstruction.
Investigations: Upper GI endoscopy to assess ulceration, narrowing, and pyloric stenosis.
Treatment Outline:

- **First-line:** Proton pump inhibitors (PPIs) and H. pylori eradication therapy.
- **Second-line:** Surgery in cases of severe stenosis or complications like bleeding.
  Correct Answer: D. Hypokalaemia ✅

MCQ 206

A 36-year-old man presented to the Emergency Department with a history of right iliac fossa pain for 9 days associated with fever and vomiting. Physical examination revealed tender abdominal mass in the right iliac fossa with sluggish bowel sounds (see report). Temperature 39 °C CT scan: Appendicular abscess, 15 x 18 cm in size, in the right iliac fossa. Which of the following is a characteristic pathophysiology?

A. Redistribution of blood flow
B. Peripheral vasoconstriction
C. Decreased cardiac index
D. Bradycardia

Explanation:
The correct answer is **A. Redistribution of blood flow**.
An **appendicular abscess** is a **complication of acute appendicitis**. In response to infection and inflammation, the body exhibits a **systemic inflammatory response** leading to **redistribution of blood flow**. This often results in **hypoperfusion** of non-essential organs as blood flow is prioritized to vital organs like the brain and heart.
Appendicular Abscess
Definition: An infected collection of pus near the appendix, typically after perforation or an unresolved appendix.
Key Clinical Presentation: Right lower quadrant abdominal pain, fever, and vomiting.
Investigations: CT scan of the abdomen to identify the size and location of the abscess.
Treatment Outline:

- **First-line:** Antibiotic therapy, drainage of the abscess.

- **Second-line:** Surgery (appendectomy) if necessary.
  Correct Answer: A. Redistribution of blood flow ✅

MCQ 207

A 38-year-old man has suffered from a severe hypovolemic shock due to traumatic splenic injury, and underwent splenectomy. He received 8 units of packed red blood cells during resuscitation and surgery. He was admitted in the intensive care unit. On the 3rd post-operative day, he was febrile (see report). Gram stain of blood: Positive for gram negative bacilli. Which of the following is the most likely source of the bacteraemia?

A. Contaminated blood
B. Urinary tract infection
C. Respiratory tract infection
D. Translocation from the bowel

Explanation:
The correct answer is **D. Translocation from the bowel**.
After **splenectomy**, the patient is **highly susceptible to infections**, particularly from **gram-negative bacteria**. The **gut flora** is the most likely source of infection due to **translocation** of bacteria across the intestinal mucosa. The spleen normally helps to filter out these bacteria, so its removal increases the risk of **systemic infections**.
Splenectomy
Definition: Surgical removal of the spleen, often due to trauma or disease.
Key Clinical Presentation: Fever, hypotension, and signs of infection.
Investigations: Blood cultures, imaging to rule out other sources of infection.
Treatment Outline:

- **First-line:** Antibiotic prophylaxis, supportive care.
- **Second-line:** Sepsis management if infection develops.
  Correct Answer: D. Translocation from the bowel ✅

MCQ 208:

A 41-year-old man underwent blood transfusion for postoperative bleeding. He developed pain at the site of the infusion with fever and chest tightness after few minutes of starting the transfusion. Temperature 38 °C Which of the following is the most likely diagnosis?

A. Febrile non-haemolytic reaction
B. Bacterial blood contamination
C. Haemolytic reaction
D. Allergic reaction

Explanation:
The correct answer is **A. Febrile non-haemolytic reaction**.
A **febrile non-haemolytic transfusion reaction** is the most common type of reaction that occurs within **minutes to hours** of transfusion. It is typically caused by the body's immune response to **white blood cells** in the transfused blood. Symptoms include **fever, chills, and chest discomfort**, without **hemolysis** or **bacterial infection**.
Blood Transfusion Reactions
Definition: A reaction that occurs after a blood transfusion, ranging from mild to severe.
Key Clinical Presentation: Fever, chills, chest tightness, pain at the infusion site.
Investigations: Clinical diagnosis, possibly with blood cultures to rule out bacterial contamination.
Treatment Outline:

- **First-line:** Symptomatic management (acetaminophen for fever).
- **Second-line:** Blood transfusion with leukoreduced products or washed red blood cells.
  Correct Answer: A. Febrile non-haemolytic reaction ✅

MCQ 209:

A 37-year-old man with a history of chronic duodenal ulcer, presented to the Emergency Department complaining of frequent postprandial vomiting since the last 2 weeks. On physical examination, he was dehydrated with positive succession splash when shaking the abdomen. Which of the following is the most expected acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
The correct answer is **B. Metabolic alkalosis**.
Frequent **postprandial vomiting** leads to the loss of **hydrochloric acid**, which results in **metabolic alkalosis**. The patient may also present with dehydration due to vomiting, which can exacerbate the alkalosis. **Succussion splash** suggests a possible **gastric outlet obstruction**, which

is also commonly seen in patients with peptic ulcers, and can further contribute to the development of **metabolic alkalosis**.
Chronic Duodenal Ulcer
Definition: Ulcers that form in the duodenum, often due to Helicobacter pylori infection or chronic NSAID use.
Key Clinical Presentation: Epigastric pain, nausea, vomiting, weight loss, and symptoms of obstruction.
Investigations: Upper gastrointestinal endoscopy, barium studies.
Treatment Outline:

- **First-line:** Proton pump inhibitors (PPIs), H. pylori eradication.
- **Second-line:** Surgery if complications like stenosis occur.
  Correct Answer: B. Metabolic alkalosis ✅

MCQ 210:

A 32-year-old man who is known to have ulcerative colitis, presented to the Emergency Department with a history of frequent attacks of watery diarrhea for 10 days. On examination, he was severely dehydrated with a non-distended, soft abdomen. Which of the following is the most expected acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
The correct answer is **A. Metabolic acidosis**.
In patients with **severe diarrhea**, there is loss of **bicarbonate** through the gastrointestinal tract, leading to **metabolic acidosis**. This condition can also be associated with dehydration, as seen in this patient, which further exacerbates the acidosis.
Ulcerative Colitis
Definition: A chronic inflammatory bowel disease that causes ulceration in the colon and rectum, leading to bloody diarrhea.
Key Clinical Presentation: Diarrhea, abdominal pain, weight loss, rectal bleeding.
Investigations: Colonoscopy, stool tests, imaging.
Treatment Outline:

- **First-line:** Anti-inflammatory drugs (aminosalicylates), immunosuppressants.

- **Second-line:** Biologics for refractory cases.
  Correct Answer: A. Metabolic acidosis ✅

MCQ 211

A 67-year-old man, known to have advanced antral carcinoma with liver metastasis, presented to the Emergency Department with a history of frequent vomiting for 10 days. On examination, he was jaundiced and dehydrated (see report). ECG: Flat T-wave. Which of the following is most expected finding in urinalysis?

A. Aciduria
B. High sodium
C. High potassium
D. Alkaline urine

Explanation:
The correct answer is **C. High potassium**.
In patients with **vomiting**, **dehydration**, and **jaundice**, there is often a loss of **potassium** due to vomiting and inadequate intake. As the body compensates for the loss, **hyperkalemia** may develop, which could present as **high potassium** levels in the blood and urine. Additionally, **ECG findings** such as **flat T-waves** suggest **hypokalemia**, which is commonly associated with **vomiting** and **dehydration**.
Advanced Antral Carcinoma
Definition: A cancer that arises from the antrum of the stomach and often metastasizes to the liver.
Key Clinical Presentation: Jaundice, weight loss, vomiting, and signs of malnutrition.
Investigations: Imaging studies, biopsy, liver function tests.
Treatment Outline:

- **First-line:** Chemotherapy, palliative care.
- **Second-line:** Surgical resection if feasible.
  Correct Answer: C. High potassium ✅

MCQ 211:

A 35-year-old man with a history of open appendectomy 5 years ago presented with abdominal pain, distension, vomiting, and constipation for 3 days. On physical examination, he was dehydrated, abdomen was distended with marked tenderness in the right iliac fossa,

and exaggerated bowel sounds. He was scheduled for emergency exploratory laparotomy (see report). Plain abdominal X-ray: Distended small bowel, multiple air-fluid levels with collapsed colon. Which of the following is contraindicated?

A. Propofol
B. Ketamine
C. Sevoflurane
D. Nitrous oxide

Explanation:
The correct answer is **D. Nitrous oxide**.
In patients with **bowel obstruction**, the use of **nitrous oxide** is contraindicated because it can **increase the size of air pockets within the bowel**, leading to a **bowel perforation**. **Nitrous oxide** is an anesthetic that may accumulate in gas-filled spaces, which can exacerbate the risk of rupture or worsen the obstruction.
Bowel Obstruction
Definition: A blockage in the intestines that prevents the normal flow of contents.
Key Clinical Presentation: Abdominal pain, distension, vomiting, constipation, and tenderness.
Investigations: Abdominal X-ray, CT scan.
Treatment Outline:

- **First-line:** Surgery for emergent cases, fluid resuscitation.
- **Second-line:** Conservative management for non-complicated cases.
  Correct Answer: D. Nitrous oxide ✅

MCQ 212

A 45-year-old man involved in a motor vehicle accident with an isolated head injury, remains in a coma for 5 days after the accident. Which of the following is the most appropriate feeding method in the early stage of management?

A. Gastrostomy tube feeding
B. Nasogastric tube feeding
C. Central parenteral feeding
D. Peripheral parenteral feeding

Explanation:
The correct answer is **B. Nasogastric tube feeding**.
In a patient with a **head injury** who is in a **coma**, **nasogastric tube feeding** is the most

appropriate early method of nutritional support. **Enteral feeding** (through the nasogastric tube) is preferred over parenteral feeding as it is associated with fewer complications such as **infection** and **gut atrophy**.

Head Injury

Definition: Trauma to the skull and brain, potentially leading to coma, seizures, or other neurological deficits.

Key Clinical Presentation: Coma, altered consciousness, signs of head trauma (e.g., swelling, bruising).

Investigations: CT scan, neurological examination.

Treatment Outline:

- **First-line:** Sedation, monitoring, enteral nutrition.
- **Second-line:** Surgery for any intracranial hemorrhage or skull fractures.
  Correct Answer: B. Nasogastric tube feeding ✅

MCQ 213:

A 64-year-old man with emphysema was admitted with hematemesis. On endoscopy, a bleeding duodenal ulcer was diagnosed and controlled, but the patient became confused and agitated (see lab results).

Test Results:

- ABG PO2: 75 (Normal range: 4.7-6.0 kPa)
- pH: 7.23 (Normal range: 7.36-7.45)
- ABG PO2: 42 (Normal range: 10.6-14.2 kPa)

Which of the following is the most appropriate therapy?

A. Dexamethasone
B. Sodium bicarbonate
C. High flow O2 therapy
D. Intubate and hyperventilate

Explanation:

The correct answer is **C. High flow O2 therapy**.

The patient's **hypoxemia (PO2 of 42)** and **acidosis (pH 7.23)** indicate significant respiratory distress, likely due to underlying emphysema and acute respiratory failure. The most appropriate immediate therapy would be **high flow oxygen** to correct hypoxemia and support ventilation. **Intubation** may be required if the patient doesn't respond to oxygen therapy, but the first step is to optimize oxygenation.

Chronic Emphysema

Definition: Chronic obstructive pulmonary disease (COPD) characterized by progressive airway obstruction, often due to smoking.

Key Clinical Presentation: Shortness of breath, chronic cough, and wheezing.

Investigations: ABG, chest X-ray, pulmonary function tests.

Treatment Outline:

- **First-line:** Oxygen therapy, bronchodilators.
- **Second-line:** Corticosteroids, mechanical ventilation if necessary.
  Correct Answer: C. High flow O2 therapy ✅

MCQ 214:

An 18-year-old woman developed urticaria and wheezing following intravenous injection of contrast for abdominal CT scan. Blood Pressure: 90/50 mmHg, Heart rate: 110 /min, Temperature: 37.2 °C. Which of the following is the most appropriate immediate therapy?

A. Steroids
B. Intubation
C. Epinephrine
D. Beta blocker

Explanation:

The correct answer is **C. Epinephrine**.

This patient is presenting with **anaphylaxis**, which is a life-threatening allergic reaction triggered by the contrast injection. The most appropriate immediate treatment for **anaphylaxis** is **epinephrine**, which acts to rapidly reverse bronchospasm, vasodilation, and improve blood pressure. **Steroids** and **antihistamines** are used in supportive care, but **epinephrine** should be administered first. **Intubation** may be necessary if the patient's airway is compromised, but the priority is the administration of epinephrine.

Anaphylaxis

Definition: Severe, systemic allergic reaction causing respiratory, cardiovascular, and dermatological symptoms.

Key Clinical Presentation: Urticaria, wheezing, hypotension, tachycardia, and potential airway compromise.

Investigations: Clinical diagnosis based on symptoms.
Treatment Outline:

- **First-line:** Epinephrine intramuscularly.
- **Second-line:** Antihistamines, steroids, bronchodilators.
  Correct Answer: C. Epinephrine ✅

MCQ 215

A 40-year-old woman presented with progressive redness over the dorsum of the left hand associated with pain and fever. She gave a history of a knife prick at the affected area 3 days ago. Temperature: 37.9 °C. Which of the following is the most likely diagnosis?

A. Cellulitis
B. Carbuncle
C. Gas gangrene
D. Necrotizing fasciitis

Explanation:
The correct answer is **A. Cellulitis**.
This patient's presentation with **redness, pain, and fever** following a **puncture wound** is characteristic of **cellulitis**, which is a skin and soft tissue infection caused by bacteria, most commonly **Streptococcus** or **Staphylococcus** species. The absence of severe systemic signs and symptoms like crepitus or rapid progression of infection rules out **gas gangrene** and **necrotizing fasciitis**. **Carbuncle** is a deeper infection of multiple furuncles, which doesn't fit the clinical picture here.
Cellulitis
Definition: Bacterial skin infection, commonly caused by **Streptococcus** or **Staphylococcus** species.
Key Clinical Presentation: Red, swollen, and painful skin, often with fever, following trauma.
Investigations: Clinical diagnosis; cultures may be performed.
Treatment Outline:

- **First-line:** Oral antibiotics (e.g., dicloxacillin, cephalexin).
- **Second-line:** IV antibiotics for severe cases or if no improvement.
  Correct Answer: A. Cellulitis ✅

MCQ 216;

A 50-year-old man who is known to have non-insulin dependent diabetes, presented with a progressive painful swelling on the back of the neck for 5 days. On physical examination, there is a 5x6 cm swelling and redness with multiple discharging openings. Which of the following is the most likely diagnosis?

A. Abscess
B. Cellulitis
C. Furuncle
D. Carbuncle

Explanation:
The correct answer is **D. Carbuncle**.
The presence of a **painful, swollen mass with multiple discharging openings** suggests a **carbuncle**, which is a collection of furuncles (boils) that are deeper and more extensive. This condition is more common in patients with diabetes. **Abscess** could also be a consideration, but the characteristic multiple drainage points point towards a carbuncle.
Carbuncle
Definition: A large, deep skin abscess formed by the coalescence of multiple furuncles, typically associated with **Staphylococcus aureus** infection.
Key Clinical Presentation: Painful, red, swollen mass with multiple drainage points, often with fever.
Investigations: Clinical diagnosis; drainage and culture of pus.
Treatment Outline:

- **First-line:** Incision and drainage, followed by antibiotics (e.g., clindamycin, cephalexin).
- **Second-line:** Surgical drainage for severe cases.
  Correct Answer: D. Carbuncle ✅

MCQ 217

Which of the following is a contraindication for liver transplantation?

A. Acute liver failure
B. Liver cirrhosis with active alcohol
C. End stage liver disease with ascites
D. End stage liver disease with encephalopathy

Explanation:
The correct answer is **B. Liver cirrhosis with active alcohol**.
Active alcohol consumption in patients with liver cirrhosis is considered a contraindication for liver transplantation, as ongoing alcohol use undermines the potential success of the transplant and increases the risk of organ rejection or complications. Patients must show a sustained period of abstinence from alcohol (usually 6 months) to be considered eligible for a transplant.
Liver Transplantation
Definition: A surgical procedure to replace a failing liver with a healthy liver from a donor.
Key Clinical Presentation: End-stage liver disease, including cirrhosis, hepatitis, or acute liver failure.
Investigations: Liver function tests, imaging studies, and biopsy in some cases.
Treatment Outline:

- **First-line:** Liver transplantation for patients with terminal liver failure.
- **Second-line:** Medical management to optimize liver function prior to transplantation.
  Correct Answer: B. Liver cirrhosis with active alcohol ✅

MCQ 218:

A patient presented with frequent vomiting due to gastric outlet obstruction. Which of the following is the most expected abnormality?

A. Metabolic acidosis with hyperkalemia
B. Metabolic alkalosis with hypokalemia
C. Metabolic acidosis with hypokalemia
D. Metabolic alkalosis with hyperkalemia

Explanation:
The correct answer is **B. Metabolic alkalosis with hypokalemia**.
Gastric outlet obstruction leads to frequent vomiting, which causes a loss of gastric acid (HCl) and results in **metabolic alkalosis**. The loss of potassium in the vomit leads to **hypokalemia**, and dehydration further exacerbates the condition.
Gastric Outlet Obstruction
Definition: A blockage of the stomach's pyloric outlet, leading to impaired gastric emptying.
Key Clinical Presentation: Vomiting, epigastric pain, bloating, and weight loss.
Investigations: Upper GI endoscopy, imaging studies like CT scan or barium swallow.
Treatment Outline:

- **First-line:** Decompression via nasogastric tube, fluid replacement, and surgical intervention if necessary.
- **Second-line:** Pyloric dilation or surgery to correct the obstruction.
  Correct Answer: B. Metabolic alkalosis with hypokalemia ✅

MCQ 219

Which of the following hormones reduces the water loss through the kidneys?

A. Vasopressin
B. Angiotensin
C. Cortisone
D. Renin

Explanation:
The correct answer is **A. Vasopressin**.
Vasopressin (ADH) is a hormone produced by the pituitary gland that acts on the kidneys to **increase water reabsorption**, reducing water loss. It does this by making the kidneys' collecting ducts more permeable to water.
Renal Regulation
Definition: Hormonal control of kidney function, primarily affecting fluid balance, blood pressure, and electrolyte levels.
Key Clinical Presentation: Symptoms related to dehydration or overhydration, altered urine output.
Investigations: Urine osmolality, ADH levels, and renal function tests.
Treatment Outline:

- **First-line:** Vasopressin or other water-retaining agents for conditions like diabetes insipidus.
- **Second-line:** Angiotensin inhibitors for blood pressure regulation.
  Correct Answer: A. Vasopressin ✅

MCQ 220

A patient presented with severe diarrhea. Which of the following is the most expected abnormality?

A. Metabolic acidosis with hypokalemia
B. Metabolic acidosis with hyperkalemia
C. Metabolic alkalosis with hypokalemia
D. Metabolic alkalosis with hyperkalemia

Explanation:
The correct answer is **A. Metabolic acidosis with hypokalemia**.
Severe diarrhea leads to the loss of **bicarbonate** in the stool, causing **metabolic acidosis**. Additionally, the loss of potassium in the stool results in **hypokalemia**.
Diarrhea and Electrolyte Imbalance
Definition: Excessive loss of fluids and electrolytes from the gastrointestinal tract, leading to dehydration and metabolic abnormalities.
Key Clinical Presentation: Frequent loose stools, abdominal discomfort, dehydration, and signs of electrolyte imbalance.
Investigations: Serum electrolytes, ABG, stool analysis.
Treatment Outline:

- **First-line:** Fluid and electrolyte replacement.
- **Second-line:** Antidiarrheal medications for symptom relief.
  Correct Answer: A. Metabolic acidosis with hypokalemia ✅

MCQ 221

A 56-year-old man with schizophrenia is started on neuroleptic medications to treat his psychotic symptoms. Which of the following side effects is the patient at greatest risk for?

A. Seizures
B. Akathisia
C. Hyperthermia
D. Myocardial infarction

Explanation:
The correct answer is **B. Akathisia**.
Akathisia is a common side effect of **antipsychotic medications**, particularly **neuroleptics**. It involves restlessness, an inability to sit still, and agitation, and is a frequent side effect of dopamine-blocking medications used in the treatment of schizophrenia.
Neuroleptic Medications and Side Effects
Definition: Antipsychotic drugs used to manage symptoms of schizophrenia and other psychotic disorders.

Key Clinical Presentation: Restlessness, tremors, tardive dyskinesia, or other motor symptoms.
Investigations: Monitoring for side effects, especially in long-term treatment.
Treatment Outline:

- **First-line:** Adjust the medication dose or switch to an atypical antipsychotic.
- **Second-line:** Use of beta-blockers or benzodiazepines for symptom relief.
  Correct Answer: B. Akathisia ✅

MCQ 222:

A patient presented with severe maxillofacial injury with mandibular fracture dislocation and profuse oral bleeding. Which of the following is the most appropriate way to secure the airway?

A. Laryngeal mask
B. Orotracheal tube
C. Nasotracheal tube
D. Cricothyroidotomy

Explanation:
The correct answer is **B. Orotracheal tube**.
In cases of severe maxillofacial injury, an **oro-tracheal tube** is often preferred to secure the airway. This is because the mouth and upper airway are the primary routes for intubation, and it provides a stable airway while maintaining access to the oral cavity. A **laryngeal mask** (Option A) can be an option in less severe cases, but for significant trauma with the risk of obstruction, endotracheal intubation via the oro-tracheal route is the best approach.
Maxillofacial Injury and Airway Management
Definition: Trauma to the face and jaw, often resulting in fractures, bleeding, and airway compromise.
Key Clinical Presentation: Difficulty breathing, facial deformities, and bleeding.
Investigations: Imaging (e.g., X-ray, CT) to assess fractures.
Treatment Outline:

- **First-line:** Secure airway with oro-tracheal intubation.
- **Second-line:** Nasotracheal intubation if oro-tracheal intubation is not possible.
  Correct Answer: B. Orotracheal tube ✅

MCQ 223:

Which of the following is the most common cause of fever in the first 24-hours after abdominal surgery?

A. Atelectasis
B. Wound infection
C. Blood transfusion
D. Intra-abdominal collection

Explanation:
The correct answer is **A. Atelectasis**.
Atelectasis is the most common cause of fever in the first 24 hours after abdominal surgery. It occurs due to reduced lung expansion, often related to anesthesia and post-surgical pain, leading to alveolar collapse. Other causes of post-surgical fever, such as **wound infection** and **intra-abdominal collection**, typically manifest later, and **blood transfusion**-related fever is uncommon in the early postoperative period.
Postoperative Fever Etiology
Definition: Fever that occurs following surgical procedures, often due to a variety of causes.
Key Clinical Presentation: Fever within the first 24-48 hours after surgery.
Investigations: Chest X-ray, blood cultures, or CT imaging to rule out other causes.
Treatment Outline:

- **First-line:** Chest physiotherapy, incentive spirometry, and early mobilization.
- **Second-line:** Antibiotics or drainage if infection or collection is suspected.
  Correct Answer: A. Atelectasis ✅

MCQ 224:

A 21-year-old man was brought to the Emergency Department following a motorbike crash, complaining of difficulty in breathing. On examination, he was conscious, oriented, dyspnoeic with reduced breath sounds and tympanic percussion over the left hemithorax. Blood pressure 90/55 mmHg Heart rate 110 /min. Which of the following is the most appropriate next step in management?

A. Plain chest X-ray
B. Endotacheal intubation and ventilation
C. Needle decompression of left hemithorax
D. Chest tube insertion in the left hemithorax

Explanation:
The correct answer is **C. Needle decompression of left hemithorax**.
This patient presents with signs of a **tension pneumothorax**, as evidenced by **dyspnea**, **decreased breath sounds**, **tympanic percussion**, and **hypotension**. The most appropriate initial management is **needle decompression** to quickly relieve the pressure buildup in the chest and prevent further cardiovascular collapse. A **chest tube insertion** (Option D) is also a treatment for pneumothorax but is usually performed after decompression.
Pneumothorax Management
Definition: Accumulation of air in the pleural space causing lung collapse and mediastinal shift.
Key Clinical Presentation: Dyspnea, hypotension, tracheal deviation, and decreased breath sounds.
Investigations: Chest X-ray or ultrasound (after initial decompression).
Treatment Outline:

- **First-line:** Needle decompression followed by chest tube insertion.
- **Second-line:** Surgery for persistent or recurrent pneumothorax.
  Correct Answer: C. Needle decompression of left hemithorax ✅

MCQ 225

A 20-year-old man was brought to the Emergency Department, 20 minutes after a motor vehicle accident. On examination, his extremities were warm, he was conscious, oriented with a clear chest and normal abdomen. Blood pressure 90/40 mmHg Heart rate 70 /min. Which of the following is the most likely type of shock?

A. Septic
B. Neurogenic
C. Cardiogenic
D. Hypovolemic

Explanation:
The correct answer is **D. Hypovolemic**.
In this case, the patient is presenting with **hypotension** and **tachycardia** following a **trauma** (motor vehicle accident), but there are no signs of infection, neurological injury, or heart failure. This combination of factors points toward **hypovolemic shock** due to **hemorrhage** or fluid loss, which is common following trauma.
Shock Types
Definition: A condition where there is inadequate perfusion of tissues, leading to organ

dysfunction.
Key Clinical Presentation: Hypotension, tachycardia, and signs of inadequate perfusion.
Investigations: Clinical assessment, vital signs, and possible imaging to identify the source of blood loss.
Treatment Outline:

- **First-line:** Fluid resuscitation, blood products if necessary.
- **Second-line:** Surgical intervention to control bleeding if indicated.
  Correct Answer: D. Hypovolemic ✅

MCQ 226

A 25-year-old man was admitted through the Emergency Department complaining of a 5-day history of right iliac fossa pain. Physical examination confirmed a tender mass in the right iliac fossa with sluggish bowel sounds. CT scan: Acute appendicitis, appendicolith, with 10x15 cm abscess, extending from the right iliac fossa to the abdominal wall. Which of the following is the most appropriate intervention?

A. Open drainage of the collection
B. Percutaneous drainage of the collection
C. Open appendectomy with drainage of the collection
D. Laparoscopic appendectomy with drainage of collection

Explanation:
The correct answer is **C. Open appendectomy with drainage of the collection**.
In this case, the patient presents with **acute appendicitis** complicated by an **abscess**. The presence of an appendicolith and a large abscess requires drainage to control infection. **Open appendectomy with drainage** is preferred because it allows thorough inspection of the abdominal cavity and the removal of the appendix along with drainage of the abscess.
Laparoscopic appendectomy (Option D) may not be appropriate due to the large abscess and potential difficulty in accessing the affected area via laparoscopy.
Appendicitis with Abscess Management
Definition: Acute inflammation of the appendix, often complicated by perforation and abscess formation.
Key Clinical Presentation: Abdominal pain, fever, and a palpable mass in the right iliac fossa.
Investigations: CT scan to confirm appendicitis and abscess formation.
Treatment Outline:

- **First-line:** Open appendectomy with drainage of abscess.

- **Second-line:** Percutaneous drainage if surgery is not immediately feasible.
  Correct Answer: C. Open appendectomy with drainage of the collection ✅

MCQ 227

A 33-year-old woman presented with a 1-day history of right upper quadrant abdominal pain associated with fever, nausea, and vomiting. Physical examination confirmed marked tenderness in the right upper quadrant of the abdomen. Ultrasound: Thick wall gallbladder, pericholecystic fluids, and multiple gallstones. Which of the following is the most appropriate management?

A. Early open cholecystectomy
B. Early laparoscopic cholecystectomy
C. Delayed cholecystectomy in 2-3 months
D. Emergency cholecystectomy tube insertion

Explanation:
The correct answer is **B. Early laparoscopic cholecystectomy**.
The patient presents with **acute cholecystitis**, as evidenced by right upper quadrant pain, fever, and ultrasound findings of gallbladder wall thickening and pericholecystic fluid. Early **laparoscopic cholecystectomy** is the treatment of choice, ideally within 24 to 72 hours of symptom onset, to avoid complications such as perforation. **Open cholecystectomy** (Option A) may be considered in certain cases, but laparoscopic surgery is preferred due to its minimally invasive nature. Delaying surgery (Option C) is not recommended in the acute setting. **Cholecystectomy tube insertion** (Option D) is generally reserved for patients who are not candidates for surgery initially.
Acute Cholecystitis Management
Definition: Inflammation of the gallbladder due to obstruction, typically by gallstones.
Key Clinical Presentation: Right upper quadrant pain, fever, and nausea.
Investigations: Ultrasound showing thickened gallbladder wall, pericholecystic fluid, and gallstones.
Treatment Outline:

- **First-line:** Laparoscopic cholecystectomy.

- **Second-line:** Percutaneous cholecystostomy if surgery is contraindicated.
  Correct Answer: B. Early laparoscopic cholecystectomy ✅

MCQ 228

A 28-year-old woman was admitted through the Emergency Department with a 12-hour history of epigastric pain. Physical examination confirmed epigastric tenderness. Over the next 3 days during the hospitalization, the patient completely recovered clinically and biochemically. Ultrasound: Multiple gallstones with mild extrahepatic biliary dilatation. Which of the following is the most appropriate intervention?

A. Abdominal computed tomography
B. Cholecystectomy before discharge
C. Interval cholecystectomy in 6-8 weeks
D. Endoscopic retrograde cholangiopancreatography

Explanation:
The correct answer is **C. Interval cholecystectomy in 6-8 weeks**.
This patient has had **biliary colic** or **acute cholecystitis** that resolved clinically, with **gallstones** identified on ultrasound. In cases of mild biliary colic without severe complications, it is typically recommended to perform an **interval cholecystectomy** (usually 6-8 weeks after the initial episode) once the patient has recovered. **Cholecystectomy before discharge** (Option B) is generally not required unless there is ongoing or recurrent acute cholecystitis. **CT scan** (Option A) is not necessary unless complications are suspected. **Endoscopic retrograde cholangiopancreatography** (ERCP, Option D) is reserved for cases with bile duct obstruction or choledocholithiasis.
Biliary Colic Management
Definition: Abdominal pain caused by the passage of gallstones through the biliary tract.
Key Clinical Presentation: Epigastric or right upper quadrant pain, often postprandial.
Investigations: Ultrasound showing gallstones and mild biliary dilatation.
Treatment Outline:

- **First-line:** Interval cholecystectomy after resolution of acute symptoms.
- **Second-line:** ERCP if choledocholithiasis is suspected.
  Correct Answer: C. Interval cholecystectomy in 6-8 weeks ✅

MCQ 229

A 38-year-old man with a history of recurrent acute pancreatitis over the last 10 years, presents with a 12-hour history of frequent episodes of vomiting fresh blood. Intravenous fluid resuscitation was initiated. Upper gastrointestinal endoscopy: Isolated bleeding fundal

varices, controlled by injection sclerotherapy. Ultrasound: Splenomegaly, normal portal vein, and thrombosed splenic vein. Which of the following is the most appropriate management?

A. Splenectomy
B. Portacaval shunt
C. Distal splenorenal shunt
D. Sengstaken-Blakemore tube

Explanation:
The correct answer is **C. Distal splenorenal shunt**.
This patient with **recurrent acute pancreatitis** and **splenomegaly** has developed **esophageal varices** likely due to portal hypertension secondary to splenic vein thrombosis. A **distal splenorenal shunt** (DSS) is the treatment of choice for managing variceal bleeding in the setting of portal hypertension, as it decompresses the portal venous system and reduces variceal pressure. **Portacaval shunt** (Option B) is another option but is less commonly used than DSS due to its potential complications. **Sengstaken-Blakemore tube** (Option D) may be used temporarily to control bleeding, but it is not a definitive treatment.
Portal Hypertension Management
Definition: Increased pressure in the portal venous system, often resulting from cirrhosis or splenic vein thrombosis.
Key Clinical Presentation: Hematemesis, splenomegaly, and esophageal varices.
Investigations: Ultrasound showing splenomegaly and thrombosed splenic vein.
Treatment Outline:

- **First-line:** Distal splenorenal shunt.
- **Second-line:** Temporary Sengstaken-Blakemore tube for stabilization.
  Correct Answer: C. Distal splenorenal shunt ✅

MCQ 230:

A 28-year-old woman presented to the Emergency Department with 18-hour history of severe epigastric pain associated with vomiting. She is known to have multiple small gallstones. Physical examination confirmed upper abdominal tenderness.

Test Results:

- WBC: 13.4 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- Alkaline phosphatase: 289 IU/L (Normal: 39-117 IU/L)

- Amylase: 488 IU/L (Normal: 24-151 IU/L)
- Direct bilirubin: 19 µmol/L (Normal: 1.5-6.5 µmol/L)
- Total bilirubin: 34 µmol/L (Normal: 3.5-16.5 µmol/L)

Which of the following is the most appropriate initial management?

A. Abdominal ultrasound
B. Intravenous crystalloid infusion
C. Intravenous broad spectrum antibiotics
D. Endoscopic retrograde cholangiopancreatography

Explanation:
The correct answer is **B. Intravenous crystalloid infusion**.
This patient presents with **gallstone pancreatitis** (evidenced by elevated amylase and bilirubin, and abdominal pain), which is an inflammatory condition of the pancreas due to obstruction of the pancreatic duct by gallstones. The most important first step in the management of acute pancreatitis is **fluid resuscitation** with **intravenous crystalloid infusion** to correct dehydration, support circulatory function, and prevent organ failure. **Abdominal ultrasound** (Option A) is important for confirming the diagnosis of gallstones but is not the immediate next step in this acute setting. **Broad-spectrum antibiotics** (Option C) are not routinely used unless there is evidence of infection. **ERCP** (Option D) may be required in the case of **bile duct obstruction**, but it is not the immediate first-line intervention in uncomplicated pancreatitis.
Acute Gallstone Pancreatitis Management
Definition: Inflammation of the pancreas due to obstruction by gallstones.
Key Clinical Presentation: Severe epigastric pain, vomiting, elevated amylase and bilirubin.
Investigations: Abdominal ultrasound to detect gallstones; blood tests for elevated amylase and bilirubin.
Treatment Outline:

- **First-line:** Intravenous crystalloid infusion to stabilize fluid balance.
- **Second-line:** ERCP for biliary obstruction or choledocholithiasis.
  Correct Answer: B. Intravenous crystalloid infusion ✅

MCQ 231

A 19-year-old woman complains of a 4-month history of a left breast mass. Breast examination confirmed a well-circumscribed, firm, smooth mobile lump, 2x2 cm in size, in the lateral upper quadrant of the left breast. Which of the following is the most likely diagnosis?

A. Sclerosis
B. Galactocele
C. Duct ectasia
D. Fibroadenoma

Explanation:
The correct answer is **D. Fibroadenoma**.
This patient's presentation with a well-circumscribed, firm, smooth, and mobile breast mass is most consistent with **fibroadenoma**, a benign tumor common in young women. **Fibroadenomas** are the most frequent benign tumors of the breast, and they typically present as painless, well-defined, mobile lumps. **Galactocele** (Option B) is a benign cyst filled with milk that can occur in lactating women. **Duct ectasia** (Option C) is more common in older women and usually presents with nipple discharge and ductal dilation rather than a palpable mass. **Sclerosis** (Option A) refers to fibrotic changes in the breast tissue but does not present as a discrete, mobile mass.
Fibroadenoma Management
Definition: Benign breast tumor composed of fibrous and glandular tissue.
Key Clinical Presentation: Painless, mobile, firm, well-circumscribed mass.
Investigations: Ultrasound and mammogram to confirm diagnosis and assess characteristics.
Treatment Outline:

- **First-line:** Observation or excisional biopsy if symptomatic or enlarging.
- **Second-line:** Surgical excision if needed for symptomatic relief.
  Correct Answer: D. Fibroadenoma ✅

MCQ 232

A 25-year-old breastfeeding mother presented with right breast pain for 24-hours. Examination confirmed a tender swelling with skin redness in the outer lower quadrant of the right breast, with warm skin, but no fluctuation. Which of the following is the most appropriate initial management?

A. Aspiration
B. Flucloxacillin
C. Prolactin antagonist
D. Incision and drainage

Explanation:
The correct answer is **B. Flucloxacillin**.
This patient presents with **mastitis**, likely due to a bacterial infection of the breast tissue.

Flucloxacillin, a **penicillin** antibiotic, is the first-line treatment for **mastitis** caused by **Staphylococcus aureus**, which is the most common pathogen. **Aspiration** (Option A) is generally used for abscesses, not for early-stage mastitis without fluctuance. **Prolactin antagonists** (Option C) are not used in the management of mastitis. **Incision and drainage** (Option D) is indicated if an abscess has formed and fluctuance is present, but it is not appropriate for early mastitis.

Mastitis Management

Definition: Infection of the breast tissue, typically in lactating women.

Key Clinical Presentation: Painful, tender breast with erythema and warmth, usually in the outer quadrant.

Investigations: Diagnosis is clinical, supported by ultrasound if an abscess is suspected.

Treatment Outline:

- **First-line:** Oral antibiotics (flucloxacillin).
- **Second-line:** Aspiration or drainage if an abscess forms.
  Correct Answer: B. Flucloxacillin ✅

MCQ 233

A 65-year-old man presented to the Emergency Department with an 8-hour history of severe, sudden central abdominal pain associated with vomiting. He has a history of cardiomyopathy. He reports having passed loose stools once. On examination, the abdomen was soft with exaggerated bowel sounds. Which of the following is the most likely diagnosis?

A. Acute pancreatitis
B. Small bowel obstruction
C. Dissecting aortic aneurysm
D. Mesenteric vascular occlusion

Explanation:

The correct answer is **D. Mesenteric vascular occlusion**.

This patient with sudden, severe central abdominal pain, vomiting, and risk factors such as **cardiomyopathy** is highly likely to have **mesenteric ischemia** (or mesenteric vascular occlusion). The patient's presentation, with exaggerated bowel sounds, is characteristic of an ischemic event affecting the mesenteric arteries. **Acute pancreatitis** (Option A) typically presents with upper abdominal pain and elevated amylase, not central abdominal pain. **Small bowel obstruction** (Option B) is a possibility, but the sudden onset and severity of pain point more toward ischemia. **Dissecting aortic aneurysm** (Option C) would present with more localized pain

and a different set of associated symptoms.
Mesenteric Vascular Occlusion Management
Definition: Occlusion of the mesenteric vessels leading to bowel ischemia.
Key Clinical Presentation: Sudden onset of severe abdominal pain, vomiting, and risk factors like heart disease.
Investigations: CT angiography or ultrasound for diagnosis.
Treatment Outline:

- **First-line:** Surgical intervention to restore blood flow.
- **Second-line:** Endovascular therapy if available.
  Correct Answer: D. Mesenteric vascular occlusion ✅

MCQ 234

A 58-year-old woman complains of a 4-month history of a right breast lump, increasing in size, associated with bloody nipple discharge. Physical examination confirmed a 4cm central hard mass with ill-defined edges. The right nipple was inverted with enlarged axillary lymph nodes.

Mammogram:

- Sub-areolar dense irregular density with microcalcification and thickening of overlying skin.

Which of the following is the most appropriate next step in management?

A. Excisional biopsy
B. Needle core biopsy
C. Discharge cytology
D. Fine needle aspiration

Explanation:
The correct answer is **B. Needle core biopsy**.
This patient's presentation of a **bloody nipple discharge**, **inverted nipple**, **palpable breast mass**, and **enlarged axillary lymph nodes** with associated **microcalcifications** on mammography suggests the possibility of **breast cancer**, most likely **invasive ductal carcinoma**. The appropriate next step in the management of suspected breast cancer is a **needle core biopsy**, which is more accurate and reliable for confirming the diagnosis than fine needle aspiration. **Excisional biopsy** (Option A) is not the first step; it would be done later if other biopsy methods fail. **Discharge cytology** (Option C) is not indicated for diagnosing the cause of a breast mass with suspicious characteristics. **Fine needle aspiration** (Option D) is less accurate than core biopsy, particularly

when the lesion is large or involves lymph nodes.
Breast Cancer Management
Definition: Malignant tumor arising from the epithelial tissue of the breast, most commonly from the ducts or lobules.
Key Clinical Presentation: Lump in the breast, bloody discharge, nipple retraction, axillary lymphadenopathy.
Investigations: Mammogram, ultrasound, core biopsy.
Treatment Outline:

- **First-line:** Needle core biopsy for histological diagnosis.
- **Second-line:** Surgical excision, chemotherapy, and radiotherapy based on stage and type.
  Correct Answer: B. Needle core biopsy ✅

MCQ 235

A 68-year-old man was admitted through the Emergency Department with a 24-hour history of sudden severe lower abdominal pain, associated with progressive abdominal distension and constipation. On examination, the abdomen was hugely distended, tympanic with tenderness over the left iliac fossa. Digital rectal examination confirmed an empty rectum.

Test Results:

- WBC: 22 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Abdominal plain X-ray:

- Y-shaped shadow surrounded by a hugely distended colon arising from the pelvis.

Which of the following is the most appropriate intervention?

A. Fleet enema
B. Colonoscopy
C. CT with double contrast
D. Sigmoid colectomy with end colostomy

Explanation:
The correct answer is **C. CT with double contrast**.
The patient's symptoms of **severe lower abdominal pain**, **abdominal distension**, and **constipation**, along with an **X-ray showing a Y-shaped shadow**, are suggestive of **volvulus**, a condition where the colon twists upon itself, leading to bowel obstruction. The **most**

appropriate next step is **CT with double contrast** to confirm the diagnosis and evaluate the extent of the obstruction and any potential bowel ischemia. **Fleet enema** (Option A) is not recommended in the case of suspected bowel ischemia or perforation. **Colonoscopy** (Option B) is generally not used in the acute setting of suspected volvulus because it can exacerbate perforation risk. **Sigmoid colectomy** (Option D) may be necessary later if bowel necrosis or irreducibility is found, but CT is the first diagnostic step.

Volvulus Management

Definition: Twisting of the colon, leading to obstruction and potentially ischemia.

Key Clinical Presentation: Severe abdominal pain, distension, constipation, vomiting.

Investigations: Plain abdominal X-ray, CT scan with contrast.

Treatment Outline:

- **First-line:** CT with double contrast for diagnosis.
- **Second-line:** Surgery (colectomy) if necessary to relieve ischemia or perforation.
  Correct Answer: C. CT with double contrast ✅

MCQ 236

A 28-year-old man presented with a 2-month history of frequent attacks of bloody stools. Physical examination was unremarkable. Proctoscopic examination showed multiple rectal polyps.

Colonoscopy:

- Colon and rectum were carpeted with polyps, multiple biopsies were taken.

Which of the following is the most likely diagnosis?

A. Ulcerative colitis
B. Diverticulosis coli
C. Familial adenomatous polyposis
D. Human Papilloma virus proctitis

Explanation:

The correct answer is **C. Familial adenomatous polyposis (FAP)**.

The patient's presentation of **frequent bloody stools** and **multiple polyps in the colon** with **rectal polyps** is strongly suggestive of **Familial adenomatous polyposis**, a genetic condition characterized by the development of hundreds to thousands of adenomatous polyps in the colon, which, if untreated, can progress to colorectal cancer. **Ulcerative colitis** (Option A) can cause bloody stools but is typically characterized by inflammation and ulceration, not polyps.

Diverticulosis coli (Option B) involves the formation of diverticula but does not present with polyps. **HPV proctitis** (Option D) usually presents with anal or rectal symptoms due to infection but is not associated with the formation of polyps.
Familial Adenomatous Polyposis (FAP) Management
Definition: An inherited condition characterized by numerous adenomatous polyps in the colon.
Key Clinical Presentation: Bloody stools, multiple polyps on colonoscopy.
Investigations: Colonoscopy, genetic testing for APC gene mutation.
Treatment Outline:

- **First-line:** Prophylactic colectomy or surveillance.
- **Second-line:** Regular colonoscopies and removal of polyps if indicated.
  Correct Answer: C. Familial adenomatous polyposis ✅

MCQ 237

A 72-year-old diabetic presented to the Emergency Department with a 2-day history of lower abdominal pain, nausea, vomiting, and bloody diarrhea. Physical examination confirmed tenderness in the left lower quadrant.

Test Results:

- WBC: 17.4 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Plain abdominal X-ray:

- Thickened descending and sigmoid colon with thumb-printing appearance.

Which of the following is the most likely diagnosis?

A. Crohn's disease
B. Ischemic colitis
C. Ulcerative colitis
D. Acute diverticulitis

Explanation:
The correct answer is **B. Ischemic colitis**.
This patient's history of **diabetes** and **abdominal pain**, along with **bloody diarrhea** and the **thumb-printing appearance** on abdominal X-ray, strongly suggests **ischemic colitis**, which occurs when blood flow to the colon is impaired, leading to ischemia and inflammation. **Crohn's disease** (Option A) can cause abdominal pain and diarrhea but typically affects the terminal ileum and has a different radiological appearance. **Ulcerative colitis** (Option C) typically involves

continuous colonic inflammation starting in the rectum, but is not usually associated with a thumb-printing appearance. **Acute diverticulitis** (Option D) typically causes localized pain, usually in the left lower quadrant, but would show colonic diverticula rather than the thumb-printing pattern on X-ray.

Ischemic Colitis Management

Definition: Inflammation and injury to the colon due to impaired blood supply.

Key Clinical Presentation: Abdominal pain, bloody diarrhea, risk factors such as diabetes or cardiovascular disease.

Investigations: CT abdomen, angiography, X-ray.

Treatment Outline:

- **First-line:** Supportive care, IV fluids, and bowel rest.
- **Second-line:** Surgery if bowel necrosis is present.
  Correct Answer: B. Ischemic colitis ✅

MCQ 238

A 28-year-old man presented with a 4-day history of abdominal pain, associated with frequent bloody diarrhoea and vomiting. Physical examination confirmed he was dehydrated, had a distended abdomen, tenderness mainly over the lower abdomen with sluggish bowel sounds. He is known to have ulcerative colitis and is on medical treatment.

Test Results:

- WBC: 21.3 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- Plain abdominal X-ray: Dilated transverse colon (15 cm) with loss of haustrations.

Which of the following is the most appropriate management?

A. High dose systemic steroid
B. High dose systemic infliximab
C. Proctocolectomy with ileal pouch
D. Total colectomy with end ileostomy

Explanation:

The correct answer is **A. High dose systemic steroid**.

This patient with **ulcerative colitis (UC)** and **toxic megacolon** (suggested by the **dilated transverse colon** on X-ray, **loss of haustrations**, and clinical signs of **severe inflammation** such as **fever**, **diarrhoea**, and **abdominal distension**) requires **high dose systemic steroids** as the first-line treatment. **Steroids** are used to control acute flare-ups of UC, including in cases of toxic megacolon, before considering more invasive interventions. **Infliximab** (Option B) is an option for refractory cases but is typically reserved for patients who do not respond to steroids. **Proctocolectomy with ileal pouch** (Option C) and **total colectomy with end ileostomy** (Option D) are reserved for patients who have failed medical therapy or have significant complications such as perforation.

Ulcerative Colitis Management

Definition: Chronic inflammatory bowel disease affecting the colon, characterized by intermittent episodes of mucosal inflammation.

Key Clinical Presentation: Bloody diarrhoea, abdominal pain, weight loss, fever, and urgency.

Investigations: Stool cultures, colonoscopy, X-ray (in this case).

Treatment Outline:

- **First-line:** High dose systemic steroids for acute flare-ups.
- **Second-line:** Immunosuppressive drugs or biologics (e.g., infliximab).
- **Definitive:** Surgery (proctocolectomy) for refractory cases or complications.
  Correct Answer: A. High dose systemic steroid ✅

MCQ 239

A 66-year-old healthy man complains of rectal bleeding for 3 months. Anorectal examination confirmed 3rd degree haemorrhoids.

Which of the following is the most appropriate next step in management?

A. Colonoscopy
B. Haemorrhoidectomy
C. Abdominal computed tomography
D. Submucosal injection of sclerosant

Explanation:
The correct answer is **A. Colonoscopy**.
Although the patient has **3rd degree haemorrhoids** (which are prolapsed and may require **surgical intervention**), **rectal bleeding** needs to be investigated further to rule out other **colorectal pathology**, such as **colorectal cancer**. **Colonoscopy** is the **gold standard** for evaluating **rectal bleeding** and can help to evaluate the severity of hemorrhoids, as well as rule out other causes. **Haemorrhoidectomy** (Option B) may be considered if the hemorrhoids are causing significant symptoms, but the bleeding must first be evaluated. **Abdominal CT** (Option C) is not indicated in this case, as colonoscopy is more appropriate. **Submucosal injection**

(Option D) is used for **internal hemorrhoids**, but further diagnostic evaluation is needed first.
Rectal Bleeding Management
Definition: Bleeding from the rectum or anus, potentially from hemorrhoids, anal fissures, or colorectal malignancy.
Key Clinical Presentation: Bright red blood on stool or toilet paper, rectal discomfort, or prolapsed tissue.
Investigations: Colonoscopy, anoscopy.
Treatment Outline:

- **First-line:** Colonoscopy for diagnostic evaluation.
- **Second-line:** Surgical treatment (e.g., hemorrhoidectomy) if appropriate.
  Correct Answer: A. Colonoscopy ✅

MCQ 240

A 52-year-old man presented to the Emergency Department complaining of a 5-day history of right upper quadrant abdominal pain associated with fever, and yellowish discoloration of sclera. On examination, he was jaundiced with mild right upper quadrant abdominal tenderness.

Test Results:

- Alkaline phosphatase: 389 IU/L (Normal: 39-117 IU/L)
- Alanine aminotransferase: 85 IU/L (Normal: 5-40 IU/L)
- Aspartate aminotransferase: 78 IU/L (Normal: 12-40 IU/L)
- Direct bilirubin: 55 µmol/L (Normal: 1.5-6.5 µmol/L)
- Total bilirubin: 66 µmol/L (Normal: 3.5-16.5 µmol/L)

Which of the following is the most appropriate next investigation?

A. Abdominal computed tomography
B. Trans-abdominal ultrasonography
C. Magnetic resonance cholangiopancreatography
D. Endoscopic retrograde cholangiopancreatography

Explanation:
The correct answer is **B. Trans-abdominal ultrasonography**.
This patient's presentation of **right upper quadrant pain**, **jaundice**, and elevated **bilirubin** and

alkaline phosphatase strongly suggests a diagnosis of **cholangitis** or **biliary obstruction**. **Trans-abdominal ultrasonography** is the **first-line** investigation to assess the biliary tree, identify **gallstones**, and detect any **biliary obstruction** or **dilatation**. **CT imaging** (Option A) can be used later if needed but is not the first choice for biliary tract evaluation. **Magnetic resonance cholangiopancreatography (MRCP)** (Option C) is highly useful for evaluating the biliary tree, but **ultrasound** is the most accessible and cost-effective initial imaging modality. **Endoscopic retrograde cholangiopancreatography (ERCP)** (Option D) is the gold standard for treating obstructive conditions like **choledocholithiasis** or **cholangitis**, but it is invasive and should only be performed if therapeutic intervention is required.

Biliary Obstruction Management

Definition: Blockage in the biliary tract, often due to gallstones or tumors, leading to jaundice and liver dysfunction.

Key Clinical Presentation: Right upper quadrant pain, jaundice, fever, elevated liver enzymes.

Investigations: Abdominal ultrasound, MRCP, CT scan.

Treatment Outline:

- **First-line:** Ultrasound for initial evaluation.
- **Second-line:** ERCP for therapeutic intervention if needed.
  Correct Answer: B. Trans-abdominal ultrasonography ✅

MCQ 241:

A 38-year-old woman presented to the Emergency Department complaining of epigastric pain and post-prandial vomiting. She was conservatively treated for acute pancreatitis 4 weeks ago. Physical examination confirmed tenderness and fullness in the epigastric region.

Test Results:

- **Amylase:** 413 IU/L (Normal: 24-151 IU/L)
- **WBC:** 15.9 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Abdominal ultrasound:

- Fluid collection surrounded by thick wall in the lesser sac, 15 x 17 cm in size.

Which of the following is the most likely diagnosis?

A. Abscess
B. Phlegmon

C. Pseudocyst
D. Walled off necrosis

Explanation:
The correct answer is **D. Walled off necrosis**.
The patient's history of **acute pancreatitis**, **elevated amylase**, and a **fluid collection in the lesser sac** with a **thick wall** is consistent with **walled-off necrosis** (WON). WON is a **complication of acute pancreatitis** that forms when pancreatic necrosis becomes encapsulated by granulation tissue. This is often seen 4-6 weeks after acute pancreatitis. **Abscesses** (Option A) may also be a complication of pancreatitis but are typically more diffuse and less organized. **Phlegmon** (Option B) refers to an inflammatory mass without well-defined borders. **Pseudocysts** (Option C) are fluid collections that do not have the solid wall characteristic of WON.
Pancreatitis Complications
Definition: A complication of pancreatitis characterized by localized inflammation or fluid collection.
Key Clinical Presentation: Severe epigastric pain, vomiting, elevated amylase, and radiological evidence of fluid collections.
Investigations: Abdominal ultrasound, CT scan, MRCP.
Treatment Outline:

- **First-line:** Conservative management, antibiotics if infected.
- **Second-line:** Surgery or drainage if complications develop (e.g., WON).
  Correct Answer: D. Walled off necrosis ✅

MCQ 242

A 42-year-old man presented to the Emergency Department complaining of a 12-hour history of severe epigastric pain. Physical examination confirmed upper abdominal tenderness.

Test Results:

- **WBC:** 14.5 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- **Amylase:** 254 IU/L (Normal: 24-151 IU/L)

Erect chest X-ray:

- Normal

Which of the following is the most appropriate next step in management?

A. Abdominal CT scan
B. Abdominal ultrasound
C. Exploratory laparotomy
D. Urine analysis for amylase

Explanation:
The correct answer is **A. Abdominal CT scan**.
This patient presents with **severe epigastric pain**, elevated **amylase**, and **leukocytosis**, which are concerning for **acute pancreatitis** or another abdominal pathology. An **abdominal CT scan** is the most appropriate next step in management to assess the severity of pancreatitis, look for complications (e.g., pancreatic pseudocyst or abscess), and rule out other potential causes of abdominal pain. **Abdominal ultrasound** (Option B) is useful for evaluating the biliary tree but does not provide as much detail for pancreatitis. **Exploratory laparotomy** (Option C) is not indicated without a more specific diagnosis. **Urine analysis for amylase** (Option D) is unnecessary, as **serum amylase** already suggests pancreatic involvement.
Acute Pancreatitis Management
Definition: Inflammation of the pancreas, often due to alcohol use or gallstones.
Key Clinical Presentation: Epigastric pain, vomiting, elevated amylase, leukocytosis.
Investigations: CT scan, ultrasound, MRI.
Treatment Outline:

- **First-line:** Conservative management (fluids, pain control).
- **Second-line:** Surgery if complications (e.g., necrosis, pseudocyst) arise.
  Correct Answer: A. Abdominal CT scan ✅

MCQ 243

A 67-year-old man presented with a 2-month history of yellowish discoloration of his sclera and skin with dark urine and pale stool. On examination, he was jaundiced with a palpable distended gallbladder.

Test Results:

- **Alkaline phosphatase:** 421 IU/L (Normal: 39-117 IU/L)
- **Direct bilirubin:** 122.3 µmol/L (Normal: 1.5-6.5 µmol/L)
- **Total bilirubin:** 134.5 µmol/L (Normal: 3.5-16.5 µmol/L)

Ultrasound:

- Dilated intra and extrahepatic bile ducts with a hugely distended gallbladder.

Which of the following is the most likely diagnosis?

A. Klatskin tumour
B. Pancreatic cancer
C. Mirizzi syndrome
D. Common bile duct stone

Explanation:
The correct answer is **D. Common bile duct stone**.
This patient's presentation with **jaundice**, **dark urine**, **pale stool**, and **dilated intra and extrahepatic bile ducts** on ultrasound is consistent with **biliary obstruction**, likely due to a **common bile duct (CBD) stone**. The **distended gallbladder** and the absence of a mass on imaging further support this diagnosis. **Klatskin tumours** (Option A) are cholangiocarcinomas located at the junction of the left and right hepatic ducts, but they typically cause obstructive jaundice without the dilated gallbladder seen here. **Pancreatic cancer** (Option B) can cause obstructive jaundice but is usually associated with a **palpable mass** and **weight loss**. **Mirizzi syndrome** (Option C) involves an extrinsic compression of the common hepatic duct by a stone in the cystic duct, but the clinical findings are more suggestive of a straightforward CBD stone.
Biliary Obstruction Management
Definition: Obstruction of the biliary tract, often due to stones or tumors, causing jaundice and liver dysfunction.
Key Clinical Presentation: Jaundice, dark urine, pale stool, elevated bilirubin and alkaline phosphatase.
Investigations: Ultrasound, MRCP, CT scan.
Treatment Outline:

- **First-line:** Endoscopic removal of the stone.
- **Second-line:** Surgery if endoscopic intervention fails.
  Correct Answer: D. Common bile duct stone ✅

MCQ 244

A 25-year-old patient presented to the Emergency Department with a 6-hour history of severe periumbilical colicky abdominal pain associated with frequent vomiting, and abdominal distension. Physical examination confirmed diffuse abdominal distention with exaggerated bowel sounds.

Which of the following is the most appropriate initial step in management?

A. Ultrasound abdomen
B. Diagnostic laparoscopy
C. Examination of the groin
D. Abdominal computed tomography

Explanation:
The correct answer is **D. Abdominal computed tomography**.
This patient is presenting with signs of **bowel obstruction**, suggested by **abdominal distension**, **colicky pain**, and **vomiting**. The most appropriate initial diagnostic test is **abdominal CT**, which provides detailed information about the cause and location of the obstruction (e.g., hernia, adhesions, or masses). **Ultrasound abdomen** (Option A) can be helpful for evaluating other abdominal conditions but is less effective for assessing bowel obstruction. **Diagnostic laparoscopy** (Option B) is an invasive procedure and is not the first step unless there is concern for surgical intervention. **Examination of the groin** (Option C) is important if there is suspicion of a **hernia**, but CT is a better initial test for a comprehensive evaluation.
Bowel Obstruction Management
Definition: Blockage in the intestines that causes a build-up of food, fluids, and gas.
Key Clinical Presentation: Abdominal distension, pain, vomiting, and constipation.
Investigations: CT scan, X-ray, ultrasound.
Treatment Outline:

- **First-line:** CT scan for diagnosis.
- **Second-line:** Surgery if obstruction is caused by a hernia or other mechanical cause.
  Correct Answer: D. Abdominal computed tomography ✅

MCQ 245:

A 28-year-old is 4-days post-exploratory laparotomy for a perforated acute appendicitis and is complaining of generalized abdominal discomfort and vomiting. Physical examination confirmed a distended abdomen, mild generalized tenderness, and sluggish bowel sounds.

Test Results:

- **WBC:** 12.5 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Abdominal X-ray:

- Multiple air-fluid levels.

Which of the following is the most likely diagnosis?

A. Paralytic ileus
B. Volvulus of small bowel
C. Intra-abdominal collection
D. Adhesive small bowel obstruction

Explanation:
The correct answer is **D. Adhesive small bowel obstruction**.
The patient's history of **recent laparotomy for perforated appendicitis** followed by symptoms of **abdominal distension**, **vomiting**, and the finding of **multiple air-fluid levels** on **abdominal X-ray** suggests **small bowel obstruction (SBO)**. Given the recent surgery, the most likely cause of this obstruction is **adhesions**, which commonly form after abdominal surgery. **Paralytic ileus** (Option A) can also present similarly but is less likely in this post-surgical setting with an acute onset. **Volvulus** (Option B) would typically present with a different clinical and radiographic pattern. **Intra-abdominal collection** (Option C) could be considered if there was evidence of infection or abscess but is less likely without other signs of sepsis or fluid collection.

Small Bowel Obstruction (SBO) Management
Definition: A blockage in the small intestine causing impaired transit of intestinal contents.
Key Clinical Presentation: Abdominal pain, distension, vomiting, and air-fluid levels on X-ray.
Investigations: Abdominal X-ray, CT scan.
Treatment Outline:

- **First-line:** Conservative management (NG tube, IV fluids, bowel rest).
- **Second-line:** Surgery if no improvement.
  Correct Answer: D. Adhesive small bowel obstruction ✅

MCQ 246

A 28-year-old man presented with a 2-year history of reducible right inguinal hernia that became painful for 16 hours. It is associated with progressive abdominal distension, and vomiting. Physical examination confirmed a distended abdomen, exaggerated bowel sounds, and a tender right inguinal swelling with red discoloration of the overlying skin.

Test Results:

- **WBC:** 9.8 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Abdominal X-ray:

- Multiple air fluid levels.

Which of the following is the most likely hernia complication?

A. Irreducible
B. Obstructed
C. Incarcerated
D. Strangulated

Explanation:
The correct answer is **D. Strangulated**.
The clinical presentation of an **inguinal hernia** with **pain**, **abdominal distension**, and **discoloration of the skin over the hernia** suggests **strangulation**, which occurs when the blood supply to the hernia contents is compromised, leading to ischemia. **Incarcerated hernia** (Option C) refers to a hernia that cannot be reduced but does not necessarily have vascular compromise. **Obstructed hernia** (Option B) can cause vomiting and distension but typically does not cause skin discoloration or pain to the extent seen here. **Irreducible** (Option A) just refers to a hernia that cannot be pushed back into the abdomen and does not describe a specific complication.

Hernia Complications
Definition: A hernia that causes compromised blood supply (strangulation) or bowel obstruction.
Key Clinical Presentation: Painful hernia, abdominal distension, skin changes, vomiting.
Investigations: Clinical diagnosis, imaging if necessary.
Treatment Outline:

- **First-line:** Emergency surgery for strangulated hernia.
- **Second-line:** Surgical repair for incarcerated or obstructed hernia.
  Correct Answer: D. Strangulated ✅

MCQ 247

A 42-year-old woman presented with a 4-month history of an anterior neck swelling. Physical examination confirmed a solitary nodule in the right lobe of thyroid.

Test Results:

- **Thyroid-Stimulating Hormone:** 0.05 µU/mL (Normal: 0.4 - 5.0 µU/mL)
- **Thyroxine (T4 free serum):** 37 pmol/L (Normal: 8.5 - 15.2 pmol/L)

Ultrasound:

- Right thyroid lobe solitary solid nodule 2 x 3 cm.

Which of the following is the most appropriate next step in management?

A. Thyroid CT scan
B. Right hemithyroidectomy
C. Thyroid isotope scintigraphy
D. Fine needle aspiration cytology

Explanation:
The correct answer is **D. Fine needle aspiration cytology**.
The patient has a **solitary thyroid nodule** with signs of **hyperthyroidism** (low TSH, high T4). The next appropriate step in evaluation is **fine needle aspiration cytology (FNAC)** to assess whether the nodule is benign or malignant, as **FNAC** provides tissue for histological examination. **Thyroid CT scan** (Option A) is not needed unless there are signs of local invasion or metastasis. **Right hemithyroidectomy** (Option B) may be considered if malignancy is confirmed, but the first step is cytology. **Thyroid isotope scintigraphy** (Option C) is not the first diagnostic step for a solitary nodule but is used in cases of multinodular goiter.

Thyroid Nodule Management
Definition: A solitary nodule in the thyroid that may be benign or malignant.
Key Clinical Presentation: Solitary thyroid nodule, hyperthyroidism signs (low TSH, high T4).
Investigations: Fine needle aspiration (FNAC), ultrasound, possibly scintigraphy.
Treatment Outline:

- **First-line:** FNAC for cytology.
- **Second-line:** Surgical management if malignancy is confirmed.
  Correct Answer: D. Fine needle aspiration cytology ✅

MCQ 248

A 33-year-old woman presented with a lump in her neck. Physical examination confirmed enlarged right cervical lymph nodes with normal thyroid gland. She underwent percutaneous biopsy of the lymph nodes.

Histopathology:

- Normal follicular thyroid cells.

Which of the following is the most likely diagnosis?

A. Ectopic thyroid
B. Thyroid lymphoma
C. Papillary thyroid cancer
D. Follicular thyroid cancer

Explanation:
The correct answer is **A. Ectopic thyroid**.
The patient has **enlarged cervical lymph nodes**, but the **histopathology shows normal follicular thyroid cells**. This suggests that the **cervical lymph nodes** are **reactive or involved in an ectopic thyroid tissue**. **Thyroid lymphoma** (Option B) is characterized by a lymphoma of the thyroid gland, not normal thyroid cells in lymph nodes. **Papillary** (Option C) and **follicular thyroid cancer** (Option D) would typically present with a **thyroid mass**, and they do not explain the finding of **normal thyroid cells in lymph nodes**.

Cervical Lymph Node Mass Management
Definition: Enlargement of cervical lymph nodes that may be reactive or metastatic.
Key Clinical Presentation: Lymphadenopathy, normal thyroid, biopsy findings of normal thyroid cells.
Investigations: FNAC, ultrasound, CT scan.
Treatment Outline:

- **First-line:** Investigation with FNAC for definitive diagnosis.
- **Second-line:** Surgical management if cancer is confirmed.
  Correct Answer: A. Ectopic thyroid ✅

MCQ 249

A 52-year-old woman presented with a 3-month history of a progressive anterior neck swelling with stridor. She has a 12-year history of Hashimoto's thyroiditis. Physical examination confirmed a diffuse, large, non-tender thyroid swelling with retrosternal extension. A core needle biopsy was ordered.

Ultrasound:

- Diffuse thyroid enlargement, encasing bilateral carotid sheaths.

Histopathology:

- Malignant thyroid tumour.

Which of the following is the most likely tumour?

A. Follicular
B. Medullary
C. Anaplastic
D. Lymphoma

Explanation:
The correct answer is **C. Anaplastic**.
This patient's **history of Hashimoto's thyroiditis**, along with the **progressive neck swelling, stridor**, and **malignant thyroid tumor** seen on biopsy, suggests **anaplastic thyroid cancer**. **Anaplastic thyroid cancer** is an aggressive and rapidly growing tumor that is often associated with a **history of Hashimoto's thyroiditis**. **Follicular** (Option A) and **medullary** thyroid cancer (Option B) are more indolent and would typically present with different clinical features. **Lymphoma** (Option D) can involve the thyroid but typically presents with a painless, rapidly growing mass and would not explain the associated stridor.

Thyroid Cancer Management
Definition: Malignant growth in the thyroid gland, including papillary, follicular, medullary, and anaplastic types.
Key Clinical Presentation: Progressive neck mass, stridor, dysphagia, history of Hashimoto's thyroiditis.
Investigations: Fine needle aspiration, ultrasound, CT scan.
Treatment Outline:

- **First-line:** Surgery, typically total thyroidectomy.
- **Second-line:** Radioactive iodine, chemotherapy for anaplastic cancer.
  Correct Answer: C. Anaplastic ✅

MCQ 250

A 75-year-old man presented with a 12-hour history of sudden lower abdominal pain, associated with progressive abdominal distension, nausea, vomiting, and absolute constipation. Physical examination confirmed a grossly distended abdomen, generalized discomfort with normal bowel sounds, and an empty rectum.

Erect abdominal X-ray:

- Massively distended bowel loop pointing to the right upper quadrant.

Which of the following is the most likely diagnosis?

A. Sigmoid volvulus
B. Rectosigmoid cancer
C. Closed loop obstruction
D. Acute sigmoid diverticulitis

Explanation:
The correct answer is **C. Closed loop obstruction**.
The **abdominal pain**, **distension**, and **vomiting**, along with **massive bowel dilation on X-ray**, are classic signs of a **closed-loop obstruction**. This occurs when a loop of bowel is obstructed at both ends, leading to a **distended bowel**. **Sigmoid volvulus** (Option A) can present similarly but would not typically show the characteristic distension pattern. **Rectosigmoid cancer** (Option B) may cause obstruction, but the absence of a palpable mass and the acute nature of this presentation make it less likely. **Acute sigmoid diverticulitis** (Option D) would present with localized tenderness and typically would not cause the massive distension seen here.

Bowel Obstruction Management
Definition: A blockage of the intestines preventing the normal passage of food and liquids.
Key Clinical Presentation: Abdominal pain, distension, vomiting, constipation.
Investigations: Abdominal X-ray, CT scan.
Treatment Outline:

- **First-line:** Decompression (NG tube, IV fluids).
- **Second-line:** Surgery for closed-loop or complicated obstructions.
  Correct Answer: C. Closed loop obstruction ✅

MCQ 251

A 58-year-old man presented with a 4-month history of generalized fatigability and shortness of breath. Physical examination confirmed a non-tender mass in the right iliac fossa.

Test Results:

- **Hb:** 66 g/L (Normal: 130-170 g/L)

Which of the following is the most important step in management?

A. Colonoscopy
B. Abdominal ultrasound
C. Percutaneous biopsy of the mass
D. Abdominal computed tomography

Explanation:
The correct answer is **D. Abdominal computed tomography (CT)**.
Given the patient's **anemia** (low hemoglobin) and a **non-tender mass** in the **right iliac fossa**, the most likely diagnosis is a **colorectal mass**. **CT scan** provides detailed imaging to assess the mass's size, location, and potential involvement of surrounding structures, as well as any metastasis. **Colonoscopy** (Option A) is important for definitive diagnosis but would typically be performed after initial imaging. **Ultrasound** (Option B) is less useful for solid abdominal masses. **Percutaneous biopsy** (Option C) is not the first step as imaging is needed first.

Colorectal Mass Management
Definition: A mass located in the colon or rectum, potentially malignant.
Key Clinical Presentation: Anemia, abdominal mass, fatigue, shortness of breath.
Investigations: Abdominal CT, colonoscopy.
Treatment Outline:

- **First-line:** Imaging (CT scan).
- **Second-line:** Biopsy, surgery for malignant tumors.
  Correct Answer: D. Abdominal computed tomography ✅

MCQ 252

A 45-year-old woman presents to the Emergency Department with severe continuous mid-abdominal pain for 16 hours, associated with nausea and vomiting. She is known to have cholelithiasis. Physical examination confirmed epigastric tenderness.

Test Results:

- **WBC:** 13 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- **Amylase:** 650 IU/L (Normal: 24-151 IU/L)

Which of the following is the most likely diagnosis?

A. Hepatitis
B. Acute pancreatitis
C. Acute cholecystitis
D. Perforated peptic ulcer disease

Explanation:
The correct answer is **B. Acute pancreatitis**.
The patient's **epigastric pain**, **nausea**, **vomiting**, and **elevated amylase** point to **acute**

pancreatitis, especially in the setting of **cholelithiasis**, which can cause **biliary pancreatitis**. **Acute cholecystitis** (Option C) could present with similar symptoms but would typically show a **distended gallbladder** on imaging. **Hepatitis** (Option A) would cause more generalized symptoms, and **perforated peptic ulcer** (Option D) would typically present with peritonitis and a rigid abdomen.

Pancreatitis Management
Definition: Inflammation of the pancreas, often due to gallstones or alcohol use.
Key Clinical Presentation: Severe epigastric pain, vomiting, elevated amylase.
Investigations: Abdominal X-ray, CT scan, ultrasound.
Treatment Outline:

- **First-line:** Supportive care (IV fluids, pain control).
- **Second-line:** Address the underlying cause (e.g., gallstone removal).
  Correct Answer: B. Acute pancreatitis ✅

MCQ 253

A 58-year-old man presented with a 3-month history of vague epigastric pain associated with weight loss and loss of appetite. Physical examination was normal.

Which of the following is the most appropriate investigation?

A. Plain abdominal CT
B. Gastroduodenoscopy
C. Abdominal ultrasound
D. Gastrografin swallow

Explanation:
The correct answer is **B. Gastroduodenoscopy**.
Given the patient's **vague epigastric pain** with **weight loss** and **loss of appetite**, **gastroduodenoscopy (endoscopy)** is the most appropriate initial investigation. It allows direct visualization of the upper gastrointestinal tract and can help identify **gastric ulcers** or **gastric cancer**. **Plain abdominal CT** (Option A) can be useful but would not provide as direct information on the stomach lining. **Abdominal ultrasound** (Option C) is useful for liver and gallbladder evaluation but is less specific for stomach pathology. **Gastrografin swallow** (Option D) is useful for evaluating esophageal issues but not the stomach.

Gastrointestinal Malignancy Investigation
Definition: Weight loss and abdominal pain may be indicative of gastrointestinal malignancy.

Key Clinical Presentation: Epigastric pain, weight loss, loss of appetite.
Investigations: Gastroduodenoscopy, abdominal CT.
Treatment Outline:

- **First-line:** Endoscopy for biopsy.
- **Second-line:** CT scan, surgery if cancer is confirmed.
  Correct Answer: B. Gastroduodenoscopy ✅

MCQ 254

A 33-year-old man presented to the Emergency Department with a 24-hour history of epigastric pain. Physical examination confirmed tenderness in the epigastric area with normal bowel sounds.

Blood pressure: 100/60 mmHg
Heart rate: 110/min
Temperature: 37.7°C

Test Results:

- **WBC:** 14 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- **Amylase:** 223 IU/L (Normal: 24-151 IU/L)

Erect chest X-ray:

- Free air under the diaphragm.

Which of the following is the most appropriate next step in management?

A. Gastrografin swallow
B. Gastroduodenoscopy
C. Exploratory laparotomy
D. Conservative treatment

Explanation:
The correct answer is **C. Exploratory laparotomy**.
The patient's **epigastric pain**, **vomiting**, and **elevated amylase**, along with the **free air under the diaphragm** on chest X-ray, are suggestive of a **perforated peptic ulcer**, which is a surgical emergency. **Exploratory laparotomy** is needed to confirm the diagnosis and treat the perforation. **Gastrografin swallow** (Option A) is used to evaluate for leaks but is not the first step in suspected perforation. **Gastroduodenoscopy** (Option B) is useful for diagnosing ulcers

but would not be appropriate in an emergency setting with free air, as surgery is indicated. **Conservative treatment** (Option D) is inappropriate given the seriousness of the condition.

Perforated Peptic Ulcer Management
Definition: A peptic ulcer that has eroded through the gastric or duodenal wall causing leakage of gastric contents into the peritoneum.
Key Clinical Presentation: Acute severe abdominal pain, vomiting, free air under diaphragm.
Investigations: Chest X-ray, abdominal X-ray, CT scan.
Treatment Outline:

- **First-line:** Surgical intervention (exploratory laparotomy).
- **Second-line:** Antacids and proton pump inhibitors post-surgery.
  Correct Answer: C. Exploratory laparotomy ✅

MCQ 255

A 27-year-old obese woman presented to the Emergency Department with a 6-day history of right iliac fossa pain associated with nausea, vomiting, and loss of appetite. Physical examination confirmed tenderness in the right iliac fossa.

Test Result:

- **WBC:** 13 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Which of the following is the most appropriate next step in management?

A. Open surgery
B. Abdominal CT scan
C. Abdominal ultrasound
D. Diagnostic laparoscopy

Explanation:
The correct answer is **C. Abdominal ultrasound**.
This patient's clinical presentation, along with **elevated WBC**, suggests possible **acute appendicitis**. The **most appropriate next step** is **abdominal ultrasound**, which is a non-invasive first-line investigation, especially in women of childbearing age to rule out gynecological causes. **Abdominal CT scan** (Option B) can also be useful but is typically not the first line in this scenario, especially with concern for radiation. **Diagnostic laparoscopy** (Option D) is invasive and should be reserved for cases where imaging results are inconclusive. **Open surgery** (Option A) is indicated after confirming the diagnosis.

Acute Appendicitis Management
Definition: Inflammation of the appendix, typically due to obstruction of the lumen.
Key Clinical Presentation: Right iliac fossa pain, fever, elevated WBC.
Investigations: Ultrasound, CT scan.
Treatment Outline:

- **First-line:** Appendectomy (open or laparoscopic).
- **Second-line:** Antibiotics for perforated appendicitis.
  Correct Answer: C. Abdominal ultrasound ✅

MCQ 256

A 34-year-old man who underwent an open repair of a right inguinal hernia 3 years ago, presented with a 3-month history of a right groin swelling that is associated with dragging pain. Physical examination confirmed a right groin reducible swelling with positive cough impulse, reaching the neck of scrotum.

Which of the following is the most appropriate next step in management?

A. Open repair
B. CT abdomen
C. Scrotal support
D. Laparoscopic mesh repair

Explanation:
The correct answer is **A. Open repair**.
The patient's presentation is typical of an **inguinal hernia**, with a **positive cough impulse** and **reducible swelling**. Given the chronicity of the hernia and the size of the swelling, **open repair** is the most appropriate treatment. **Laparoscopic mesh repair** (Option D) is an alternative technique but may not be the first option in this patient. **CT abdomen** (Option B) is not necessary for diagnosis, as the diagnosis is clinical. **Scrotal support** (Option C) might provide temporary relief but does not address the hernia itself.

Inguinal Hernia Management
Definition: Protrusion of intra-abdominal contents through the inguinal canal.
Key Clinical Presentation: Groin swelling, cough impulse, reducibility.
Investigations: Clinical diagnosis.
Treatment Outline:

- **First-line:** Surgical repair (open or laparoscopic).

- **Second-line:** Watchful waiting in asymptomatic hernias.
  Correct Answer: A. Open repair ✅

MCQ 257

A 48-year-old woman presented with a 12-hour history of a painful swelling in the right groin. Physical examination confirmed an irreducible tender swelling in the groin area that is 3x4 cm. Surgical exploration confirmed a strangulated hernia and the neck is lateral and inferior to the pubic tubercle.

Which of the following is the most likely hernia?

A. Femoral
B. Obturator
C. Direct inguinal
D. Indirect inguinal

Explanation:
The correct answer is **A. Femoral**.
A **femoral hernia** typically presents with a **painful, irreducible swelling** below the inguinal ligament, often **lateral and inferior to the pubic tubercle**. It is more likely to become strangulated due to its narrow neck. **Obturator hernia** (Option B) is less common and usually presents with **vague abdominal pain**. **Direct inguinal** (Option C) and **indirect inguinal** (Option D) hernias typically present above the inguinal ligament and are less likely to be strangulated.

Femoral Hernia Management
Definition: Hernia that protrudes through the femoral canal, located below the inguinal ligament.
Key Clinical Presentation: Irreducible, tender groin mass, typically below the inguinal ligament.
Investigations: Clinical diagnosis, occasionally confirmed by imaging.
Treatment Outline:

- **First-line:** Surgical repair, usually via open surgery.
- **Second-line:** Watchful waiting if asymptomatic.
  Correct Answer: A. Femoral ✅

MCQ 258

A 25-year-old presented to the Emergency Department with a 2-day history of diffuse colicky abdominal pain, associated with abdominal distension, vomiting, and constipation. Physical examination confirmed diffuse abdominal discomfort with exaggerated bowel sounds.

CT scan:

- Distended proximal small bowel with complete obstruction at the proximal ileum.

Exploratory laparotomy:

- 3 strictures over 60 cm of the ileum, the proximal one was the cause of small bowel obstruction.

Which of the following is the most likely diagnosis?

A. TB enteritis
B. Crohn's disease
C. Small bowel lymphoma
D. Gastrointestinal stromal tumour

Explanation:
The correct answer is **B. Crohn's disease**.
The presence of **multiple strictures** in the **ileum** and the history of **recurrent abdominal pain** with **obstruction** points to **Crohn's disease**. **TB enteritis** (Option A) could cause strictures, but it is less common in developed countries and typically affects the ileocecal region with a more insidious onset. **Small bowel lymphoma** (Option C) can present with obstruction but is less likely to cause multiple strictures in the ileum. **Gastrointestinal stromal tumours** (Option D) are rare and typically present with a mass, rather than multiple strictures.

Crohn's Disease Management
Definition: Chronic inflammatory bowel disease with transmural inflammation affecting any part of the gastrointestinal tract, most commonly the ileum.
Key Clinical Presentation: Abdominal pain, weight loss, diarrhea, and sometimes obstruction due to strictures.
Investigations: Colonoscopy, CT scan, biopsy.
Treatment Outline:

- **First-line:** Anti-inflammatory drugs (e.g., corticosteroids, immunosuppressants).
- **Second-line:** Surgical intervention for complications like strictures and obstructions.
  Correct Answer: B. Crohn's disease ✅

MCQ 259

A 29-year-old patient presented with a 3-year history of frequent attacks of bloody diarrhea. On examination, he was pale with a normal abdominal examination.

Barium enema:

- Featureless "lead-pipe" colon.

Colonoscopy:

- Diffusely inflamed colon with pseudopolyps, featureless terminal ileum, and backwash ileitis.

Which of the following is the most likely type of colitis?

A. Crohn's
B. Amebic
C. Ischemic
D. Ulcerative

Explanation:
The correct answer is **D. Ulcerative colitis**.
The **featureless "lead-pipe" colon** on **barium enema**, along with **pseudopolyps** and **backwash ileitis** seen on colonoscopy, is characteristic of **ulcerative colitis**. **Crohn's disease** (Option A) typically shows segmental inflammation and **skip lesions**. **Amebic colitis** (Option B) is less likely in this context as it typically presents with **flask-shaped ulcers**. **Ischemic colitis** (Option C) usually affects the **watershed areas** of the colon and would not show the characteristic findings of **pseudopolyps** and **backwash ileitis**.

Ulcerative Colitis Management
Definition: Chronic inflammatory bowel disease affecting the colon and rectum, characterized by continuous mucosal inflammation.
Key Clinical Presentation: Bloody diarrhea, abdominal pain, tenesmus, and weight loss.
Investigations: Colonoscopy, biopsy, barium enema.
Treatment Outline:

- **First-line:** Aminosalicylates and corticosteroids.
- **Second-line:** Immunosuppressive drugs or biologics (e.g., TNF inhibitors).
  Correct Answer: D. Ulcerative colitis ✅

MCQ 260

A 19-year-old man presented with a 3-day history of frequent attacks of per-rectal fresh bleeding. Physical examination confirmed clotted blood in the rectum. Nasogastric tube was draining greenish-colored fluids.

Test Results:

- **Hb:** 85 g/L (Normal: 130-170 g/L for male)

Colonoscopy:

- No colorectal pathology.

Which of the following is the most appropriate diagnostic tool to localize the source of bleeding?

A. Diagnostic laparoscopy
B. Gastrografin follow-through
C. Upper gastrointestinal endoscopy
D. 99mTc-labelled sodium pertechnetate

Explanation:
The correct answer is **C. Upper gastrointestinal endoscopy**.
The presence of **greenish-colored fluid** in the nasogastric tube suggests a **gastrointestinal source** of bleeding, likely from the **upper gastrointestinal tract**. **Upper gastrointestinal endoscopy** is the best next step to identify potential causes of **upper GI bleeding**, such as peptic ulcers, varices, or Mallory-Weiss tears. **Gastrografin follow-through** (Option B) is not suitable for detecting bleeding. **Diagnostic laparoscopy** (Option A) is not appropriate for this scenario, and **99mTc-labelled sodium pertechnetate** (Option D) is used for localized **gastrointestinal bleeding** but not in cases of fresh rectal bleeding.

Upper GI Bleeding Management
Definition: Bleeding originating from the **esophagus, stomach**, or **duodenum**.
Key Clinical Presentation: Hematemesis, coffee ground vomitus, and melena.
Investigations: Upper endoscopy, angiography, CT scan.
Treatment Outline:

- **First-line:** Endoscopic hemostasis, proton pump inhibitors.
- **Second-line:** Surgery for refractory cases.
  Correct Answer: C. Upper gastrointestinal endoscopy ✅

MCQ 261

A 55-year-old man complains of a 2-month history of central abdominal pain. Physical examination was normal.

Test Results:

- **24-hour 5-HIAA:** 21 mg (Normal: 2-7 mg/24 h)

CT scan:

- Small bowel mass 4x5 cm, with some calcifications.

Which of the following is the most likely small bowel tumour?

A. Carcinoid
B. Lymphoma
C. Schwannoma
D. Gastrointestinal stroma

Explanation:
The correct answer is **A. Carcinoid**.
Carcinoid tumors are well-known to produce **5-HIAA**, a metabolite of serotonin. The **elevated 5-HIAA** level, along with the presence of a **mass with calcifications** in the **small bowel**, suggests **carcinoid** tumor, which often arises in the **small intestine** and can lead to **carcinoid syndrome**.
Lymphoma (Option B) can cause a mass but typically does not lead to elevated **5-HIAA**.
Schwannomas (Option C) are rare tumors that arise from nerve tissue, and **gastrointestinal stromal tumors (GIST)** (Option D) are typically **CD117 positive** on immunohistochemistry and do not produce **5-HIAA**.

Carcinoid Tumor Management
Definition: A slow-growing neoplasm most commonly found in the small bowel, often producing serotonin.
Key Clinical Presentation: Abdominal pain, flushing, diarrhea, and heart failure (carcinoid syndrome).
Investigations: Elevated 5-HIAA, CT scan, biopsy.
Treatment Outline:

- **First-line:** Surgical resection of the tumor.
- **Second-line:** Somatostatin analogs (e.g., octreotide).
  Correct Answer: A. Carcinoid ✅

MCQ 262:

A 32-year-old man presented to the Emergency Department with pain and swelling of his right thigh after a trivial injury. Physical examination confirmed swelling, deformity of his right thigh with tenderness.

Test Results:

- **Parathyroid hormone (intact PTH levels):** 8.9 pmol/L (Normal: 1.1-5.3 pmol/L)
- **Calcium:** 4.5 mmol/L (Normal: 2.15-2.62 mmol/L)

X-ray:

- Spiral fracture of right femur with subperiosteal resorption.

Which of the following is the most likely diagnosis?

A. Parathyroid adenoma
B. Parathyroid carcinoma
C. Tertiary hyperparathyroidism
D. Secondary hyperparathyroidism

Explanation:
The correct answer is **A. Parathyroid adenoma**.
Subperiosteal resorption seen on the X-ray along with **elevated calcium** and **parathyroid hormone (PTH)** levels strongly suggest **primary hyperparathyroidism**, which is most commonly caused by **parathyroid adenoma**. This condition leads to excess PTH secretion, resulting in elevated calcium levels and bone resorption. **Parathyroid carcinoma** (Option B) is less common and usually presents with more severe symptoms and higher calcium levels. **Tertiary hyperparathyroidism** (Option C) occurs in patients with long-standing **secondary hyperparathyroidism** (due to chronic kidney disease), but it is not associated with the findings in this patient. **Secondary hyperparathyroidism** (Option D) would show low calcium levels with high PTH.

Primary Hyperparathyroidism Management
Definition: Excessive secretion of PTH, leading to hypercalcemia and bone resorption.
Key Clinical Presentation: Bone pain, fractures, kidney stones, abdominal pain, and psychiatric symptoms.
Investigations: Serum calcium, PTH, and imaging studies for parathyroid adenoma.
Treatment Outline:

- **First-line:** Surgical removal of the parathyroid adenoma.
- **Second-line:** Medical management in non-surgical candidates.
  Correct Answer: A. Parathyroid adenoma ✅

MCQ 263

A 28-year-old married woman presented to the Emergency Department complaining of a 6-hour history of right iliac fossa pain associated with drowsiness. Physical examination confirmed tenderness over the lower abdomen with sluggish bowel sounds.

Which of the following is the most likely diagnosis?

A. Septic shock
B. Ectopic pregnancy
C. Ruptured ovarian cyst
D. Pelvic inflammatory disease

Explanation:
The correct answer is **B. Ectopic pregnancy**.
Right iliac fossa pain in a **young, married woman** with **drowsiness** strongly suggests **ectopic pregnancy**, especially if she is sexually active. This condition is a common cause of **acute abdominal pain** and can present with **shock** if there is **rupture**. **Septic shock** (Option A) could cause drowsiness but is not likely given the localized pain in the right iliac fossa. **Ruptured ovarian cyst** (Option C) could cause abdominal pain but is typically associated with a more localized sharp pain and less likely to present with drowsiness. **Pelvic inflammatory disease** (Option D) causes lower abdominal pain and fever, but the absence of **fever** and the localization of pain make it less likely in this case.

Ectopic Pregnancy Management
Definition: Pregnancy that occurs outside the uterine cavity, most commonly in the fallopian tube.
Key Clinical Presentation: Abdominal pain, vaginal bleeding, and amenorrhea.
Investigations: Urine pregnancy test, transvaginal ultrasound, and serum beta-hCG.
Treatment Outline:

- **First-line:** Methotrexate (for early, non-ruptured cases) or surgical intervention for rupture.
  Correct Answer: B. Ectopic pregnancy ✅

MCQ 264

A 32-year-old man presented to the Emergency Department with a 6-hour history of severe sudden epigastric pain. Physical examination confirmed diffuse severe abdominal tenderness with sluggish bowel sounds.

Test Results:

- **WBC:** 13.8 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)
- **Amylase:** 300 IU/L (Normal: 24-151 IU/L)

Which of the following is the most appropriate next step in management?

A. Erect chest X-ray
B. Abdominal X-ray
C. CT scan abdomen
D. Ultrasound abdomen

Explanation:
The correct answer is **C. CT scan abdomen**.
The patient has a **classic presentation of acute pancreatitis**, evidenced by **epigastric pain** and **elevated amylase**. The **next best step** in management is to perform a **CT scan** to assess the extent of the pancreatitis, complications (e.g., pseudocysts or abscesses), and other possible causes. **Erect chest X-ray** (Option A) could be helpful in identifying **diaphragmatic rupture** or **pleural effusion**, but it will not help with pancreatitis. **Abdominal X-ray** (Option B) is not sufficient for confirming acute pancreatitis. **Ultrasound abdomen** (Option D) can help identify gallstones but is not as sensitive for assessing complications of pancreatitis.

Acute Pancreatitis Management
Definition: Acute inflammation of the pancreas, often due to gallstones or alcohol consumption.
Key Clinical Presentation: Severe epigastric pain, nausea, vomiting, and elevated amylase/lipase.
Investigations: Abdominal CT scan, ultrasound, and blood tests.
Treatment Outline:

- **First-line:** Supportive care (IV fluids, pain management, and NPO status).
- **Second-line:** Address underlying cause (e.g., cholecystectomy for gallstones).
  Correct Answer: C. CT scan abdomen ✅

MCQ 265

A 53-year-old man complains of a 3-month history of vague epigastric pain and weight loss. Physical examination was normal.

Upper GIT endoscopy:

- Antral gastric mass. Multiple deep biopsies were taken.

Histopathology:

- Mucosa-associated lymphoid tissue lymphoma and positive for Helicobacter pylori.

Which of the following is the most appropriate initial management?

A. Surgery
B. Antibiotics
C. Radiotherapy
D. Chemotherapy

Explanation:
The correct answer is **B. Antibiotics**.
This patient has **MALT lymphoma** (Mucosa-Associated Lymphoid Tissue lymphoma), which is often associated with **H. pylori infection**. The **first-line treatment** is **eradication of H. pylori** using **antibiotics** (e.g., clarithromycin, amoxicillin, and proton pump inhibitors). **Surgery** (Option A) and **chemotherapy** (Option D) are not first-line treatments unless the lymphoma persists or is non-responsive to antibiotics. **Radiotherapy** (Option C) may be used in cases where antibiotics fail but is not the initial treatment.

MALT Lymphoma Management
Definition: A low-grade lymphoma associated with H. pylori infection, often affecting the stomach.
Key Clinical Presentation: Epigastric pain, weight loss, nausea, and vomiting.
Investigations: Endoscopy with biopsy, H. pylori testing.
Treatment Outline:

- **First-line:** Antibiotics to eradicate H. pylori.
- **Second-line:** Chemotherapy or radiation if antibiotics are ineffective.
  Correct Answer: B. Antibiotics ✅

MCQ 266:

A 31-year-old patient presented with a 12-hour history of severe epigastric abdominal pain. Physical examination confirmed a tender abdomen with sluggish bowel sounds.

Erect chest X-ray:

- Air under the diaphragm.

Exploratory laparotomy:

- A 5x5 mm perforation in the anterior aspect of the 1st part of duodenum with peritoneal soiling.

Which of the following is the most appropriate management?

A. Distal gastrectomy
B. Pyloroplasty with truncal vagotomy
C. Simple closure with omental patching
D. Pyloroplasty with high selective vagotomy

Explanation:
The correct answer is **C. Simple closure with omental patching**.
This patient has a **perforated duodenal ulcer** (most commonly due to H. pylori or NSAID use). The standard management of a perforated duodenal ulcer is to **close the perforation** with an **omentum patch** to cover the perforation and allow healing. **Distal gastrectomy** (Option A) and **pyloroplasty with truncal vagotomy** (Option B) are more complex procedures used for specific ulcer complications but are not required here. **Pyloroplasty with high selective vagotomy** (Option D) is also not appropriate as the ulcer is perforated, not just causing obstruction.

Duodenal Ulcer Perforation Management
Definition: A complication of peptic ulcer disease in which an ulcer erodes through the wall of the duodenum, leading to peritonitis.
Key Clinical Presentation: Severe abdominal pain, peritonitis, shock, and free air under the diaphragm.
Investigations: Abdominal X-ray (for free air), CT scan, and exploratory laparotomy.
Treatment Outline:

- **First-line:** Simple closure of the perforation with omental patch.
  Correct Answer: C. Simple closure with omental patching ✅

MCQ 267;

A 55-year-old man presented with a 4-month history of vague epigastric pain associated with 15 kg weight loss. Physical examination confirmed enlarged left supraclavicular lymph nodes and gross ascites.

Which of the following is the most appropriate next step in management?

A. Abdominal ultrasound
B. Diagnostic laparoscopy
C. Abdominal paracentesis
D. Upper gastrointestinal endoscopy

Explanation:
The correct answer is **C. Abdominal paracentesis**.
This patient has **signs of advanced malignancy** (weight loss, ascites, and supraclavicular lymphadenopathy). **Abdominal paracentesis** is the most appropriate next step to analyze the ascitic fluid, which will help in diagnosing whether the ascites is due to malignancy, infection, or other causes. **Abdominal ultrasound** (Option A) can be useful to assess the ascites and organs, but **paracentesis** is crucial to guide further management. **Diagnostic laparoscopy** (Option B) may be considered later to evaluate peritoneal involvement, but paracentesis is first-line. **Upper gastrointestinal endoscopy** (Option D) is not appropriate without first evaluating the ascites.

Management of Ascites
Definition: Fluid accumulation in the peritoneal cavity, often associated with liver disease or malignancy.
Key Clinical Presentation: Abdominal distension, weight loss, and ascitic fluid.
Investigations: Abdominal paracentesis for fluid analysis.
Treatment Outline:

- **First-line:** Paracentesis for diagnostic purposes.
- **Second-line:** Treat underlying cause (e.g., chemotherapy for malignancy).
  Correct Answer: C. Abdominal paracentesis ✅

MCQ 268

A 47-year-old man reported frequent vomiting over the past 24-hours due to food poisoning. He presented to the Emergency Department complaining of a 2-hour history of vomiting fresh blood. Physical examination confirmed mild epigastric tenderness. A nasogastric tube was inserted and was draining bloody stained fluids.

Which of the following is the most likely diagnosis?

A. Gastritis
B. Dieulafoy’s lesion
C. Peptic ulcer disease
D. Mallory-Weiss syndrome

Explanation:
The correct answer is **D. Mallory-Weiss syndrome**.
Mallory-Weiss syndrome is characterized by **tears in the mucosa at the junction of the stomach and esophagus** due to **violent vomiting**. The **fresh blood** in the vomit and history of **vomiting** following food poisoning point towards this condition. **Gastritis** (Option A) can cause upper GI bleeding but is less likely to cause a specific tear in the mucosa. **Dieulafoy's lesion** (Option B) involves an abnormal artery that erodes the mucosa and causes bleeding, but it is less common. **Peptic ulcer disease** (Option C) can cause upper GI bleeding but typically causes ulcers in the stomach or duodenum, not mucosal tears.

Mallory-Weiss Syndrome Management
Definition: Mucosal tear at the gastroesophageal junction, often due to vomiting.
Key Clinical Presentation: Hematemesis following vomiting, often in the context of alcohol use or food poisoning.
Investigations: Endoscopy to visualize the tear.
Treatment Outline:

- **First-line:** Supportive care (IV fluids), endoscopic hemostasis if needed.
  Correct Answer: D. Mallory-Weiss syndrome ✅

MCQ 269

A 62-year-old man presented with a 3-month history of early satiety and 11 kg weight loss. Physical examination was normal.

Upper GIT endoscopy:

- Ulcerating mass in the antrum. Multiple biopsies were taken.

Histopathology:

- Invasive adenocarcinoma.

Which of the following is the most appropriate next investigation?

A. Abdominal ultrasound
B. Endoscopic ultrasound
C. Diagnostic laparoscopy
D. CT scan chest, abdomen and pelvis

Explanation:
The correct answer is **D. CT scan chest, abdomen and pelvis**.

Given the diagnosis of **gastric adenocarcinoma**, the next step is to assess the **extent of disease** and check for metastasis. A **CT scan of the chest, abdomen, and pelvis** is the most appropriate to evaluate the primary tumor, lymph node involvement, and potential distant metastases. **Endoscopic ultrasound** (Option B) may be used for **staging** and evaluating the depth of tumor invasion, but a **CT scan** is more comprehensive. **Diagnostic laparoscopy** (Option C) is used to evaluate peritoneal metastasis but is not the first step. **Abdominal ultrasound** (Option A) is less helpful in assessing the full extent of gastric cancer.

Gastric Adenocarcinoma Management
Definition: A malignant tumor originating from the gastric epithelium.
Key Clinical Presentation: Weight loss, early satiety, nausea, and vomiting.
Investigations: Endoscopy with biopsy, CT scan for staging.
Treatment Outline:

- **First-line:** Surgical resection (gastrectomy) if feasible.
- **Second-line:** Chemotherapy or palliative care if advanced.
  Correct Answer: D. CT scan chest, abdomen and pelvis ✅

MCQ 270

A 29-year-old woman presented to the Emergency Department with an 8-hour history of severe epigastric pain radiating to the back associated with vomiting. She underwent a sleeve gastrectomy 3 months ago for morbid obesity. Physical examination confirmed mild epigastric tenderness.

Test Result Normal Values

- WBC 9.8 (4.5-10.5 x 10^9/L)
- Amylase 700 (24-151 IU/L)

Ultrasound:

- Biliary sludge but no gallstone. Normal bile ducts.

Which of the following is the most appropriate intervention?

A. Endoscopic ultrasound
B. Endoscopic sphincterotomy
C. Laparoscopic cholecystectomy
D. Percutaneous cholecystostomy

Explanation:
The correct answer is **C. Laparoscopic cholecystectomy**.
This patient presents with symptoms consistent with **biliary colic** or **gallbladder dysfunction**. Given the **history of sleeve gastrectomy** and the **presence of biliary sludge** on ultrasound, she may be at risk for **cholelithiasis** or **cholecystitis**. The most appropriate management is **laparoscopic cholecystectomy** to remove the gallbladder and prevent further complications such as **cholecystitis** or **cholelithiasis**. **Endoscopic ultrasound** (Option A) could be considered to investigate bile duct pathology but is not the first-line treatment for gallbladder issues. **Endoscopic sphincterotomy** (Option B) and **percutaneous cholecystostomy** (Option D) are typically used in specific situations such as bile duct obstruction or in patients who cannot undergo surgery.

Management of Biliary Disease
Definition: A range of conditions involving the gallbladder, including biliary colic, cholelithiasis, and cholecystitis.
Key Clinical Presentation: Epigastric pain, vomiting, fever, jaundice.
Investigations: Ultrasound, CT scan, endoscopic ultrasound if needed.
Treatment Outline:

- **First-line:** Laparoscopic cholecystectomy.
  Correct Answer: C. Laparoscopic cholecystectomy ✅

MCQ 271

A 58-year-old woman presented to the Emergency Department with a 2-day history of central abdominal pain, associated with abdominal distension, vomiting, and constipation. She has no history of abdominal surgery. Physical examination confirmed a distended, tympanic abdomen, with exaggerated bowel sounds.

Erect abdominal X-ray:

- Multiple air-fluid levels with pneumobilia.

Which of the following is the most likely diagnosis?

A. Gallstone ileus
B. Intussusception
C. Small bowel volvulus
D. Small bowel ischemia

Explanation:
The correct answer is **A. Gallstone ileus**.
Gallstone ileus occurs when a **gallstone** passes into the **small intestine** through a fistula (often from a gallbladder perforation). The presence of **pneumobilia** (air in the bile ducts) on X-ray, along with **air-fluid levels**, is highly suggestive of **gallstone ileus**, which occurs when the stone obstructs the bowel. **Intussusception** (Option B) typically presents in children and is unlikely in this adult patient. **Small bowel volvulus** (Option C) presents with similar symptoms but would not show pneumobilia on X-ray. **Small bowel ischemia** (Option D) can present with bowel obstruction but is not associated with pneumobilia.

Gallstone Ileus Management
Definition: Mechanical bowel obstruction caused by a gallstone.
Key Clinical Presentation: Abdominal pain, distension, vomiting, and constipation.
Investigations: X-ray showing air-fluid levels and pneumobilia.
Treatment Outline:

- **First-line:** Surgical removal of the gallstone, usually via laparotomy.
  Correct Answer: A. Gallstone ileus ✅

MCQ 272

A 27-year-old man presented with a 10-day history of right upper quadrant abdominal pain associated with fever. Physical examination confirmed tenderness in the right upper quadrant.

Enzyme-linked immunosorbent assay:

- Positive for **Entamoeba histolytica**.

CT scan:

- Large (10 x15 cm), septated abscess in the right lobe of the liver with poorly defined margin.

Which of the following is the most appropriate initial management?

A. Cephalosporin
B. Metronidazole
C. Percutaneous drainage
D. Percutaneous aspiration

Explanation:
The correct answer is **B. Metronidazole**.
This patient presents with a **liver abscess** caused by **Entamoeba histolytica**, a protozoan infection. The first-line treatment for **amoebic liver abscess** is **metronidazole**, which is effective against the organism. **Cephalosporin** (Option A) is used for bacterial infections, but not for amoebic abscesses. **Percutaneous drainage** (Option C) and **percutaneous aspiration** (Option D) can be considered for large abscesses or if there is failure of medical management, but **metronidazole** is the first-line therapy.

Amoebic Liver Abscess Management
Definition: Abscess formation in the liver due to Entamoeba histolytica infection.
Key Clinical Presentation: Fever, right upper quadrant pain, and positive serology for Entamoeba histolytica.
Investigations: Ultrasound, CT scan, and positive serology.
Treatment Outline:

- **First-line:** Metronidazole (Flagyl) therapy.
- **Second-line:** Drainage if abscess is large or there is failure of medical treatment.
  Correct Answer: B. Metronidazole ✅

MCQ 273

A 65-year-old man was admitted and ventilated in the Intensive Care Unit following a major burn injury. He was referred to surgery due to a 24-hour history of right upper quadrant abdominal pain. Physical examination confirmed tenderness in the right upper quadrant.

Erect chest X-ray:

- Normal.

Ultrasound:

- Distended gallbladder with thick wall and pericholecystic fluid but no gallstone.

Which of the following is the most appropriate next step?

A. Open cholecystectomy
B. Cholecystostomy tube insertion
C. Abdominal computed tomography
D. Magnetic resonant cholangiopancreatography

Explanation:
The correct answer is **B. Cholecystostomy tube insertion**.
This patient likely has **acute acalculous cholecystitis**, which is commonly seen in critically ill patients, especially those with burn injuries. The most appropriate next step is **cholecystostomy tube insertion** to drain the gallbladder and relieve the obstruction. **Open cholecystectomy** (Option A) is generally avoided in critically ill patients, and **CT** (Option C) or **MRCP** (Option D) are not first-line investigations for suspected acalculous cholecystitis.

Acute Acalculous Cholecystitis Management
Definition: Inflammation of the gallbladder without the presence of gallstones.
Key Clinical Presentation: Fever, right upper quadrant pain, and signs of sepsis in critically ill patients.
Investigations: Ultrasound, CT scan.
Treatment Outline:

- **First-line:** Cholecystostomy tube for drainage.
- **Second-line:** Surgery if the patient improves.
  Correct Answer: B. Cholecystostomy tube insertion ✅

MCQ 274

A 65-year-old man presented with a 3-month history of vague upper abdominal pain associated with progressive yellowish discoloration of his sclera, dark urine, and weight loss. On examination, he was jaundiced with a palpable, distended non-tender gallbladder.

Test Result Normal Values

- Alkaline phosphatase: 320 (39-117 IU/L)
- Direct bilirubin: 77 (1.5-6.5 µmol/L)
- Total bilirubin: 88 (3.5-16.5 µmol/L)

Which of the following is the most likely diagnosis?

A. Klatskin tumour
B. Periampullary tumour
C. Common bile duct stone
D. Hepatocellular carcinoma

Explanation:
The correct answer is **A. Klatskin tumour**.

The clinical presentation and imaging findings suggest **obstructive jaundice** with **biliary tree dilation** and a non-tender distended gallbladder. A **Klatskin tumour**, which is a **cholangiocarcinoma** arising at the junction of the right and left hepatic bile ducts, is often associated with such features. **Periampullary tumours** (Option B) can also cause biliary obstruction, but they are more commonly associated with **pancreatic** or **ampullary** malignancies. **Common bile duct stones** (Option C) and **hepatocellular carcinoma** (Option D) are less likely to present with this combination of findings.

Cholangiocarcinoma Management
Definition: A cancer of the bile ducts, often presenting with obstructive jaundice.
Key Clinical Presentation: Jaundice, weight loss, and abdominal pain with findings on ultrasound of **biliary dilation**.
Investigations: Ultrasound, CT scan, and possibly **biopsy**.
Treatment Outline:

- **First-line:** Surgery if resectable.
- **Second-line:** Chemotherapy for advanced stages.
  Correct Answer: A. Klatskin tumour ✅

MCQ 275

A 58-year-old man presented to the Emergency Department complaining of right iliac fossa pain for 5 days. Physical examination confirmed a tender mass in the right iliac fossa. The patient was admitted, treated non-operatively, and discharged from the hospital after 7 days with complete recovery.

Ultrasound:

- Appendiceal mass without collection.

Which of the following is the most appropriate next step?

A. No further intervention
B. Colonoscopy in 6 weeks
C. Open appendectomy in 12 weeks
D. Laparoscopic appendectomy in 12 weeks

Explanation:
The correct answer is **A. No further intervention**.
This patient likely has **phlegmonous appendicitis** (an inflamed, non-perforated appendix with surrounding inflammation). In the absence of perforation or abscess formation, **non-operative**

management is often successful. **Appendectomy** (Options C and D) is typically performed after the acute inflammatory process resolves, but in this case, it is unnecessary in the short term, especially when the patient is stable. **Colonoscopy** (Option B) is not indicated in the diagnosis of appendicitis.

Management of Phlegmonous Appendicitis
Definition: Inflammation of the appendix without perforation, often associated with localized peritoneal inflammation.
Key Clinical Presentation: Abdominal pain, fever, and localized tenderness in the right lower quadrant.
Investigations: Ultrasound or CT scan.
Treatment Outline:

- **First-line:** Non-operative management with antibiotics.
- **Second-line:** Appendectomy after resolution of acute inflammation.
  Correct Answer: A. No further intervention ✅

MCQ 276

A 36-year-old man presented to the Emergency Department with a 24-hour history of central abdominal pain associated with abdominal distension, vomiting, and constipation. He had an appendectomy 7 years ago. Physical examination confirmed diffuse abdominal distension, mild abdominal tenderness with exaggerated bowel sounds.

Test Result Normal Values

- Amylase: 288 (24-151 IU/L)
- WBC: 11.4 (4.5-10.5 x 10^9/L)

Abdominal X-ray:

- Multiple air-fluid levels with dilated small bowel.

Which of the following is the most appropriate investigation?

A. Abdominal ultrasound
B. Exploratory laparotomy
C. Abdominal computed tomography
D. Laparoscopic abdominal exploration

Explanation:
The correct answer is **C. Abdominal computed tomography**.
This patient presents with symptoms and signs of **small bowel obstruction**. The next appropriate step is **CT scan** to confirm the diagnosis, assess the location of the obstruction, and rule out other potential causes such as volvulus or hernias. **Ultrasound** (Option A) may be useful for certain types of abdominal conditions, but CT is more appropriate here. **Exploratory laparotomy** (Option B) and **laparoscopic exploration** (Option D) are more invasive and should be considered after imaging confirms the diagnosis.

Management of Small Bowel Obstruction
Definition: Obstruction of the small bowel, often due to adhesions, hernias, or tumors.
Key Clinical Presentation: Abdominal pain, distension, vomiting, and constipation.
Investigations: CT scan or abdominal X-ray.
Treatment Outline:

- **First-line:** Surgical intervention for complete obstructions or after failure of conservative measures.
  Correct Answer: C. Abdominal computed tomography ✅

MCQ 277

A 25-year-old man underwent an exploratory laparotomy 5-days ago for a perforated peptic ulcer. He continues to have abdominal distension and vomiting. Physical examination confirmed a distended non-tender abdomen with sluggish bowel sounds.

Abdominal X-ray:

- Dilated small and large bowel with multiple air fluid levels.

Which of the following is the most appropriate next step in management?

A. Abdominal ultrasound
B. Gastrografin follow through
C. Check serum potassium level
D. Abdominal computed tomography

Explanation:
The correct answer is **D. Abdominal computed tomography**.
This patient, post-perforation surgery, presents with symptoms suggestive of **postoperative bowel obstruction** (e.g., distension, vomiting). The next step is to **rule out any complications such as an intra-abdominal collection or adhesions** by performing a **CT scan**. **Gastrografin**

follow-through (Option B) is sometimes used to assess for leaks but is not as informative in this context. **Serum potassium** (Option C) is useful to check in cases of dehydration but does not address the cause of the obstruction.

Postoperative Small Bowel Obstruction Management
Definition: Obstruction of the small bowel following abdominal surgery, often due to adhesions.
Key Clinical Presentation: Abdominal pain, distension, vomiting, and failure to pass flatus or stool.
Investigations: CT scan to assess for adhesions or other complications.
Treatment Outline:

- **First-line:** Conservative management with IV fluids, nasogastric tube, and monitoring.
- **Second-line:** Surgical intervention if conservative management fails.
  Correct Answer: D. Abdominal computed tomography ✅

MCQ 278:

A 41-year-old woman presented to the Emergency Department complaining of a 2-week history of upper abdominal pain associated with vomiting and fever. History reveals she underwent conservative treatment of acute hyperlipidaemia-induced pancreatitis 6 weeks ago.

Physical examination confirmed tenderness in the epigastric region.

- **Blood pressure:** 110/70 mmHg
- **Heart rate:** 100 /min
- **Temperature:** 38.1 °C

Test Result Normal Values

- WBC: 20.2 (4.5-10.5 x 10^9/L)
- Amylase: 245 (24-151 IU/L)

Ultrasound:

- Well-circumscribed collection (15 x 17 cm) in the lesser sac.

Which of the following is the most appropriate initial method of drainage?

A. Endoscopic
B. Laparoscopic
C. Percutaneous
D. Open surgical

Explanation:
The most appropriate management for a **pancreatic pseudocyst** (as suspected by imaging) is **C. Percutaneous drainage**.
A **pseudocyst** is a common complication of **pancreatitis**, and if it is symptomatic (causing pain, fever, or obstruction), **percutaneous drainage** is often preferred as the first-line method to alleviate symptoms and avoid surgery. **Endoscopic drainage** (Option A) can also be effective, but **percutaneous** is less invasive. **Laparoscopic** or **open surgery** (Options B and D) are reserved for larger or more complicated cases, such as those with infected pseudocysts or when other methods fail.

Management of Pancreatic Pseudocyst

- **Definition:** A collection of pancreatic fluid surrounded by a fibrous wall, usually following pancreatitis.
- **Key Clinical Presentation:** Abdominal pain, fever, and palpable mass.
- **Investigations:** Ultrasound or CT scan.
- **Treatment Outline:**
    - **First-line:** Percutaneous or endoscopic drainage.
    - **Second-line:** Surgery if complications arise.
    Correct Answer: C. Percutaneous ✅

MCQ 279

A 45-year-old presented to the Emergency Department with a 12-hour history of vomiting fresh blood. He has a 10-year history of recurrent peptic ulcers.

Physical examination was normal.
Upper GIT endoscopy:

- Esophagitis, multiple non-bleeding gastroduodenal ulcers, the distal one is in the 4th part of the duodenum.

Which of the following has the most diagnostic value?

A. Calcium
B. Parathormone
C. Fasting serum gastrin
D. Vasoactive intestinal polypeptide

Explanation:
The most diagnostic value in this case is **C. Fasting serum gastrin**.
The patient's history of recurrent peptic ulcers and gastrointestinal bleeding raises suspicion for **Zollinger-Ellison syndrome**, a condition characterized by **gastrin-secreting tumors** (gastrinoma) leading to excess gastric acid production. **Fasting serum gastrin** is the most useful test for diagnosing this condition. **Calcium** (Option A) and **parathormone** (Option B) are not typically involved in diagnosing this condition, and **vasoactive intestinal polypeptide** (Option D) is more relevant for other types of tumors like **VIPomas**.

Management of Zollinger-Ellison Syndrome

- **Definition:** A gastrin-secreting tumor, leading to excessive gastric acid secretion and peptic ulcers.
- **Key Clinical Presentation:** Recurrent ulcers, diarrhea, and gastrointestinal bleeding.
- **Investigations:** Fasting serum gastrin levels and imaging studies (e.g., CT or MRI).
- **Treatment Outline:**
  - **First-line:** Proton pump inhibitors to control acid secretion.
  - **Second-line:** Surgical resection of the tumor.
    Correct Answer: C. Fasting serum gastrin ✅

MCQ 280

A 37-year-old complained of a right breast lump for 3 months. She has a family history of breast cancer. Physical examination confirmed a hard lump in the upper outer quadrant of the right breast with ill-defined edges, skin tethering, and no palpable axillary lymph node.

Which of the following is the most appropriate next step in management?

A. MRI breast
B. Core needle biopsy

C. Bilateral mammography
D. Fine needle aspiration cytology

Explanation:
The most appropriate next step is **B. Core needle biopsy**.
The patient has a suspicious breast lump, particularly with **family history of breast cancer**. **Core needle biopsy** is the gold standard for evaluating breast masses as it allows for tissue diagnosis, including assessing for malignancy. **Mammography** (Option C) can be helpful, but biopsy is the definitive next step for diagnosis. **MRI breast** (Option A) is used in high-risk patients or for assessing the extent of disease, but not as the first diagnostic tool. **Fine needle aspiration cytology** (Option D) can be used, but core needle biopsy provides a more accurate and reliable sample.

MCQ 281

A 32-year-old woman presented with a 2-month history of left breast pain associated with bloody nipple discharge. Physical examination confirmed a normal left breast and axilla.

Which of the following is the most appropriate initial breast investigation?

A. MRI
B. CT scan
C. Ultrasound
D. Mammogram

Explanation:
The most appropriate initial investigation is **D. Mammogram**.
Mammography is the gold standard imaging test for breast evaluation, particularly for **bloody nipple discharge**, as it can help detect underlying pathology such as **ductal carcinoma in situ (DCIS)** or **intraductal papilloma**. **Ultrasound** (Option C) may be helpful in younger women or when mammography is inconclusive but is usually used as a complement. **MRI** (Option A) is typically reserved for high-risk patients or to assess the extent of known disease. **CT scan** (Option B) is not indicated for breast evaluation.

MCQ 282

A 44-year-old patient presented with an anterior neck swelling for 2 months. Physical examination confirmed a hard lump in the left side of the anterior triangle of the neck moving up with swallowing but no palpable cervical lymph nodes.

Ultrasound:

- Mildly enlarged thyroid, with 1.5 cm solid nodule in the left lobe.
  Fine needle aspiration cytology:

- Medullary thyroid cancer.

Which of the following is the most appropriate procedure?

A. Nodule enucleation
B. Total thyroidectomy
C. Subtotal thyroidectomy
D. Left hemithyroidectomy

Explanation:
The most appropriate procedure is **B. Total thyroidectomy**.
Medullary thyroid cancer is a **neuroendocrine malignancy** arising from the parafollicular C-cells of the thyroid. It often requires **total thyroidectomy** because the tumor is usually multifocal and may involve both thyroid lobes. **Left hemithyroidectomy** (Option D) and **subtotal thyroidectomy** (Option C) are inadequate because medullary thyroid cancer often spreads beyond the affected lobe. **Nodule enucleation** (Option A) is not sufficient for malignancy.

MCQ 283

A 32-year-old woman complained of an anterior neck lump noticed 3 months ago. Physical examination confirmed a firm lump in the right side of the anterior triangle of the neck moving up with swallowing but no palpable cervical lymph nodes.

Blood pressure 120/70 mmHg
Heart rate 115 /min
Temperature 36.7 °C

Test Result Normal Values

- **TSH:** 0.1 (0.4 - 5.0 µU/mL)

- **Free T3:** 11 (3.5 - 6.5 pmol/L)

- **Total T4:** 277 (64-155 nmol/L)

Ultrasound:

- Lump in the right thyroid lobe 2x3 cm.
  Thyroid isotope scan:

- Hot nodule in the right lobe with the remainder of the gland being cold.

Which of the following is the most appropriate initial step in management?

A. Anti-thyroid drugs
B. Total thyroidectomy
C. Right thyroid lobectomy
D. Radioactive iodine ablation

Explanation:
The most appropriate next step is **C. Right thyroid lobectomy**.
The patient has a **hot nodule** on the thyroid isotope scan, indicating a **functioning benign nodule** (likely a **toxic adenoma**). The best treatment option is to perform a **lobectomy** of the affected thyroid lobe. **Radioactive iodine ablation** (Option D) is used for diffuse hyperthyroid conditions like **Graves' disease**, not for solitary toxic nodules. **Anti-thyroid drugs** (Option A) are used in hyperthyroidism but are not curative for toxic adenomas. **Total thyroidectomy** (Option B) is more appropriate for malignant cases or diffuse thyroid disease.

MCQ 284

A 25-year-old man presented to the Emergency Department with severe pain during and after defecation for 3 days associated with the passage of a small amount of fresh blood after defecation. Physical examination confirmed an acute posterior anal fissure. Digital and proctoscopic examination were not performed due to the anal pain.

Which of the following is the most appropriate management?

A. Examination under anesthesia
B. Lateral internal anal sphincterotomy
C. Chemical sphincterotomy with diltiazem
D. Botulinus toxin paralysis of anal sphincter

Explanation:
The most appropriate initial management is **C. Chemical sphincterotomy with diltiazem**.
Acute anal fissures are typically treated conservatively with **topical treatments** such as **diltiazem**, a calcium channel blocker that helps reduce **sphincter spasm** and improve healing. If the fissure persists or becomes chronic, **surgical sphincterotomy** (Option B) can be considered. **Examination under anesthesia** (Option A) may be necessary for chronic or complicated fissures but is not routinely required for acute cases. **Botulinum toxin** (Option D) is reserved for refractory fissures.

MCQ 285:

A 36-year-old woman complaining of per-rectal bleeding and prolapsed tissue masses through the anal verge. Anorectal examination confirmed grade IV haemorrhoids at 3, 7, and 11 o'clock positions.

Which of the following is the most appropriate management?

A. Rubber banding
B. Open haemorrhoidectomy
C. Infrared photocoagulation
D. Submucosal injection of sclerosing agent

Explanation:
The most appropriate management for **grade IV haemorrhoids** is **B. Open haemorrhoidectomy**. Grade IV haemorrhoids are **irreducible** and require **surgical intervention**. **Haemorrhoidectomy** is the most effective and definitive treatment. **Rubber banding** (Option A) is usually used for grade II or III haemorrhoids. **Infrared photocoagulation** (Option C) and **submucosal injection of sclerosing agent** (Option D) are less effective for severe, prolapsed haemorrhoids.

MCQ 286

A 65-year-old man presented to the outpatient department complaining of alternating constipation and diarrhoea for 4 months. Physical examination was normal.

Rigid Sigmoidoscopy:

- Circumferential, ulcerating mass in the distal part of the sigmoid colon. Multiple biopsies were taken.
  Histopathology:
- Adenocarcinoma.

Which of the following is the most appropriate next step in management?

A. Abdominal CT scan
B. Sigmoidectomy
C. Colonoscopy
D. Pelvic MRI

Explanation:
The most appropriate next step is **A. Abdominal CT scan**.
The patient has a **sigmoid colon adenocarcinoma** with **ulcerating mass**, and the next step is to evaluate the extent of the disease and check for distant metastasis. **CT scan** is the most useful imaging modality to assess the local and distant spread of the cancer. **Sigmoidectomy** (Option B) may be performed later if surgical intervention is indicated. **Colonoscopy** (Option C) would have already been done for diagnosis. **Pelvic MRI** (Option D) is used in some cases to evaluate local spread but is not typically the first-line imaging in this scenario.

MCQ 287

A 48-year-old man complains of right upper quadrant abdominal pain for 4 months. Physical examination was normal.

Serology:

- Echinococcus granulosus was positive.
  Ultrasound:

- Large cystic lesion (10x15 cm) in the right lobe of the liver with multiple intra-cystic daughter cysts, but no calcifications.

Which of the following is the most appropriate management?

A. Surgical de-roofing
B. Percutaneous aspiration
C. Long-term albendazole therapy
D. Percutaneous sclerosant

Explanation:
The most appropriate management is **C. Long-term albendazole therapy**.
This patient likely has a **hydatid cyst** caused by **Echinococcus granulosus**. The treatment of choice is **albendazole** to treat the parasite. **Surgical de-roofing** (Option A) can be an option in some cases, but **albendazole therapy** is typically the first-line treatment. **Percutaneous aspiration** (Option B) is not recommended for hydatid cysts due to the risk of cyst rupture and anaphylaxis. **Percutaneous sclerosant** (Option D) is not an appropriate treatment for hydatid cysts.

MCQ 288

A 29-year-old woman presented to the Emergency Department with diffuse abdominal pain for 1 day associated with progressive abdominal distension, vomiting, and constipation. Physical examination confirmed abdominal distension, diffuse mild tenderness, and exaggerated bowel sounds.

CT scan:

- Distended small bowel with ileocecal intussusception causing complete small bowel obstruction.

Which of the following is the most appropriate procedure?

A. Colonoscopy
B. Surgical resection
C. Surgical reduction
D. Reduction by barium enema

Explanation:
The most appropriate procedure is **C. Surgical reduction**.
Ileocecal intussusception is causing a complete **small bowel obstruction**, and the most appropriate intervention is **surgical reduction**. This is typically required for **adult intussusception**, which is usually managed surgically. **Colonoscopy** (Option A) can sometimes reduce intussusception in pediatric cases, but not in adults. **Surgical resection** (Option B) is only needed if there is bowel necrosis. **Reduction by barium enema** (Option D) is more commonly used in pediatric cases but is not appropriate in this adult case.

MCQ 289

A 32-year-old woman presented to the Emergency Department with severe pain in the right upper quadrant of the abdomen following a fatty meal 6 hours ago. This pain has decreased in severity over the last 2 hours. The pain was radiating to the right subscapular area associated with vomiting. Physical examination was normal.

Test Result Normal Values

- WBC: 8.2 (4.5-10.5 x 10^9/L)
- Amylase: 134 (24-151 IU/L)
- Ultrasound: Multiple gallstones with normal gallbladder wall.

Which of the following is the most appropriate management?

A. Observation
B. Ursodeoxycholic
C. Laparoscopic cholecystectomy
D. Magnetic resonance cholangiopancreatography

Explanation:
The patient has **biliary colic** (due to **gallstones**), which is usually self-limiting and does not require immediate surgical intervention. Therefore, the most appropriate initial management is **A. Observation**.

- **Laparoscopic cholecystectomy** (Option C) is typically done for symptomatic gallstones, but this patient is not acutely ill.
- **Ursodeoxycholic acid** (Option B) is used to dissolve small gallstones, but it's not the first choice for managing biliary colic.
- **Magnetic resonance cholangiopancreatography** (Option D) is more appropriate for evaluating bile duct stones or other ductal issues, but not necessary in this case.

MCQ 290

A 36-year-old presented to the Emergency Department with a 9-day history of right iliac fossa pain, diarrhea, and fever. He was diagnosed with Crohn's disease 10 years ago. Physical examination confirmed a tender mass in the right iliac fossa.

Test Result Normal Value

- WBC: 17.6 (4.5-10.5 x 10^9/L)
- CT scan: Collection in the right iliac fossa (12x15 cm) with jejune-ileal fistula.

Which of the following is the most appropriate initial treatment?

A. Laparoscopic drainage
B. Percutaneous drainage
C. Open surgical drainage
D. Open drainage with resection of fistula

Explanation:
The most appropriate initial treatment is **B. Percutaneous drainage**.

- The presence of a **collection** and **fistula** suggests a **complicated Crohn's disease** with a **paracolonic abscess**. Percutaneous drainage is less invasive and effective for draining collections in this scenario.
- **Laparoscopic drainage** (Option A) can be used for certain cases, but percutaneous drainage is preferred for larger abscesses.
- **Open surgical drainage** (Option C) and **resection** (Option D) may be needed later if complications persist or if there is a bowel obstruction, but they are not the initial step.

MCQ 291

A 33-year-old woman presented with a lump in her neck. Physical examination confirmed enlarged right cervical lymph nodes with normal thyroid gland. She underwent percutaneous biopsy of the lymph nodes.

Histopathology:

- Normal follicular thyroid cells.

Which of the following is the most appropriate next step?

A. Reassurance
B. Surgical referral
C. Lymph node excision
D. 3 months follow up ultrasound

Explanation:
The most appropriate next step is **B. Surgical referral**.

- The patient has **enlarged cervical lymph nodes** with no obvious thyroid pathology. Given the lymph node involvement, it is important to rule out any **malignancy** such as **metastatic thyroid cancer**.
- **Lymph node excision** (Option C) may be considered, but a surgical referral is more appropriate for further evaluation and biopsy.

MCQ 292

A 42-year-old woman presented with a right breast mass that has been present for 3 months and is not associated with other symptoms. Physical examination confirmed a 2 x 2 cm firm partially movable mass with associated skin tethering and an enlarged axillary lymph node.

Bilateral breast mammogram and ultrasound:

- Ill-defined lobulated dense mass that is hypoechoic on ultrasound, measuring 2 x 2 cm, not associated with speculation or suspicious microcalcifications. Coded as BI-RADS IV.

Which of the following is the most appropriate next step in management?

A. Excision biopsy
B. Needle core biopsy
C. Fine needle aspiration
D. Short interval follow-up imaging

Explanation:
The most appropriate next step is **B. Needle core biopsy**.

- The **BI-RADS IV** classification suggests a **high suspicion for malignancy**, and the next step is to obtain tissue diagnosis via **core needle biopsy**.
- **Fine needle aspiration** (Option C) is less accurate for some breast lesions, especially those that are more heterogeneous.
- **Excision biopsy** (Option A) is not necessary upfront, and **short interval follow-up** (Option D) is not appropriate given the high suspicion for cancer.

MCQ 293

A 54-year-old woman complains of a left breast mass that has been present for 9 months. Physical examination confirmed a 3 x 3 cm firm fixed mass at the left retro-areolar area with associated nipple retraction.

Bilateral breast mammogram:

- Speculated mass at left retro-areolar region associated with suspicious clustered microcalcifications. There are multiple pathological left axillary lymph nodes. Findings coded as BI-RADS V (Probably Malignant).

Which of the following is the most appropriate next step in management?

A. Excision biopsy
B. Needle core biopsy
C. Staging CT and bone scan
D. Modified radical mastectomy

Explanation:
The most appropriate next step is **B. Needle core biopsy**.

- The **BI-RADS V** finding strongly suggests **malignant breast cancer**, and obtaining tissue through a **needle core biopsy** is the next step for histological diagnosis.
- **Modified radical mastectomy** (Option D) is a treatment option but would only be considered after biopsy confirmation of malignancy.
- **Staging CT and bone scan** (Option C) might be necessary after confirmation of malignancy but is not the first step.

MCQ 294

A 40-year-old woman presented for her first screening mammography. She has no past medical history, and no family history of cancer.

Bilateral breast mammogram:

- Bilateral dense breast with multiple fibro-glandular opacities, no suspicious microcalcifications or speculated lesions.

Breast ultrasound:

- Bilateral multiple variable sizes cysts. Findings coded as BI-RADS III (Probably benign).

Which of the following is the most appropriate next step in management?

A. Annual screening
B. Needle core biopsy
C. Short interval follow-up imaging
D. Magnetic resonance mammography

Explanation:
The most appropriate next step is **C. Short interval follow-up imaging**.

- **BI-RADS III** indicates **probably benign** findings, and short-term follow-up imaging (typically in 6 months) is the appropriate management.
- **Needle core biopsy** (Option B) is not needed as the lesion is likely benign.
- **MRI** (Option D) is not routinely needed in BI-RADS III cases.

MCQ 295

A 32-year-old woman, 28 weeks pregnant, presents with an enlarging right breast mass that has been present for 4 months and is associated with mild pain. Physical examination confirmed a 3 x 2 cm firm, movable mass, not associated with skin changes or enlarged axillary lymph nodes.

Which of the following is the most appropriate next step in management?

A. Fine needle aspiration
B. Bilateral breast ultrasound
C. Bilateral diagnostic mammography
D. Reassure and reassess after delivery

Explanation:
The most appropriate next step in management is **B. Bilateral breast ultrasound**.

- **Bilateral breast ultrasound** is the preferred initial imaging modality for pregnant women, as it is safe during pregnancy and helps to characterize the mass.
- **Fine needle aspiration** (Option A) could be considered if the ultrasound findings are suspicious, but it is not the first step unless clinically indicated.
- **Bilateral diagnostic mammography** (Option C) is typically avoided in pregnant women unless absolutely necessary.
- **Reassure and reassess after delivery** (Option D) is inappropriate as the mass should be evaluated and appropriately managed during pregnancy to rule out malignancy.

MCQ 296

A 60-year-old woman complains of left nipple discharge that is spontaneous, blood-stained, and recurrent for the past 6 months. Physical examination was normal.

Bilateral Breast Mammogram:

- Normal.

Breast Ultrasound:

- Bilateral ductal dilatation, with a 12 mm left intraductal lesion, suggestive of intraductal papilloma.

Which of the following is the most appropriate next step in management?

A. Discharge fluid cytology
B. Left central duct excision
C. Image-guided core biopsy
D. Short interval follow-up imaging

Explanation:
The most appropriate next step is **C. Image-guided core biopsy**.

- The patient has a **bloody nipple discharge** with an intraductal lesion on ultrasound, most likely an **intraductal papilloma**, which should be biopsied to rule out malignancy and confirm the diagnosis.
- **Discharge fluid cytology** (Option A) is not the best next step for an intraductal lesion.
- **Left central duct excision** (Option B) may be considered later if the biopsy shows malignancy, but it's not the immediate step.
- **Short interval follow-up imaging** (Option D) would not be appropriate as the lesion is suspicious and warrants further investigation.

MCQ 297

A 28-year-old woman presents with a neck mass that has been present for 1 year and is not associated with pressure symptoms. Physical examination confirmed diffuse thyroid enlargement.

Blood pressure 120/70 mmHg
Heart rate 90 /min
Temperature 36.7 °C

Test Result Normal Values

- **Thyroid-Stimulating Hormone:** 0.001 (0.4 - 5.0 µU/mL)
- **Thyroxine (T4 free serum):** 21 (8.5 - 15.2 pmol/L)

Ultrasound thyroid:

- Enlarged both thyroid lobes, with a solid nodule on the left side measuring 1 x 2 cm and another nodule on the right side measuring 4 x 3 cm.

Which of the following is the most appropriate next step in management?

A. Thyroid scan
B. Total thyroidectomy
C. Fine needle aspiration of both nodules
D. Fine needle aspiration of the larger nodule

Explanation:
The most appropriate next step is **D. Fine needle aspiration of the larger nodule**.

- The **TSH suppression** (low TSH and elevated T4) suggests a **toxic nodule**, and the larger nodule should be investigated further with **fine needle aspiration** (FNA) to rule out malignancy and evaluate its functional status.
- **Thyroid scan** (Option A) is not indicated as FNA is the gold standard for evaluating nodules.
- **Total thyroidectomy** (Option B) is not necessary without confirmation of malignancy or specific functional problems.
- **FNA of both nodules** (Option C) is not required for both, as the larger nodule is the priority.

MCQ 298

A 24-year-old man presented with neck pain for 4 months. Physical examination of the neck was normal.

Test Result Normal Values

- **Thyroid-Stimulating Hormone:** 3.6 (0.4 - 5.0 µU/mL)
- **Thyroxine (T4 free serum):** 11 (8.5 - 15.2 pmol/L)

Neck Ultrasound:

- Normal size thyroid gland with a solitary solid left thyroid nodule measuring 9 x 7 mm. Benign-looking bilateral cervical lymph nodes.

Which of the following is the most appropriate management?

A. Thyroid scan
B. Follow-up imaging
C. Fine needle aspiration
D. Left thyroid lobectomy

Explanation:
The most appropriate management is **B. Follow-up imaging**.

- The nodule is **small** and the ultrasound findings do not suggest malignancy. The patient has **benign-looking lymph nodes**, so **follow-up imaging** in 6 months is the best option.
- **Thyroid scan** (Option A) is not necessary as the nodule is likely benign.
- **Fine needle aspiration** (Option C) is unnecessary given the benign characteristics of the nodule.
- **Left thyroid lobectomy** (Option D) is not needed unless malignancy is suspected, which is not the case here.

MCQ 299

A 36-year-old patient presented with a neck mass that has been present for 3 years. Physical examination confirmed a palpable left thyroid gland with no palpable cervical lymph nodes.

Neck Ultrasound:

- Left thyroid nodule occupying most of the left lobe, measuring 3.4 x 3.8 cm, with mixed cystic and solid components. The right lobe is normal.

Fine Needle Aspiration:

- Finding coded as Bethesda IV (suspicious for follicular neoplasm).

Which of the following is the most appropriate management?

A. Thyroid scan
B. Total thyroidectomy
C. Left hemithyroidectomy
D. Repeat fine needle aspiration

Explanation:
The most appropriate management is **C. Left hemithyroidectomy**.

- A **Bethesda IV** result indicates a **suspicious follicular neoplasm**, which requires surgical intervention. Given that the nodule is localized to the left thyroid lobe, a **left hemithyroidectomy** is the appropriate approach.

- **Thyroid scan** (Option A) is not necessary since the nodule is already identified and suspicious for malignancy.
- **Total thyroidectomy** (Option B) is typically reserved for cases where there is a high suspicion of malignancy or if the disease involves both lobes.
- **Repeat fine needle aspiration** (Option D) is not required as the diagnosis has already been made with the current biopsy.

MCQ 300

A 29-year-old patient underwent a right hemithyroidectomy for a symptomatic right thyroid nodule.

Histopathology:

- A 3 x 4 cm colloid nodule with a focal well-differentiated papillary carcinoma measured 8 mm away from the main lesion. There was no extrathyroidal extension or lymphovascular invasion.

Which of the following is the most appropriate management?

A. Follow-up
B. Whole body iodine scan
C. Radioactive iodine ablation
D. Completion of thyroidectomy

Explanation:
The most appropriate management is **D. Completion of thyroidectomy**.

- The presence of **papillary carcinoma** (albeit small) within the thyroid suggests that **completion of thyroidectomy** should be performed to remove the remaining thyroid tissue and prevent recurrence.
- **Follow-up** (Option A) is not sufficient since the patient has malignancy that requires definitive treatment.
- **Whole body iodine scan** (Option B) and **radioactive iodine ablation** (Option C) are used for certain cases of papillary carcinoma, but the most appropriate first step is to complete the thyroidectomy.

MCQ 301

A 23-year-old woman presented with symptoms of hyperthyroidism. Physical examination confirmed a non-palpable thyroid gland, and bilateral exophthalmos was noted. She has been on anti-thyroid medication for 10 months.

Test Results Normal Values:

- **Thyroid-Stimulating Hormone (TSH):** 0.001 (0.4 - 5.0 µU/mL)
- **Thyroxine (T4 free serum):** 29 (8.5 - 15.2 pmol/L)

Neck Ultrasound:

- Mild diffuse enlargement of the thyroid gland.

Thyroid scan:

- Diffuse uptake of the thyroid gland consistent with toxic multi-nodular goiter.

Which of the following is the most appropriate management?

A. Subtotal thyroidectomy
B. Near total thyroidectomy
C. Radioactive iodine ablation
D. Continue on medical therapy

Explanation:
The most appropriate management is **C. Radioactive iodine ablation**.

- The patient has **toxic multi-nodular goiter** and **long-standing hyperthyroidism**. Radioactive iodine ablation is a common and effective treatment for this condition.
- **Subtotal thyroidectomy** (Option A) and **near total thyroidectomy** (Option B) may be considered in some cases, but radioactive iodine therapy is preferred for its minimal invasiveness.
- **Continue on medical therapy** (Option D) is not ideal for long-term management, especially since the patient has persistent symptoms despite anti-thyroid medication.

MCQ 302

A morbidly obese 35-year-old with no relevant past medical history consults for weight reduction.

Which of the following is the most useful to determine the most appropriate procedure?

A. Barium swallow
B. Upper GI endoscopy
C. CT scan of abdomen
D. Ultrasound abdomen

Explanation:
The most useful investigation is **B. Upper GI endoscopy**.

- **Upper GI endoscopy** (Option B) is typically performed to assess for any **gastrointestinal pathology** that might impact bariatric surgery, such as reflux disease or peptic ulcers. It provides a comprehensive look at the upper GI tract and is helpful in evaluating the patient's suitability for surgical procedures.
- **Barium swallow** (Option A) and **CT scan of abdomen** (Option C) are not routinely used to assess bariatric surgery candidacy.
- **Ultrasound abdomen** (Option D) may be useful for evaluating the liver and gallbladder but is not the primary investigation for planning weight loss surgery.

MCQ 303

A morbidly obese 23-year-old man complains of gastroesophageal reflux disease (GERD).

Upper GI endoscopy:

- Large hiatus hernia with grade III reflux.

Which of the following is the most appropriate weight reduction procedure?

A. Sleeve gastrectomy
B. Intra-gastric balloon
C. Billo-pancreatic bypass
D. Roux-en-y gastric bypass

Explanation:
The most appropriate procedure is **D. Roux-en-Y gastric bypass**.

- **Roux-en-Y gastric bypass (RYGB)** is often preferred for morbidly obese patients with **GERD** and **hiatal hernia**. It not only helps with weight loss but also has a beneficial effect on reflux by creating a smaller gastric pouch and bypassing the duodenum.
- **Sleeve gastrectomy** (Option A) may also help with weight loss but can worsen GERD due to the increased intra-abdominal pressure.

- **Intra-gastric balloon** (Option B) is less effective for long-term weight loss in morbid obesity and does not address the reflux symptoms.
- **Bilio-pancreatic bypass** (Option C) is typically not recommended unless there is significant metabolic disease such as diabetes.

MCQ 304

A 26-year-old man presented 3 days after a sleeve gastrectomy with left upper abdominal pain. Physical examination confirmed only mild upper abdominal tenderness with no sign of peritonitis.

Blood pressure: 100/70 mmHg
Heart rate: 116 /min

Which of the following is the most likely explanation?

A. Dehydration
B. Gastric leak
C. Pulmonary embolism
D. Inadequate analgesia

Explanation:
The most likely explanation is **A. Dehydration**.

- After **sleeve gastrectomy**, dehydration is common, especially if the patient is not consuming enough fluids due to postoperative pain or reduced gastric capacity. Mild tenderness and a normal examination, with a relatively elevated heart rate and low blood pressure, suggest dehydration.
- **Gastric leak** (Option B) is a serious complication, but it would likely present with signs of peritonitis or more severe symptoms.
- **Pulmonary embolism** (Option C) is less likely in the absence of respiratory symptoms or significant risk factors.
- **Inadequate analgesia** (Option D) is possible but would typically cause pain rather than the hemodynamic changes observed.

MCQ 305

A 46-year-old man underwent an ERCP for biliary obstruction; the procedure was prolonged and difficult. 4 hours later, he developed extensive surgical emphysema of the abdomen, chest, and neck.

Where is the most likely injury?

A. Gastric
B. Tracheal
C. Duodenal
D. Esophageal

Explanation:
The most likely injury is **D. Esophageal**.

- **Surgical emphysema** in the abdomen, chest, and neck after ERCP is typically caused by **esophageal perforation**. During ERCP, especially if it's difficult or prolonged, the esophagus can be perforated, leading to air leakage into the surrounding tissues.
- **Gastric** (Option A), **tracheal** (Option B), and **duodenal** (Option C) injuries are less likely to cause the widespread emphysema observed here.

MCQ 306

A 22-year-old man underwent an open left inguinal hernia mesh repair 2 weeks ago, and presented with continuous discharge from the surgical wound. Examination confirmed a 2 x 3 cm pus-discharging wound with part of the mesh exposed.

Which of the following is the most appropriate management?

A. Systemic antibiotic
B. Wound debridement
C. Wound drainage and mesh removal
D. Open entire wound for daily dressing

Explanation:
The most appropriate management is **C. Wound drainage and mesh removal**.

- The presence of **pus discharge** and **exposure of mesh** suggests an **infection** involving the mesh, which may require removal of the infected mesh, drainage of the wound, and appropriate antibiotic therapy.
- **Systemic antibiotics** (Option A) alone will not address the source of infection, which is the exposed mesh.

- **Wound debridement** (Option B) may be necessary, but removal of the mesh is the most critical step in this case.
- **Opening the wound for daily dressing** (Option D) is not an appropriate first-line intervention for mesh infection.

MCQ 307

Which of the following is the most common cause of biliary colic?

A. Cystic duct stone
B. Gall bladder stone
C. Gall bladder sludge
D. Choledocholithiasis

Explanation:
The most common cause is **B. Gall bladder stone**.

- **Gall bladder stones** are the most common cause of **biliary colic** as they can intermittently obstruct the cystic duct, causing pain.
- **Cystic duct stones** (Option A) can also cause biliary colic but are less common.
- **Gall bladder sludge** (Option C) is a precursor to gallstones but does not usually cause symptoms unless it forms stones.
- **Choledocholithiasis** (Option D), which refers to stones in the common bile duct, typically causes more severe symptoms, such as jaundice, and is less likely to be the cause of typical biliary colic.

MCQ 308

Which of the following is most commonly associated with biliary colic?

A. Bilirubin
B. Amylase
C. Cholecystokinase
D. Alkaline phosphatase

Explanation:
The most commonly associated enzyme is **D. Alkaline phosphatase**.

- **Alkaline phosphatase (ALP)** is often elevated in cases of biliary obstruction or gallbladder disease, including **biliary colic**, which typically arises from obstruction of the cystic duct due to gallstones.
- **Bilirubin** (Option A) is elevated in jaundice, but it's not specifically associated with biliary colic.
- **Amylase** (Option B) is related to **pancreatitis** and is less commonly associated with biliary colic.
- **Cholecystokinase** (Option C) is a hormone involved in gallbladder contraction and digestive processes but is not used as a diagnostic marker for biliary colic.

MCQ 309

Which of the following is the most preferred mesh placement for a ventral hernia repair?

A. Inlay
B. Onlay
C. Sublay
D. Underlay

Explanation:
The most preferred mesh placement is **D. Underlay**.

- **Underlay** mesh placement, where the mesh is placed behind the abdominal wall, is the most common and preferred approach for **ventral hernia repair** as it provides better support and reduces the risk of recurrence.
- **Inlay** (Option A) involves placing the mesh on top of the defect, which is less preferred because it has a higher recurrence rate.
- **Onlay** (Option B) places the mesh on the surface of the abdominal wall, which is typically avoided due to higher risk of complications.
- **Sublay** (Option C) is a less common option that places the mesh in a plane just below the muscle layer but is not as commonly used for ventral hernia repair as **underlay**.

MCQ 310

A 22-year-old woman presented 1-week after a laparoscopic appendectomy with mild abdominal pain not associated with nausea or vomiting, she reports passing infrequent loose

stool. Physical examination confirmed a soft abdomen with mild tenderness at the wound site.

CT scan of abdomen:

- 2 x 2 cm retrocaccal collection.

Which of the following is the most appropriate next step in management?

A. Percutaneous drainage
B. Open surgical drainage
C. Diagnostic laparoscopy
D. Conservative management

Explanation:
The most appropriate next step is **A. Percutaneous drainage**.

- The patient has a **postoperative retrocaccal collection**, which suggests the possibility of a **subclinical abscess** following appendectomy.
- **Percutaneous drainage** is the preferred initial treatment for localized abdominal collections and abscesses.
- **Open surgical drainage** (Option B) is typically reserved for cases with large or unresponsive abscesses.
- **Diagnostic laparoscopy** (Option C) may not be needed if the abscess is confirmed on imaging.
- **Conservative management** (Option D) might be appropriate for small collections, but drainage is usually the first-line approach.

MCQ 311

A 52-year-old complains of left lower abdominal pain associated with passing loose stool and a loss of appetite. Examination confirmed left iliac fossa guarding and rebound tenderness.

Blood pressure 110/70 mmHg
Heart rate 98 /min
Test Result Normal Value
WBC 19.4 (5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Diverticulitis
B. Pyelonephritis
C. Sigmoid malignancy
D. Ruptured ovarian cyst

Explanation:
The most likely diagnosis is **A. Diverticulitis**.

- The patient has **left lower quadrant abdominal pain**, **rebound tenderness**, and **elevated white blood cells**, which are classic signs of **diverticulitis**—inflammation or infection of diverticula in the colon, particularly in the sigmoid colon.
- **Pyelonephritis** (Option B) would usually present with fever and flank pain, and it is less likely in this scenario.
- **Sigmoid malignancy** (Option C) could cause similar symptoms but is less likely to present with such an acute inflammatory process.
- **Ruptured ovarian cyst** (Option D) typically presents with pelvic pain, and is less likely to cause rebound tenderness and elevated white blood cells in the setting of this presentation.

MCQ 312

What structure passes through the deep inguinal ring?

A. Ilioinguinal nerve
B. Inferior epigastric artery
C. Medial umbilical ligament
D. Round ligament of the uterus

Explanation:
The structure that passes through the deep inguinal ring is **D. Round ligament of the uterus**.

- The **round ligament of the uterus** passes through the deep inguinal ring in females, contributing to the formation of the inguinal canal.
- **Ilioinguinal nerve** (Option A) does pass through the inguinal canal, but it enters the canal at a different point.
- **Inferior epigastric artery** (Option B) is located outside the inguinal canal and runs superior to the deep inguinal ring.

- **Medial umbilical ligament** (Option C) is located in the abdominal wall and is not involved in the inguinal canal.

MCQ 313

A 36-year-old patient presented with a left neck mass. Examination confirmed a 2 x 2 cm mass at the posterior triangle just below the angle of the mandible. Ultrasound of the neck showed a normal thyroid gland, large left cervical lymph node with cystic changes. Fine needle aspiration: The entire smear consists of follicular thyroid cells. Which of the following is the most likely diagnosis?

A. Thyroglossal cyst
B. Ectopic thyroid gland
C. Apparent thyroid gland
D. Metastatic thyroid cancer

Explanation:
The most likely diagnosis is **D. Metastatic thyroid cancer**.

- The presence of **follicular thyroid cells** on fine needle aspiration (FNA) from a cervical lymph node suggests a **metastatic thyroid cancer**, specifically **papillary** or **follicular thyroid carcinoma**.
- **Thyroglossal cyst** (Option A) typically presents as a midline neck mass, not at the angle of the mandible.
- **Ectopic thyroid gland** (Option B) usually presents in a different location, often in the midline or base of the tongue.
- **Apparent thyroid gland** (Option C) is not a common clinical term and seems to be an incorrect choice.

MCQ 314

A 45-year-old patient complains of frequent anal itching, and post-defecation pain. Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Perianal hematoma
D. Internal hemorrhoids

Explanation:
The most likely diagnosis is **A. Anal fissure**.

- **Anal fissure** causes **pain** during and after defecation, especially when passing hard stools. The **itching** can also occur due to irritation. The **pain** is typically sharp and localized.
- **Perianal fistula** (Option B) would typically cause recurrent drainage, but not significant itching or pain after defecation.
- **Perianal hematoma** (Option C) would cause a localized lump and **pain**, not the typical anal itching.
- **Internal hemorrhoids** (Option D) typically cause **rectal bleeding** and discomfort, but pain is less pronounced unless there is thrombosis.

MCQ 315

A 50-year-old man underwent an open mesh repair of a left inguinal hernia. He presented 2 months later because he noticed that the left testes had become smaller in size. Which of the following is the most likely cause?

A. Ligation of testicular artery
B. Mesh migration to the scrotum
C. Thrombosis of pampiniform plexus
D. Tightening of external inguinal ring

Explanation:
The most likely cause is **A. Ligation of testicular artery**.

- **Ligation of the testicular artery** during hernia repair can lead to **testicular atrophy**, as it compromises the blood supply to the testicle.
- **Mesh migration to the scrotum** (Option B) is uncommon but could cause irritation, not atrophy.
- **Thrombosis of the pampiniform plexus** (Option C) could lead to **varicocele** but not testicular atrophy.
- **Tightening of external inguinal ring** (Option D) could affect blood flow, but **ligation of the testicular artery** is a more likely cause.

MCQ 316

A 30-year-old underwent a laparoscopic appendectomy 2 weeks ago. Histology: Carcinoid tumor at the tip of the appendix measuring 0.5 cm. Which of the following is the most appropriate management?

A. Colonoscopy
B. No further action needed
C. Limited right hemicolectomy
D. CT scan of chest abdomen and pelvis

Explanation:
The most appropriate management is **B. No further action needed**.

- **Appendiceal carcinoids** smaller than **2 cm** typically do not require further treatment beyond appendectomy if the tumor is confined to the appendix and has no evidence of metastasis or vascular invasion.
- **Colonoscopy** (Option A) is not needed unless there is suspicion of involvement of other parts of the colon.
- **Limited right hemicolectomy** (Option C) is usually reserved for larger or more invasive appendiceal carcinoid tumors.
- **CT scan of chest, abdomen, and pelvis** (Option D) might be needed if there is concern for metastasis, but for tumors under 2 cm and with no other concerning features, no further imaging is typically required.

MCQ 318

A 38-year-old man presented with a 4-hour history of sudden severe abdominal pain. He reports frequent use of non-steroidal anti-inflammatory drugs for joint pain. Examination confirmed generalized abdominal tenderness and guarding with absent bowel sounds. Temperature 37.8 °C. Test Result: WBC 13 (Normal: 4.5-10.5 x 10^9/L).

Which of the following is the most appropriate initial imaging?

A. Erect chest X-ray
B. CT scan abdomen
C. Ultrasound abdomen
D. Supine abdominal X-ray

Explanation:
The most appropriate initial imaging is **B. CT scan abdomen**.

- The patient has signs of **acute abdomen**, possibly related to a **perforated viscera** or **acute abdominal pathology**, which can be best evaluated with a **CT scan**. CT is the gold standard for evaluating conditions like **perforated peptic ulcer**, **diverticulitis**, or **bowel obstruction**.
- **Erect chest X-ray** (A) may show free air if there's perforation, but a **CT scan** provides more detailed information about the underlying pathology.
- **Ultrasound abdomen** (C) is more useful for liver or biliary problems but not the first choice for suspected perforation or generalized abdominal issues.
- **Supine abdominal X-ray** (D) is useful for detecting obstruction or free air but less detailed than a CT scan for this situation.

MCQ 319

A 47-year-old man presented with a 6-month history of right groin swelling. The swelling recently increased in size. Examination confirmed an inguino-scrotal swelling with positive cough impulse and separable from the testis.

Which of the following is the most likely diagnosis?

A. Spermatocele
B. Epididymal cyst
C. Vaginal hydrocele
D. Indirect inguinal hernia

Explanation:
The most likely diagnosis is **D. Indirect inguinal hernia**.

- The **inguino-scrotal** swelling with a **positive cough impulse** and separation from the testis is characteristic of an **indirect inguinal hernia**.
- **Spermatocele** (A) would be a **painless**, fluid-filled mass in the **epididymis**, not connected to the inguinal canal.
- **Epididymal cyst** (B) also presents as a painless mass, but it's not **inguino-scrotal** in location.

- **Vaginal hydrocele** (C) is a fluid collection around the testicle but is typically not separated from the testis, and it does not have a **cough impulse**.

MCQ 320

A 38-year-old patient developed a painful swelling in the right lower quadrant of his abdomen after moving a heavy object. Examination confirmed a localized tender swelling in the right lower abdominal quadrant with negative cough impulse; it persisted even with contraction of abdominal muscles.

Which of the following is the most likely diagnosis?

A. Ventral hernia
B. Appendicular mass
C. Extraperitoneal lipoma
D. Rectus sheath hematoma

Explanation:
The most likely diagnosis is **D. Rectus sheath hematoma**.

- The **painful swelling** in the **right lower quadrant** with **negative cough impulse** and associated with **recent trauma** (moving a heavy object) suggests a **rectus sheath hematoma**, a condition where blood collects in the sheath of the **rectus abdominis** muscle.
- **Ventral hernia** (A) would cause a swelling with a **positive cough impulse**, and the swelling would be more palpable on cough.
- **Appendicular mass** (B) could present with tenderness, but it typically doesn't cause localized swelling outside of the abdomen.
- **Extraperitoneal lipoma** (C) would present as a painless, soft mass, not associated with **painful swelling** and **muscle contraction**.

MCQ 321

A 60-year-old patient presented with a 6-month history of a dull achey abdominal swelling associated with weight loss. Examination confirmed a 20x20 cm firm, non-tender, intra-abdominal mass.

Abdominal CT scan:
Lobulated soft tissue mass arising from retro-peritoneum, with multiple hypodense liver lesions.

Which of the following is the most likely diagnosis?

A. Liposarcoma
B. Lymphosarcoma
C. Germ cell tumour
D. Peripheral nerve tumour

Explanation:
The most likely diagnosis is **A. Liposarcoma**.

- **Liposarcoma** is a **malignant soft tissue tumour** that arises from **fat cells** in the **retroperitoneum** and can present with an **intra-abdominal mass**. It is commonly associated with **hypodense liver lesions** due to metastasis.
- **Lymphosarcoma** (B) may involve the retroperitoneum but typically involves **lymph nodes** rather than a solitary mass.
- **Germ cell tumour** (C) is rare in the retroperitoneum, and typically presents in younger individuals.
- **Peripheral nerve tumour** (D) would present with neurological symptoms and would not typically present as a large abdominal mass with liver metastasis.

MCQ 322

Which of following is associated with the highest risk of malignant transformation?

A. Villous adenoma
B. Tubular adenoma
C. Hyperplastic polyp
D. Tubulovillous adenoma

Explanation:
The answer is **A. Villous adenoma**.

- **Villous adenomas** are known for their **high malignant potential**. They have a higher risk of becoming **colorectal cancer** compared to other types of adenomas.
- **Tubular adenoma** (B) has a lower risk of malignancy but still requires monitoring for potential transformation.

- **Hyperplastic polyps** (C) are generally **benign** and have little to no risk of malignant transformation.
- **Tubulovillous adenoma** (D) has a higher risk of malignancy than tubular adenomas but lower than villous adenomas.

MCQ 323:

A 30-year-old presented with intermittent perianal pain associated with purulent discharge. Examination confirmed a low-lying perianal fistula with an external opening posteriorly close to the anal verge.

Which of the following is the most appropriate management?

A. MRI pelvis
B. Fistulogram
C. Fistulotomy
D. Lateral sphincterotomy

Explanation:
The most appropriate management is **C. Fistulotomy**.

- A **fistulotomy** is the standard treatment for **low-lying perianal fistulas**. It involves cutting open the fistula tract to allow healing. This is effective in most cases, especially when the external opening is near the anal verge.
- **MRI pelvis** (A) is useful for high anal fistulas to evaluate the sphincter involvement but not the first step for low-lying fistulas.
- **Fistulogram** (B) is also used to visualize the fistula tract, but it is less commonly performed now due to the availability of MRI and clinical examination.
- **Lateral sphincterotomy** (D) is used in cases of **anal fissures** or **high anal fistulas** but not typically for low perianal fistulas.

MCQ 324

A 30-year-old presented with a 3-day history of painful perianal swelling. Examination confirmed a 1x1 cm tender swelling close to the anal verge.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Prolapsed pile
C. Perianal abscess
D. Perianal hematoma

Explanation:
The most likely diagnosis is **C. Perianal abscess**.

- A **perianal abscess** presents with a **tender swelling** near the **anal verge** and often occurs after infection in the anal glands.
- **Anal fissure** (A) usually presents with **pain during defecation** and bleeding, but not a swelling or abscess.
- **Prolapsed pile** (B) is typically a **painless swelling** that is related to hemorrhoids.
- **Perianal hematoma** (D) could present with a **swelling**, but it is typically related to **painful swelling** after straining or injury and may not have the classic signs of infection.

MCQ 325

A 14-year-old patient presented with a painless neck swelling. Examination confirmed a midline neck swelling below the hyoid bone that moves up and down with swallowing.

Which of the following is the most likely diagnosis?

A. Cystic hygroma
B. Thyroglossal cyst
C. Midline dermoid cyst
D. Submental lymphadenopathy

Explanation:
The most likely diagnosis is **B. Thyroglossal cyst**.

- A **thyroglossal cyst** is a **midline neck swelling** that is **movable with swallowing** and typically occurs **below the hyoid bone**. It results from the remnant of the **thyroglossal duct** during embryological development.
- **Cystic hygroma** (A) typically presents as a **soft, cystic mass** in the **posterior triangle** of the neck, not in the midline.

- **Midline dermoid cyst** (C) is usually located along the midline but would typically present with **a firm, non-mobile mass** and not associated with swallowing.
- **Submental lymphadenopathy** (D) usually presents as a **firm, immobile** mass and is not associated with movement during swallowing.

MCQ 326

A 42-year-old man complains of chest pain after being involved in a motor vehicle accident. Examination confirmed a patent airway, equal air entry on both sides of the chest, and distended neck veins.

Which of the following is the most likely diagnosis?

A. Flail chest
B. Cardiac tamponade

Explanation:
The most likely diagnosis is **B. Cardiac tamponade**.

- **Cardiac tamponade** is characterized by **distended neck veins**, **hypotension**, and **tachycardia** (often seen after chest trauma) due to **fluid accumulation in the pericardial sac**, leading to **impaired cardiac output**.
- **Flail chest** (A) would usually present with **paradoxical movement** of the chest wall and **difficulty breathing** but not specifically with **distended neck veins** or signs of shock.

MCQ 327

A 42-year-old man complains of chest pain after being involved in a motor vehicle accident. Examination confirmed a patent airway, equal air entry on both sides of the chest, and distended neck veins. Blood pressure 88/45 mmHg, Heart rate 110 /min, Respiratory rate 30 /min.

Which of the following is the most likely diagnosis?

A. Flail chest
B. Cardiac tamponade
C. Tension pneumothorax
D. Simple pneumothorax

Explanation:
The most likely diagnosis is **B. Cardiac tamponade**.

- **Cardiac tamponade** can occur after chest trauma and presents with **distended neck veins**, **hypotension**, and **tachycardia**, all seen in this patient. The **distended neck veins** suggest pressure on the heart, preventing proper filling.
- **Flail chest** (A) presents with **paradoxical movement** of the chest wall but is not typically associated with distended neck veins or hypotension unless there is significant associated trauma.
- **Tension pneumothorax** (C) typically presents with **tracheal deviation** and **absent breath sounds** on one side, but this patient has equal air entry.
- **Simple pneumothorax** (D) would typically present with **unilateral breath sounds** and not the signs of shock or distended neck veins.

MCQ 328

A 25-year-old was involved in a head-on motor vehicle accident. On examination, he was unresponsive, had distended neck veins, absent breath sounds, and hyper-resonant percussion on the right side. His GCS score was 8.

Which of the following is the most urgent initial management?

A. Pericardiocentesis
B. Neck immobilization
C. Endotracheal intubation
D. Needle right thoracocentesis

Explanation:
The most urgent initial management is **D. Needle right thoracocentesis**.

- The clinical signs, including **hyper-resonant percussion** and **absent breath sounds**, suggest a **tension pneumothorax**. This is a life-threatening emergency that requires **needle decompression** to relieve pressure on the lung and heart.
- **Pericardiocentesis** (A) is indicated if **cardiac tamponade** is suspected, but the signs here suggest pneumothorax.
- **Neck immobilization** (B) is important for spinal injuries, but airway management and decompressing the pneumothorax take precedence.

- **Endotracheal intubation** (C) is important if the airway is compromised, but **needle thoracocentesis** takes priority to relieve the life-threatening tension pneumothorax.

MCQ 329

A 25-year-old man was brought to the Emergency Department after being involved in a motor vehicle accident. He opened his eyes spontaneously and responded appropriately to verbal commands. His respiration was shallow, and he had left chest wall contusion. He was able to shrug his shoulders but unable to move his elbows or lower limbs. Blood pressure 80/40 mmHg, Heart rate 70 /min, Respiratory rate 30 /min.

Which of the following is the most likely cause of the hypotension?

A. Cardiac tamponade
B. Internal hemorrhage
C. High spinal cord injury
D. Tension pneumothorax

Explanation:
The most likely cause of the hypotension is **C. High spinal cord injury**.

- This patient's **inability to move below the shoulders** and **hypotension** suggests **high spinal cord injury**, particularly in the cervical region, which can impair sympathetic nervous system function and lead to **neurogenic shock**.
- **Cardiac tamponade** (A) could cause hypotension, but it would usually present with **distended neck veins** and **muffled heart sounds**, which are not described here.
- **Internal hemorrhage** (B) is a possible cause but is less likely without signs of active bleeding.
- **Tension pneumothorax** (D) would typically present with **tracheal deviation** and **absent breath sounds**, which are not mentioned.

MCQ 330

Which of the following is the most likely source?

A. Hair follicles
B. Sebaceous glands

C. Eccrine sweat glands
D. Apocrine sweat glands

Explanation:
The most likely source is **B. Sebaceous glands**.

- Sebaceous glands are responsible for producing **sebum**, which can lead to conditions like **acne** or other **skin disorders**. These glands are commonly found near **hair follicles**.
- **Apocrine sweat glands** (D) are primarily found in areas like the **armpits** and **groin**, and they are involved in **body odor** production, not commonly in the conditions described here.
- **Eccrine sweat glands** (C) are responsible for producing **sweat** but are not typically linked to the conditions described here.
- **Hair follicles** (A) are related to **hair growth** and **sebaceous glands**, but they are not the primary source in the context given.

MCQ 331

A 50-year-old diabetic man presented with a 5-year history of an unhealed foot ulcer. Physical examination confirmed a 2x2 cm irregular-shaped ulcer at the dorsal aspect of the first toe with pearly white discoloration. Biopsy of the ulcer reported as pseudoepitheliomatous hyperplasia.

Which of the following is the most appropriate management?

A. Repeat biopsy
B. First toe amputation
C. Surgical debridement
D. Negative pressure dressing

Explanation:
The most appropriate management is **C. Surgical debridement**.

- **Pseudoepitheliomatous hyperplasia** suggests **chronic inflammation** that can occur in diabetic ulcers. **Surgical debridement** is typically the first step to remove non-viable tissue and promote wound healing.
- **Repeat biopsy** (A) is not necessary unless malignancy is suspected.

- **First toe amputation** (B) would only be indicated if there were significant **infection** or **necrosis**, which is not the case here.
- **Negative pressure dressing** (D) could be useful in wound management, but **debridement** is the first necessary step to manage the ulcer.

MCQ332

A 32-year-old man had excision and primary closure of a skin contracture that was 5x3 cm on the forearm, 1 year ago. He presented with an overgrowth at the scar site that extends beyond the surgical scar margins, it started a few weeks after the surgery and gradually increased in size.

Which of the following most accurately describes the lesion?

A. Keloid
B. Scar fibrosis
C. Desmoid tumor
D. Scar hypertrophy

Explanation:
The most accurate description of this lesion is **A. Keloid**.

- **Keloids** are characterized by excessive **scar tissue** that grows beyond the original wound boundaries and can continue to grow even after healing.
- **Scar fibrosis** (B) refers to the formation of scar tissue but does not involve the overgrowth beyond the scar.
- **Desmoid tumors** (C) are rare, benign tumors arising from fibrous tissue, typically in the abdominal wall or extremities, but they are less common in post-surgical scars.
- **Scar hypertrophy** (D) is similar to keloid but does not extend beyond the wound margin and typically regresses over time.

MCQ 333

A 58-year-old woman presented with a right thigh mass that measured 5 x 6 cm (see report).

Core biopsy: High-grade sarcoma.

Which of the following is the most appropriate staging image?

A. Bone scan
B. MRI abdomen
C. Chest CT scan
D. Right thigh plain X-ray

Explanation:
The most appropriate staging image is **C. Chest CT scan**.

- **Chest CT scan** is essential in sarcoma staging to evaluate for potential **pulmonary metastasis**, which is the most common site for sarcoma spread.
- **Bone scan** (A) is useful for detecting bone metastases but not routine for initial staging of soft tissue sarcomas.
- **MRI abdomen** (B) is not the primary imaging choice for staging sarcomas unless there are indications of abdominal involvement.
- **Right thigh plain X-ray** (D) may be helpful for identifying bone involvement but does not give the comprehensive staging information needed for soft tissue sarcoma.

MCQ 334

Which of the following is the most appropriate biopsy for a limb sarcoma?

A. Incisional
B. Excisional
C. Core needle
D. Fine needle aspiration

Explanation:
The most appropriate biopsy is **C. Core needle**.

- **Core needle biopsy** (C) is the most accurate method for diagnosing sarcomas, as it allows for a large enough tissue sample to assess tumor architecture and grade.
- **Incisional biopsy** (A) involves removing part of the mass and is typically used when the mass is too large to remove entirely or when the diagnosis is uncertain.
- **Excisional biopsy** (B) would be inappropriate as it could lead to the tumor's seeding and also would be premature before confirming the diagnosis.

- **Fine needle aspiration** (D) can be used in some cases but is generally less accurate than core needle biopsy for sarcomas because it doesn't provide enough tissue for histological assessment of tumor type.

MCQ:334

A 36-year-old man presents with a 2-day history of severe right upper quadrant abdominal pain after consuming a fatty meal. The pain radiates to the right shoulder and is associated with nausea and vomiting. On examination, he has a normal temperature, and there is tenderness over the right upper abdomen. Ultrasound shows multiple gallstones, and the gallbladder wall appears normal. What is the most appropriate next step in management?

A. Observation
B. Ursodeoxycholic acid
C. Laparoscopic cholecystectomy
D. Magnetic Resonance Cholangiopancreatography (MRCP)

Explanation:
The patient is presenting with symptoms consistent with **biliary colic** (intermittent pain due to gallstones). The absence of any acute signs (e.g., fever or peritoneal signs) and the fact that the gallbladder wall is normal on ultrasound suggests no acute cholecystitis. In patients with symptomatic gallstones, the definitive treatment is **laparoscopic cholecystectomy** to prevent further episodes.

Cholelithiasis (Gallstones)

Definition:
The presence of gallstones in the gallbladder, often leading to symptoms such as biliary colic, and in some cases, complications like cholecystitis, pancreatitis, or cholangitis.

Key Clinical Presentation:

- Intermittent right upper quadrant pain after meals (especially fatty foods).
- Nausea and vomiting associated with the pain.

- No fever or jaundice in uncomplicated cases.

Investigations:

- Best initial: Abdominal ultrasound (shows gallstones and gallbladder wall).
- Confirmatory: MRCP (for suspected complications such as common bile duct stones).

Treatment Outline:

- First-line: **Laparoscopic cholecystectomy** (definitive management in symptomatic patients).
- Second-line: Observation and management of symptoms if surgery is contraindicated.
- Definitive: Cholecystectomy to prevent recurrent biliary colic or complications.

Correct Answer:
C. Laparoscopic cholecystectomy ✅

MCQ 335:

A 44-year-old man presented with a 6-month history of vague central abdominal pain.
CT scan of abdomen:
Presence of 20x10 cm retroperitoneal mass. No enlarged mesenteric or para-aortic lymph nodes noted.
Which of the following is the most likely diagnosis?

A. Lymphoma
B. Liposarcoma
C. Germ cell tumor
D. Metastatic testicular tumor

Explanation:
The patient has a large retroperitoneal mass without any associated lymphadenopathy. **Liposarcoma**, a malignant tumor of adipose tissue, is the most likely diagnosis in this scenario. It commonly occurs in the retroperitoneum and is often characterized by large, well-defined

masses on imaging. Other options like lymphoma or metastasis typically present with enlarged lymph nodes.

Retroperitoneal Masses

Definition:
A mass located in the retroperitoneal space, which is the anatomical area behind the peritoneum, includes various types of tumors such as liposarcomas, lymphoma, and metastatic cancers.

Key Clinical Presentation:

- Non-specific abdominal pain.
- Large mass, often with vague symptoms or weight loss.
- May be asymptomatic for a long period.

Investigations:

- Best initial: CT scan (provides detailed imaging of the retroperitoneal mass).
- Confirmatory: Biopsy for histopathological confirmation.

Treatment Outline:

- First-line: Surgical resection if feasible.
- Second-line: Chemotherapy and radiotherapy for malignant tumors.
- Definitive: Surgical excision with histological analysis.

Correct Answer:
B. Liposarcoma ✅

MCQ 336

A 38-year-old multiparous woman presented with a 3-year history of abdominal bulging. Physical examination confirmed a diffuse smooth midline bulge on attempting to lean forward, with negative cough impulse (see report).
CT scan of abdomen:

No intra or extra-abdominal mass lesion, and abdominal fascia is intact with no visible defect. Which of the following is the most likely diagnosis?

A. Spigelian hernia
B. Incisional hernia
C. Diastasis recti
D. Transversalis muscle weakness

Explanation:
The patient has a midline bulge with no defect or hernia on imaging. **Diastasis recti**, a condition where the rectus abdominis muscles separate, is the most likely diagnosis. This condition is common in multiparous women and can cause a bulging in the midline, especially when attempting activities that increase intra-abdominal pressure.

Diastasis Recti

Definition:
A condition where the two sides of the rectus abdominis muscle are separated, often resulting in a bulging midline.

Key Clinical Presentation:

- Bulging along the midline of the abdomen.
- No palpable hernia or defect.
- Worsens with activities that increase intra-abdominal pressure (e.g., sitting up, bending forward).

Investigations:

- Best initial: Physical examination (often visible bulge).
- Confirmatory: Imaging, but diagnosis is typically clinical.

Treatment Outline:

- First-line: Physical therapy, core strengthening exercises.
- Second-line: Surgical correction if symptomatic or persistent.

Correct Answer:

C. Diastasis recti ✅

MCQ 337

A 29-year-old man underwent left open inguinal hernia mesh repair 3 weeks ago. He presents today with continuous intractable groin pain radiating to the left upper thigh and associated with persistent hyperesthesia. Physical examination excluded any recurrence or surgical site infection.
Which of the following is the most appropriate treatment?

A. Nerve block
B. Mesh staples removal
C. Systemic anti-inflammatory
D. Neuroctomy and mesh removal

Explanation:
The patient likely has **postoperative neuralgia**, a common complication following inguinal hernia surgery. The most appropriate management is a **nerve block**, which can help alleviate pain without the need for invasive procedures. Removal of mesh staples or nerve excision would be excessive and unnecessary.

Postoperative Complications

Definition:
Postoperative complications following inguinal hernia repair can include pain due to nerve irritation or entrapment, known as postoperative neuralgia.

Key Clinical Presentation:

- Persistent pain following surgery.
- Radiating pain to the thigh.
- Hyperesthesia over the groin area.

Investigations:

- Best initial: Clinical examination.

- Confirmatory: Nerve block or imaging if necessary.

Treatment Outline:

- First-line: Nerve block and conservative pain management.
- Second-line: Surgery if nerve entrapment is confirmed.

Correct Answer:
A. Nerve block ✅

MCQ 338

A 35-year-old man post total thyroidectomy for multinodular goiter presented with persistent hypocalcemia despite adequate replacement with oral and intravenous calcium.
Which of the following is the most appropriate next step in management?

A. Start diuretics
B. Start recombinant PTH
C. Check serum magnesium level
D. IV bolus of potassium chloride

Explanation:
Persistent hypocalcemia after thyroidectomy may be due to **hypomagnesemia**, which impairs the release of parathyroid hormone (PTH). The most appropriate next step is to **check serum magnesium levels** and correct it if necessary, as magnesium deficiency can cause resistance to calcium supplementation.

Hypocalcemia after Thyroidectomy

Definition:
Hypocalcemia following thyroidectomy is often due to accidental damage or removal of the parathyroid glands.

Key Clinical Presentation:

- Symptoms of hypocalcemia (e.g., numbness, tetany, seizures).
- Persistent low calcium despite supplementation.

Investigations:

- Best initial: Serum calcium and magnesium levels.
- Confirmatory: Serum magnesium level if hypocalcemia persists.

Treatment Outline:

- First-line: Magnesium replacement.
- Second-line: Adjust calcium supplementation after magnesium correction.

Correct Answer:
C. Check serum magnesium level ✅

MCQ 339

A 40-year-old woman underwent major bowel resection and developed high-output fistula. Which of the following is the most expected consequence?

A. Fast loss of body fat
B. Decrease total body weight
C. Slow loss of lean body mass
D. Expansion of the extracellular fluid

Explanation:
In the setting of high-output fistula, fluid loss and malnutrition occur primarily due to protein and electrolyte loss, which leads to **decrease in lean body mass**. This occurs over time as the body loses proteins and muscle mass despite caloric intake. Body fat may be lost as well, but the main consequence of a high-output fistula is the loss of lean body mass.

Postoperative Nutrition and Fluid Loss

Definition:
High-output fistula is a complication of gastrointestinal surgery where excessive fluid loss occurs through the fistula, leading to malnutrition and dehydration.

Key Clinical Presentation:

- Persistent abdominal fluid loss.
- Weight loss with signs of malnutrition.
- Electrolyte imbalances and dehydration.

Investigations:

- Best initial: Serum albumin, electrolytes, and fluid balance.
- Confirmatory: Nutritional assessments, stool analysis.

Treatment Outline:

- First-line: Parenteral nutrition, fluid and electrolyte management.
- Second-line: Surgical repair of the fistula if possible.

Correct Answer:
C. Slow loss of lean body mass ✅

MCQ 340

A patient with obstructive jaundice developed septic cholangitis.
Blood pressure 90/50 mmHg
Heart rate 105 /min
Respiratory rate 18 /min
Temperature 39.2 °C
Which of the following is most likely to be present?

A. Hypoglycaemia
B. Increased cardiac output
C. Generalized vasoconstriction
D. Increased peripheral resistance

Explanation:
In septic cholangitis, the body reacts with **generalized vasodilation** rather than vasoconstriction. This leads to low peripheral resistance and shock, which is characteristic of sepsis. The increased heart rate and fever are signs of systemic inflammation.

Septic Cholangitis

Definition:
Infection of the biliary tract due to obstruction, often caused by gallstones, leading to systemic infection.

Key Clinical Presentation:

- Fever, chills, jaundice (Charcot's triad).
- Hypotension and tachycardia.
- Elevated liver enzymes and bilirubin.

Investigations:

- Best initial: Blood cultures, liver function tests.
- Confirmatory: Abdominal ultrasound, cholangiography.

Treatment Outline:

- First-line: IV antibiotics, drainage of the biliary system (ERCP).
- Second-line: Surgical drainage if needed.

Correct Answer:
B. Increased cardiac output due to sepsis)

MCQ 341

A 60-year-old patient who underwent a cardiac bypass presents with central abdominal pain. Examination confirmed high cardiac output and cardiac index with low peripheral vascular resistance.
Blood pressure: 90/60 mmHg
Heart rate: 120 /min
Which of the following is the most likely diagnosis?

A. Septic shock
B. Unstable angina
C. Cardiogenic shock
D. Hypovolemic shock

Explanation:
The combination of low blood pressure, high cardiac output, and low peripheral vascular resistance is characteristic of **septic shock**. The patient has a high cardiac output with low resistance, which is typical of septic shock. Other causes of shock, like cardiogenic or hypovolemic, would present with low cardiac output, not high.

Shock Types

Definition:
Septic shock is characterized by persistent hypotension despite adequate fluid resuscitation, with high cardiac output and low systemic vascular resistance due to infection.

Key Clinical Presentation:

- Hypotension despite fluid administration.
- Tachycardia, fever.
- Evidence of systemic infection (e.g., elevated WBC, blood cultures).

Investigations:

- Best initial: Blood cultures, lactate levels.
- Confirmatory: Imaging or source control procedures.

Treatment Outline:

- First-line: IV fluids, broad-spectrum antibiotics.
- Second-line: Vasopressors for refractory hypotension.

Correct Answer:
A. Septic shock ✅

MCQ 342

A patient in the Intensive Care Unit who is post-major surgery received 15 units of blood. The patient developed bleeding from the incision, nasogastric tube, and vein puncture sites. Which of the following is the most likely cause?

A. Hypocalcaemia
B. Thrombocytopenia
C. Transfusion reaction
D. Von Willebrand disease

Explanation:
Hypocalcaemia is a known complication of massive blood transfusion, particularly due to citrate toxicity (which is used as an anticoagulant in blood bags). The calcium chelation can cause bleeding tendencies and other systemic effects. This is a typical complication after large-volume transfusions.

Transfusion Reactions

Definition:
Massive blood transfusion leading to transfusion-associated hypocalcaemia due to the citrate used as an anticoagulant.

Key Clinical Presentation:

- Bleeding from multiple sites (e.g., incision, tubes).
- Symptoms of hypocalcaemia (e.g., muscle spasms, tetany).

Investigations:

- Best initial: Serum calcium levels.
- Confirmatory: Monitoring of transfusion-related electrolyte disturbances.

Treatment Outline:

- First-line: Calcium supplementation (IV calcium gluconate).
- Second-line: Reducing the volume of transfusions and citrate exposure.

Correct Answer:
A. Hypocalcaemia ✅

MCQ 343

A patient that has been in the Intensive Care Unit for 2 weeks is receiving parenteral nutrition (see results).

Test Result Normal Values

Platelet count: 250 (150-400 x 10^9/L)

APTT: 37 (30-40 sec)

INR: 2 (0.8-1.2)

Total bilirubin: 12 (3.5-16.5 µmol/L)

Which of the following is the most likely cause?

A. Disseminated intravascular coagulopathy
B. Vitamin K deficiency
C. Factor VII deficiency
D. Chronic liver disease

Explanation:
An elevated INR in a patient on parenteral nutrition suggests **vitamin K deficiency**, which is common in patients receiving long-term parenteral nutrition due to lack of fat-soluble vitamins, including vitamin K. This leads to impaired coagulation factor synthesis, which results in an elevated INR.

Vitamin K Deficiency in Parenteral Nutrition

Definition:
Vitamin K deficiency due to prolonged parenteral nutrition without adequate supplementation.

Key Clinical Presentation:

- Prolonged INR, increased bleeding risk.
- Jaundice and abnormal liver function tests may also be present.

Investigations:

- Best initial: Serum vitamin K levels, PT/INR.
- Confirmatory: Clinical improvement after vitamin K supplementation.

Treatment Outline:

- First-line: Administer vitamin K (IV or subcutaneous).
- Second-line: Modify parenteral nutrition formula to include vitamin K.

Correct Answer:
B. Vitamin K deficiency ✅

MCQ 344

An immunocompromised patient presented with an acute painful perineal swelling.
Examination confirmed erythematous fluctuant area with crepitus over it and a foul-smelling discharge.
Which of the following is the most appropriate management?

A. Penicillin G infusion
B. Surgical debridement
C. Aspiration of collection
D. Topical polymycin ointment

Explanation:
The description of a painful, erythematous, fluctuant perineal swelling with crepitus and foul-smelling discharge in an immunocompromised patient suggests **necrotizing fasciitis** or a severe soft tissue infection. **Surgical debridement** is the most appropriate treatment for such infections to remove necrotic tissue and control the infection. Antibiotics like Penicillin G are also used, but surgical intervention is crucial in managing the infection.

Necrotizing Fasciitis (Soft Tissue Infection)

Definition:
Severe, life-threatening soft tissue infection, often caused by mixed bacterial flora, including anaerobes and gram-positive organisms.

Key Clinical Presentation:

- Rapidly progressing pain, swelling, erythema, and crepitus.
- Foul-smelling discharge, systemic signs of sepsis.
- Often occurs in immunocompromised patients.

Investigations:

- Best initial: Clinical diagnosis.

- Confirmatory: Surgical exploration and tissue culture.

Treatment Outline:

- First-line: Surgical debridement, broad-spectrum IV antibiotics.
- Second-line: Hyperbaric oxygen therapy if available.

Correct Answer:
B. Surgical debridement ✅

MCQ 345

Which of the following is the most likely pathogen to infect a polytetrafluoroethylene vascular graft?

A. Klebsiella pneumonia
B. Staphylococcus aureus
C. Pseudomonas aeruginosa
D. Staphylococcus epidermidis

Explanation:
Staphylococcus aureus is the most common pathogen to infect vascular grafts, especially synthetic ones like polytetrafluoroethylene (PTFE). This pathogen is virulent and can lead to early graft infection and complications.

Vascular Graft Infection

Definition:
Infection of a synthetic vascular graft, commonly due to skin flora or hematogenous spread.

Key Clinical Presentation:

- Fever, localized redness, tenderness, and swelling around the graft site.
- Late infection may present with graft dysfunction or abscess formation.

Investigations:

- Best initial: Blood cultures, wound cultures, graft biopsy.

- Confirmatory: Graft imaging or surgical exploration.

Treatment Outline:

- First-line: Antibiotics based on culture sensitivity (usually MRSA coverage).
- Second-line: Surgical removal and replacement of the graft if infected.

Correct Answer:
B. Staphylococcus aureus ✅

MCQ 346

An unstable trauma patient has confirmed multiple liver lacerations.
Which of the following is the most appropriate management?

A. Right hepatectomy
B. Perinepatic packing
C. Right hepatic artery ligation
D. Individual ligation of the bleeding vessels

Explanation:
In trauma patients with **liver lacerations**, **perinepatic packing** is the most appropriate initial management to control bleeding temporarily. If packing fails, further interventions such as ligation of bleeding vessels or hepatectomy may be considered.

Liver Trauma Management

Definition:
Liver lacerations or contusions caused by blunt or penetrating trauma, resulting in hemorrhage.

Key Clinical Presentation:

- Hypotension, tachycardia, abdominal distension.
- Right upper quadrant tenderness, bruising.

Investigations:

- Best initial: FAST ultrasound or CT scan.

- Confirmatory: CT angiography or exploratory laparotomy.

Treatment Outline:

- First-line: Hemodynamic stabilization, perinepatic packing, or selective embolization.
- Second-line: Surgical resection or arterial ligation if necessary.

Correct Answer:
B. Perinepatic packing ✅

MCQ 347

5 days post-appendectomy, a patient developed diffuse abdominal distension, constipation, and absent bowel sounds.
Which of the following is the most appropriate next step in management?

A. Insert a rectal tube
B. Check serum electrolyte levels
C. Re-exploration of the abdomen
D. Abdominal computed tomography

Explanation:
Abdominal CT scan is the most appropriate next step to rule out complications like **intra-abdominal abscess**, bowel obstruction, or other causes of post-operative ileus. Re-exploration may be needed if the CT scan suggests complications.

Postoperative Ileus and Complications

Definition:
Failure of bowel motility after surgery, which can result in bloating, distension, and discomfort.

Key Clinical Presentation:

- Abdominal distension, constipation, and absent bowel sounds after surgery.
- May progress to vomiting or signs of sepsis.

Investigations:

- Best initial: Abdominal CT scan.
- Confirmatory: Abdominal ultrasound, clinical assessment.

Treatment Outline:

- First-line: Conservative management, electrolyte correction.
- Second-line: Re-exploration if signs of peritonitis or abscess formation.

Correct Answer:
D. Abdominal computed tomography ✅

MCQ 348

Which of the following is the most definitive treatment for a post-renal transplant lymphocele?

A. Percutaneous aspiration
B. Observation and follow-up
C. Fenestration into the peritoneal cavity
D. Percutaneous injection of sclerotherapy

Explanation:
Fenestration into the peritoneal cavity is considered the most definitive treatment for post-renal transplant lymphoceles. It allows the lymph to drain into the peritoneal cavity, thereby preventing recurrence.

Renal Transplant Lymphocele

Definition:
A collection of lymph fluid that can form after renal transplantation due to disruption of lymphatic vessels.

Key Clinical Presentation:

- Abdominal or flank swelling.
- May be asymptomatic or present with discomfort or pressure effects.

Investigations:

- Best initial: Ultrasound or CT scan to confirm lymphocele.
- Confirmatory: Contrast studies or lymphangiography.

Treatment Outline:

- First-line: Conservative management with observation.
- Second-line: Percutaneous aspiration if symptomatic.
- Definitive: Surgical fenestration or drainage into the peritoneal cavity.

Correct Answer:
C. Fenestration into the peritoneal cavity ✅

MCQ 349:

Which of the following is most likely to cause small bowel distention?

A. Nitrous oxide
B. Muscle relaxants
C. Depolarizing agents
D. Esophageal intubation

Explanation:
Nitrous oxide is known to cause small bowel distention, especially when used in high amounts during surgery. It can lead to bowel distension by accumulating in the bowel and causing gas expansion.

Small Bowel Distention

Definition:
Accumulation of gas in the small bowel, leading to swelling and distension, often associated with impaired motility or obstruction.

Key Clinical Presentation:

- Abdominal distension, bloating, nausea, vomiting.
- May be associated with surgical or anesthetic interventions.

Investigations:

- Best initial: Abdominal X-ray or CT scan.
- Confirmatory: Abdominal ultrasound or MRI.

Treatment Outline:

- First-line: Conservative management with decompression.
- Second-line: Surgery if underlying obstruction is identified.

Correct Answer:
A. Nitrous oxide ✅

MCQ 350

A hypothyroid patient is scheduled to undergo an elective surgery.
Which of the following is the most appropriate management?

A. Perform surgery then start thyroxine
B. Start thyroxine and proceed with surgery
C. Perform surgery if clinically asymptomatic
D. Delay surgery until the patient is euthyroid

Explanation:
In a hypothyroid patient, it is essential to **start thyroxine** and **achieve euthyroidism** before proceeding with elective surgery to avoid complications such as poor wound healing, impaired metabolism, and cardiovascular instability. If the patient is clinically stable, surgery can proceed after thyroxine administration.

Hypothyroidism in Surgery

Definition:
Underactive thyroid gland, leading to low levels of thyroid hormones, which can affect metabolism, cardiovascular function, and wound healing.

Key Clinical Presentation:

- Fatigue, cold intolerance, constipation, bradycardia, weight gain.

- May have delayed reflexes, skin changes, and coarse hair.

Investigations:

- Best initial: Serum TSH and free T4.
- Confirmatory: Thyroid function tests.

Treatment Outline:

- First-line: Start levothyroxine before surgery to normalize thyroid levels.
- Second-line: Adjust dose based on TSH and clinical response.

Correct Answer:
B. Start thyroxine and proceed with surgery ✅

MCQ 351

A heavy smoker man is going for a major surgical procedure.
Which of the following is the most sensitive predictor of postoperative respiratory complications?

A. Chest X-ray
B. Bronchoscopy
C. Lung function test
D. Arterial blood gases

Explanation:
Lung function tests (such as spirometry and assessment of lung volumes) are the most sensitive predictors of postoperative respiratory complications, especially in smokers. They assess pulmonary function and help identify risks for complications like pneumonia and atelectasis.

Lung Function in Surgical Risk

Definition:
Tests that evaluate the capacity and function of the lungs, helping to assess a patient's risk for respiratory complications.

Key Clinical Presentation:

- Chronic cough, shortness of breath, history of smoking.
- Potential increased risk for postoperative pulmonary complications.

Investigations:

- Best initial: Pulmonary function tests (spirometry, FEV1, etc.).
- Confirmatory: Arterial blood gases if needed.

Treatment Outline:

- First-line: Pulmonary rehabilitation if indicated.
- Second-line: Oxygen therapy and postoperative monitoring.

Correct Answer:
C. Lung function test ✅

MCQ 352

A man presented with diarrhea and persistent peptic ulcer refractory to proton pump inhibitors.
Secretin stimulation test is positive.
Which of the following is the most likely diagnosis?

A. VIPoma
B. Gastrinoma
C. Glucagonoma
D. Carcinoid tumor

Explanation:
A **gastrinoma** is a tumor that secretes excessive amounts of gastrin, leading to Zollinger-Ellison syndrome, characterized by peptic ulcers and diarrhea. The secretin stimulation test is positive in gastrinomas because secretin usually stimulates gastrin release from these tumors.

Gastrinoma (Zollinger-Ellison Syndrome)

Definition:
A gastrin-secreting tumor, most commonly found in the pancreas or duodenum, causing severe peptic ulcers and gastrointestinal symptoms.

Key Clinical Presentation:

- Refractory peptic ulcers, diarrhea, abdominal pain.
- Positive secretin stimulation test.

Investigations:

- Best initial: Serum gastrin level, secretin stimulation test.
- Confirmatory: Abdominal CT scan or endoscopic ultrasound.

Treatment Outline:

- First-line: Proton pump inhibitors for acid suppression.
- Second-line: Surgical resection if the tumor is localized.
- Definitive: Surgical removal of the gastrinoma or hepatic metastases.

Correct Answer:
B. Gastrinoma ✅

MCQ 353

A 50-year-old woman post-modified radical mastectomy, presented with numbness and a burning sensation on the inner aspect of the upper arm.
Which of the following nerves is most likely injured?

A. Thorcodorsal
B. Long thoracic
C. Medial pectoral
D. Intercostobrachial

Explanation:
The **intercostobrachial nerve**, which arises from the second intercostal nerve, provides

sensation to the medial aspect of the upper arm. Injury to this nerve is common after axillary surgery, such as a modified radical mastectomy.

Intercostobrachial Nerve Injury

Definition:
Damage to the intercostobrachial nerve, often due to axillary surgery, leading to sensory loss or pain in the medial upper arm.

Key Clinical Presentation:

- Numbness or burning sensation along the inner aspect of the upper arm.
- Often occurs after axillary lymph node dissection.

Investigations:

- Best initial: Clinical examination.
- Confirmatory: Electromyography (EMG) or nerve conduction studies.

Treatment Outline:

- First-line: Conservative management with analgesics.
- Second-line: Nerve blocks or physical therapy if persistent.

Correct Answer:
D. Intercostobrachial ✅

MCQ 354

Post laparoscopic cholecystectomy, transection of the common bile duct above the cystic duct is confirmed.
Which of the following is the most appropriate treatment?

A. Hepatico-jejunostomy
B. Hepatico-duodenostomy
C. Choledocho-jejunostomy
D. Choledocho-duodenostomy

Explanation:
Hepatico-jejunostomy is the most appropriate treatment for a transected common bile duct, as it creates a new bile duct drainage pathway to the jejunum. This procedure is done to bypass the damaged bile duct and restore bile flow.

Common Bile Duct Injury

Definition:
Injury to the common bile duct during surgery, often during laparoscopic cholecystectomy, leading to bile leakage or obstruction.

Key Clinical Presentation:

- Jaundice, dark urine, pale stools, and elevated liver enzymes.
- Possible biliary peritonitis or sepsis if untreated.

Investigations:

- Best initial: Liver function tests, abdominal ultrasound.
- Confirmatory: MRCP or ERCP.

Treatment Outline:

- First-line: Hepatico-jejunostomy or choledocho-jejunostomy if biliary leakage persists.
- Second-line: Surgical repair or stent placement if indicated.

Correct Answer:
A. Hepatico-jejunostomy ✅

MCQ 355

A 36-year-old woman was undergoing a caesarean section, and just before closure, the surgeon noticed profound bleeding coming from the upper abdomen.
Which of the following is the most likely source of bleeding?

A. Splenic aneurysm
B. Liver hemangioma
C. Mesenteric aneurysm
D. Perforated peptic ulcer

Explanation:
A **splenic aneurysm** is a known source of bleeding that can be encountered in abdominal surgery, especially in cases where the spleen is manipulated or in close proximity to the surgical field. It can present as sudden and severe bleeding, especially in the context of trauma or surgery. Other causes, such as liver hemangioma or mesenteric aneurysm, are less common and would not typically present with this pattern of bleeding during a caesarean section.

Splenic Aneurysm

Definition:
An abnormal dilation of the splenic artery, often asymptomatic, but can cause life-threatening hemorrhage if ruptured.

Key Clinical Presentation:

- Abdominal pain, hypotension, and signs of shock if ruptured.
- Bleeding during surgery, especially involving the upper abdomen.

Investigations:

- Best initial: Abdominal ultrasound.
- Confirmatory: CT angiography or MRI.

Treatment Outline:

- First-line: Surgical repair if ruptured.
- Second-line: Endovascular embolization if appropriate.

Correct Answer:
A. Splenic aneurysm ✅

MCQ 356

Which of the following is the strongest indicator for surgery in Graves' disease?

A. Paediatric patient
B. Presence of eye symptoms

C. Presence of anti-TSH antibodies
D. Large goiter

Explanation:
The **presence of eye symptoms** in Graves' disease is the strongest indicator for surgery, especially if medical management has failed or if symptoms are severe. Eye symptoms such as proptosis, diplopia, and optic nerve compression can significantly affect the patient's quality of life and may require surgical intervention for correction. Large goiter or pediatric age alone are not direct indications for surgery.

Graves' Disease

Definition:
An autoimmune disorder characterized by hyperthyroidism, goiter, and often, eye involvement (Graves' orbitopathy).

Key Clinical Presentation:

- Hyperthyroid symptoms: weight loss, tremor, heat intolerance, palpitations.
- Goiter and exophthalmos (eye bulging), possibly with optic nerve compression.

Investigations:

- Best initial: Serum TSH, T4, T3 levels, and anti-TSH receptor antibodies.
- Confirmatory: Radioactive iodine uptake scan or ultrasound.

Treatment Outline:

- First-line: Antithyroid medications (methimazole, propylthiouracil).
- Second-line: Radioactive iodine therapy.
- Definitive: Surgery (thyroidectomy) if there are significant complications or resistance to therapy.

Correct Answer:
B. Presence of eye symptoms ✅

MCQ 357

A young man received 4 litres of blood in the Emergency Department following a road traffic accident.
Which of the following is the most likely complication?

A. Hypokalaemia
B. Citrate toxicity
C. Hypocalcaemia
D. Hyperalbunemia

Explanation:
Hypocalcaemia is a common complication after massive blood transfusions due to the citrate used as an anticoagulant in the blood. Citrate binds to calcium, lowering the levels in the blood, and can cause symptoms such as tetany, arrhythmias, or muscle cramps. Citrate toxicity can also lead to metabolic alkalosis and symptoms like hypotension.

Blood Transfusion Complications

Definition:
Complications that arise after receiving a blood transfusion, often related to the volume and rapidity of administration.

Key Clinical Presentation:

- Hypocalcemia: Symptoms of muscle cramps, tingling, and seizures.
- Citrate toxicity: Hypotension, arrhythmias, and alkalosis.

Investigations:

- Best initial: Serum calcium levels.
- Confirmatory: Ionized calcium levels for precise measurement.

Treatment Outline:

- First-line: Calcium supplementation (IV calcium gluconate).
- Second-line: Adjust transfusion rate if possible and manage underlying cause.

Correct Answer:

C. Hypocalcaemia ✅

MCQ 358

A young man comes to the Emergency Department after being involved in a road traffic accident. A Focused Assessment with Sonography for Trauma (FAST) was ordered.
Which of the following is the most likely finding?

A. Pneumothorax
B. Vascular injury
C. Subdural hematoma
D. Intraperitoneal free fluid

Explanation:
Intraperitoneal free fluid is the most common finding in trauma patients undergoing a FAST exam. This typically indicates internal bleeding, most commonly from solid organ injuries like the spleen or liver. FAST is used to identify free fluid in the peritoneal cavity, which helps guide further management in trauma cases.

Focused Assessment with Sonography for Trauma (FAST)

Definition:
A rapid ultrasound examination performed in trauma patients to detect free intraperitoneal fluid (usually blood) or signs of organ injury.

Key Clinical Presentation:

- Trauma with hypotension or signs of internal bleeding.
- Abdominal distension, tenderness, or signs of shock.

Investigations:

- Best initial: FAST exam.
- Confirmatory: CT abdomen or diagnostic laparoscopy if necessary.

Treatment Outline:

- First-line: Surgery (laparotomy) or interventional radiology for bleeding control.

- Second-line: Conservative management in stable patients with minor injuries.

Correct Answer:
D. Intraperitoneal free fluid ✅

MCQ 359

A 45-year-old diabetic man presents with a non-healing wound, which he developed after incision and drainage for diabetic foot 2 months ago.
Which of the following is the most likely etiology?

A. Poor blood supply
B. Underlying malignancy
C. Peripheral neuropathy
D. Non-compliance with treatment

Explanation:
Poor blood supply is the most likely cause for a non-healing wound in diabetic patients. Diabetic foot ulcers often occur due to peripheral vascular disease, which impairs blood supply to the extremities, leading to delayed wound healing and increased risk of infection. Peripheral neuropathy also plays a role in trauma but is not the direct cause of non-healing wounds.

Diabetic Foot Ulcer

Definition:
A wound or ulceration of the lower extremities in a diabetic patient, often due to a combination of neuropathy, poor circulation, and infection.

Key Clinical Presentation:

- Non-healing wound, usually on the foot.
- Often associated with infection, gangrene, and ischemia.

Investigations:

- Best initial: Doppler ultrasound for assessing blood flow.
- Confirmatory: Ankle-brachial index (ABI) to assess for peripheral artery disease.

Treatment Outline:

- First-line: Glycemic control, debridement, and revascularization if necessary.
- Second-line: Wound care with topical agents and offloading pressure.
- Definitive: Amputation in severe cases.

Correct Answer:
A. Poor blood supply ✅

MCQ 360

An elderly man is brought to the Emergency Department with an altered level of consciousness after he fell down the stairs.
CT scan: Suggestive of an extradural hematoma.
Which of the following arteries is most likely involved?

A. Basilar
B. Pontine
C. Anterior cerebral
D. Middle meningeal

Explanation:
The **middle meningeal artery** is the most common artery involved in extradural hematomas (EDH). EDH typically results from trauma to the skull that causes a rupture of the middle meningeal artery, leading to a collection of blood between the dura mater and the skull. This can cause rapid neurological deterioration if not treated promptly.

Extradural Hematoma (EDH)

Definition:
A collection of blood between the inner surface of the skull and the dura mater, often resulting from trauma.

Key Clinical Presentation:

- Loss of consciousness followed by a lucid interval.
- Progressive neurological deterioration.

- Can present with symptoms of raised intracranial pressure.

Investigations:

- Best initial: CT scan of the brain.
- Confirmatory: MRI brain.

Treatment Outline:

- First-line: Surgical evacuation of the hematoma.
- Second-line: Medical management for small, stable hematomas.

Correct Answer:
D. Middle meningeal ✅

MCQ 361

A 25-year-old man underwent a thyroidectomy due to multinodular goiter. Post-operatively, he was unable to maintain high-pitched sounds.
Which of the following nerves is most likely injured?

A. Vagus
B. Inferior laryngeal
C. Superior laryngeal
D. Recurrent laryngeal

Explanation:
The **superior laryngeal nerve** is responsible for innervating the cricothyroid muscle, which is important for tensing the vocal cords, particularly for high-pitched sounds. Injury to the superior laryngeal nerve during thyroid surgery can result in hoarseness or an inability to produce high-pitched sounds, which is consistent with the patient's symptoms.

Superior Laryngeal Nerve Injury

Definition:
Damage to the superior laryngeal nerve, often occurring during thyroidectomy or neck surgery.

Key Clinical Presentation:

- Inability to produce high-pitched sounds.
- Hoarseness or weak voice.
- Difficulty with speech in high vocal registers.

Investigations:

- Best initial: Laryngoscopy to assess vocal cord movement.
- Confirmatory: Electromyography of the laryngeal muscles.

Treatment Outline:

- First-line: Voice therapy and rest.
- Second-line: Surgical re-innervation or botulinum toxin injections if symptoms persist.

Correct Answer:
C. Superior laryngeal ✅

MCQ 362

A 42-year-old woman complains of left-sided chest and abdominal pain that is worse on inspiration. She underwent a splenectomy a few days ago. Examination confirms decreased air entry at the left lung base, dullness to percussion, and left upper quadrant tenderness. Temperature 38.6 °C.
Which of the following is the most likely diagnosis?

A. Gastric stasis
B. Subphrenic abscess
C. Lower lobe pneumonia
D. Overwhelming post-splenectomy sepsis

Explanation:
A **subphrenic abscess** is a common complication after splenectomy, especially in the early postoperative period. It can cause pain in the left upper quadrant and referred pain to the chest, along with fever and signs of localized infection. The physical exam findings of dullness to percussion and decreased air entry are consistent with a subphrenic abscess.

Subphrenic Abscess

Definition:
A collection of pus in the space between the diaphragm and the peritoneal organs, commonly occurring post-surgery, including after splenectomy.

Key Clinical Presentation:

- Fever and abdominal pain (especially in the upper left quadrant).
- Respiratory symptoms, such as decreased breath sounds or pleuritic chest pain.
- Positive signs of localized peritonitis.

Investigations:

- Best initial: Abdominal ultrasound or CT scan.
- Confirmatory: Aspiration or drainage for microbiology culture.

Treatment Outline:

- First-line: Percutaneous drainage and antibiotics.
- Second-line: Surgical drainage if percutaneous drainage is unsuccessful.

Correct Answer:
B. Subphrenic abscess ✅

MCQ 363

A 40-year-old diabetic patient with good glycemic control is admitted electively for an inguinal hernia repair.
What is the most accurate American Society of Anesthesiologists (ASA) physical status class?

A. I
B. II
C. III
D. IV

Explanation:
This patient is a **Class II** according to the ASA classification. Class II is assigned to patients with mild systemic disease or well-controlled chronic conditions, such as diabetes, that do not limit the patient's activity. This patient's diabetes is well-controlled, making ASA II the most appropriate classification.

ASA Physical Status Classification

Definition:
A system used by anesthesiologists to assess and communicate the fitness of patients before surgery.

Key Clinical Presentation:

- Class I: Healthy, no systemic disease.
- Class II: Mild systemic disease, well-controlled.
- Class III: Severe systemic disease that limits activity.
- Class IV: Severe systemic disease that is a constant threat to life.

Investigations:

- Best initial: Preoperative evaluation to assess comorbidities.
- Confirmatory: ASA classification is based on clinical judgment.

Treatment Outline:

- First-line: General anesthesia or regional anesthesia, based on patient condition.
- Second-line: Adjust anesthesia technique based on comorbidities.

Correct Answer:
B. II ✅

MCQ 364

A 20-year-old woman with no past medical history is admitted with a perforated appendix and is in septic shock. On arrival, she was intubated. She requires high inotropic support to maintain

minimum recordable blood pressure.
What is the most accurate American Society of Anesthesiologists (ASA) physical status class?

A. II
B. III
C. IV
D. V

Explanation:
This patient would be classified as **ASA Class V** due to the presence of septic shock and the need for high inotropic support to maintain blood pressure. Class V is used for patients who are not expected to survive without the procedure and are in critical condition.

ASA Physical Status Classification

Definition:
A system used to assess the overall health and surgical risk of patients based on systemic health and disease severity.

Key Clinical Presentation:

- Class I: Healthy, no systemic disease.
- Class II: Mild systemic disease, well-controlled.
- Class III: Severe systemic disease that limits activity.
- Class IV: Severe systemic disease that is a constant threat to life.
- Class V: Moribund patient not expected to survive without the surgery.

Investigations:

- Best initial: Preoperative assessment of vital signs and organ function.
- Confirmatory: ASA classification based on clinical condition.

Treatment Outline:

- First-line: Critical care management and surgery to stabilize the patient.
- Second-line: Supportive therapy as needed in the ICU.

Correct Answer:
D. V ✅

MCQ 365

A 40-year-old man presents with an advanced colonic malignancy.
What is the most likely site of metastasis?

A. Lung
B. Liver
C. Thyroid
D. Prostate

Explanation:
The **liver** is the most common site of metastasis for colonic carcinoma. Colorectal cancer commonly spreads to the liver through the portal circulation. The lung is also a common site of metastasis, but liver metastasis is far more common.

Colorectal Cancer Metastasis

Definition:
Cancer originating in the colon or rectum that can spread to distant organs, most commonly to the liver.

Key Clinical Presentation:

- Symptoms of advanced cancer: weight loss, fatigue, rectal bleeding, and abdominal pain.
- Liver dysfunction signs: jaundice, hepatomegaly.

Investigations:

- Best initial: Colonoscopy with biopsy.
- Confirmatory: CT scan for staging and liver metastasis detection.

Treatment Outline:

- First-line: Surgery (resection of primary tumor) + chemotherapy for metastatic disease.
- Second-line: Chemotherapy regimens like FOLFOX or FOLFIRI.

Correct Answer:
B. Liver ✅

MCQ 366

A young man is hypoxic 24 hours after major upper abdominal surgery.
Which of the following is the most likely cause?

A. Pneumonia
B. Basal atelectasis
C. Pulmonary embolism
D. Intra-abdominal bleeding

Explanation:
Basal atelectasis is the most common cause of hypoxia 24 hours after abdominal surgery. It occurs due to shallow breathing, reduced mobility, and the effects of anesthesia, leading to partial lung collapse, especially in the basal regions. It typically presents as hypoxia and can resolve with appropriate respiratory physiotherapy.

Postoperative Pulmonary Complications

Definition:
Respiratory issues that arise following surgery, commonly due to atelectasis, pneumonia, or pulmonary embolism.

Key Clinical Presentation:

- Hypoxia, tachypnea, and respiratory distress.
- Reduced breath sounds over lung bases.
- History of recent abdominal or thoracic surgery.

Investigations:

- Best initial: Chest X-ray.
- Confirmatory: CT scan if concerned about pulmonary embolism or other complications.

Treatment Outline:

- First-line: Oxygen therapy, chest physiotherapy, incentive spirometry.
- Second-line: Positive pressure ventilation or CPAP if severe.

Correct Answer:
B. Basal atelectasis ✅

MCQ 367

Which of the following is the most likely cause of hypoxia 24 hours after major pelvic surgery for ovarian cancer?

A. Pneumonia
B. Pelvic abscess
C. Basal atelectasis
D. Pulmonary embolism

Explanation:
Basal atelectasis is the most common cause of hypoxia in the postoperative period, including following pelvic surgery. This condition arises due to shallow breathing and limited mobility after surgery, leading to collapse of the lower lobes of the lungs. It is usually self-limiting and improves with deep breathing exercises and physiotherapy.

Postoperative Pulmonary Complications

Definition:
Respiratory complications post-pelvic surgery, including atelectasis and pneumonia.

Key Clinical Presentation:

- Tachypnea, hypoxia, and dyspnea 24 hours post-surgery.
- Reduced breath sounds at the lung bases.

Investigations:

- Best initial: Chest X-ray.

- Confirmatory: CT scan if pneumonia or embolism is suspected.

Treatment Outline:

- First-line: Oxygen therapy, chest physiotherapy, and incentive spirometry.
- Second-line: Mechanical ventilation for severe cases.

Correct Answer:
C. Basal atelectasis ✅

MCQ 368

A 30-year-old woman sustained a burn from an oven grill. 8 days later, she noticed greenish discharge in the wound.
Which of the following is the most likely cause?

A. E. coli
B. Klebsiella
C. Pseudomonas
D. Streptococcus pyogenes

Explanation:
Pseudomonas aeruginosa is the most common pathogen associated with burn wound infections, especially when there is greenish discharge. It produces a distinctive greenish pigment (pyocyanin) that can color the exudate. Pseudomonas is a common colonizer in moist, necrotic tissue and can cause serious infections in burn wounds.

Burn Wound Infection

Definition:
Infection of a burn wound, commonly caused by opportunistic pathogens like Pseudomonas aeruginosa.

Key Clinical Presentation:

- Greenish or foul-smelling discharge.
- Pain, redness, and swelling around the wound.

- Fever and signs of systemic infection if severe.

Investigations:

- Best initial: Wound swab for culture.
- Confirmatory: Blood cultures if septicemia is suspected.

Treatment Outline:

- First-line: Topical antibiotics (silver sulfadiazine, polymyxin B).
- Second-line: Systemic antibiotics if the infection is severe.

Correct Answer:
C. Pseudomonas ✅

MCQ 369

A 55-year-old butcher accidentally cut himself at work. Later that night, he noticed some red lines radiating from the wound.
Which of the following is the most likely cause?

A. Brucella
B. Klebsiella
C. Pseudomonas
D. Streptococcus pyogenes

Explanation:
Streptococcus pyogenes is the most likely pathogen in this case, causing **lymphangitis**, which presents as red streaks radiating from a wound. This is a hallmark of **infection with Streptococcus pyogenes** and is often associated with cellulitis or more severe soft tissue infections.

Soft Tissue Infections and Lymphangitis

Definition:
Infections of the skin and soft tissues, often leading to lymphangitis, where the infection spreads along lymphatic channels.

Key Clinical Presentation:

- Red streaks (lymphangitis) radiating from the site of injury.
- Swelling, warmth, and erythema at the wound site.
- Systemic signs of infection (fever).

Investigations:

- Best initial: Wound culture.
- Confirmatory: Blood cultures if septicemia is suspected.

Treatment Outline:

- First-line: Oral or intravenous antibiotics (penicillin or cephalosporins).
- Second-line: Surgery if an abscess or necrotizing fasciitis develops.

Correct Answer:
D. Streptococcus pyogenes ✅

MCQ 370

Where is the most common primary location of basal cell carcinoma?

A. Face
B. Breast
C. Lower limb
D. Upper limb

Explanation:
The face is the most common location for **basal cell carcinoma (BCC)**, particularly the **nasal** area, followed by the **cheeks** and **ears**. BCC typically develops in sun-exposed areas, and the face, being the most exposed, is the most common site for this type of skin cancer.

Basal Cell Carcinoma (BCC)

Definition:
The most common type of skin cancer, originating from the basal cells of the epidermis.

Key Clinical Presentation:

- Painless, slow-growing, pearly nodule.
- May have ulcerated, crusted centers and rolled borders.
- Often seen on sun-exposed areas, especially the face.

Investigations:

- Best initial: Clinical examination and dermoscopy.
- Confirmatory: Biopsy for histopathological examination.

Treatment Outline:

- First-line: Surgical excision or Mohs micrographic surgery.
- Second-line: Cryotherapy, topical treatments, or radiation therapy for non-surgical candidates.

Correct Answer:
A. Face ✅

MCQ 371:

A young man presents with a lump on the right mid-thigh. On examination, the overlying skin is normal, the mass is soft, movable with a positive fluctuation test.
Which of the following is the most likely diagnosis?

A. Lipoma
B. Sarcoma
C. Aneurysm
D. Sebaceous cyst

Explanation:
A **sebaceous cyst** (also known as an epidermoid cyst) is a common diagnosis for a soft, movable, fluctuant mass that is not associated with skin changes. These cysts arise from sebaceous glands or hair follicles and often present with a positive fluctuation test. They are benign and often require simple excision for removal.

Sebaceous Cyst

Definition:
A benign, closed sac under the skin filled with a cheesy, keratinous material, often arising from hair follicles or sebaceous glands.

Key Clinical Presentation:

- Soft, movable lump, commonly on the face, neck, or trunk.
- Positive fluctuation test.
- No overlying skin changes unless infected.

Investigations:

- Best initial: Clinical examination.
- Confirmatory: Aspiration or biopsy in atypical cases.

Treatment Outline:

- First-line: Surgical excision.
- Second-line: Incision and drainage if infected.

Correct Answer:
D. Sebaceous cyst ✅

MCQ 372

A 40-year-old is admitted with a 3-day history of diffuse abdominal pain. On examination, there is diffuse abdominal tenderness along with **Grey Turner's sign**.
Which of the following is the most likely diagnosis?

A. Acute porphyria
B. Mesenteric ischemia
C. Necrotizing pancreatitis
D. Ruptured aortic aneurysm

Explanation:
Grey Turner's sign, which refers to bruising of the flanks, is a classic sign of **necrotizing pancreatitis**, which often presents with severe abdominal pain, tenderness, and systemic signs of sepsis. This sign occurs due to retroperitoneal hemorrhage associated with severe pancreatic inflammation.

Necrotizing Pancreatitis

Definition:
A severe form of pancreatitis characterized by pancreatic necrosis and possible retroperitoneal bleeding.

Key Clinical Presentation:

- Severe abdominal pain radiating to the back.
- Abdominal tenderness and guarding.
- Signs of systemic shock and sepsis.
- **Grey Turner's sign** (flank bruising) and **Cullen's sign** (periumbilical bruising) in severe cases.

Investigations:

- Best initial: Serum amylase and lipase.
- Confirmatory: Contrast-enhanced CT of the abdomen.

Treatment Outline:

- First-line: Supportive care, fluids, pain management.
- Second-line: Surgical debridement if necrosis is extensive.

Correct Answer:
C. Necrotizing pancreatitis ✅

MCQ 373:

An elderly man presents for follow-up after he sustained a road traffic accident 3 days ago. On examination, **"Battle Sign"** is noted.
Which of the following is the most likely cause?

A. Le Forte fracture
B. Mandibular fracture
C. Fracture base of skull
D. Fracture middle cranial fossa

Explanation:
Battle Sign (bruising over the mastoid process) is a classic sign of **a basilar skull fracture**, particularly from trauma to the temporal bone, which may lead to bleeding in the middle ear and bruising behind the ear.

Basilar Skull Fracture

Definition:
Fracture of the bones at the base of the skull, often associated with significant trauma.

Key Clinical Presentation:

- Battle's sign (bruising behind the ear).
- Raccoon eyes (periorbital bruising).
- Cerebrospinal fluid (CSF) leak from the ears or nose.

Investigations:

- Best initial: CT scan of the head.
- Confirmatory: MRI for further detail of injury.

Treatment Outline:

- First-line: Conservative management and monitoring.
- Second-line: Surgical intervention if there is associated brain injury or large CSF leak.

Correct Answer:

C. Fracture base of skull ✅

MCQ 374

A 50-year-old diabetic man presents with a non-healing wound, which he developed after incision and drainage for a diabetic foot ulcer 2 months ago.
Which of the following is the most likely etiology?

A. Poor blood supply
B. Underlying malignancy
C. Peripheral neuropathy
D. Non-compliance with treatment

Explanation:
The most common etiology for a non-healing diabetic foot ulcer is **poor blood supply**, which is a result of peripheral arterial disease in diabetic patients. This impairs wound healing and increases the risk of infection.

Diabetic Foot Ulcers

Definition:
Chronic ulcers on the feet of diabetic patients, often due to poor circulation and neuropathy.

Key Clinical Presentation:

- Non-healing ulcers, usually on the plantar aspect of the feet.
- Can be associated with neuropathy, poor sensation, and ischemia.

Investigations:

- Best initial: Ankle-brachial index (ABI) for peripheral vascular disease.
- Confirmatory: Doppler ultrasound or angiography for vascular assessment.

Treatment Outline:

- First-line: Proper diabetic management, debridement, and off-loading.
- Second-line: Revascularization or surgical intervention if necessary.

Correct Answer:

A. Poor blood supply ✅

MCQ 375

A 20-year-old man was hit by a car 2 hours ago. On examination, his abdomen is distended, tense, and tender.
Blood pressure 100/70 mmHg
Heart rate 120 /min
Respiratory rate 30 /min
Temperature 35 °C
Oxygen saturation 90 %
What type of shock is the most likely diagnosis?

A. Spinal
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:
The patient's hypotension, tachycardia, and abdominal distention after trauma suggest **hemorrhagic shock**, likely due to internal bleeding, such as from solid organ injury or retroperitoneal hemorrhage.

Hemorrhagic Shock

Definition:
A type of shock resulting from significant blood loss, leading to inadequate tissue perfusion.

Key Clinical Presentation:

- Hypotension, tachycardia, and tachypnea.
- Cold extremities and delayed capillary refill.
- Abdominal distention and tenderness, particularly after trauma.

Investigations:

- Best initial: FAST (focused assessment with sonography for trauma) or CT scan of the abdomen.
- Confirmatory: Blood tests (e.g., hemoglobin and hematocrit) and crossmatch.

Treatment Outline:

- First-line: IV fluids, blood transfusion, and surgery if necessary.
- Second-line: Vasopressors if fluid resuscitation is insufficient.

Correct Answer:
C. Haemorrhagic ✅

MCQ 376:

A 30-year-old man was hit by a truck 4 hours ago. On examination, he has raised jugular venous pressure and diminished heart sounds.
Blood pressure 90/60 mmHg
Heart rate 140 /min
Respiratory rate 32 /min
Temperature 36.6 °C
Oxygen saturation 85 %

What type of shock is the most likely diagnosis?

A. Septic
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:
This patient presents with **raised jugular venous pressure**, **diminished heart sounds**, **hypotension**, and **tachycardia**, which are suggestive of **cardiac tamponade** due to trauma. **Cardiogenic shock** in this context is most likely due to tamponade, where blood accumulates in the pericardial sac, preventing the heart from filling properly, leading to poor cardiac output.

Cardiogenic Shock

Definition:
A state of shock due to inadequate cardiac output, typically caused by heart failure, but in this case, likely due to **cardiac tamponade** from trauma.

Key Clinical Presentation:

- Hypotension, tachycardia, and tachypnea.
- Raised jugular venous pressure.
- Diminished heart sounds (pericardial effusion).
- Pulsus paradoxus (decrease in blood pressure during inspiration).

Investigations:

- Best initial: Clinical examination, FAST (Focused Assessment with Sonography for Trauma).
- Confirmatory: Echocardiogram showing pericardial effusion.

Treatment Outline:

- First-line: Pericardiocentesis to relieve pressure on the heart.
- Second-line: Surgical intervention if required for severe cases.

Correct Answer:
B. Cardiogenic ✅

MCQ 377

A 20-year-old patient complains of painless fresh bleeding post defecation over the last 6 months.
Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal fistula
D. Perianal abscess

Explanation:
Haemorrhoids are the most likely diagnosis in this patient. They commonly present with **painless fresh bleeding**, often seen during or after defecation. In contrast, **anal fissures** usually present with **painful bleeding**, and **perianal abscesses** or **fistulas** are typically associated with pain and discharge.

Haemorrhoids

Definition:
Swollen veins in the lower rectum and anus, often caused by straining during bowel movements, pregnancy, or prolonged sitting.

Key Clinical Presentation:

- Painless rectal bleeding, often noticed on toilet paper.
- Lump or swelling near the anus.
- Possible itching or discomfort.

Investigations:

- Best initial: Clinical examination.
- Confirmatory: Anoscopy or proctoscopy.

Treatment Outline:

- First-line: Conservative treatment (dietary fiber, topical agents).
- Second-line: Rubber band ligation, sclerotherapy.
- Definitive: Surgical excision in severe cases.

Correct Answer:
B. Haemorrhoids ✅

MCQ 378

A 40-year-old diabetic patient complains of a painful perianal swelling with fever for the last 3 days.
Blood pressure 110/70 mmHg

Heart rate 110 /min
Respiratory rate 20 /min
Temperature 38 °C

Test Result:

- Hb 135 (130-170 g/L)
- WBC 16.4 (4.5-10.5 x 10^9/L)
- Neutrophils 80% (40-60%)
- Random Glucose 6.5 (3.9-5.5 mmol/L)

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:
This patient with **fever**, **painful perianal swelling**, and **elevated WBC count** likely has a **perianal abscess**. The presence of **increased neutrophils** and systemic signs of infection support this diagnosis. An **anal fissure** is typically not associated with fever, and **haemorrhoids** would not present with an abscess-like swelling or systemic symptoms.

Perianal Abscess

Definition:
A collection of pus in the tissues around the anus, usually due to infection of the anal glands.

Key Clinical Presentation:

- Painful, swollen area near the anus.
- Fever, chills, and erythema in the perianal area.
- Possible drainage of pus.

Investigations:

- Best initial: Clinical examination.

- Confirmatory: Ultrasound or MRI if deeper abscess is suspected.

Treatment Outline:

- First-line: Incision and drainage.
- Second-line: Antibiotics if infection is extensive or systemic symptoms are present.

Correct Answer:
C. Perianal abscess ✅

MCQ 379

A 50-year-old patient underwent a total thyroidectomy 5 hours ago. He has developed a large neck swelling associated with stridor and shortness of breath.
Blood pressure 165/90 mmHg
Heart rate 130 /min
Respiratory rate 24 /min
Temperature 37 °C

Which of the following is the most appropriate next step in management?

A. Tracheostomy
B. Close observation
C. Percutaneous drainage
D. Bedside wound exploration

Explanation:
The patient most likely has a **post-thyroidectomy hematoma**, which can compress the airway, causing **stridor** and **respiratory distress**. The most appropriate next step is **bedside wound exploration** to evacuate the hematoma and relieve the airway obstruction.

Post-thyroidectomy hematoma

Definition:
A collection of blood in the surgical site following thyroidectomy, which can compress the airway.

Key Clinical Presentation:

- Neck swelling, pain, and difficulty breathing.
- Stridor and respiratory distress.
- Symptoms occur within hours of surgery.

Investigations:

- Best initial: Clinical examination and urgent bedside exploration.
- Confirmatory: CT scan of the neck if needed.

Treatment Outline:

- First-line: Immediate wound exploration to evacuate the hematoma.
- Second-line: Tracheostomy if the airway cannot be secured.

Correct Answer:
D. Bedside wound exploration ✅

MCQ 380

A 60-year-old man had a laparoscopic inguinal hernia repair 5 days ago. Today, he noticed a swelling at the wound area. Examination confirmed a swelling at the site of surgery that was not red, or hot, but mildly tender. There was no cough impulse.
Blood pressure 135/80 mmHg
Heart rate 80 /min
Respiratory rate 24 /min
Temperature 37.2 °C

Which of the following is the most likely diagnosis?

A. Seroma
B. Hematoma
C. Recurrence
D. Wound infection

Explanation:
The patient is presenting with **mild tenderness**, **no redness**, and **no cough impulse**. These features are more suggestive of a **seroma**, which is a collection of fluid that commonly occurs post-operatively. **Hematoma** typically presents with bruising and more significant tenderness. **Recurrence** would typically present with a cough impulse and would appear later. **Wound infection** would be associated with redness, heat, and increased tenderness, which this patient does not show.

Seroma

Definition:
A seroma is a collection of serous fluid within a tissue cavity, typically seen after surgery.

Key Clinical Presentation:

- Painless or mildly tender swelling.
- No redness, warmth, or signs of infection.
- Common after surgeries involving tissue dissection, such as hernia repair.

Investigations:

- Best initial: Clinical examination.
- Confirmatory: Ultrasound can be used to assess fluid collection.

Treatment Outline:

- First-line: Conservative management (e.g., compression dressings, observation).
- Second-line: Aspiration or drainage if symptomatic.
- Definitive: Rarely needed; usually resolves with conservative measures.

Correct Answer:

A. Seroma ✅

MCQ 381

A 32-year-old man underwent a hemorrhoidectomy for a prolapsed pile. 5 hours later, he complained of severe supra-pubic pain and was unable to pass urine.
Which of the following is the most likely cause?

A. Dehydration
B. Urethral injury
C. Inadequate analgesia
D. Post-operative hematoma

Explanation:
Inadequate analgesia is the most likely cause in this patient who is experiencing **severe supra-pubic pain** and **urinary retention** shortly after the procedure. **Urethral injury** is uncommon in hemorrhoidectomy and typically presents with blood at the urethral meatus. **Dehydration** and **post-operative hematoma** may contribute to discomfort but would not be as likely to cause urinary retention directly after surgery.

Inadequate Analgesia

Definition:
Inadequate pain control leading to post-operative discomfort and possible complications such as urinary retention.

Key Clinical Presentation:

- Severe pain following surgery.
- Urinary retention due to bladder discomfort and muscle spasms.

Investigations:

- Best initial: Clinical evaluation, voiding trial.

Treatment Outline:

- First-line: Adequate analgesia (e.g., opioids, nerve blocks).
- Second-line: Catheterization if retention persists.
- Definitive: Pain management during recovery.

Correct Answer:

C. Inadequate analgesia ✅

MCQ 382

A 52-year-old woman presented with a recent rapid enlargement of a neck swelling associated with weight loss. She was diagnosed 25 years ago with Hashimoto's thyroiditis.
Which of the following is the most likely diagnosis?

A. Hemorrhagic goiter
B. Thyroid lymphoma
C. Papillary carcinoma
D. Sub-acute thyroiditis

Explanation:
Given the patient's history of **Hashimoto's thyroiditis** and **rapid enlargement** of the neck swelling, the most likely diagnosis is **thyroid lymphoma**. This can occur as a complication of Hashimoto's thyroiditis, with rapid enlargement of the thyroid. **Hemorrhagic goiter** typically presents with a painful, enlarged thyroid but is less likely to cause weight loss. **Papillary carcinoma** presents with a firm, non-tender mass, and **sub-acute thyroiditis** would be painful and associated with thyroid dysfunction.

Thyroid Lymphoma

Definition:
A rare malignancy of the thyroid, often associated with Hashimoto's thyroiditis. It presents as rapidly enlarging, non-tender thyroid masses.

Key Clinical Presentation:

- Rapidly enlarging neck mass.
- History of Hashimoto's thyroiditis.
- Weight loss and systemic symptoms.

Investigations:

- Best initial: Fine needle aspiration biopsy.

- Confirmatory: Imaging (CT or ultrasound) and biopsy.

Treatment Outline:

- First-line: Chemotherapy and/or radiation therapy.
- Second-line: Surgical intervention if possible.

Correct Answer:
B. Thyroid lymphoma ✅

MCQ 383

A 56-year-old woman presented with generalized muscle and bone ache associated with abdominal cramps that is relieved by drinking cold milk (see lab results).
Test Result Normal Values

- Calcium: 2.85 (2.15-2.62 mmol/L)
- Calcium ionized: 1.9 (1.1-1.3 mmol/L)
- Phosphate, inorganic: 0.71 (0.82-1.51 mmol/L)
- Parathyroid hormone (intact PTH levels): 5.5 (1.1-5.3 pmol/L)
- 25-Hydroxy Vitamin D3 (Women): 15 (25-90 nmol/L)

Which of the following is the most likely diagnosis?

A. Milk alkali syndrome
B. Idiopathic hypercalcemia
C. Primary hyperparathyroidism
D. Secondary hyperparathyroidism

Explanation:
The patient's symptoms of **bone ache**, **muscle pain**, and **abdominal cramps** relieved by drinking milk, combined with **elevated calcium levels**, point to **milk alkali syndrome**. This syndrome is caused by excessive calcium and alkali intake, leading to hypercalcemia, suppressed parathyroid hormone (PTH) levels, and low phosphate. **Primary hyperparathyroidism** and **secondary hyperparathyroidism** would typically show elevated PTH levels.

Milk Alkali Syndrome

Definition:
A condition caused by excessive intake of calcium and alkali, leading to hypercalcemia, metabolic alkalosis, and renal impairment.

Key Clinical Presentation:

- Hypercalcemia.
- Abdominal pain, bone pain, muscle ache.
- Symptoms relieved by milk.

Investigations:

- Best initial: Serum calcium, phosphate, PTH, and vitamin D levels.
- Confirmatory: History of excessive calcium and alkali intake.

Treatment Outline:

- First-line: Discontinuation of calcium and alkali intake.
- Second-line: Hydration and loop diuretics if renal function is impaired.

Correct Answer:
A. Milk alkali syndrome ✅

MCQ 384:

A 36-year-old man sustained a knife stab to the left side of his neck, below the cricoid cartilage (Zone I). Examination confirmed diffuse neck subcutaneous emphysema.
Blood pressure: 110/70 mmHg
Heart rate: 90 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C

Which of the following is the most appropriate next step in management?

A. Neck exploration
B. Close observation

C. CT of the neck and chest
D. Interventional radiology embolization

Explanation:
The presence of **subcutaneous emphysema** after a **knife wound** in Zone I (the area below the cricoid cartilage) requires immediate **imaging** to assess for any involvement of the **airway**, **vascular structures**, or **esophagus**. The **most appropriate next step** is to perform a **CT scan of the neck and chest** to assess the extent of injury and rule out major vascular or esophageal involvement.
Neck exploration may be indicated later, but initial imaging is critical. Observation is inappropriate given the potential severity of neck trauma, and **embolization** is unnecessary without confirmation of vascular injury.

Neck Trauma

Definition:
Injury to the neck structures, which can involve the airway, vascular structures, and esophagus. Subcutaneous emphysema is often a sign of air leakage from these structures following trauma.

Key Clinical Presentation:

- Stab wound or blunt trauma to the neck.
- Subcutaneous emphysema.
- Pain, difficulty swallowing, or respiratory distress in severe cases.

Investigations:

- Best initial: CT of the neck and chest.
- Confirmatory: Imaging to evaluate for vascular, tracheal, or esophageal injury.

Treatment Outline:

- First-line: Imaging (CT scan) to assess the extent of the injury.
- Second-line: Surgical exploration and repair if necessary.
- Definitive: Based on findings, repair of the damaged structures (vessels, trachea, esophagus).

Correct Answer:

C. CT of the neck and chest ✅

MCQ 385

A 57-year-old hypertensive complained of non-specific lower abdominal pain. Hormonal studies for adrenal glands are within normal values. A CT scan was ordered (see report).
CT scan:
5 cm rounded hypodense left adrenal mass, described as lipid-rich lesion.

Which of the following is the most appropriate management?

A. Adrenal MRI
B. No intervention is indicated
C. Laparoscopic adrenalectomy
D. Image-guided adrenal biopsy

Explanation:
The **lipid-rich** nature of the adrenal mass on CT scan is most consistent with a **benign adrenal adenoma**, which is common in patients with no significant symptoms and normal hormonal studies. In the absence of any other alarming features (such as functional activity or malignancy), **no intervention** is usually required.
Adrenal MRI is often used for further characterization, but **benign adenomas** do not require surgery unless symptomatic or functional. **Laparoscopic adrenalectomy** is indicated for symptomatic or suspicious masses, while **image-guided biopsy** is not typically performed unless there is suspicion of malignancy.

Adrenal Adenoma

Definition:
A benign tumor of the adrenal gland, often asymptomatic, and typically diagnosed incidentally on imaging.

Key Clinical Presentation:

- Often asymptomatic.
- May present with symptoms related to hormone excess (e.g., Cushing's syndrome or pheochromocytoma).

Investigations:

- Best initial: CT scan or MRI for further characterization.
- Confirmatory: Hormonal evaluation (if functional).

Treatment Outline:

- First-line: Observation for non-functional adenomas.
- Second-line: Surgery (adrenalectomy) if symptomatic or if the mass is suspected to be malignant.
- Definitive: Surgical removal for functional tumors or masses with suspicious characteristics.

Correct Answer:
B. No intervention is indicated ✅

MCQ 386

A 37-year-old patient presented to the Emergency Department with a 2-day history of right upper quadrant abdominal pain associated with vomiting. Physical examination confirmed rebound tenderness at right hypochondrium (see lab result and report).
Blood pressure: 125/70 mmHg
Heart rate: 90 /min
Respiratory rate: 20 /min
Temperature: 38.8 °C
Test Result Normal Values
WBC: 16.5 (4.5-10.5 x 10^9/L)
Ultrasound abdomen:
Edematous, thick gallbladder wall, pericholecystic fluid, and multiple gallstones.

Which of the following is the most appropriate management?

A. Laparoscopic cholecystectomy
B. Percutaneous cholecystostomy drain
C. Exploratory laparotomy with cholecystectomy
D. Endoscopic retrograde cholangiopancreatography

Explanation:
The patient is presenting with **right upper quadrant pain**, **fever**, and **rebound tenderness** with ultrasound findings of **gallbladder wall thickening** and **gallstones**, consistent with **acute cholecystitis**. The appropriate management is **laparoscopic cholecystectomy**, which is the gold standard treatment for this condition. In certain patients (e.g., high surgical risk), **percutaneous cholecystostomy** can be performed as an initial approach. **Exploratory laparotomy** is not necessary unless there are signs of more extensive peritonitis. **Endoscopic retrograde cholangiopancreatography (ERCP)** is indicated if there is concern for **biliary duct obstruction**.

Acute Cholecystitis

Definition:
Acute inflammation of the gallbladder, usually due to gallstones, leading to obstruction of the cystic duct.

Key Clinical Presentation:

- Right upper quadrant pain.
- Fever.
- Positive Murphy's sign (tenderness over the gallbladder).

Investigations:

- Best initial: Ultrasound of the abdomen.
- Confirmatory: HIDA scan if ultrasound is inconclusive.

Treatment Outline:

- First-line: Laparoscopic cholecystectomy within 24-72 hours of diagnosis.
- Second-line: Percutaneous cholecystostomy in high-risk surgical patients.
- Definitive: Surgical removal of the gallbladder.

Correct Answer:
A. Laparoscopic cholecystectomy ✅

MCQ 387

A 65-year-old man presented with a 3-month history of right upper quadrant abdominal pain, associated with a loss of appetite and weight loss. He has hepatitis C (see lab result and report).
Test Result Normal Value
Alpha fetoprotein: 1160 (0-40 ng/mL)
CT scan abdomen:
Hypervascular lesion at the right lobe of the liver.

Which of the following is the most likely diagnosis?

A. Hamartoma
B. Cholangiocarcinoma
C. Hepatocellular adenoma
D. Hepatocellular carcinoma

Explanation:
The patient has **hepatitis C**, **weight loss**, **right upper quadrant pain**, and an **elevated alpha-fetoprotein (AFP)** level. These findings are most suggestive of **hepatocellular carcinoma (HCC)**, a common complication of chronic liver disease, particularly in patients with cirrhosis or hepatitis C. **Hamartomas** are benign and not typically associated with elevated AFP. **Cholangiocarcinoma** may present with similar symptoms but does not typically have elevated AFP. **Hepatocellular adenoma** is usually asymptomatic and does not present with such a high AFP level.

Hepatocellular Carcinoma (HCC)

Definition:
Primary malignant tumor of the liver, commonly associated with cirrhosis and chronic liver disease (e.g., hepatitis B, C).

Key Clinical Presentation:

- Right upper quadrant pain.
- Weight loss.
- Jaundice.
- Elevated AFP.

Investigations:

- Best initial: Ultrasound of the abdomen.
- Confirmatory: CT scan or MRI of the liver.
- Diagnostic: Biopsy (if uncertain).

Treatment Outline:

- First-line: Surgical resection or liver transplant if feasible.
- Second-line: Ablation therapies (radiofrequency, transarterial chemoembolization).
- Definitive: Transplantation for patients with cirrhosis and HCC.

Correct Answer:
D. Hepatocellular carcinoma ✅

MCQ 388

A 37-year-old man presented to the Emergency Department with a 10-day history of frequent vomiting. He was dehydrated, and his electrocardiogram showed flattening of the T-wave. Which of the following is the most likely acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
The patient's prolonged vomiting is most likely to result in **metabolic alkalosis**, which is characterized by **loss of gastric acid**. Vomiting leads to the loss of hydrogen ions, increasing the pH of the body. The **flattening of the T-wave** on ECG is indicative of **hypokalemia**, which is a common finding in metabolic alkalosis due to potassium shift into cells in response to the alkalosis.
Metabolic alkalosis is the most likely acid-base disturbance in this patient.

Acid-Base Disorders

Definition:
Metabolic alkalosis is an increase in blood pH due to excessive loss of hydrogen ions or excess of bicarbonate.

Key Clinical Presentation:

- Vomiting
- Dehydration
- Hypokalemia
- Flattening of T-waves on ECG

Investigations:

- Best initial: Serum electrolyte levels (especially potassium and bicarbonate).
- Confirmatory: Arterial blood gas.

Treatment Outline:

- First-line: Correction of underlying cause (e.g., fluid replacement with potassium).
- Second-line: Acidifying agents in severe cases.

Correct Answer:
B. Metabolic alkalosis ✅

MCQ 389

A 32-year-old man presented with a 2-week history of frequent attacks of watery diarrhea. On examination, he was severely dehydrated.
Blood pressure: 90/45 mmHg
Heart rate: 130 /min
Respiratory rate: 18 /min
Temperature: 37.2 °C
Which of the following is the most likely acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
Severe dehydration and frequent watery diarrhea typically lead to **metabolic acidosis** due to the loss of bicarbonate in stool. The body compensates for this acidosis by increasing the respiratory rate to eliminate CO_2. **Hypovolemia** and **electrolyte imbalances** are also common findings. This patient's clinical presentation is consistent with **metabolic acidosis**.

Acid-Base Disorders

Definition:
Metabolic acidosis is a condition in which there is a decrease in pH due to excessive accumulation of acid or loss of bicarbonate.

Key Clinical Presentation:

- Diarrhea
- Dehydration
- Tachycardia and hypotension

Investigations:

- Best initial: Serum electrolytes and arterial blood gas.
- Confirmatory: Urine analysis for anion gap.

Treatment Outline:

- First-line: Fluid resuscitation and correction of electrolyte imbalances.
- Second-line: Sodium bicarbonate in severe cases.

Correct Answer:
A. Metabolic acidosis ✅

MCQ 390

A 43-year-old woman who recovered from acute pancreatitis 5 weeks ago, presented with epigastric pain and postprandial vomiting. Physical examination confirmed tenderness and fullness in the epigastric region.

Blood pressure: 127/70 mmHg
Heart rate: 70 /min
Respiratory rate: 16 /min
Temperature: 36.6 °C
Test Result Normal Values
WBC: 6.4 (4.5-10.5 x 10^9/L)
Amylase: 430 (24-151 IU/L)
CT scan of abdomen:
Large cystic collection abutting the posterior wall of the stomach.

Which of the following is the most likely pancreatic pathology?

A. Cancer
B. Abscess
C. Necrosis
D. Pseudocyst

Explanation:
The patient has a history of **acute pancreatitis** and now presents with a **cystic collection** seen on CT scan, which is suggestive of a **pseudocyst**. Pancreatic pseudocysts are common sequelae of pancreatitis, occurring weeks to months after the initial episode. They typically contain pancreatic fluid, and the absence of infection or necrosis suggests a benign pseudocyst rather than abscess or necrosis.

Pancreatic Disorders

Definition:
A pseudocyst is a collection of fluid rich in pancreatic enzymes, surrounded by a fibrous capsule, often resulting from pancreatitis.

Key Clinical Presentation:

- Epigastric pain
- Vomiting
- Abdominal fullness
- History of pancreatitis

Investigations:

- Best initial: CT scan of the abdomen.
- Confirmatory: Ultrasound or endoscopic ultrasound if needed.

Treatment Outline:

- First-line: Observation for asymptomatic pseudocysts.
- Second-line: Drainage (percutaneous or surgical) if symptomatic.
- Definitive: Surgical drainage or resection if complications arise.

Correct Answer:
D. Pseudocyst ✅

MCQ 391

A 50-year-old patient underwent a total thyroidectomy 5 hours ago. He has developed a large neck swelling associated with stridor and shortness of breath.
Blood pressure: 165/90 mmHg
Heart rate: 130 /min
Respiratory rate: 24 /min
Temperature: 37 °C

Which of the following is the most appropriate next step in management?

A. Tracheostomy
B. Close observation
C. Percutaneous drainage
D. Bedside wound exploration

Explanation:
The patient has developed a **neck swelling** associated with **stridor** and **shortness of breath** after thyroid surgery, which is concerning for **postoperative hematoma** compressing the airway. **Immediate exploration** of the surgical site is needed to evacuate the hematoma and relieve airway obstruction. **Tracheostomy** is not required unless there is failure to secure the airway. Close observation is insufficient given the symptoms of airway compromise.

Thyroid Surgery Complications

Definition:
Post-thyroidectomy hematoma can occur as a complication, causing compression of the airway and leading to respiratory distress.

Key Clinical Presentation:

- Neck swelling
- Stridor
- Difficulty breathing after thyroid surgery

Investigations:

- Best initial: Clinical evaluation and immediate examination of the surgical site.
- Confirmatory: No imaging is needed if symptoms are suggestive.

Treatment Outline:

- First-line: Immediate exploration and evacuation of the hematoma.
- Second-line: Airway management (intubation, tracheostomy if needed).

Correct Answer:
D. Bedside wound exploration ✅

MCQ 392

Which of the following is most commonly cured by having a splenectomy?

A. β-thalassemia
B. a-thalassemia
C. Sickle cell trait
D. Idiopathic thrombocytopenia purpura

Explanation:
Splenectomy is most commonly performed for **idiopathic thrombocytopenia purpura (ITP)**, as it helps improve platelet count by removing the spleen, which is responsible for platelet destruction in this condition. It is not typically used for thalassemia or sickle cell trait.

Hematologic Disorders

Definition:
Idiopathic thrombocytopenic purpura (ITP) is an autoimmune disorder characterized by low platelet count due to immune-mediated destruction in the spleen.

Key Clinical Presentation:

- Bleeding tendency
- Easy bruising
- Petechiae

Investigations:

- Best initial: Blood counts and peripheral smear.
- Confirmatory: Clinical diagnosis after ruling out other causes of thrombocytopenia.

Treatment Outline:

- First-line: Steroids, immunoglobulin.
- Second-line: Splenectomy for refractory cases.

Correct Answer:
D. Idiopathic thrombocytopenia purpura ✅

MCQ 393

A 25-year-old man is undergoing appendectomy for acute appendicitis. Upon entering the peritoneal cavity, the surgeon could not find the appendix.
Which of the following is the most effective way to localize the appendix?

A. Following the terminal ileum
B. Palpating the ileocaecal valve
C. Looking at the confluence of the taenia coli
D. Following the course of the right colic artery

Explanation:
The **most effective way to localize the appendix** is by **following the terminal ileum**. The appendix is typically located at the **junction of the ileum and cecum**, so following the terminal ileum allows for easier identification.

Abdominal Surgery

Definition:
Appendectomy is a surgical procedure to remove the appendix, typically performed in cases of appendicitis.

Key Clinical Presentation:

- Right lower quadrant pain
- Fever
- Nausea and vomiting

Investigations:

- Best initial: Clinical examination and ultrasound or CT scan.
- Confirmatory: Exploration during surgery.

Treatment Outline:

- First-line: Appendectomy (laparoscopic or open).
- Second-line: Conservative management with antibiotics for uncomplicated cases.

Correct Answer:
A. Following the terminal ileum ✅

MCQ 394

A 29-year-old woman underwent a left lower parathyroidectomy for primary hyperparathyroidism. Four months later, she presented with generalized fatigue, muscle and bone ache, and constipation (see lab results).
Which of the following is the most likely cause?

A. New adenoma
B. Missed adenoma
C. Parathyroid cancer
D. Parathyroid hyperplasia

Explanation:
The most likely cause in this patient is a **missed adenoma**. Parathyroidectomy is usually performed to remove an adenoma causing primary hyperparathyroidism, but sometimes the adenoma is missed during the initial surgery, leading to persistent symptoms. **Parathyroid hyperplasia** is a less common cause.

Endocrine Disorders

Definition:
Primary hyperparathyroidism is caused by excessive parathyroid hormone (PTH) secretion, often due to a parathyroid adenoma.

Key Clinical Presentation:

- Hypercalcemia
- Fatigue
- Bone pain
- Constipation

Investigations:

- Best initial: Serum calcium and PTH levels.
- Confirmatory: Imaging of the parathyroid glands (ultrasound, Sestamibi scan).

Treatment Outline:

- First-line: Parathyroidectomy.
- Second-line: Medical management with bisphosphonates or calcimimetics for refractory cases.

Correct Answer:

B. Missed adenoma ✅

MCQ 395

A 39-year-old woman presented with a 3-day history of central abdominal pain associated with nausea and vomiting for 1 day. She underwent a laparoscopic sleeve gastrectomy for obesity 6 years ago.
Examination confirmed a dehydrated patient with abdominal distention and exaggerated bowel sounds.
Which of the following is the most likely cause?

A. Adhesions
B. Internal hernia
C. Intussusception
D. Incisional hernia

Explanation:
The most likely cause in this patient is an **internal hernia**. After bariatric surgery such as sleeve gastrectomy, there is a risk of internal hernias developing, where part of the small intestine becomes trapped and causes bowel obstruction. Adhesions are also common but are more likely after a longer duration.

Bariatric Surgery Complications

Definition:
Internal hernia is a potential complication following bariatric surgery, where the bowel is trapped within mesenteric defects.

Key Clinical Presentation:

- Abdominal pain
- Nausea and vomiting
- Bowel obstruction symptoms

Investigations:

- Best initial: CT scan of the abdomen.

- Confirmatory: Laparoscopy.

Treatment Outline:

- First-line: Laparoscopic repair of the hernia.
- Second-line: Open surgical approach for complicated cases.

Correct Answer:
B. Internal hernia ✅

MCQ 5:

Which of the following is the most effective way to minimize excessive scar formation?

A. Pre-operative steroid
B. Elimination of wound tension
C. Post-operative topical steroid
D. Closure with subcuticular suture

Explanation:
Eliminating wound tension is the most effective method for minimizing excessive scar formation. This can be achieved by proper surgical technique, tension-free closure, and sometimes the use of sutures that minimize tension on the wound edges.

Wound Healing

Definition:
Excessive scar formation, or hypertrophic scarring, can occur due to tension on the wound edges, infection, or poor healing conditions.

Key Clinical Presentation:

- Raised, red scars
- Poor wound healing
- Keloid formation

Investigations:

- Best initial: Clinical evaluation.
- Confirmatory: Biopsy if excessive scarring or infection is suspected.

Treatment Outline:

- First-line: Tension-free closure, possible use of steroids or silicone sheets for scar prevention.
- Second-line: Steroid injections for existing scars.

Correct Answer:
B. Elimination of wound tension ✅

MCQ 396

Which of the following is most important factor for small bowel adaptation?

A. Prolonged TPN
B. Enteral feeding
C. Parietal cell vagotomy
D. Reversal of jejunal limb

Explanation:
Enteral feeding is the most important factor in promoting small bowel adaptation after surgery. The presence of nutrients in the bowel stimulates the growth and function of the enterocytes, aiding in bowel adaptation and recovery. Prolonged TPN is not as effective in promoting adaptation.

Small Bowel Disorders

Definition:
Small bowel adaptation is the process by which the remaining small intestine compensates for resected portions to maintain nutrient absorption.

Key Clinical Presentation:

- Diarrhea

- Malabsorption
- Weight loss

Investigations:

- Best initial: Nutritional assessment and electrolyte balance.
- Confirmatory: Imaging if anatomical changes are suspected.

Treatment Outline:

- First-line: Enteral feeding, nutritional support.
- Second-line: Surgical interventions if necessary.

Correct Answer:
B. Enteral feeding ✅

MCQ 397:

A 62-year-old man presented with sudden onset epigastric pain radiating to the back of 6 hours' duration. He has ischemic heart disease, on plavix and aspirin. Examination confirmed an ill looking and dehydrated patient, with generalized abdominal tenderness and absent bowel sounds (see report).
Erect chest X-ray: Air under the left hemidiaphragm.
Which of the following is the most likely diagnosis?

A. Gall stone disease
B. Acute pancreatitis
C. Perforated peptic ulcer
D. Gastric outlet obstruction

Explanation:
The **presence of air under the left hemidiaphragm** on an erect chest X-ray is a classic finding for **perforated peptic ulcer**, where the perforation allows air to escape into the peritoneal cavity. This condition presents with sudden epigastric pain, which often radiates to the back, and is associated with tenderness, guarding, and absent bowel sounds.

Gastrointestinal Emergencies

Definition:
Perforated peptic ulcer occurs when an ulcer in the stomach or duodenum breaches the wall, leading to leakage of gastric contents into the peritoneal cavity.

Key Clinical Presentation:

- Sudden onset severe abdominal pain
- Epigastric pain radiating to the back
- Tenderness, guarding, and absent bowel sounds

Investigations:

- Best initial: Abdominal X-ray (air under diaphragm)
- Confirmatory: CT scan if needed

Treatment Outline:

- First-line: Surgical repair (laparotomy and closure)
- Second-line: Medical management and observation if the patient is stable (rare)

Correct Answer:
C. Perforated peptic ulcer ✅

MCQ 398

A 23-year-old man sustained a closed right mid-shaft tibia-fibula fracture in a motor vehicle accident. He was alert, awake, complained of severe pain and numbness in his right lower extremity. Examination confirmed non-palpable dorsalis pedis and posterior tibial pulses.
Blood pressure 120/70 mmHg
Heart rate 85 /min
Respiratory rate 20 /min
Temperature 36.6 °C
Which of the following is the most appropriate management?

A. Fasciotomy
B. Angiography

C. Leg elevation and splint
D. Anticoagulation therapy

Explanation:
The patient presents with a **closed tibia-fibula fracture** along with signs of **vascular compromise**, as indicated by the non-palpable pulses in the lower extremity. This suggests possible **compartment syndrome** or arterial injury. The most appropriate management is **fasciotomy** to relieve the pressure in the compartment and restore blood flow, preventing long-term tissue damage.

Orthopedic Emergencies

Definition:
Compartment syndrome is a condition in which increased pressure within a muscle compartment impairs blood flow and tissue perfusion, potentially leading to irreversible damage.

Key Clinical Presentation:

- Pain out of proportion to the injury
- Numbness, loss of pulses, and swelling in the affected limb
- Absence of pulses

Investigations:

- Best initial: Clinical diagnosis
- Confirmatory: Pressure measurement within the compartment

Treatment Outline:

- First-line: Fasciotomy
- Second-line: Surgical repair of arterial injury if present

Correct Answer:
A. Fasciotomy ✅

MCQ 399

A 28-year-old woman was undergoing a laparoscopic cholecystectomy for gallstones. After creating pneumo-peritoneum and inserting the trocars, her heart rate was 40 beats/min. Which of the following is the most likely cause?

A. Overhydration
B. Cold gas insufflation
C. Increased venous return
D. Rapid stretching of parietal peritoneum

Explanation:
Cold gas insufflation during laparoscopic surgery is a well-known cause of **vagal response**, leading to bradycardia. The sudden stretching of the peritoneum and cooling effect of the insufflation can trigger the vagus nerve, causing a **decrease in heart rate** (known as vagal bradycardia).

Surgical Complications

Definition:
Vagal bradycardia can occur during laparoscopic procedures due to stimulation of the vagus nerve by cold insufflation or peritoneal stretching.

Key Clinical Presentation:

- Bradycardia during laparoscopic procedures
- Normal vital signs otherwise

Investigations:

- Best initial: Clinical observation during surgery
- Confirmatory: Electrocardiogram to confirm bradycardia

Treatment Outline:

- First-line: Adjusting gas temperature, use of atropine if needed
- Second-line: Monitoring and discontinuation of insufflation if needed

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Correct Answer:

B. Cold gas insufflation ✅

MCQ 400

A 35-year-old patient was brought to a small primary care hospital with no surgical facility. The patient's primary survey confirmed a right tension pneumothorax, and a compound right femoral fracture. After insertion of a chest tube and immobilization of the right lower limb, the treating doctor decided to transfer the patient to a trauma center. During the transfer to the ambulance, the patient's oxygen saturation suddenly dropped and he developed tachycardia and tachypnea.
Which of the following is the most appropriate management?

A. Proceed with the transfer
B. Perform a rapid sequence intubation
C. Check the femoral deformity for bleeding
D. Check the chest tube for place and patency

Explanation:
The sudden drop in oxygen saturation and the development of tachycardia and tachypnea suggests a **dislodged chest tube** or **pneumothorax recurrence**. The most appropriate immediate management is to **check the chest tube for place and patency** to ensure it is functioning correctly and preventing air from accumulating in the pleural space.

Trauma Management

Definition:
A tension pneumothorax occurs when air accumulates in the pleural space and increases pressure on the lungs and heart, impairing circulation and oxygenation.

Key Clinical Presentation:

- Tachypnea
- Tachycardia
- Decreased oxygen saturation
- Unilateral chest pain and absent breath sounds

Investigations:

- Best initial: Clinical assessment, chest X-ray
- Confirmatory: Ultrasound for lung sliding or CT chest

Treatment Outline:

- First-line: Chest tube insertion
- Second-line: Needle decompression if chest tube insertion is not immediately possible

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Correct Answer:
D. Check the chest tube for place and patency ✅

MCQ 401

A 50-year-old patient underwent a total thyroidectomy 5 hours ago. He has developed a large neck swelling associated with stridor and shortness of breath.
Blood pressure 165/90 mmHg
Heart rate 130 /min
Respiratory rate 24 /min
Temperature 37 °C
Which of the following is the most appropriate next step in management?

A. Tracheostomy
B. Close observation
C. Percutaneous drainage
D. Bedside wound exploration

Explanation:
The patient is showing signs of a **postoperative hematoma** in the neck, which is compressing the airway. The most appropriate management is **bedside wound exploration** to evacuate the hematoma and relieve the airway obstruction.

Surgical Complications

Definition:
Post-thyroidectomy hematoma is a common complication that can cause airway obstruction, particularly if it occurs in the first few hours after surgery.

Key Clinical Presentation:

- Neck swelling
- Stridor or difficulty breathing
- Hypoxia and tachycardia

Investigations:

- Best initial: Clinical examination
- Confirmatory: Ultrasound or CT neck if needed

Treatment Outline:

- First-line: Bedside exploration and evacuation of hematoma
- Second-line: Tracheostomy if airway is compromised

Correct Answer:
D. Bedside wound exploration ✅

MCQ 402

A 25-year-old woman presented with a right breast lump, which has been increasing in size over the past year. Physical examination confirmed a 10x12 cm firm, mobile mass involving the upper half of the right breast with no palpable axillary lymph nodes (see report).
Core needle biopsy: Malignant phyllodes tumour.
Which of the following is the most appropriate next step?

A. Wide local excision
B. Whole body PET scan
C. Contrast CT scan of chest
D. Immunohistochemistry for hormone receptors

Explanation:
Phyllodes tumors are fibroepithelial tumors that are typically benign but can also present with malignant characteristics, as in this case. The treatment of choice for malignant phyllodes tumors is **wide local excision** to ensure clear margins and reduce the risk of recurrence. **Lymph node evaluation** may be considered if there is evidence of nodal involvement. Other imaging studies like PET or CT scans are not routinely indicated unless there is suspicion of metastatic disease.

Breast Cancer and Tumors

Definition:
Phyllodes tumors are rare fibroepithelial tumors of the breast that can be benign, borderline, or malignant. Malignant forms require wide excision to avoid recurrence.

Key Clinical Presentation:

- A rapidly growing breast mass, typically larger than typical fibroadenomas
- Absence of axillary lymphadenopathy in many cases

Investigations:

- Best initial: Core needle biopsy
- Confirmatory: Surgical excision with histopathological assessment

Treatment Outline:

- First-line: Wide local excision
- Second-line: Mastectomy if the tumor is large or poorly localized

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Correct Answer:
A. Wide local excision ✅

MCQ 403

A 47-year-old single woman with a positive family history of breast cancer presented with a lesion found on routine mammography at the lower outer quadrant of the right breast (see report).

Image guided core needle biopsy: Atypical ductal hyperplasia.
Which of the following is the most appropriate next step in management?

A. Simple mastectomy
B. Wire localization excision
C. Chemoprevention therapy
D. 6-month follow-up mammography

Explanation:
Atypical ductal hyperplasia (ADH) is considered a premalignant lesion and is associated with an increased risk of breast cancer. The next step in managing ADH is **surgical excision** to ensure that there is no associated carcinoma. **Chemoprevention therapy** may be considered after excision, especially in high-risk individuals, but is not the next step in management for ADH. **Follow-up mammography** would not be sufficient without removal of the lesion.

Breast Cancer Risk Management

Definition:
Atypical ductal hyperplasia is characterized by abnormal cell growth in the milk ducts, which increases the risk of developing invasive breast cancer.

Key Clinical Presentation:

- Detected as an incidental finding on biopsy or mammography
- Typically asymptomatic

Investigations:

- Best initial: Core needle biopsy or excisional biopsy for confirmation

Treatment Outline:

- First-line: Surgical excision with margin evaluation
- Second-line: Consideration of chemoprevention if excised tissue shows residual atypia

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Correct Answer:
B. Wire localization excision ✅

MCQ 404

A 65-year-old woman presented with sudden onset epigastric pain radiating to the back for the past 4 hours. She has ischemic heart disease with a coronary stent, and is taking plavix and aspirin. She was diagnosed with a perforated viscus and was taken for an exploratory laparotomy, which confirmed a perforated anterior duodenal ulcer.
Which of the following is the most appropriate management?

A. Partial gastrectomy
B. Gastrojejunostomy
C. Closure with omental patch
D. Selective truncal vagotomy

Explanation:
A perforated duodenal ulcer is managed by **surgical closure with an omental patch** to seal the perforation. **Gastrectomy** and **gastrojejunostomy** are not indicated unless there is significant resection or complications. **Vagotomy** is typically reserved for refractory peptic ulcer disease and is not the first-line management for a perforated ulcer.

Peptic Ulcer Disease

Definition:
A perforated duodenal ulcer occurs when an ulcer in the duodenum erodes through the wall, causing leakage of gastric contents into the peritoneum.

Key Clinical Presentation:

- Sudden, severe epigastric pain, often radiating to the back
- Abdominal tenderness and signs of peritonitis

Investigations:

- Best initial: Abdominal X-ray (air under diaphragm) or CT scan for confirmation
- Confirmatory: Surgical exploration during laparotomy

Treatment Outline:

- First-line: Surgical closure of the perforation with omental patch
- Second-line: Vagotomy or other surgical options if necessary in case of recurrent ulcers

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Correct Answer:

C. Closure with omental patch ✅

404

MCQ:

A 38-year-old man sustained a pelvic fracture in a motor vehicle accident. There was blood per urethra (see report).
Retrograde urethrogram: Retroperitoneal urethral injury.
Which of the following is the most appropriate next step?

A. Urethroscopy
B. Foley's catheter
C. Laparoscopic repair
D. Suprapubic cystostomy

Correct Answer:

D. Suprapubic cystostomy ✅

A 38-year-old man sustained a pelvic fracture in a motor vehicle accident. There was blood per urethra (see report). Retrograde urethrogram: Retroperitoneal urethral injury. Which of the following is the most appropriate next step? A. Urethroscopy B. Foley's catheter C. Laparoscopic repair D. Suprapubic cystostomy

Explanation:

In cases of **retroperitoneal urethral injury** following a pelvic fracture, the most appropriate next step is **D. Suprapubic cystostomy**.

Rationale:

- A **suprapubic cystostomy** is the preferred method of urinary diversion in patients with urethral injury, particularly when there is concern about further injury to the urethra. This approach avoids further trauma to the urethra and allows for healing while providing a means of drainage.

- **Urethroscopy** (A) may be indicated later to assess the extent of the injury but is not typically the first step in managing such injuries. Early assessment through imaging (such as retrograde urethrogram) usually guides the next step.
- **Foley's catheter** (B) would not be appropriate in the case of urethral injury with blood at the meatus, as insertion of a catheter might exacerbate the injury, particularly in the case of a retroperitoneal urethral injury.
- **Laparoscopic repair** (C) is typically not indicated unless there are associated intra-abdominal injuries requiring laparoscopic surgery, but for isolated urethral injuries, a suprapubic catheter is preferred for urinary diversion.

Correct Answer:
D. Suprapubic cystostomy ✅

MCQ 406

A 25-year-old patient sustained blunt trauma to the right chest wall. There is paradoxical movement of the right side of the chest. His vital signs are as follows:

- Heart rate: 96 /min
- Respiratory rate: 35 /min
- Oxygen saturation: 84%

Chest X-ray shows fractures of the 2nd through 4th ribs at more than 1 site. What is the most appropriate initial management?

A. Fluid resuscitation
B. Assisted ventilation
C. Needle decompression
D. Thoracostomy chest tube

Explanation:

The most appropriate initial management for **paradoxical movement** and **rib fractures** indicating **flail chest** is **D. Thoracostomy chest tube**. Flail chest occurs when multiple rib fractures lead to a segment of the chest wall becoming detached from the rest of the thoracic cage, impairing breathing. The key management in this case is to manage the underlying pneumothorax or hemothorax with thoracostomy, to prevent respiratory compromise.

Topic Section:

Flail Chest Management

Definition:
Flail chest refers to the fracture of two or more ribs in two or more places, resulting in paradoxical chest wall movement.

Key Clinical Presentation:

- Paradoxical chest wall movement
- Respiratory distress
- Tachypnea, tachycardia, hypotension

Investigations:

- Best initial: **Chest X-ray**
- Confirmatory: **CT scan** (if indicated for further detail)

Treatment Outline:

- First-line: **Thoracostomy chest tube** (for managing pneumothorax/hemothorax)
- Second-line: **Ventilatory support** (e.g., assisted ventilation if required)
- Definitive: Surgical fixation in severe cases

Correct Answer:
D. Thoracostomy chest tube ✅

MCQ 407

A 9-year-old child presented with blunt abdominal trauma. His vital signs are:

- Blood pressure: 110/70 mmHg

- Heart rate: 90 /min
- Respiratory rate: 22 /min
- Temperature: 36.6 °C

Test results:

- Hb: 110 g/L (Normal range: 130-170 g/L)

CT scan abdomen reveals a 2 cm splenic laceration with perispheric fluid. What is the most appropriate management?

A. Non-operative
B. Laparoscopic splenectomy
C. Laparotomy and splenorrhaphy
D. Transfuse 2 units of packed RBCs

Explanation:

For a **splenic laceration** with **stable vital signs** and no signs of significant hemodynamic instability or large bleeding, **A. Non-operative management** is the most appropriate. Splenic injuries in children are often managed conservatively, with close monitoring and supportive care, including IV fluids and blood transfusions if needed.

Topic Section:

Management of Splenic Trauma

Definition:
Traumatic injury to the spleen, commonly from blunt trauma, leading to lacerations or rupture.

Key Clinical Presentation:

- Abdominal pain (typically in the left upper quadrant)
- Decreased hemoglobin in the case of bleeding

Investigations:

- Best initial: **CT scan**
- Confirmatory: **Ultrasound** for hemodynamic monitoring

Treatment Outline:

- First-line: **Non-operative management** (close observation)
- Second-line: **Surgical intervention** (splenorrhaphy or splenectomy if necessary)
- Definitive: In severe cases, **splenectomy** (if not manageable conservatively)

Correct Answer:
A. Non-operative ✅

MCQ 408

A 28-year-old patient underwent resection of the terminal ileum for inflammatory bowel disease. Which of the following nutrients is most important to replace parenterally?

A. Iron
B. Zinc
C. Bile salts
D. Vitamin B12

Explanation:

After **resection of the terminal ileum**, **D. Vitamin B12** is the most important nutrient to replace parenterally. The terminal ileum is the site of absorption for **Vitamin B12**, and its loss can lead to deficiency, resulting in megaloblastic anemia. This requires lifelong B12 supplementation, especially in those with significant resections.

Topic Section:

Vitamin B12 Deficiency after Ileal Resection

Definition:
Vitamin B12 deficiency caused by the loss of ileal absorption due to surgery or disease.

Key Clinical Presentation:

- Fatigue

- Anemia (megaloblastic)
- Neurological symptoms (e.g., paresthesia, cognitive impairment)

Investigations:

- Best initial: **Serum vitamin B12 levels**
- Confirmatory: **Schilling test** (historical use)

Treatment Outline:

- First-line: **Parenteral Vitamin B12 supplementation**
- Second-line: **Oral B12 supplementation** (if part of the ileum remains functional)

Correct Answer:
D. Vitamin B12 ✅

MCQ 409

A 48-year-old woman presented with vague central abdominal pain. Upper GI endoscopy reveals a 5 cm rounded mass at the posterior wall of the stomach. Biopsy confirms a gastrointestinal stromal tumor (GIST). What is the most appropriate management?

A. Total gastrectomy
B. Wide local excision
C. Partial gastrectomy
D. Enucleation of the lesion

Explanation:

Gastrointestinal stromal tumors (GISTs) are typically treated with **C. Partial gastrectomy**. These tumors are often resected with clear margins, and partial gastrectomy is the most appropriate approach to ensure complete removal while preserving gastric function.

Topic Section:

Gastrointestinal Stromal Tumor (GIST)

Definition:
A mesenchymal tumor of the GI tract, often arising from the stomach, that may present as a mass with bleeding or obstruction.

Key Clinical Presentation:

- Abdominal pain
- Palpable mass
- GI bleeding (less common)

Investigations:

- Best initial: **Endoscopy and biopsy**
- Confirmatory: **CT scan or MRI** for staging

Treatment Outline:

- First-line: **Surgical resection** (partial gastrectomy)
- Second-line: **Tyrosine kinase inhibitors** (e.g., imatinib) for advanced disease

---

Correct Answer:
C. Partial gastrectomy ✅

MCQ 410

A 30-year-old man presents with severe, increasing pain in the right lower limb. Examination reveals a tense, swollen calf with reduced range of motion. The compartment pressure measurement confirms compartment syndrome. The earliest symptom of this condition is pain. Which of the following best describes the mechanism of pain in compartment syndrome?

A. Skin hypoxia
B. Bone ischemia
C. Nerve hypoxia
D. Muscle engorgement

Explanation:

In **compartment syndrome**, **C. Nerve hypoxia** is the primary mechanism of pain. The elevated pressure within a confined anatomical space reduces blood flow to muscles and nerves, causing ischemia and hypoxia of the nerve tissue, leading to pain. The pain is out of proportion to the injury and worsens with passive movement of the affected muscles.

Topic Section:

Compartment Syndrome

Definition:
A condition where increased pressure within a muscle compartment reduces blood flow, leading to ischemia of muscles and nerves.

Key Clinical Presentation:

- Severe pain out of proportion to injury
- Swelling and tightness of the affected area
- Reduced pulses, paresthesia, and paralysis in severe cases

Investigations:

- Best initial: **Clinical evaluation and compartment pressure measurement**
- Confirmatory: **CT or MRI** (in doubtful cases)

Treatment Outline:

- First-line: **Fasciotomy** (surgical release of compartment pressure)
- Second-line: Supportive care (pain management, hydration)

Correct Answer:
C. Nerve hypoxia ✅

MCQ 411

Which of the following has the highest diagnostic value for inhalation injuries in a trauma patient?

A. Tachypnea
B. Voice hoarseness
C. Low oxygen saturation
D. Carbonaceous sputum

Explanation:

D. Carbonaceous sputum is the most specific finding for inhalation injury. The presence of soot or carbon particles in the sputum strongly suggests that the patient has inhaled smoke or hot gases, indicating damage to the respiratory tract, typically from a fire or similar event.

Topic Section:

Inhalation Injury

Definition:
Inhalation injury refers to damage to the respiratory tract due to inhaling smoke, toxic gases, or chemical fumes.

Key Clinical Presentation:

- Tachypnea, hoarseness, cough, wheezing
- Carbonaceous sputum (black or dark sputum due to inhalation of soot)

Investigations:

- Best initial: **Clinical assessment (including signs of respiratory distress)**
- Confirmatory: **Bronchoscopy** (visualization of airway damage), **CT scan** (in some cases)

Treatment Outline:

- First-line: **Oxygen therapy** (humidified, if needed), **Bronchodilators**, **Corticosteroids**
- Second-line: **Mechanical ventilation** in severe cases
- Definitive: **Intensive care management** (for airway protection and oxygenation)

Correct Answer:
D. Carbonaceous sputum ✅

MCQ 412

A 67-year-old man with ischemic heart disease presents with a 24-hour history of severe central abdominal pain. His lab results show the following:

- WBC: 28 (normal range: 4.5-10.5 x 10^9/L)
- Amylase: 600 (normal range: 24-151 IU/L)
- ABG: pH 7.2 (normal range: 7.36-7.45)
- Plain X-ray: Dilated small bowel with thickened wall

Which of the following is the most likely diagnosis?

A. Acute pancreatitis
B. Intestinal obstruction
C. Perforated duodenal ulcer
D. Mesenteric vascular occlusion

Explanation:

D. Mesenteric vascular occlusion is the most likely diagnosis. This condition leads to ischemia of the intestines, resulting in abdominal pain, elevated white blood cells (WBC), and abnormal amylase levels. The X-ray showing dilated bowel loops with a thickened wall is indicative of bowel ischemia.

Topic Section:

Mesenteric Ischemia

Definition:
A condition caused by reduced blood flow to the intestines, often due to embolic or thrombotic occlusion of the mesenteric vessels.

Key Clinical Presentation:

- Severe, sudden abdominal pain (often out of proportion to physical findings)
- Vomiting, diarrhea, and possibly hematochezia
- Abdominal distension

Investigations:

- Best initial: **CT angiography** (to visualize vascular occlusion)
- Confirmatory: **Mesenteric angiography**

Treatment Outline:

- First-line: **Surgical resection** of necrotic bowel if present
- Second-line: **Endovascular therapy** (in some cases)
- Definitive: **Revascularization** (surgical or endovascular)

Correct Answer:
D. Mesenteric vascular occlusion ✅

MCQ 413

A 20-year-old man presents with a chronic infection of both axillae. On examination, multiple discharging sinuses are observed. Which of the following is the most likely source of the infection?

A. Hair follicles
B. Sebaceous glands
C. Eccrine sweat glands
D. Apocrine sweat glands

Explanation:

D. Apocrine sweat glands are the most common source of chronic infections in the axillae. These glands are found in the axillary, genital, and perineal areas, and their ducts can become blocked, leading to infections such as **hidradenitis suppurativa**, which presents as multiple draining sinuses.

Topic Section:

Hidradenitis Suppurativa (HS)

Definition:
A chronic skin condition characterized by inflamed, painful lumps or abscesses, primarily in areas with apocrine sweat glands.

Key Clinical Presentation:

- Recurrent, painful lumps in areas such as the axillae, groin, and buttocks
- Draining sinus tracts with foul-smelling discharge
- Possible scarring over time

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Skin biopsy** (if diagnosis is unclear)

Treatment Outline:

- First-line: **Topical or oral antibiotics**, **Steroids**, and **Drainage**
- Second-line: **Incision and drainage** for abscesses, **Biologic therapies** (e.g., adalimumab)
- Definitive: **Surgical excision** of affected areas (in severe cases)

Correct Answer:
D. Apocrine sweat glands ✅

MCQ 415

A 50-year-old diabetic man presents with a 5-year history of an unhealed foot ulcer. On examination, a 2x2 cm irregular-shaped ulcer is noted at the dorsal aspect of the first toe with pearly white discoloration. Biopsy of the ulcer reveals pseudoepitheliomatous hyperplasia. Which of the following is the most appropriate management?

A. Repeat biopsy
B. First toe amputation
C. Surgical debridement
D. Negative pressure dressing

Explanation:

C. Surgical debridement is the most appropriate management for an unhealed diabetic foot ulcer with pseudoepitheliomatous hyperplasia. This technique helps to remove necrotic tissue, promote wound healing, and prevent further complications like infection.

Topic Section:

Diabetic Foot Ulcers

Definition:
Chronic ulcers that occur in diabetic patients, often due to neuropathy, poor circulation, and infection.

Key Clinical Presentation:

- Non-healing ulcers on weight-bearing surfaces (e.g., feet)
- Presence of neuropathy and vascular compromise
- Risk of infection and gangrene

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Wound biopsy** (to assess for infection and malignancy)

Treatment Outline:

- First-line: **Surgical debridement**, **Pressure relief** (e.g., offloading), **Antibiotics** (if infected)
- Second-line: **Hyperbaric oxygen therapy**
- Definitive: **Amputation** (if necessary in advanced cases)

Correct Answer:
C. Surgical debridement ✅

MCQ 416

A 32-year-old man had excision and primary closure of a skin contracture that was 5x3 cm on the forearm, 1 year ago. He presented with an overgrowth at the scar site that extends beyond the surgical scar margins, which started a few weeks after the surgery and gradually increased in size. Which of the following most accurately describes the lesion?

A. Keloid
B. Scar fibrosis
C. Desmoid tumor
D. Scar hypertrophy

Explanation:

A. Keloid is the most likely diagnosis. Keloids are characterized by excessive collagen formation that extends beyond the original scar margins, often growing progressively. They are more common in individuals with a genetic predisposition and often occur after trauma or surgery. This patient's history of a skin contracture excision and the overgrowth beyond the surgical scar margin are characteristic of keloid formation.

Topic Section:

Keloid Formation

Definition:
A keloid is an overgrowth of fibrous tissue that occurs at the site of skin injury, extending beyond the original wound boundaries.

Key Clinical Presentation:

- Raised, smooth, shiny, and often itchy scar
- May extend beyond the original wound site
- Commonly occurs after surgery, trauma, or burns

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Biopsy** (if diagnosis is unclear)

Treatment Outline:

- First-line: **Intralesional corticosteroids**
- Second-line: **Laser therapy**, **Cryotherapy**, or **Silicone sheets**
- Definitive: **Surgical excision** (with high recurrence risk)

Correct Answer:

A. Keloid ✅

MCQ 417

A 53-year-old man presented with sudden onset of severe right lower abdominal pain. On examination, there is a 3x3 cm right lower abdominal tender mass, which becomes more prominent with abdominal wall contraction. The patient has been on anticoagulants for atrial fibrillation for the past 4 years (see lab results and report).

Test Result Normal Values
INR: 1.5 (0.8-1.2)
Prothrombin time: 11 (10-13 sec)
CT scan: 3x3 cm enhancing mass at the right rectus muscle.

Which of the following is the most appropriate management?

A. Surgical excision
B. Rest and analgesia
C. Fresh frozen plasma
D. Angiography with embolization

Explanation:

B. Rest and analgesia is the most appropriate management for a **rectus sheath hematoma** in this patient. The patient is on anticoagulants, which predisposes him to hematomas, and the clinical presentation (tender mass in the abdominal wall) along with the CT findings suggest a rectus sheath hematoma. Conservative management with rest, analgesia, and monitoring is the first-line treatment, unless there is significant hemodynamic instability or ongoing bleeding.

Topic Section:

Rectus Sheath Hematoma

Definition:
A rectus sheath hematoma is a collection of blood within the rectus sheath, often caused by trauma or anticoagulant use.

Key Clinical Presentation:

- Localized pain and swelling in the abdominal wall
- May present with a mass or visible bruising
- Pain exacerbated by abdominal wall movement

Investigations:

- Best initial: **CT scan** or **Ultrasound**
- Confirmatory: **CT** (if needed to assess the size and location)

Treatment Outline:

- First-line: **Conservative management** with rest and analgesia
- Second-line: **Surgical evacuation** (if large or symptomatic)
- Definitive: **Control of anticoagulation therapy** (if applicable)

Correct Answer:
B. Rest and analgesia ✅

MCQ 418

A 58-year-old woman presented with a right thigh mass that measured 5x6 cm. Core biopsy: High-grade sarcoma. Which of the following is the most appropriate staging image?

A. Bone scan
B. MRI abdomen
C. Chest CT scan
D. Right thigh plain X-ray

Explanation:

C. Chest CT scan is the most appropriate staging image for **high-grade sarcoma**. Sarcomas are prone to metastasize to the lungs, and a chest CT scan is essential to evaluate for pulmonary metastases. Bone scans and MRI of the abdomen are not routinely used in staging of soft tissue sarcomas unless there is specific suspicion of bone involvement or abdominal involvement.

Topic Section:

High-Grade Sarcoma

Definition:
A high-grade sarcoma is a malignant tumor originating from connective tissues, characterized by rapid growth and a high potential for metastasis.

Key Clinical Presentation:

- Palpable mass, often painless initially
- Rapid growth of the tumor
- Can be associated with systemic symptoms in advanced stages

Investigations:

- Best initial: **CT scan** of the chest (to assess lung metastasis)
- Confirmatory: **MRI** of the primary site (to assess size and spread)

Treatment Outline:

- First-line: **Surgical excision** with clear margins
- Second-line: **Chemotherapy** and/or **Radiation**
- Definitive: **Surgical resection** (as the main treatment for localized disease)

Correct Answer:
C. Chest CT scan ✅

MCQ 419

Which of the following is the most appropriate biopsy for a limb sarcoma?

A. Incisional
B. Excisional
C. Core needle
D. Fine needle aspiration

Explanation:

A. Incisional biopsy is the most appropriate biopsy for a **limb sarcoma**. Excisional biopsy should be avoided for sarcomas as it can result in seeding of the tumor. Incisional biopsy, where only a portion of the tumor is removed, is preferred to obtain a definitive diagnosis without compromising surgical resection.

Topic Section:

Limb Sarcoma

Definition:
A limb sarcoma is a malignant tumor arising from connective tissues, often requiring biopsy to confirm the diagnosis.

Key Clinical Presentation:

- Painless, firm mass
- May present with rapid growth
- Can be associated with pain if pressing on adjacent structures

Investigations:

- Best initial: **Incisional biopsy**
- Confirmatory: **Core needle biopsy** or **Fine needle aspiration** (if applicable)

Treatment Outline:

- First-line: **Surgical excision** with wide margins
- Second-line: **Chemotherapy** or **Radiotherapy** (in advanced or non-resectable cases)

Correct Answer:
A. Incisional biopsy ✅

MCQ 420

A 44-year-old man presented with a 6-month history of vague central abdominal pain. CT scan of abdomen: Presence of a 20x10 cm retroperitoneal mass. No enlarged mesenteric or para-aortic lymph nodes noted. Which of the following is the most likely diagnosis?

A. Lymphoma
B. Liposarcoma
C. Germ cell tumor
D. Metastatic testicular tumor

Explanation:

B. Liposarcoma is the most likely diagnosis. A large, well-defined retroperitoneal mass, without enlarged lymph nodes, suggests **liposarcoma**, which is a common soft tissue tumor in this area. Lymphoma typically presents with enlarged lymph nodes, and germ cell tumors and metastatic testicular tumors are less likely to present as large retroperitoneal masses without other clinical signs.

Topic Section:

Liposarcoma

Definition:
Liposarcoma is a malignant tumor of adipose tissue, most commonly arising in the retroperitoneum or extremities.

Key Clinical Presentation:

- Painless, slowly enlarging mass
- Often located in the retroperitoneum
- May cause symptoms due to compression of surrounding structures

Investigations:

- Best initial: **CT scan** or **MRI** (to assess the size and location)
- Confirmatory: **Biopsy** (for histological confirmation)

Treatment Outline:

- First-line: **Surgical resection**
- Second-line: **Radiotherapy** or **Chemotherapy** (if inoperable or recurrent)
- Definitive: **Surgical excision** with clear margins

Correct Answer:

B. Liposarcoma ✅

MCQ 421

A 30-year-old man was hit by a truck 4 hours ago. On examination, he has raised jugular venous pressure and diminished heart sounds.

Blood pressure: 90/60 mmHg
Heart rate: 140 /min
Respiratory rate: 32 /min
Temperature: 36.6 °C
Oxygen saturation: 85 %

What type of shock is the most likely diagnosis?

A. Septic
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:

B. Cardiogenic shock is the most likely diagnosis. In this patient with trauma and diminished heart sounds, **cardiac tamponade** is highly suggested, a complication of trauma leading to fluid accumulation in the pericardial sac, which impairs cardiac output. This leads to raised jugular venous pressure, hypotension, tachycardia, and diminished heart sounds (Beck's triad), characteristic of cardiogenic shock.

Topic Section:

Cardiogenic Shock

Definition:
Cardiogenic shock occurs when the heart is unable to pump sufficient blood to meet the body's needs, often due to severe cardiac dysfunction or mechanical obstruction.

Key Clinical Presentation:

- Hypotension, tachycardia
- Raised jugular venous pressure

- Diminished heart sounds
- Cold, clammy skin
- Organ hypoperfusion (e.g., oliguria)

Investigations:

- Best initial: **Echocardiography** or **CT scan** (if trauma-related)
- Confirmatory: **Cardiac catheterization**

Treatment Outline:

- First-line: **IV fluids**, **inotropic support** (e.g., dopamine, dobutamine)
- Second-line: **Mechanical circulatory support** (e.g., intra-aortic balloon pump)
- Definitive: **Treat underlying cause** (e.g., pericardiocentesis for tamponade, surgery for structural damage)

Correct Answer:
B. Cardiogenic ✅

MCQ 422

A 20-year-old patient complains of painless fresh bleeding post defecation over the last 6 months.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal fistula
D. Perianal abscess

Explanation:

B. Haemorrhoids are the most likely diagnosis in this patient. Painless, fresh rectal bleeding that occurs during or after defecation is characteristic of **haemorrhoids**, particularly internal hemorrhoids. The patient's complaint of long-term bleeding with no pain supports this diagnosis.

Topic Section:

Haemorrhoids

Definition:
Haemorrhoids are swollen and inflamed veins in the rectum and anus, causing symptoms such as bleeding, pain, and itching.

Key Clinical Presentation:

- Painless rectal bleeding (bright red)
- May prolapse or cause itching and discomfort
- Blood typically seen on toilet paper or in stool

Investigations:

- Best initial: **Clinical examination** (digital rectal exam, anoscopy)
- Confirmatory: **Anoscopy** for internal hemorrhoids

Treatment Outline:

- First-line: **Conservative measures** (high-fiber diet, stool softeners)
- Second-line: **Rubber band ligation**, **sclerotherapy**
- Definitive: **Surgical excision** for large or prolapsed hemorrhoids

---

Correct Answer:
B. Haemorrhoids ✅

MCQ 423

A 40-year-old diabetic patient complains of a painful perianal swelling with fever for the last 3 days (see lab results).

Test Result Normal Values
Hb: 135 (130-170 g/L)
WBC: 16.4 (4.5-10.5 x 10^9/L)

Neutrophils: 80% (40-60%)
Random Glucose: 6.5 (3.9-5.5 mmol/L)

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:

C. Perianal abscess is the most likely diagnosis. The combination of **painful perianal swelling**, **fever**, and **leukocytosis** is typical of a **perianal abscess**. This condition often occurs in patients with diabetes and presents with localized tenderness and swelling around the anus.

Topic Section:

Perianal Abscess

Definition:
A perianal abscess is a collection of pus in the perianal or anal area due to infection of the anal glands.

Key Clinical Presentation:

- Painful, fluctuating mass in the perianal region
- Fever, erythema, and warmth over the swelling
- Possible drainage of pus

Investigations:

- Best initial: **Clinical examination** (digital rectal exam)
- Confirmatory: **Ultrasound** or **CT scan** if needed for abscess localization

Treatment Outline:

- First-line: **Incision and drainage**
- Second-line: **Antibiotics** if cellulitis or systemic infection is present

- Definitive: **Excision or marsupialization** in chronic cases or for recurrent abscesses

---

Correct Answer:
C. Perianal abscess ✅

MCQ 424

A 20-year-old primi-gravida complains of severe perianal pain post defecation along with fresh bleeding per rectum for the last 4 days. She describes the pain to be sharp and persists long after the passage of stool. She denies any history of associated swelling or fever.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:

A. Anal fissure is the most likely diagnosis. The patient's description of **sharp pain** post defecation that **persists long after the stool** suggests an **anal fissure**. Fissures are commonly associated with bright red blood, particularly after passing hard stools.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a small tear in the skin of the anal canal, typically caused by the passage of hard stools or trauma to the anal region.

Key Clinical Presentation:

- Severe, sharp pain during or after defecation
- Bright red blood on the stool or toilet paper
- May be associated with itching or spasms in the anal sphincter

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Anoscopy** if needed

Treatment Outline:

- First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**
- Second-line: **Topical nitroglycerin** or **calcium channel blockers** for sphincter relaxation
- Definitive: **Surgical lateral internal sphincterotomy** in chronic cases

---

Correct Answer:
A. Anal fissure ✅

MCQ 425

A 50-year-old hypertensive diabetic complains of a 2-month history of perianal swelling with discharge. 2 months ago, he noticed a small swelling, which ruptured.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Perianal abscess
D. Thrombosed piles

Explanation:

B. Perianal fistula is the most likely diagnosis. A perianal fistula is a chronic condition that often follows the rupture of a perianal abscess, leading to the formation of an abnormal passage that discharges pus. The patient's history of swelling, rupture, and ongoing discharge strongly supports this diagnosis.

Topic Section:

Perianal Fistula

Definition:
A perianal fistula is a small tunnel that forms between the skin and the anus, often following an abscess.

Key Clinical Presentation:

- Painless or mildly painful perianal discharge
- History of a prior perianal abscess
- Recurrent drainage of pus or serosanguinous fluid from the anal area

Investigations:

- Best initial: **Clinical examination** and **digital rectal exam**
- Confirmatory: **Anoscopy** or **MRI** for complex fistulas

Treatment Outline:

- First-line: **Fistulotomy** or **Seton placement** for draining fistulas
- Second-line: **Antibiotics** for infection control if necessary
- Definitive: **Surgical fistulotomy** or **fistula repair** for recurrent or complex cases

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Correct Answer:
B. Perianal fistula ✅

MCQ 426

A 30-year-old multigravida mother complains of severe perianal pain, swelling, and absolute constipation for the last 2 days. On examination, swollen anal cushions, which are irreducible and tender to touch, are noted.

Blood pressure: 120/90 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 37 °C

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Anal carcinoma

C. Perianal abscess
D. Thrombosed piles

Explanation:

D. Thrombosed piles is the most likely diagnosis. The patient presents with **irreducible**, **tender anal cushions** and symptoms of **perianal pain**, **swelling**, and **constipation**, which are typical signs of **thrombosed hemorrhoids** (thrombosed piles). The condition occurs when a clot forms inside the hemorrhoid, causing swelling and pain.

Topic Section:

Thrombosed Hemorrhoids

Definition:
Thrombosed hemorrhoids occur when blood clots form inside the veins of the hemorrhoidal plexus, causing painful swelling.

Key Clinical Presentation:

- Sudden onset of **severe perianal pain**
- **Irreducible, tender anal mass**
- Associated with **constipation** and **straining**

Investigations:

- Best initial: **Clinical examination** (digital rectal exam)
- Confirmatory: **Anoscopy** for internal hemorrhoids

Treatment Outline:

- First-line: **Conservative treatment** (fiber, stool softeners, pain relief)
- Second-line: **Surgical excision** of the thrombosed hemorrhoid if severe
- Definitive: **Excision** for persistent or recurrent cases

Correct Answer:
D. Thrombosed piles ✅

MCQ 427

A 36-year-old man presents with a 2-day history of constipation. On examination, he looks nervous and severe pain was experienced during parting of the perianal region. Linear cracks were noted at both 6 and 12 O'clock positions.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Solitary rectal ulcer
D. Intersphincteric abscess

Explanation:

A. Anal fissure is the most likely diagnosis. The presence of **linear cracks** at the **6 o'clock** and **12 o'clock** positions is characteristic of an **anal fissure**. Fissures are typically caused by trauma (e.g., hard stools) and cause sharp pain during and after defecation.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a tear in the skin of the anal canal, often caused by the passage of hard stools or trauma to the anal area.

Key Clinical Presentation:

- **Sharp, severe pain** during and after defecation
- **Linear crack** at the **6 o'clock** or **12 o'clock** position
- Bright red blood on the stool or toilet paper

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Anoscopy**

Treatment Outline:

- First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**
- Second-line: **Topical nitroglycerin** or **calcium channel blockers**
- Definitive: **Lateral internal sphincterotomy** in chronic cases

---

Correct Answer:
A. Anal fissure ✅

MCQ 428

A 65-year-old diabetic presents with a 4-day history of severe perianal pain with high-grade fever. On examination, there is no external swelling. On digital rectal examination, there is posterior bogginess and tenderness with slight purulent discharge coming from the anal canal.

Temperature: 38 °C

Which of the following is the most likely diagnosis?

A. Perianal fistula
B. Ulcerative colitis
C. Intersphincteric abscess
D. Condylomata acuminate

Explanation:

C. Intersphincteric abscess is the most likely diagnosis. The patient's history of **severe perianal pain** and **fever** with **bogginess** and **purulent discharge** on digital rectal examination suggests an **intersphincteric abscess**, a common cause of perianal abscesses.

Topic Section:

Intersphincteric Abscess

Definition:
An intersphincteric abscess is a collection of pus between the internal and external anal sphincters, often presenting with pain, swelling, and fever.

Key Clinical Presentation:

- **Severe perianal pain**
- **Fever**, often high-grade
- **Boggy, tender mass** on digital rectal exam
- **Purulent discharge** from the anal canal

Investigations:

- Best initial: **Digital rectal exam** and **clinical examination**
- Confirmatory: **Anoscopy** or **MRI** for complex cases

Treatment Outline:

- First-line: **Incision and drainage**
- Second-line: **Antibiotics** if cellulitis or systemic infection is present
- Definitive: **Excision** or **marsupialization** in recurrent or chronic cases

---

Correct Answer:
C. Intersphincteric abscess ✅

MCQ 429

A 30-year-old patient complains of a 2-year history of perianal itching and discomfort. On perianal examination, there are cauliflower-like multiple lesions clustered together. They are pink to grey in color and tender to touch.

Which of the following is the most likely diagnosis?

A. Anal carcinoma
B. Rectal carcinoma
C. Fibroepithelial polyps
D. Condylomata acuminate

Explanation:

D. Condylomata acuminate is the most likely diagnosis. The presence of **cauliflower-like lesions** that are **tender to touch** and **pink to grey in color** suggests **condylomata acuminate** (genital warts), commonly caused by **human papillomavirus (HPV)** infection.

Topic Section:

Condylomata Acuminate (Genital Warts)

Definition:
Condylomata acuminate are benign, proliferative warts caused by the human papillomavirus (HPV), commonly affecting the genital or anal area.

Key Clinical Presentation:

- **Cauliflower-like lesions** in the perianal or genital region
- **Pink to grey** in color
- Often tender to touch
- May be associated with **itching** and **discomfort**

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **HPV DNA testing** or **biopsy** in cases of uncertainty

Treatment Outline:

- First-line: **Topical treatments** (e.g., imiquimod, podophyllin)
- Second-line: **Cryotherapy** or **laser ablation**
- Definitive: **Excision** in persistent or large lesions

Correct Answer:
D. Condylomata acuminate ✅

MCQ 429

A 65-year-old complains of a 6-month change in bowel habits. On examination, he is pale, thin, and emaciated. His abdomen is slightly distended. On digital rectal examination, there is

a lesion 2 cm from the anal verge. On proctoscopy, the lesion appears to be a cauliflower-like mass, which is friable to touch.

Which of the following is the most likely diagnosis?

A. Anal carcinoma
B. Rectal carcinoma
C. Fibroepithelial polyps
D. Condylomata acuminate

Explanation:

A. Anal carcinoma is the most likely diagnosis. The patient's presentation with **weight loss**, **anorexia**, and a **friable cauliflower-like mass** near the anal verge strongly suggests **anal carcinoma**.

Topic Section:

Anal Carcinoma

Definition:
Anal carcinoma is a malignant tumor arising from the anal canal or perianal skin, often associated with HPV infection.

Key Clinical Presentation:

- **Change in bowel habits**, often with **rectal bleeding**
- **Weight loss** and **emaciation**
- **Friable, cauliflower-like mass** on proctoscopy
- **Palpable mass** on digital rectal exam

Investigations:

- Best initial: **Proctoscopy** and **biopsy**
- Confirmatory: **CT scan** or **MRI** for staging

Treatment Outline:

- First-line: **Chemoradiation** for early-stage disease

- Second-line: **Surgical resection** for locally advanced disease
- Definitive: **Chemoradiation** followed by **surgical resection** for advanced cases

---

Correct Answer:
A. Anal carcinoma ✅

MCQ 430

A 25-year-old presents with a 2-week history of severe perianal pain and an inability to pass stool for the last 2 days. History confirmed a haemorrhoidectomy 12 months ago. On examination, he is uneasy and in distress. On rectal examination, the finger could not be passed beyond the anal verge due to severe pain.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Anal stenosis
C. Anal carcinoma
D. Intersphincteric abscess

Explanation:

A. Anal fissure is the most likely diagnosis. The patient's history of **severe perianal pain** and **inability to pass stool**, with **severe pain on rectal examination**, suggests **anal fissure**, especially after **haemorrhoidectomy**. The acute onset of pain and difficulty with bowel movements is characteristic of this condition.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a tear or crack in the skin around the anus, often caused by trauma, constipation, or straining during bowel movements.

Key Clinical Presentation:

- **Severe perianal pain** during and after defecation
- **Bleeding**, typically bright red, on the stool or toilet paper
- Painful **spasm** during rectal examination

Investigations:

- Best initial: **Clinical examination** and **digital rectal examination**
- Confirmatory: **Anoscopy** for internal fissures

Treatment Outline:

- First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**
- Second-line: **Topical nitroglycerin** or **calcium channel blockers**
- Definitive: **Lateral internal sphincterotomy** for chronic cases

---

Correct Answer:
A. Anal fissure ✅

MCQ 431

A 20-year-old weight-lifter complains of a 2-day history of bilious vomiting and abdominal distention. Examination revealed swelling in the left inguinal region, which is firm and tender.

Blood pressure: 100/70 mmHg
Heart rate: 116 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C

X-ray abdomen:
Multiple air-fluid levels with no air in the large bowel.

What type of hernia is the most likely diagnosis?

A. Obstructed
B. Irreducible
C. Incarcerated
D. Strangulated

Explanation:

D. Strangulated hernia is the most likely diagnosis. The **bilious vomiting** and **abdominal distention**, along with **tender swelling** in the inguinal region, suggest a **strangulated hernia**, where the blood supply to the hernia contents is compromised, potentially leading to bowel necrosis. The **X-ray showing air-fluid levels** and absence of air in the large bowel also points to bowel obstruction due to a hernia.

Topic Section:

Strangulated Hernia

Definition:
A strangulated hernia occurs when the blood supply to the hernia contents (typically bowel) is compromised, leading to ischemia and possible necrosis.

Key Clinical Presentation:

- **Severe abdominal pain** with **vomiting** (bilious)
- **Tender swelling** at the hernia site
- **Absence of bowel sounds** or **abdominal distention**

Investigations:

- Best initial: **Abdominal X-ray** showing air-fluid levels and bowel obstruction
- Confirmatory: **CT scan** or **ultrasound** for hernia and vascular compromise

Treatment Outline:

- First-line: **Emergency surgical exploration** and reduction of hernia
- Second-line: **Resection of necrotic bowel** if needed
- Definitive: **Herniorrhaphy** to repair the defect

Correct Answer:
D. Strangulated ✅

MCQ 432

A 25-year-old man was brought to the Emergency Department with a penetrating chest injury 45 minutes ago. He was initially fine in the ambulance but then collapsed. Emergency response personnel started resuscitation. On arrival, he had a weak thread pulse, raised JVP, and bilateral equal breath sounds.

Blood pressure: 90/60 mmHg
Heart rate: 140 /min
Respiratory rate: 30 /min
Temperature: 36.6 °C
Oxygen saturation: 85 %

Which of the following is the most likely diagnosis?

A. Flail segment
B. Pneumothorax
C. Cardiac tamponade
D. Pulmonary contusions

Explanation:

C. Cardiac tamponade is the most likely diagnosis. The patient's **raised JVP**, **weak thread pulse**, and **collapse** suggest **cardiac tamponade**, where blood accumulates in the pericardial sac, compressing the heart. This is a common consequence of blunt or penetrating chest trauma.

Topic Section:

Cardiac Tamponade

Definition:
Cardiac tamponade is the compression of the heart due to the accumulation of fluid (e.g., blood) in the pericardial sac, impairing heart function.

Key Clinical Presentation:

- **Hypotension** with **weak pulse** (pulsus paradoxus)
- **Elevated jugular venous pressure (JVP)**
- **Tachycardia**

- **Muffled heart sounds**
- **Dyspnea**

Investigations:

- Best initial: **Clinical examination** with **echocardiography** or **focused assessment with sonography for trauma (FAST)**
- Confirmatory: **Chest X-ray** and **echocardiogram**

Treatment Outline:

- First-line: **Pericardiocentesis** to relieve pressure
- Second-line: **Pericardial window** if persistent tamponade
- Definitive: **Surgical repair** if trauma-related

Correct Answer:
C. Cardiac tamponade ✅

MCQ 435

A young man was brought to the Emergency Department after he was hit in the chest while playing football. On examination, he was dyspnoeic and drowsy with GCS 11/15. Tracheal shifting to the left and raised JVP were noted. He has barely audible breath sounds on the right side.

Blood pressure: 80/50 mmHg
Heart rate: 130 /min
Respiratory rate: 32 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most likely diagnosis?

A. Flail segment
B. Cardiac tamponade
C. Pulmonary contusions
D. Tension pneumothorax

Explanation:

D. Tension pneumothorax is the most likely diagnosis. The combination of **tracheal shift**, **raised JVP**, **hypotension**, and **diminished breath sounds on the right** suggests a **tension pneumothorax**, where air is trapped in the pleural space, causing lung collapse and mediastinal shift.

Topic Section:

Tension Pneumothorax

Definition:
Tension pneumothorax occurs when air enters the pleural space and is unable to escape, leading to increased pressure on the lung and mediastinal structures, impairing respiratory and circulatory function.

Key Clinical Presentation:

- **Dyspnea** and **hypoxia**
- **Tachycardia** and **hypotension**
- **Tracheal deviation** away from the affected side
- **Diminished breath sounds** on the affected side

Investigations:

- Best initial: **Clinical examination** with **needle decompression**
- Confirmatory: **Chest X-ray** or **ultrasound**

Treatment Outline:

- First-line: **Needle decompression** followed by **chest tube insertion**
- Second-line: **Chest tube** for ongoing air leak
- Definitive: **Surgical intervention** if persistent

Correct Answer:
D. Tension pneumothorax ✅

MCQ 436

A 70-year-old alcoholic complains of a 6-month history of gradual progressive dysphagia and weight loss. On examination, he is cachectic and pale. His abdomen is soft and non-tender. Digital rectal examination confirms altered coloured stool in the rectum.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Values:
WBC: 9.5 (4.5-10.5 x 109/L)
Hb: 90 (130-170 g/L)
Urea: 8 (2.75-7.4 mmol/L)
Creatinine: 120 (44-115 µmol/L)

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Oesophageal cancer
C. Acid peptic disease
D. Duodenal perforation

Explanation:

B. Oesophageal cancer is the most likely diagnosis given the patient's **gradual progressive dysphagia** and **weight loss**, which are hallmark symptoms of esophageal cancer. The patient's history of **alcoholism** is a significant risk factor for this condition.

Topic Section:

Esophageal Cancer

Definition:
Esophageal cancer is a malignant tumor that originates in the esophagus. It typically presents with symptoms related to obstruction or invasion of nearby structures.

Key Clinical Presentation:

- **Progressive dysphagia**
- **Weight loss**

- **Anorexia**
- **Altered colored stool** (could indicate bleeding)

Investigations:

- Best initial: **Endoscopy** with **biopsy**
- Confirmatory: **CT scan** of the chest and abdomen

Treatment Outline:

- First-line: **Surgical resection** if feasible
- Second-line: **Chemotherapy** and/or **radiotherapy**
- Definitive: **Multidisciplinary approach** with surgery, chemotherapy, and radiotherapy

---

Correct Answer:
B. Oesophageal cancer ✅

MCQ 437

A 40-year-old man, alcoholic and smoker, is brought to the Emergency Department with a 4-hour history of severe upper abdominal pain radiating to the back. On examination, his abdomen is distended, tense, and tender. Digital Rectal examination confirms a collapsed empty rectum.

Blood pressure: 100/70 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 37 °C

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Pneumothorax
C. Duodenal perforation
D. Acalculous cholecystitis

Explanation:

A. Pancreatitis is the most likely diagnosis. The patient's **severe upper abdominal pain**, radiating to the back, along with **distension** and **tenderness**, points to **acute pancreatitis**, especially given the history of **alcohol consumption**. The collapsed rectum suggests no distal bowel obstruction, which is typical for pancreatitis.

Topic Section:

Acute Pancreatitis

Definition:
Acute pancreatitis is inflammation of the pancreas, often caused by gallstones or alcohol use, leading to digestive enzyme activation and pancreatic injury.

Key Clinical Presentation:

- **Severe upper abdominal pain** radiating to the back
- **Nausea** and **vomiting**
- **Abdominal distension**
- **Tenderness** on palpation
- **Raised amylase** levels

Investigations:

- Best initial: **Serum amylase** and **lipase**
- Confirmatory: **CT scan** of the abdomen

Treatment Outline:

- First-line: **Supportive care** (hydration, analgesia, fasting)
- Second-line: **ERCP** if gallstones are the cause
- Definitive: **Surgical management** for complications or if the cause is due to pancreatitis-related complications

---

Correct Answer:
A. Pancreatitis ✅

MCQ 438

A 55-year-old man was admitted with a 2-year history of recurrent upper abdominal pain attacks. History confirmed a recurrent colicky pain aggravated by morphine or its analogues.

Blood pressure: 110/70 mmHg
Heart rate: 96 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Values:
Indirect bilirubin: 5 (3.2-12.1 µmol/L)
Direct bilirubin: 10 (1.5-6.5 µmol/L)
Total bilirubin: 15 (3.5-16.5 µmol/L)
Alkaline phosphatase: 250 (39-117 IU/L)
Amylase: 150 (24-151 IU/L)

Ultrasound abdomen:
Prominent common bile duct along with minimal dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Biliary colic
C. Cholecystitis
D. Sphincter of Oddi dysfunction

Explanation:

D. Sphincter of Oddi dysfunction is the most likely diagnosis. The patient's **recurrent colicky pain**, **increased bilirubin**, and the history of morphine triggering pain point to **Sphincter of Oddi dysfunction**. This condition is characterized by impaired function of the sphincter of Oddi, which can cause intermittent biliary obstruction.

Topic Section:

Sphincter of Oddi Dysfunction

Definition:
Sphincter of Oddi dysfunction is a disorder of the biliary system caused by impaired relaxation or contraction of the sphincter of Oddi, leading to bile flow abnormalities.

Key Clinical Presentation:

- **Colicky abdominal pain**, often postprandial
- **Pain exacerbated by morphine**
- **Jaundice** (if biliary obstruction occurs)

Investigations:

- Best initial: **Ultrasound** or **MRCP**
- Confirmatory: **Manometry of the sphincter of Oddi**

Treatment Outline:

- First-line: **Endoscopic sphincterotomy**
- Second-line: **Medication for pain relief**
- Definitive: **Surgical intervention** if other treatments fail

---

Correct Answer:
D. Sphincter of Oddi dysfunction ✅

MCQ 439

A 20-year-old woman recently had an infant complains that she has been having recurrent episodes of upper abdominal pain since the third trimester. The pain radiates to her back and is associated with nausea. On examination, her abdomen is soft and non-tender. On deep palpation, there is a mass measuring 8 x 8 cm in the epigastric region, which appears to be non-pulsatile and non-tender.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Value:
Lipase: 250 (0-160 IU/L)

Which of the following is the most likely diagnosis?

A. Hydatid cyst
B. Gall bladder mucocele
C. Pancreatic pseudocyst
D. Gastrointestinal stromal tumour

Explanation:

C. Pancreatic pseudocyst is the most likely diagnosis. The patient has **upper abdominal pain**, **nausea**, and a **non-pulsatile mass** in the epigastric region. The elevated **lipase** supports a pancreatic origin, making **pancreatic pseudocyst** the most likely diagnosis.

Topic Section:

Pancreatic Pseudocyst

Definition:
A pancreatic pseudocyst is a fluid-filled collection in or around the pancreas, usually following acute pancreatitis, that does not have an epithelial lining.

Key Clinical Presentation:

- **Epigastric pain**, nausea
- **Mass in the epigastric region**
- **Elevated lipase/amylase**
- **Non-tender and non-pulsatile mass**

Investigations:

- Best initial: **Abdominal ultrasound** or **CT scan**
- Confirmatory: **Endoscopic ultrasound** or **MRI**

Treatment Outline:

- First-line: **Conservative management** with observation
- Second-line: **Percutaneous drainage** for symptomatic pseudocysts
- Definitive: **Surgical resection** for large or complicated pseudocysts

---

Correct Answer:

C. Pancreatic pseudocyst ✅

MCQ 440

A 30-year-old patient presents to the Emergency Department with an 8-hour history of severe upper abdominal pain that is not relieved by oral medications. History confirmed recurrent similar events over the last year that were relieved by anti-spasmodic and anti-inflammatory drugs. Examination revealed jaundice, with a soft abdomen and marked tenderness in the right hypochondriac region.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 37.9 °C

Test Result Normal Values:
WBC: 16 (4.5-10.5 x 109/L)
Indirect bilirubin: 5 (3.2-12.1 µmol/L)
Direct bilirubin: 15 (1.5-6.5 µmol/L)
Total bilirubin: 20 (3.5-16.5 µmol/L)
Alkaline phosphatase: 400 (39-117 IU/L)
Amylase: 115 (24-151 IU/L)

Ultrasound:
Minimal CBD of 1.1 cm with dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Hepatitis
B. Cholangitis
C. Pancreatitis
D. Cholecystitis

Explanation:

B. Cholangitis is the most likely diagnosis given the patient's **jaundice**, **right upper quadrant tenderness**, and **dilated intrahepatic ducts** seen on ultrasound. The elevated **bilirubin** and **alkaline phosphatase** suggest obstruction of the biliary tree, which is characteristic of **cholangitis**.

Topic Section:

Cholangitis

Definition:
Cholangitis is an infection of the biliary system, often due to bile duct obstruction, typically caused by gallstones, strictures, or malignancy.

Key Clinical Presentation:

- **Fever**
- **Jaundice**
- **Right upper quadrant pain**
- **Tachycardia**
- **Tachypnea**

Investigations:

- Best initial: **Ultrasound**
- Confirmatory: **ERCP** or **MRCP**

Treatment Outline:

- First-line: **IV antibiotics** (e.g., piperacillin-tazobactam)
- Second-line: **Endoscopic retrograde cholangiopancreatography (ERCP)** for biliary drainage
- Definitive: **Surgical intervention** if necessary

Correct Answer:
B. Cholangitis ✅

MCQ 441

A 45-year-old patient complains of recurrent episodes of upper abdominal pain for the past 2 weeks with fever. On examination, the patient is jaundiced, abdomen is soft with marked tenderness all over the abdomen.

Blood pressure: 90/60 mmHg
Heart rate: 130 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Test Result Normal Values:
Indirect bilirubin: 5.3 (3.2-12.1 µmol/L)
Direct bilirubin: 20.1 (1.5-6.5 µmol/L)
Total bilirubin: 25.3 (3.5-16.5 µmol/L)
Alkaline phosphatase: 450 (39-117 IU/L)
Amylase: 1400 (24-151 IU/L)

Ultrasound:
CBD of 1.4 cm with dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Hepatitis
B. Cholangitis
C. Pancreatitis
D. Acute Cholecystitis

Explanation:

B. Cholangitis is the most likely diagnosis based on the patient's **jaundice**, **fever**, and **elevated bilirubin**. The **dilated common bile duct** and elevated **amylase** point toward a **biliary obstruction**, which could be complicated by infection.

Topic Section:

Cholangitis

Definition:
Cholangitis is an infection of the bile ducts, often caused by bacterial infections secondary to biliary obstruction.

Key Clinical Presentation:

- **Fever**
- **Jaundice**

- **Right upper quadrant pain**
- **Tachycardia**
- **Tachypnea**

Investigations:

- Best initial: **Ultrasound**
- Confirmatory: **ERCP** or **MRCP**

Treatment Outline:

- First-line: **IV antibiotics** (e.g., piperacillin-tazobactam)
- Second-line: **Endoscopic retrograde cholangiopancreatography (ERCP)** for biliary drainage
- Definitive: **Surgical intervention** if necessary

Correct Answer:
B. Cholangitis ✅

MCQ 442

A 20-year-old woman presents with a lump in the right lower quadrant of her breast that has been present for the last 2 years. The lump was initially small but over the past 6 months has constantly been growing in size. On examination, the mass measures 8 x 10 cm, is mobile and not attached to underlying muscles. Overlying skin is thinned out due to pressure effect of the mass. There are no signs of surrounding inflammation or infection.

Which of the following is the most likely diagnosis?

A. Mastitis
B. Phyllodes
C. Galactocele
D. Fibroadenoma

Explanation:

B. Phyllodes is the most likely diagnosis given the **rapid growth**, **large size**, and **mobile mass** in a young woman. **Phyllodes tumors** are fibro-epithelial lesions that can grow rapidly and are often benign but may have malignant potential.

Topic Section:

Phyllodes Tumor

Definition:
Phyllodes tumors are rare fibroepithelial breast tumors that can be benign or malignant, characterized by rapid growth and a leaf-like architecture on histology.

Key Clinical Presentation:

- **Painless mass**
- **Rapid growth**
- **Non-tender, mobile**
- **May have overlying skin changes**
- **No inflammatory signs**

Investigations:

- Best initial: **Ultrasound**
- Confirmatory: **Core needle biopsy** or **excision biopsy**

Treatment Outline:

- First-line: **Wide excision**
- Second-line: **Lumpectomy** or **mastectomy** for larger tumors or malignant variants

---

Correct Answer:
B. Phyllodes ✅

MCQ 443

A 30-year-old patient was brought to the Emergency Department after being shot in the chest 1 hour ago. A tube thoracostomy on the left side was done for a pneumothorax. After 15 minutes, the underwater seal bottle is filled with 2 liters of blood.

Blood pressure: 90/60 mmHg
Heart rate: 120 /min
Respiratory rate: 22 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. Thoracotomy
B. CT scan chest
C. Tube thoracostomy
D. Angiographic embolization

Explanation:

A. Thoracotomy is the most appropriate management in this case of **massive hemothorax** with **persistent blood drainage** from the chest tube. A **thoracotomy** allows for control of bleeding and evacuation of the hemothorax.

Topic Section:

Massive Hemothorax

Definition:
Massive hemothorax is the accumulation of a large amount of blood in the pleural cavity, typically due to trauma.

Key Clinical Presentation:

- **Hypotension**
- **Tachycardia**
- **Breathlessness**
- **Cyanosis**
- **Shock**

Investigations:

- Best initial: **Chest X-ray**
- Confirmatory: **CT scan** of the chest

Treatment Outline:

- First-line: **Tube thoracostomy** for blood drainage
- Second-line: **Thoracotomy** for ongoing bleeding
- Definitive: **Surgical intervention** (e.g., embolization, thoracotomy)

Correct Answer:
A. Thoracotomy ✅

MCQ 444

A 5-year-old child is brought to the Emergency Department after being stabbed in the right lower chest. His abdomen is distended. A focused abdominal sonogram for trauma is suggestive of free fluid in the abdomen.

Blood pressure: 100/80 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. Thoracotomy
B. Tube thoracostomy
C. Exploratory laparotomy
D. Angiographic embolization

Explanation:

C. Exploratory laparotomy is the most appropriate management for a child with **blunt abdominal trauma** and **free fluid** seen on sonogram, as it indicates possible **intra-abdominal injury**, requiring surgical exploration.

Topic Section:

Trauma Management in Children

Definition:
Trauma management in children involves rapid assessment and stabilization, followed by targeted diagnostic imaging and appropriate surgical intervention if needed.

Key Clinical Presentation:

- **Trauma-related symptoms**
- **Abdominal distension**
- **Shock**
- **Tachycardia, tachypnea, hypotension**

Investigations:

- Best initial: **Focused abdominal sonography** or **CT scan**
- Confirmatory: **Exploratory laparotomy** if free fluid or solid organ injury is suspected

Treatment Outline:

- First-line: **Initial resuscitation** (IV fluids, oxygen)
- Second-line: **Exploratory laparotomy** or **conservative management**
- Definitive: **Surgical repair** or **interventional radiology**

Correct Answer:
C. Exploratory laparotomy ✅

MCQ 445

A 66-year-old man recently underwent an anterior resection of the rectum and is making a slow recovery. He complains that in the middle of the night he has pain in the left calf. On examination, he has a tender left calf, which is slightly swollen and red.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min

Temperature: 36.6 °C
Oxygen saturation: 90 %

Ultrasound Doppler:
Deep vein thrombosis in the left popliteal region extending up to the common femoral junction.

Which of the following is the most appropriate management?

A. Heparin
B. Warfarin
C. IVC filter
D. Enoxaparin

Explanation:

D. Enoxaparin is the most appropriate initial treatment for **deep vein thrombosis (DVT)** in this patient. **Enoxaparin** is a low molecular weight heparin (LMWH), commonly used for the initial treatment of DVT to prevent further clot formation. This is the first-line treatment until transition to oral anticoagulants like warfarin, if necessary.

Topic Section:

Deep Vein Thrombosis (DVT)

Definition:
DVT refers to the formation of a thrombus within the deep veins, most commonly in the legs, which can cause complications such as pulmonary embolism.

Key Clinical Presentation:

- **Pain, swelling, redness, and warmth** in the affected leg
- **Tenderness** over the affected vein
- **Positive Homan's sign** (not diagnostic)

Investigations:

- Best initial: **Ultrasound Doppler**
- Confirmatory: **CT pulmonary angiography** (if pulmonary embolism is suspected)

Treatment Outline:

- First-line: **Enoxaparin** or **unfractionated heparin** for anticoagulation
- Second-line: **Warfarin** for long-term anticoagulation
- Definitive: **IVC filter** (in cases of contraindications to anticoagulation)

Correct Answer:
D. Enoxaparin ✅

MCQ 446

A 12-year-old boy presents to the Emergency Department after a fall that led to a supra-condylar fracture. Presently his distal pulses are not palpable.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min

Which of the following is the most appropriate management?

A. Early K-wire
B. Hand elevation
C. Watchful waiting
D. Surgical exploration

Explanation:

D. Surgical exploration is the most appropriate management for a **supra-condylar fracture** with **absent distal pulses**. In this case, the absence of pulses suggests potential vascular injury, and surgical exploration is required to evaluate and repair any damage to the arteries or nerves.

Topic Section:

Supracondylar Fracture of the Humerus

Definition:
A supracondylar fracture is a common elbow fracture in children, typically occurring due to a fall on an outstretched hand.

Key Clinical Presentation:

- **Swelling and deformity** around the elbow
- **Absent or diminished distal pulses**
- **Pain and limited movement** of the elbow joint

Investigations:

- Best initial: **X-ray** (AP and lateral views of the elbow)
- Confirmatory: **CT scan** (if displacement or complex fractures are suspected)

Treatment Outline:

- First-line: **Surgical exploration** for vascular or nerve injury
- Second-line: **Closed reduction and percutaneous pinning** (if stable fracture)
- Definitive: **Open reduction and internal fixation** for severe fractures or complications

---

Correct Answer:
D. Surgical exploration ✅

MCQ 447

A 5-month-old baby was admitted with an inability to tolerate food and non-billous vomiting over the past 2-days. On examination, he appears to be dehydrated. There is an olive shaped mass palpable in the epigastric region measuring 2 x 2 cm.

Which of the following is the most appropriate management?

A. Pyloromyotomy
B. Balloon dilatation
C. Gastrojejunostomy

Explanation:

A. Pyloromyotomy is the treatment of choice for **pyloric stenosis** in infants. The presence of an **olive-shaped mass** in the epigastric region along with **non-bilious vomiting** is characteristic of pyloric stenosis, and **pyloromyotomy** is the definitive surgical treatment to relieve the obstruction.

Topic Section:

Pyloric Stenosis

Definition:
Pyloric stenosis is a condition in infants where the **pyloric sphincter** becomes thickened, leading to **gastric outlet obstruction**.

Key Clinical Presentation:

- **Non-bilious vomiting** (often after feeding)
- **Olive-shaped mass** in the epigastric region
- **Dehydration and weight loss**

Investigations:

- Best initial: **Ultrasound** (shows thickened pylorus)
- Confirmatory: **Upper GI contrast study** (shows delayed gastric emptying)

Treatment Outline:

- First-line: **Pyloromyotomy** (surgical division of the hypertrophied pyloric muscle)
- Second-line: **Balloon dilatation** (if surgery is not possible)

Correct Answer:
A. Pyloromyotomy ✅

MCQ 448

A 45-year-old patient complains of pain and bilateral groin swelling over the last 6-months. On examination, there are small bilateral inguino-scrotal swellings. Cough impulse is positive, and his abdomen is soft and non-tender.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Oxygen saturation: 95 %
BMI: 30 kg/m²

Which of the following is the most appropriate management?

A. Herniotomy
B. Lichtenstein repair
C. Laparoscopic mesh repair
D. Biological mesh placement

Explanation:

B. Lichtenstein repair is the most appropriate management for an **inguinal hernia**. This **open surgical repair** involves the placement of a **mesh** to reinforce the abdominal wall and prevent recurrence.

Topic Section:

Inguinal Hernia

Definition:
An inguinal hernia occurs when a portion of the intestine or fat protrudes through the abdominal wall or into the inguinal canal.

Key Clinical Presentation:

- **Groin swelling** (that becomes more prominent with coughing or straining)
- **Cough impulse positive**
- **Painful or painless mass**

Investigations:

- Best initial: **Physical examination**
- Confirmatory: **Ultrasound** or **CT scan** (if complications or doubts)

Treatment Outline:

- First-line: **Hernia repair** (Lichtenstein or laparoscopic mesh repair)
- Second-line: **Watchful waiting** in asymptomatic patients

- Definitive: **Surgical intervention** to prevent complications

---

Correct Answer:
B. Lichtenstein repair ✅

MCQ 449

A 5-month-old baby is presented with bilateral groin swelling that has been present since birth. On examination, there are small bilateral inguino-scrotal swellings, which become prominent when the child cries. His abdomen is soft and non-tender.

Which of the following is the most appropriate management?

A. Herniotomy
B. Lichtenstein repair
C. Laparoscopic mesh repair
D. Biological mesh placement

Explanation:

A. Herniotomy is the most appropriate management for **congenital inguinal hernia** in an infant. The hernia needs to be repaired surgically, and **herniotomy** involves the removal of the hernial sac.

Topic Section:

Inguinal Hernia in Children

Definition:
An inguinal hernia in children is a common congenital condition where a portion of the intestine or fat protrudes through the inguinal canal.

Key Clinical Presentation:

- **Bilateral groin swelling** that becomes more prominent with crying
- **Soft, non-tender abdomen**
- **Visible mass in the scrotum or groin area**

Investigations:

- Best initial: **Physical examination**
- Confirmatory: **Ultrasound** (to confirm contents of the hernia)

Treatment Outline:

- First-line: **Herniotomy** (surgical removal of the hernial sac)
- Second-line: **Laparoscopic repair** (in specific cases, e.g., recurrent hernias)

Correct Answer:
A. Herniotomy ✅

MCQ 450

A 42-year-old woman complains of recurrent episodes of upper abdominal pain over the past 2 weeks with associated high-grade fever. On examination, she looks icteric. Her abdomen is soft with marked tenderness in the right hypochondriac region.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Test Result Normal Values:
Indirect bilirubin 10 3.2-12.1 µmol/L
Direct bilirubin 20 1.5-6.5 µmol/L
Total bilirubin 30 3.5-16.5 µmol/L
Alkaline phosphatase 250 39-117 IU/L
Amylase 100 24-151 IU/L
WBC 20 4.5-10.5 x 109/L

Which of the following is the most appropriate management?

A. CBD exploration
B. Laparoscopic cholecystectomy
C. Percutaneous cholecystostomy
D. Endoscopic retrograde cholangiopancreatography

Explanation:

D. Endoscopic retrograde cholangiopancreatography (ERCP) is the most appropriate management in this patient with **cholangitis**, as indicated by the presence of jaundice, fever, elevated bilirubin, and white blood cells. **ERCP** is both diagnostic and therapeutic, allowing for the removal of bile duct stones and drainage.

Topic Section:

Cholangitis

Definition:
Cholangitis is the inflammation of the bile ducts, commonly caused by bacterial infection due to a biliary obstruction.

Key Clinical Presentation:

- **Right upper quadrant pain, fever, and jaundice** (Charcot's triad)
- **Elevated bilirubin and alkaline phosphatase**

Investigations:

- Best initial: **Ultrasound abdomen**
- Confirmatory: **ERCP** for bile duct stone extraction and drainage

Treatment Outline:

- First-line: **Antibiotics** (IV broad-spectrum) and **ERCP**
- Second-line: **Cholecystectomy** (after infection resolution)
- Definitive: **Cholecystectomy** + **CBD exploration** (if the source of obstruction is identified)

Correct Answer:
D. Endoscopic retrograde cholangiopancreatography ✅

MCQ 451

A 45-year-old patient complains of perianal swelling, fresh bleeding per-rectum, and pain over the last 3 months. On examination, there is a mass 1 cm from the anal verge. She also has obstructive symptoms.

Blood pressure: 110/70 mmHg
Heart rate: 96 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Biopsy:
Adenocarcinoma

MRI abdomen:
Localized lesion with cranio-caudal extension of 3 cm with associated lymphadenopathy

CT scan chest:
No evidence of metastasis

Which of the following is the most appropriate treatment?

A. Diversion colostomy
B. Low anterior resection
C. Concurrent chemoradiation
D. Abdominoperineal resection

Explanation:

C. Concurrent chemoradiation is the most appropriate treatment for locally advanced **anal adenocarcinoma**. In cases where surgical resection is not immediately appropriate due to the extent of disease or lymph node involvement, chemoradiation is often used first to downstage the tumor before surgery.

Topic Section:

Anal Cancer (Adenocarcinoma)

Definition:
Anal carcinoma is a malignant tumor arising from the anal canal, most commonly **squamous cell carcinoma**, though **adenocarcinomas** can also occur.

Key Clinical Presentation:

- **Perianal mass**, bleeding, and pain
- **Obstructive symptoms**

- **Lymphadenopathy**

Investigations:

- Best initial: **Proctoscopy and biopsy**
- Confirmatory: **MRI abdomen** (for local extension and lymph node involvement)

Treatment Outline:

- First-line: **Concurrent chemoradiation** for anal adenocarcinoma
- Second-line: **Low anterior resection** (if suitable for surgery post-chemoradiation)
- Definitive: **Abdominoperineal resection** (in cases of extensive local disease)

Correct Answer:
C. Concurrent chemoradiation ✅

MCQ 452

A 30-year-old woman presents with a mass in the right upper quadrant of her breast for the last 4 years. According to the patient, the mass was initially small but over the past 2 years, it has been constantly growing in size. On examination, the mass measures 15 x 15 cm, is mobile and not attached to underlying muscle. Overlying skin is thinned out due to the mass effect.

Ultrasound breast:
Solid mass containing single or multiple, round or cleft-like cystic spaces and demonstrating posterior acoustic enhancement.

Which of the following is the most appropriate management?

A. Radiation therapy
B. Simple mastectomy
C. Neoadjuvant chemotherapy
D. Modified radical mastectomy

Explanation:

B. Simple mastectomy is the most appropriate management for a **breast mass** identified as likely **fibroadenoma**. The described **solid mass** with cystic components and acoustic enhancement on ultrasound points to this benign condition, and excision is typically curative.

Topic Section:

Fibroadenoma of the Breast

Definition:
Fibroadenomas are the most common benign tumors of the breast, usually affecting women in their 20s and 30s.

Key Clinical Presentation:

- **Painless, firm, well-defined, mobile mass**
- **Overlying skin not affected**
- **No lymphadenopathy**

Investigations:

- Best initial: **Ultrasound**
- Confirmatory: **Core needle biopsy** (to differentiate from other lesions)

Treatment Outline:

- First-line: **Excisional biopsy or simple mastectomy** for large or symptomatic fibroadenomas
- Second-line: **Observation** if small and asymptomatic

Correct Answer:
B. Simple mastectomy ✅

MCQ 453

A young woman underwent a laparoscopic cholecystectomy for symptomatic gallstones 1 day ago. Post-operatively, there are decreased breath sounds in the lower chest on auscultation.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. IV antibiotics
B. Bronchodilators
C. Ultrasound chest
D. Incentive spirometry

Explanation:

D. Incentive spirometry is the most appropriate management for **atelectasis** following surgery. It is a common post-operative complication, particularly after laparoscopic procedures, where shallow breathing can lead to **lung collapse** in the lower lung zones. **Incentive spirometry** helps improve lung expansion and oxygenation.

Topic Section:

Post-operative Atelectasis

Definition:
Atelectasis refers to the collapse of part or all of a lung due to alveolar deflation, which can occur after surgery, especially in patients with shallow breathing.

Key Clinical Presentation:

- **Decreased breath sounds** on auscultation, particularly in the lower lobes
- **Shallow breathing** and **tachypnea**
- **Hypoxia**

Investigations:

- Best initial: **Chest X-ray**
- Confirmatory: **CT chest** (if there is suspicion of a more severe process)

Treatment Outline:

- First-line: **Incentive spirometry**
- Second-line: **Postural drainage** and **deep breathing exercises**
- Definitive: **Chest physiotherapy** and **oxygen therapy** if required

---

Correct Answer:
D. Incentive spirometry ✅

MCQ 454

A 25-year-old man underwent an open appendectomy for a perforated appendix. He was discharged on the 2nd post-operative day uneventfully. He returned on the 8th post-operative day complaining of pain and redness around the wound site. On examination, there is purulent discharge from the wound site. His abdomen is soft with tenderness over the wound site. Digital rectal examination was unremarkable.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate management?

A. IV antibiotics
B. Open drainage
C. CT scan abdomen
D. Exploratory laparotomy

Explanation:

A. IV antibiotics are the most appropriate management for **post-surgical wound infection** in this case. The patient presents with signs of a wound infection (pain, redness, purulent discharge) without signs of intra-abdominal pathology. **IV antibiotics** should be started while considering wound care and monitoring for further signs of complications.

Topic Section:

Post-operative Infections

Definition:
Infections that occur after surgery, typically due to bacterial colonization of the wound or internal sites.

Key Clinical Presentation:

- Redness, pain, and purulent discharge at the wound site
- Fever
- Tachycardia

Investigations:

- Best initial: **Wound culture**
- Confirmatory: **CT scan abdomen** (if concerned about abscess formation or intra-abdominal complications)

Treatment Outline:

- First-line: **IV antibiotics**
- Second-line: **Wound care** and **drainage** if indicated
- Definitive: **Exploratory surgery** if there are signs of abscess formation or further complications

---

Correct Answer:
A. IV antibiotics ✅

MCQ 456

A 55-year-old man underwent an open appendectomy for a perforated appendix. He presents to the Emergency Department on the 6th post-operative day complaining of severe abdominal pain, distension, and an inability to pass stool. On examination, there is feculent discharge from the wound site with signs of peritonitis.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Oxygen saturation: 95 %

Test Result Normal Values:
ABG HCO3- 10 22-28 mmol/L
ABG PCO2 4.8 4.7-6.0 kPa
pH 7.25 7.36-7.45
ABG PO2 11 10.6-14.2 kPa
Base access -12 -2 to 2 mmol/L
Ultrasound abdomen:
Moderate ascites with echoes and septations.

Which of the following is the most appropriate management?

A. Wound care
B. IV antibiotics
C. CT scan abdomen
D. Exploratory laparotomy

Explanation:

D. Exploratory laparotomy is the most appropriate management. The patient presents with signs of **peritonitis** (feculent discharge, abdominal distension, and pain) and a significant metabolic disturbance (low bicarbonate, acidosis). These findings are consistent with a **severe intra-abdominal infection** or **peritonitis**, necessitating an urgent **exploratory laparotomy** to identify and treat the source of the infection, such as a leaking anastomosis or abscess formation.

Topic Section:

Post-operative Peritonitis

Definition:
Infection and inflammation of the peritoneal cavity following surgery, often due to leakage or bacterial contamination.

Key Clinical Presentation:

- **Severe abdominal pain, distension, and feculent discharge** from a surgical wound
- **Signs of systemic infection** (tachycardia, fever)
- **Metabolic acidosis** (low bicarbonate, base deficit)

- **Peritonitis** on clinical examination

Investigations:

- Best initial: **Ultrasound abdomen**
- Confirmatory: **CT scan abdomen** or **exploratory laparotomy**

Treatment Outline:

- First-line: **IV antibiotics** and **supportive care**
- Second-line: **Exploratory laparotomy** to identify and manage the source of infection
- Definitive: **Surgical intervention** (e.g., reoperation to repair leaks or drain abscesses)

Correct Answer:
D. Exploratory laparotomy ✅

MCQ 457

A 45-year-old patient complains of a high-grade fever with chills and multiple spikes on the 4th post-operative day after a diverticular perforation surgery. On examination, his wound is healthy and his chest is clear to auscultation. On Digital rectal examination, there is bogginess anteriorly.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate treatment?

A. IV antibiotics
B. IV paracetamol
C. Abdominal cavity washout
D. Ultrasound guided drainage

Explanation:

A. IV antibiotics is the most appropriate treatment for **post-surgical infection** suspected to be intra-abdominal. The patient has a post-surgical fever, and the presence of **bogginess** on digital rectal examination suggests possible **intra-abdominal infection** or **abscess formation**. IV antibiotics should be started immediately, and further imaging or drainage may be required based on clinical progression.

Topic Section:

Post-surgical Infections (Intra-abdominal)

Definition:
Infections that arise after surgery, most commonly from leakage or contamination of the abdominal cavity.

Key Clinical Presentation:

- **Fever and chills** after surgery
- **Boggy areas** on digital examination or tenderness
- **Tachycardia** and systemic signs of infection

Investigations:

- Best initial: **Ultrasound abdomen** or **CT scan abdomen**
- Confirmatory: **Drainage and culture** of peritoneal fluid if needed

Treatment Outline:

- First-line: **IV antibiotics** and **supportive care**
- Second-line: **Intra-abdominal drainage** (either percutaneous or surgical)
- Definitive: **Exploratory laparotomy** or **washout** if needed to manage the infection source

---

Correct Answer:
A. IV antibiotics ✅

MCQ 458

A 55-year-old man with no known co-morbidities underwent a Graham's patch repair for a perforated duodenal ulcer 10 days ago. On examination, his abdomen is soft and non-tender. On auscultation, there is decreased breath sounds on the right lower chest. Blood cultures obtained from the central and peripheral line of the patient are growing gram-positive cocci.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate treatment?

A. Chest physiotherapy
B. Removal of central line
C. Abdominal cavity washout
D. Ultrasound guided drainage

Explanation:

B. Removal of central line is the most appropriate treatment for suspected **central venous catheter-related infection** with **gram-positive cocci**. The patient has a **central venous line infection**, which is suggested by the growth of bacteria from both the central and peripheral blood cultures. The first step in management is to **remove the central line** and initiate appropriate **antibiotic therapy**.

Topic Section:

Central Line Infections

Definition:
Infection associated with the use of a central venous catheter, often caused by bacterial colonization at the insertion site or within the catheter.

Key Clinical Presentation:

- **Fever and chills** after central line insertion
- Positive blood cultures from **both central and peripheral lines**
- **Tachycardia** and systemic signs of infection

Investigations:

- Best initial: **Blood cultures** from both central and peripheral lines
- Confirmatory: **Catheter tip culture** and imaging for potential complications

Treatment Outline:

- First-line: **Removal of the central line** and **IV antibiotics**
- Second-line: **Ultrasound or CT** if abscess formation is suspected
- Definitive: **Surgical removal** if infection persists or complications arise

---

Correct Answer:
B. Removal of central line ✅

MCQ 459

A patient that underwent a cholecystectomy earlier has pyrexia with temperature spikes. On examination, the wound is healthy. His abdomen is soft with mild tenderness in the right hypochondriac region. There is decreased air entry on the right side of the chest on auscultation.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 20 /min
Temperature: 38.5 °C

Ultrasound: Collection of 10 x 15 cm in the gall bladder fossa.

Which of the following is the most appropriate management?

A. IV antibiotics
B. Chest physiotherapy
C. Ultrasound guided drainage
D. Endoscopic retrograde cholangiopancreatography

Explanation:

C. Ultrasound guided drainage is the most appropriate management. The patient has a **post-surgical collection** in the gall bladder fossa, most likely a **bile collection** or abscess. The management of this is **ultrasound-guided drainage**, which helps relieve the collection and prevent further complications. **IV antibiotics** may be started if there is suspicion of infection but drainage is key for proper resolution.

Topic Section:

Post-operative Abscess Management

Definition:
Abscess formation in the abdominal cavity following surgery, commonly after procedures like cholecystectomy or appendectomy.

Key Clinical Presentation:

- **Fever** and **temperature spikes**
- **Localized tenderness** and **decreased breath sounds** (if associated with pleural involvement)
- Post-operative history of surgery (e.g., cholecystectomy)

Investigations:

- Best initial: **Ultrasound abdomen**
- Confirmatory: **CT scan abdomen** if further information is required

Treatment Outline:

- First-line: **Ultrasound-guided drainage**
- Second-line: **IV antibiotics** if infection is suspected
- Definitive: **Surgical drainage** if percutaneous drainage fails or complications arise

---

Correct Answer:
C. Ultrasound guided drainage ✅

MCQ 460

A 25-year-old underwent an open appendectomy for a perforated appendix 8 days ago and returns for follow-up. On examination, there is a seroma formation at the wound site, which is draining freely through a gaping wound. There are no signs of surrounding inflammation.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Which of the following is the most appropriate treatment?

A. Antibiotics
B. CT scan abdomen
C. Regular wound dressings
D. Ultrasound guided drainage

Explanation:

C. Regular wound dressings is the most appropriate treatment. This patient has **seroma formation** following appendectomy, which is a common post-operative complication. **Regular wound dressings** with proper hygiene will allow the seroma to resolve. There are no signs of infection or further complications, so antibiotics or drainage are unnecessary.

Topic Section:

Post-operative Seroma Management

Definition:
A **seroma** is a collection of **serous fluid** in a tissue cavity, usually occurring after surgery, particularly in areas with potential space or low tissue adherence.

Key Clinical Presentation:

- **Swelling** at the surgical site
- **Drainage of clear fluid** (serous) from the wound
- No signs of infection (e.g., redness, fever)

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Ultrasound** if there is concern about abscess or other collections

Treatment Outline:

- First-line: **Regular wound dressings**
- Second-line: **Compression bandage** or **aspiration** if the seroma persists
- Definitive: **Surgical intervention** if seroma recurs or persists

---

Correct Answer:
C. Regular wound dressings ✅

MCQ 461

Which of the following is most likely involved in neuro-hormonal stress response?

A. Androgen
B. Parathyroid
C. Thyroid stimulating
D. Corticotrophin releasing

Explanation:

D. Corticotrophin releasing is the correct answer. The **neuro-hormonal stress response** involves the **hypothalamic-pituitary-adrenal (HPA) axis**, where the release of **corticotrophin-releasing hormone (CRH)** from the hypothalamus triggers the release of **ACTH** from the pituitary, leading to the secretion of **cortisol** from the adrenal glands. Cortisol is crucial in managing the body's stress response.

Topic Section:

Neuro-hormonal Stress Response

Definition:
The body's physiological response to stress involves the activation of the **sympathoadrenal system** and the **hypothalamic-pituitary-adrenal (HPA) axis**, leading to increased production of cortisol, catecholamines, and other hormones.

Key Clinical Presentation:

- **Increased heart rate**, **blood pressure**

- **Elevated blood glucose**
- **Suppression of non-essential functions** (e.g., reproduction, digestion)

Investigations:

- Best initial: **Serum cortisol level**
- Confirmatory: **ACTH stimulation test** for adrenal insufficiency

Treatment Outline:

- First-line: **Supportive care** in acute stress
- Second-line: **Steroid therapy** if chronic adrenal insufficiency (Addison's disease) is suspected

---

Correct Answer:
D. Corticotrophin releasing ✅

MCQ 462

A 30-year-old woman underwent an exploratory laparotomy for a perforated duodenal ulcer. Post-operatively she developed septic shock syndrome.

Which of the following is the most common physiological response?

A. Hypokalaemia
B. Respiratory acidosis
C. Cellular anaerobic respiration
D. Increase glomerular filtration rate

Explanation:

C. Cellular anaerobic respiration is the correct answer. In **septic shock**, tissue hypoperfusion leads to **anaerobic metabolism**, resulting in the production of **lactic acid** and metabolic **acidosis**. This process reflects inadequate oxygen delivery to tissues, and the body shifts from aerobic to anaerobic pathways.

Topic Section:

Septic Shock

Definition:
A life-threatening condition caused by widespread infection, resulting in systemic inflammation and **impaired perfusion** to vital organs.

Key Clinical Presentation:

- **Hypotension**, **tachycardia**, and **oliguria**
- **Fever**, **tachypnea**, and **altered mental status**

Investigations:

- Best initial: **Blood cultures**
- Confirmatory: **Lactate levels** to assess severity

Treatment Outline:

- First-line: **Fluid resuscitation** and **broad-spectrum antibiotics**
- Second-line: **Vasopressors** if blood pressure does not respond to fluids
- Definitive: **Source control** (e.g., drainage of abscess)

Correct Answer:
C. Cellular anaerobic respiration ✅

MCQ 463

A 65-year-old patient is intubated in the Intensive Care Unit because of septic shock.

Which of the following will best indicate adequate systemic perfusion?

A. Cardiac index
B. Central venous pressure
C. Mixed venous oxygen saturation
D. Pulmonary capillary wedge pressure

Explanation:

C. Mixed venous oxygen saturation is the best indicator of **adequate systemic perfusion** in critically ill patients, especially those in shock. **Mixed venous oxygen saturation** (SvO2) reflects the balance between oxygen delivery and oxygen consumption. A low SvO2 indicates insufficient oxygen delivery to tissues, which is a hallmark of inadequate perfusion.

Topic Section:

Monitoring Shock and Perfusion

Definition:
In critically ill patients, monitoring **perfusion status** is crucial for managing **shock** and preventing organ failure. SvO2, along with other clinical indicators, helps guide therapy.

Key Clinical Presentation:

- **Hypotension**, **tachycardia**, and **altered mental status**
- **Decreased urine output** and **cool extremities**

Investigations:

- Best initial: **Central venous oxygen saturation** (ScvO2 or SvO2)
- Confirmatory: **Lactate levels**, **cardiac output measurement**

Treatment Outline:

- First-line: **Fluid resuscitation** and **vasopressors**
- Second-line: **Inotropes** if cardiac output is low
- Definitive: **Address the underlying cause** (e.g., source control in sepsis)

Correct Answer:
C. Mixed venous oxygen saturation ✅

MCQ 464

A 55-year-old is scheduled to undergo laparoscopic cholecystectomy.

Which of the following is the most appropriate prophylactic antibiotic regimen?

A. Regular dose for 5 days
B. 3 doses postoperatively

C. Single dose with anaesthesia induction
D. Single dose 1 hour before skin incision

Explanation:

D. Single dose 1 hour before skin incision is the most appropriate prophylactic antibiotic regimen. For **laparoscopic cholecystectomy**, a **single dose of antibiotics** administered **1 hour before the skin incision** is the standard practice to reduce the risk of infection, especially **biliary infections**. Prolonged use or multiple doses are typically not necessary unless there are specific risk factors.

Topic Section:

Surgical Prophylaxis in Cholecystectomy

Definition:
Prophylactic antibiotics are administered before surgery to prevent postoperative infections.

Key Clinical Presentation:

- **Elective surgery** (e.g., cholecystectomy)
- **High-risk patients** (e.g., immunocompromised)

Investigations:

- Best initial: **Clinical evaluation**
- Confirmatory: Not needed for routine procedures

Treatment Outline:

- First-line: **Single dose of antibiotics** (1 hour before incision)
- Second-line: **Additional doses** if high risk for infection or extended surgeries

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Correct Answer:
D. Single dose 1 hour before skin incision ✅

MCQ 465

A 33-year-old man was brought to the Emergency Department after a high velocity trauma. He had distended neck veins and markedly decreased breath sounds on the right side.

Heart rate: 105 /min
Blood Pressure: 90/60 mmHg
Oxygen saturation: 85%

Which of the following is the most likely cause?

A. Flail chest
B. Right hemothorax
C. Pericardial tamponade
D. Right tension pneumothorax

Explanation:

D. Right tension pneumothorax is the most likely cause. The clinical features, such as **distended neck veins** and **decreased breath sounds** on one side, suggest **tension pneumothorax**. This condition occurs when air enters the pleural space and cannot escape, causing **increased intrathoracic pressure**, leading to **hypoxia**, **hemodynamic instability**, and **distended neck veins**.

Topic Section:

Tension Pneumothorax

Definition:
A life-threatening condition where air accumulates in the pleural cavity and cannot escape, leading to **increased intrathoracic pressure** and **cardiovascular compromise**.

Key Clinical Presentation:

- **Hypotension**
- **Distended neck veins**
- **Decreased breath sounds** on the affected side
- **Tracheal deviation** (late finding)

Investigations:

- Best initial: **Clinical diagnosis** (e.g., physical examination, imaging if available)

- Confirmatory: **Chest X-ray** or **ultrasound**

Treatment Outline:

- First-line: **Needle decompression** followed by **chest tube insertion**
- Second-line: **Surgical intervention** if necessary

Correct Answer:
D. Right tension pneumothorax ✅

MCQ 466

Which of the following is the most likely cerebral perfusion pressure in mmHg?

A. 40
B. 50
C. 60
D. 70

Explanation:

C. 60 is the most likely cerebral perfusion pressure (CPP) in mmHg. **Cerebral perfusion pressure (CPP)** is typically between **60-80 mmHg** in healthy adults. A **CPP of 60 mmHg** is generally considered adequate for maintaining **cerebral blood flow**.

Topic Section:

Cerebral Perfusion Pressure

Definition:
Cerebral perfusion pressure (CPP) is the pressure gradient driving blood to the brain, defined as the **mean arterial pressure (MAP)** minus the **intracranial pressure (ICP)**.

Key Clinical Presentation:

- **Decreased consciousness**
- **Signs of brain injury** (e.g., post-traumatic brain injury)

- **Hypotension**

Investigations:

- Best initial: **Intracranial pressure (ICP) monitoring**
- Confirmatory: **Head CT scan** to assess structural brain damage

Treatment Outline:

- First-line: **Optimization of blood pressure**
- Second-line: **Surgical decompression** if necessary

Correct Answer:
C. 60 ✅

MCQ 467

A 35-year-old woman presented with partial thickness burns on her arms and leg due to hot water spillage 1 day ago, with wound edema and congestion.

Which of the following is the most likely responsible mediator?

A. Serotonin
B. Bradykinin
C. Prostaglandin I2
D. Thromboxane A2

Explanation:

B. Bradykinin is the most likely mediator responsible for the edema and congestion seen in **partial thickness burns**. **Bradykinin** plays a major role in the **inflammatory response**, contributing to **vasodilation**, **increased vascular permeability**, and **edema**.

Topic Section:

Burns and Inflammation

Definition:
Partial thickness burns involve the **epidermis and part of the dermis**, leading to **edema**, **pain**, and **congestion** due to **inflammatory mediators** like bradykinin.

Key Clinical Presentation:

- **Painful**, **red, blistered skin**
- **Edema** and **congestion** around the burn site

Investigations:

- Best initial: **Clinical assessment**
- Confirmatory: **Burn depth and severity assessment** (e.g., **Lund-Browder chart**)

Treatment Outline:

- First-line: **Fluid resuscitation** and **wound care**
- Second-line: **Topical antimicrobial agents**
- Definitive: **Surgical debridement** if necessary

Correct Answer:
B. Bradykinin ✅

MCQ 468

A 55-year-old diabetic woman with steroid-dependent bronchial asthma underwent a right hemicolectomy for carcinoma of the caecum. Postoperatively, she was kept in the Intensive Care Unit (see lab results).

Blood pressure: 80/40 mmHg
Heart rate: 140 /min
Respiratory rate: 25 /min
Temperature: 35.1 °C
Oxygen saturation: 95 %

Test Result Normal Values
Sodium: 131 134-146 mmol/L
Potassium: 6.0 3.5-5.1 mmol/L

Bicarbonate: 18 21-28 mmol/L
Random Glucose: 2.3 3.9-5.5 mmol/L

Which of the following is the most likely clinical condition?

A. Thyroid storm
B. Adrenal insufficiency
C. Diabetic ketoacidosis
D. Septic shock syndrome

Explanation:

B. Adrenal insufficiency is the most likely diagnosis. The patient is showing signs of **hypotension**, **electrolyte imbalances (elevated potassium, low sodium)**, **hypoglycemia**, and **fever**. These are suggestive of **adrenal insufficiency**, especially in a post-operative setting with steroid dependence, which can precipitate **acute adrenal crisis**.

Topic Section:

Adrenal Insufficiency (Adrenal Crisis)

Definition:
A potentially life-threatening condition caused by **insufficient cortisol production**, often due to **acute stress** or **steroid withdrawal**.

Key Clinical Presentation:

- **Hypotension**, **shock**
- **Electrolyte abnormalities** (e.g., **hyperkalemia**, **hyponatremia**)
- **Hypoglycemia**, **fever**

Investigations:

- Best initial: **Serum cortisol level**
- Confirmatory: **ACTH stimulation test** (low cortisol response)

Treatment Outline:

- First-line: **IV hydrocortisone** and **fluids**
- Second-line: **Electrolyte correction** (e.g., potassium, sodium)

- Definitive: **Chronic steroid replacement** (e.g., hydrocortisone)

Correct Answer:
B. Adrenal insufficiency ✅

MCQ 468

A 35-year-old man sustained a right tibia fracture in a road traffic accident. He is being managed by closed reduction with cast but he develops swelling in his toes and pain.

Which of the following is the earliest alarming sign?

A. Pain
B. Foot drop
C. Diminished pulses
D. Paraesthesia between toes

Explanation:

D. Paraesthesia between toes is the earliest alarming sign of **compartment syndrome**. **Compartment syndrome** occurs when there is increased pressure within a closed compartment, affecting blood flow and nerve function. **Paraesthesia** (numbness and tingling) is an early sign of nerve compression, which precedes more severe symptoms like **pain** and **diminished pulses**.

Topic Section:

Compartment Syndrome

Definition:
Increased pressure within a confined anatomical space, leading to tissue ischemia and potential nerve and muscle damage.

Key Clinical Presentation:

- **Severe pain** disproportionate to injury
- **Swelling** and **tightness** in the affected compartment
- **Paraesthesia** and **diminished pulses** (late signs)

Investigations:

- Best initial: **Clinical assessment**
- Confirmatory: **Intracompartmental pressure measurement**

Treatment Outline:

- First-line: **Fasciotomy**
- Second-line: **Elevate limb** to reduce pressure
- Definitive: Surgical release to relieve pressure

Correct Answer:
D. Paraesthesia between toes ✅

MCQ 469

A 35-year-old woman is concerned about her mother's history of invasive breast cancer and sister with ovarian cancer. Her clinical examination was unremarkable.

Which of the following is the most appropriate breast cancer screening?

A. Ultrasound
B. Mammogram
C. BRCA gene test
D. Breast self-examination

Explanation:

C. BRCA gene test is the most appropriate breast cancer screening for this patient with a **family history of breast cancer** and ovarian cancer. The **BRCA1** and **BRCA2** mutations significantly increase the risk of both **breast and ovarian cancers**, so genetic testing is indicated for patients with **strong family histories** of these cancers.

Topic Section:

Breast Cancer Screening

Definition:
Screening for breast cancer to detect early-stage malignancy, especially in high-risk individuals (family history, genetic predisposition).

Key Clinical Presentation:

- **Family history of breast or ovarian cancer**
- **Unremarkable physical examination**
- Concern for **genetic risk factors** (e.g., BRCA mutations)

Investigations:

- Best initial: **BRCA gene test**
- Confirmatory: **Mammogram** (in older patients) or **Ultrasound** (in younger patients)

Treatment Outline:

- First-line: **Genetic counseling** and **BRCA gene testing**
- Second-line: **Regular screening** (e.g., mammograms, breast MRI)
- Definitive: **Preventive measures** (e.g., prophylactic mastectomy, oophorectomy)

Correct Answer:
C. BRCA gene test ✅

MCQ 470

A 55-year-old man presents to the Emergency Department with severe abdominal pain and absolute constipation for 2 days associated with nausea and vomiting. He has a significant past history of constipation requiring use of laxatives. On examination, his abdomen is distended, tense, tender all over with a palpable mass in the left lumbar and hypochondrium (see image).

Which of the following is the most likely site of volvulus?

A. Caecum
B. Rectum
C. Sigmoid colon
D. Transverse colon

Explanation:

C. Sigmoid colon is the most common site for **volvulus**, especially in older adults with a history of constipation. **Sigmoid volvulus** occurs when the **sigmoid colon twists on its mesentery**, causing **obstruction** and **ischemia**. This is common in patients with **chronic constipation**.

Topic Section:

Colonic Volvulus

Definition:
A condition where a segment of the colon twists around its mesenteric axis, leading to bowel obstruction and ischemia.

Key Clinical Presentation:

- **Severe abdominal pain** and **distention**
- **Constipation** and **vomiting**
- **Tense, tender abdomen** with a palpable mass (commonly in the **left lower abdomen**)

Investigations:

- Best initial: **Abdominal X-ray** (showing distended bowel loops)
- Confirmatory: **CT scan** or **contrast enema** to identify the site of the volvulus

Treatment Outline:

- First-line: **Endoscopic detorsion**
- Second-line: **Surgical intervention** (if failed detorsion or recurrent volvulus)

Correct Answer:
C. Sigmoid colon ✅

MCQ 471

A 19-year-old married woman complains of a 6-hour history of right lower abdominal pain and nausea. On examination, she is tender in her right lower abdomen with rebound tenderness. Her LMP was 2 weeks back (see lab results).

Which of the following is the most likely diagnosis?

A. Ureterocele
B. Ovarian torsion
C. Honeymoon cystitis
D. Pelvic inflammatory disease

Explanation:

B. Ovarian torsion is the most likely diagnosis. **Ovarian torsion** causes **acute onset lower abdominal pain** and can present with nausea and vomiting. The **absence of fever** and the timing of the patient's **LMP** makes **pelvic inflammatory disease** less likely. **Ovarian torsion** often occurs in women of reproductive age and can be caused by ovarian cysts or **abnormal ovarian mobility**.

Topic Section:

Ovarian Torsion

Definition:
Twisting of the ovary around its supporting ligaments, causing ischemia and pain.

Key Clinical Presentation:

- **Acute onset** lower abdominal pain
- **Nausea**, **vomiting**, and **fever** (less common)
- **Pelvic tenderness**, **rebound tenderness**
- **Absence of fever** and **LMP history** important clues

Investigations:

- Best initial: **Pelvic ultrasound** with **Doppler** to assess ovarian blood flow
- Confirmatory: **Laproscopy** for definitive diagnosis

Treatment Outline:

- First-line: **Surgical detorsion** (ovarian sparing)
- Second-line: **Oophorectomy** if the ovary is nonviable

Correct Answer:
B. Ovarian torsion ✅

MCQ 472

A 75-year-old diabetic hypertensive man underwent an exploratory laparotomy for a perforated duodenal ulcer. Postoperatively, he was kept in the Intensive Care Unit with ventilator support. His cardiac output is 2.5L/min and pulmonary capillary wedge pressure is elevated to 20 mm Hg.

Which of the following is the most likely type of shock?

A. Septic
B. Cardiogenic
C. Hypovolemic
D. Anaphylactic

Explanation:

B. Cardiogenic shock is the most likely diagnosis. The patient's **low cardiac output** and **elevated pulmonary capillary wedge pressure** (a marker of left ventricular dysfunction) are characteristic of **cardiogenic shock**. **Hypovolemic shock** would typically show **low preload**, and **septic shock** is often associated with **decreased systemic vascular resistance**.

Topic Section:

Cardiogenic Shock

Definition:
A condition where the heart fails to pump sufficient blood, leading to **hypoperfusion** of organs.

Key Clinical Presentation:

- **Low cardiac output** and **elevated preload** (PCWP)

- **Hypotension** and **poor perfusion** despite adequate volume status
- **Tachycardia**, **weak pulses**, **cool extremities**

Investigations:

- Best initial: **Echocardiography** to assess heart function
- Confirmatory: **Cardiac catheterization** for coronary assessment

Treatment Outline:

- First-line: **Inotropic support** (e.g., **dobutamine**)
- Second-line: **Mechanical circulatory support** (e.g., intra-aortic balloon pump)
- Definitive: Treat underlying cause (e.g., **revascularization** in case of **myocardial infarction**)

Correct Answer:
B. Cardiogenic ✅

MCQ 473

A 13-year-old boy presents for elective surgery. He has a significant past history of bleeding tendency and has contracture of right knee and hemarthrosis.

Which of the following is the most likely underlying clinical disorder?

A. Haemophilia
B. Aplastic anaemia
C. Wilson's disease
D. Henoch-Schönlein purpura

Explanation:

A. Haemophilia is the most likely diagnosis. **Haemophilia** is an inherited bleeding disorder due to deficiency of clotting factors (factor VIII or IX). It commonly presents with **hemarthrosis** (bleeding into joints) and spontaneous bleeding, especially in **children**. The history of **bleeding tendency** and **contracture** (due to repeated joint bleeds) points towards this diagnosis.

Topic Section:

Haemophilia

Definition:
An X-linked recessive bleeding disorder due to deficiency of clotting factors (Hemophilia A - factor VIII deficiency; Hemophilia B - factor IX deficiency).

Key Clinical Presentation:

- **Spontaneous bleeding** or **after minor trauma**
- **Hemarthrosis** (bleeding into joints), especially knees, elbows, and ankles
- **Contractures** and deformities due to repeated joint bleeds

Investigations:

- Best initial: **Coagulation profile** (prolonged aPTT)
- Confirmatory: **Factor VIII or IX assay**

Treatment Outline:

- First-line: **Factor replacement therapy** (IV infusion of missing clotting factor)
- Second-line: **Desmopressin** (for mild Hemophilia A)
- Definitive: **Gene therapy** (emerging treatment)

Correct Answer:
A. Haemophilia ✅

MCQ 474

A 15-year-old girl underwent an exploratory laparotomy for a perforated appendix. On the 7th postoperative day, she remained febrile and developed respiratory distress requiring mechanical ventilation. On examination, her abdomen was distended, tense, and tender. A contrast-enhanced computed tomography (CT) was ordered. Post-CT, she developed bleeding from her mouth, trachea, and I/V puncture site.

Which of the following is the most likely diagnosis?

A. Haemophilia
B. Anaphylactic contrast reaction

C. Idiopathic thrombocytopenic purpura
D. Disseminated intravascular coagulation

Explanation:

D. Disseminated intravascular coagulation (DIC) is the most likely diagnosis. **DIC** is a condition characterized by **widespread clotting** and **bleeding**, often seen in critically ill patients, including those with sepsis or infection. The **fever, respiratory distress**, and **post-contrast bleeding** are typical of **DIC**, which is a **complication** that can occur after **severe infections** or **trauma**. The **CT scan findings** of abdominal tenderness and distension are consistent with a **complicated postoperative infection**.

Topic Section:

Disseminated Intravascular Coagulation (DIC)

Definition:
A life-threatening condition involving systemic activation of the coagulation cascade, leading to both **thrombosis** and **bleeding**.

Key Clinical Presentation:

- **Bleeding** from multiple sites (e.g., mouth, puncture sites, mucosal membranes)
- **Fever**, **respiratory distress**, and **organ dysfunction**
- **Abdominal distension** and **tenderness** (in cases of sepsis)

Investigations:

- Best initial: **D-dimer** and **PT/aPTT** (prolonged clotting times)
- Confirmatory: **Fibrinogen levels** (low) and **platelet count** (low)

Treatment Outline:

- First-line: **Supportive care**, including **anticoagulation** and **platelet transfusion**
- Second-line: **Treatment of underlying cause** (e.g., infection, sepsis)
- Definitive: **Addressing the underlying disorder** (e.g., sepsis, trauma)

Correct Answer:

D. Disseminated intravascular coagulation ✅

MCQ 475

A 55-year-old woman complains of abdominal pain, distention, and bilious vomiting. On examination, she is pale, dehydrated, and her abdomen is soft, distended, with hyperdynamic gut sounds. She was diagnosed with gallstone disease 10 years ago (see report).

Which of the following is the most likely diagnosis?

A. Cholangitis
B. Pancreatitis
C. Cholecystitis
D. Gallstone ileus

Explanation:

D. Gallstone ileus is the most likely diagnosis. **Gallstone ileus** occurs when a gallstone passes through a fistula between the gallbladder and the intestine, leading to bowel obstruction. The **history of gallstone disease**, **bilious vomiting**, and **abdominal distension** with **hyperdynamic gut sounds** suggest an obstruction. The **CT scan findings** of air in the biliary system further confirm this diagnosis.

Topic Section:

Gallstone Ileus

Definition:
A rare complication of **cholelithiasis** where a gallstone passes into the intestinal tract, causing **bowel obstruction**.

Key Clinical Presentation:

- **Abdominal pain**, **distention**, and **bilious vomiting**
- **History of gallstone disease**
- **Hyperdynamic gut sounds** and signs of bowel obstruction

Investigations:

- Best initial: **Abdominal X-ray** (showing air in the biliary system)
- Confirmatory: **CT scan abdomen**

Treatment Outline:

- First-line: **Surgical removal of the stone**
- Second-line: **Endoscopic or percutaneous intervention** if surgery is not feasible
- Definitive: **Cholecystectomy** to prevent recurrence

Correct Answer:
D. Gallstone ileus ✅

MCQ 476

A 55-year-old hypertensive diabetic with ischemic heart disease woman presented with sudden onset of left-sided lower abdominal pain associated with vomiting for 6 hours. On examination, she is pale, dehydrated, and flushed. Peripheral pulses are regularly irregular. Her abdomen is distended, tender all over (see lab results).

Which of the following is the most likely diagnosis?

A. Diverticulitis
B. Ulcerative colitis
C. Perforated viscus
D. Mesenteric ischemia

Explanation:

D. Mesenteric ischemia is the most likely diagnosis. **Mesenteric ischemia** presents with **acute abdominal pain**, particularly in patients with **cardiac risk factors** such as **ischemic heart disease**, **hypertension**, and **diabetes**. The **sudden onset of pain**, **abdominal distention**, and **dehydration** with **irregular peripheral pulses** are indicative of **poor perfusion to the intestines**.

Topic Section:

Mesenteric Ischemia

Definition:
A condition caused by reduced blood flow to the intestines, leading to **ischemic damage** and **potential bowel infarction**.

Key Clinical Presentation:

- **Acute abdominal pain**, often **severe and sudden onset**
- **Dehydration** and **abdominal distention**
- **Risk factors**: **Cardiac disease**, **hypertension**, and **diabetes**

Investigations:

- Best initial: **CT angiography** of the mesenteric vessels
- Confirmatory: **Mesenteric angiography**

Treatment Outline:

- First-line: **Surgical exploration** to assess bowel viability
- Second-line: **Embolectomy or bypass** for mesenteric artery blockage
- Definitive: **Revascularization** and addressing the underlying cause (e.g., atrial fibrillation)

Correct Answer:
D. Mesenteric ischemia ✅

MCQ 476

A 65-year-old diabetic man complains of intermittent claudication of his right leg. The foot and sole are cold and numb. He underwent right lower limb angiography showing diffuse disease. 2 days later he developed a painful swelling in his right groin. On examination, there is a firm swelling below the inguinal crease, partially mobile, irreducible, non-expansile, and tender with transmissible pulse.

Which of the following is the most likely diagnosis?

A. Psoas abscess
B. Saphena varix
C. Femoral hernia
D. Pseudoaneurysm

Explanation:

D. Pseudoaneurysm is the most likely diagnosis. The description of a **firm, irreducible, non-expansile swelling** with **transmissible pulse** and the patient's history of **angiography** indicates a **pseudoaneurysm**. This occurs when the arterial wall is damaged, causing blood to leak out and form a hematoma that communicates with the artery, creating a false aneurysm.

Topic Section:

Pseudoaneurysm

Definition:
A **false aneurysm** is the collection of blood between the layers of the arterial wall due to a breach, communicating with the arterial lumen.

Key Clinical Presentation:

- **Pulsatile swelling** (transmissible pulse)
- **Irreducible, non-expansile mass**
- Associated with **trauma** or **vascular interventions** (angiography, catheterization)

Investigations:

- Best initial: **Ultrasound Doppler**
- Confirmatory: **CT angiography**

Treatment Outline:

- First-line: **Ultrasound-guided compression** or **endovascular repair**
- Second-line: **Surgical repair** if the lesion is large or refractory

---

Correct Answer:
D. Pseudoaneurysm ✅

MCQ 477

A 33-year-old woman underwent surgery for her right leg varicose veins. On the 2nd postoperative day, she complained of pain and paraesthesia on the medial aspect of her leg and foot. On examination, there is sensory loss of fine touch and pain in her right leg and foot, but no motor loss. Wounds are healthy.

Which of the following nerves is most likely injured?

A. Sciatic
B. Femoral
C. Obturator
D. Saphenous

Explanation:

D. Saphenous nerve is most likely injured. The **saphenous nerve**, which is a branch of the femoral nerve, supplies sensation to the medial aspect of the leg and foot. The description of **sensory loss** on the **medial side** with **no motor loss** suggests **saphenous nerve injury**, a common complication after varicose vein surgery.

Topic Section:

Saphenous Nerve Injury

Definition:
A sensory nerve injury often associated with **surgical procedures** around the **femoral artery** or **saphenous vein**, leading to loss of sensation along the medial leg and foot.

Key Clinical Presentation:

- **Sensory loss** along the **medial leg** and **foot**
- **No motor involvement**

Investigations:

- Best initial: **Clinical examination** and **nerve conduction study** if needed

Treatment Outline:

- First-line: **Conservative management** (rest, analgesia, compression)
- Second-line: **Nerve block** or **surgical repair** if symptoms persist

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Correct Answer:
D. Saphenous ✅

MCQ 478

A 25-year-old complains of an inability to extend her fingers and numbness over her left wrist. On examination, there is paraesthesia of the skin over the left wrist and anatomical snuffbox, and an inability to extend her metacarpophalangeal joint. She also demonstrates wrist drop. Clinical injury to radial nerve is suspected.

Which of the following is the most likely site of injury?

A. In the axilla
B. Carpal tunnel
C. Olecranon process
D. Spiral groove of humerus

Explanation:

D. Spiral groove of humerus is the most likely site of injury. The **radial nerve** is most commonly injured in the **spiral groove** of the **humerus**, leading to **wrist drop** and **sensory deficits** in the **anatomical snuffbox**. The other options are less likely to cause this specific clinical presentation.

Topic Section:

Radial Nerve Injury

Definition:
Radial nerve injury can lead to **wrist drop** and **sensory loss** in the **dorsal hand** and **forearm**.

Key Clinical Presentation:

- **Wrist drop** (inability to extend wrist and fingers)
- **Paraesthesia** over the **anatomical snuffbox**
- Injury in the **spiral groove** of the **humerus**

Investigations:

- Best initial: **Clinical examination**
- Confirmatory: **Nerve conduction studies**

Treatment Outline:

- First-line: **Conservative management** (splinting, physical therapy)
- Second-line: **Surgical exploration** if conservative management fails

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Correct Answer:
D. Spiral groove of humerus ✅

MCQ 479

A 25-year-old woman is brought to the Emergency Department after being rescued from a burning house after 2 hours. She is drowsy, and covered in black soot. On examination, she has 35% burns and nostrils and mouth are filled with soot. After initial resuscitation, her carboxyhemoglobin levels are raised (40%).

Which of the following is the most appropriate management?

A. Exchange transfusion
B. Hyperbaric oxygen therapy
C. Carbonic anhydrase inhibitor therapy
D. Intubation and ventilation with 100% oxygen

Topic Section:

Carbon Monoxide Poisoning

Definition:
Carbon monoxide (CO) poisoning is caused by inhaling CO gas, leading to hypoxia due to CO binding more strongly to hemoglobin than oxygen.

Key Clinical Presentation:

- **Drowsiness**, **confusion**, **headache**, or **coma**
- **Soot in the mouth and nose** (after fire inhalation)
- **Elevated carboxyhemoglobin levels** (typically >10%)

Investigations:

- Best initial: **Carboxyhemoglobin levels**
- Confirmatory: **Clinical history** of exposure (e.g., house fire, carbon monoxide source)

Treatment Outline:

- First-line: **Intubation and ventilation with 100% oxygen**
- Second-line: **Hyperbaric oxygen therapy** (for severe cases or when symptoms persist despite 100% oxygen)

Correct Answer:
D. Intubation and ventilation with 100% oxygen ✅

MCQ 480

A 40-year-old presents with a right breast lump that has been present for 2 years. On examination, there is a firm lump about 4 x 5 cm in the inner quadrant, overlying skin is normal, and no palpable axillary nodes (see report).
Trucut biopsy: Cystosarcoma Phylloides.
Which of the following is the most appropriate management?

A. Chemotherapy
B. Radiation therapy
C. Wide local excision
D. Simple mastectomy

Explanation:
Cystosarcoma phylloides, a rare breast tumor, requires surgical excision. The preferred approach is **wide local excision**, with clear margins to minimize the risk of recurrence. In some cases, **mastectomy** may be necessary, but it is typically reserved for larger or more aggressive tumors.

Correct Answer: C. Wide local excision ✅

Topic Section:

Cystosarcoma Phylloides (Cystosarcoma Phylloides)

Definition:
A rare fibroepithelial breast tumor that can be benign, borderline, or malignant. Characterized by the presence of epithelial and stromal components.

Key Clinical Presentation:

- Firm, painless, mobile breast lump
- Rapidly enlarging mass (though slow-growing)
- No signs of skin involvement or axillary lymphadenopathy

Investigations:

- **Best initial**: Clinical examination, ultrasound, biopsy (Trucut or core needle)
- **Confirmatory**: Histopathology

Treatment Outline:

- **First-line**: Wide local excision
- **Second-line**: Mastectomy if margins cannot be achieved or the tumor is large/aggressive
- **Definitive**: Close follow-up for recurrence, particularly if borderline or malignant

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MCQ 481

A 55-year-old man complains of dark stool and weakness. He has peptic ulcer disease and endoscopy investigations have confirmed multiple ulcers in the antrum. He has been kept on medical therapy. On examination, his abdomen is soft and non-tender. Repeat endoscopy confirms multiple ulcers in the antrum (see lab results).
Test Result Normal Values
Hb 58 130-170 g/L
HCT 0.18 0.42-0.52
Which of the following is the most appropriate management?

A. Total gastrectomy
B. Partial distal gastrectomy
C. Endoscopic laser photocoagulation
D. Truncal vagotomy and pyloroplasty

Explanation:
This patient is presenting with **gastric bleeding** due to peptic ulcer disease and has significant

anemia. The most appropriate management in this case would be **truncal vagotomy and pyloroplasty**, which addresses the underlying cause (gastric acid secretion) and prevents further ulcer formation and bleeding.

Correct Answer: D. Truncal vagotomy and pyloroplasty ✅

Topic Section:

Peptic Ulcer Disease (PUD)

Definition:
A condition characterized by the formation of ulcers in the stomach or duodenum due to acid or pepsin digestion of the mucosal lining.

Key Clinical Presentation:

- Epigastric pain, typically relieved by food or antacids
- Dark stools (melena) due to bleeding
- Fatigue and weakness from anemia (as seen in this patient)

Investigations:

- **Best initial**: Endoscopy
- **Confirmatory**: Histology and biopsies if malignancy is suspected

Treatment Outline:

- **First-line**: Proton pump inhibitors (PPIs), H. pylori eradication
- **Second-line**: Surgery (if complications such as bleeding or perforation occur)
- **Definitive**: Truncal vagotomy and pyloroplasty in refractory or complicated cases

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MCQ 481

A 22-year-old complains of a 2-day history of right-sided chest pain and dyspnea. It started during landing from a long flight. On examination, he is tall, breath sounds are absent on the right side and percussion note is resonant (see report).
CXR: Right pneumothorax.
Which of the following is the most appropriate initial management?

A. Thoracentesis
B. Tube thoracostomy
C. Thoracoscopic blebectomy
D. Conservative management

Explanation:
The initial management of **simple pneumothorax** in a patient with stable vital signs is **conservative management**, including observation and oxygen therapy. In cases of large pneumothorax or symptoms worsening, **tube thoracostomy** may be required.

Correct Answer: D. Conservative management ✅

Topic Section:

Pneumothorax

Definition:
The presence of air in the pleural space causing lung collapse. Can be spontaneous, traumatic, or secondary to underlying lung disease.

Key Clinical Presentation:

- Sudden-onset pleuritic chest pain
- Dyspnea
- Reduced breath sounds on the affected side

Investigations:

- **Best initial**: Chest X-ray
- **Confirmatory**: CT chest (in uncertain or complicated cases)

Treatment Outline:

- **First-line**: Observation for small pneumothoraces
- **Second-line**: Chest tube placement (tube thoracostomy) for larger pneumothoraces
- **Definitive**: Surgery or thoracoscopic blebectomy for recurrent or persistent pneumothorax

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MCQ 482

A 70-year-old woman is bed bound due to basal ganglia bleed and is having difficulty swallowing and has experienced significant weight loss. On examination, she is emaciated, and lethargic. Weak muscles of mastication and an absent gag reflex.
Which of the following is the most appropriate option to establish feeding?

A. Gastrostomy tube
B. Jejunostomy tube
C. Parenteral nutrition
D. Nasogastric tube feeding

Explanation:
In this case, **gastrostomy tube** is the most appropriate option for long-term feeding in a patient with difficulty swallowing due to neurological impairment. It provides a safe and reliable route for nutrition in patients with chronic swallowing dysfunction.

Correct Answer: A. Gastrostomy tube ✅

Topic Section:

Dysphagia and Feeding Tubes

Definition:
Dysphagia refers to difficulty swallowing, which can be due to a variety of causes, including neurological impairment, esophageal motility disorders, or structural abnormalities.

Key Clinical Presentation:

- Difficulty swallowing solid or liquid foods
- Weight loss due to poor intake
- Aspiration risk, particularly in neurologically impaired patients

Investigations:

- **Best initial**: Clinical assessment, swallowing assessment, and endoscopy if required
- **Confirmatory**: Video fluoroscopy or manometry

Treatment Outline:

- **First-line**: Nutritional support via feeding tubes (e.g., gastrostomy tube for long-term use)
- **Second-line**: Parenteral nutrition (for short-term support in the hospital setting)

- **Definitive**: Surgical intervention if necessary, such as correction of anatomical defects or therapy for underlying conditions

MCQ 483

A 37-year-old woman complains of abdominal fullness, epigastric discomfort, and loss of appetite. She is recovering from an episode of acute pancreatitis 5 weeks ago. On examination, her abdomen is distended, and there is vague fullness in the epigastrium (see report).
CT scan:
Cyst about 24 x 18 cm in the lesser sac.
Which of the following is the most appropriate management?

A. Excision of cyst
B. Surgical drainage
C. Image-guided aspiration
D. Endoscopic Internal drainage

Explanation:
The presence of a large pancreatic cyst (likely a pseudocyst) in a patient recovering from acute pancreatitis suggests that it needs to be managed with **surgical drainage** if symptomatic. **Image-guided aspiration** is a temporary solution and can be used if there are concerns about the cyst's nature or complications. Surgical intervention, such as drainage, is the most appropriate in this setting.

Correct Answer: B. Surgical drainage ✅

Topic Section:

Pancreatic Pseudocyst (Pancreatic Pseudocyst)

Definition:
A fluid collection encapsulated by a wall of fibrous tissue, often a sequelae of acute pancreatitis, characterized by a lack of epithelial lining.

Key Clinical Presentation:

- Abdominal pain
- Nausea, vomiting
- Abdominal fullness and distension

Investigations:

- **Best initial**: CT scan or ultrasound for cyst size and location
- **Confirmatory**: Endoscopic ultrasound or MRI for further characterization

Treatment Outline:

- **First-line**: Surgical drainage, especially in symptomatic cases or large cysts
- **Second-line**: Percutaneous drainage or endoscopic drainage in select cases
- **Definitive**: Surgical resection if recurrent or complicated

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MCQ 484

A 65-year-old smoker with diabetes mellitus presents with a long history of intermittent buttock pain and claudication that has worsened over the past few weeks. On examination, there is atrophy of buttock muscles with decreased power and painful movements. The femoral pulses are weak (see report).
Angiogram:
Aortolilac insufficiency.
Which of the following is the most appropriate management?

A. Vasodilator therapy
B. Aortofemoral bypass
C. Axillobifemoral bypass
D. Lumbar sympathectomy

Explanation:
In a patient with **aortoiliac insufficiency**, the most appropriate management is **aortofemoral bypass** to restore blood flow to the lower extremities. This procedure is typically indicated for patients with significant symptoms or non-healing wounds due to peripheral arterial disease.

Correct Answer: B. Aortofemoral bypass ✅

Topic Section:

Aortoiliac Disease (Aortoiliac Disease)

Definition:
Atherosclerotic narrowing or blockage of the aorta and iliac arteries, leading to impaired blood flow to the lower limbs.

Key Clinical Presentation:

- Intermittent claudication, particularly in the buttocks and thighs
- Weak or absent femoral pulses
- Muscle atrophy in the affected leg

Investigations:

- **Best initial**: Doppler ultrasound, angiogram
- **Confirmatory**: CT angiography or MRI angiography

Treatment Outline:

- **First-line**: Surgical intervention (bypass surgery, angioplasty)
- **Second-line**: Medical therapy with vasodilators and antiplatelet agents
- **Definitive**: Bypass surgery for critical cases

MCQ 485

A 45-year-old presents after a 50 feet fall. On examination, he looks pale, peripheral pulses are feeble, and his chest has decreased breath sounds on the left side (see image and report).
Blood pressure 100/60 mmHg
Heart rate 100 /min
Respiratory rate 20 /min
Temperature 36.6 °C
X-ray trauma series:
Left dislocated hip and calcaneal fracture.
Which of the following is the most appropriate management?

A. Left tube thoracostomy
B. Emergency thoracotomy
C. Aortography and stenting
D. Intubation and ventilation

Explanation:
Given the mechanism of trauma and presentation with **decreased breath sounds**, the most likely diagnosis is **hemothorax**. The **initial management** involves **tube thoracostomy** to evacuate blood and air from the pleural space, thereby restoring normal lung expansion and preventing respiratory distress.

Correct Answer: A. Left tube thoracostomy ✅

Topic Section:

Hemothorax (Hemothorax)

Definition:
Collection of blood in the pleural space, commonly caused by trauma, leading to impaired lung expansion.

Key Clinical Presentation:

- Dyspnea
- Decreased breath sounds on the affected side
- Tachypnea, hypotension if severe

Investigations:

- **Best initial**: Chest X-ray
- **Confirmatory**: CT scan if needed for further evaluation

Treatment Outline:

- **First-line**: Tube thoracostomy for drainage
- **Second-line**: Surgical intervention if bleeding persists or is massive
- **Definitive**: Control of the underlying cause, including surgery if needed

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MCQ 486

A 25-year-old woman that is taking steroids for inflammatory bowel disease complains of abdominal pain and bilious vomiting. On examination, her abdomen is distended and tender in the right iliac fossa. She had a colonoscopy 2 weeks ago, which was normal. Contrast barium study was ordered (see report).
Blood pressure 110/70 mmHg
Heart rate 100 /min
Barium study:
Single stricture at terminal ileum, 1 cm from ileocecal valve.
Which of the following is the most appropriate management?

A. Stricturoplasty
B. Right hemicolectomy
C. Conservative medical management
D. Segmental resection with ileostomy

Explanation:
The **stricture** seen at the **terminal ileum** is most likely a **result of Crohn's disease**, especially in a patient with **inflammatory bowel disease**. **Stricturoplasty** is the treatment of choice for non-complicated strictures, as it preserves bowel length and function.

Correct Answer: A. Stricturoplasty ✅

Topic Section:

Crohn's Disease (Crohn's Disease)

Definition:
Chronic inflammatory bowel disease that can affect any part of the gastrointestinal tract, with a predilection for the ileum.

Key Clinical Presentation:

- Abdominal pain and cramping
- Diarrhea, weight loss, malnutrition
- Possible fistulas or strictures

Investigations:

- **Best initial**: Colonoscopy with biopsy
- **Confirmatory**: Imaging (CT, MRI) for complications

Treatment Outline:

- **First-line**: Medical management with corticosteroids and immunosuppressants
- **Second-line**: Surgical intervention (stricturoplasty or resection) for complications or refractory disease
- **Definitive**: Surgery for complications such as obstruction or perforation

MCQ 487

A 24-year-old woman complains of abdominal pain, sweating, and palpitations. She was recently diagnosed with hypertension and has had a low response to medical therapy. On examination, her abdomen is soft and non-tender (see lab result and report).
Blood pressure: 215/110 mmHg
Heart rate: 124/min
Respiratory rate: 21/min
Temperature: 38.6 °C
Test Result: Catecholamines 2100 (<591 nmol/day)
Ultrasound: 4 cm mass in the right adrenal cortex.
Which of the following is the most appropriate management for the hypertension?

A. Beta blockers
B. ACE inhibitors
C. Alpha blockers
D. Calcium channel blocker

Explanation:
The presence of a mass in the right adrenal cortex and markedly elevated catecholamine levels suggests **pheochromocytoma**, a catecholamine-secreting tumor. **Alpha blockers** (e.g., phenoxybenzamine) are the most appropriate initial management in pheochromocytoma to control hypertension before surgery.

Correct Answer: C. Alpha blockers ✅

Topic Section:

Pheochromocytoma (Pheochromocytoma)

Definition:
A rare catecholamine-secreting tumor usually arising from the adrenal medulla, causing episodic or sustained hypertension.

Key Clinical Presentation:

- Severe hypertension (episodic or sustained)
- Abdominal pain, sweating, palpitations
- Headache, tremor

Investigations:

- **Best initial**: Measurement of plasma metanephrines or 24-hour urinary catecholamines

- **Confirmatory**: CT or MRI of the abdomen for tumor localization

Treatment Outline:

- **First-line**: Alpha blockers (e.g., phenoxybenzamine) to control blood pressure
- **Second-line**: Beta blockers after alpha blockade for tachycardia control
- **Definitive**: Surgical resection of the tumor

MCQ 487

A 5-year-old child complains of a painful left testicular swelling. On examination, there is a firm swelling in his left hemi-scrotum, it is irreducible, with a negative transillumination test.
Which of the following is the most appropriate management?

A. Herniotomy
B. Orchidopexy
C. Herniorrhaphy
D. Tension-free repair

Explanation:
The child most likely has an **inguinal hernia** with incarcerated bowel or fat. **Herniotomy** (surgical repair of the hernia) is the most appropriate management for an irreducible inguinal hernia, as it involves removing the hernial sac and repairing the defect.

Correct Answer: A. Herniotomy ✅

Topic Section:

Inguinal Hernia (Inguinal Hernia)

Definition:
Protrusion of abdominal contents through the inguinal canal, often presenting as a swelling in the groin or scrotum.

Key Clinical Presentation:

- Swelling or bulge in the groin or scrotum
- Pain, especially with coughing or straining
- Irreducible swelling, sometimes associated with bowel obstruction

Investigations:

- **Best initial**: Physical examination and hernia reduction test
- **Confirmatory**: Ultrasound or CT scan if the diagnosis is uncertain

Treatment Outline:

- **First-line**: Herniotomy (surgical repair)
- **Second-line**: Watchful waiting in asymptomatic cases
- **Definitive**: Surgical repair with mesh or suture techniques

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MCQ 488

A 27-year-old woman complains of a 3-month history of right upper quadrant pain, fever, and malaise. On examination, she is febrile, her abdomen is soft and tender in the right upper quadrant. Indirect hemagglutination titer for *Echinococcus granulosus* is positive (see image).
Blood pressure: 110/70 mmHg
Heart rate: 95 /min
Temperature: 37.4 °C
Which of the following is the most appropriate management?

A. Albendazole therapy
B. Percutaneous drainage
C. Open cyst deroofing
D. Laparoscopic pericystectomy

Explanation:
The patient most likely has a **hydatid cyst** caused by *Echinococcus granulosus*. The most appropriate treatment for hydatid cysts involves **albendazole therapy** to reduce the risk of cyst rupture, along with surgical treatment in severe cases.

Correct Answer: A. Albendazole therapy ✅

Topic Section:

Hydatid Disease (Hydatid Disease)

Definition:
A parasitic infection caused by *Echinococcus granulosus*, commonly affecting the liver, lungs, or other organs.

Key Clinical Presentation:

- Abdominal pain, nausea, vomiting
- Jaundice in cases involving the liver
- Fever and malaise in acute infections

Investigations:

- **Best initial**: Serologic testing for *Echinococcus granulosus* antibodies
- **Confirmatory**: Ultrasound or CT scan to visualize cysts

Treatment Outline:

- **First-line**: Albendazole therapy for cyst management
- **Second-line**: Percutaneous drainage or surgery for cyst removal
- **Definitive**: Surgery (cyst deroofing or resection) in complicated cases

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MCQ 489

A 55-year-old patient is diagnosed with a gastrointestinal stromal tumor of the stomach (4 x 3 cm), confined to the body of the stomach, no surrounding infiltration, no distant metastases, or lymphadenopathy.
Which of the following is the most appropriate management?

A. Gastrectomy
B. Endoscopic ablation
C. Systemic chemo-radiation
D. Wedge resection with clear margins

Explanation:
For **gastrointestinal stromal tumors (GISTs)** that are localized, **wedge resection** is the most appropriate management, as it allows for removal of the tumor with clear margins. Systemic therapy is reserved for high-risk or metastatic disease.

Correct Answer: D. Wedge resection with clear margins ✅

Topic Section:

Gastrointestinal Stromal Tumor (GIST)

Definition:
A type of tumor arising from the interstitial cells of Cajal in the gastrointestinal tract, most commonly in the stomach.

Key Clinical Presentation:

- Abdominal pain, bloating
- Vomiting or weight loss in larger tumors
- Bleeding in some cases

Investigations:

- **Best initial**: Endoscopy, CT scan
- **Confirmatory**: Biopsy and immunohistochemistry for CD117 (KIT)

Treatment Outline:

- **First-line**: Surgical resection (wedge resection)
- **Second-line**: Imatinib (tyrosine kinase inhibitor) for advanced or metastatic disease
- **Definitive**: Complete surgical resection with clear margins

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MCQ 490

A 35-year-old man with gastroesophageal reflux disease presents to discuss his family history of oesophageal and gastric carcinoma.
Which of the following is the main risk factor for oesophageal carcinoma?

A. Smoking
B. Idiopathic
C. Barrett's oesophagus
D. Long-standing achalasia

Explanation:
Barrett's oesophagus, a condition in which the squamous epithelium of the esophagus is

replaced by columnar epithelium, is a major risk factor for **oesophageal adenocarcinoma**. It is caused by chronic acid reflux and is strongly linked to an increased risk of esophageal cancer.

Correct Answer: C. Barrett's oesophagus ✅

Topic Section:

Esophageal Carcinoma (Esophageal Carcinoma)

Definition:
Malignancy of the esophagus, which can be either squamous cell carcinoma or adenocarcinoma.

Key Clinical Presentation:

- Dysphagia (progressive)
- Weight loss
- Odynophagia (painful swallowing)

Investigations:

- **Best initial**: Endoscopy with biopsy
- **Confirmatory**: CT scan or endoscopic ultrasound for staging

Treatment Outline:

- **First-line**: Surgical resection (esophagectomy) for early-stage disease
- **Second-line**: Chemotherapy and radiation for advanced disease
- **Definitive**: Chemoradiation or surgery depending on the stage

MCQ 491

A 26-year-old woman complains of right upper quadrant pain and nausea. She is 20 weeks pregnant. On examination, her abdomen is soft and non-tender.
Ultrasound: Cholelithiasis.
Which of the following is the most appropriate management?

A. Cholecystostomy tube
B. Open cholecystectomy
C. Conservative treatment
D. Laparoscopic cholecystectomy

Explanation:
In a pregnant woman with symptomatic cholelithiasis, the first-line treatment is usually **conservative management** with monitoring and medical therapy. If symptoms persist or there are complications, **laparoscopic cholecystectomy** may be considered in the second trimester, as it is the safest during this time.

Correct Answer: C. Conservative treatment ✅

Topic Section:

Cholelithiasis in Pregnancy (Cholelithiasis)

Definition:
Presence of gallstones in the gallbladder, which may or may not cause symptoms.

Key Clinical Presentation:

- Right upper quadrant pain, nausea, vomiting
- Jaundice (in severe cases)
- Often presents during pregnancy or in patients with obesity, diabetes, or certain medications.

Investigations:

- **Best initial**: Ultrasound of the abdomen
- **Confirmatory**: Cholescintigraphy (if diagnosis is uncertain)

Treatment Outline:

- **First-line**: Conservative management (pain control, observation, diet modifications)
- **Second-line**: Laparoscopic cholecystectomy (safe during the second trimester)
- **Definitive**: Surgery in symptomatic cases or if complications like cholecystitis or pancreatitis occur

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MCQ 492

A 20-year-old woman presents with fever and pain in her right leg. On examination, her right leg is swollen, tender with multiple red streaks, and palpable inguinal nodes.
Which of the following is the most likely causative organism?

A. Bacteroides
B. Enterococcus
C. Klebsiella pneumoniae
D. Streptococcus pyogenes

Explanation:
The clinical presentation of fever, leg pain, redness, and streaks with inguinal lymphadenopathy is suggestive of **cellulitis**, which is most commonly caused by **Streptococcus pyogenes** (Group A Streptococcus).

Correct Answer: D. Streptococcus pyogenes ✅

Topic Section:

Cellulitis (Cellulitis)

Definition:
A common, often superficial skin infection, typically caused by **Streptococcus pyogenes** or **Staphylococcus aureus**.

Key Clinical Presentation:

- Red, swollen, warm, and painful area of skin
- Fever and malaise in severe cases
- Lymphangitis or lymphadenopathy (if the infection spreads)

Investigations:

- **Best initial**: Clinical diagnosis based on physical examination
- **Confirmatory**: Blood cultures (if systemic infection is suspected)

Treatment Outline:

- **First-line**: Antibiotics targeting Gram-positive organisms (e.g., penicillin or cephalexin)
- **Second-line**: Vancomycin or clindamycin for MRSA
- **Definitive**: Drainage of abscesses, if present

MCQ 493

Which of the following is most commonly associated with retroperitoneal sarcoma?

A. Retroperitoneal bleeding
B. Systemic lymphadenopathy
C. Invasion into surrounding tissues
D. Compression of surrounding tissues

Explanation:
Retroperitoneal sarcomas typically **compress surrounding tissues**, as they grow in the confined space of the retroperitoneum. These tumors are often asymptomatic in the early stages and can present with nonspecific symptoms such as abdominal pain or a palpable mass.

Correct Answer: D. Compression of surrounding tissues ✅

Topic Section:

Retroperitoneal Sarcoma (Retroperitoneal Sarcoma)

Definition:
A rare malignant tumor arising from the mesenchymal tissues of the retroperitoneum.

Key Clinical Presentation:

- Often asymptomatic early on
- Abdominal pain, fullness, or mass in the abdomen
- Nonspecific symptoms (e.g., weight loss)

Investigations:

- **Best initial**: CT or MRI of the abdomen
- **Confirmatory**: Biopsy of the mass

Treatment Outline:

- **First-line**: Surgical resection with clear margins
- **Second-line**: Radiation therapy (if the tumor is unresectable)
- **Definitive**: Complete surgical resection

MCQ 494

A 45-year-old patient complains of chills and rigors. He underwent a common bile duct exploration a few hours ago (see lab results).
Blood pressure: 110/70 mmHg
Heart rate: 80 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Test Result:

- Hb: 130 (Normal)
- WBC: 9.4 (Normal)
- Neutrophils: 50% (Normal)
  Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
Post-procedure fever with chills and rigors can indicate **sepsis**, which is an infection that has spread to the bloodstream and is causing a systemic inflammatory response. This can be a complication after procedures like bile duct exploration.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
A systemic inflammatory response to infection, leading to organ dysfunction and potential organ failure.

Key Clinical Presentation:

- Fever, chills, and rigors
- Tachycardia, hypotension
- Organ dysfunction (e.g., respiratory distress, renal failure)

Investigations:

- **Best initial**: Blood cultures, urinalysis, chest X-ray
- **Confirmatory**: Cultures of the suspected infection source

Treatment Outline:

- **First-line**: Broad-spectrum antibiotics (based on the suspected source)
- **Second-line**: Source control (e.g., drainage of abscesses, removal of infected devices)
- **Definitive**: Supportive care in an ICU setting if organ failure occurs

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MCQ 495

A 45-year-old man is complaining of chills and rigors. He underwent an elective subtotal colectomy for ulcerative colitis 12 hours ago (see lab results).
Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Test Result:

- Hb: 110
- HCT: 0.40
- WBC: 15
- Neutrophils: 60%
  Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
This patient is presenting with **SIRS (Systemic Inflammatory Response Syndrome)** following major surgery. The elevated white blood cell count, fever, and tachycardia are indicative of SIRS, a precursor to sepsis. While this patient is not in septic shock yet, these signs suggest he is at risk for sepsis.

Correct Answer: D. Systemic inflammatory response syndrome ✅

Topic Section:

Systemic Inflammatory Response Syndrome (SIRS)

Definition:
A systemic response to a variety of insults (e.g., infection, trauma, surgery) characterized by inflammation, tachycardia, fever, and leukocytosis.

Key Clinical Presentation:

- Fever or hypothermia
- Tachycardia, tachypnea
- Leukocytosis or leukopenia

Investigations:

- **Best initial**: Clinical diagnosis based on physical signs (e.g., fever, tachycardia)
- **Confirmatory**: None (diagnosis is based on clinical criteria)

Treatment Outline:

- **First-line**: Treat the underlying cause (e.g., infection, surgical complications)
- **Second-line**: Supportive care (e.g., fluids, oxygen therapy)
- **Definitive**: Management of the underlying cause (e.g., antibiotics for infection, surgical intervention for complications)

MCQ 496

A 65-year-old patient complains of chills and rigors 12 hours after undergoing removal of an infected ureteric stent. Prior to the cystoscopy, her urine culture was positive for E. Coli (see lab result).
Blood pressure: 110/70 mmHg
Heart rate: 96/min
Respiratory rate: 18/min
Temperature: 38.6 °C
Test Result:
WBC: 15.4 (Normal range: 4.5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
This patient has chills and rigors following the removal of an infected ureteric stent. The positive urine culture for **E. coli**, along with fever and elevated WBC, suggest **sepsis** (a systemic infection with potential organ dysfunction). While systemic inflammatory response syndrome (SIRS) can be present in the early stages of infection, the confirmed infection with E. coli makes **sepsis** the most likely diagnosis.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key Clinical Presentation:

- Fever, chills, and rigors
- Tachycardia, hypotension
- Organ dysfunction (e.g., respiratory failure, renal failure)
- Positive blood cultures

Investigations:

- **Best initial**: Blood cultures, urinalysis, chest X-ray
- **Confirmatory**: Cultures of the suspected infection source

Treatment Outline:

- **First-line**: Broad-spectrum antibiotics
- **Second-line**: Source control (e.g., drainage of abscesses, removal of infected devices)
- **Definitive**: ICU support if needed for organ failure

MCQ 497

A 65-year-old required intubation and transfer to the Intensive Care Unit after collapsing in the ward. 2 days previously, she underwent an operation for empyema of the gallbladder (see lab result).
Blood pressure: 110/70 mmHg
Heart rate: 110/min
Respiratory rate: 24/min
Temperature: 38.7 °C
Test Result:
WBC: 19.4 (Normal range: 4.5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
The patient, who has recently undergone surgery for empyema, is presenting with fever, tachycardia, and leukocytosis (elevated WBC). These findings, combined with a history of recent surgery, suggest **sepsis**. Since the patient has collapsed and required ICU transfer, this may indicate severe sepsis, but the most likely diagnosis given the available information is **sepsis** at this point.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key Clinical Presentation:

- Fever, chills, and rigors
- Tachycardia, hypotension

- Organ dysfunction (e.g., respiratory failure, renal failure)
- Positive blood cultures

Investigations:

- **Best initial**: Blood cultures, urinalysis, chest X-ray
- **Confirmatory**: Cultures of the suspected infection source

Treatment Outline:

- **First-line**: Broad-spectrum antibiotics
- **Second-line**: Source control (e.g., drainage of abscesses, removal of infected devices)
- **Definitive**: ICU support if needed for organ failure

MCQ 498

A 24-month-old baby is brought in with a right-sided red inflamed hemiscrotum. On exploration, the right testis is viable, however, the cord is thick and oedematous.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orthitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a red, inflamed hemiscrotum with a thick and oedematous cord suggests **testicular torsion**, a surgical emergency that typically presents with severe pain and swelling of the scrotum. The viability of the testis and the oedema of the cord are characteristic of this condition, which requires immediate surgical intervention.

Correct Answer: A. Testicular torsion ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency characterized by the twisting of the spermatic cord, cutting off the blood supply to the testis.

Key Clinical Presentation:

- Sudden onset of severe scrotal pain and swelling
- Nausea and vomiting
- Red, swollen scrotum
- Absent cremasteric reflex

Investigations:

- **Best initial**: Clinical diagnosis based on physical examination
- **Confirmatory**: Doppler ultrasound (to assess blood flow)

Treatment Outline:

- **First-line**: Immediate surgical exploration and detorsion
- **Second-line**: Orchidopexy to prevent recurrence
- **Definitive**: Surgery to salvage the testis and prevent future torsion

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MCQ 499

A 7-month-old baby is brought in with a red and inflamed left hemiscrotum. Examination confirms a firm, tender, irreducible lump in the scrotum extending to the groin. The left testis cannot be palpated.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orthitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a **firm, tender, irreducible lump** in the scrotum that extends into the groin, with the testis not palpable, is most likely an **incarcerated inguinal hernia**. This occurs when a portion of the intestine or other abdominal contents protrude through the inguinal canal and becomes trapped, leading to swelling and tenderness.

Correct Answer: C. Incarcerated inguinal hernia ✅

Topic Section:

Incarcerated Inguinal Hernia (Inguinal Hernia)

Definition:
A condition in which part of the intestine or other abdominal contents becomes trapped in the inguinal canal, causing swelling, pain, and risk of strangulation.

Key Clinical Presentation:

- Swelling or lump in the groin or scrotum
- Pain and tenderness, especially when coughing or straining
- Inability to reduce the lump manually
- Vomiting and signs of bowel obstruction in severe cases

Investigations:

- **Best initial**: Clinical examination
- **Confirmatory**: Ultrasound or CT scan (if needed for complex cases)

Treatment Outline:

- **First-line**: Immediate surgical intervention (herniorrhaphy)
- **Second-line**: Conservative management if reducible (but not for incarcerated cases)
- **Definitive**: Herniorrhaphy (surgical repair)

MCQ 500

A 12-year-old presents with left-sided lower abdominal pain. Examination confirms a slightly elevated, tender left testis with redness and oedema of the scrotum.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a tender, swollen, and red scrotum with an elevated testis in a young patient is most consistent with **epididymo-orchitis**, an infection of the epididymis and testis, commonly caused by a bacterial infection such as **E. coli** or **Chlamydia**. This condition is often associated with fever, dysuria, and urinary symptoms.

Correct Answer: B. Epididymo-orchitis ✅

Topic Section:

Epididymo-orchitis (Epididymo-orchitis)

Definition:
Infection of the epididymis and testis, often caused by bacterial pathogens such as **E. coli**, **Chlamydia**, and **Gonorrhea**.

Key Clinical Presentation:

- Scrotal pain, swelling, and redness
- Elevated testis with tenderness over the epididymis
- Urinary symptoms such as dysuria or frequency
- Fever

Investigations:

- **Best initial**: Clinical examination and urinalysis
- **Confirmatory**: Doppler ultrasound (to evaluate blood flow and exclude torsion)

Treatment Outline:

- **First-line**: Antibiotics (e.g., ciprofloxacin, doxycycline)
- **Second-line**: Pain management and scrotal elevation
- **Definitive**: Surgical drainage (if abscess formation)

MCQ 501

A 12-year-old presents with a 2-day history of right-sided scrotal pain. Examination confirms right testis with vertical orientation. There is an area of tenderness around the superior pole

and the scrotum is red and oedematous.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
This patient's symptoms of scrotal pain, a vertically oriented testis, redness, and oedema are most indicative of **testicular torsion**, a surgical emergency. The testis rotates around its spermatic cord, leading to compromised blood flow and acute pain. Immediate intervention is necessary to salvage the testis.

Correct Answer: A. Testicular torsion ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency caused by the twisting of the spermatic cord, leading to impaired blood flow to the testis.

Key Clinical Presentation:

- Sudden onset of severe scrotal pain
- Vomiting and nausea
- Swollen, tender scrotum
- Horizontal or vertical orientation of the testis
- Absence of the cremasteric reflex

Investigations:

- **Best initial**: Clinical examination and Doppler ultrasound (to assess blood flow)
- **Confirmatory**: Surgical exploration (if diagnosis is unclear)

Treatment Outline:

- **First-line**: Immediate surgical exploration and detorsion

- **Second-line**: Orchidopexy to prevent recurrence
- **Definitive**: Surgery to preserve the testis

MCQ 502

A 24-month-old is presented with redness of both hemi-scrotum. Examination confirms non-tender, redness and oedema of the scrotum extending to both groins.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Idiopathic scrotal oedema
D. Torsion of a testicular appendage

Explanation:
The **non-tender**, **non-painful**, and **bilateral** redness and oedema of the scrotum are characteristic of **idiopathic scrotal oedema**. This condition is benign and typically resolves on its own, unlike other causes of scrotal swelling, which are usually painful.

Correct Answer: C. Idiopathic scrotal oedema ✅

Topic Section:

Idiopathic Scrotal Oedema (Idiopathic scrotal oedema)

Definition:
A benign condition characterized by non-painful, bilateral swelling and redness of the scrotum, often resolving without intervention.

Key Clinical Presentation:

- Non-tender, bilateral scrotal swelling
- Redness and oedema without underlying testicular pathology
- Often resolves spontaneously

Investigations:

- **Best initial**: Clinical examination
- **Confirmatory**: None needed, diagnosis based on clinical features

Treatment Outline:

- **First-line**: Observation and reassurance
- **Second-line**: Supportive care (e.g., scrotal elevation) if needed
- **Definitive**: None required, condition is self-limiting

MCQ 503

A 12-year-old patient presents with a 4-hour history of right-sided lower abdominal pain. Examination confirms a slightly elevated, tender right testis. The lie of the testis is horizontal with no redness or oedema of the scrotum.
Which of the following is the most appropriate management?

A. Oral analgesics
B. Oral antibiotics
C. Doppler ultrasound
D. Scrotal exploration

Explanation:
The **horizontal orientation** of the testis suggests **testicular torsion**, a surgical emergency. **Doppler ultrasound** is the most appropriate initial investigation to confirm the diagnosis, as it can assess blood flow to the testis. If torsion is confirmed, immediate surgical exploration is required.

Correct Answer: C. Doppler ultrasound ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency caused by the twisting of the spermatic cord, leading to impaired blood flow to the testis.

Key Clinical Presentation:

- Sudden onset of severe scrotal pain
- Vomiting and nausea
- Swollen, tender scrotum

- Horizontal or vertical orientation of the testis
- Absence of the cremasteric reflex

Investigations:

- **Best initial**: Clinical examination and Doppler ultrasound (to assess blood flow)
- **Confirmatory**: Surgical exploration (if diagnosis is unclear)

Treatment Outline:

- **First-line**: Immediate surgical exploration and detorsion
- **Second-line**: Orchidopexy to prevent recurrence
- **Definitive**: Surgery to preserve the testis

MCQ 504

A 45-year-old sustained a knife injury to his left wrist. Examination confirms a clean wound with division of the underlying tendons and median nerve.
Which of the following is the most appropriate management?

A. Debridement and primary closure
B. Primary repair of injured structures

Explanation:
The appropriate management for a clean knife injury with division of tendons and a nerve is **primary repair of the injured structures**. This includes repairing both the tendons and median nerve as early as possible to minimize the risk of complications like functional loss and nerve damage.

Correct Answer: B. Primary repair of injured structures ✅

Topic Section:

Tendon and Nerve Injury (Tendon and Nerve Injury)

Definition:
Injury to tendons and nerves caused by trauma, resulting in loss of function or sensation.

Key Clinical Presentation:

- Trauma with visible lacerations or wounds
- Tenderness and loss of function in the affected area
- Decreased sensation or motor loss (in case of nerve injury)

Investigations:

- **Best initial**: Clinical examination
- **Confirmatory**: Imaging or exploration during surgery

Treatment Outline:

- **First-line**: Primary repair of tendons and nerves as soon as possible
- **Second-line**: Rehabilitation and therapy post-repair

MCQ 505

A 72-year-old underwent a subtotal colectomy 5 days ago. His urine output over the past 8 hours has been 7 ml/hr.
Blood pressure 110/70 mmHg
Heart rate 90 /min
Temperature 36.6 °C
Which of the following is most appropriate to administer intravenously?

A. Diuretics
B. Inotropes
C. Antibiotics
D. 500 NS fluid challenge

Explanation:
The most appropriate management for a patient with low urine output post-surgery is to administer a **500 ml normal saline (NS) fluid challenge** to assess for possible hypovolemia. The low urine output in this case suggests possible fluid depletion rather than an intrinsic renal issue.

Correct Answer: D. 500 NS fluid challenge ✅

Topic Section:

Postoperative Acute Kidney Injury (Postoperative Acute Kidney Injury)

Definition:
Acute kidney injury (AKI) occurring after surgery, often due to hypovolemia, sepsis, or medication effects.

Key Clinical Presentation:

- Decreased urine output post-surgery
- Elevated serum creatinine or urea levels (if available)
- Possible signs of fluid imbalance (e.g., hypotension)

Investigations:

- **Best initial**: Urine output monitoring
- **Confirmatory**: Serum creatinine and BUN levels

Treatment Outline:

- **First-line**: Fluid resuscitation with normal saline or Ringer's lactate
- **Second-line**: Diuretics if fluid overload is confirmed
- **Definitive**: Dialysis if necessary

MCQ 506

A 75-year-old man complains of severe diarrhoea for the past 2 days. He has been taking antibiotics for the past 2 weeks for a lower respiratory tract infection. Examination confirms generalized distension and tenderness.
Blood pressure 100/60 mmHg
Heart rate 110 /min
Respiratory rate 22 /min
Temperature 38 °C
Which of the following is the most appropriate management?

A. CT scan of chest
B. Diagnostic laparoscopy
C. Exploratory laparotomy
D. Stool for clostridium difficile toxins

Explanation:
Given the patient's recent antibiotic use and current symptoms of severe diarrhea, **Clostridium**

difficile infection (C. difficile) is the most likely diagnosis. The most appropriate test to confirm this diagnosis is stool testing for C. difficile toxins.

Correct Answer: D. Stool for clostridium difficile toxins ✅

Topic Section:

Clostridium Difficile Infection (Clostridium Difficile Infection)

Definition:
A bacterial infection caused by Clostridium difficile, typically following antibiotic use, leading to diarrhea, fever, and abdominal pain.

Key Clinical Presentation:

- Watery diarrhea, often with blood
- Abdominal cramps and tenderness
- Recent history of antibiotic use

Investigations:

- **Best initial**: Stool culture or toxin test for C. difficile
- **Confirmatory**: PCR for C. difficile toxins

Treatment Outline:

- **First-line**: Oral vancomycin or fidaxomicin
- **Second-line**: Metronidazole
- **Definitive**: Surgical intervention if severe complications (e.g., megacolon)

MCQ 507

A 60-year-old woman presents with a 12-hour history of severe pain in the epigastrium radiating to her back. It is associated with nausea and 2 episodes of vomiting. She has a history of cholelithiasis.
Abdominal examination confirms generalized tenderness in abdomen.
Which of the following is the most likely diagnosis?

A. Biliary colic
B. Acute pancreatitis
C. Acute cholecystitis
D. Empyema gallbladder

Explanation:
The patient's presentation of severe epigastric pain radiating to the back, nausea, vomiting, and a history of cholelithiasis is most consistent with **acute pancreatitis**. The pain is typically more severe and radiates to the back, distinguishing it from biliary colic or acute cholecystitis.

Correct Answer: B. Acute pancreatitis ✅

Topic Section:

Acute Pancreatitis (Acute Pancreatitis)

Definition:
Inflammation of the pancreas, often caused by alcohol use, gallstones, or other factors, leading to abdominal pain, nausea, vomiting, and elevated pancreatic enzymes.

Key Clinical Presentation:

- Severe epigastric pain radiating to the back
- Nausea and vomiting
- History of risk factors (e.g., alcohol, gallstones)

Investigations:

- **Best initial**: Serum amylase and lipase
- **Confirmatory**: CT scan if diagnosis is uncertain or complications suspected

Treatment Outline:

- **First-line**: Supportive care, IV fluids, and pain management
- **Second-line**: Nutritional support (oral or enteral feeding)
- **Definitive**: Surgical management if complications (e.g., abscess, necrosis)

MCQ 508

A 7-day-old is presented for circumcision. Examination confirms urethral opening on the dorsal surface.
Which of the following is the most likely diagnosis?

A. Epispadias
B. Coronal hypospadias
C. Glandular hypospadias
D. Mid-penile hypospadias

Explanation:
The urethral opening on the dorsal surface of the penis is characteristic of **epispadias**. This congenital condition involves the malformation of the urethra, with the opening located on the dorsal aspect of the penis, as opposed to the ventral location in hypospadias.

Correct Answer: A. Epispadias ✅

Topic Section:

Epispadias (Epispadias)

Definition:
A congenital defect where the urethral opening is located on the dorsal surface of the penis.

Key Clinical Presentation:

- Urethral opening on the dorsal side of the penis
- May be associated with other genital anomalies

Investigations:

- **Best initial**: Physical examination
- **Confirmatory**: Surgical evaluation

Treatment Outline:

- **First-line**: Surgical repair, often in infancy
- **Second-line**: Follow-up care for any associated abnormalities

---

MCQ 509

A 3-year-old uncircumcised boy has a history of foreskin ballooning during micturition. Examination confirms a ring of white scarred tissue and an inability to retract foreskin. Which of the following is the most likely diagnosis?

A. Chordee
B. Phimosis
C. Hypospadias
D. Paraphimosis

Explanation:
The inability to retract the foreskin with a history of ballooning during micturition suggests **phimosis**, a condition where the foreskin is too tight to be retracted over the glans, often due to scarring or congenital tightness.

Correct Answer: B. Phimosis ✅

Topic Section:

Phimosis (Phimosis)

Definition:
A condition where the foreskin of the penis cannot be retracted over the glans due to tightness or scarring.

Key Clinical Presentation:

- Difficulty retracting the foreskin
- Ballooning of the foreskin during urination
- Scarring or fibrotic ring at the tip of the foreskin

Investigations:

- **Best initial**: Physical examination
- **Confirmatory**: If necessary, referral for urology evaluation

Treatment Outline:

- **First-line**: Topical steroid application to loosen the foreskin
- **Second-line**: Surgical intervention (circumcision) if conservative methods fail

---

MCQ 510

A 3-day-old is presented for circumcision. Examination confirms chordee and hooded foreskin. Urethral opening is at mid-penile level.
Which of the following is the most appropriate management?

A. Open circumcision
B. Plastibell circumcision
C. Circumcision using gumco
D. Referral to pediatric surgeon

Explanation:
Given the presence of **chordee** (a curvature of the penis) and **hooded foreskin**, along with a mid-penile urethral opening, **referral to a pediatric surgeon** is the most appropriate course of action. Chordee often requires correction alongside any other associated anomalies, and this type of repair is best done by a specialist.

Correct Answer: D. Referral to pediatric surgeon ✅

Topic Section:

Chordee (Chordee)

Definition:
A congenital condition characterized by a downward curvature of the penis, often associated with other anomalies such as hypospadias.

Key Clinical Presentation:

- Curvature of the penis during erection
- Possible associated hypospadias or other genital anomalies

Investigations:

- **Best initial**: Physical examination
- **Confirmatory**: Surgical evaluation and correction

Treatment Outline:

- **First-line**: Surgical correction, often in infancy or early childhood
- **Second-line**: Follow-up care for any other associated abnormalities

---

MCQ 511

What color of nipple discharge is most commonly associated with intraductal papilloma of the breast?

A. Red
B. Black
C. White
D. Green

Explanation:
Intraductal papillomas often present with **serous or blood-tinged nipple discharge**, and **red** is the most common color of discharge associated with this benign condition.

Correct Answer: A. Red ✅

Topic Section:

Intraductal Papilloma (Intraductal Papilloma)

Definition:
A benign tumor that forms within the milk ducts of the breast, commonly presenting with nipple discharge.

Key Clinical Presentation:

- Blood-tinged or serous nipple discharge
- A palpable mass or no palpable mass

Investigations:

- **Best initial**: Mammography or ultrasound
- **Confirmatory**: Core biopsy for diagnosis

Treatment Outline:

- **First-line**: Surgical excision
- **Second-line**: Monitoring or ductal excision if necessary

---

MCQ 512

A 60-year-old woman presents with a mass in her right breast. Clinical examination confirms a hard, non-tender irregular lump with skin tethering.
Which of the following is the most likely diagnosis?

A. Breast cyst
B. Duct ectasia
C. Fibroadenoma
D. Carcinoma of breast

Explanation:
A hard, irregular mass with skin tethering in an older woman is highly suggestive of **breast carcinoma**, particularly invasive ductal carcinoma.

Correct Answer: D. Carcinoma of breast ✅

Topic Section:

Breast Cancer (Breast Cancer)

Definition:
Malignant tumor arising from the cells of the breast tissue, commonly presenting as a lump.

Key Clinical Presentation:

- Hard, irregular mass in the breast
- Skin changes such as dimpling or tethering
- Possible nipple discharge or retraction

Investigations:

- **Best initial**: Mammography or ultrasound
- **Confirmatory**: Biopsy for histopathological diagnosis

Treatment Outline:

- **First-line**: Surgical excision (lumpectomy or mastectomy)
- **Second-line**: Chemotherapy, radiation, and hormone therapy
- **Definitive**: Based on tumor stage and hormone receptor status

---

MCQ 513

A 35-year-old woman presents with difficulty in breastfeeding her baby. Examination confirms unilateral nipple inversion. Gentle pressure exerts the nipple and it is transversely slitted. Which of the following is the most likely diagnosis?

A. Breast cyst
B. Duct ectasia
C. Fibroadenoma
D. Carcinoma of breast

Explanation:
Nipple inversion with a transverse slit appearance on pressure suggests **duct ectasia**, a benign condition involving the dilatation of mammary ducts. It is commonly associated with nipple inversion and occurs in women in their 30s and 40s.

Correct Answer: B. Duct ectasia ✅

Topic Section:

Duct Ectasia (Duct Ectasia)

Definition:
Dilatation of the breast ducts, commonly affecting the subareolar ducts, often presenting with nipple inversion, discharge, and pain.

Key Clinical Presentation:

- Nipple inversion or retraction
- Nipple discharge (often green or brown)
- Pain or tenderness in the breast

Investigations:

- **Best initial**: Physical examination
- **Confirmatory**: Ultrasound or mammography

Treatment Outline:

- **First-line**: Conservative management, including warm compresses and anti-inflammatory drugs
- **Second-line**: Surgical intervention (excision of affected ducts) if symptoms persist

---

MCQ 514

A 28-year-old lactating mother presents with pain in her right breast. Examination confirms a 3 cm tender swelling with red and hot overlying skin. Temperature 37.9 °C.
Which of the following is the most likely diagnosis?

A. Duct ectasia
B. Fibroadenosis
C. Fibroadenoma
D. Breast Abscess

Explanation:
The combination of tenderness, redness, and warmth over the affected area, along with fever, points towards a **breast abscess**, a common complication of breastfeeding due to blocked milk ducts or mastitis.

Correct Answer: D. Breast Abscess ✅

Topic Section:

Breast Abscess (Breast Abscess)

Definition:
A localized collection of pus in the breast tissue, commonly occurring in lactating women due to infection of the milk ducts.

Key Clinical Presentation:

- Painful, swollen, and red area on the breast
- Fever and malaise
- Tenderness with fluctuation

Investigations:

- **Best initial**: Physical examination

- **Confirmatory**: Ultrasound to confirm the presence of an abscess

Treatment Outline:

- **First-line**: Antibiotics and drainage (needle aspiration or incision and drainage)
- **Second-line**: Surgery for recurrent or complicated cases

---

MCQ 515

A 36-year-old woman presents with pain in both breasts associated with multiple lumps that become prominent during her premenstrual phase. Examination confirms multiple nodules of variable size and consistency. Examination of the axillae is normal. Ultrasound: Multiple bilateral cysts of variable sizes, and 2 solid oval shape lesion in the left breast.
Which of the following is the most likely diagnosis?

A. Fibroadenomas
B. Chronic mastitis
C. Fibrocystic disease
D. Carcinoma of breast

Explanation:
The most likely diagnosis is **fibrocystic disease**, which is characterized by multiple cysts and fibrous tissue changes in the breast, often exacerbated by hormonal fluctuations (e.g., premenstrual phase). This is a benign condition.

Correct Answer: C. Fibrocystic disease ✅

Topic Section:

Fibrocystic Disease (Fibrocystic Breast Disease)

Definition:
A benign condition involving the development of multiple cysts and fibrous tissue in the breast, often related to hormonal changes.

Key Clinical Presentation:

- Multiple lumps in the breast that vary in size and consistency
- Pain or tenderness, especially around the menstrual cycle
- No axillary lymph node involvement

Investigations:

- **Best initial**: Physical examination and ultrasound
- **Confirmatory**: Fine needle aspiration (FNA) if necessary

Treatment Outline:

- **First-line**: Conservative management with pain relief (NSAIDs, warm compresses)
- **Second-line**: Hormonal treatments if symptoms are severe

---

MCQ 516

A 24-year-old lactating mother presents with pain in her right breast. Examination confirms a 5 cm tender swelling. The overlying skin is red, inflamed, and thinned out. Temperature 38.7 °C. What is the most appropriate management?

A. Lump excision
B. Needle aspiration
C. Incision and drainage
D. Conservative management with antibiotics

Explanation:
This presentation, along with fever, suggests a **breast abscess**, which is typically treated with incision and drainage (I&D) if an abscess has formed. Antibiotics are also necessary.

Correct Answer: C. Incision and drainage ✅

Topic Section:

Breast Abscess (Breast Abscess)

Definition:
A localized infection in the breast that leads to the formation of a pus-filled cavity, often seen in lactating women due to blocked ducts.

Key Clinical Presentation:

- Painful, swollen, warm, and erythematous breast
- Fever and chills
- Fluctuant mass

Investigations:

- **Best initial**: Physical examination
- **Confirmatory**: Ultrasound to confirm the presence of an abscess

Treatment Outline:

- **First-line**: Antibiotics and incision & drainage (I&D)
- **Second-line**: Surgical drainage for recurrent or severe cases

---

MCQ 517

A 25-year-old man underwent surgery for inflammatory bowel disease. He requires regular replacement of fat-soluble vitamins parenterally. Which part of the bowel was most likely removed?

A. Distal ileum
B. Mid jejunum
C. Transverse colon
D. Descending colon

Explanation:
The **distal ileum** is responsible for absorbing fat-soluble vitamins (A, D, E, K). Removal of this section, commonly in surgeries for inflammatory bowel disease (e.g., Crohn's disease), can lead to malabsorption of these vitamins.

Correct Answer: A. Distal ileum ✅

Topic Section:

Ileal Resection (Ileal Resection)

Definition:
Surgical removal of the ileum, which may occur in conditions like Crohn's disease or other inflammatory bowel diseases.

Key Clinical Presentation:

- Malabsorption of nutrients, particularly fat-soluble vitamins
- Diarrhea or weight loss in some cases

- Fatigue or other signs of vitamin deficiency

Investigations:

- **Best initial**: Nutritional assessment
- **Confirmatory**: Lab tests for vitamin deficiency (e.g., serum vitamin levels)

Treatment Outline:

- **First-line**: Vitamin supplementation (fat-soluble vitamins)
- **Second-line**: Management of malabsorption with dietary modifications or medications

---

MCQ 518

A 75-year-old woman presents with a 7-day history of severe pain in the epigastrium. The pain is relieved by vomiting and becomes worse when eating. History confirms that she has been taking medication for joint pain for the last 2 months. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Boerhaave's syndrome
D. Perforated peptic ulcer

Explanation:
This patient's history of recent medication use (non-steroidal anti-inflammatory drugs) and epigastric pain relieved by vomiting points towards **acute gastritis**. NSAIDs are commonly associated with causing gastritis, which can present with the described symptoms.

Correct Answer: B. Acute gastritis ✅

Topic Section:

Acute Gastritis (Acute Gastritis)

Definition:
Inflammation of the gastric mucosa, often triggered by factors like NSAIDs, alcohol, infections, or stress.

Key Clinical Presentation:

- Epigastric pain, often relieved by vomiting
- Nausea and vomiting
- Use of NSAIDs or alcohol

Investigations:

- **Best initial**: Clinical evaluation and history
- **Confirmatory**: Endoscopy for mucosal appearance

Treatment Outline:

- **First-line**: Proton pump inhibitors (PPIs), H2 blockers, antacids
- **Second-line**: Discontinuation of the offending agent (e.g., NSAIDs)
- **Definitive**: Endoscopic intervention if complications arise (e.g., bleeding)

MCQ 519

A 60-year-old patient complains of sudden onset of upper abdominal pain after a few bouts of vomiting. Examination confirms a sick patient with tenderness in the epigastrium and supraclavicular subcutaneous emphysema. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Perforated peptic ulcer
D. Boerhaave’s syndrome

Explanation:
The presence of **supraclavicular subcutaneous emphysema** following vomiting, with severe upper abdominal pain, strongly suggests **Boerhaave’s syndrome**, a spontaneous rupture of the esophagus, often due to forceful vomiting.

Correct Answer: D. Boerhaave’s syndrome ✅

Topic Section:

Boerhaave's Syndrome (Boerhaave's Syndrome)

Definition:
Spontaneous rupture of the esophagus, typically following forceful vomiting, leading to mediastinal air leakage and subcutaneous emphysema.

Key Clinical Presentation:

- Severe chest or epigastric pain following vomiting
- Subcutaneous emphysema in the neck or chest
- Shock or sepsis signs

Investigations:

- **Best initial**: Chest X-ray (may show pneumomediastinum or subcutaneous emphysema)
- **Confirmatory**: Contrast esophagram or CT scan

Treatment Outline:

- **First-line**: Emergency surgery or drainage
- **Second-line**: Broad-spectrum antibiotics
- **Definitive**: Surgical repair of the esophageal tear

MCQ 520

A 25-year-old man presents with a 6-month history of heartburn. The pain becomes worse at night after lying down. On lifting weight, he experiences a bitter taste in mouth. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Boerhaave's syndrome
D. Perforated peptic ulcer

Explanation:
This patient's history of heartburn and the symptom of bitter taste on lifting weight points toward **gastroesophageal reflux disease (GERD)**, which causes acid reflux, leading to symptoms such as heartburn, regurgitation, and a bitter taste in the mouth.

Correct Answer: A. Esophagitis ✅

Topic Section:

Gastroesophageal Reflux Disease (GERD)

Definition:
A condition where stomach acid or bile irritates the esophagus, causing symptoms like heartburn and regurgitation.

Key Clinical Presentation:

- Heartburn, especially after eating or lying down
- Regurgitation and bitter taste
- Sore throat or cough in some cases

Investigations:

- **Best initial**: Clinical evaluation
- **Confirmatory**: Endoscopy if needed

Treatment Outline:

- **First-line**: Proton pump inhibitors (PPIs)
- **Second-line**: H2 receptor antagonists or antacids
- **Definitive**: Lifestyle modifications (e.g., weight loss, avoiding triggers)

MCQ 521

A 55-year-old man presents with sudden, severe and constant pain in his epigastrium for the past 2 hours. History confirms that he has been on medication for joint pain for the last 2 years. Examination confirms a very sick patient with sunken eyes. Abdomen is tender and rigid to palpation. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Perforated peptic ulcer
D. Boerhaave's syndrome

Explanation:
This patient's severe, sudden epigastric pain, along with rigidity on abdominal examination,

suggests **perforated peptic ulcer**, which occurs due to erosion of the gastric or duodenal lining leading to free perforation into the peritoneal cavity.

Correct Answer: C. Perforated peptic ulcer ✅

Topic Section:

Perforated Peptic Ulcer (Perforated Peptic Ulcer)

Definition:
A complication of peptic ulcer disease where an ulcer erodes through the stomach or duodenal wall, leading to leakage of gastric contents into the peritoneum.

Key Clinical Presentation:

- Sudden, severe epigastric pain
- Abdominal rigidity (signs of peritonitis)
- Recent history of NSAID use or gastritis

Investigations:

- **Best initial**: X-ray (free air under diaphragm)
- **Confirmatory**: CT scan or laparoscopy

Treatment Outline:

- **First-line**: Surgery (laparotomy with ulcer repair)
- **Second-line**: Broad-spectrum antibiotics for infection
- **Definitive**: Surgical closure with omental patch and acid suppression

---

MCQ 522

A 50-year-old man presents with a long-standing history of indigestion and epigastric discomfort. He has poor appetite and has lost 10 kg recently. Examination confirms wasting of facial muscles, palpable supraclavicular lymph nodes and dull epigastric mass. Which of the following is the most likely diagnosis?

A. Liver metastasis
B. Chronic peptic ulcer

C. Stomach carcinoma
D. Chronic pancreatitis

Explanation:
This patient's symptoms, weight loss, and epigastric mass, combined with palpable supraclavicular lymph nodes, suggest **stomach carcinoma**, a malignancy commonly associated with weight loss and systemic signs of cancer.

Correct Answer: C. Stomach carcinoma ✅

Topic Section:

Stomach Carcinoma (Gastric Cancer)

Definition:
Malignancy of the stomach, often presenting with vague upper abdominal symptoms, weight loss, and sometimes lymphadenopathy.

Key Clinical Presentation:

- Epigastric pain or discomfort
- Weight loss and anorexia
- Palpable supraclavicular lymph nodes (Virchow's node)

Investigations:

- **Best initial**: Endoscopy with biopsy
- **Confirmatory**: CT scan for staging

Treatment Outline:

- **First-line**: Surgical resection (gastrectomy)
- **Second-line**: Chemotherapy (for metastases)
- **Definitive**: Chemoradiation therapy if applicable

MCQ 523

A 65-year-old man presents with a 72-hour history of severe pain in the epigastrium radiating to his back with associated nausea and 2 episodes of vomiting. History confirms that he is a known

case of cholelithiasis. Abdominal examination confirms generalized tenderness in the abdomen. US-guided aspiration of the intra-abdominal fluid grows E.coli. Blood pressure 100/70 mmHg, Heart rate 110 /min, Respiratory rate 22 /min, Temperature 38.6 °C. Test Result:
Amylase: 576 (24-151 IU/L)
Lipase: 300 (0-160 IU/L)
C-reactive peptide: 135 (<8.2 mg/L).
What is the most likely explanation for the bacterial growth?

A. Bowel perforation
B. Bacterial translocation
C. Haematogenous spread
D. Exogenous contamination

Explanation:
This patient's history and presentation of pain with nausea, vomiting, and raised inflammatory markers, along with the bacterial growth of E. coli from intra-abdominal fluid, are consistent with **bacterial translocation**. This occurs when bacteria from the gastrointestinal tract, often due to intestinal ischemia or perforation, move to the peritoneal cavity and cause infection.

Correct Answer: B. Bacterial translocation ✅

Topic Section:

Bacterial Translocation (Bacterial Translocation)

Definition:
The movement of bacteria from the gastrointestinal tract to sterile areas like the peritoneal cavity, bloodstream, or lymphatic system.

Key Clinical Presentation:

- Abdominal pain and tenderness
- Nausea and vomiting
- Fever and signs of sepsis

Investigations:

- **Best initial**: Blood and fluid cultures
- **Confirmatory**: CT scan if perforation or abscess is suspected

Treatment Outline:

- **First-line**: Antibiotics targeting E. coli and other common peritoneal pathogens
- **Second-line**: Surgical management if there is a source of perforation or abscess
- **Definitive**: Surgical intervention for any underlying pathology (e.g., bowel perforation)

MCQ 524

A 25-year-old woman presents with a longstanding history of lower abdominal and right hypochondrial pain. Diagnostic laparoscopy confirms pelvic and perihepatic adhesions. Which of the following is most likely responsible?

A. E.coli
B. Chlamydia
C. Bacteroides
D. Streptococci

Explanation:
Chronic pelvic pain with adhesions, particularly in the setting of previous infections, is most commonly caused by **Chlamydia**, a common sexually transmitted infection that can lead to pelvic inflammatory disease (PID) and subsequent adhesions in the pelvic and perihepatic regions.

Correct Answer: B. Chlamydia ✅

Topic Section:

Pelvic Inflammatory Disease (PID)

Definition:
An infection of the female reproductive organs, commonly caused by sexually transmitted infections, leading to inflammation and the formation of adhesions.

Key Clinical Presentation:

- Chronic pelvic pain
- Abnormal vaginal discharge
- History of sexually transmitted infections

Investigations:

- **Best initial**: Pelvic ultrasound or laparoscopy
- **Confirmatory**: Culture for Chlamydia or Gonorrhea

Treatment Outline:

- **First-line**: Antibiotics targeting Chlamydia and Gonorrhea
- **Second-line**: Pain management and surgical removal of adhesions if needed
- **Definitive**: Prevention and treatment of STIs

MCQ 525

A 25-year-old woman presents with a 6-hour history of lower abdominal pain associated with nausea and 1 episode of vomiting. Examination confirms minimal tenderness in the suprapubic and left iliac fossa region. Urine dipstick and pregnancy tests are normal. Blood pressure 110/70 mmHg, Heart rate 76 /min, Temperature 36.6 °C. Test Result:
Hb 110 (130-170 g/L Male, 120-160 Female),
WBC 9 (4.5-10.5 x 10^9/L),
Neutrophils 50 (40-60%).
Which of the following is the most appropriate management?

A. US abdomen
B. Abdominal X-ray
C. Diagnostic laparoscopy
D. Discharge on oral analgesics

Explanation:
This patient's mild symptoms and lack of alarming signs suggest a likely benign condition, such as a minor gastrointestinal issue. **Ultrasound abdomen** is the most appropriate initial investigation, especially to rule out gynecological or urinary tract issues like ovarian cysts or UTI.

Correct Answer: A. US abdomen ✅

Topic Section:

Abdominal Pain (Abdominal Pain)

Definition:
Pain arising from the abdominal organs, often requiring clinical evaluation and imaging to identify the underlying cause.

Key Clinical Presentation:

- Abdominal pain with associated symptoms such as vomiting, diarrhea, or tenderness
- Clinical history is essential in differentiating between gastrointestinal, gynecological, or other causes

Investigations:

- **Best initial**: Abdominal ultrasound for gynecological and gastrointestinal pathology
- **Confirmatory**: CT scan if needed for further evaluation

Treatment Outline:

- **First-line**: Symptomatic treatment for benign conditions
- **Second-line**: Surgical intervention if indicated
- **Definitive**: Address the underlying cause based on diagnosis

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MCQ 526

A 55-year-old is admitted for an elective laparoscopic cholecystectomy. He is diabetic and hypertensive for 5 years. History confirms a recent admission in Intensive Care Unit for anterior wall MI 6 weeks ago. What is the most appropriate management?

A. Open cholecystectomy during this admission
B. Laparoscopic cholecystectomy after 6 months
C. Laparoscopic Cholecystectomy during this admission
D. Per-cutaneous cholecystostomy during this admission

Explanation:
For a patient who recently had a myocardial infarction (MI), it is essential to **delay elective surgery** for at least 6 weeks after MI to allow the heart to heal. **Laparoscopic cholecystectomy after 6 months** is the most appropriate approach for this patient.

Correct Answer: B. Laparoscopic cholecystectomy after 6 months ✅

Topic Section:

Laparoscopic Cholecystectomy (Laparoscopic Cholecystectomy)

Definition:
A minimally invasive procedure to remove the gallbladder, typically performed for symptomatic gallstones or cholecystitis.

Key Clinical Presentation:

- Right upper quadrant pain
- History of cholelithiasis or cholecystitis

Investigations:

- **Best initial**: Abdominal ultrasound
- **Confirmatory**: Liver function tests or CT scan if needed

Treatment Outline:

- **First-line**: Laparoscopic cholecystectomy for symptomatic gallstones
- **Second-line**: Percutaneous cholecystostomy for high-risk patients
- **Definitive**: Surgical resection if indicated

MCQ 527

A 65-year-old patient is admitted for elective ventral hernia repair. He is diabetic and hypertensive for 10 years. Examination confirms raised JVP, bilateral basal crepitation on chest auscultation and pedal edema. What is the most appropriate next step in management?

A. Open repair as scheduled
B. Laparoscopic repair as scheduled
C. Surgery after optimization at later date
D. No surgery unless presents with obstruction

Explanation:
This patient shows signs of **congestive heart failure** (elevated JVP, basal crepitations, pedal edema). **Surgery should be delayed** until the patient is optimized to avoid perioperative complications related to heart failure.

Correct Answer: C. Surgery after optimization at later date ✅

Topic Section:

Congestive Heart Failure (CHF)

Definition:
A condition in which the heart is unable to pump blood effectively, leading to fluid accumulation in the lungs and peripheral tissues.

Key Clinical Presentation:

- Shortness of breath, edema, orthopnea, elevated JVP
- History of hypertension, diabetes, or coronary artery disease

Investigations:

- **Best initial**: Chest X-ray and echocardiogram
- **Confirmatory**: BNP levels and cardiac MRI

Treatment Outline:

- **First-line**: Diuretics, ACE inhibitors, beta-blockers
- **Second-line**: Digoxin, aldosterone antagonists
- **Definitive**: Lifestyle modifications, heart transplantation if needed

MCQ 528

A 45-year-old woman complains of dysphagia to liquids, retrosternal pain on swallowing, and regurgitation of food. Which of the following has the highest diagnostic value?

A. Barium swallow
B. CT with contrast
C. Upper gastrointestinal endoscopy
D. Lower oesophageal sphincter manometry

Explanation:
The patient’s symptoms of dysphagia, retrosternal pain, and regurgitation suggest a possible esophageal motility disorder or mechanical obstruction. **Upper gastrointestinal endoscopy** is the most appropriate diagnostic tool, allowing direct visualization of the esophagus, and can help diagnose conditions like esophagitis, strictures, or malignancy.

Correct Answer: C. Upper gastrointestinal endoscopy ✅

Topic Section:

Esophageal Disorders (Esophageal Disorders)

Definition:
Disorders affecting the esophagus that can cause dysphagia, regurgitation, and retrosternal pain, including motility disorders and mechanical obstructions.

Key Clinical Presentation:

- Dysphagia (difficulty swallowing)
- Regurgitation of food
- Retrosternal pain

Investigations:

- **Best initial**: Upper gastrointestinal endoscopy
- **Confirmatory**: Manometry for motility disorders, Barium swallow

Treatment Outline:

- **First-line**: Medical therapy (e.g., proton pump inhibitors for reflux) or surgical intervention for mechanical obstruction
- **Second-line**: Endoscopic procedures for strictures or dilation
- **Definitive**: Surgery or long-term management depending on underlying cause

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MCQ 529

Which of the following is the most objective method for identifying high-risk patients for peri-operative morbidity and mortality?

A. Metabolic equivalent tasks
B. Revised cardiac risk index of Lee
C. Cardiopulmonary exercise testing
D. American society of anaesthesiologist scoring

Explanation:
The **Revised cardiac risk index of Lee** is the most objective and widely used tool for identifying high-risk patients for perioperative complications, particularly in relation to cardiovascular risk. It takes into account factors such as age, heart disease history, and type of surgery.

Correct Answer: B. Revised cardiac risk index of Lee ✅

Topic Section:

Perioperative Risk Assessment (Perioperative Risk Assessment)

Definition:
Assessment of a patient's risk of morbidity and mortality during or after surgery, taking into account comorbidities and the type of surgical procedure.

Key Clinical Presentation:

- History of cardiovascular disease
- Age and overall physical condition
- Type of surgery

Investigations:

- **Best initial**: Revised cardiac risk index of Lee, American Society of Anesthesiologists (ASA) score
- **Confirmatory**: Cardiopulmonary exercise testing if needed

Treatment Outline:

- **First-line**: Preoperative optimization (e.g., managing diabetes, hypertension, or smoking cessation)
- **Second-line**: Close monitoring during and after surgery
- **Definitive**: Surgical approach tailored to the patient's risk profile

---

MCQ 530

A 45-year-old is brought to the Emergency Department following a high-speed motor vehicle accident. Examination confirms seat belt sign. CT: Chance fracture of lumbar vertebrae. Which of the following is the most likely finding on an abdominal CT?

A. Injury to vena cava
B. Stomach perforation
C. Duodenal perforation
D. Isolated jejunal blowout

Explanation:
In trauma with a seat belt sign and Chance fracture (a type of vertebral fracture), the most likely abdominal injury is a **duodenal perforation** due to the compression and shearing forces during impact. The duodenum is particularly vulnerable in these types of trauma.

Correct Answer: C. Duodenal perforation ✅

Topic Section:

Abdominal Trauma (Abdominal Trauma)

Definition:
Injury to the abdomen, often resulting from blunt trauma, that may involve solid organs, hollow viscera, or vascular structures.

Key Clinical Presentation:

- Abdominal pain or tenderness
- Positive signs of trauma (e.g., seatbelt sign)
- Abdominal distension and guarding

Investigations:

- **Best initial**: CT scan of the abdomen, Focused Assessment with Sonography for Trauma (FAST)
- **Confirmatory**: Diagnostic peritoneal lavage or exploratory laparotomy if needed

Treatment Outline:

- **First-line**: Conservative management for minor injuries
- **Second-line**: Surgical intervention for major injuries (e.g., bowel perforation)
- **Definitive**: Surgical repair for injuries involving hollow organs or vascular structures

---

MCQ 531

A 25-year-old is brought to the Emergency Department after being rescued from a burning building. Examination confirms a slightly confused patient with singed facial and nasal hair. Which of the following is the most appropriate next step in management?

A. Elective intubation
B. Observation in ICU
C. Avoid medico-legal issues
D. Advice on pain management

Explanation:
For a patient who has been exposed to smoke inhalation and burns, **elective intubation** is critical to secure the airway, especially with signs of inhalation injury (e.g., singed facial hair). Intubation should be performed before any further deterioration occurs.

Correct Answer: A. Elective intubation ✅

Topic Section:

Burns and Smoke Inhalation Injury (Burns and Smoke Inhalation Injury)

Definition:
Injury from exposure to heat or chemicals causing damage to the skin, airways, and other tissues, often with a risk of inhalation injury from smoke.

Key Clinical Presentation:

- Singed facial or nasal hair
- Respiratory distress, confusion
- Burns over a large body surface area

Investigations:

- **Best initial**: Clinical evaluation, arterial blood gases (ABG), chest X-ray
- **Confirmatory**: Bronchoscopy for assessing the extent of airway injury

Treatment Outline:

- **First-line**: Airway management, oxygen supplementation, and intubation if needed
- **Second-line**: Fluid resuscitation (e.g., Parkland formula)
- **Definitive**: Burn care and long-term management of inhalation injuries

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MCQ 532

A 45-year-old is presented following poly-trauma. He requires intubation to protect airway obstruction. After resuscitation and management, he is transferred to ICU. Which of the following is the most appropriate method to clear the cervical spine?

A. CT scan
B. Clinical judgement
C. MRI of cervical spine
D. AP and Lateral view cervical spine x-ray

Explanation:
In a poly-trauma patient, especially when there is concern for cervical spine injury, **CT scan** is the most appropriate method for clearing the cervical spine. It is highly sensitive for detecting fractures and dislocations of the cervical spine, and it can be performed rapidly, making it the best initial imaging choice in trauma cases.

Correct Answer: A. CT scan ✅

Topic Section:

Cervical Spine Injury Management (Cervical Spine Injury Management)

Definition:
Injuries to the cervical spine can lead to severe neurological compromise, and their management is crucial in poly-trauma patients.

Key Clinical Presentation:

- Trauma to the neck or head
- Mechanism of injury suggesting potential cervical spine involvement
- Neurological deficits such as weakness, sensory loss, or paralysis

Investigations:

- **Best initial**: CT scan for rapid assessment of fractures or dislocations
- **Confirmatory**: MRI for soft tissue damage or ligamentous injuries

Treatment Outline:

- **First-line**: Immobilization of the neck, and if injury is suspected, CT scan of the cervical spine
- **Second-line**: MRI if soft tissue injury or spinal cord involvement is suspected

- **Definitive**: Surgical intervention for fractures or instability

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MCQ 533

Which of the following is the most appropriate strategy to avoid air bag injury in children under the age of 12 years?

A. Restrain in back seats
B. Restrain in front seats
C. Rear facing in front seats
D. Booster seats in front seats

Explanation:
The most appropriate strategy for children under the age of 12 years to avoid airbag injury is to **restrain in back seats**. The back seat is the safest place for children, especially when they are under 12 years old, as it reduces the risk of injury from airbags and is generally safer in the event of a collision.

Correct Answer: A. Restrain in back seats ✅

Topic Section:

Child Passenger Safety (Child Passenger Safety)

Definition:
Ensuring the safety of children during vehicular transport through appropriate restraint systems.

Key Clinical Presentation:

- Child safety concerns during travel
- Risk of injury due to improper seat placement or restraint system

Investigations:

- **Best initial**: Evaluate the child's current restraint system and seating position
- **Confirmatory**: Review of safety guidelines from organizations like NHTSA

Treatment Outline:

- **First-line**: Proper restraint in back seat for children under 12 years

- **Second-line**: Proper booster seat for children who outgrow car seats but are too small for standard seat belts
- **Definitive**: Use of age-appropriate restraints for all children

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MCQ 534

A 50-year-old woman comes to the clinic with numbness in the lower part of the ear, lower face, and upper neck. History reveals, she underwent a lateral neck dissection. Examination shows, loss of sensation over both surfaces of the lower part of the pinna and skin overlying the angle of the mandible on the same side of the operation. Which of the following nerves is most likely injured?

A. Third occipital
B. Great auricular
C. Lesser occipital
D. Greater occipital

Explanation:
The **Great auricular nerve** is responsible for sensory innervation to the lower part of the ear, the lower face, and the upper neck, which matches the clinical presentation of this patient after lateral neck dissection. This nerve is commonly injured during such procedures.

Correct Answer: B. Great auricular ✅

Topic Section:

Nerve Injuries in Neck Surgeries (Nerve Injuries in Neck Surgeries)

Definition:
Nerve injuries that occur as a complication of surgical procedures in the neck, such as lateral neck dissection.

Key Clinical Presentation:

- Numbness or sensory loss in the ear, face, or neck after surgery
- Specific patterns of sensory loss based on the affected nerve

Investigations:

- **Best initial**: Clinical examination and history of surgery

- **Confirmatory**: Nerve conduction studies if necessary

Treatment Outline:

- **First-line**: Conservative management with observation and physiotherapy
- **Second-line**: Surgical repair if function does not improve or if significant symptoms persist
- **Definitive**: Nerve grafting or other surgical interventions in severe cases

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MCQ 535

A 25-year-old man is brought to the Emergency Room after a motor vehicle accident. Examination shows, he has a severed nerve in the right upper limb, resulting in a loss of adduction of all the digits of the right hand. Which of the following is the most likely affected nerve?

A. Ulnar
B. Median
C. Musculocutaneous
D. Upper subscapular

Explanation:
The **ulnar nerve** innervates the muscles responsible for adduction of the fingers (including the interossei muscles). A lesion to this nerve would result in an inability to adduct the fingers, as described in this patient.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Nerve Injuries (Ulnar Nerve Injuries)

Definition:
Damage to the ulnar nerve, leading to loss of function in the hand, particularly affecting the small and ring fingers and the ability to adduct the digits.

Key Clinical Presentation:

- Loss of finger adduction
- Weakness or atrophy of the intrinsic hand muscles

- Sensory loss along the ulnar aspect of the hand

Investigations:

- **Best initial**: Clinical evaluation
- **Confirmatory**: Nerve conduction studies or electromyography

Treatment Outline:

- **First-line**: Conservative management or splinting
- **Second-line**: Surgical exploration and repair if necessary
- **Definitive**: Nerve repair or grafting in cases of severe injury

---

MCQ 536

A 26-year-old man is brought to the Emergency Room after sustaining a bullet injury to his arm. Examination shows his median nerve was damaged. Which of the following is the most likely clinical finding?

A. Ape hand
B. Claw hand
C. Wrist drop
D. Waiter's tip hand

Explanation:
Damage to the **median nerve** results in **Ape hand** deformity, characterized by the inability to oppose the thumb, leading to a loss of thumb opposition and flattening of the thenar eminence.

Correct Answer: A. Ape hand ✅

Topic Section:

Median Nerve Injuries (Median Nerve Injuries)

Definition:
Injury to the median nerve, typically resulting in loss of function in the thumb and the hand, affecting the ability to oppose the thumb and flex the wrist.

Key Clinical Presentation:

- Weakness in thumb opposition

- Flattening of the thenar eminence (Ape hand)
- Sensory loss along the radial side of the palm and fingers

Investigations:

- **Best initial**: Clinical evaluation and history of injury
- **Confirmatory**: Nerve conduction studies or electromyography

Treatment Outline:

- **First-line**: Conservative management with splinting
- **Second-line**: Surgical exploration and repair if necessary
- **Definitive**: Nerve repair or grafting for severe injuries

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MCQ 537

A 19-year-old patient comes to the clinic because he is unable to bring the fork to his mouth to feed himself. History reveals the patient had a closed head injury in a motor vehicle accident 10 days ago. Which of the following is the most likely involved area?

A. Cerebellum
B. Parietal lobe
C. Occipital lobe
D. Temporal lobe

Explanation:
The **parietal lobe** is primarily responsible for the integration of sensory information and motor control, particularly for tasks such as feeding oneself. Damage to this area can lead to difficulty with motor coordination (i.e., difficulty bringing a fork to the mouth).

Correct Answer: B. Parietal lobe ✅

Topic Section:

Parietal Lobe Injuries (Parietal Lobe Injuries)

Definition:
Damage to the parietal lobe, which is involved in sensory processing, spatial orientation, and motor control.

Key Clinical Presentation:

- Difficulty with motor tasks (e.g., feeding oneself, dressing)
- Impaired sensory processing
- Hemispatial neglect

Investigations:

- **Best initial**: CT scan or MRI of the brain
- **Confirmatory**: Neuropsychological testing

Treatment Outline:

- **First-line**: Physical therapy and occupational therapy to improve motor coordination
- **Second-line**: Cognitive rehabilitation if necessary

MCQ 538

A 24-year-old woman comes to the clinic after noticing a lump in her breast. Examination shows a firm, painless, and freely movable 3 cm mass in the left breast. The mass has grown slowly over the past year and does not change during the menstrual cycle. Which of the following is the most likely diagnosis?

A. Fat cyst
B. Fibroadenoma
C. Fibro-cystic changes
D. Intraductal papilloma

Explanation:
The **fibroadenoma** is the most likely diagnosis. It is a benign, firm, painless, and freely movable mass that grows slowly and is typically non-cyclical in nature. Fibroadenomas are common in young women and are not influenced by hormonal cycles.

Correct Answer: B. Fibroadenoma ✅

Topic Section:

Fibroadenoma of the Breast (Fibroadenoma of the Breast)

Definition:
A benign tumor of the breast made up of both glandular and stromal tissue.

Key Clinical Presentation:

- Firm, painless, mobile mass
- Often seen in younger women (typically 20-30 years old)
- Does not change during the menstrual cycle

Investigations:

- **Best initial**: Clinical examination and mammography or ultrasound
- **Confirmatory**: Core needle biopsy if needed

Treatment Outline:

- **First-line**: Observation if asymptomatic
- **Second-line**: Surgical excision if symptomatic or increasing in size

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MCQ 539

A 21-year-old patient presents with an abnormal left testicle. On examination, the area adjacent to the left testicle feels like a bag of worms, which gets larger with a valsalva maneuver. Which of the following is the most likely diagnosis?

A. Hydrocele
B. Varicocele
C. Spermatocele
D. Left inguinal hernia

Explanation:
A **varicocele** is the most likely diagnosis. It is characterized by a "bag of worms" appearance on examination and is caused by dilated veins within the pampiniform plexus of the scrotum, often becoming more prominent with a valsalva maneuver.

Correct Answer: B. Varicocele ✅

Topic Section:

Varicocele (Varicocele)

Definition:
Enlargement of the veins within the scrotum, typically affecting the left side.

Key Clinical Presentation:

- "Bag of worms" feeling on palpation
- Symptoms may include dull aching or heaviness in the scrotum
- Can be exacerbated by standing or valsalva maneuver

Investigations:

- **Best initial**: Clinical examination
- **Confirmatory**: Doppler ultrasound

Treatment Outline:

- **First-line**: Observation if asymptomatic
- **Second-line**: Surgical intervention or embolization if causing symptoms or infertility

---

MCQ 540

A 35-year-old man patient comes to the clinic with a painless ulcer on his penis. History reveals, he had unprotected sexual encounter 3 weeks ago. Examination shows that the lesion has a clean base with sharp margins, and there are non-tender palpable regional lymph nodes. Which of the following is the most likely diagnosis?

A. Syphilis
B. Chancroid
C. Gonorrhoea
D. Granuloma inguinale

Explanation:
The presence of a painless ulcer with clean base and sharp margins, along with non-tender regional lymphadenopathy, suggests **syphilis**. This is the characteristic presentation of primary syphilis, caused by *Treponema pallidum*.

Correct Answer: A. Syphilis ✅

Topic Section:

Syphilis (Syphilis)

Definition:
A sexually transmitted infection caused by *Treponema pallidum* that presents in stages, with primary syphilis characterized by a painless chancre (ulcer).

Key Clinical Presentation:

- Painless ulcer (chancre) with clean base and sharp margins
- Non-tender regional lymphadenopathy
- Resolves on its own but requires treatment to prevent progression

Investigations:

- **Best initial**: Serologic testing (VDRL, RPR)
- **Confirmatory**: Dark field microscopy or PCR for *Treponema pallidum*

Treatment Outline:

- **First-line**: Penicillin G (IM)
- **Second-line**: Doxycycline for penicillin-allergic patients

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MCQ 541

A 53-year-old man presents to the hospital with fever and pain in the right knee for 2 days. On physical examination, the right knee was swollen and red with a limited range of motion. A knee joint aspiration was performed. Then, the patient started empirically on Cloxacillin. On day 3 after admission, a further report was received: Gram stain showed numerous white blood cells and gram-positive cocci in clusters. Bacteriology (3 days later): *Staphylococcus aureus* that was resistant to cefoxitin. Which of the following is the most appropriate action?

A. Add gentamicin
B. Start vancomycin
C. Continue cloxacillin
D. Arthroscopic washout

Explanation:
Since *Staphylococcus aureus* resistant to cefoxitin (suggesting methicillin resistance), **vancomycin** is the most appropriate choice for resistant infections, especially for osteoarticular infections.

Correct Answer: B. Start vancomycin ✅

Topic Section:

Methicillin-resistant Staphylococcus aureus (MRSA) (Methicillin-resistant Staphylococcus aureus)

Definition:
A strain of *Staphylococcus aureus* that is resistant to beta-lactam antibiotics, including methicillin, and requires alternative antibiotics for treatment.

Key Clinical Presentation:

- Localized infection (e.g., skin, bone, joint)
- Can cause septic arthritis, osteomyelitis, or soft tissue infections
- Fever, redness, and swelling at the site of infection

Investigations:

- **Best initial**: Gram stain and culture of aspirated fluid
- **Confirmatory**: PCR or antimicrobial susceptibility testing

Treatment Outline:

- **First-line**: Vancomycin or clindamycin for MRSA
- **Second-line**: Linezolid or daptomycin for severe cases
- **Definitive**: Drainage of infected joints or tissue if required

MCQ 542

A 75-year-old man presents to the Outpatient Department, complaining of back pain and increasing difficulty with passing urine. Test Result: Prostate-specific antigen 84 (0-4 μg/L), Alkaline phosphatase 410 (39-117 IU/L), Albumin 41 (34-56 g/L), Gamma glutamyltransferase 19 (6 to 37 IU/L). Which of the following is the most likely diagnosis?

A. Prostatitis
B. Prostatic cancer
C. Urinary bladder cancer
D. Benign prostate hyperplasia

Explanation:
The elevated prostate-specific antigen (PSA) levels and alkaline phosphatase are indicative of **prostatic cancer**, particularly when accompanied by symptoms of back pain and urinary difficulty. Elevated PSA in combination with bone pain can be a sign of metastatic prostate cancer, making it the most likely diagnosis.

Correct Answer: B. Prostatic cancer ✅

Topic Section:

Prostatic Cancer (Prostate Cancer)

Definition:
A malignant tumor originating from the prostate gland, often characterized by elevated PSA levels and symptoms of urinary obstruction.

Key Clinical Presentation:

- Back pain (especially if metastasis to bone occurs)
- Urinary symptoms (difficulty passing urine, frequency, urgency)
- Elevated PSA levels
- Weight loss in advanced stages

Investigations:

- **Best initial**: Prostate-specific antigen (PSA)
- **Confirmatory**: Transrectal ultrasound-guided biopsy of the prostate

Treatment Outline:

- **First-line**: Radical prostatectomy, radiation therapy, or androgen deprivation therapy (ADT)
- **Second-line**: Chemotherapy for metastatic disease
- **Definitive**: Surgery or radiation based on disease staging

---

MCQ 543

A 60-year-old man presents to the clinic with pain in the left flank, weight loss, and haematuria. Examination confirms a firm mass in the lumbar region. Blood pressure 158/98 mmHg, Heart rate 76 /min, Respiratory rate 18 /min, Temperature 36.6°C. Which of the following imaging is the most diagnostic?

A. MRI
B. CT scan
C. Ultrasound
D. Radionuclide scan

Explanation:
A **CT scan** is the most diagnostic for kidney masses, particularly when evaluating for potential renal cell carcinoma (RCC), which can present with flank pain, weight loss, and haematuria. CT scan offers detailed imaging for detecting tumors and metastases, as well as for staging.

Correct Answer: B. CT scan ✅

Topic Section:

Renal Cell Carcinoma (RCC) (Renal Cell Carcinoma)

Definition:
A malignant tumor originating from the renal epithelium, often presenting with flank pain, hematuria, and a palpable abdominal mass.

Key Clinical Presentation:

- Flank pain
- Hematuria
- Weight loss
- Palpable abdominal mass in some cases

Investigations:

- **Best initial**: Ultrasound for initial evaluation
- **Confirmatory**: Contrast-enhanced CT or MRI of the abdomen

Treatment Outline:

- **First-line**: Nephrectomy (partial or radical) for localized disease

- **Second-line**: Targeted therapy (e.g., tyrosine kinase inhibitors) for metastatic disease

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MCQ 544

A 30-year-old patient presents to the clinic with a pelvic fracture, due to blunt trauma. A retrograde urethrography demonstrates disruption of the membranous urethra. Which of the following is the most appropriate first step in management?

A. Perineal repair
B. Retropubic repair
C. Suprapubic catheter
D. Transurethral catheter

Explanation:
The most appropriate initial step is to place a **suprapubic catheter** to decompress the bladder and avoid further injury to the urethra. This is particularly important when there is urethral injury, as inserting a transurethral catheter may worsen the injury.

Correct Answer: C. Suprapubic catheter ✅

Topic Section:

Urethral Injury (Urethral Injury)

Definition:
Injury to the urethra, often resulting from blunt trauma or pelvic fractures. This can lead to urinary retention and possible infection if not managed appropriately.

Key Clinical Presentation:

- Blood at the urethral meatus
- Difficulty passing urine
- Pelvic fracture history
- Tenderness over the pelvic region

Investigations:

- **Best initial**: Retrograde urethrography
- **Confirmatory**: Cystoscopy or further imaging as needed

Treatment Outline:

- **First-line**: Suprapubic catheter insertion to avoid worsening urethral injury
- **Second-line**: Surgical repair based on the type of injury
- **Definitive**: Urethral repair (perineal or retropubic approach depending on injury site)

MCQ 545

A 2-year-old boy is brought to the clinic with a large left-sided abdominal mass. IV pyelogram shows, the mass appears to arise from the left kidney and distort and displace the collecting system. Chest X-ray: Multiple pulmonary nodules. Which of the following is the most likely diagnosis?

A. Wilm’s tumour
B. Neuroblastoma
C. Rhabdomyosarcoma
D. Non-Hodgkin lymphoma

Explanation:
Wilm’s tumor is the most likely diagnosis. It is the most common renal malignancy in children and typically presents as an abdominal mass with or without symptoms. The presence of pulmonary nodules suggests metastatic spread, which is common in advanced stages of Wilm’s tumor.

Correct Answer: A. Wilm’s tumour ✅

Topic Section:

Wilm's Tumor (Wilms' Tumor)

Definition:
A malignant kidney tumor commonly seen in children, typically presenting as a painless abdominal mass.

Key Clinical Presentation:

- Abdominal mass
- Hematuria (less common)
- Hypertension in some cases

- Pulmonary metastasis in advanced stages

Investigations:

- **Best initial**: Abdominal ultrasound
- **Confirmatory**: CT scan or MRI for staging, biopsy if necessary

Treatment Outline:

- **First-line**: Nephrectomy and chemotherapy
- **Second-line**: Radiation for advanced or recurrent disease
- **Definitive**: Surgery (nephrectomy) followed by chemotherapy based on staging

---

MCQ 546

A 30-year-old patient presents with recurrent urinary tract infections, polyuria, and electrolyte imbalance. Ultrasound: Bilaterally enlarged kidneys with multiple, variably-sized and thin-walled cysts distributed throughout the renal parenchyma. Which of the following is the most likely diagnosis?

A. Carcinoma bladder
B. Renal calculi disease
C. Medullary sponge kidney
D. Polycystic kidney disease

Explanation:
Polycystic kidney disease (PKD) is the most likely diagnosis. It is characterized by multiple cysts in the kidneys, leading to renal enlargement and dysfunction. It is often associated with recurrent urinary tract infections and electrolyte imbalances as the disease progresses.

Correct Answer: D. Polycystic kidney disease ✅

Topic Section:

Polycystic Kidney Disease (Polycystic Kidney Disease)

Definition:
A genetic disorder characterized by the growth of numerous cysts in the kidneys, leading to renal enlargement and progressive renal failure.

Key Clinical Presentation:

- Enlarged kidneys (palpable in some cases)
- Recurrent urinary tract infections
- Hypertension
- Electrolyte imbalances

Investigations:

- **Best initial**: Ultrasound to detect renal cysts
- **Confirmatory**: Genetic testing (ADPKD1 and ADPKD2 mutations)

Treatment Outline:

- **First-line**: Blood pressure control (ACE inhibitors or ARBs), pain management
- **Second-line**: Dialysis and kidney transplant for advanced renal failure
- **Definitive**: Kidney transplant in end-stage disease

MCQ 547

A 80-year-old man presents with dull aching pain in the loins. Test Result: Creatinine 280 (44-115 µmol/L), Urea 12.5 (2.75-7.4 mmol/L). Ultrasound abdomen: Bilateral hydronephrosis. Which of the following is the most likely diagnosis?

A. Stricture of urethral meatus
B. Prostatic enlargement
C. Neoplasm of bladder
D. Pelvic cancer

Explanation:
Bilateral hydronephrosis with elevated creatinine and urea is suggestive of obstructive uropathy. The most likely cause in this elderly male is **prostatic enlargement**, which can lead to urinary retention and bilateral hydronephrosis by obstructing the flow of urine.

Correct Answer: B. Prostatic enlargement ✅

Topic Section:

Prostatic Enlargement (Benign Prostatic Hyperplasia)

Definition:
Benign enlargement of the prostate gland commonly affecting older men, leading to urinary obstruction and other lower urinary tract symptoms.

Key Clinical Presentation:

- Urinary frequency, especially at night (nocturia)
- Difficulty initiating urination
- Weak urinary stream
- Sensation of incomplete bladder emptying
- Pain in the loins or lower abdomen

Investigations:

- **Best initial**: Digital rectal exam (DRE) and PSA measurement
- **Confirmatory**: Ultrasound of the prostate, cystoscopy if needed

Treatment Outline:

- **First-line**: Alpha-blockers (e.g., tamsulosin)
- **Second-line**: 5-alpha-reductase inhibitors (e.g., finasteride)
- **Definitive**: Surgical options like transurethral resection of the prostate (TURP) for refractory cases

---

MCQ 548

A 65-year-old male comes to the clinic with a mild intermittent urinary flow reduction. Rectal examination, urinalysis and prostate specific antigen studies are normal. Ultrasound prostate: Enlarged median lobe. Which of the following is the best way to investigate?

A. Annual renal function monitoring
B. Periodic PSA measurement
C. Beta-blocker therapy
D. Cystoscopy

Explanation:
The normal PSA, rectal exam, and urinalysis suggest benign prostatic hypertrophy (BPH) rather than prostate cancer. The **best next step** for further investigation would be **periodic PSA**

measurement to monitor the condition, as well as symptomatic treatment with alpha-blockers if required.

Correct Answer: B. Periodic PSA measurement ✅

Topic Section:

Benign Prostatic Hyperplasia (BPH)

Definition:
Non-cancerous enlargement of the prostate gland, leading to lower urinary tract symptoms (LUTS).

Key Clinical Presentation:

- Increased frequency of urination, particularly nocturia
- Weak stream, difficulty initiating urination
- Sensation of incomplete emptying
- No fever, pain, or hematuria

Investigations:

- **Best initial**: Digital rectal exam (DRE) and PSA
- **Confirmatory**: Ultrasound of prostate, cystoscopy if necessary

Treatment Outline:

- **First-line**: Alpha-blockers (e.g., tamsulosin)
- **Second-line**: 5-alpha-reductase inhibitors (e.g., finasteride)
- **Definitive**: Surgical options like TURP

MCQ 549

A 82-year-old man presents to the clinic with inability to urinate. The patient complains of severe pain and bloating in the lower abdomen, painful and urgent need to urinate. Which of the following is the most appropriate management?

A. Semi-urgent prostatectomy
B. Antibiotic for a urinary infection

C. Foley catheterisation and culture urine
D. Cystoscopy and Transurethral Resection Prostatectomy

Explanation:
The patient is presenting with acute urinary retention, likely due to benign prostatic hyperplasia (BPH) or other obstruction. The most immediate and appropriate intervention is **Foley catheterization** to relieve urinary retention, followed by a urine culture to rule out infection.

Correct Answer: C. Foley catheterisation and culture urine ✅

Topic Section:

Acute Urinary Retention (Acute Urinary Retention)

Definition:
Sudden inability to urinate, often due to obstruction such as benign prostatic hyperplasia, bladder stones, or other conditions.

Key Clinical Presentation:

- Severe pain and bloating in the lower abdomen
- Inability to void, with a strong urge to urinate
- Firm, distended bladder on palpation

Investigations:

- **Best initial**: Clinical examination and bladder ultrasound
- **Confirmatory**: Urine culture, cystoscopy, and prostate examination

Treatment Outline:

- **First-line**: Foley catheterization to relieve the retention
- **Second-line**: Treat underlying cause (e.g., alpha-blockers for BPH)
- **Definitive**: Surgery for persistent obstruction or cause

---

MCQ 550

A 70-year-old, non-smoking man presents with increased urinary frequency, especially at night. The patient feels that the bladder is not fully empty after voiding, and the urine stream

has slightly diminished. The patient's high blood pressure is well controlled with a calcium channel blocker. There is no associated fever, weight loss, haematuria, or pain with voiding. The digital rectal examination was significant for a moderate enlargement of the prostate, but is otherwise normal. Results of urinalysis and kidney function tests were normal. PSA antigen 1 (< 4 ng/ml). Which of the following is the most appropriate initial management?

A. Cystoscopy
B. Open prostatectomy
C. Alpha-blocker therapy
D. Transurethral Resection Prostatectomy

Explanation:
The patient likely has **benign prostatic hyperplasia (BPH)**, as indicated by urinary frequency, nocturia, and weak stream with normal prostate exam findings. **Alpha-blocker therapy** (e.g., tamsulosin) is the first-line treatment for symptomatic BPH, improving urine flow and relieving symptoms.

Correct Answer: C. Alpha-blocker therapy ✅

Topic Section:

Benign Prostatic Hyperplasia (BPH)

Definition:
Enlargement of the prostate gland, often leading to urinary obstruction and lower urinary tract symptoms.

Key Clinical Presentation:

- Increased frequency of urination
- Nocturia
- Weak stream, difficulty starting urination
- Sensation of incomplete emptying of the bladder

Investigations:

- **Best initial**: Digital rectal exam (DRE) and PSA level
- **Confirmatory**: Ultrasound of prostate and cystoscopy if needed

Treatment Outline:

- **First-line**: Alpha-blockers (e.g., tamsulosin)
- **Second-line**: 5-alpha-reductase inhibitors (e.g., finasteride)
- **Definitive**: Surgical options like TURP for refractory cases

---

MCQ 551

A 56-year-old multiparous patient on oral oestrogen, complains of painless urine incontinence when coughing, sneezing, or laughing for 1 year. On examination, there was mild vaginal atrophy and constrictor muscle laxity. Urinary incontinence is demonstrated in the lithotomy position with Valsalva manoeuvre. Urinalysis: Normal. Which of the following would be the recommended management?

A. Beta blocker
B. Kegel exercises
C. Periurethral bulking
D. Post-coital antibiotics

Explanation:
The patient's symptoms are indicative of **stress urinary incontinence** (SUI), which is commonly seen in multiparous women, particularly with vaginal atrophy and laxity of the pelvic floor muscles. The most effective initial treatment is **Kegel exercises**, which strengthen the pelvic floor muscles and help manage stress incontinence.

Correct Answer: B. Kegel exercises ✅

Topic Section:

Stress Urinary Incontinence (SUI)

Definition:
Involuntary leakage of urine during activities that increase intra-abdominal pressure (e.g., coughing, sneezing, laughing, or exercising), due to weakening of the pelvic floor muscles or damage to the urethral sphincter.

Key Clinical Presentation:

- Urinary incontinence associated with coughing, sneezing, or physical activity
- Palpably distended bladder, especially after voiding

- Vaginal atrophy or pelvic floor laxity may be present in older women

Investigations:

- **Best initial**: Clinical examination and history
- **Confirmatory**: Urodynamic testing if needed

Treatment Outline:

- **First-line**: Kegel exercises, lifestyle changes (weight loss, fluid management)
- **Second-line**: Periurethral bulking agents, vaginal pessaries
- **Definitive**: Surgery, such as sling procedures, in refractory cases

---

MCQ 552

A 37-year-old man presents to the clinic with flank pain, haematuria and burning sensation while urinating. The pain became more severe and constant in the last several days. Which of the following is the most appropriate investigation?

A. Nuclear renal scan
B. Ultrasound of Kidneys
C. CT of Abdomen without contrast
D. CT of Abdomen with oral and IV contrast

Explanation:
Given the patient's symptoms of flank pain, hematuria, and dysuria, the most likely diagnosis is a **urinary stone**. A **CT of the abdomen with contrast** is the most appropriate investigation as it provides detailed imaging to detect stones in the kidneys, ureters, and bladder, including those that may not be visible on ultrasound.

Correct Answer: D. CT of Abdomen with oral and IV contrast ✅

Topic Section:

Urinary Tract Stones (Urolithiasis)

Definition:
Formation of stones in the urinary tract, commonly in the kidneys, ureters, or bladder, resulting in pain, hematuria, and sometimes obstruction.

Key Clinical Presentation:

- Severe flank pain (renal colic), hematuria, and burning sensation on urination
- Pain often radiates to the groin or lower abdomen
- Symptoms may worsen with movement or voiding

Investigations:

- **Best initial**: Non-contrast CT scan or ultrasound
- **Confirmatory**: CT of abdomen with contrast for detailed stone imaging

Treatment Outline:

- **First-line**: Conservative management with hydration and analgesia for small stones
- **Second-line**: Extracorporeal shock wave lithotripsy (ESWL) for larger stones
- **Definitive**: Surgical intervention (ureteroscopy or nephrolithotomy) for large or impacted stones

---

MCQ 553

A 70-year-old man presents with urinary incontinence. The patient's bladder is palpably distended after voicing. The patient stated that he voids frequently, although he has difficulty initiating the urine stream. Which of the following is the most likely type of incontinence?

A. Urge
B. Stress
C. Reflex
D. Overflow

Explanation:
The patient's presentation of urinary retention, difficulty initiating urination, and a palpable distended bladder suggests **overflow incontinence**, which occurs when the bladder cannot empty fully, leading to frequent, incomplete voiding and leakage due to overflow.

Correct Answer: D. Overflow ✅

Topic Section:

Overflow Urinary Incontinence

Definition:
Involuntary leakage of urine due to an overfull bladder, often caused by incomplete bladder emptying, leading to urine retention and leakage.

Key Clinical Presentation:

- Palpably distended bladder, frequent small voids with difficulty initiating urination
- Leakage of urine, especially at night
- May be associated with neurological conditions or bladder outlet obstruction

Investigations:

- **Best initial**: Post-void residual volume measurement (bladder scan)
- **Confirmatory**: Urodynamic studies to assess bladder function

Treatment Outline:

- **First-line**: Bladder drainage via intermittent catheterization or indwelling catheter
- **Second-line**: Alpha-blockers or surgical intervention for bladder outlet obstruction
- **Definitive**: Surgery for bladder outlet obstruction or neurogenic bladder management

---

MCQ 554

A 40-year-old man comes to the clinic with pain in the lower back, and blood in the urine. IVP shows a non-opaque filling defects in the renal pelvis. Ultrasound revealed dense echoes and acoustic shadowing. Which of the following is the most likely diagnosis?

A. Tumour
B. Blood clot
C. Uric acid stone
D. Sloughed renal papilla

Explanation:
The findings of non-opaque filling defects on IVP, along with dense echoes and acoustic shadowing on ultrasound, are highly suggestive of a **urinary stone**, particularly a **uric acid stone**, which is radiolucent and may present with these features.

Correct Answer: C. Uric acid stone ✅

Topic Section:

Uric Acid Stones

Definition:
A type of kidney stone that forms in acidic urine, often seen in conditions like gout, dehydration, or metabolic syndrome.

Key Clinical Presentation:

- Severe flank pain (renal colic), hematuria, dysuria
- Pain often radiates to the groin or lower abdomen

Investigations:

- **Best initial**: Non-contrast CT or ultrasound for stone detection
- **Confirmatory**: Urinary pH measurement and stone analysis (if available)

Treatment Outline:

- **First-line**: Hydration, alkalinization of urine (e.g., potassium citrate)
- **Second-line**: Lithotripsy or ureteroscopy for larger stones
- **Definitive**: Surgical removal for large or obstructing stones

---

MCQ 555

A 74-year-old man is referred to the thoracic surgeon with dysphagia of solid food for six weeks and marked weight loss for 2 months. Oesophageal endoscopy shows a stenotic mass in the mid oesophagus. Chest X-ray: Normal. Which of the following is the most likely diagnosis?

A. Lymphoma
B. Adenocarcinoma
C. Squamous cell carcinoma
D. Extension from neuroendocrine cancer

Explanation:
This patient presents with dysphagia, weight loss, and a stenotic mass seen on oesophageal endoscopy. The most likely diagnosis is **squamous cell carcinoma** of the esophagus, which is more commonly located in the mid to upper oesophagus, particularly in patients with risk factors such as smoking and alcohol use.

Correct Answer: C. Squamous cell carcinoma ✅

Topic Section:

Oesophageal Squamous Cell Carcinoma

Definition:
A malignant tumor originating in the squamous epithelial cells of the esophagus, commonly associated with smoking, alcohol, and other environmental factors.

Key Clinical Presentation:

- Dysphagia, initially to solids, progressing to liquids
- Unintentional weight loss
- Painful swallowing (odynophagia)
- Regurgitation of food

Investigations:

- **Best initial**: Oesophagoscopy with biopsy for histopathology
- **Confirmatory**: Endoscopic ultrasound for staging

Treatment Outline:

- **First-line**: Chemoradiotherapy for localized disease
- **Second-line**: Esophagectomy with lymph node dissection
- **Definitive**: Combination of surgery, radiation, and chemotherapy for advanced cases

---

MCQ 556

An army soldier comes to the Outpatient Clinic complaining of painful flat feet after several years of service, which included thousands of kilometers of marching in the desert. The pain was acute in the medial aspect of his sole. Which of the following structures is the most likely strained?

A. Tendon achilles
B. Spring ligament

C. Flexor retinaculum
D. Extensor retinaculum

Explanation:
The pain on the medial aspect of the sole with prolonged marching in a soldier suggests strain on the **spring ligament**, which supports the medial arch of the foot. The spring ligament is crucial in maintaining the arch, and its injury can lead to pain in the medial foot.

Correct Answer: B. Spring ligament ✅

Topic Section:

Spring Ligament (Plantar Calcaneonavicular Ligament)

Definition:
A ligament in the foot that supports the medial longitudinal arch, attaching the calcaneus to the navicular bone.

Key Clinical Presentation:

- Pain in the medial arch, especially with prolonged standing or activity
- Tenderness over the spring ligament

Investigations:

- **Best initial**: Clinical examination and history
- **Confirmatory**: MRI if conservative management fails

Treatment Outline:

- **First-line**: Rest, ice, and orthotic support
- **Second-line**: Physical therapy, NSAIDs
- **Definitive**: Surgical repair if conservative measures fail

---

MCQ 557

A 65-year-old woman comes to the Outpatient Clinic complaining of pain in her left hand and fingers from typing on the keyboard for a while. Investigations reveal insufficient blood flow into the superficial palmar arch. Occlusion of which of the following arteries is the most likely cause?

A. Ulnar
B. Radial
C. Anterior interosseous
D. Posterior interosseous

Explanation:
The **ulnar artery** is the primary source of blood supply to the superficial palmar arch, and occlusion here can result in insufficient blood flow to the hand and fingers, particularly in the context of repetitive activities like typing.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Artery Occlusion

Definition:
Occlusion of the ulnar artery, which can lead to decreased blood flow to the hand, especially the medial side.

Key Clinical Presentation:

- Pain, numbness, or tingling in the hand and fingers
- Decreased blood flow in the affected hand
- May be exacerbated by repetitive activities (e.g., typing)

Investigations:

- **Best initial**: Doppler ultrasound to assess blood flow
- **Confirmatory**: Angiography in cases of persistent symptoms

Treatment Outline:

- **First-line**: Conservative management with rest and anti-inflammatory medications
- **Second-line**: Surgical intervention in cases of persistent ischemia or claudication
- **Definitive**: Arterial bypass or thrombectomy for severe cases

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MCQ 558

As part of physical examination to evaluate the lower limb function of a football player, the physician asked him to stand on his tiptoes. Which of the following nerves was the physician testing?

A. Tibial
B. Femoral
C. Deep fibular
D. Superficial fibular

Explanation:
The ability to stand on tiptoes tests the function of the **tibial nerve**, which innervates the gastrocnemius and soleus muscles, responsible for plantar flexion at the ankle.

Correct Answer: A. Tibial ✅

Topic Section:

Tibial Nerve Function

Definition:
The tibial nerve is responsible for motor innervation to muscles of the posterior compartment of the leg, including the gastrocnemius and soleus, which control plantar flexion of the foot.

Key Clinical Presentation:

- Inability to plantar flex the foot or stand on tiptoes when the tibial nerve is damaged
- Weakness in foot inversion and toe flexion

Investigations:

- **Best initial**: Clinical examination of foot movements
- **Confirmatory**: Electromyography (EMG) if necessary

Treatment Outline:

- **First-line**: Physical therapy and conservative management
- **Definitive**: Surgical intervention for nerve decompression or repair in severe cases

---

MCQ 559

A 25-year-old man is brought to the Emergency Room after a motor vehicle accident. Examination shows, he has a severed nerve in the right upper limb, resulting in a loss of adduction of all the digits of the right hand. Which of the following is the most likely affected nerve?

A. Ulnar
B. Median
C. Musculocutaneous
D. Upper subscapular

Explanation:
The **ulnar nerve** innervates the intrinsic muscles of the hand responsible for adducting the fingers, including the palmar interossei and the adductor pollicis. Damage to the ulnar nerve results in loss of adduction of the digits.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Nerve Injury

Definition:
Damage to the ulnar nerve can result in impaired hand function, particularly the loss of fine motor control and grip strength, including the ability to adduct the fingers.

Key Clinical Presentation:

- Weakness in hand grip and finger adduction
- Claw hand deformity (if prolonged damage)
- Numbness along the ulnar side of the hand and forearm

Investigations:

- **Best initial**: Clinical examination for hand movements and grip strength
- **Confirmatory**: Electromyography (EMG) to assess nerve function

Treatment Outline:

- **First-line**: Conservative management with hand therapy
- **Second-line**: Splinting or bracing for claw hand deformity

- **Definitive**: Nerve repair or decompression if necessary

MCQ 560

A 74-year-old man is referred to the thoracic surgeon with dysphagia of solid food for six weeks and marked weight loss for 2 months. Oesophageal endoscopy shows a stenotic mass in the mid oesophagus. Chest X-ray: Normal. Which of the following is the most likely diagnosis?

A. Lymphoma
B. Adenocarcinoma
C. Squamous cell carcinoma
D. Extension from neuroendocrine cancer

Explanation:
The most likely diagnosis for a patient presenting with dysphagia to solid foods and weight loss with a stenotic mass in the mid-oesophagus is **squamous cell carcinoma**. It is commonly seen in the middle third of the oesophagus, particularly in older patients with a history of smoking and alcohol use.

Correct Answer: C. Squamous cell carcinoma ✅

Topic Section:

Oesophageal Squamous Cell Carcinoma

Definition:
A malignant tumour originating in the squamous epithelium of the oesophagus, typically in the mid-to-lower part of the oesophagus.

Key Clinical Presentation:

- Progressive dysphagia (initially to solids, then to liquids)
- Unintended weight loss
- Odynophagia (painful swallowing)
- Hoarseness (if the recurrent laryngeal nerve is involved)

Investigations:

- **Best initial**: Oesophagoscopy with biopsy for histopathological examination

- **Confirmatory**: Endoscopic ultrasound (EUS) for staging

Treatment Outline:

- **First-line**: Chemotherapy and radiation for localized disease
- **Second-line**: Surgery, if resectable (esophagectomy)
- **Definitive**: Combined modality treatment (surgery, chemotherapy, and radiation) for advanced stages

---

MCQ 561

An army soldier comes to the Outpatient Clinic complaining of painful flat feet after several years of service, which included thousands of kilometres of marching in the desert. The pain was acute in the medial aspect of his sole. Which of the following structures is the most likely strained?

A. Tendon achilles
B. Spring ligament
C. Flexor retinaculum
D. Extensor retinaculum

Explanation:
The **spring ligament** is responsible for supporting the medial arch of the foot, and strain or injury to it can cause pain in the medial sole. This condition is often exacerbated by activities like marching, which put strain on the arch.

Correct Answer: B. Spring ligament ✅

Topic Section:

Spring Ligament Injury (Plantar Calcaneonavicular Ligament)

Definition:
A ligament that supports the medial longitudinal arch of the foot, attaching the calcaneus to the navicular bone.

Key Clinical Presentation:

- Pain in the medial arch, particularly with activity
- Tenderness along the spring ligament

- Possible swelling in the arch area

Investigations:

- **Best initial**: Clinical examination and imaging (X-ray or MRI)
- **Confirmatory**: MRI to assess ligament integrity

Treatment Outline:

- **First-line**: Rest, orthotics, and anti-inflammatory medications
- **Second-line**: Physical therapy to strengthen foot muscles
- **Definitive**: Surgical repair if conservative measures fail

---

MCQ 562

A 65-year-old woman comes to the Outpatient Clinic complaining of pain in her left hand and fingers from typing on the keyboard for a while. Investigations reveal insufficient blood flow into the superficial palmar arch. Occlusion of which of the following arteries is the most likely cause?

A. Ulnar
B. Radial
C. Anterior interosseous
D. Posterior interosseous

Explanation:
The **ulnar artery** is the primary source of blood to the superficial palmar arch. Occlusion of the ulnar artery is the most likely cause of insufficient blood flow to the hand and fingers, especially in cases where repetitive activity like typing is involved.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Artery Occlusion

Definition:
Occlusion of the ulnar artery, which causes reduced blood flow to the hand, particularly the ulnar side.

Key Clinical Presentation:

- Numbness, tingling, and pain in the hand, particularly in the ulnar distribution
- Weakness in hand grip and difficulty with fine motor control
- Coldness or pallor in the affected hand

Investigations:

- **Best initial**: Doppler ultrasound to assess blood flow
- **Confirmatory**: Angiography for detailed vascular mapping

Treatment Outline:

- **First-line**: Conservative management with rest and anti-inflammatory medications
- **Second-line**: Physical therapy or vasodilators
- **Definitive**: Surgical revascularization (bypass or thrombectomy) for severe cases

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MCQ 564

As part of physical examination to evaluate the lower limb function of a football player, the physician asked him to stand on his tiptoes. Which of the following nerves was the physician testing?

A. Tibial
B. Femoral
C. Deep fibular
D. Superficial fibular

Explanation:
Standing on tiptoes tests the function of the **tibial nerve**, which innervates the gastrocnemius and soleus muscles responsible for plantar flexion of the foot.

Correct Answer: A. Tibial ✅

Topic Section:

Tibial Nerve Function

Definition:
The tibial nerve innervates the posterior muscles of the leg, including the gastrocnemius and soleus muscles, which are responsible for plantar flexion of the foot.

Key Clinical Presentation:

- Inability to stand on tiptoes or plantar flex the foot if the tibial nerve is injured
- Weakness in foot inversion and toe flexion
- Loss of sensation in the sole of the foot

Investigations:

- **Best initial**: Physical examination (observe the patient's ability to stand on tiptoes)
- **Confirmatory**: Nerve conduction studies if necessary

Treatment Outline:

- **First-line**: Rest and physiotherapy for mild injury
- **Second-line**: Foot orthotics or splints
- **Definitive**: Nerve repair or decompression surgery in severe cases

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MCQ 565

A 25-year-old man is brought to the Emergency Room after a motor vehicle accident. Examination shows, he has a severed nerve in the right upper limb, resulting in a loss of adduction of all the digits of the right hand. Which of the following is the most likely affected nerve?

A. Ulnar
B. Median
C. Musculocutaneous
D. Upper subscapular

Explanation:
The **ulnar nerve** innervates the intrinsic hand muscles responsible for adducting the fingers, such as the palmar interossei and adductor pollicis. Injury to this nerve results in the loss of finger adduction.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Nerve Injury

Definition:
The ulnar nerve provides motor innervation to the intrinsic muscles of the hand responsible for fine motor control, including finger adduction and flexion.

Key Clinical Presentation:

- Weakness in the hand, especially with grip strength and finger adduction
- Claw hand deformity (if untreated)
- Numbness along the ulnar side of the hand and forearm

Investigations:

- **Best initial**: Clinical examination for motor and sensory deficits
- **Confirmatory**: Electromyography (EMG) to assess nerve damage

Treatment Outline:

- **First-line**: Conservative management with splints or bracing
- **Second-line**: Hand therapy to improve function
- **Definitive**: Nerve repair or decompression if necessary

MCQ 566

A 20-year-old man comes to the clinic complaining of knee pain. History reveals he received a severe blow on the inferolateral side of the left knee joint while playing football. Radiographic examination revealed a fracture of the head and neck of the fibula. Which of the following nerves is the most likely to be damaged?

A. Tibial
B. Deep peroneal
C. Common peroneal
D. Superficial peroneal

Explanation:
The **common peroneal nerve** is most likely to be damaged in fractures of the fibula, particularly at the head and neck. This nerve passes around the fibular head and supplies the muscles responsible for dorsiflexion and eversion of the foot. Damage to this nerve causes foot drop and sensory loss in the dorsum of the foot.

Correct Answer: C. Common peroneal ✅

Topic Section:

Common Peroneal Nerve Injury

Definition:
Injury to the common peroneal nerve, often resulting from trauma to the fibular head or neck, leading to motor and sensory deficits.

Key Clinical Presentation:

- Foot drop (inability to dorsiflex the foot)
- Loss of sensation on the dorsum of the foot and lateral aspect of the lower leg
- Weakness in eversion of the foot

Investigations:

- **Best initial**: Clinical examination to assess motor and sensory function
- **Confirmatory**: Electromyography (EMG) or nerve conduction studies

Treatment Outline:

- **First-line**: Conservative management with ankle-foot orthotics
- **Second-line**: Physical therapy to improve foot function
- **Definitive**: Nerve repair or decompression if necessary

MCQ 567

A 25-year-old man is brought to the Emergency Room after a severe road traffic accident. Examination shows he has an unstable left knee joint and it was possible to pull the tibia abnormally forwards on the femur. Which of the following ligaments is most likely injured?

A. Fibular collateral
B. Tibial collateral
C. Anterior cruciate
D. Posterior cruciate

Explanation:
The **anterior cruciate ligament (ACL)** is most likely injured in cases of instability and abnormal forward movement (anterior drawer sign) of the tibia relative to the femur. This is a common injury in trauma, especially in road traffic accidents and sports.

Correct Answer: C. Anterior cruciate ✅

Topic Section:

Anterior Cruciate Ligament Injury

Definition:
A tear or sprain of the anterior cruciate ligament (ACL), which stabilizes the knee by preventing anterior displacement of the tibia relative to the femur.

Key Clinical Presentation:

- Instability or "giving way" of the knee
- Pain and swelling shortly after injury
- Positive anterior drawer test or Lachman test

Investigations:

- **Best initial**: Physical examination with Lachman and anterior drawer tests
- **Confirmatory**: MRI to confirm ligament rupture

Treatment Outline:

- **First-line**: Conservative management with physical therapy (for minor sprains)
- **Second-line**: ACL reconstruction surgery for significant tears
- **Definitive**: Surgical intervention for severe ACL injuries with persistent instability

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MCQ 568

A 35-year-old man is brought to the Emergency Room after a road traffic accident. X-ray shows a displaced fracture of the midshaft of the humerus. Which of the following is the most likely clinical complication?

A. Wrist drop
B. Claw hand
C. Waiter's tip hand
D. Klumpke's paralysis

Explanation:
The **radial nerve** is at risk in fractures of the midshaft of the humerus. Damage to the radial nerve can result in **wrist drop**, characterized by the inability to extend the wrist and fingers.

Correct Answer: A. Wrist drop ✅

Topic Section:

Radial Nerve Injury

Definition:
Injury to the radial nerve, typically caused by midshaft humeral fractures, leading to motor and sensory deficits.

Key Clinical Presentation:

- Inability to extend the wrist and fingers (wrist drop)
- Sensory loss on the dorsum of the hand and forearm

Investigations:

- **Best initial**: Clinical examination for motor and sensory loss
- **Confirmatory**: Nerve conduction studies or EMG

Treatment Outline:

- **First-line**: Conservative management, including wrist splints and physical therapy
- **Second-line**: Surgical intervention for severe or persistent cases
- **Definitive**: Surgical repair or decompression in cases of nerve compression or severe injury

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MCQ 569

A 24-year-old man came to the hospital with pain, swelling, and difficulty in moving the arm. Examination shows he has a humeral fracture. X-ray: The fracture was at the level of the surgical neck. Which of the following nerves was most likely injured?

A. Ulnar
B. Radial
C. Axillary
D. Median

Explanation:
The **axillary nerve** is most likely injured in fractures of the surgical neck of the humerus. The axillary nerve innervates the deltoid and teres minor muscles and is responsible for shoulder abduction. Injury to this nerve results in shoulder weakness and sensory loss over the deltoid area.

Correct Answer: C. Axillary ✅

Topic Section:

Axillary Nerve Injury

Definition:
Damage to the axillary nerve, often caused by fractures of the surgical neck of the humerus, leading to motor and sensory deficits.

Key Clinical Presentation:

- Weakness in shoulder abduction
- Sensory loss over the deltoid region
- Muscle atrophy of the deltoid muscle

Investigations:

- **Best initial**: Physical examination for shoulder movement and sensory loss
- **Confirmatory**: Electromyography (EMG) and nerve conduction studies

Treatment Outline:

- **First-line**: Conservative management, including physical therapy to strengthen the shoulder

- **Second-line**: Surgical intervention for persistent weakness
- **Definitive**: Surgical repair if nerve is severed or severely compressed

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MCQ 569

A 41-year-old woman developed atrophy of the thenar eminence, but the sensation over it is intact. Which of the following nerves is most likely damaged?

A. Ulnar
B. Radial
C. Axillary
D. Median

Explanation:
The **median nerve** innervates the thenar eminence muscles (responsible for thumb movements). Atrophy of the thenar eminence with preserved sensation suggests median nerve injury, which commonly occurs in conditions like carpal tunnel syndrome.

Correct Answer: D. Median ✅

Topic Section:

Median Nerve Injury

Definition:
Injury to the median nerve, which provides motor innervation to the thenar muscles and sensation to the lateral palm and fingers.

Key Clinical Presentation:

- Atrophy of the thenar eminence (e.g., "ape hand" deformity)
- Weakness in thumb opposition and flexion
- Sensory loss in the lateral palm and digits

Investigations:

- **Best initial**: Clinical examination for motor function and sensory loss
- **Confirmatory**: Nerve conduction studies or electromyography (EMG)

Treatment Outline:

- **First-line**: Conservative management (splints, physical therapy) for mild cases
- **Second-line**: Surgical decompression (e.g., carpal tunnel release)
- **Definitive**: Surgical repair if the nerve is severely compressed or severed

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MCQ 570

A 27-year-old pianist with known carpal tunnel syndrome experiences difficulty in finger movements. Which of the following muscles is most likely paralyzed?

A. Thenar muscles
B. Dorsal interossei
C. Palmar interossei
D. Medial two lumbricals

Explanation:
The **thenar muscles** are most likely to be paralyzed in carpal tunnel syndrome, as the median nerve innervates the muscles responsible for thumb opposition and flexion. This condition leads to weakness and atrophy of the thenar eminence.

Correct Answer: A. Thenar muscles ✅

Topic Section:

Carpal Tunnel Syndrome

Definition:
Compression of the median nerve at the wrist, leading to motor and sensory deficits in the hand, particularly affecting the thenar muscles.

Key Clinical Presentation:

- Numbness, tingling, and weakness in the thumb and first two fingers
- Atrophy of the thenar eminence (e.g., "ape hand" deformity)
- Positive Tinel's sign or Phalen's maneuver

Investigations:

- **Best initial**: Physical examination for weakness and sensory loss
- **Confirmatory**: Nerve conduction studies

Treatment Outline:

- **First-line**: Wrist splints, rest, and anti-inflammatory medications
- **Second-line**: Corticosteroid injections or physical therapy
- **Definitive**: Surgical decompression (carpal tunnel release) for severe cases

MCQ 571

A 52-year-old woman slipped and fell, and she complains of being unable to extend her leg at the knee joint. Which of the following muscles is most likely paralyzed?

A. Sartorius
B. Biceps femoris
C. Semitendinosus
D. Quadriceps femoris

Explanation:
The **quadriceps femoris** is responsible for extending the knee. If this muscle is paralyzed, the patient will be unable to extend her leg at the knee joint. The quadriceps is innervated by the femoral nerve.

Correct Answer: D. Quadriceps femoris ✅

Topic Section:

Quadriceps Femoris Muscle

Definition:
The quadriceps femoris is a large muscle group that includes four muscles: rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius. It is responsible for knee extension.

Key Clinical Presentation:

- Inability to extend the knee joint
- Weakness or atrophy in the anterior thigh
- Inability to perform squatting or rising from a seated position

Investigations:

- **Best initial:** Clinical examination and neurological assessment
- **Confirmatory:** MRI or EMG to assess nerve or muscle damage

Treatment Outline:

- **First-line:** Physical therapy and rehabilitation
- **Definitive:** Surgery if there is nerve damage or significant muscle injury

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MCQ 572

A 45-year-old man was brought to the Emergency Department after a motor vehicle accident with severe pain in his left arm. X-rays confirmed left humeral and ulnar fractures. After an open reduction and internal fixation, the patient was unable to extend his forearm, wrist, and fingers. Which of the following is the most likely damaged?

A. Ulnar nerve at medial epicondyle
B. Median nerve at the cubital fossa
C. Radial nerve above the spiral groove
D. Median nerve at the lateral epicondyle

Explanation:
The **radial nerve** is most commonly damaged in humeral fractures, particularly at the spiral groove. Injury to the radial nerve results in an inability to extend the forearm, wrist, and fingers, commonly referred to as "wrist drop."

Correct Answer: C. Radial nerve above the spiral groove ✅

Topic Section:

Radial Nerve Injury

Definition:
The radial nerve innervates the muscles responsible for extension of the arm, wrist, and fingers. Damage to the nerve, especially at the humeral level, leads to weakness in wrist and finger extension.

Key Clinical Presentation:

- Inability to extend the wrist and fingers (wrist drop)
- Sensory loss on the dorsal aspect of the hand and forearm

Investigations:

- **Best initial:** Clinical examination to assess motor and sensory function

- **Confirmatory:** Nerve conduction studies or EMG

Treatment Outline:

- **First-line:** Conservative management with wrist splints and physical therapy
- **Second-line:** Surgical decompression or nerve repair if necessary
- **Definitive:** Surgical repair for severe cases

MCQ 573

A 25-year-old man presents to the hospital with an open fracture of his right leg, after a motorbike accident on a farm. Later, he develops severe pain, necrosis, and blue discoloration of the skin, and there is gas in the deep tissues of the leg. Which of the following is the most likely causing agent?

A. Actinomyces israelii
B. Staphylococcus aureus
C. Clostridium perfringens
D. Mycobacterium ulcerans

Explanation:
Clostridium perfringens is the most likely causing agent in this case, as it is associated with gas gangrene, a severe soft tissue infection that causes tissue necrosis and gas production. This is typically seen in trauma involving open fractures.

Correct Answer: C. Clostridium perfringens ✅

Topic Section:

Gas Gangrene

Definition:
Gas gangrene is a life-threatening soft tissue infection caused primarily by **Clostridium perfringens**. It leads to tissue necrosis, sepsis, and gas formation in tissues.

Key Clinical Presentation:

- Severe pain and swelling in the affected area
- Skin discoloration (blue, black)

- Gas production in tissues (crepitus on palpation)
- Rapid progression of necrosis

Investigations:

- **Best initial:** Clinical examination and X-rays to detect gas in the tissues
- **Confirmatory:** Cultures from the infected tissue for **Clostridium perfringens**

Treatment Outline:

- **First-line:** Surgical debridement and high-dose IV antibiotics (penicillin + clindamycin)
- **Second-line:** Hyperbaric oxygen therapy
- **Definitive:** Surgical removal of necrotic tissue and supportive care

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MCQ 574

A 26-year-old man complained of pain in his buttocks, lower back, and stiffness which improved after moving around and doing exercises. The patient had similar complaints a year ago. Which of the following is the most likely diagnosis?

A. Hyperuricemia
B. Psoriatic arthritis
C. Reactive arthritis
D. Ankylosing spondylitis

Explanation:
Ankylosing spondylitis is the most likely diagnosis, as it typically presents with chronic low back pain and stiffness that improves with movement. It commonly affects young men and is associated with sacroiliac joint involvement.

Correct Answer: D. Ankylosing spondylitis ✅

Topic Section:

Ankylosing Spondylitis

Definition:
Ankylosing spondylitis is a chronic inflammatory arthritis primarily affecting the spine and sacroiliac joints, often leading to fusion of the joints.

Key Clinical Presentation:

- Chronic low back pain and stiffness (worse in the morning or after rest)
- Improvement with physical activity
- Sacroiliac joint tenderness
- Reduced spinal mobility

Investigations:

- **Best initial:** Clinical examination and history
- **Confirmatory:** MRI of the sacroiliac joints or HLA-B27 testing

Treatment Outline:

- **First-line:** Nonsteroidal anti-inflammatory drugs (NSAIDs)
- **Second-line:** Disease-modifying antirheumatic drugs (DMARDs) like TNF inhibitors
- **Definitive:** Long-term physical therapy and disease management

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MCQ 575

A 36-year-old man has unilateral mild pain in the neck, which radiates to the right interscapular and shoulder regions. The pain is followed by sharp, electric shock-like pain, which goes down the arm with numbness, weakness, and loss of tendon reflexes. Which of the following is the most likely diagnosis?

A. Whiplash injury
B. Cervical disc prolapse
C. Rotator cuff tendonitis
D. Polymyalgia rheumatica

Explanation:
Cervical disc prolapse is the most likely diagnosis, as it can cause neck pain radiating to the shoulder and arm with associated neurological symptoms (numbness, weakness, loss of tendon reflexes).

Correct Answer: B. Cervical disc prolapse ✅

Topic Section:

Cervical Disc Prolapse

Definition:
Cervical disc prolapse occurs when a disc in the cervical spine herniates, pressing on adjacent nerve roots, causing pain, numbness, weakness, and loss of reflexes.

Key Clinical Presentation:

- Neck pain radiating to the shoulder or arm
- Numbness and tingling in the arm or hand
- Weakness in the upper limbs
- Loss of tendon reflexes

Investigations:

- **Best initial:** Clinical examination and neurological assessment
- **Confirmatory:** MRI of the cervical spine

Treatment Outline:

- **First-line:** Conservative management with pain control, physical therapy, and rest
- **Second-line:** Epidural steroid injections for refractory cases
- **Definitive:** Surgery (discectomy or fusion) for severe or persistent symptoms

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MCQ 576

A 58-year-old healthy patient has a six-month history of central lower back pain. The pain is present on awakening and, without treatment, resolves after about 30 minutes. Acetaminophen seemed to help it resolve sooner. Morning examination showed mild parasplinal muscle spasm with no neurological changes. MRI Lumbar: Mild lumbar spinal stenosis. Which of the following is the most appropriate management?

A. Biofeedback
B. Physical therapy
C. Lumbar laminectomy
D. Epidural steroid injection

Explanation:
The most appropriate management for mild lumbar spinal stenosis is **physical therapy**.

Conservative measures like physical therapy and lifestyle modifications are effective in managing symptoms in mild cases.

Correct Answer: B. Physical therapy ✅

Topic Section:

Lumbar Spinal Stenosis

Definition:
Lumbar spinal stenosis is the narrowing of the spinal canal in the lumbar region, often leading to compression of the spinal nerves.

Key Clinical Presentation:

- Central lower back pain, often worse on waking
- Pain relieved by standing or walking
- Numbness or weakness in the legs in severe cases
- Mild parasplinal muscle spasm on examination

Investigations:

- **Best initial:** Clinical examination
- **Confirmatory:** MRI of the lumbar spine

Treatment Outline:

- **First-line:** Physical therapy, weight management, and nonsteroidal anti-inflammatory drugs (NSAIDs)
- **Second-line:** Epidural steroid injections for refractory cases
- **Definitive:** Surgical intervention (laminectomy) for severe symptoms or nerve compression

577:

A 55-year-old patient presents to the clinic with shortness of breath and a cough productive of yellow sputum for the past 2 days. There has also been blood-stained sputum for the last 3 hours. The patient has smoked 2 packs of cigarettes a day for the past 35 years. On examination, there are rhonchi, wheezing, and crepitations heard over the right hemothorax.

Blood pressure: 110/70 mmHg
Heart rate: 84 /min
Respiratory rate: 22 /min
Temperature: 38.8°C
Oxygen saturation: 95% on room air
Chest X-ray: It shows a dense infiltrate in the right lower lobe.

Which of the following is the most likely diagnosis?

A. Community acquired pneumonia

B. Acute bronchitis

C. Tuberculosis

D. COPD

Explanation:

The patient's presentation of shortness of breath, cough with yellow and blood-stained sputum, and the chest X-ray showing a dense infiltrate in the right lower lobe is most consistent with **community acquired pneumonia (CAP)**. CAP is a bacterial infection of the lung parenchyma, typically presenting with fever, productive cough, and infiltrates on chest X-ray. The patient's significant smoking history increases the risk of respiratory infections like pneumonia.

Community Acquired Pneumonia (CAP)

Definition: Acute infection of the lung parenchyma, typically caused by bacteria, in individuals who acquire the infection outside of a healthcare setting.

Key Clinical Presentation: Productive cough, fever, pleuritic chest pain, dyspnea, blood-stained sputum, and abnormal breath sounds (rhonchi, crepitations).

Investigations:

Best initial: Chest X-ray (shows consolidation or infiltrates).

Confirmatory: Sputum culture or blood culture to identify the causative organism.

Treatment Outline:

First-line: Empiric antibiotic therapy, including a macrolide (e.g., azithromycin) or fluoroquinolone (e.g., levofloxacin) for typical pathogens like *Streptococcus pneumoniae*.

Second-line: Consider broader spectrum antibiotics if hospitalized or suspected resistant organisms.

Definitive: Tailored therapy based on culture results.

Correct Answer: A. Community acquired pneumonia ✅

578:

A 27-year-old woman with asthma has not had a severe attack for the last 3 months. However, she has a history of cough and wheezing 2-3 times each week. Which of the following is the most appropriate management? A. Ipratropium bromide B. Inhaled corticosteroid C. Salbutamol D. Salmeterol

MCQ:

A 27-year-old woman with asthma has not had a severe attack for the last 3 months. However, she has a history of cough and wheezing 2-3 times each week.

Which of the following is the most appropriate management?

A. Ipratropium bromide
B. Inhaled corticosteroid
C. Salbutamol
D. Salmeterol

Explanation:

For a patient with asthma who experiences intermittent symptoms such as cough and wheezing 2-3 times per week but does not have severe attacks, the **most appropriate management** is an **inhaled corticosteroid** (option B). Inhaled corticosteroids (ICS) are the cornerstone of asthma management for patients with persistent symptoms, as they reduce inflammation and prevent symptom exacerbation. ICS are used for daily control of asthma symptoms and help reduce the frequency of attacks.

Asthma Management

Definition: Chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Wheezing, cough, shortness of breath, and chest tightness, often exacerbated by triggers like allergens, exercise, or respiratory infections.

Investigations:

Best initial: Spirometry or peak flow measurement to assess lung function.

Confirmatory: Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge test).

Treatment Outline:

First-line: Inhaled corticosteroids for daily control of inflammation (e.g., fluticasone, budesonide).

Second-line: Long-acting beta-agonists (LABAs) added if needed for additional symptom control.

Rescue (for acute symptoms): Short-acting beta-agonists (e.g., salbutamol) for quick relief during exacerbations.

Correct Answer: B. Inhaled corticosteroid ✅

MCQ 579

A 21-year-old woman comes to the clinic with episodes of shortness of breath on exertion for one year. She has to stop after 15-20 minutes of exercise, because of dyspnoea and coughing, which get better after about one hour of rest. There are no symptoms at rest. Physical examination and spirometry are normal.

Blood pressure 125/75 mmHg
Heart rate 80 /min
Respiratory rate 12 /min
Temperature 37.0° C

Which of the following is the most appropriate inhaled treatment before exercise?

A. Salmeterol
B. Salbutamol
C. Budesonide
D. Ipratropium

Explanation:

This patient presents with **exercise-induced asthma** (EIA), characterized by shortness of breath and cough triggered by exercise, which improves with rest. The most appropriate inhaled treatment to prevent symptoms before exercise is a **short-acting beta-agonist (SABA)** like **salbutamol** (option B). Salbutamol is commonly used as a **rescue inhaler** and can be used 15-30 minutes before exercise to prevent bronchoconstriction.

Exercise-Induced Asthma

Definition: A form of asthma triggered by physical activity, leading to reversible airway obstruction, wheezing, cough, and shortness of breath during or after exercise.

Key Clinical Presentation: Dyspnoea and coughing during exercise, relieved by rest, with normal symptoms at rest.

Investigations:

Best initial: Spirometry or peak flow measurement during or after exercise.

Confirmatory: Methacholine challenge if needed.

Treatment Outline:

First-line: Short-acting beta-agonists (e.g., salbutamol) used before exercise to prevent bronchoconstriction.

Second-line: Long-acting beta-agonists (e.g., salmeterol) or leukotriene inhibitors for persistent symptoms.

Correct Answer: B. Salbutamol ✅

MCQ 580

A 42-year-old man with asthma presents with daily symptoms despite regularly using a long-acting β-agonist inhaler and an inhaled high-dose corticosteroid. He also uses a short-acting β-agonist inhaler as "rescue" medication. These episodes also wake him from sleep 2-3 times a week. Physical examination shows bilateral wheezing.

Blood pressure 140/85 mmHg
Heart rate 90 /min
Respiratory rate 18 /min
Temperature 36.7° C
Forced Expiratory Volume in 1 Second:
Before bronchodilator: 65% (25-75%)
After bronchodilators: 91%

Which of the following is the most appropriate next step in management?

A. Add theophylline
B. Add a leukotriene inhibitor
C. Observe the patient using his inhaler
D. Treat for gastroesophageal reflux disease

Explanation:

This patient has poorly controlled asthma, as evidenced by daily symptoms, nighttime awakenings, and an FEV1 of 65% (indicating moderate asthma). The most appropriate next step is to **add a leukotriene inhibitor** (option B). Leukotriene inhibitors (e.g., montelukast) are effective for patients with asthma that is not adequately controlled by inhaled corticosteroids and long-acting beta-agonists. These medications can help reduce inflammation and improve asthma control, particularly for patients with nighttime symptoms.

Asthma Management

Definition: Chronic inflammatory disorder of the airways, leading to episodes of wheezing, breathlessness, chest tightness, and coughing.

Key Clinical Presentation: Daily symptoms, nocturnal awakenings, use of short-acting beta-agonists for relief, and spirometry showing reduced FEV1.

Investigations:

Best initial: Spirometry for lung function.

Confirmatory: Assessment of bronchial hyperresponsiveness (e.g., methacholine challenge).

Treatment Outline:

First-line: Inhaled corticosteroids (ICS) and long-acting beta-agonists (LABAs).

Second-line: Leukotriene inhibitors, theophylline, or biologic agents for more severe asthma.

Rescue: Short-acting beta-agonists (SABA) for acute symptom relief.

Correct Answer: B. Add a leukotriene inhibitor ✅

MCQ 581

A 67-year-old patient presents to the Emergency Department with chest pain, shortness of breath, and haemoptysis following a car trip for more than 5 hours (see report). ECG: Right axis deviation, an S1-Q3-T3 pattern, and a right bundle branch block.

Which of the following is the most likely diagnosis?

A. Acute myocardial infarction
B. Bronchogenic carcinoma
C. Pulmonary embolism
D. Pericarditis

Explanation:

The patient's clinical presentation with chest pain, shortness of breath, haemoptysis, and prolonged immobility (after a long car trip) raises suspicion for **pulmonary embolism (PE)** (option C). The ECG showing **S1-Q3-T3 pattern** and **right axis deviation** are classic signs of PE. A right bundle branch block may also be seen in PE due to strain on the right side of the heart from obstructed pulmonary circulation.

Pulmonary Embolism (PE)

Definition: Blockage of the pulmonary arteries by a thrombus, often originating from deep vein thrombosis (DVT).

Key Clinical Presentation: Sudden onset of chest pain, shortness of breath, haemoptysis, tachycardia, hypoxia, and signs of right heart strain (e.g., S1-Q3-T3 pattern on ECG).

Investigations:

Best initial: CT pulmonary angiography (CTPA)

Confirmatory: Pulmonary angiography (gold standard), D-dimer

Treatment Outline:

First-line: Anticoagulation with heparin or low molecular weight heparin (LMWH)

Second-line: Thrombolysis or surgical embolectomy for massive PE

Definitive: Ongoing anticoagulation for 3–6 months, depending on risk factors.

Correct Answer: C. Pulmonary embolism ✅

MCQ 582

A 55-year-old smoker presents to a clinic with progressive shortness of breath (see report). Chest X-ray: It reveals a massive right pleural effusion.

Which of the following is the best next step in management?

A. Thoracentesis
B. High resolution CT scan
C. Intercostal tube under water seal
D. Repeated bronchodilator inhalations

Explanation:

The presence of a **massive pleural effusion** in a smoker raises concern for potential **malignant pleural effusion**, which could be caused by lung cancer or other malignancies. The best next step in management is to perform **thoracentesis** (option A), which allows for both diagnostic and therapeutic relief. Thoracentesis will help determine the etiology of the effusion by analyzing the pleural fluid (e.g., cytology for malignancy, cell count, pH, and other markers).

Pleural Effusion

Definition: Accumulation of fluid in the pleural space, which can be transudative or exudative, depending on the underlying cause.

Key Clinical Presentation: Progressive shortness of breath, pleuritic chest pain, decreased breath sounds, dullness to percussion, and egophony on examination.

Investigations:

Best initial: Chest X-ray to identify the effusion.

Confirmatory: Thoracentesis to obtain pleural fluid for analysis (cell count, protein, LDH, pH, and cytology).

Treatment Outline:

First-line: Thoracentesis for diagnostic and therapeutic purposes

Second-line: Chest tube drainage or pleurodesis if the effusion is recurrent or malignant

Definitive: Management of the underlying cause (e.g., chemotherapy for malignancy, diuretics for heart failure).

Correct Answer: A. Thoracentesis ✅

MCQ 583

A 28-year-old patient develops the sudden onset of dyspnoea and tachycardia 5 days post knee surgery.

Which of the following is the most likely diagnosis?

A. Pulmonary embolism
B. Myocardial infarction
C. Ventricular tachycardia
D. Pulmonary emphysema

Explanation:

The patient's **sudden onset of dyspnoea** and **tachycardia** 5 days post-surgery, especially knee surgery, raises strong suspicion for **pulmonary embolism (PE)** (option A). PE commonly occurs in post-surgical patients, particularly after orthopedic surgeries like knee replacements, due to deep vein thrombosis (DVT) embolizing to the lungs. This condition can present with shortness of breath, tachycardia, and other signs of right heart strain.

Pulmonary Embolism (PE)

Definition: Blockage of a pulmonary artery or one of its branches by a clot, typically arising from a deep vein thrombosis.

Key Clinical Presentation: Sudden onset of dyspnoea, tachycardia, pleuritic chest pain, hypoxia, and, if severe, signs of right heart failure.

Investigations:

Best initial: D-dimer, followed by CT pulmonary angiography (CTPA) for diagnosis.

Confirmatory: Pulmonary angiography (gold standard).

Treatment Outline:

First-line: Anticoagulation therapy with heparin or low molecular weight heparin (LMWH).

Second-line: Thrombolytic therapy or surgical embolectomy in cases of massive PE.

Definitive: Long-term anticoagulation therapy depending on the cause.

Correct Answer: A. Pulmonary embolism ✅

584:

An 18-year-old man presents to the Emergency Department with 1-week history of euphoria, increased blood pressure, tachycardia, nausea, weight loss, psychomotor agitation, chills, sweating, and visual hallucinations. Which of the following is the most likely diagnosis?

A. Amphetamine intoxication
B. Cannabis intoxication

C. Acute schizophrenia
D. Cocaine withdrawal

Explanation:
The most likely diagnosis is **Amphetamine intoxication** (option A). The symptoms of euphoria, increased blood pressure, tachycardia, and psychomotor agitation are consistent with stimulant use, specifically amphetamines.

Substance Use and Intoxication
Definition: The use of substances leading to acute psychological and physical symptoms.
Key Clinical Presentation: Euphoria, tachycardia, hypertension, psychomotor agitation, and possible hallucinations.
Investigations: Urine toxicology screen for amphetamines.
Treatment Outline:

First-line: Supportive care, hydration, and monitoring.

Second-line: Benzodiazepines for agitation, antipsychotics for hallucinations.

Correct Answer: A. Amphetamine intoxication ✅

MCQ 585

Which of the following clinical signs is caused by a lesion in Wernicke's area?

A. Can understand but unable to say words easily
B. Cannot understand spoken or written words
C. Loss of ability to recognize faces
D. Loss of short-term memory

Explanation:
The correct answer is **B. Cannot understand spoken or written words**.
A lesion in Wernicke's area (located in the left temporal lobe) causes **Wernicke's aphasia**, which is characterized by **impaired comprehension** of both spoken and written language, while speech production remains fluent but nonsensical (without proper meaning). This contrasts with Broca's aphasia, where speech production is impaired but comprehension remains intact.

Wernicke's Aphasia
Definition: A language disorder caused by damage to Wernicke's area, resulting in fluent but meaningless speech and poor comprehension.
Key Clinical Presentation: Difficulty understanding spoken and written language, fluent but nonsensical speech.

Investigations: Neuroimaging (CT/MRI), clinical language assessment.
Treatment Outline:

First-line: Speech and language therapy, supportive care.

Second-line: Antidepressants or antipsychotics if associated psychiatric symptoms occur.

Correct Answer: B. Cannot understand spoken or written words ✅

MCQ 586:

A 55-year-old woman presents to the Primary Health Care Centre complaining of loss of interest and pleasure in daily activities for 3 weeks. She also complains that she has difficulty concentrating and sleeping. Further, she has also expressed that she feels that she is worthless in the world and therefore wants to end her life. Which of the following biochemical substance deficiency is the most likely cause?

A. Aspartate
B. Serotonin
C. Glutamate
D. Acetylcholine

Explanation:
The correct answer is **B. Serotonin**.
The patient's symptoms — including **loss of interest**, **difficulty concentrating**, **sleep disturbances**, and **suicidal ideation** — are indicative of **major depressive disorder (MDD)**, which is most commonly associated with a **serotonin deficiency** in the brain. Serotonin plays a crucial role in mood regulation, and its deficiency is linked to mood disorders like depression.

Major Depressive Disorder (MDD)
Definition: A mood disorder characterized by persistent sadness, loss of interest, sleep disturbances, and suicidal thoughts.
Key Clinical Presentation: Depressed mood, anhedonia, concentration difficulty, suicidal ideation.
Investigations: Clinical evaluation, screening tools (e.g., PHQ-9), serotonin level assessment (if available).
Treatment Outline:

First-line: SSRIs (Fluoxetine, Sertraline).

Second-line: SNRIs (Duloxetine), psychotherapy.

Correct Answer: B. Serotonin ✅

MCQ 587:

Which of the following reuptake inhibitors is the best treatment for depression?

A. DOPA
B. GABA
C. Serotonin
D. Acetylcholine

Explanation:
The correct answer is **C. Serotonin**.
Selective serotonin reuptake inhibitors (SSRIs), which work by increasing serotonin levels in the brain, are the **first-line treatment for depression**. Serotonin reuptake inhibitors block the reabsorption (reuptake) of serotonin in the brain, making more serotonin available to improve mood and emotional well-being.

Antidepressant Treatment
Definition: Medications that treat depression by regulating neurotransmitters like serotonin.
Key Clinical Presentation: Low mood, lack of interest, sleep disturbances, appetite changes.
Investigations: Clinical evaluation for depression, response to antidepressant therapy.
Treatment Outline:

First-line: SSRIs (Fluoxetine, Sertraline).

Second-line: SNRIs (Duloxetine) or other classes like TCAs.

Correct Answer: C. Serotonin ✅

MCQ 588

Which of the following drugs used in Alzheimer's disease is most likely to cause hepatotoxicity?

A. Tacrine
B. Donepezil
C. Rivastigmine
D. Galantamine

Explanation:
The correct answer is **A. Tacrine**.
Tacrine, an acetylcholinesterase inhibitor, was used in the past for treating Alzheimer's disease but was discontinued due to its **hepatotoxicity**. It can cause significant liver damage, which is not seen with other acetylcholinesterase inhibitors like donepezil, rivastigmine, and galantamine.

Alzheimer's Disease Treatment
Definition: A neurodegenerative condition characterized by cognitive decline, memory loss, and behavioral changes.
Key Clinical Presentation: Memory loss, confusion, difficulty with communication.
Investigations: Clinical assessment, brain imaging, cognitive testing.
Treatment Outline:

First-line: Acetylcholinesterase inhibitors (Donepezil, Rivastigmine).

Second-line: NMDA receptor antagonists (Memantine).

Correct Answer: A. Tacrine ✅

MCQ 589:

A 35-year-old depressed housewife was started with antidepressants. After 2 weeks of taking the medication, the patient started complaining of constipation. Which of the following groups of antidepressants is the most likely cause?

A. Tricyclic antidepressants
B. Monoamine oxidase inhibitors
C. Selective serotonin reuptake inhibitors
D. Selective nor-adrenaline and serotonin inhibitors

Explanation:
The correct answer is **A. Tricyclic antidepressants**.
Tricyclic antidepressants (TCAs) such as **amitriptyline** are known to cause **anticholinergic side effects**, including **constipation**, dry mouth, and urinary retention. These side effects are related to their **muscarinic receptor antagonism**. Other antidepressant classes (SSRIs, SNRIs, and MAOIs) are less likely to cause such significant gastrointestinal symptoms.

Antidepressant Treatment Side Effects
Definition: Medications used to treat depression, each with potential side effects.
Key Clinical Presentation: Constipation, dry mouth, blurred vision (due to anticholinergic

effects).
Investigations: Clinical evaluation of side effects, patient history.
Treatment Outline:

First-line: For depressive treatment, SSRIs or SNRIs.

Second-line: If TCA-related constipation occurs, consider switching to an SSRI.

Correct Answer: A. Tricyclic antidepressants ✅

MCQ 590:

Which of the following conditions is the most likely cause of dementia?

A. Spinocerebellar ataxia
B. Vitamin B12 deficiency
C. Idiopathic parkinsonism
D. Amyotrophic lateral sclerosis

Explanation:
The correct answer is **B. Vitamin B12 deficiency**.
Vitamin B12 deficiency can cause **subacute combined degeneration** of the spinal cord, leading to cognitive decline and dementia. It can also cause **peripheral neuropathy** and **psychiatric symptoms** like depression and memory impairment. This is a reversible cause of dementia when treated appropriately.

Dementia
Definition: A cognitive disorder characterized by memory loss, language difficulties, and impaired executive function.
Key Clinical Presentation: Memory impairment, confusion, mood changes, difficulty performing daily tasks.
Investigations: B12 levels, MRI of the brain, cognitive testing.
Treatment Outline:

First-line: Vitamin B12 supplementation (oral or injections).

Second-line: Cognitive rehabilitation if needed.

Correct Answer: B. Vitamin B12 deficiency ✅

MCQ 590

A 15-year-old girl who had a serious disagreement with her friend. She did not want to meet her friend or talk to her anymore. If the girl saw her friend coming towards her or sitting in a room, she would move away and not go in that direction. Which of the following defense mechanisms is the most likely explanation for this behaviour?

A. Denial
B. Acting out
C. Avoidance
D. Repression

Explanation:
The correct answer is **C. Avoidance**.
Avoidance is a defense mechanism where an individual actively **avoids situations, people, or thoughts** that are emotionally distressing or anxiety-provoking. In this case, the girl avoids her friend because of the emotional discomfort caused by the disagreement. **Denial**, **acting out**, and **repression** are other defense mechanisms, but they do not directly fit this scenario.

Defense Mechanisms
Definition: Psychological strategies used by individuals to cope with reality and maintain self-image.
Key Clinical Presentation: Active avoidance of distressing stimuli.
Investigations: Clinical evaluation of the patient's emotional response.
Treatment Outline:

First-line: Therapy (Cognitive Behavioral Therapy) to address avoidance behavior.

Second-line: Psychodynamic therapy if avoidance is a recurring problem.

Correct Answer: C. Avoidance ✅

MCQ 591:

A 65-year-old man with depression presents to the clinic for a follow-up visit. The physician did his consultation using a bio-psycho-social model of treatment. Which of the following is a characteristic of this treatment model?

A. Evidence based medicine
B. Traditional allopathy
C. Geriatric medicine
D. Holistic medicine

Explanation:
The correct answer is **D. Holistic medicine**.
The **bio-psycho-social model** takes a **holistic approach** to treatment, considering not just the biological aspects (e.g., neurochemistry) but also the psychological and social factors (e.g., family dynamics, lifestyle). This model is comprehensive and focuses on treating the whole person, not just the disease.

Bio-psycho-social Model
Definition: A model that integrates biological, psychological, and social factors in understanding and treating illness.
Key Clinical Presentation: Focus on the whole person and all contributing factors.
Investigations: Comprehensive evaluation of mental health, lifestyle, and social context.
Treatment Outline:

First-line: Multidisciplinary treatment involving medication, therapy, and social support.

Second-line: Alternative therapies, including complementary medicine if appropriate.

Correct Answer: D. Holistic medicine ✅

MCQ 592:

A 36-year-old diabetic woman presents to the clinic for follow-up visit, and after an initial assessment, the physician wished to perform a physical examination. He first asked her permission, which was granted. Which of the following ethical measures was the physician adopting?

A. Obtain informed consent
B. Respect for the patient
C. Fulfil his responsibility
D. Show his efficiency

Explanation:
The correct answer is **A. Obtain informed consent**.
Obtaining **informed consent** is an essential ethical practice where the patient is provided with all the relevant information regarding the procedure or treatment, and their agreement is sought before proceeding. This is an important ethical principle in healthcare, especially when touching or examining a patient.

Informed Consent
Definition: The process of ensuring that the patient understands and agrees to the treatment or

procedure.
Key Clinical Presentation: Asking for permission before performing procedures or examinations.
Investigations: Assessing the patient's capacity to understand the information provided.
Treatment Outline:

First-line: Clear communication and documentation of consent.

Second-line: If patient is unable to consent, involve family or legal guardians.

Correct Answer: A. Obtain informed consent ✅

MCQ 593:

A 40-year-old diabetic man was treated appropriately with small doses of insulin. On next follow-up visit, he bought a very expensive watch as a gesture of gratitude to his physician. Which of the following is the most appropriate response?

A. Accept and advise not to repeat
B. Refuse and give a warning
C. Politely refuse the gift
D. Accept from the rich

Explanation:
The correct answer is **C. Politely refuse the gift**.
It is ethically inappropriate for healthcare providers to accept gifts from patients as it may compromise professional boundaries and objectivity. The most appropriate response is to politely refuse the gift while acknowledging the patient's gratitude. Accepting gifts could lead to conflicts of interest, real or perceived.
Professional Ethics
Definition: The set of ethical standards and principles that govern the conduct of healthcare providers.
Key Clinical Presentation: Patient offers a gift or token of appreciation.
Investigations: Assess if the acceptance of the gift could compromise professional objectivity.
Treatment Outline:

First-line: Politely refuse the gift to maintain professional boundaries.

Second-line: Educate the patient on alternative ways of showing appreciation.

Correct Answer: C. Politely refuse the gift ✅

MCQ 594:

A 36-year-old newly diagnosed hypertensive man presents for follow-up visit. After a thorough assessment, the physician discussed with him his diagnosis, the etiology of his illness, the probable side effects of the prescribed medicine, and the prognosis of his illness and finally the restrictions the patient might experience due to the illness. Which of the following professional skills is the physician using?

A. Developing rapport
B. Improving communication
C. Providing informational care
D. Developing bonding with the patient

Explanation:
The correct answer is **C. Providing informational care**.
In this scenario, the physician is offering **informational care** by providing the patient with detailed information about his condition, treatment options, and potential restrictions. This is an essential part of the physician's role in ensuring that patients understand their health status and treatment plan.
Professional Skills in Medical Practice
Definition: Providing patients with relevant, accurate, and timely information to aid in decision-making.
Key Clinical Presentation: Giving detailed explanations about diagnosis, treatment, and prognosis.
Investigations: Evaluate how well the patient comprehends the information provided.
Treatment Outline:

First-line: Provide clear, empathetic, and evidence-based information.

Second-line: Use additional educational resources if the patient requires more clarification.

Correct Answer: C. Providing informational care ✅

MCQ 595:

Which of the following is the most useful communication technique when taking a patient history?

A. Humour
B. Silent listening

C. Specific questions
D. Open-ended questions

Explanation:
The correct answer is **D. Open-ended questions**.
Open-ended questions encourage the patient to express themselves more freely and provide a more comprehensive history. These types of questions help in building rapport and allowing the patient to share important details that may not come out with more closed, specific questions.
Communication Techniques
Definition: Methods used by healthcare professionals to gather information from patients effectively.
Key Clinical Presentation: Eliciting a comprehensive patient history.
Investigations: Evaluate the patient's responses and use open-ended questions to explore further.
Treatment Outline:

First-line: Use open-ended questions to facilitate communication and gather comprehensive information.

Second-line: Ask specific questions if needed to clarify or narrow down the patient's responses.

Correct Answer: D. Open-ended questions ✅

MCQ 596:

Which of the following is an example of the open-ended questions in consultation?

A. "Tell me about the pain."
B. "Tell me about the pain in your chest."
C. "Point to the area of pain in your chest."
D. "Have you seen a doctor in the past 6 months?"

Explanation:
The correct answer is **A. "Tell me about the pain."**
This is an open-ended question because it invites the patient to provide a detailed response. Open-ended questions allow patients to describe their symptoms and experiences in their own words, providing richer information for the clinician.
Open-Ended Questions
Definition: Questions that allow the patient to give a full, descriptive answer.
Key Clinical Presentation: Eliciting detailed patient responses.

Investigations: Use open-ended questions in the early stages of the consultation to gain a comprehensive understanding of the patient's concerns.
Treatment Outline:

First-line: Ask open-ended questions to start the consultation and gather essential information.

Second-line: Use follow-up questions to clarify or expand upon the initial responses.

Correct Answer: A. "Tell me about the pain." ✅

MCQ 597:

A 6-year-old boy is brought to the clinic by his mother. She noticed that the child has been eating clay and paper for the last 4 months. Which of the following is the most appropriate initial treatment?

A. Fluoxetine
B. Imipramine
C. Clonazepam
D. Behavioural therapy

Explanation:
The correct answer is **D. Behavioural therapy**.
The behavior described, eating clay and paper, is a form of **pica**, which can be addressed through **behavioral therapy**. This therapy aims to modify the child's behavior by reinforcing positive actions and discouraging the ingestion of non-food items. Medications like **fluoxetine** or **imipramine** are not first-line treatments for pica.
Behavioral Therapy for Pica
Definition: A form of therapy focused on changing maladaptive behaviors.
Key Clinical Presentation: Consumption of non-food items such as clay or paper.
Investigations: Psychological evaluation to understand the underlying causes of pica.
Treatment Outline:

First-line: Behavioral therapy to address pica.

Second-line: Address any underlying psychological or nutritional deficiencies.

Correct Answer: D. Behavioural therapy ✅

MCQ 598:

A 34-year-old woman presents to the clinic with a 6-month history of extreme fear of going to the park, the zoo, and sporting events. This has worsened to the point where she has stopped going out of the house. Which of the following is the most likely diagnosis?

A. Conversion disorder
B. Schizophrenia
C. Social phobia
D. Agoraphobia

Explanation:
The correct answer is **D. Agoraphobia**.
Agoraphobia is characterized by the fear of being in situations where escape might be difficult or help unavailable, such as in public spaces. The patient is avoiding specific places and has stopped leaving the house altogether, which is indicative of agoraphobia.
Anxiety Disorders
Definition: Agoraphobia involves intense fear of being in places where escape might be difficult or help unavailable.
Key Clinical Presentation: Fear of specific places, avoiding leaving the house.
Investigations: Clinical interview and symptom assessment.
Treatment Outline:

First-line: Cognitive-behavioral therapy (CBT)

Second-line: Medication (SSRI or benzodiazepine for acute anxiety).

Correct Answer: D. Agoraphobia ✅

MCQ 599:

A 45-year-old man presents in an anxious, tearful state. He complains of chronic, upper abdominal pain and is certain that he has stomach cancer but no one believes him. He has seen 6 different physicians in the past year and has had a GI series and barium enema performed, an ultrasound of the abdomen twice, and a CT scan. The patient was thinking of next going to a Herbal Clinic. Which of the following is the most likely diagnosis?

A. Somatoform pain disorder
B. Somatization disorder
C. Conversion disorder
D. Hypochondriasis

Explanation:
The correct answer is **D. Hypochondriasis** (now referred to as **Illness Anxiety Disorder**).
Hypochondriasis is characterized by a preoccupation with having or acquiring a serious illness despite medical reassurance and negative test results. The patient has excessive concern about his health and seeks numerous medical opinions without accepting reassurance from doctors.
Somatoform Disorders
Definition: Hypochondriasis involves excessive worry about having a serious disease, even after negative tests.
Key Clinical Presentation: Preoccupation with illness, seeking multiple doctors' opinions.
Investigations: Multiple unnecessary medical tests.
Treatment Outline:

First-line: Cognitive-behavioral therapy (CBT)

Second-line: Medication (SSRIs or anxiolytics).

Correct Answer: D. Hypochondriasis ✅

MCQ 600:

A 43-year-old woman delivered a full-term infant. On examination, the infant has epicanthic folds, delayed milestones, a flat facial profile, simian palmar creases, and an atrial septal defect. Which of the following is the most likely diagnosis?

A. Trisomy 12
B. Trisomy 21
C. Di-George syndrome
D. Robertsonian translocation

Explanation:
The correct answer is **B. Trisomy 21** (Down syndrome).
Trisomy 21 is characterized by features such as epicanthic folds, a flat facial profile, simian palmar creases, developmental delay, and congenital heart defects like an atrial septal defect. These findings are consistent with Down syndrome.
Genetic Disorders
Definition: Trisomy 21 is a genetic condition caused by an extra copy of chromosome 21, leading to developmental and physical abnormalities.
Key Clinical Presentation: Epicanthic folds, simian crease, developmental delay, congenital heart defects.

Investigations: Karyotyping to confirm the trisomy 21 diagnosis.
Treatment Outline:

First-line: Early intervention, including physical therapy and developmental support.

Second-line: Medical management for associated conditions like congenital heart defects.

Correct Answer: B. Trisomy 21 ✅

MCQ 601:

A 16-year-old girl is brought by her father to the Emergency Department with complaints of tachypnoea, light-headedness, paraesthesia, and palpitations. Her father gives the history that she has recently failed her math test. Which of the following is the most likely diagnosis?

A. Oppositional defiant disorder
B. Hyperventilation syndrome
C. Munchausen syndrome
D. Anxiety disorder

Explanation:
The correct answer is **B. Hyperventilation syndrome**.
Hyperventilation syndrome often occurs in response to anxiety or stress, which is evident in this patient who has experienced a stressful event (failing a test). Symptoms like tachypnoea, light-headedness, and paraesthesia are common in hyperventilation.
Anxiety Disorders
Definition: Hyperventilation syndrome is caused by rapid, shallow breathing, leading to respiratory alkalosis and symptoms such as light-headedness and paraesthesia.
Key Clinical Presentation: Tachypnoea, light-headedness, anxiety-related symptoms.
Investigations: Diagnosis is clinical, with improvement after controlled breathing.
Treatment Outline:

First-line: Breathing exercises to reduce hyperventilation.

Second-line: Stress management techniques.

Correct Answer: B. Hyperventilation syndrome ✅

MCQ 602:

A 23-year-old woman with frequent panic attacks, presents to the clinic complaining that during her last attack 3 days ago, she felt that she was disconnected from the world, as if the environment is strange. Which of the following would best describe her feelings?

A. Illusion
B. Derealisation
C. Dull perception
D. Depersonalization

Explanation:
The correct answer is **B. Derealisation**.
Derealisation refers to the feeling that the external world is unreal or strange, which is described by the patient in this scenario. It is commonly associated with anxiety disorders and panic attacks.
Disorders of Perception
Definition: Derealisation is the alteration in perception where the surroundings seem strange or distant, while **depersonalization** refers to the feeling of being detached from oneself.
Key Clinical Presentation: A feeling that the world around the person is strange or unreal, often occurring during panic attacks.
Investigations: Diagnosis is clinical, based on patient's description of symptoms.
Treatment Outline:

First-line: Cognitive-behavioral therapy (CBT) and relaxation techniques for panic disorder.

Second-line: Medications like SSRIs or benzodiazepines for acute relief.

Correct Answer: B. Derealisation ✅

MCQ 603:

A 30-year-old man has a personality disorder, which is characterized by being aggressive and violent. Which of the following is most likely associated with violence?

A. Low levels of serotonin
B. High levels of serotonin
C. Low levels of endorphins
D. High levels of endorphins

Explanation:
The correct answer is **A. Low levels of serotonin**.
Low serotonin levels have been linked to increased aggression and violent behavior. This is

commonly seen in patients with personality disorders, particularly those with impulsive aggression.
Personality Disorders
Definition: Personality disorders often involve long-standing patterns of behavior that deviate from societal expectations, and these can include aggression and impulsivity.
Key Clinical Presentation: Aggression, impulsivity, violent outbursts.
Investigations: Psychological evaluation, possible hormone and neurotransmitter levels.
Treatment Outline:

First-line: Psychotherapy (e.g., Dialectical behavior therapy for Borderline Personality Disorder).

Second-line: Medications (e.g., SSRIs for impulsivity and aggression).

Correct Answer: A. Low levels of serotonin ✅

MCQ 604:

Which of the following is associated with dyslexia?

A. Abnormal gait
B. Brainstem defects
C. Verbal language difficulties
D. Defect in visual or hearing acuity

Explanation:
The correct answer is **C. Verbal language difficulties**.
Dyslexia is a specific learning disability that primarily affects reading, spelling, and sometimes writing, due to difficulties with phonological processing. It is not associated with deficits in visual or hearing acuity but rather with verbal language and processing difficulties.
Learning Disabilities
Definition: Dyslexia is a learning disorder marked by difficulties with word recognition, decoding, and spelling, despite normal intelligence.
Key Clinical Presentation: Difficulty reading, spelling, and writing, with normal intelligence.
Investigations: Diagnosis is clinical, often confirmed through reading and language assessments.
Treatment Outline:

First-line: Special education services and reading interventions.

Second-line: Behavioral therapy, phonics-based approaches.

Correct Answer: C. Verbal language difficulties ✅

MCQ 605:

Which of the following is the most likely clinical presentation of delusional patients?

A. Disturbance in perception
B. Rare in manic disorders
C. Found only in schizophrenia
D. Disturbance of thought content

Explanation:
The correct answer is **D. Disturbance of thought content**.
Delusions are fixed, false beliefs that are not in line with reality, which can affect the content of a person's thinking. Patients with delusions typically experience disturbances in thought content rather than in perception (which would be seen in hallucinations).
Psychotic Disorders
Definition: Delusions are false beliefs that are resistant to reasoning or contrary evidence, often seen in psychotic disorders.
Key Clinical Presentation: Beliefs not aligned with reality, such as paranoia, grandiosity, or persecution.
Investigations: Psychiatric evaluation and mental status examination.
Treatment Outline:

First-line: Antipsychotic medications (e.g., haloperidol, risperidone).

Second-line: Cognitive-behavioral therapy for delusions.

Correct Answer: D. Disturbance of thought content ✅

MCQ 606:

A 7-year-old boy is brought by his mother because he was unable to sit in school without interrupting the class, wandering around the classroom or poking his neighbours. On examination, he has normal growth parameters. Which of the following is the best management?

A. Bupropion
B. Paroxetine
C. Guanfacine
D. Dextroamphetamine

Explanation:
The correct answer is **D. Dextroamphetamine**.
The child exhibits symptoms consistent with **Attention Deficit Hyperactivity Disorder (ADHD)**, which includes difficulty staying seated, impulsivity, and inattentiveness. **Dextroamphetamine** is a stimulant medication commonly used to treat ADHD.
Attention Deficit Hyperactivity Disorder (ADHD)
Definition: A neurodevelopmental disorder characterized by inattention, hyperactivity, and impulsivity.
Key Clinical Presentation: Inability to sit still, distractibility, impulsive behavior.
Investigations: Diagnosis is clinical, based on behavior assessment and rating scales from parents and teachers.
Treatment Outline:

First-line: Stimulant medications (e.g., dextroamphetamine, methylphenidate).

Second-line: Non-stimulants (e.g., guanfacine).
Correct Answer: D. Dextroamphetamine ✅

MCQ 607:

A 30-year-old woman presents with a complaint of lower back pain for 1 year. During the last 2 years, she had presented with recurrent abdominal pain, bloating, headache, and migratory joint pain. Her menstrual periods are irregular and heavy. She was terminated twice from the job due to frequent absence from work. She is upset that proper diagnosis or treatment was not given to her. She had multiple laboratory tests done during the last 2 years and all were normal. The physical examination is unremarkable. Which of the following is the most likely diagnosis?

A. Malingering
B. Hypochondriasis
C. Factitious disorder
D. Somatization disorder

Explanation:
The correct answer is **D. Somatization disorder**.
Somatization disorder involves the presence of multiple physical complaints (e.g., abdominal pain, back pain, headaches) without an identifiable medical cause, which is characteristic of the patient's symptoms. This disorder is often marked by frequent healthcare visits and unexplained physical symptoms.

Somatization Disorder
Definition: A psychiatric condition characterized by multiple unexplained physical symptoms that are not explained by medical tests.
Key Clinical Presentation: Multiple physical symptoms with no medical cause, often leading to frequent healthcare visits.
Investigations: Diagnosis is clinical, often based on exclusion of medical causes and assessment of symptom patterns.
Treatment Outline:

First-line: Cognitive-behavioral therapy (CBT) and psychotherapy.

Second-line: Medications for symptom management (e.g., antidepressants).
Correct Answer: D. Somatization disorder ✅

MCQ 608:

A 23-year-old man with bronchial asthma presents with a 5-year history of severe anxiety in social gatherings. He often has sweating and palpitations when he leads discussions at work. He prefers to sit alone at home. He notices that his presenting complaint is provoked by salbutamol. His physical examination is unremarkable. Which of the following is the most appropriate therapy?

A. Sertraline
B. Bupropion
C. Alprazolam
D. Propranolol

Explanation:
The correct answer is **A. Sertraline**.
This patient most likely has **social anxiety disorder (SAD)**, which is characterized by an intense fear of being judged negatively in social situations. **Sertraline**, a selective serotonin reuptake inhibitor (SSRI), is the most appropriate first-line treatment for this condition.
Social Anxiety Disorder
Definition: A marked fear or anxiety about one or more social situations where the individual is exposed to possible scrutiny.
Key Clinical Presentation: Fear of embarrassment in social situations, avoidance of public speaking, physical symptoms like sweating and palpitations.
Investigations: Diagnosis is clinical, often based on the patient's reported symptoms and clinical

interview.
Treatment Outline:

First-line: SSRIs (e.g., sertraline, fluoxetine) or CBT.

Second-line: Benzodiazepines (e.g., alprazolam) for short-term use.
Correct Answer: A. Sertraline ✅

MCQ 609:

A 28-year-old woman with bipolar disorder, treated for 6 months, presents for follow-up visit. On examination, she has 8 kg weight gain, generalized weakness, and hair loss. Which of the following is the most likely cause?

A. Hypothalamic disorder
B. Hypothyroidism
C. Verapamil
D. Lithium

Explanation:
The correct answer is **D. Lithium**.
Lithium is a common medication used to treat bipolar disorder, and weight gain, weakness, and hair loss are known side effects of lithium.
Bipolar Disorder
Definition: A mood disorder characterized by periods of mania and depression.
Key Clinical Presentation: Episodes of elevated mood (mania) and low mood (depression).
Investigations: Diagnosis is clinical, based on the pattern of mood episodes.
Treatment Outline:

First-line: Lithium, anticonvulsants (e.g., valproate, lamotrigine).

Second-line: Antipsychotics for acute mania.
Correct Answer: D. Lithium ✅

MCQ 610:

Which of the following is the most likely benefit of using antipsychotic therapy in schizophrenic patients?

A. Improve apathy
B. Reduce delusions
C. Improve thinking ability
D. Increase social interaction

Explanation:
The correct answer is **B. Reduce delusions**.
Antipsychotic medications are primarily used to manage the **positive symptoms of schizophrenia**, including **delusions** and **hallucinations**. These medications have a significant effect on reducing delusions, which are false beliefs that are not in accordance with reality.
Schizophrenia
Definition: A chronic, severe mental health disorder characterized by psychosis, which includes delusions, hallucinations, disorganized thinking, and negative symptoms (e.g., social withdrawal, apathy).
Key Clinical Presentation: Delusions, auditory hallucinations, disorganized speech, and thought disorder.
Investigations: Diagnosis is clinical, based on symptom history and exclusion of other causes.
Treatment Outline:

First-line: Antipsychotic medications (e.g., risperidone, olanzapine)

Second-line: Cognitive-behavioral therapy, supportive therapy.
Correct Answer: B. Reduce delusions ✅

MCQ 611:

A 23-year-old man is brought to the psychiatrist clinic because for the first time he started to hear some voices talking to him with unknown sources. On examination, he was apathetic and emotionally withdrawn. Which of the following is the best management?

A. Olanzapine
B. Fluoxetine
C. Cognitive behavioral therapy
D. Antipsychotic and cognitive behavioral therapy

Explanation:
The correct answer is **D. Antipsychotic and cognitive behavioral therapy**.
This patient is presenting with **first-episode psychosis**, likely indicating **schizophrenia** or another psychotic disorder. The best treatment involves **antipsychotic medication** (such as **olanzapine**) to manage psychotic symptoms like **auditory hallucinations**. **Cognitive behavioral therapy** (CBT)

is also beneficial for addressing cognitive distortions and promoting recovery.
Schizophrenia
Definition: A mental health disorder characterized by persistent psychotic symptoms such as delusions, hallucinations, and disorganized behavior.
Key Clinical Presentation: Hallucinations, delusions, emotional withdrawal, apathy.
Investigations: Diagnosis is clinical and may involve imaging and lab work to rule out other causes.
Treatment Outline:

First-line: Antipsychotic medications (e.g., olanzapine, risperidone)

Second-line: Cognitive behavioral therapy, family therapy.
Correct Answer: D. Antipsychotic and cognitive behavioral therapy ✅

MCQ 612:

A 45-year-old man with recurrent episodes of schizophrenia is on maintenance therapy to maintain suppression of symptoms, prevent relapse, improve quality of life, and support engagement in psychosocial therapy. Which of the following is the best outcome following the maintenance therapy?

A. Complete resolution
B. 5% will achieve remission
C. 70% will have a satisfactory quality of life
D. 33% obtain long-lasting symptom reduction

Explanation:
The correct answer is **D. 33% obtain long-lasting symptom reduction**.
Schizophrenia is a chronic condition, and **maintenance therapy** aims to control symptoms, prevent relapse, and improve the patient's quality of life. A **significant percentage** of patients may achieve **long-lasting symptom reduction**, but complete resolution is rare.
Schizophrenia
Definition: A chronic psychiatric disorder that affects thinking, emotional regulation, and behavior.
Key Clinical Presentation: Positive symptoms (delusions, hallucinations) and negative symptoms (apathy, withdrawal).
Investigations: Clinical diagnosis based on symptoms and exclusion of other causes.
Treatment Outline:

First-line: Antipsychotics (maintenance treatment).

Second-line: Psychosocial therapy, cognitive-behavioral therapy.
Correct Answer: D. 33% obtain long-lasting symptom reduction ✅

MCQ 613:

Which of the following selective serotonin reuptake inhibitors is the most toxic in overdose?

A. Sertraline
B. Citalopram
C. Paroxetine
D. Fluoxetine

Explanation:
The correct answer is **B. Citalopram**.
Citalopram has been associated with a higher risk of **QT prolongation** and **cardiotoxicity** compared to other SSRIs, making it **more toxic in overdose**.
Selective Serotonin Reuptake Inhibitors (SSRIs)
Definition: A class of drugs commonly used to treat depression and anxiety disorders by increasing serotonin levels in the brain.
Key Clinical Presentation: Used to treat depression, anxiety, OCD, and PTSD.
Investigations: Monitoring for adverse effects, particularly cardiac effects in overdose situations.
Treatment Outline:

First-line: SSRIs (e.g., fluoxetine, sertraline, citalopram).

Second-line: SNRIs, tricyclic antidepressants.
Correct Answer: B. Citalopram ✅

MCQ 614;

A 23-year-old woman is newly diagnosed with paranoid schizophrenia based on presence of both negative and positive symptoms of schizophrenia. Which of the following is the most effective medication?

A. Clozapine
B. Trazodone
C. Fluphenazine
D. Chlorpromazine

Explanation:
The correct answer is **A. Clozapine**.
Clozapine is the most effective **atypical antipsychotic** for **treatment-resistant schizophrenia**, especially when the patient has not responded to other antipsychotics. It is also known to reduce **both positive and negative symptoms** of schizophrenia.
Schizophrenia
Definition: A chronic psychiatric disorder that leads to disturbances in thinking, perception, emotions, and behavior.
Key Clinical Presentation: Hallucinations, delusions, disorganized behavior.
Investigations: Diagnosis is based on clinical criteria, ruling out other medical causes.
Treatment Outline:

First-line: Antipsychotic medications (e.g., risperidone, clozapine).

Second-line: Cognitive-behavioral therapy, social skills training.
Correct Answer: A. Clozapine ✅

MCQ 615:

A 35-year-old man with a history of open appendectomy 5 years ago presented with abdominal pain, distension, vomiting, and constipation for 3 days. On physical examination, he was dehydrated, abdomen was distended with marked tenderness in the right iliac fossa, and exaggerated bowel sounds. He was scheduled for emergency exploratory laparotomy (see report). Plain abdominal X-ray: Distended small bowel, multiple air-fluid levels with collapsed colon. Which of the following is contraindicated?

A. Propofol
B. Ketamine
C. Sevoflurane
D. Nitrous oxide

Explanation:
The correct answer is **D. Nitrous oxide**.
In patients with **bowel obstruction**, the use of **nitrous oxide** is contraindicated because it can **increase the size of air pockets within the bowel**, leading to a **bowel perforation**. **Nitrous oxide** is an anesthetic that may accumulate in gas-filled spaces, which can exacerbate the risk of rupture or worsen the obstruction.
Bowel Obstruction

Definition: A blockage in the intestines that prevents the normal flow of contents.
Key Clinical Presentation: Abdominal pain, distension, vomiting, constipation, and tenderness.
Investigations: Abdominal X-ray, CT scan.
Treatment Outline:

First-line: Surgery for emergent cases, fluid resuscitation.

Second-line: Conservative management for non-complicated cases.
Correct Answer: D. Nitrous oxide ✅

MCQ 616:

A 45-year-old man involved in a motor vehicle accident with an isolated head injury, remains in a coma for 5 days after the accident. Which of the following is the most appropriate feeding method in the early stage of management?

A. Gastrostomy tube feeding
B. Nasogastric tube feeding
C. Central parenteral feeding
D. Peripheral parenteral feeding

Explanation:
The correct answer is **B. Nasogastric tube feeding**.
In a patient with a **head injury** who is in a **coma**, **nasogastric tube feeding** is the most appropriate early method of nutritional support. **Enteral feeding** (through the nasogastric tube) is preferred over parenteral feeding as it is associated with fewer complications such as **infection** and **gut atrophy**.
Head Injury
Definition: Trauma to the skull and brain, potentially leading to coma, seizures, or other neurological deficits.
Key Clinical Presentation: Coma, altered consciousness, signs of head trauma (e.g., swelling, bruising).
Investigations: CT scan, neurological examination.
Treatment Outline:

First-line: Sedation, monitoring, enteral nutrition.

Second-line: Surgery for any intracranial hemorrhage or skull fractures.
Correct Answer: B. Nasogastric tube feeding ✅

MCQ 616:

Which of the following is a contraindication for liver transplantation?

A. Acute liver failure
B. Liver cirrhosis with active alcohol
C. End stage liver disease with ascites
D. End stage liver disease with encephalopathy

Explanation:
The correct answer is **B. Liver cirrhosis with active alcohol**.
Active alcohol consumption in patients with liver cirrhosis is considered a contraindication for liver transplantation, as ongoing alcohol use undermines the potential success of the transplant and increases the risk of organ rejection or complications. Patients must show a sustained period of abstinence from alcohol (usually 6 months) to be considered eligible for a transplant.
Liver Transplantation
Definition: A surgical procedure to replace a failing liver with a healthy liver from a donor.
Key Clinical Presentation: End-stage liver disease, including cirrhosis, hepatitis, or acute liver failure.
Investigations: Liver function tests, imaging studies, and biopsy in some cases.
Treatment Outline:

First-line: Liver transplantation for patients with terminal liver failure.

Second-line: Medical management to optimize liver function prior to transplantation.
Correct Answer: B. Liver cirrhosis with active alcohol ✅

MCQ 617:

A patient presented with severe maxillofacial injury with mandibular fracture dislocation and profuse oral bleeding. Which of the following is the most appropriate way to secure the airway?

A. Laryngeal mask
B. Orotracheal tube
C. Nasotracheal tube
D. Cricothyroidotomy

Explanation:
The correct answer is **B. Orotracheal tube**.
In cases of severe maxillofacial injury, an **oro-tracheal tube** is often preferred to secure the

airway. This is because the mouth and upper airway are the primary routes for intubation, and it provides a stable airway while maintaining access to the oral cavity. A **laryngeal mask** (Option A) can be an option in less severe cases, but for significant trauma with the risk of obstruction, endotracheal intubation via the oro-tracheal route is the best approach.

Maxillofacial Injury and Airway Management

Definition: Trauma to the face and jaw, often resulting in fractures, bleeding, and airway compromise.

Key Clinical Presentation: Difficulty breathing, facial deformities, and bleeding.

Investigations: Imaging (e.g., X-ray, CT) to assess fractures.

Treatment Outline:

First-line: Secure airway with oro-tracheal intubation.

Second-line: Nasotracheal intubation if oro-tracheal intubation is not possible.

Correct Answer: B. Orotracheal tube ✅

MCQ 618:

A 28-year-old woman presented to the Emergency Department with 18-hour history of severe epigastric pain associated with vomiting. She is known to have multiple small gallstones. Physical examination confirmed upper abdominal tenderness.

Test Results:

WBC: 13.4 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Alkaline phosphatase: 289 IU/L (Normal: 39-117 IU/L)

Amylase: 488 IU/L (Normal: 24-151 IU/L)

Direct bilirubin: 19 µmol/L (Normal: 1.5-6.5 µmol/L)

Total bilirubin: 34 µmol/L (Normal: 3.5-16.5 µmol/L)

Which of the following is the most appropriate initial management?

A. Abdominal ultrasound
B. Intravenous crystalloid infusion
C. Intravenous broad spectrum antibiotics
D. Endoscopic retrograde cholangiopancreatography

Explanation:

The correct answer is **B. Intravenous crystalloid infusion**.

This patient presents with **gallstone pancreatitis** (evidenced by elevated amylase and bilirubin, and abdominal pain), which is an inflammatory condition of the pancreas due to obstruction of the pancreatic duct by gallstones. The most important first step in the management of acute pancreatitis is **fluid resuscitation** with **intravenous crystalloid infusion** to correct dehydration, support circulatory function, and prevent organ failure. **Abdominal ultrasound** (Option A) is important for confirming the diagnosis of gallstones but is not the immediate next step in this acute setting. **Broad-spectrum antibiotics** (Option C) are not routinely used unless there is evidence of infection. **ERCP** (Option D) may be required in the case of **bile duct obstruction**, but it is not the immediate first-line intervention in uncomplicated pancreatitis.

Acute Gallstone Pancreatitis Management

Definition: Inflammation of the pancreas due to obstruction by gallstones.

Key Clinical Presentation: Severe epigastric pain, vomiting, elevated amylase and bilirubin.

Investigations: Abdominal ultrasound to detect gallstones; blood tests for elevated amylase and bilirubin.

Treatment Outline:

First-line: Intravenous crystalloid infusion to stabilize fluid balance.

Second-line: ERCP for biliary obstruction or choledocholithiasis.

Correct Answer: B. Intravenous crystalloid infusion ✅

Thyroid Cancer Management

Definition: Malignant growth in the thyroid gland, including papillary, follicular, medullary, and anaplastic types.

Key Clinical Presentation: Progressive neck mass, stridor, dysphagia, history of Hashimoto’s thyroiditis.

Investigations: Fine needle aspiration, ultrasound, CT scan.

Treatment Outline:

First-line: Surgery, typically total thyroidectomy.

Second-line: Radioactive iodine, chemotherapy for anaplastic cancer.

Correct Answer: C. Anaplastic ✅

MCQ 619:

A 75-year-old man presented with a 12-hour history of sudden lower abdominal pain, associated with progressive abdominal distension, nausea, vomiting, and absolute constipation. Physical examination confirmed a grossly distended abdomen, generalized discomfort with normal bowel sounds, and an empty rectum.

Erect abdominal X-ray:

Massively distended bowel loop pointing to the right upper quadrant.

Which of the following is the most likely diagnosis?

A. Sigmoid volvulus
B. Rectosigmoid cancer
C. Closed loop obstruction
D. Acute sigmoid diverticulitis

Explanation:
The correct answer is **C. Closed loop obstruction**.
The **abdominal pain**, **distension**, and **vomiting**, along with **massive bowel dilation on X-ray**, are classic signs of a **closed-loop obstruction**. This occurs when a loop of bowel is obstructed at both ends, leading to a **distended bowel**. **Sigmoid volvulus** (Option A) can present similarly but would not typically show the characteristic distension pattern. **Rectosigmoid cancer** (Option B) may cause obstruction, but the absence of a palpable mass and the acute nature of this presentation make it less likely. **Acute sigmoid diverticulitis** (Option D) would present with localized tenderness and typically would not cause the massive distension seen here.

Bowel Obstruction Management
Definition: A blockage of the intestines preventing the normal passage of food and liquids.
Key Clinical Presentation: Abdominal pain, distension, vomiting, constipation.
Investigations: Abdominal X-ray, CT scan.
Treatment Outline:

First-line: Decompression (NG tube, IV fluids).

Second-line: Surgery for closed-loop or complicated obstructions.
Correct Answer: C. Closed loop obstruction ✅

MCQ 620:

A 58-year-old man presented with a 4-month history of generalized fatigability and shortness of breath. Physical examination confirmed a non-tender mass in the right iliac fossa.

Test Results:

Hb: 66 g/L (Normal: 130-170 g/L)

Which of the following is the most important step in management?

A. Colonoscopy
B. Abdominal ultrasound
C. Percutaneous biopsy of the mass
D. Abdominal computed tomography

Explanation:
The correct answer is **D. Abdominal computed tomography (CT)**.
Given the patient's **anemia** (low hemoglobin) and a **non-tender mass** in the **right iliac fossa**, the most likely diagnosis is a **colorectal mass**. **CT scan** provides detailed imaging to assess the mass's size, location, and potential involvement of surrounding structures, as well as any metastasis. **Colonoscopy** (Option A) is important for definitive diagnosis but would typically be performed after initial imaging. **Ultrasound** (Option B) is less useful for solid abdominal masses. **Percutaneous biopsy** (Option C) is not the first step as imaging is needed first.

Colorectal Mass Management
Definition: A mass located in the colon or rectum, potentially malignant.
Key Clinical Presentation: Anemia, abdominal mass, fatigue, shortness of breath.
Investigations: Abdominal CT, colonoscopy.
Treatment Outline:

First-line: Imaging (CT scan).

Second-line: Biopsy, surgery for malignant tumors.
Correct Answer: D. Abdominal computed tomography ✅

MCQ 621

A 33-year-old man presented to the Emergency Department with a 24-hour history of epigastric pain. Physical examination confirmed tenderness in the epigastric area with normal bowel sounds.

Blood pressure: 100/60 mmHg
Heart rate: 110/min
Temperature: 37.7°C

Test Results:

WBC: 14 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Amylase: 223 IU/L (Normal: 24-151 IU/L)

Erect chest X-ray:

Free air under the diaphragm.

Which of the following is the most appropriate next step in management?

A. Gastrografin swallow
B. Gastroduodenoscopy
C. Exploratory laparotomy
D. Conservative treatment

Explanation:
The correct answer is **C. Exploratory laparotomy**.
The patient's **epigastric pain**, **vomiting**, and **elevated amylase**, along with the **free air under the diaphragm** on chest X-ray, are suggestive of a **perforated peptic ulcer**, which is a surgical emergency. **Exploratory laparotomy** is needed to confirm the diagnosis and treat the perforation. **Gastrografin swallow** (Option A) is used to evaluate for leaks but is not the first step in suspected perforation. **Gastroduodenoscopy** (Option B) is useful for diagnosing ulcers but would not be appropriate in an emergency setting with free air, as surgery is indicated. **Conservative treatment** (Option D) is inappropriate given the seriousness of the condition.

Perforated Peptic Ulcer Management
Definition: A peptic ulcer that has eroded through the gastric or duodenal wall causing leakage of gastric contents into the peritoneum.
Key Clinical Presentation: Acute severe abdominal pain, vomiting, free air under diaphragm.
Investigations: Chest X-ray, abdominal X-ray, CT scan.
Treatment Outline:

First-line: Surgical intervention (exploratory laparotomy).

Second-line: Antacids and proton pump inhibitors post-surgery.
Correct Answer: C. Exploratory laparotomy ✅

MCQ 622:

A 27-year-old obese woman presented to the Emergency Department with a 6-day history of right iliac fossa pain associated with nausea, vomiting, and loss of appetite. Physical examination confirmed tenderness in the right iliac fossa.

Test Result:

WBC: 13 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Which of the following is the most appropriate next step in management?

A. Open surgery
B. Abdominal CT scan
C. Abdominal ultrasound
D. Diagnostic laparoscopy

Explanation:
The correct answer is **C. Abdominal ultrasound**.
This patient's clinical presentation, along with **elevated WBC**, suggests possible **acute appendicitis**. The **most appropriate next step** is **abdominal ultrasound**, which is a non-invasive first-line investigation, especially in women of childbearing age to rule out gynecological causes. **Abdominal CT scan** (Option B) can also be useful but is typically not the first line in this scenario, especially with concern for radiation. **Diagnostic laparoscopy** (Option D) is invasive and should be reserved for cases where imaging results are inconclusive. **Open surgery** (Option A) is indicated after confirming the diagnosis.

Acute Appendicitis Management
Definition: Inflammation of the appendix, typically due to obstruction of the lumen.
Key Clinical Presentation: Right iliac fossa pain, fever, elevated WBC.
Investigations: Ultrasound, CT scan.
Treatment Outline:

First-line: Appendectomy (open or laparoscopic).

Second-line: Antibiotics for perforated appendicitis.
Correct Answer: C. Abdominal ultrasound ✅

MCQ 623:

A 34-year-old man who underwent an open repair of a right inguinal hernia 3 years ago, presented with a 3-month history of a right groin swelling that is associated with dragging pain. Physical examination confirmed a right groin reducible swelling with positive cough impulse, reaching the neck of scrotum.

Which of the following is the most appropriate next step in management?

A. Open repair
B. CT abdomen
C. Scrotal support
D. Laparoscopic mesh repair

Explanation:
The correct answer is **A. Open repair**.
The patient's presentation is typical of an **inguinal hernia**, with a **positive cough impulse** and **reducible swelling**. Given the chronicity of the hernia and the size of the swelling, **open repair** is the most appropriate treatment. **Laparoscopic mesh repair** (Option D) is an alternative technique but may not be the first option in this patient. **CT abdomen** (Option B) is not necessary for diagnosis, as the diagnosis is clinical. **Scrotal support** (Option C) might provide temporary relief but does not address the hernia itself.

Inguinal Hernia Management
Definition: Protrusion of intra-abdominal contents through the inguinal canal.
Key Clinical Presentation: Groin swelling, cough impulse, reducibility.
Investigations: Clinical diagnosis.
Treatment Outline:

First-line: Surgical repair (open or laparoscopic).

Second-line: Watchful waiting in asymptomatic hernias.
Correct Answer: A. Open repair ✅

MCQ 624:

A 48-year-old woman presented with a 12-hour history of a painful swelling in the right groin. Physical examination confirmed an irreducible tender swelling in the groin area that is 3x4 cm. Surgical exploration confirmed a strangulated hernia and the neck is lateral and inferior to the pubic tubercle.

Which of the following is the most likely hernia?

A. Femoral
B. Obturator
C. Direct inguinal
D. Indirect inguinal

Explanation:
The correct answer is **A. Femoral**.
A **femoral hernia** typically presents with a **painful, irreducible swelling** below the inguinal ligament, often **lateral and inferior to the pubic tubercle**. It is more likely to become strangulated due to its narrow neck. **Obturator hernia** (Option B) is less common and usually presents with **vague abdominal pain**. **Direct inguinal** (Option C) and **indirect inguinal** (Option D) hernias typically present above the inguinal ligament and are less likely to be strangulated.

Femoral Hernia Management
Definition: Hernia that protrudes through the femoral canal, located below the inguinal ligament.
Key Clinical Presentation: Irreducible, tender groin mass, typically below the inguinal ligament.
Investigations: Clinical diagnosis, occasionally confirmed by imaging.
Treatment Outline:

First-line: Surgical repair, usually via open surgery.

Second-line: Watchful waiting if asymptomatic.
Correct Answer: A. Femoral ✅

MCQ 625:

A 25-year-old presented to the Emergency Department with a 2-day history of diffuse colicky abdominal pain, associated with abdominal distension, vomiting, and constipation. Physical examination confirmed diffuse abdominal discomfort with exaggerated bowel sounds.

CT scan:

Distended proximal small bowel with complete obstruction at the proximal ileum.

Exploratory laparotomy:

3 strictures over 60 cm of the ileum, the proximal one was the cause of small bowel obstruction.

Which of the following is the most likely diagnosis?

A. TB enteritis
B. Crohn's disease
C. Small bowel lymphoma
D. Gastrointestinal stromal tumour

Explanation:
The correct answer is **B. Crohn's disease**.

The presence of **multiple strictures** in the **ileum** and the history of **recurrent abdominal pain** with **obstruction** points to **Crohn's disease**. **TB enteritis** (Option A) could cause strictures, but it is less common in developed countries and typically affects the ileocecal region with a more insidious onset. **Small bowel lymphoma** (Option C) can present with obstruction but is less likely to cause multiple strictures in the ileum. **Gastrointestinal stromal tumours** (Option D) are rare and typically present with a mass, rather than multiple strictures.

Crohn's Disease Management
Definition: Chronic inflammatory bowel disease with transmural inflammation affecting any part of the gastrointestinal tract, most commonly the ileum.
Key Clinical Presentation: Abdominal pain, weight loss, diarrhea, and sometimes obstruction due to strictures.
Investigations: Colonoscopy, CT scan, biopsy.
Treatment Outline:

First-line: Anti-inflammatory drugs (e.g., corticosteroids, immunosuppressants).

Second-line: Surgical intervention for complications like strictures and obstructions.
Correct Answer: B. Crohn's disease ✅

MCQ 626:

A 29-year-old patient presented with a 3-year history of frequent attacks of bloody diarrhea. On examination, he was pale with a normal abdominal examination.

Barium enema:

Featureless "lead-pipe" colon.

Colonoscopy:

Diffusely inflamed colon with pseudopolyps, featureless terminal ileum, and backwash ileitis.

Which of the following is the most likely type of colitis?

A. Crohn's
B. Amebic
C. Ischemic
D. Ulcerative

Explanation:
The correct answer is **D. Ulcerative colitis**.
The **featureless "lead-pipe" colon** on **barium enema**, along with **pseudopolyps** and **backwash**

ileitis seen on colonoscopy, is characteristic of **ulcerative colitis**. **Crohn's disease** (Option A) typically shows segmental inflammation and **skip lesions**. **Amebic colitis** (Option B) is less likely in this context as it typically presents with **flask-shaped ulcers**. **Ischemic colitis** (Option C) usually affects the **watershed areas** of the colon and would not show the characteristic findings of **pseudopolyps** and **backwash ileitis**.

Ulcerative Colitis Management
Definition: Chronic inflammatory bowel disease affecting the colon and rectum, characterized by continuous mucosal inflammation.
Key Clinical Presentation: Bloody diarrhea, abdominal pain, tenesmus, and weight loss.
Investigations: Colonoscopy, biopsy, barium enema.
Treatment Outline:

First-line: Aminosalicylates and corticosteroids.

Second-line: Immunosuppressive drugs or biologics (e.g., TNF inhibitors).
Correct Answer: D. Ulcerative colitis ✅

MCQ 627:

A 19-year-old man presented with a 3-day history of frequent attacks of per-rectal fresh bleeding. Physical examination confirmed clotted blood in the rectum. Nasogastric tube was draining greenish-colored fluids.

Test Results:

Hb: 85 g/L (Normal: 130-170 g/L for male)

Colonoscopy:

No colorectal pathology.

Which of the following is the most appropriate diagnostic tool to localize the source of bleeding?

A. Diagnostic laparoscopy
B. Gastrografin follow-through
C. Upper gastrointestinal endoscopy
D. 99mTc-labelled sodium pertechnetate

Explanation:
The correct answer is **C. Upper gastrointestinal endoscopy**.
The presence of **greenish-colored fluid** in the nasogastric tube suggests a **gastrointestinal**

source of bleeding, likely from the **upper gastrointestinal tract**. **Upper gastrointestinal endoscopy** is the best next step to identify potential causes of **upper GI bleeding**, such as peptic ulcers, varices, or Mallory-Weiss tears. **Gastrografin follow-through** (Option B) is not suitable for detecting bleeding. **Diagnostic laparoscopy** (Option A) is not appropriate for this scenario, and **99mTc-labelled sodium pertechnetate** (Option D) is used for localized **gastrointestinal bleeding** but not in cases of fresh rectal bleeding.

Upper GI Bleeding Management
Definition: Bleeding originating from the **esophagus, stomach**, or **duodenum**.
Key Clinical Presentation: Hematemesis, coffee ground vomitus, and melena.
Investigations: Upper endoscopy, angiography, CT scan.
Treatment Outline:

First-line: Endoscopic hemostasis, proton pump inhibitors.

Second-line: Surgery for refractory cases.
Correct Answer: C. Upper gastrointestinal endoscopy ✅

MCQ 628:

A 55-year-old man complains of a 2-month history of central abdominal pain. Physical examination was normal.

Test Results:

24-hour 5-HIAA: 21 mg (Normal: 2-7 mg/24 h)

CT scan:

Small bowel mass 4x5 cm, with some calcifications.

Which of the following is the most likely small bowel tumour?

A. Carcinoid
B. Lymphoma
C. Schwannoma
D. Gastrointestinal stroma

Explanation:
The correct answer is **A. Carcinoid**.
Carcinoid tumors are well-known to produce **5-HIAA**, a metabolite of serotonin. The **elevated 5-HIAA** level, along with the presence of a **mass with calcifications** in the **small bowel**, suggests **carcinoid** tumor, which often arises in the **small intestine** and can lead to **carcinoid syndrome**.

Lymphoma (Option B) can cause a mass but typically does not lead to elevated **5-HIAA**. **Schwannomas** (Option C) are rare tumors that arise from nerve tissue, and **gastrointestinal stromal tumors (GIST)** (Option D) are typically **CD117 positive** on immunohistochemistry and do not produce **5-HIAA**.

Carcinoid Tumor Management
Definition: A slow-growing neoplasm most commonly found in the small bowel, often producing serotonin.
Key Clinical Presentation: Abdominal pain, flushing, diarrhea, and heart failure (carcinoid syndrome).
Investigations: Elevated 5-HIAA, CT scan, biopsy.
Treatment Outline:

First-line: Surgical resection of the tumor.

Second-line: Somatostatin analogs (e.g., octreotide).
Correct Answer: A. Carcinoid ✅

MCQ 629:

A 32-year-old man presented to the Emergency Department with pain and swelling of his right thigh after a trivial injury. Physical examination confirmed swelling, deformity of his right thigh with tenderness.

Test Results:

Parathyroid hormone (intact PTH levels): 8.9 pmol/L (Normal: 1.1-5.3 pmol/L)

Calcium: 4.5 mmol/L (Normal: 2.15-2.62 mmol/L)

X-ray:

Spiral fracture of right femur with subperiosteal resorption.

Which of the following is the most likely diagnosis?

A. Parathyroid adenoma
B. Parathyroid carcinoma
C. Tertiary hyperparathyroidism
D. Secondary hyperparathyroidism

Explanation:
The correct answer is **A. Parathyroid adenoma**.

Subperiosteal resorption seen on the X-ray along with **elevated calcium** and **parathyroid hormone (PTH)** levels strongly suggest **primary hyperparathyroidism**, which is most commonly caused by **parathyroid adenoma**. This condition leads to excess PTH secretion, resulting in elevated calcium levels and bone resorption. **Parathyroid carcinoma** (Option B) is less common and usually presents with more severe symptoms and higher calcium levels. **Tertiary hyperparathyroidism** (Option C) occurs in patients with long-standing **secondary hyperparathyroidism** (due to chronic kidney disease), but it is not associated with the findings in this patient. **Secondary hyperparathyroidism** (Option D) would show low calcium levels with high PTH.

Primary Hyperparathyroidism Management
Definition: Excessive secretion of PTH, leading to hypercalcemia and bone resorption.
Key Clinical Presentation: Bone pain, fractures, kidney stones, abdominal pain, and psychiatric symptoms.
Investigations: Serum calcium, PTH, and imaging studies for parathyroid adenoma.
Treatment Outline:

First-line: Surgical removal of the parathyroid adenoma.

Second-line: Medical management in non-surgical candidates.
Correct Answer: A. Parathyroid adenoma ✅

MCQ 630

A 28-year-old married woman presented to the Emergency Department complaining of a 6-hour history of right iliac fossa pain associated with drowsiness. Physical examination confirmed tenderness over the lower abdomen with sluggish bowel sounds.

Which of the following is the most likely diagnosis?

A. Septic shock
B. Ectopic pregnancy
C. Ruptured ovarian cyst
D. Pelvic inflammatory disease

Explanation:
The correct answer is **B. Ectopic pregnancy**.
Right iliac fossa pain in a **young, married woman** with **drowsiness** strongly suggests **ectopic pregnancy**, especially if she is sexually active. This condition is a common cause of **acute abdominal pain** and can present with **shock** if there is **rupture**. **Septic shock** (Option A) could

cause drowsiness but is not likely given the localized pain in the right iliac fossa. **Ruptured ovarian cyst** (Option C) could cause abdominal pain but is typically associated with a more localized sharp pain and less likely to present with drowsiness. **Pelvic inflammatory disease** (Option D) causes lower abdominal pain and fever, but the absence of **fever** and the localization of pain make it less likely in this case.

Ectopic Pregnancy Management
Definition: Pregnancy that occurs outside the uterine cavity, most commonly in the fallopian tube.
Key Clinical Presentation: Abdominal pain, vaginal bleeding, and amenorrhea.
Investigations: Urine pregnancy test, transvaginal ultrasound, and serum beta-hCG.
Treatment Outline:

First-line: Methotrexate (for early, non-ruptured cases) or surgical intervention for rupture.
Correct Answer: B. Ectopic pregnancy ✅

MCQ 631;

A 32-year-old man presented to the Emergency Department with a 6-hour history of severe sudden epigastric pain. Physical examination confirmed diffuse severe abdominal tenderness with sluggish bowel sounds.

Test Results:

WBC: 13.8 x 10^9/L (Normal: 4.5-10.5 x 10^9/L)

Amylase: 300 IU/L (Normal: 24-151 IU/L)

Which of the following is the most appropriate next step in management?

A. Erect chest X-ray
B. Abdominal X-ray
C. CT scan abdomen
D. Ultrasound abdomen

Explanation:
The correct answer is **C. CT scan abdomen**.
The patient has a **classic presentation of acute pancreatitis**, evidenced by **epigastric pain** and **elevated amylase**. The **next best step** in management is to perform a **CT scan** to assess the extent of the pancreatitis, complications (e.g., pseudocysts or abscesses), and other possible causes. **Erect chest X-ray** (Option A) could be helpful in identifying **diaphragmatic rupture** or **pleural effusion**, but it will not help with pancreatitis. **Abdominal X-ray** (Option B) is not

sufficient for confirming acute pancreatitis. **Ultrasound abdomen** (Option D) can help identify gallstones but is not as sensitive for assessing complications of pancreatitis.

Acute Pancreatitis Management
Definition: Acute inflammation of the pancreas, often due to gallstones or alcohol consumption.
Key Clinical Presentation: Severe epigastric pain, nausea, vomiting, and elevated amylase/lipase.
Investigations: Abdominal CT scan, ultrasound, and blood tests.
Treatment Outline:

First-line: Supportive care (IV fluids, pain management, and NPO status).

Second-line: Address underlying cause (e.g., cholecystectomy for gallstones).
Correct Answer: C. CT scan abdomen ✅

MCQ 632;

A 53-year-old man complains of a 3-month history of vague epigastric pain and weight loss. Physical examination was normal.

Upper GIT endoscopy:

Antral gastric mass. Multiple deep biopsies were taken.

Histopathology:

Mucosa-associated lymphoid tissue lymphoma and positive for Helicobacter pylori.

Which of the following is the most appropriate initial management?

A. Surgery
B. Antibiotics
C. Radiotherapy
D. Chemotherapy

Explanation:
The correct answer is **B. Antibiotics**.
This patient has **MALT lymphoma** (Mucosa-Associated Lymphoid Tissue lymphoma), which is often associated with **H. pylori infection**. The **first-line treatment** is **eradication of H. pylori** using **antibiotics** (e.g., clarithromycin, amoxicillin, and proton pump inhibitors). **Surgery** (Option A) and **chemotherapy** (Option D) are not first-line treatments unless the lymphoma persists or is non-responsive to antibiotics. **Radiotherapy** (Option C) may be used in cases where antibiotics fail but is not the initial treatment.

MALT Lymphoma Management
Definition: A low-grade lymphoma associated with H. pylori infection, often affecting the stomach.
Key Clinical Presentation: Epigastric pain, weight loss, nausea, and vomiting.
Investigations: Endoscopy with biopsy, H. pylori testing.
Treatment Outline:

First-line: Antibiotics to eradicate H. pylori.

Second-line: Chemotherapy or radiation if antibiotics are ineffective.
Correct Answer: B. Antibiotics ✅

MCQ 633:

A 31-year-old patient presented with a 12-hour history of severe epigastric abdominal pain. Physical examination confirmed a tender abdomen with sluggish bowel sounds.

Erect chest X-ray:

Air under the diaphragm.

Exploratory laparotomy:

A 5x5 mm perforation in the anterior aspect of the 1st part of duodenum with peritoneal soiling.

Which of the following is the most appropriate management?

A. Distal gastrectomy
B. Pyloroplasty with truncal vagotomy
C. Simple closure with omental patching
D. Pyloroplasty with high selective vagotomy

Explanation:
The correct answer is **C. Simple closure with omental patching**.
This patient has a **perforated duodenal ulcer** (most commonly due to H. pylori or NSAID use). The standard management of a perforated duodenal ulcer is to **close the perforation** with an **omentum patch** to cover the perforation and allow healing. **Distal gastrectomy** (Option A) and **pyloroplasty with truncal vagotomy** (Option B) are more complex procedures used for specific ulcer complications but are not required here. **Pyloroplasty with high selective vagotomy** (Option D) is also not appropriate as the ulcer is perforated, not just causing obstruction.

Duodenal Ulcer Perforation Management
Definition: A complication of peptic ulcer disease in which an ulcer erodes through the wall of the duodenum, leading to peritonitis.
Key Clinical Presentation: Severe abdominal pain, peritonitis, shock, and free air under the diaphragm.
Investigations: Abdominal X-ray (for free air), CT scan, and exploratory laparotomy.
Treatment Outline:

First-line: Simple closure of the perforation with omental patch.
Correct Answer: C. Simple closure with omental patching ✅

MCQ 634:

A 55-year-old man presented with a 4-month history of vague epigastric pain associated with 15 kg weight loss. Physical examination confirmed enlarged left supraclavicular lymph nodes and gross ascites.

Which of the following is the most appropriate next step in management?

A. Abdominal ultrasound
B. Diagnostic laparoscopy
C. Abdominal paracentesis
D. Upper gastrointestinal endoscopy

Explanation:
The correct answer is **C. Abdominal paracentesis**.
This patient has **signs of advanced malignancy** (weight loss, ascites, and supraclavicular lymphadenopathy). **Abdominal paracentesis** is the most appropriate next step to analyze the ascitic fluid, which will help in diagnosing whether the ascites is due to malignancy, infection, or other causes. **Abdominal ultrasound** (Option A) can be useful to assess the ascites and organs, but **paracentesis** is crucial to guide further management. **Diagnostic laparoscopy** (Option B) may be considered later to evaluate peritoneal involvement, but paracentesis is first-line. **Upper gastrointestinal endoscopy** (Option D) is not appropriate without first evaluating the ascites.

Management of Ascites
Definition: Fluid accumulation in the peritoneal cavity, often associated with liver disease or malignancy.
Key Clinical Presentation: Abdominal distension, weight loss, and ascitic fluid.
Investigations: Abdominal paracentesis for fluid analysis.
Treatment Outline:

First-line: Paracentesis for diagnostic purposes.

Second-line: Treat underlying cause (e.g., chemotherapy for malignancy).
Correct Answer: C. Abdominal paracentesis ✅

MCQ 635:

A 47-year-old man reported frequent vomiting over the past 24-hours due to food poisoning. He presented to the Emergency Department complaining of a 2-hour history of vomiting fresh blood. Physical examination confirmed mild epigastric tenderness. A nasogastric tube was inserted and was draining bloody stained fluids.

Which of the following is the most likely diagnosis?

A. Gastritis
B. Dieulafoy's lesion
C. Peptic ulcer disease
D. Mallory-Weiss syndrome

Explanation:
The correct answer is **D. Mallory-Weiss syndrome**.
Mallory-Weiss syndrome is characterized by **tears in the mucosa at the junction of the stomach and esophagus** due to **violent vomiting**. The **fresh blood** in the vomit and history of **vomiting** following food poisoning point towards this condition. **Gastritis** (Option A) can cause upper GI bleeding but is less likely to cause a specific tear in the mucosa. **Dieulafoy's lesion** (Option B) involves an abnormal artery that erodes the mucosa and causes bleeding, but it is less common. **Peptic ulcer disease** (Option C) can cause upper GI bleeding but typically causes ulcers in the stomach or duodenum, not mucosal tears.

Mallory-Weiss Syndrome Management
Definition: Mucosal tear at the gastroesophageal junction, often due to vomiting.
Key Clinical Presentation: Hematemesis following vomiting, often in the context of alcohol use or food poisoning.
Investigations: Endoscopy to visualize the tear.
Treatment Outline:

First-line: Supportive care (IV fluids), endoscopic hemostasis if needed.
Correct Answer: D. Mallory-Weiss syndrome ✅

MCQ 636:

A 62-year-old man presented with a 3-month history of early satiety and 11 kg weight loss. Physical examination was normal.

Upper GIT endoscopy:

Ulcerating mass in the antrum. Multiple biopsies were taken.

Histopathology:

Invasive adenocarcinoma.

Which of the following is the most appropriate next investigation?

A. Abdominal ultrasound
B. Endoscopic ultrasound
C. Diagnostic laparoscopy
D. CT scan chest, abdomen and pelvis

Explanation:
The correct answer is **D. CT scan chest, abdomen and pelvis**.
Given the diagnosis of **gastric adenocarcinoma**, the next step is to assess the **extent of disease** and check for metastasis. A **CT scan of the chest, abdomen, and pelvis** is the most appropriate to evaluate the primary tumor, lymph node involvement, and potential distant metastases. **Endoscopic ultrasound** (Option B) may be used for **staging** and evaluating the depth of tumor invasion, but a **CT scan** is more comprehensive. **Diagnostic laparoscopy** (Option C) is used to evaluate peritoneal metastasis but is not the first step. **Abdominal ultrasound** (Option A) is less helpful in assessing the full extent of gastric cancer.

Gastric Adenocarcinoma Management
Definition: A malignant tumor originating from the gastric epithelium.
Key Clinical Presentation: Weight loss, early satiety, nausea, and vomiting.
Investigations: Endoscopy with biopsy, CT scan for staging.
Treatment Outline:

First-line: Surgical resection (gastrectomy) if feasible.

Second-line: Chemotherapy or palliative care if advanced.
Correct Answer: D. CT scan chest, abdomen and pelvis ✅

MCQ 637:

A 29-year-old woman presented to the Emergency Department with an 8-hour history of severe epigastric pain radiating to the back associated with vomiting. She underwent a sleeve gastrectomy 3 months ago for morbid obesity. Physical examination confirmed mild epigastric tenderness.

Test Result Normal Values

WBC 9.8 (4.5-10.5 x 10^9/L)

Amylase 700 (24-151 IU/L)

Ultrasound:

Biliary sludge but no gallstone. Normal bile ducts.

Which of the following is the most appropriate intervention?

A. Endoscopic ultrasound
B. Endoscopic sphincterotomy
C. Laparoscopic cholecystectomy
D. Percutaneous cholecystostomy

Explanation:
The correct answer is **C. Laparoscopic cholecystectomy**.
This patient presents with symptoms consistent with **biliary colic** or **gallbladder dysfunction**. Given the **history of sleeve gastrectomy** and the **presence of biliary sludge** on ultrasound, she may be at risk for **cholelithiasis** or **cholecystitis**. The most appropriate management is **laparoscopic cholecystectomy** to remove the gallbladder and prevent further complications such as **cholecystitis** or **cholelithiasis**. **Endoscopic ultrasound** (Option A) could be considered to investigate bile duct pathology but is not the first-line treatment for gallbladder issues. **Endoscopic sphincterotomy** (Option B) and **percutaneous cholecystostomy** (Option D) are typically used in specific situations such as bile duct obstruction or in patients who cannot undergo surgery.

Management of Biliary Disease
Definition: A range of conditions involving the gallbladder, including biliary colic, cholelithiasis, and cholecystitis.
Key Clinical Presentation: Epigastric pain, vomiting, fever, jaundice.
Investigations: Ultrasound, CT scan, endoscopic ultrasound if needed.
Treatment Outline:

First-line: Laparoscopic cholecystectomy.

Correct Answer: C. Laparoscopic cholecystectomy ✅

MCQ 638:

A 37-year-old complained of a right breast lump for 3 months. She has a family history of breast cancer. Physical examination confirmed a hard lump in the upper outer quadrant of the right breast with ill-defined edges, skin tethering, and no palpable axillary lymph node.

Which of the following is the most appropriate next step in management?

A. MRI breast
B. Core needle biopsy
C. Bilateral mammography
D. Fine needle aspiration cytology

Explanation:
The most appropriate next step is **B. Core needle biopsy**.
The patient has a suspicious breast lump, particularly with **family history of breast cancer**. **Core needle biopsy** is the gold standard for evaluating breast masses as it allows for tissue diagnosis, including assessing for malignancy. **Mammography** (Option C) can be helpful, but biopsy is the definitive next step for diagnosis. **MRI breast** (Option A) is used in high-risk patients or for assessing the extent of disease, but not as the first diagnostic tool. **Fine needle aspiration cytology** (Option D) can be used, but core needle biopsy provides a more accurate and reliable sample.

MCQ 639:

A 32-year-old woman presented with a 2-month history of left breast pain associated with bloody nipple discharge. Physical examination confirmed a normal left breast and axilla.

Which of the following is the most appropriate initial breast investigation?

A. MRI
B. CT scan
C. Ultrasound
D. Mammogram

Explanation:
The most appropriate initial investigation is **D. Mammogram**.

Mammography is the gold standard imaging test for breast evaluation, particularly for **bloody nipple discharge**, as it can help detect underlying pathology such as **ductal carcinoma in situ (DCIS)** or **intraductal papilloma**. **Ultrasound** (Option C) may be helpful in younger women or when mammography is inconclusive but is usually used as a complement. **MRI** (Option A) is typically reserved for high-risk patients or to assess the extent of known disease. **CT scan** (Option B) is not indicated for breast evaluation.

MCQ 640:

A 44-year-old patient presented with an anterior neck swelling for 2 months. Physical examination confirmed a hard lump in the left side of the anterior triangle of the neck moving up with swallowing but no palpable cervical lymph nodes.

Ultrasound:

Mildly enlarged thyroid, with 1.5 cm solid nodule in the left lobe.
Fine needle aspiration cytology:

Medullary thyroid cancer.

Which of the following is the most appropriate procedure?

A. Nodule enucleation
B. Total thyroidectomy
C. Subtotal thyroidectomy
D. Left hemithyroidectomy

Explanation:
The most appropriate procedure is **B. Total thyroidectomy**.
Medullary thyroid cancer is a **neuroendocrine malignancy** arising from the parafollicular C-cells of the thyroid. It often requires **total thyroidectomy** because the tumor is usually multifocal and may involve both thyroid lobes. **Left hemithyroidectomy** (Option D) and **subtotal thyroidectomy** (Option C) are inadequate because medullary thyroid cancer often spreads beyond the affected lobe. **Nodule enucleation** (Option A) is not sufficient for malignancy.

MCQ 641:

A 32-year-old woman complained of an anterior neck lump noticed 3 months ago. Physical examination confirmed a firm lump in the right side of the anterior triangle of the neck moving up with swallowing but no palpable cervical lymph nodes.

Blood pressure 120/70 mmHg
Heart rate 115 /min
Temperature 36.7 °C

Test Result Normal Values

TSH: 0.1 (0.4 - 5.0 µU/mL)

Free T3: 11 (3.5 - 6.5 pmol/L)

Total T4: 277 (64-155 nmol/L)

Ultrasound:

Lump in the right thyroid lobe 2x3 cm.
Thyroid isotope scan:

Hot nodule in the right lobe with the remainder of the gland being cold.

Which of the following is the most appropriate initial step in management?

A. Anti-thyroid drugs
B. Total thyroidectomy
C. Right thyroid lobectomy
D. Radioactive iodine ablation

Explanation:
The most appropriate next step is **C. Right thyroid lobectomy**.
The patient has a **hot nodule** on the thyroid isotope scan, indicating a **functioning benign nodule** (likely a **toxic adenoma**). The best treatment option is to perform a **lobectomy** of the affected thyroid lobe. **Radioactive iodine ablation** (Option D) is used for diffuse hyperthyroid conditions like **Graves' disease**, not for solitary toxic nodules. **Anti-thyroid drugs** (Option A) are used in hyperthyroidism but are not curative for toxic adenomas. **Total thyroidectomy** (Option B) is more appropriate for malignant cases or diffuse thyroid disease.

MCQ 642:

A 25-year-old man presented to the Emergency Department with severe pain during and after defecation for 3 days associated with the passage of a small amount of fresh blood after defecation. Physical examination confirmed an acute posterior anal fissure. Digital and proctoscopic examination were not performed due to the anal pain.

Which of the following is the most appropriate management?

A. Examination under anesthesia
B. Lateral internal anal sphincterotomy
C. Chemical sphincterotomy with diltiazem
D. Botulinus toxin paralysis of anal sphincter

Explanation:
The most appropriate initial management is **C. Chemical sphincterotomy with diltiazem**. **Acute anal fissures** are typically treated conservatively with **topical treatments** such as **diltiazem**, a calcium channel blocker that helps reduce **sphincter spasm** and improve healing. If the fissure persists or becomes chronic, **surgical sphincterotomy** (Option B) can be considered. **Examination under anesthesia** (Option A) may be necessary for chronic or complicated fissures but is not routinely required for acute cases. **Botulinum toxin** (Option D) is reserved for refractory fissures.

MCQ 642:

A 36-year-old woman complaining of per-rectal bleeding and prolapsed tissue masses through the anal verge. Anorectal examination confirmed grade IV haemorrhoids at 3, 7, and 11 o'clock positions.

Which of the following is the most appropriate management?

A. Rubber banding
B. Open haemorrhoidectomy
C. Infrared photocoagulation
D. Submucosal injection of sclerosing agent

Explanation:
The most appropriate management for **grade IV haemorrhoids** is **B. Open haemorrhoidectomy**. Grade IV haemorrhoids are **irreducible** and require **surgical intervention**. **Haemorrhoidectomy** is the most effective and definitive treatment. **Rubber banding** (Option A) is usually used for grade II or III haemorrhoids. **Infrared photocoagulation** (Option C) and **submucosal injection of sclerosing agent** (Option D) are less effective for severe, prolapsed haemorrhoids.

MCQ 643:

A 65-year-old man presented to the outpatient department complaining of alternating constipation and diarrhoea for 4 months. Physical examination was normal.

Rigid Sigmoidoscopy:

Circumferential, ulcerating mass in the distal part of the sigmoid colon. Multiple biopsies were taken.

Histopathology:

Adenocarcinoma.

Which of the following is the most appropriate next step in management?

A. Abdominal CT scan
B. Sigmoidectomy
C. Colonoscopy
D. Pelvic MRI

Explanation:
The most appropriate next step is **A. Abdominal CT scan**.
The patient has a **sigmoid colon adenocarcinoma** with **ulcerating mass**, and the next step is to evaluate the extent of the disease and check for distant metastasis. **CT scan** is the most useful imaging modality to assess the local and distant spread of the cancer. **Sigmoidectomy** (Option B) may be performed later if surgical intervention is indicated. **Colonoscopy** (Option C) would have already been done for diagnosis. **Pelvic MRI** (Option D) is used in some cases to evaluate local spread but is not typically the first-line imaging in this scenario.

MCQ 644:

A 48-year-old man complains of right upper quadrant abdominal pain for 4 months. Physical examination was normal.

Serology:

Echinococcus granulosus was positive.

Ultrasound:

Large cystic lesion (10x15 cm) in the right lobe of the liver with multiple intra-cystic daughter cysts, but no calcifications.

Which of the following is the most appropriate management?

A. Surgical de-roofing
B. Percutaneous aspiration
C. Long-term albendazole therapy
D. Percutaneous sclerosant

Explanation:
The most appropriate management is **C. Long-term albendazole therapy**.
This patient likely has a **hydatid cyst** caused by **Echinococcus granulosus**. The treatment of choice is **albendazole** to treat the parasite. **Surgical de-roofing** (Option A) can be an option in some cases, but **albendazole therapy** is typically the first-line treatment. **Percutaneous aspiration** (Option B) is not recommended for hydatid cysts due to the risk of cyst rupture and anaphylaxis. **Percutaneous sclerosant** (Option D) is not an appropriate treatment for hydatid cysts.

MCQ 645:

A 29-year-old woman presented to the Emergency Department with diffuse abdominal pain for 1 day associated with progressive abdominal distension, vomiting, and constipation. Physical examination confirmed abdominal distension, diffuse mild tenderness, and exaggerated bowel sounds.

CT scan:

Distended small bowel with ileocecal intussusception causing complete small bowel obstruction.

Which of the following is the most appropriate procedure?

A. Colonoscopy
B. Surgical resection
C. Surgical reduction
D. Reduction by barium enema

Explanation:
The most appropriate procedure is **C. Surgical reduction**.
Ileocecal intussusception is causing a complete **small bowel obstruction**, and the most appropriate intervention is **surgical reduction**. This is typically required for **adult intussusception**, which is usually managed surgically. **Colonoscopy** (Option A) can sometimes reduce intussusception in pediatric cases, but not in adults. **Surgical resection** (Option B) is only needed if there is bowel necrosis. **Reduction by barium enema** (Option D) is more commonly used in pediatric cases but is not appropriate in this adult case.

MCQ 646:

A 32-year-old woman presented to the Emergency Department with severe pain in the right upper quadrant of the abdomen following a fatty meal 6 hours ago. This pain has decreased in severity over the last 2 hours. The pain was radiating to the right subscapular area associated with vomiting. Physical examination was normal.

Test Result Normal Values

WBC: 8.2 (4.5-10.5 x 10^9/L)

Amylase: 134 (24-151 IU/L)

Ultrasound: Multiple gallstones with normal gallbladder wall.

Which of the following is the most appropriate management?

A. Observation
B. Ursodeoxycholic
C. Laparoscopic cholecystectomy
D. Magnetic resonance cholangiopancreatography

Explanation:
The patient has **biliary colic** (due to **gallstones**), which is usually self-limiting and does not require immediate surgical intervention. Therefore, the most appropriate initial management is **A. Observation**.

Laparoscopic cholecystectomy (Option C) is typically done for symptomatic gallstones, but this patient is not acutely ill.

Ursodeoxycholic acid (Option B) is used to dissolve small gallstones, but it's not the first choice for managing biliary colic.

Magnetic resonance cholangiopancreatography (Option D) is more appropriate for evaluating bile duct stones or other ductal issues, but not necessary in this case.

MCQ 647:

A 36-year-old presented to the Emergency Department with a 9-day history of right iliac fossa pain, diarrhea, and fever. He was diagnosed with Crohn's disease 10 years ago. Physical examination confirmed a tender mass in the right iliac fossa.

Test Result Normal Value

WBC: 17.6 (4.5-10.5 x 10^9/L)

CT scan: Collection in the right iliac fossa (12x15 cm) with jejune-ileal fistula.

Which of the following is the most appropriate initial treatment?

A. Laparoscopic drainage
B. Percutaneous drainage
C. Open surgical drainage
D. Open drainage with resection of fistula

Explanation:
The most appropriate initial treatment is **B. Percutaneous drainage**.

The presence of a **collection** and **fistula** suggests a **complicated Crohn's disease** with a **paracolonic abscess**. Percutaneous drainage is less invasive and effective for draining collections in this scenario.

Laparoscopic drainage (Option A) can be used for certain cases, but percutaneous drainage is preferred for larger abscesses.

Open surgical drainage (Option C) and **resection** (Option D) may be needed later if complications persist or if there is a bowel obstruction, but they are not the initial step.

MCQ 648:

A 33-year-old woman presented with a lump in her neck. Physical examination confirmed enlarged right cervical lymph nodes with normal thyroid gland. She underwent percutaneous biopsy of the lymph nodes.

Histopathology:

Normal follicular thyroid cells.

Which of the following is the most appropriate next step?

A. Reassurance
B. Surgical referral
C. Lymph node excision
D. 3 months follow up ultrasound

Explanation:
The most appropriate next step is **B. Surgical referral**.

The patient has **enlarged cervical lymph nodes** with no obvious thyroid pathology. Given the lymph node involvement, it is important to rule out any **malignancy** such as **metastatic thyroid cancer**.

Lymph node excision (Option C) may be considered, but a surgical referral is more appropriate for further evaluation and biopsy.

MCQ 649:

A 42-year-old woman presented with a right breast mass that has been present for 3 months and is not associated with other symptoms. Physical examination confirmed a 2 x 2 cm firm partially movable mass with associated skin tethering and an enlarged axillary lymph node.

Bilateral breast mammogram and ultrasound:

Ill-defined lobulated dense mass that is hypoechoic on ultrasound, measuring 2 x 2 cm, not associated with speculation or suspicious microcalcifications. Coded as BI-RADS IV.

Which of the following is the most appropriate next step in management?

A. Excision biopsy
B. Needle core biopsy
C. Fine needle aspiration
D. Short interval follow-up imaging

Explanation:
The most appropriate next step is **B. Needle core biopsy**.

The **BI-RADS IV** classification suggests a **high suspicion for malignancy**, and the next step is to obtain tissue diagnosis via **core needle biopsy**.

Fine needle aspiration (Option C) is less accurate for some breast lesions, especially those that are more heterogeneous.

Excision biopsy (Option A) is not necessary upfront, and **short interval follow-up** (Option D) is not appropriate given the high suspicion for cancer.

MCQ 650:

A 54-year-old woman complains of a left breast mass that has been present for 9 months. Physical examination confirmed a 3 x 3 cm firm fixed mass at the left retro-areolar area with associated nipple retraction.

Bilateral breast mammogram:

Speculated mass at left retro-areolar region associated with suspicious clustered microcalcifications. There are multiple pathological left axillary lymph nodes. Findings coded as BI-RADS V (Probably Malignant).

Which of the following is the most appropriate next step in management?

A. Excision biopsy
B. Needle core biopsy
C. Staging CT and bone scan
D. Modified radical mastectomy

Explanation:
The most appropriate next step is **B. Needle core biopsy**.

The **BI-RADS V** finding strongly suggests **malignant breast cancer**, and obtaining tissue through a **needle core biopsy** is the next step for histological diagnosis.

Modified radical mastectomy (Option D) is a treatment option but would only be considered after biopsy confirmation of malignancy.

Staging CT and bone scan (Option C) might be necessary after confirmation of malignancy but is not the first step.

MCQ 651:

A 40-year-old woman presented for her first screening mammography. She has no past medical history, and no family history of cancer.

Bilateral breast mammogram:

Bilateral dense breast with multiple fibro-glandular opacities, no suspicious microcalcifications or speculated lesions.

Breast ultrasound:

Bilateral multiple variable sizes cysts. Findings coded as BI-RADS III (Probably benign).

Which of the following is the most appropriate next step in management?

A. Annual screening
B. Needle core biopsy

C. Short interval follow-up imaging
D. Magnetic resonance mammography

Explanation:
The most appropriate next step is **C. Short interval follow-up imaging**.

BI-RADS III indicates **probably benign** findings, and short-term follow-up imaging (typically in 6 months) is the appropriate management.

Needle core biopsy (Option B) is not needed as the lesion is likely benign.

MRI (Option D) is not routinely needed in BI-RADS III cases.

MCQ 652:

A 32-year-old woman, 28 weeks pregnant, presents with an enlarging right breast mass that has been present for 4 months and is associated with mild pain. Physical examination confirmed a 3 x 2 cm firm, movable mass, not associated with skin changes or enlarged axillary lymph nodes.

Which of the following is the most appropriate next step in management?

A. Fine needle aspiration
B. Bilateral breast ultrasound
C. Bilateral diagnostic mammography
D. Reassure and reassess after delivery

Explanation:
The most appropriate next step in management is **B. Bilateral breast ultrasound**.

Bilateral breast ultrasound is the preferred initial imaging modality for pregnant women, as it is safe during pregnancy and helps to characterize the mass.

Fine needle aspiration (Option A) could be considered if the ultrasound findings are suspicious, but it is not the first step unless clinically indicated.

Bilateral diagnostic mammography (Option C) is typically avoided in pregnant women unless absolutely necessary.

Reassure and reassess after delivery (Option D) is inappropriate as the mass should be evaluated and appropriately managed during pregnancy to rule out malignancy.

MCQ 653:

A 60-year-old woman complains of left nipple discharge that is spontaneous, blood-stained, and recurrent for the past 6 months. Physical examination was normal.

Bilateral Breast Mammogram:

Normal.

Breast Ultrasound:

Bilateral ductal dilatation, with a 12 mm left intraductal lesion, suggestive of intraductal papilloma.

Which of the following is the most appropriate next step in management?

A. Discharge fluid cytology
B. Left central duct excision
C. Image-guided core biopsy
D. Short interval follow-up imaging

Explanation:
The most appropriate next step is **C. Image-guided core biopsy**.

The patient has a **bloody nipple discharge** with an intraductal lesion on ultrasound, most likely an **intraductal papilloma**, which should be biopsied to rule out malignancy and confirm the diagnosis.

Discharge fluid cytology (Option A) is not the best next step for an intraductal lesion.

Left central duct excision (Option B) may be considered later if the biopsy shows malignancy, but it’s not the immediate step.

Short interval follow-up imaging (Option D) would not be appropriate as the lesion is suspicious and warrants further investigation.

MCQ 654:

A 28-year-old woman presents with a neck mass that has been present for 1 year and is not associated with pressure symptoms. Physical examination confirmed diffuse thyroid enlargement.

Blood pressure 120/70 mmHg
Heart rate 90 /min
Temperature 36.7 °C

Test Result Normal Values

Thyroid-Stimulating Hormone: 0.001 (0.4 - 5.0 µU/mL)

Thyroxine (T4 free serum): 21 (8.5 - 15.2 pmol/L)

Ultrasound thyroid:

Enlarged both thyroid lobes, with a solid nodule on the left side measuring 1 x 2 cm and another nodule on the right side measuring 4 x 3 cm.

Which of the following is the most appropriate next step in management?

A. Thyroid scan
B. Total thyroidectomy
C. Fine needle aspiration of both nodules
D. Fine needle aspiration of the larger nodule

Explanation:
The most appropriate next step is **D. Fine needle aspiration of the larger nodule**.

The **TSH suppression** (low TSH and elevated T4) suggests a **toxic nodule**, and the larger nodule should be investigated further with **fine needle aspiration** (FNA) to rule out malignancy and evaluate its functional status.

Thyroid scan (Option A) is not indicated as FNA is the gold standard for evaluating nodules.

Total thyroidectomy (Option B) is not necessary without confirmation of malignancy or specific functional problems.

FNA of both nodules (Option C) is not required for both, as the larger nodule is the priority.

MCQ 655:

A 24-year-old man presented with neck pain for 4 months. Physical examination of the neck was normal.

Test Result Normal Values

Thyroid-Stimulating Hormone: 3.6 (0.4 - 5.0 µU/mL)

Thyroxine (T4 free serum): 11 (8.5 - 15.2 pmol/L)

Neck Ultrasound:

Normal size thyroid gland with a solitary solid left thyroid nodule measuring 9 x 7 mm. Benign-looking bilateral cervical lymph nodes.

Which of the following is the most appropriate management?

A. Thyroid scan
B. Follow-up imaging
C. Fine needle aspiration
D. Left thyroid lobectomy

Explanation:
The most appropriate management is **B. Follow-up imaging**.

The nodule is **small** and the ultrasound findings do not suggest malignancy. The patient has **benign-looking lymph nodes**, so **follow-up imaging** in 6 months is the best option.

Thyroid scan (Option A) is not necessary as the nodule is likely benign.

Fine needle aspiration (Option C) is unnecessary given the benign characteristics of the nodule.

Left thyroid lobectomy (Option D) is not needed unless malignancy is suspected, which is not the case here.

MCQ 656:

A 36-year-old patient presented with a neck mass that has been present for 3 years. Physical examination confirmed a palpable left thyroid gland with no palpable cervical lymph nodes.

Neck Ultrasound:

Left thyroid nodule occupying most of the left lobe, measuring 3.4 x 3.8 cm, with mixed cystic and solid components. The right lobe is normal.

Fine Needle Aspiration:

Finding coded as Bethesda IV (suspicious for follicular neoplasm).

Which of the following is the most appropriate management?

A. Thyroid scan
B. Total thyroidectomy

C. Left hemithyroidectomy
D. Repeat fine needle aspiration

Explanation:
The most appropriate management is **C. Left hemithyroidectomy**.

A **Bethesda IV** result indicates a **suspicious follicular neoplasm**, which requires surgical intervention. Given that the nodule is localized to the left thyroid lobe, a **left hemithyroidectomy** is the appropriate approach.

Thyroid scan (Option A) is not necessary since the nodule is already identified and suspicious for malignancy.

Total thyroidectomy (Option B) is typically reserved for cases where there is a high suspicion of malignancy or if the disease involves both lobes.

Repeat fine needle aspiration (Option D) is not required as the diagnosis has already been made with the current biopsy.

MCQ 657:

A 29-year-old patient underwent a right hemithyroidectomy for a symptomatic right thyroid nodule.

Histopathology:

A 3 x 4 cm colloid nodule with a focal well-differentiated papillary carcinoma measured 8 mm away from the main lesion. There was no extrathyroidal extension or lymphovascular invasion.

Which of the following is the most appropriate management?

A. Follow-up
B. Whole body iodine scan
C. Radioactive iodine ablation
D. Completion of thyroidectomy

Explanation:
The most appropriate management is **D. Completion of thyroidectomy**.

The presence of **papillary carcinoma** (albeit small) within the thyroid suggests that **completion of thyroidectomy** should be performed to remove the remaining thyroid tissue and prevent recurrence.

Follow-up (Option A) is not sufficient since the patient has malignancy that requires definitive treatment.

Whole body iodine scan (Option B) and **radioactive iodine ablation** (Option C) are used for certain cases of papillary carcinoma, but the most appropriate first step is to complete the thyroidectomy.

MCQ 658:

A 23-year-old woman presented with symptoms of hyperthyroidism. Physical examination confirmed a non-palpable thyroid gland, and bilateral exophthalmos was noted. She has been on anti-thyroid medication for 10 months.

Test Results Normal Values:

Thyroid-Stimulating Hormone (TSH): 0.001 (0.4 - 5.0 µU/mL)

Thyroxine (T4 free serum): 29 (8.5 - 15.2 pmol/L)

Neck Ultrasound:

Mild diffuse enlargement of the thyroid gland.

Thyroid scan:

Diffuse uptake of the thyroid gland consistent with toxic multi-nodular goiter.

Which of the following is the most appropriate management?

A. Subtotal thyroidectomy
B. Near total thyroidectomy
C. Radioactive iodine ablation
D. Continue on medical therapy

Explanation:
The most appropriate management is **C. Radioactive iodine ablation**.

The patient has **toxic multi-nodular goiter** and **long-standing hyperthyroidism**. Radioactive iodine ablation is a common and effective treatment for this condition.

Subtotal thyroidectomy (Option A) and **near total thyroidectomy** (Option B) may be considered in some cases, but radioactive iodine therapy is preferred for its minimal invasiveness.

Continue on medical therapy (Option D) is not ideal for long-term management, especially since the patient has persistent symptoms despite anti-thyroid medication.

MCQ 659:

A morbidly obese 35-year-old with no relevant past medical history consults for weight reduction.

Which of the following is the most useful to determine the most appropriate procedure?

A. Barium swallow
B. Upper GI endoscopy
C. CT scan of abdomen
D. Ultrasound abdomen

Explanation:
The most useful investigation is **B. Upper GI endoscopy**.

Upper GI endoscopy (Option B) is typically performed to assess for any **gastrointestinal pathology** that might impact bariatric surgery, such as reflux disease or peptic ulcers. It provides a comprehensive look at the upper GI tract and is helpful in evaluating the patient's suitability for surgical procedures.

Barium swallow (Option A) and **CT scan of abdomen** (Option C) are not routinely used to assess bariatric surgery candidacy.

Ultrasound abdomen (Option D) may be useful for evaluating the liver and gallbladder but is not the primary investigation for planning weight loss surgery.

MCQ 660:

A morbidly obese 23-year-old man complains of gastroesophageal reflux disease (GERD).

Upper GI endoscopy:

Large hiatus hernia with grade III reflux.

Which of the following is the most appropriate weight reduction procedure?

A. Sleeve gastrectomy
B. Intra-gastric balloon
C. Billo-pancreatic bypass
D. Roux-en-y gastric bypass

Explanation:
The most appropriate procedure is **D. Roux-en-Y gastric bypass**.

Roux-en-Y gastric bypass (RYGB) is often preferred for morbidly obese patients with **GERD** and **hiatal hernia**. It not only helps with weight loss but also has a beneficial effect on reflux by creating a smaller gastric pouch and bypassing the duodenum.

Sleeve gastrectomy (Option A) may also help with weight loss but can worsen GERD due to the increased intra-abdominal pressure.

Intra-gastric balloon (Option B) is less effective for long-term weight loss in morbid obesity and does not address the reflux symptoms.

Bilio-pancreatic bypass (Option C) is typically not recommended unless there is significant metabolic disease such as diabetes.

MCQ 661:

A 26-year-old man presented 3 days after a sleeve gastrectomy with left upper abdominal pain. Physical examination confirmed only mild upper abdominal tenderness with no sign of peritonitis.

Blood pressure: 100/70 mmHg
Heart rate: 116 /min

Which of the following is the most likely explanation?

A. Dehydration
B. Gastric leak
C. Pulmonary embolism
D. Inadequate analgesia

Explanation:
The most likely explanation is **A. Dehydration**.

After **sleeve gastrectomy**, dehydration is common, especially if the patient is not consuming enough fluids due to postoperative pain or reduced gastric capacity. Mild tenderness and a normal examination, with a relatively elevated heart rate and low blood pressure, suggest dehydration.

Gastric leak (Option B) is a serious complication, but it would likely present with signs of peritonitis or more severe symptoms.

Pulmonary embolism (Option C) is less likely in the absence of respiratory symptoms or significant risk factors.

Inadequate analgesia (Option D) is possible but would typically cause pain rather than the hemodynamic changes observed.

MCQ 662:

A 46-year-old man underwent an ERCP for biliary obstruction; the procedure was prolonged and difficult. 4 hours later, he developed extensive surgical emphysema of the abdomen, chest, and neck.

Where is the most likely injury?

A. Gastric
B. Tracheal
C. Duodenal
D. Esophageal

Explanation:
The most likely injury is **D. Esophageal**.

Surgical emphysema in the abdomen, chest, and neck after ERCP is typically caused by **esophageal perforation**. During ERCP, especially if it's difficult or prolonged, the esophagus can be perforated, leading to air leakage into the surrounding tissues.

Gastric (Option A), **tracheal** (Option B), and **duodenal** (Option C) injuries are less likely to cause the widespread emphysema observed here.

MCQ 663:

A 22-year-old man underwent an open left inguinal hernia mesh repair 2 weeks ago, and presented with continuous discharge from the surgical wound. Examination confirmed a 2 x 3 cm pus-discharging wound with part of the mesh exposed.

Which of the following is the most appropriate management?

A. Systemic antibiotic
B. Wound debridement
C. Wound drainage and mesh removal
D. Open entire wound for daily dressing

Explanation:
The most appropriate management is **C. Wound drainage and mesh removal**.

The presence of **pus discharge** and **exposure of mesh** suggests an **infection** involving the mesh, which may require removal of the infected mesh, drainage of the wound, and appropriate antibiotic therapy.

Systemic antibiotics (Option A) alone will not address the source of infection, which is the exposed mesh.

Wound debridement (Option B) may be necessary, but removal of the mesh is the most critical step in this case.

Opening the wound for daily dressing (Option D) is not an appropriate first-line intervention for mesh infection.

MCQ 664:

Which of the following is the most common cause of biliary colic?

A. Cystic duct stone
B. Gall bladder stone
C. Gall bladder sludge
D. Choledocholithiasis

Explanation:
The most common cause is **B. Gall bladder stone**.

Gall bladder stones are the most common cause of **biliary colic** as they can intermittently obstruct the cystic duct, causing pain.

Cystic duct stones (Option A) can also cause biliary colic but are less common.

Gall bladder sludge (Option C) is a precursor to gallstones but does not usually cause symptoms unless it forms stones.

Choledocholithiasis (Option D), which refers to stones in the common bile duct, typically causes more severe symptoms, such as jaundice, and is less likely to be the cause of typical biliary colic.

MCQ 665:

Which of the following is most commonly associated with biliary colic?

A. Bilirubin
B. Amylase
C. Cholecystokinase
D. Alkaline phosphatase

Explanation:
The most commonly associated enzyme is **D. Alkaline phosphatase**.

Alkaline phosphatase (ALP) is often elevated in cases of biliary obstruction or gallbladder disease, including **biliary colic**, which typically arises from obstruction of the cystic duct due to gallstones.

Bilirubin (Option A) is elevated in jaundice, but it's not specifically associated with biliary colic.

Amylase (Option B) is related to **pancreatitis** and is less commonly associated with biliary colic.

Cholecystokinase (Option C) is a hormone involved in gallbladder contraction and digestive processes but is not used as a diagnostic marker for biliary colic.

MCQ 666:

A 22-year-old woman presented 1-week after a laparoscopic appendectomy with mild abdominal pain not associated with nausea or vomiting, she reports passing infrequent loose stool. Physical examination confirmed a soft abdomen with mild tenderness at the wound site.

CT scan of abdomen:

2 x 2 cm retrocaccal collection.

Which of the following is the most appropriate next step in management?

A. Percutaneous drainage
B. Open surgical drainage
C. Diagnostic laparoscopy
D. Conservative management

Explanation:
The most appropriate next step is **A. Percutaneous drainage**.

The patient has a **postoperative retrocaccal collection**, which suggests the possibility of a **subclinical abscess** following appendectomy.

Percutaneous drainage is the preferred initial treatment for localized abdominal collections and abscesses.

Open surgical drainage (Option B) is typically reserved for cases with large or unresponsive abscesses.

Diagnostic laparoscopy (Option C) may not be needed if the abscess is confirmed on imaging.

Conservative management (Option D) might be appropriate for small collections, but drainage is usually the first-line approach.

MCQ 667:

A 52-year-old complains of left lower abdominal pain associated with passing loose stool and a loss of appetite. Examination confirmed left iliac fossa guarding and rebound tenderness.

Blood pressure 110/70 mmHg
Heart rate 98 /min
Test Result Normal Value
WBC 19.4 (5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Diverticulitis
B. Pyelonephritis
C. Sigmoid malignancy
D. Ruptured ovarian cyst

Explanation:
The most likely diagnosis is **A. Diverticulitis**.

The patient has **left lower quadrant abdominal pain**, **rebound tenderness**, and **elevated white blood cells**, which are classic signs of **diverticulitis**—inflammation or infection of diverticula in the colon, particularly in the sigmoid colon.

Pyelonephritis (Option B) would usually present with fever and flank pain, and it is less likely in this scenario.

Sigmoid malignancy (Option C) could cause similar symptoms but is less likely to present with such an acute inflammatory process.

Ruptured ovarian cyst (Option D) typically presents with pelvic pain, and is less likely to cause rebound tenderness and elevated white blood cells in the setting of this presentation.

MCQ 668:

What structure passes through the deep inguinal ring?

A. Ilioinguinal nerve
B. Inferior epigastric artery
C. Medial umbilical ligament
D. Round ligament of the uterus

Explanation:
The structure that passes through the deep inguinal ring is **D. Round ligament of the uterus**.

The **round ligament of the uterus** passes through the deep inguinal ring in females, contributing to the formation of the inguinal canal.

Ilioinguinal nerve (Option A) does pass through the inguinal canal, but it enters the canal at a different point.

Inferior epigastric artery (Option B) is located outside the inguinal canal and runs superior to the deep inguinal ring.

Medial umbilical ligament (Option C) is located in the abdominal wall and is not involved in the inguinal canal.

MCQ 669:

A 36-year-old patient presented with a left neck mass. Examination confirmed a 2 x 2 cm mass at the posterior triangle just below the angle of the mandible. Ultrasound of the neck showed a normal thyroid gland, large left cervical lymph node with cystic changes. Fine needle aspiration: The entire smear consists of follicular thyroid cells. Which of the following is the most likely diagnosis?

A. Thyroglossal cyst
B. Ectopic thyroid gland
C. Apparent thyroid gland
D. Metastatic thyroid cancer

Explanation:
The most likely diagnosis is **D. Metastatic thyroid cancer**.

The presence of **follicular thyroid cells** on fine needle aspiration (FNA) from a cervical lymph node suggests a **metastatic thyroid cancer**, specifically **papillary** or **follicular thyroid carcinoma**.

Thyroglossal cyst (Option A) typically presents as a midline neck mass, not at the angle of the mandible.

Ectopic thyroid gland (Option B) usually presents in a different location, often in the midline or base of the tongue.

Apparent thyroid gland (Option C) is not a common clinical term and seems to be an incorrect choice.

MCQ 670:

A 45-year-old patient complains of frequent anal itching, and post-defecation pain. Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Perianal hematoma
D. Internal hemorrhoids

Explanation:
The most likely diagnosis is **A. Anal fissure**.

Anal fissure causes **pain** during and after defecation, especially when passing hard stools. The **itching** can also occur due to irritation. The **pain** is typically sharp and localized.

Perianal fistula (Option B) would typically cause recurrent drainage, but not significant itching or pain after defecation.

Perianal hematoma (Option C) would cause a localized lump and **pain**, not the typical anal itching.

Internal hemorrhoids (Option D) typically cause **rectal bleeding** and discomfort, but pain is less pronounced unless there is thrombosis.

MCQ 671:

A 50-year-old man underwent an open mesh repair of a left inguinal hernia. He presented 2 months later because he noticed that the left testes had become smaller in size. Which of the following is the most likely cause?

A. Ligation of testicular artery
B. Mesh migration to the scrotum
C. Thrombosis of pampiniform plexus
D. Tightening of external inguinal ring

Explanation:
The most likely cause is **A. Ligation of testicular artery**.

Ligation of the testicular artery during hernia repair can lead to **testicular atrophy**, as it compromises the blood supply to the testicle.

Mesh migration to the scrotum (Option B) is uncommon but could cause irritation, not atrophy.

Thrombosis of the pampiniform plexus (Option C) could lead to **varicocele** but not testicular atrophy.

Tightening of external inguinal ring (Option D) could affect blood flow, but **ligation of the testicular artery** is a more likely cause.

MCQ 672:

A 30-year-old underwent a laparoscopic appendectomy 2 weeks ago. Histology: Carcinoid tumor at the tip of the appendix measuring 0.5 cm. Which of the following is the most appropriate management?

A. Colonoscopy
B. No further action needed
C. Limited right hemicolectomy
D. CT scan of chest abdomen and pelvis

Explanation:
The most appropriate management is **B. No further action needed**.

Appendiceal carcinoids smaller than **2 cm** typically do not require further treatment beyond appendectomy if the tumor is confined to the appendix and has no evidence of metastasis or vascular invasion.

Colonoscopy (Option A) is not needed unless there is suspicion of involvement of other parts of the colon.

Limited right hemicolectomy (Option C) is usually reserved for larger or more invasive appendiceal carcinoid tumors.

CT scan of chest, abdomen, and pelvis (Option D) might be needed if there is concern for metastasis, but for tumors under 2 cm and with no other concerning features, no further imaging is typically required.

MCQ 673:

A 38-year-old man presented with a 4-hour history of sudden severe abdominal pain. He reports frequent use of non-steroidal anti-inflammatory drugs for joint pain. Examination confirmed generalized abdominal tenderness and guarding with absent bowel sounds. Temperature 37.8 °C. Test Result: WBC 13 (Normal: 4.5-10.5 x 10^9/L).

Which of the following is the most appropriate initial imaging?

A. Erect chest X-ray
B. CT scan abdomen
C. Ultrasound abdomen
D. Supine abdominal X-ray

Explanation:
The most appropriate initial imaging is **B. CT scan abdomen**.

The patient has signs of **acute abdomen**, possibly related to a **perforated viscera** or **acute abdominal pathology**, which can be best evaluated with a **CT scan**. CT is the gold standard for evaluating conditions like **perforated peptic ulcer**, **diverticulitis**, or **bowel obstruction**.

Erect chest X-ray (A) may show free air if there's perforation, but a **CT scan** provides more detailed information about the underlying pathology.

Ultrasound abdomen (C) is more useful for liver or biliary problems but not the first choice for suspected perforation or generalized abdominal issues.

Supine abdominal X-ray (D) is useful for detecting obstruction or free air but less detailed than a CT scan for this situation.

MCQ 674:

A 47-year-old man presented with a 6-month history of right groin swelling. The swelling recently increased in size. Examination confirmed an inguino-scrotal swelling with positive cough impulse and separable from the testis.

Which of the following is the most likely diagnosis?

A. Spermatocele
B. Epididymal cyst
C. Vaginal hydrocele
D. Indirect inguinal hernia

Explanation:
The most likely diagnosis is **D. Indirect inguinal hernia**.

The **inguino-scrotal** swelling with a **positive cough impulse** and separation from the testis is characteristic of an **indirect inguinal hernia**.

Spermatocele (A) would be a **painless**, fluid-filled mass in the **epididymis**, not connected to the inguinal canal.

Epididymal cyst (B) also presents as a painless mass, but it's not **inguino-scrotal** in location.

Vaginal hydrocele (C) is a fluid collection around the testicle but is typically not separated from the testis, and it does not have a **cough impulse**.

MCQ 675:

A 38-year-old patient developed a painful swelling in the right lower quadrant of his abdomen after moving a heavy object. Examination confirmed a localized tender swelling in the right lower abdominal quadrant with negative cough impulse; it persisted even with contraction of abdominal muscles.

Which of the following is the most likely diagnosis?

A. Ventral hernia
B. Appendicular mass
C. Extraperitoneal lipoma
D. Rectus sheath hematoma

Explanation:
The most likely diagnosis is **D. Rectus sheath hematoma**.

The **painful swelling** in the **right lower quadrant** with **negative cough impulse** and associated with **recent trauma** (moving a heavy object) suggests a **rectus sheath hematoma**, a condition where blood collects in the sheath of the **rectus abdominis** muscle.

Ventral hernia (A) would cause a swelling with a **positive cough impulse**, and the swelling would be more palpable on cough.

Appendicular mass (B) could present with tenderness, but it typically doesn't cause localized swelling outside of the abdomen.

Extraperitoneal lipoma (C) would present as a painless, soft mass, not associated with **painful swelling** and **muscle contraction**.

MCQ 676:

A 60-year-old patient presented with a 6-month history of a dull achey abdominal swelling associated with weight loss. Examination confirmed a 20x20 cm firm, non-tender, intra-abdominal mass.

Abdominal CT scan:
Lobulated soft tissue mass arising from retro-peritoneum, with multiple hypodense liver lesions.

Which of the following is the most likely diagnosis?

A. Liposarcoma
B. Lymphosarcoma
C. Germ cell tumour
D. Peripheral nerve tumour

Explanation:
The most likely diagnosis is **A. Liposarcoma**.

Liposarcoma is a **malignant soft tissue tumour** that arises from **fat cells** in the **retroperitoneum** and can present with an **intra-abdominal mass**. It is commonly associated with **hypodense liver lesions** due to metastasis.

Lymphosarcoma (B) may involve the retroperitoneum but typically involves **lymph nodes** rather than a solitary mass.

Germ cell tumour (C) is rare in the retroperitoneum, and typically presents in younger individuals.

Peripheral nerve tumour (D) would present with neurological symptoms and would not typically present as a large abdominal mass with liver metastasis.

MCQ 677:

Which of following is associated with the highest risk of malignant transformation?

A. Villous adenoma
B. Tubular adenoma
C. Hyperplastic polyp
D. Tubulovillous adenoma

Explanation:
The answer is **A. Villous adenoma**.

Villous adenomas are known for their **high malignant potential**. They have a higher risk of becoming **colorectal cancer** compared to other types of adenomas.

Tubular adenoma (B) has a lower risk of malignancy but still requires monitoring for potential transformation.

Hyperplastic polyps (C) are generally **benign** and have little to no risk of malignant transformation.

Tubulovillous adenoma (D) has a higher risk of malignancy than tubular adenomas but lower than villous adenomas.

MCQ 678:

A 30-year-old presented with intermittent perianal pain associated with purulent discharge. Examination confirmed a low-lying perianal fistula with an external opening posteriorly close to the anal verge.

Which of the following is the most appropriate management?

A. MRI pelvis
B. Fistulogram
C. Fistulotomy
D. Lateral sphincterotomy

Explanation:
The most appropriate management is **C. Fistulotomy**.

A **fistulotomy** is the standard treatment for **low-lying perianal fistulas**. It involves cutting open the fistula tract to allow healing. This is effective in most cases, especially when the external opening is near the anal verge.

MRI pelvis (A) is useful for high anal fistulas to evaluate the sphincter involvement but not the first step for low-lying fistulas.

Fistulogram (B) is also used to visualize the fistula tract, but it is less commonly performed now due to the availability of MRI and clinical examination.

Lateral sphincterotomy (D) is used in cases of **anal fissures** or **high anal fistulas** but not typically for low perianal fistulas.

MCQ 679:

A 30-year-old presented with a 3-day history of painful perianal swelling. Examination confirmed a 1x1 cm tender swelling close to the anal verge.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Prolapsed pile
C. Perianal abscess
D. Perianal hematoma

Explanation:
The most likely diagnosis is **C. Perianal abscess**.

A **perianal abscess** presents with a **tender swelling** near the **anal verge** and often occurs after infection in the anal glands.

Anal fissure (A) usually presents with **pain during defecation** and bleeding, but not a swelling or abscess.

Prolapsed pile (B) is typically a **painless swelling** that is related to hemorrhoids.

Perianal hematoma (D) could present with a **swelling**, but it is typically related to **painful swelling** after straining or injury and may not have the classic signs of infection.

MCQ 680:

A 14-year-old patient presented with a painless neck swelling. Examination confirmed a midline neck swelling below the hyoid bone that moves up and down with swallowing.

Which of the following is the most likely diagnosis?

A. Cystic hygroma
B. Thyroglossal cyst

C. Midline dermoid cyst
D. Submental lymphadenopathy

Explanation:
The most likely diagnosis is **B. Thyroglossal cyst**.

A **thyroglossal cyst** is a **midline neck swelling** that is **movable with swallowing** and typically occurs **below the hyoid bone**. It results from the remnant of the **thyroglossal duct** during embryological development.

Cystic hygroma (A) typically presents as a **soft, cystic mass** in the **posterior triangle** of the neck, not in the midline.

Midline dermoid cyst (C) is usually located along the midline but would typically present with **a firm, non-mobile mass** and not associated with swallowing.

Submental lymphadenopathy (D) usually presents as a **firm, immobile** mass and is not associated with movement during swallowing.

MCQ 681:

A 42-year-old man complains of chest pain after being involved in a motor vehicle accident. Examination confirmed a patent airway, equal air entry on both sides of the chest, and distended neck veins. Blood pressure 88/45 mmHg, Heart rate 110 /min, Respiratory rate 30 /min.

Which of the following is the most likely diagnosis?

A. Flail chest
B. Cardiac tamponade
C. Tension pneumothorax
D. Simple pneumothorax

Explanation:
The most likely diagnosis is **B. Cardiac tamponade**.

Cardiac tamponade can occur after chest trauma and presents with **distended neck veins**, **hypotension**, and **tachycardia**, all seen in this patient. The **distended neck veins** suggest pressure on the heart, preventing proper filling.

Flail chest (A) presents with **paradoxical movement** of the chest wall but is not typically associated with distended neck veins or hypotension unless there is significant associated trauma.

Tension pneumothorax (C) typically presents with **tracheal deviation** and **absent breath sounds** on one side, but this patient has equal air entry.

Simple pneumothorax (D) would typically present with **unilateral breath sounds** and not the signs of shock or distended neck veins.

MCQ 682:

A 25-year-old was involved in a head-on motor vehicle accident. On examination, he was unresponsive, had distended neck veins, absent breath sounds, and hyper-resonant percussion on the right side. His GCS score was 8.

Which of the following is the most urgent initial management?

A. Pericardiocentesis
B. Neck immobilization
C. Endotracheal intubation
D. Needle right thoracocentesis

Explanation:
The most urgent initial management is **D. Needle right thoracocentesis**.

The clinical signs, including **hyper-resonant percussion** and **absent breath sounds**, suggest a **tension pneumothorax**. This is a life-threatening emergency that requires **needle decompression** to relieve pressure on the lung and heart.

Pericardiocentesis (A) is indicated if **cardiac tamponade** is suspected, but the signs here suggest pneumothorax.

Neck immobilization (B) is important for spinal injuries, but airway management and decompressing the pneumothorax take precedence.

Endotracheal intubation (C) is important if the airway is compromised, but **needle thoracocentesis** takes priority to relieve the life-threatening tension pneumothorax.

MCQ 683:

A 25-year-old man was brought to the Emergency Department after being involved in a motor vehicle accident. He opened his eyes spontaneously and responded appropriately to verbal commands. His respiration was shallow, and he had left chest wall contusion. He was able to

shrug his shoulders but unable to move his elbows or lower limbs. Blood pressure 80/40 mmHg, Heart rate 70 /min, Respiratory rate 30 /min.

Which of the following is the most likely cause of the hypotension?

A. Cardiac tamponade
B. Internal hemorrhage
C. High spinal cord injury
D. Tension pneumothorax

Explanation:
The most likely cause of the hypotension is **C. High spinal cord injury**.

This patient's **inability to move below the shoulders** and **hypotension** suggests **high spinal cord injury**, particularly in the cervical region, which can impair sympathetic nervous system function and lead to **neurogenic shock**.

Cardiac tamponade (A) could cause hypotension, but it would usually present with **distended neck veins** and **muffled heart sounds**, which are not described here.

Internal hemorrhage (B) is a possible cause but is less likely without signs of active bleeding.

Tension pneumothorax (D) would typically present with **tracheal deviation** and **absent breath sounds**, which are not mentioned.

MCQ 684:

A 50-year-old diabetic man presented with a 5-year history of an unhealed foot ulcer. Physical examination confirmed a 2x2 cm irregular-shaped ulcer at the dorsal aspect of the first toe with pearly white discoloration. Biopsy of the ulcer reported as pseudoepitheliomatous hyperplasia.

Which of the following is the most appropriate management?

A. Repeat biopsy
B. First toe amputation
C. Surgical debridement
D. Negative pressure dressing

Explanation:
The most appropriate management is **C. Surgical debridement**.

Pseudoepitheliomatous hyperplasia suggests **chronic inflammation** that can occur in diabetic ulcers. **Surgical debridement** is typically the first step to remove non-viable tissue and promote wound healing.

Repeat biopsy (A) is not necessary unless malignancy is suspected.

First toe amputation (B) would only be indicated if there were significant **infection** or **necrosis**, which is not the case here.

Negative pressure dressing (D) could be useful in wound management, but **debridement** is the first necessary step to manage the ulcer.

MCQ 686:

A 32-year-old man had excision and primary closure of a skin contracture that was 5x3 cm on the forearm, 1 year ago. He presented with an overgrowth at the scar site that extends beyond the surgical scar margins, it started a few weeks after the surgery and gradually increased in size.

Which of the following most accurately describes the lesion?

A. Keloid
B. Scar fibrosis
C. Desmoid tumor
D. Scar hypertrophy

Explanation:
The most accurate description of this lesion is **A. Keloid**.

Keloids are characterized by excessive **scar tissue** that grows beyond the original wound boundaries and can continue to grow even after healing.

Scar fibrosis (B) refers to the formation of scar tissue but does not involve the overgrowth beyond the scar.

Desmoid tumors (C) are rare, benign tumors arising from fibrous tissue, typically in the abdominal wall or extremities, but they are less common in post-surgical scars.

Scar hypertrophy (D) is similar to keloid but does not extend beyond the wound margin and typically regresses over time.

MCQ 687:

A 58-year-old woman presented with a right thigh mass that measured 5 x 6 cm (see report).

Core biopsy: High-grade sarcoma.

Which of the following is the most appropriate staging image?

A. Bone scan
B. MRI abdomen
C. Chest CT scan
D. Right thigh plain X-ray

Explanation:
The most appropriate staging image is **C. Chest CT scan**.

Chest CT scan is essential in sarcoma staging to evaluate for potential **pulmonary metastasis**, which is the most common site for sarcoma spread.

Bone scan (A) is useful for detecting bone metastases but not routine for initial staging of soft tissue sarcomas.

MRI abdomen (B) is not the primary imaging choice for staging sarcomas unless there are indications of abdominal involvement.

Right thigh plain X-ray (D) may be helpful for identifying bone involvement but does not give the comprehensive staging information needed for soft tissue sarcoma.

MCQ 688

Which of the following is the most appropriate biopsy for a limb sarcoma?

A. Incisional
B. Excisional
C. Core needle
D. Fine needle aspiration

Explanation:
The most appropriate biopsy is **C. Core needle**.

Core needle biopsy (C) is the most accurate method for diagnosing sarcomas, as it allows for a large enough tissue sample to assess tumor architecture and grade.

Incisional biopsy (A) involves removing part of the mass and is typically used when the mass is too large to remove entirely or when the diagnosis is uncertain.

Excisional biopsy (B) would be inappropriate as it could lead to the tumor's seeding and also would be premature before confirming the diagnosis.

Fine needle aspiration (D) can be used in some cases but is generally less accurate than core needle biopsy for sarcomas because it doesn't provide enough tissue for histological assessment of tumor type.

MCQ:689;

A 36-year-old man presents with a 2-day history of severe right upper quadrant abdominal pain after consuming a fatty meal. The pain radiates to the right shoulder and is associated with nausea and vomiting. On examination, he has a normal temperature, and there is tenderness over the right upper abdomen. Ultrasound shows multiple gallstones, and the gallbladder wall appears normal. What is the most appropriate next step in management?

A. Observation
B. Ursodeoxycholic acid
C. Laparoscopic cholecystectomy
D. Magnetic Resonance Cholangiopancreatography (MRCP)

Explanation:
The patient is presenting with symptoms consistent with **biliary colic** (intermittent pain due to gallstones). The absence of any acute signs (e.g., fever or peritoneal signs) and the fact that the gallbladder wall is normal on ultrasound suggests no acute cholecystitis. In patients with symptomatic gallstones, the definitive treatment is **laparoscopic cholecystectomy** to prevent further episodes.

Cholelithiasis (Gallstones)

Definition:
The presence of gallstones in the gallbladder, often leading to symptoms such as biliary colic, and in some cases, complications like cholecystitis, pancreatitis, or cholangitis.

Key Clinical Presentation:

Intermittent right upper quadrant pain after meals (especially fatty foods).

Nausea and vomiting associated with the pain.

No fever or jaundice in uncomplicated cases.

Investigations:

Best initial: Abdominal ultrasound (shows gallstones and gallbladder wall).

Confirmatory: MRCP (for suspected complications such as common bile duct stones).

Treatment Outline:

First-line: **Laparoscopic cholecystectomy** (definitive management in symptomatic patients).

Second-line: Observation and management of symptoms if surgery is contraindicated.

Definitive: Cholecystectomy to prevent recurrent biliary colic or complications.

Correct Answer:
C. Laparoscopic cholecystectomy ✅

MCQ 690:

A 44-year-old man presented with a 6-month history of vague central abdominal pain.
CT scan of abdomen:
Presence of 20x10 cm retroperitoneal mass. No enlarged mesenteric or para-aortic lymph nodes noted.
Which of the following is the most likely diagnosis?

A. Lymphoma
B. Liposarcoma
C. Germ cell tumor
D. Metastatic testicular tumor

Explanation:
The patient has a large retroperitoneal mass without any associated lymphadenopathy. **Liposarcoma**, a malignant tumor of adipose tissue, is the most likely diagnosis in this scenario. It commonly occurs in the retroperitoneum and is often characterized by large, well-defined masses on imaging. Other options like lymphoma or metastasis typically present with enlarged lymph nodes.

Retroperitoneal Masses

Definition:
A mass located in the retroperitoneal space, which is the anatomical area behind the peritoneum, includes various types of tumors such as liposarcomas, lymphoma, and metastatic cancers.

Key Clinical Presentation:

Non-specific abdominal pain.

Large mass, often with vague symptoms or weight loss.

May be asymptomatic for a long period.

Investigations:

Best initial: CT scan (provides detailed imaging of the retroperitoneal mass).

Confirmatory: Biopsy for histopathological confirmation.

Treatment Outline:

First-line: Surgical resection if feasible.

Second-line: Chemotherapy and radiotherapy for malignant tumors.

Definitive: Surgical excision with histological analysis.

Correct Answer:
B. Liposarcoma ✅

MCQ 691:

A 38-year-old multiparous woman presented with a 3-year history of abdominal bulging. Physical examination confirmed a diffuse smooth midline bulge on attempting to lean forward, with negative cough impulse (see report).
CT scan of abdomen:
No intra or extra-abdominal mass lesion, and abdominal fascia is intact with no visible defect.
Which of the following is the most likely diagnosis?

A. Spigelian hernia
B. Incisional hernia
C. Diastasis recti
D. Transversalis muscle weakness

Explanation:
The patient has a midline bulge with no defect or hernia on imaging. **Diastasis recti**, a condition where the rectus abdominis muscles separate, is the most likely diagnosis. This condition is common in multiparous women and can cause a bulging in the midline, especially when attempting activities that increase intra-abdominal pressure.

Diastasis Recti

Definition:
A condition where the two sides of the rectus abdominis muscle are separated, often resulting in a bulging midline.

Key Clinical Presentation:

Bulging along the midline of the abdomen.

No palpable hernia or defect.

Worsens with activities that increase intra-abdominal pressure (e.g., sitting up, bending forward).

Investigations:

Best initial: Physical examination (often visible bulge).

Confirmatory: Imaging, but diagnosis is typically clinical.

Treatment Outline:

First-line: Physical therapy, core strengthening exercises.

Second-line: Surgical correction if symptomatic or persistent.

Correct Answer:
C. Diastasis recti ✅

MCQ 692:

A 29-year-old man underwent left open inguinal hernia mesh repair 3 weeks ago. He presents today with continuous intractable groin pain radiating to the left upper thigh and associated with persistent hyperesthesia. Physical examination excluded any recurrence or surgical site infection.
Which of the following is the most appropriate treatment?

A. Nerve block
B. Mesh staples removal
C. Systemic anti-inflammatory
D. Neuroctomy and mesh removal

Explanation:
The patient likely has **postoperative neuralgia**, a common complication following inguinal hernia surgery. The most appropriate management is a **nerve block**, which can help alleviate pain without the need for invasive procedures. Removal of mesh staples or nerve excision would be excessive and unnecessary.

Postoperative Complications

Definition:
Postoperative complications following inguinal hernia repair can include pain due to nerve irritation or entrapment, known as postoperative neuralgia.

Key Clinical Presentation:

Persistent pain following surgery.

Radiating pain to the thigh.

Hyperesthesia over the groin area.

Investigations:

Best initial: Clinical examination.

Confirmatory: Nerve block or imaging if necessary.

Treatment Outline:

First-line: Nerve block and conservative pain management.

Second-line: Surgery if nerve entrapment is confirmed.

Correct Answer:
A. Nerve block ✅

MCQ 693:

A 35-year-old man post total thyroidectomy for multinodular goiter presented with persistent hypocalcemia despite adequate replacement with oral and intravenous calcium.
Which of the following is the most appropriate next step in management?

A. Start diuretics
B. Start recombinant PTH
C. Check serum magnesium level
D. IV bolus of potassium chloride

Explanation:
Persistent hypocalcemia after thyroidectomy may be due to **hypomagnesemia**, which impairs the release of parathyroid hormone (PTH). The most appropriate next step is to **check serum magnesium levels** and correct it if necessary, as magnesium deficiency can cause resistance to calcium supplementation.

Hypocalcemia after Thyroidectomy

Definition:
Hypocalcemia following thyroidectomy is often due to accidental damage or removal of the parathyroid glands.

Key Clinical Presentation:

Symptoms of hypocalcemia (e.g., numbness, tetany, seizures).

Persistent low calcium despite supplementation.

Investigations:

Best initial: Serum calcium and magnesium levels.

Confirmatory: Serum magnesium level if hypocalcemia persists.

Treatment Outline:

First-line: Magnesium replacement.

Second-line: Adjust calcium supplementation after magnesium correction.

Correct Answer:
C. Check serum magnesium level ✅

MCQ 694:

A 40-year-old woman underwent major bowel resection and developed high-output fistula. Which of the following is the most expected consequence?

A. Fast loss of body fat
B. Decrease total body weight
C. Slow loss of lean body mass
D. Expansion of the extracellular fluid

Explanation:
In the setting of high-output fistula, fluid loss and malnutrition occur primarily due to protein and electrolyte loss, which leads to **decrease in lean body mass**. This occurs over time as the body loses proteins and muscle mass despite caloric intake. Body fat may be lost as well, but the main consequence of a high-output fistula is the loss of lean body mass.

Postoperative Nutrition and Fluid Loss

Definition:
High-output fistula is a complication of gastrointestinal surgery where excessive fluid loss occurs through the fistula, leading to malnutrition and dehydration.

Key Clinical Presentation:

Persistent abdominal fluid loss.

Weight loss with signs of malnutrition.

Electrolyte imbalances and dehydration.

Investigations:

Best initial: Serum albumin, electrolytes, and fluid balance.

Confirmatory: Nutritional assessments, stool analysis.

Treatment Outline:

First-line: Parenteral nutrition, fluid and electrolyte management.

Second-line: Surgical repair of the fistula if possible.

Correct Answer:
C. Slow loss of lean body mass ✅

MCQ 695:

An immunocompromised patient presented with an acute painful perineal swelling.
Examination confirmed erythematous fluctuant area with crepitus over it and a foul-smelling discharge.
Which of the following is the most appropriate management?

A. Penicillin G infusion
B. Surgical debridement
C. Aspiration of collection
D. Topical polymycin ointment

Explanation:
The description of a painful, erythematous, fluctuant perineal swelling with crepitus and foul-smelling discharge in an immunocompromised patient suggests **necrotizing fasciitis** or a severe soft tissue infection. **Surgical debridement** is the most appropriate treatment for such infections to remove necrotic tissue and control the infection. Antibiotics like Penicillin G are also used, but surgical intervention is crucial in managing the infection.

Necrotizing Fasciitis (Soft Tissue Infection)

Definition:
Severe, life-threatening soft tissue infection, often caused by mixed bacterial flora, including anaerobes and gram-positive organisms.

Key Clinical Presentation:

Rapidly progressing pain, swelling, erythema, and crepitus.

Foul-smelling discharge, systemic signs of sepsis.

Often occurs in immunocompromised patients.

Investigations:

Best initial: Clinical diagnosis.

Confirmatory: Surgical exploration and tissue culture.

Treatment Outline:

First-line: Surgical debridement, broad-spectrum IV antibiotics.

Second-line: Hyperbaric oxygen therapy if available.

Correct Answer:
B. Surgical debridement ✅

MCQ 696:

Which of the following is the most likely pathogen to infect a polytetrafluoroethylene vascular graft?

A. Klebsiella pneumonia
B. Staphylococcus aureus
C. Pseudomonas aeruginosa
D. Staphylococcus epidermidis

Explanation:
Staphylococcus aureus is the most common pathogen to infect vascular grafts, especially synthetic ones like polytetrafluoroethylene (PTFE). This pathogen is virulent and can lead to early graft infection and complications.

Vascular Graft Infection

Definition:
Infection of a synthetic vascular graft, commonly due to skin flora or hematogenous spread.

Key Clinical Presentation:

Fever, localized redness, tenderness, and swelling around the graft site.

Late infection may present with graft dysfunction or abscess formation.

Investigations:

Best initial: Blood cultures, wound cultures, graft biopsy.

Confirmatory: Graft imaging or surgical exploration.

Treatment Outline:

First-line: Antibiotics based on culture sensitivity (usually MRSA coverage).

Second-line: Surgical removal and replacement of the graft if infected.

Correct Answer:
B. Staphylococcus aureus ✅

MCQ 697:

An unstable trauma patient has confirmed multiple liver lacerations.
Which of the following is the most appropriate management?

A. Right hepatectomy
B. Perinepatic packing
C. Right hepatic artery ligation
D. Individual ligation of the bleeding vessels

Explanation:
In trauma patients with **liver lacerations**, **perinepatic packing** is the most appropriate initial management to control bleeding temporarily. If packing fails, further interventions such as ligation of bleeding vessels or hepatectomy may be considered.

Liver Trauma Management

Definition:
Liver lacerations or contusions caused by blunt or penetrating trauma, resulting in hemorrhage.

Key Clinical Presentation:

Hypotension, tachycardia, abdominal distension.

Right upper quadrant tenderness, bruising.

Investigations:

Best initial: FAST ultrasound or CT scan.

Confirmatory: CT angiography or exploratory laparotomy.

Treatment Outline:

First-line: Hemodynamic stabilization, perinepatic packing, or selective embolization.

Second-line: Surgical resection or arterial ligation if necessary.

Correct Answer:
B. Perihepatic packing ✅

MCQ 698:

5 days post-appendectomy, a patient developed diffuse abdominal distension, constipation, and absent bowel sounds.
Which of the following is the most appropriate next step in management?

A. Insert a rectal tube
B. Check serum electrolyte levels
C. Re-exploration of the abdomen
D. Abdominal computed tomography

Explanation:
Abdominal CT scan is the most appropriate next step to rule out complications like **intra-**

abdominal abscess, bowel obstruction, or other causes of post-operative ileus. Re-exploration may be needed if the CT scan suggests complications.

Postoperative Ileus and Complications

Definition:
Failure of bowel motility after surgery, which can result in bloating, distension, and discomfort.

Key Clinical Presentation:

Abdominal distension, constipation, and absent bowel sounds after surgery.

May progress to vomiting or signs of sepsis.

Investigations:

Best initial: Abdominal CT scan.

Confirmatory: Abdominal ultrasound, clinical assessment.

Treatment Outline:

First-line: Conservative management, electrolyte correction.

Second-line: Re-exploration if signs of peritonitis or abscess formation.

Correct Answer:
D. Abdominal computed tomography ✅

MCQ 699:

Which of the following is the most definitive treatment for a post-renal transplant lymphocele?

A. Percutaneous aspiration
B. Observation and follow-up
C. Fenestration into the peritoneal cavity
D. Percutaneous injection of sclerotherapy

Explanation:
Fenestration into the peritoneal cavity is considered the most definitive treatment for post-

renal transplant lymphoceles. It allows the lymph to drain into the peritoneal cavity, thereby preventing recurrence.

Renal Transplant Lymphocele

Definition:
A collection of lymph fluid that can form after renal transplantation due to disruption of lymphatic vessels.

Key Clinical Presentation:

Abdominal or flank swelling.

May be asymptomatic or present with discomfort or pressure effects.

Investigations:

Best initial: Ultrasound or CT scan to confirm lymphocele.

Confirmatory: Contrast studies or lymphangiography.

Treatment Outline:

First-line: Conservative management with observation.

Second-line: Percutaneous aspiration if symptomatic.

Definitive: Surgical fenestration or drainage into the peritoneal cavity.

Correct Answer:
C. Fenestration into the peritoneal cavity ✅

MCQ 700:

Which of the following is most likely to cause small bowel distention?

A. Nitrous oxide
B. Muscle relaxants
C. Depolarizing agents
D. Esophageal intubation

Explanation:
Nitrous oxide is known to cause small bowel distention, especially when used in high amounts during surgery. It can lead to bowel distension by accumulating in the bowel and causing gas expansion.

Small Bowel Distention

Definition:
Accumulation of gas in the small bowel, leading to swelling and distension, often associated with impaired motility or obstruction.

Key Clinical Presentation:

Abdominal distension, bloating, nausea, vomiting.

May be associated with surgical or anesthetic interventions.

Investigations:

Best initial: Abdominal X-ray or CT scan.

Confirmatory: Abdominal ultrasound or MRI.

Treatment Outline:

First-line: Conservative management with decompression.

Second-line: Surgery if underlying obstruction is identified.

Correct Answer:
A. Nitrous oxide ✅

MCQ 701:

A hypothyroid patient is scheduled to undergo an elective surgery.
Which of the following is the most appropriate management?

A. Perform surgery then start thyroxine
B. Start thyroxine and proceed with surgery
C. Perform surgery if clinically asymptomatic
D. Delay surgery until the patient is euthyroid

Explanation:
In a hypothyroid patient, it is essential to **start thyroxine** and **achieve euthyroidism** before proceeding with elective surgery to avoid complications such as poor wound healing, impaired metabolism, and cardiovascular instability. If the patient is clinically stable, surgery can proceed after thyroxine administration.

Hypothyroidism in Surgery

Definition:
Underactive thyroid gland, leading to low levels of thyroid hormones, which can affect metabolism, cardiovascular function, and wound healing.

Key Clinical Presentation:

Fatigue, cold intolerance, constipation, bradycardia, weight gain.

May have delayed reflexes, skin changes, and coarse hair.

Investigations:

Best initial: Serum TSH and free T4.

Confirmatory: Thyroid function tests.

Treatment Outline:

First-line: Start levothyroxine before surgery to normalize thyroid levels.

Second-line: Adjust dose based on TSH and clinical response.

Correct Answer:
B. Start thyroxine and proceed with surgery ✅

MCQ 703:

A heavy smoker man is going for a major surgical procedure.
Which of the following is the most sensitive predictor of postoperative respiratory complications?

A. Chest X-ray
B. Bronchoscopy
C. Lung function test
D. Arterial blood gases

Explanation:
Lung function tests (such as spirometry and assessment of lung volumes) are the most sensitive predictors of postoperative respiratory complications, especially in smokers. They assess pulmonary function and help identify risks for complications like pneumonia and atelectasis.

Lung Function in Surgical Risk

Definition:
Tests that evaluate the capacity and function of the lungs, helping to assess a patient's risk for respiratory complications.

Key Clinical Presentation:

Chronic cough, shortness of breath, history of smoking.

Potential increased risk for postoperative pulmonary complications.

Investigations:

Best initial: Pulmonary function tests (spirometry, FEV1, etc.).

Confirmatory: Arterial blood gases if needed.

Treatment Outline:

First-line: Pulmonary rehabilitation if indicated.

Second-line: Oxygen therapy and postoperative monitoring.

Correct Answer:
C. Lung function test ✅

MCQ 704:

A man presented with diarrhea and persistent peptic ulcer refractory to proton pump inhibitors.
Secretin stimulation test is positive.
Which of the following is the most likely diagnosis?

A. VIPoma
B. Gastrinoma
C. Glucagonoma
D. Carcinoid tumor

Explanation:
A **gastrinoma** is a tumor that secretes excessive amounts of gastrin, leading to Zollinger-Ellison syndrome, characterized by peptic ulcers and diarrhea. The secretin stimulation test is positive in gastrinomas because secretin usually stimulates gastrin release from these tumors.

Gastrinoma (Zollinger-Ellison Syndrome)

Definition:
A gastrin-secreting tumor, most commonly found in the pancreas or duodenum, causing severe peptic ulcers and gastrointestinal symptoms.

Key Clinical Presentation:

Refractory peptic ulcers, diarrhea, abdominal pain.

Positive secretin stimulation test.

Investigations:

Best initial: Serum gastrin level, secretin stimulation test.

Confirmatory: Abdominal CT scan or endoscopic ultrasound.

Treatment Outline:

First-line: Proton pump inhibitors for acid suppression.

Second-line: Surgical resection if the tumor is localized.

Definitive: Surgical removal of the gastrinoma or hepatic metastases.

Correct Answer:

B. Gastrinoma ✅

MCQ 706:

A 50-year-old woman post-modified radical mastectomy, presented with numbness and a burning sensation on the inner aspect of the upper arm.
Which of the following nerves is most likely injured?

A. Thorcodorsal
B. Long thoracic
C. Medial pectoral
D. Intercostobrachial

Explanation:
The **intercostobrachial nerve**, which arises from the second intercostal nerve, provides sensation to the medial aspect of the upper arm. Injury to this nerve is common after axillary surgery, such as a modified radical mastectomy.

Intercostobrachial Nerve Injury

Definition:
Damage to the intercostobrachial nerve, often due to axillary surgery, leading to sensory loss or pain in the medial upper arm.

Key Clinical Presentation:

Numbness or burning sensation along the inner aspect of the upper arm.

Often occurs after axillary lymph node dissection.

Investigations:

Best initial: Clinical examination.

Confirmatory: Electromyography (EMG) or nerve conduction studies.

Treatment Outline:

First-line: Conservative management with analgesics.

Second-line: Nerve blocks or physical therapy if persistent.

Correct Answer:
D. Intercostobrachial ✅

MCQ 707:

Post laparoscopic cholecystectomy, transection of the common bile duct above the cystic duct is confirmed.
Which of the following is the most appropriate treatment?

A. Hepatico-jejunostomy
B. Hepatico-duodenostomy
C. Choledocho-jejunostomy
D. Choledocho-duodenostomy

Explanation:
Hepatico-jejunostomy is the most appropriate treatment for a transected common bile duct, as it creates a new bile duct drainage pathway to the jejunum. This procedure is done to bypass the damaged bile duct and restore bile flow.

Common Bile Duct Injury

Definition:
Injury to the common bile duct during surgery, often during laparoscopic cholecystectomy, leading to bile leakage or obstruction.

Key Clinical Presentation:

Jaundice, dark urine, pale stools, and elevated liver enzymes.

Possible biliary peritonitis or sepsis if untreated.

Investigations:

Best initial: Liver function tests, abdominal ultrasound.

Confirmatory: MRCP or ERCP.

Treatment Outline:

First-line: Hepatico-jejunostomy or choledocho-jejunostomy if biliary leakage persists.

Second-line: Surgical repair or stent placement if indicated.

Correct Answer:
A. Hepatico-jejunostomy ✅

MCQ 708:

A 36-year-old woman was undergoing a caesarean section, and just before closure, the surgeon noticed profound bleeding coming from the upper abdomen.
Which of the following is the most likely source of bleeding?

A. Splenic aneurysm
B. Liver hemangioma
C. Mesenteric aneurysm
D. Perforated peptic ulcer

Explanation:
A **splenic aneurysm** is a known source of bleeding that can be encountered in abdominal surgery, especially in cases where the spleen is manipulated or in close proximity to the surgical field. It can present as sudden and severe bleeding, especially in the context of trauma or surgery. Other causes, such as liver hemangioma or mesenteric aneurysm, are less common and would not typically present with this pattern of bleeding during a caesarean section.

Splenic Aneurysm

Definition:
An abnormal dilation of the splenic artery, often asymptomatic, but can cause life-threatening hemorrhage if ruptured.

Key Clinical Presentation:

Abdominal pain, hypotension, and signs of shock if ruptured.

Bleeding during surgery, especially involving the upper abdomen.

Investigations:

Best initial: Abdominal ultrasound.

Confirmatory: CT angiography or MRI.

Treatment Outline:

First-line: Surgical repair if ruptured.

Second-line: Endovascular embolization if appropriate.

Correct Answer:
A. Splenic aneurysm ✅

MCQ 709:

Which of the following is the strongest indicator for surgery in Graves' disease?

A. Paediatric patient
B. Presence of eye symptoms
C. Presence of anti-TSH antibodies
D. Large goiter

Explanation:
The **presence of eye symptoms** in Graves' disease is the strongest indicator for surgery, especially if medical management has failed or if symptoms are severe. Eye symptoms such as proptosis, diplopia, and optic nerve compression can significantly affect the patient's quality of life and may require surgical intervention for correction. Large goiter or pediatric age alone are not direct indications for surgery.

Graves' Disease

Definition:
An autoimmune disorder characterized by hyperthyroidism, goiter, and often, eye involvement (Graves' orbitopathy).

Key Clinical Presentation:

Hyperthyroid symptoms: weight loss, tremor, heat intolerance, palpitations.

Goiter and exophthalmos (eye bulging), possibly with optic nerve compression.

Investigations:

Best initial: Serum TSH, T4, T3 levels, and anti-TSH receptor antibodies.

Confirmatory: Radioactive iodine uptake scan or ultrasound.

Treatment Outline:

First-line: Antithyroid medications (methimazole, propylthiouracil).

Second-line: Radioactive iodine therapy.

Definitive: Surgery (thyroidectomy) if there are significant complications or resistance to therapy.

Correct Answer:
B. Presence of eye symptoms ✅

MCQ 710:

A young man received 4 litres of blood in the Emergency Department following a road traffic accident.
Which of the following is the most likely complication?

A. Hypokalaemia
B. Citrate toxicity
C. Hypocalcaemia
D. Hyperalbunemia

Explanation:
Hypocalcaemia is a common complication after massive blood transfusions due to the citrate used as an anticoagulant in the blood. Citrate binds to calcium, lowering the levels in the blood, and can cause symptoms such as tetany, arrhythmias, or muscle cramps. Citrate toxicity can also lead to metabolic alkalosis and symptoms like hypotension.

Blood Transfusion Complications

Definition:
Complications that arise after receiving a blood transfusion, often related to the volume and rapidity of administration.

Key Clinical Presentation:

Hypocalcemia: Symptoms of muscle cramps, tingling, and seizures.

Citrate toxicity: Hypotension, arrhythmias, and alkalosis.

Investigations:

Best initial: Serum calcium levels.

Confirmatory: Ionized calcium levels for precise measurement.

Treatment Outline:

First-line: Calcium supplementation (IV calcium gluconate).

Second-line: Adjust transfusion rate if possible and manage underlying cause.

Correct Answer:
C. Hypocalcaemia ✅

MCQ 711:

A young man comes to the Emergency Department after being involved in a road traffic accident. A Focused Assessment with Sonography for Trauma (FAST) was ordered.
Which of the following is the most likely finding?

A. Pneumothorax
B. Vascular injury
C. Subdural hematoma
D. Intraperitoneal free fluid

Explanation:
Intraperitoneal free fluid is the most common finding in trauma patients undergoing a FAST exam. This typically indicates internal bleeding, most commonly from solid organ injuries like the spleen or liver. FAST is used to identify free fluid in the peritoneal cavity, which helps guide further management in trauma cases.

Focused Assessment with Sonography for Trauma (FAST)

Definition:
A rapid ultrasound examination performed in trauma patients to detect free intraperitoneal fluid (usually blood) or signs of organ injury.

Key Clinical Presentation:

Trauma with hypotension or signs of internal bleeding.

Abdominal distension, tenderness, or signs of shock.

Investigations:

Best initial: FAST exam.

Confirmatory: CT abdomen or diagnostic laparoscopy if necessary.

Treatment Outline:

First-line: Surgery (laparotomy) or interventional radiology for bleeding control.

Second-line: Conservative management in stable patients with minor injuries.

Correct Answer:
D. Intraperitoneal free fluid ✅

MCQ 712:

A 45-year-old diabetic man presents with a non-healing wound, which he developed after incision and drainage for diabetic foot 2 months ago.
Which of the following is the most likely etiology?

A. Poor blood supply
B. Underlying malignancy
C. Peripheral neuropathy
D. Non-compliance with treatment

Explanation:
Poor blood supply is the most likely cause for a non-healing wound in diabetic patients. Diabetic

foot ulcers often occur due to peripheral vascular disease, which impairs blood supply to the extremities, leading to delayed wound healing and increased risk of infection. Peripheral neuropathy also plays a role in trauma but is not the direct cause of non-healing wounds.

Diabetic Foot Ulcer

Definition:
A wound or ulceration of the lower extremities in a diabetic patient, often due to a combination of neuropathy, poor circulation, and infection.

Key Clinical Presentation:

Non-healing wound, usually on the foot.

Often associated with infection, gangrene, and ischemia.

Investigations:

Best initial: Doppler ultrasound for assessing blood flow.

Confirmatory: Ankle-brachial index (ABI) to assess for peripheral artery disease.

Treatment Outline:

First-line: Glycemic control, debridement, and revascularization if necessary.

Second-line: Wound care with topical agents and offloading pressure.

Definitive: Amputation in severe cases.

Correct Answer:
A. Poor blood supply ✅

MCQ 713:

An elderly man is brought to the Emergency Department with an altered level of consciousness after he fell down the stairs.
CT scan: Suggestive of an extradural hematoma.
Which of the following arteries is most likely involved?

A. Basilar
B. Pontine

C. Anterior cerebral
D. Middle meningeal

Explanation:
The **middle meningeal artery** is the most common artery involved in extradural hematomas (EDH). EDH typically results from trauma to the skull that causes a rupture of the middle meningeal artery, leading to a collection of blood between the dura mater and the skull. This can cause rapid neurological deterioration if not treated promptly.

Extradural Hematoma (EDH)

Definition:
A collection of blood between the inner surface of the skull and the dura mater, often resulting from trauma.

Key Clinical Presentation:

Loss of consciousness followed by a lucid interval.

Progressive neurological deterioration.

Can present with symptoms of raised intracranial pressure.

Investigations:

Best initial: CT scan of the brain.

Confirmatory: MRI brain.

Treatment Outline:

First-line: Surgical evacuation of the hematoma.

Second-line: Medical management for small, stable hematomas.

Correct Answer:
D. Middle meningeal ✅

MCQ 714:

A 25-year-old man underwent a thyroidectomy due to multinodular goiter. Post-operatively, he was unable to maintain high-pitched sounds.
Which of the following nerves is most likely injured?

A. Vagus
B. Inferior laryngeal
C. Superior laryngeal
D. Recurrent laryngeal

Explanation:
The **superior laryngeal nerve** is responsible for innervating the cricothyroid muscle, which is important for tensing the vocal cords, particularly for high-pitched sounds. Injury to the superior laryngeal nerve during thyroid surgery can result in hoarseness or an inability to produce high-pitched sounds, which is consistent with the patient's symptoms.

Superior Laryngeal Nerve Injury

Definition:
Damage to the superior laryngeal nerve, often occurring during thyroidectomy or neck surgery.

Key Clinical Presentation:

Inability to produce high-pitched sounds.

Hoarseness or weak voice.

Difficulty with speech in high vocal registers.

Investigations:

Best initial: Laryngoscopy to assess vocal cord movement.

Confirmatory: Electromyography of the laryngeal muscles.

Treatment Outline:

First-line: Voice therapy and rest.

Second-line: Surgical re-innervation or botulinum toxin injections if symptoms persist.

Correct Answer:

C. Superior laryngeal ✅

MCQ 715:

A 42-year-old woman complains of left-sided chest and abdominal pain that is worse on inspiration. She underwent a splenectomy a few days ago. Examination confirms decreased air entry at the left lung base, dullness to percussion, and left upper quadrant tenderness. Temperature 38.6 °C.
Which of the following is the most likely diagnosis?

A. Gastric stasis
B. Subphrenic abscess
C. Lower lobe pneumonia
D. Overwhelming post-splenectomy sepsis

Explanation:
A **subphrenic abscess** is a common complication after splenectomy, especially in the early postoperative period. It can cause pain in the left upper quadrant and referred pain to the chest, along with fever and signs of localized infection. The physical exam findings of dullness to percussion and decreased air entry are consistent with a subphrenic abscess.

Subphrenic Abscess

Definition:
A collection of pus in the space between the diaphragm and the peritoneal organs, commonly occurring post-surgery, including after splenectomy.

Key Clinical Presentation:

Fever and abdominal pain (especially in the upper left quadrant).

Respiratory symptoms, such as decreased breath sounds or pleuritic chest pain.

Positive signs of localized peritonitis.

Investigations:

Best initial: Abdominal ultrasound or CT scan.

Confirmatory: Aspiration or drainage for microbiology culture.

Treatment Outline:

First-line: Percutaneous drainage and antibiotics.

Second-line: Surgical drainage if percutaneous drainage is unsuccessful.

Correct Answer:
B. Subphrenic abscess ✅

MCQ 716:

A 40-year-old diabetic patient with good glycemic control is admitted electively for an inguinal hernia repair.
What is the most accurate American Society of Anesthesiologists (ASA) physical status class?

A. I
B. II
C. III
D. IV

Explanation:
This patient is a **Class II** according to the ASA classification. Class II is assigned to patients with mild systemic disease or well-controlled chronic conditions, such as diabetes, that do not limit the patient's activity. This patient's diabetes is well-controlled, making ASA II the most appropriate classification.

ASA Physical Status Classification

Definition:
A system used by anesthesiologists to assess and communicate the fitness of patients before surgery.

Key Clinical Presentation:

Class I: Healthy, no systemic disease.

Class II: Mild systemic disease, well-controlled.

Class III: Severe systemic disease that limits activity.

Class IV: Severe systemic disease that is a constant threat to life.

Investigations:

Best initial: Preoperative evaluation to assess comorbidities.

Confirmatory: ASA classification is based on clinical judgment.

Treatment Outline:

First-line: General anesthesia or regional anesthesia, based on patient condition.

Second-line: Adjust anesthesia technique based on comorbidities.

Correct Answer:
B. II ✅

MCQ 717:

A 20-year-old woman with no past medical history is admitted with a perforated appendix and is in septic shock. On arrival, she was intubated. She requires high inotropic support to maintain minimum recordable blood pressure.
What is the most accurate American Society of Anesthesiologists (ASA) physical status class?

A. II
B. III
C. IV
D. V

Explanation:
This patient would be classified as **ASA Class V** due to the presence of septic shock and the need for high inotropic support to maintain blood pressure. Class V is used for patients who are not expected to survive without the procedure and are in critical condition.

ASA Physical Status Classification

Definition:
A system used to assess the overall health and surgical risk of patients based on systemic health and disease severity.

Key Clinical Presentation:

Class I: Healthy, no systemic disease.

Class II: Mild systemic disease, well-controlled.

Class III: Severe systemic disease that limits activity.

Class IV: Severe systemic disease that is a constant threat to life.

Class V: Moribund patient not expected to survive without the surgery.

Investigations:

Best initial: Preoperative assessment of vital signs and organ function.

Confirmatory: ASA classification based on clinical condition.

Treatment Outline:

First-line: Critical care management and surgery to stabilize the patient.

Second-line: Supportive therapy as needed in the ICU.

Correct Answer:
D. V ✅

MCQ 717:

A 40-year-old man presents with an advanced colonic malignancy.
What is the most likely site of metastasis?

A. Lung
B. Liver
C. Thyroid
D. Prostate

Explanation:
The **liver** is the most common site of metastasis for colonic carcinoma. Colorectal cancer commonly spreads to the liver through the portal circulation. The lung is also a common site of metastasis, but liver metastasis is far more common.

Colorectal Cancer Metastasis

Definition:
Cancer originating in the colon or rectum that can spread to distant organs, most commonly to the liver.

Key Clinical Presentation:

Symptoms of advanced cancer: weight loss, fatigue, rectal bleeding, and abdominal pain.

Liver dysfunction signs: jaundice, hepatomegaly.

Investigations:

Best initial: Colonoscopy with biopsy.

Confirmatory: CT scan for staging and liver metastasis detection.

Treatment Outline:

First-line: Surgery (resection of primary tumor) + chemotherapy for metastatic disease.

Second-line: Chemotherapy regimens like FOLFOX or FOLFIRI.

Correct Answer:
B. Liver ✅

MCQ 718:

A young man is hypoxic 24 hours after major upper abdominal surgery.
Which of the following is the most likely cause?

A. Pneumonia
B. Basal atelectasis
C. Pulmonary embolism
D. Intra-abdominal bleeding

Explanation:
Basal atelectasis is the most common cause of hypoxia 24 hours after abdominal surgery. It occurs due to shallow breathing, reduced mobility, and the effects of anesthesia, leading to partial lung collapse, especially in the basal regions. It typically presents as hypoxia and can resolve with appropriate respiratory physiotherapy.

Postoperative Pulmonary Complications

Definition:
Respiratory issues that arise following surgery, commonly due to atelectasis, pneumonia, or pulmonary embolism.

Key Clinical Presentation:

Hypoxia, tachypnea, and respiratory distress.

Reduced breath sounds over lung bases.

History of recent abdominal or thoracic surgery.

Investigations:

Best initial: Chest X-ray.

Confirmatory: CT scan if concerned about pulmonary embolism or other complications.

Treatment Outline:

First-line: Oxygen therapy, chest physiotherapy, incentive spirometry.

Second-line: Positive pressure ventilation or CPAP if severe.

Correct Answer:
B. Basal atelectasis ✅

MCQ 719:

Which of the following is the most likely cause of hypoxia 24 hours after major pelvic surgery for ovarian cancer?

A. Pneumonia
B. Pelvic abscess

C. Basal atelectasis
D. Pulmonary embolism

Explanation:
Basal atelectasis is the most common cause of hypoxia in the postoperative period, including following pelvic surgery. This condition arises due to shallow breathing and limited mobility after surgery, leading to collapse of the lower lobes of the lungs. It is usually self-limiting and improves with deep breathing exercises and physiotherapy.

Postoperative Pulmonary Complications

Definition:
Respiratory complications post-pelvic surgery, including atelectasis and pneumonia.

Key Clinical Presentation:

Tachypnea, hypoxia, and dyspnea 24 hours post-surgery.

Reduced breath sounds at the lung bases.

Investigations:

Best initial: Chest X-ray.

Confirmatory: CT scan if pneumonia or embolism is suspected.

Treatment Outline:

First-line: Oxygen therapy, chest physiotherapy, and incentive spirometry.

Second-line: Mechanical ventilation for severe cases.

Correct Answer:
C. Basal atelectasis ✅

MCQ 720:

A 30-year-old woman sustained a burn from an oven grill. 8 days later, she noticed greenish discharge in the wound.
Which of the following is the most likely cause?

A. E. coli
B. Klebsiella
C. Pseudomonas
D. Streptococcus pyogenes

Explanation:
Pseudomonas aeruginosa is the most common pathogen associated with burn wound infections, especially when there is greenish discharge. It produces a distinctive greenish pigment (pyocyanin) that can color the exudate. Pseudomonas is a common colonizer in moist, necrotic tissue and can cause serious infections in burn wounds.

Burn Wound Infection

Definition:
Infection of a burn wound, commonly caused by opportunistic pathogens like Pseudomonas aeruginosa.

Key Clinical Presentation:

Greenish or foul-smelling discharge.

Pain, redness, and swelling around the wound.

Fever and signs of systemic infection if severe.

Investigations:

Best initial: Wound swab for culture.

Confirmatory: Blood cultures if septicemia is suspected.

Treatment Outline:

First-line: Topical antibiotics (silver sulfadiazine, polymyxin B).

Second-line: Systemic antibiotics if the infection is severe.

Correct Answer:
C. Pseudomonas ✅

MCQ 721:

A 55-year-old butcher accidentally cut himself at work. Later that night, he noticed some red lines radiating from the wound.
Which of the following is the most likely cause?

A. Brucella
B. Klebsiella
C. Pseudomonas
D. Streptococcus pyogenes

Explanation:
Streptococcus pyogenes is the most likely pathogen in this case, causing **lymphangitis**, which presents as red streaks radiating from a wound. This is a hallmark of **infection with Streptococcus pyogenes** and is often associated with cellulitis or more severe soft tissue infections.

Soft Tissue Infections and Lymphangitis

Definition:
Infections of the skin and soft tissues, often leading to lymphangitis, where the infection spreads along lymphatic channels.

Key Clinical Presentation:

Red streaks (lymphangitis) radiating from the site of injury.

Swelling, warmth, and erythema at the wound site.

Systemic signs of infection (fever).

Investigations:

Best initial: Wound culture.

Confirmatory: Blood cultures if septicemia is suspected.

Treatment Outline:

First-line: Oral or intravenous antibiotics (penicillin or cephalosporins).

Second-line: Surgery if an abscess or necrotizing fasciitis develops.

Correct Answer:

D. Streptococcus pyogenes ✅

MCQ 723:

Where is the most common primary location of basal cell carcinoma?

A. Face
B. Breast
C. Lower limb
D. Upper limb

Explanation:
The face is the most common location for **basal cell carcinoma (BCC)**, particularly the **nasal** area, followed by the **cheeks** and **ears**. BCC typically develops in sun-exposed areas, and the face, being the most exposed, is the most common site for this type of skin cancer.

Basal Cell Carcinoma (BCC)

Definition:
The most common type of skin cancer, originating from the basal cells of the epidermis.

Key Clinical Presentation:

Painless, slow-growing, pearly nodule.

May have ulcerated, crusted centers and rolled borders.

Often seen on sun-exposed areas, especially the face.

Investigations:

Best initial: Clinical examination and dermoscopy.

Confirmatory: Biopsy for histopathological examination.

Treatment Outline:

First-line: Surgical excision or Mohs micrographic surgery.

Second-line: Cryotherapy, topical treatments, or radiation therapy for non-surgical candidates.

Correct Answer:
A. Face ✅

MCQ 724:

A young man presents with a lump on the right mid-thigh. On examination, the overlying skin is normal, the mass is soft, movable with a positive fluctuation test.
Which of the following is the most likely diagnosis?

A. Lipoma
B. Sarcoma
C. Aneurysm
D. Sebaceous cyst

Explanation:
A **sebaceous cyst** (also known as an epidermoid cyst) is a common diagnosis for a soft, movable, fluctuant mass that is not associated with skin changes. These cysts arise from sebaceous glands or hair follicles and often present with a positive fluctuation test. They are benign and often require simple excision for removal.

Sebaceous Cyst

Definition:
A benign, closed sac under the skin filled with a cheesy, keratinous material, often arising from hair follicles or sebaceous glands.

Key Clinical Presentation:

Soft, movable lump, commonly on the face, neck, or trunk.

Positive fluctuation test.

No overlying skin changes unless infected.

Investigations:

Best initial: Clinical examination.

Confirmatory: Aspiration or biopsy in atypical cases.

Treatment Outline:

First-line: Surgical excision.

Second-line: Incision and drainage if infected.

Correct Answer:
D. Sebaceous cyst ✅

MCQ 725:

A 40-year-old is admitted with a 3-day history of diffuse abdominal pain. On examination, there is diffuse abdominal tenderness along with **Grey Turner's sign**.
Which of the following is the most likely diagnosis?

A. Acute porphyria
B. Mesenteric ischemia
C. Necrotizing pancreatitis
D. Ruptured aortic aneurysm

Explanation:
Grey Turner's sign, which refers to bruising of the flanks, is a classic sign of **necrotizing pancreatitis**, which often presents with severe abdominal pain, tenderness, and systemic signs of sepsis. This sign occurs due to retroperitoneal hemorrhage associated with severe pancreatic inflammation.

Necrotizing Pancreatitis

Definition:
A severe form of pancreatitis characterized by pancreatic necrosis and possible retroperitoneal bleeding.

Key Clinical Presentation:

Severe abdominal pain radiating to the back.

Abdominal tenderness and guarding.

Signs of systemic shock and sepsis.

Grey Turner's sign (flank bruising) and **Cullen's sign** (periumbilical bruising) in severe cases.

Investigations:

Best initial: Serum amylase and lipase.

Confirmatory: Contrast-enhanced CT of the abdomen.

Treatment Outline:

First-line: Supportive care, fluids, pain management.

Second-line: Surgical debridement if necrosis is extensive.

Correct Answer:
C. Necrotizing pancreatitis ✅

MCQ 726:

An elderly man presents for follow-up after he sustained a road traffic accident 3 days ago. On examination, **"Battle Sign"** is noted.
Which of the following is the most likely cause?

A. Le Forte fracture
B. Mandibular fracture
C. Fracture base of skull
D. Fracture middle cranial fossa

Explanation:
Battle Sign (bruising over the mastoid process) is a classic sign of **a basilar skull fracture**, particularly from trauma to the temporal bone, which may lead to bleeding in the middle ear and bruising behind the ear.

Basilar Skull Fracture

Definition:
Fracture of the bones at the base of the skull, often associated with significant trauma.

Key Clinical Presentation:

Battle's sign (bruising behind the ear).

Raccoon eyes (periorbital bruising).

Cerebrospinal fluid (CSF) leak from the ears or nose.

Investigations:

Best initial: CT scan of the head.

Confirmatory: MRI for further detail of injury.

Treatment Outline:

First-line: Conservative management and monitoring.

Second-line: Surgical intervention if there is associated brain injury or large CSF leak.

Correct Answer:
C. Fracture base of skull ✅

MCQ 730:

A 50-year-old diabetic man presents with a non-healing wound, which he developed after incision and drainage for a diabetic foot ulcer 2 months ago.
Which of the following is the most likely etiology?

A. Poor blood supply
B. Underlying malignancy
C. Peripheral neuropathy
D. Non-compliance with treatment

Explanation:
The most common etiology for a non-healing diabetic foot ulcer is **poor blood supply**, which is a result of peripheral arterial disease in diabetic patients. This impairs wound healing and increases the risk of infection.

Diabetic Foot Ulcers

Definition:
Chronic ulcers on the feet of diabetic patients, often due to poor circulation and neuropathy.

Key Clinical Presentation:

Non-healing ulcers, usually on the plantar aspect of the feet.

Can be associated with neuropathy, poor sensation, and ischemia.

Investigations:

Best initial: Ankle-brachial index (ABI) for peripheral vascular disease.

Confirmatory: Doppler ultrasound or angiography for vascular assessment.

Treatment Outline:

First-line: Proper diabetic management, debridement, and off-loading.

Second-line: Revascularization or surgical intervention if necessary.

Correct Answer:
A. Poor blood supply ✅

MCQ 721

A 20-year-old man was hit by a car 2 hours ago. On examination, his abdomen is distended, tense, and tender.
Blood pressure 100/70 mmHg
Heart rate 120 /min
Respiratory rate 30 /min
Temperature 35 °C
Oxygen saturation 90 %
What type of shock is the most likely diagnosis?

A. Spinal
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:
The patient's hypotension, tachycardia, and abdominal distention after trauma suggest **hemorrhagic shock**, likely due to internal bleeding, such as from solid organ injury or retroperitoneal hemorrhage.

Hemorrhagic Shock

Definition:
A type of shock resulting from significant blood loss, leading to inadequate tissue perfusion.

Key Clinical Presentation:

Hypotension, tachycardia, and tachypnea.

Cold extremities and delayed capillary refill.

Abdominal distention and tenderness, particularly after trauma.

Investigations:

Best initial: FAST (focused assessment with sonography for trauma) or CT scan of the abdomen.

Confirmatory: Blood tests (e.g., hemoglobin and hematocrit) and crossmatch.

Treatment Outline:

First-line: IV fluids, blood transfusion, and surgery if necessary.

Second-line: Vasopressors if fluid resuscitation is insufficient.

Correct Answer:
C. Haemorrhagic ✅

MCQ 722;

A 30-year-old man was hit by a truck 4 hours ago. On examination, he has raised jugular venous pressure and diminished heart sounds.
Blood pressure 90/60 mmHg
Heart rate 140 /min
Respiratory rate 32 /min

Temperature 36.6 °C
Oxygen saturation 85 %

What type of shock is the most likely diagnosis?

A. Septic
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:
This patient presents with **raised jugular venous pressure**, **diminished heart sounds**, **hypotension**, and **tachycardia**, which are suggestive of **cardiac tamponade** due to trauma. **Cardiogenic shock** in this context is most likely due to tamponade, where blood accumulates in the pericardial sac, preventing the heart from filling properly, leading to poor cardiac output.

Cardiogenic Shock

Definition:
A state of shock due to inadequate cardiac output, typically caused by heart failure, but in this case, likely due to **cardiac tamponade** from trauma.

Key Clinical Presentation:

Hypotension, tachycardia, and tachypnea.

Raised jugular venous pressure.

Diminished heart sounds (pericardial effusion).

Pulsus paradoxus (decrease in blood pressure during inspiration).

Investigations:

Best initial: Clinical examination, FAST (Focused Assessment with Sonography for Trauma).

Confirmatory: Echocardiogram showing pericardial effusion.

Treatment Outline:

First-line: Pericardiocentesis to relieve pressure on the heart.

Second-line: Surgical intervention if required for severe cases.

Correct Answer:

B. Cardiogenic ✅

MCQ 723:

A 20-year-old patient complains of painless fresh bleeding post defecation over the last 6 months.
Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal fistula
D. Perianal abscess

Explanation:
Haemorrhoids are the most likely diagnosis in this patient. They commonly present with **painless fresh bleeding**, often seen during or after defecation. In contrast, **anal fissures** usually present with **painful bleeding**, and **perianal abscesses** or **fistulas** are typically associated with pain and discharge.

Haemorrhoids

Definition:
Swollen veins in the lower rectum and anus, often caused by straining during bowel movements, pregnancy, or prolonged sitting.

Key Clinical Presentation:

Painless rectal bleeding, often noticed on toilet paper.

Lump or swelling near the anus.

Possible itching or discomfort.

Investigations:

Best initial: Clinical examination.

Confirmatory: Anoscopy or proctoscopy.

Treatment Outline:

First-line: Conservative treatment (dietary fiber, topical agents).

Second-line: Rubber band ligation, sclerotherapy.

Definitive: Surgical excision in severe cases.

Correct Answer:
B. Haemorrhoids ✅

MCQ 724;

A 40-year-old diabetic patient complains of a painful perianal swelling with fever for the last 3 days.
Blood pressure 110/70 mmHg
Heart rate 110 /min
Respiratory rate 20 /min
Temperature 38 °C

Test Result:

Hb 135 (130-170 g/L)

WBC 16.4 (4.5-10.5 x 10^9/L)

Neutrophils 80% (40-60%)

Random Glucose 6.5 (3.9-5.5 mmol/L)

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:
This patient with **fever**, **painful perianal swelling**, and **elevated WBC count** likely has a **perianal abscess**. The presence of **increased neutrophils** and systemic signs of infection support this

diagnosis. An **anal fissure** is typically not associated with fever, and **haemorrhoids** would not present with an abscess-like swelling or systemic symptoms.

Perianal Abscess

Definition:
A collection of pus in the tissues around the anus, usually due to infection of the anal glands.

Key Clinical Presentation:

Painful, swollen area near the anus.

Fever, chills, and erythema in the perianal area.

Possible drainage of pus.

Investigations:

Best initial: Clinical examination.

Confirmatory: Ultrasound or MRI if deeper abscess is suspected.

Treatment Outline:

First-line: Incision and drainage.

Second-line: Antibiotics if infection is extensive or systemic symptoms are present.

Correct Answer:
C. Perianal abscess ✅

MCQ 726:

A 50-year-old patient underwent a total thyroidectomy 5 hours ago. He has developed a large neck swelling associated with stridor and shortness of breath.
Blood pressure 165/90 mmHg
Heart rate 130 /min
Respiratory rate 24 /min
Temperature 37 °C

Which of the following is the most appropriate next step in management?

A. Tracheostomy
B. Close observation
C. Percutaneous drainage
D. Bedside wound exploration

Explanation:
The patient most likely has a **post-thyroidectomy hematoma**, which can compress the airway, causing **stridor** and **respiratory distress**. The most appropriate next step is **bedside wound exploration** to evacuate the hematoma and relieve the airway obstruction.

Post-thyroidectomy hematoma

Definition:
A collection of blood in the surgical site following thyroidectomy, which can compress the airway.

Key Clinical Presentation:

Neck swelling, pain, and difficulty breathing.

Stridor and respiratory distress.

Symptoms occur within hours of surgery.

Investigations:

Best initial: Clinical examination and urgent bedside exploration.

Confirmatory: CT scan of the neck if needed.

Treatment Outline:

First-line: Immediate wound exploration to evacuate the hematoma.

Second-line: Tracheostomy if the airway cannot be secured.

Correct Answer:
D. Bedside wound exploration ✅

MCQ 727:

A 60-year-old man had a laparoscopic inguinal hernia repair 5 days ago. Today, he noticed a swelling at the wound area. Examination confirmed a swelling at the site of surgery that was not red, or hot, but mildly tender. There was no cough impulse.
Blood pressure 135/80 mmHg
Heart rate 80 /min
Respiratory rate 24 /min
Temperature 37.2 °C

Which of the following is the most likely diagnosis?

A. Seroma
B. Hematoma
C. Recurrence
D. Wound infection

Explanation:
The patient is presenting with **mild tenderness**, **no redness**, and **no cough impulse**. These features are more suggestive of a **seroma**, which is a collection of fluid that commonly occurs post-operatively. **Hematoma** typically presents with bruising and more significant tenderness. **Recurrence** would typically present with a cough impulse and would appear later. **Wound infection** would be associated with redness, heat, and increased tenderness, which this patient does not show.

Seroma

Definition:
A seroma is a collection of serous fluid within a tissue cavity, typically seen after surgery.

Key Clinical Presentation:

Painless or mildly tender swelling.

No redness, warmth, or signs of infection.

Common after surgeries involving tissue dissection, such as hernia repair.

Investigations:

Best initial: Clinical examination.

Confirmatory: Ultrasound can be used to assess fluid collection.

Treatment Outline:

First-line: Conservative management (e.g., compression dressings, observation).

Second-line: Aspiration or drainage if symptomatic.

Definitive: Rarely needed; usually resolves with conservative measures.

Correct Answer:
A. Seroma ✅

MCQ 728:

A 32-year-old man underwent a hemorrhoidectomy for a prolapsed pile. 5 hours later, he complained of severe supra-pubic pain and was unable to pass urine.
Which of the following is the most likely cause?

A. Dehydration
B. Urethral injury
C. Inadequate analgesia
D. Post-operative hematoma

Explanation:
Inadequate analgesia is the most likely cause in this patient who is experiencing **severe supra-pubic pain** and **urinary retention** shortly after the procedure. **Urethral injury** is uncommon in hemorrhoidectomy and typically presents with blood at the urethral meatus. **Dehydration** and **post-operative hematoma** may contribute to discomfort but would not be as likely to cause urinary retention directly after surgery.

Inadequate Analgesia

Definition:
Inadequate pain control leading to post-operative discomfort and possible complications such as urinary retention.

Key Clinical Presentation:

Severe pain following surgery.

Urinary retention due to bladder discomfort and muscle spasms.

Investigations:

Best initial: Clinical evaluation, voiding trial.

Treatment Outline:

First-line: Adequate analgesia (e.g., opioids, nerve blocks).

Second-line: Catheterization if retention persists.

Definitive: Pain management during recovery.

Correct Answer:
C. Inadequate analgesia ✅

MCQ 729:

A 52-year-old woman presented with a recent rapid enlargement of a neck swelling associated with weight loss. She was diagnosed 25 years ago with Hashimoto's thyroiditis.
Which of the following is the most likely diagnosis?

A. Hemorrhagic goiter
B. Thyroid lymphoma
C. Papillary carcinoma
D. Sub-acute thyroiditis

Explanation:
Given the patient's history of **Hashimoto's thyroiditis** and **rapid enlargement** of the neck swelling, the most likely diagnosis is **thyroid lymphoma**. This can occur as a complication of Hashimoto's thyroiditis, with rapid enlargement of the thyroid. **Hemorrhagic goiter** typically presents with a painful, enlarged thyroid but is less likely to cause weight loss. **Papillary carcinoma** presents with a firm, non-tender mass, and **sub-acute thyroiditis** would be painful and associated with thyroid dysfunction.

Thyroid Lymphoma

Definition:
A rare malignancy of the thyroid, often associated with Hashimoto's thyroiditis. It presents as rapidly enlarging, non-tender thyroid masses.

Key Clinical Presentation:

Rapidly enlarging neck mass.

History of Hashimoto's thyroiditis.

Weight loss and systemic symptoms.

Investigations:

Best initial: Fine needle aspiration biopsy.

Confirmatory: Imaging (CT or ultrasound) and biopsy.

Treatment Outline:

First-line: Chemotherapy and/or radiation therapy.

Second-line: Surgical intervention if possible.

Correct Answer:
B. Thyroid lymphoma ✅

MCQ 730:

A 56-year-old woman presented with generalized muscle and bone ache associated with abdominal cramps that is relieved by drinking cold milk (see lab results).
Test Result Normal Values

Calcium: 2.85 (2.15-2.62 mmol/L)

Calcium ionized: 1.9 (1.1-1.3 mmol/L)

Phosphate, inorganic: 0.71 (0.82-1.51 mmol/L)

Parathyroid hormone (intact PTH levels): 5.5 (1.1-5.3 pmol/L)

25-Hydroxy Vitamin D3 (Women): 15 (25-90 nmol/L)

Which of the following is the most likely diagnosis?

A. Milk alkali syndrome
B. Idiopathic hypercalcemia
C. Primary hyperparathyroidism
D. Secondary hyperparathyroidism

Explanation:
The patient's symptoms of **bone ache**, **muscle pain**, and **abdominal cramps** relieved by drinking milk, combined with **elevated calcium levels**, point to **milk alkali syndrome**. This syndrome is caused by excessive calcium and alkali intake, leading to hypercalcemia, suppressed parathyroid hormone (PTH) levels, and low phosphate. **Primary hyperparathyroidism** and **secondary hyperparathyroidism** would typically show elevated PTH levels.

Milk Alkali Syndrome

Definition:
A condition caused by excessive intake of calcium and alkali, leading to hypercalcemia, metabolic alkalosis, and renal impairment.

Key Clinical Presentation:

Hypercalcemia.

Abdominal pain, bone pain, muscle ache.

Symptoms relieved by milk.

Investigations:

Best initial: Serum calcium, phosphate, PTH, and vitamin D levels.

Confirmatory: History of excessive calcium and alkali intake.

Treatment Outline:

First-line: Discontinuation of calcium and alkali intake.

Second-line: Hydration and loop diuretics if renal function is impaired.

Correct Answer:
A. Milk alkali syndrome ✅

MCQ 730:

A 36-year-old man sustained a knife stab to the left side of his neck, below the cricoid cartilage (Zone I). Examination confirmed diffuse neck subcutaneous emphysema.
Blood pressure: 110/70 mmHg
Heart rate: 90 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C

Which of the following is the most appropriate next step in management?

A. Neck exploration
B. Close observation
C. CT of the neck and chest
D. Interventional radiology embolization

Explanation:
The presence of **subcutaneous emphysema** after a **knife wound** in Zone I (the area below the cricoid cartilage) requires immediate **imaging** to assess for any involvement of the **airway, vascular structures**, or **esophagus**. The **most appropriate next step** is to perform a **CT scan of the neck and chest** to assess the extent of injury and rule out major vascular or esophageal involvement.
Neck exploration may be indicated later, but initial imaging is critical. Observation is inappropriate given the potential severity of neck trauma, and **embolization** is unnecessary without confirmation of vascular injury.

Neck Trauma

Definition:
Injury to the neck structures, which can involve the airway, vascular structures, and esophagus. Subcutaneous emphysema is often a sign of air leakage from these structures following trauma.

Key Clinical Presentation:

Stab wound or blunt trauma to the neck.

Subcutaneous emphysema.

Pain, difficulty swallowing, or respiratory distress in severe cases.

Investigations:

Best initial: CT of the neck and chest.

Confirmatory: Imaging to evaluate for vascular, tracheal, or esophageal injury.

Treatment Outline:

First-line: Imaging (CT scan) to assess the extent of the injury.

Second-line: Surgical exploration and repair if necessary.

Definitive: Based on findings, repair of the damaged structures (vessels, trachea, esophagus).

Correct Answer:
C. CT of the neck and chest ✅

MCQ 731:

A 57-year-old hypertensive complained of non-specific lower abdominal pain. Hormonal studies for adrenal glands are within normal values. A CT scan was ordered (see report).
CT scan:
5 cm rounded hypodense left adrenal mass, described as lipid-rich lesion.

Which of the following is the most appropriate management?

A. Adrenal MRI
B. No intervention is indicated
C. Laparoscopic adrenalectomy
D. Image-guided adrenal biopsy

Explanation:
The **lipid-rich** nature of the adrenal mass on CT scan is most consistent with a **benign adrenal adenoma**, which is common in patients with no significant symptoms and normal hormonal studies. In the absence of any other alarming features (such as functional activity or malignancy), **no intervention** is usually required.
Adrenal MRI is often used for further characterization, but **benign adenomas** do not require surgery unless symptomatic or functional. **Laparoscopic adrenalectomy** is indicated for symptomatic or suspicious masses, while **image-guided biopsy** is not typically performed unless there is suspicion of malignancy.

Adrenal Adenoma

Definition:
A benign tumor of the adrenal gland, often asymptomatic, and typically diagnosed incidentally on imaging.

Key Clinical Presentation:

Often asymptomatic.

May present with symptoms related to hormone excess (e.g., Cushing's syndrome or pheochromocytoma).

Investigations:

Best initial: CT scan or MRI for further characterization.

Confirmatory: Hormonal evaluation (if functional).

Treatment Outline:

First-line: Observation for non-functional adenomas.

Second-line: Surgery (adrenalectomy) if symptomatic or if the mass is suspected to be malignant.

Definitive: Surgical removal for functional tumors or masses with suspicious characteristics.

Correct Answer:
B. No intervention is indicated ✅

MCQ 732:

A 37-year-old patient presented to the Emergency Department with a 2-day history of right upper quadrant abdominal pain associated with vomiting. Physical examination confirmed rebound tenderness at right hypochondrium (see lab result and report).
Blood pressure: 125/70 mmHg
Heart rate: 90 /min
Respiratory rate: 20 /min
Temperature: 38.8 °C
Test Result Normal Values

WBC: 16.5 (4.5-10.5 x 10^9/L)
Ultrasound abdomen:
Edematous, thick gallbladder wall, pericholecystic fluid, and multiple gallstones.

Which of the following is the most appropriate management?

A. Laparoscopic cholecystectomy
B. Percutaneous cholecystostomy drain
C. Exploratory laparotomy with cholecystectomy
D. Endoscopic retrograde cholangiopancreatography

Explanation:
The patient is presenting with **right upper quadrant pain**, **fever**, and **rebound tenderness** with ultrasound findings of **gallbladder wall thickening** and **gallstones**, consistent with **acute cholecystitis**. The appropriate management is **laparoscopic cholecystectomy**, which is the gold standard treatment for this condition. In certain patients (e.g., high surgical risk), **percutaneous cholecystostomy** can be performed as an initial approach. **Exploratory laparotomy** is not necessary unless there are signs of more extensive peritonitis. **Endoscopic retrograde cholangiopancreatography (ERCP)** is indicated if there is concern for **biliary duct obstruction**.

Acute Cholecystitis

Definition:
Acute inflammation of the gallbladder, usually due to gallstones, leading to obstruction of the cystic duct.

Key Clinical Presentation:

Right upper quadrant pain.

Fever.

Positive Murphy's sign (tenderness over the gallbladder).

Investigations:

Best initial: Ultrasound of the abdomen.

Confirmatory: HIDA scan if ultrasound is inconclusive.

Treatment Outline:

First-line: Laparoscopic cholecystectomy within 24-72 hours of diagnosis.

Second-line: Percutaneous cholecystostomy in high-risk surgical patients.

Definitive: Surgical removal of the gallbladder.

Correct Answer:
A. Laparoscopic cholecystectomy ✅

MCQ 733:

A 65-year-old man presented with a 3-month history of right upper quadrant abdominal pain, associated with a loss of appetite and weight loss. He has hepatitis C (see lab result and report).
Test Result Normal Value
Alpha fetoprotein: 1160 (0-40 ng/mL)
CT scan abdomen:
Hypervascular lesion at the right lobe of the liver.

Which of the following is the most likely diagnosis?

A. Hamartoma
B. Cholangiocarcinoma
C. Hepatocellular adenoma
D. Hepatocellular carcinoma

Explanation:
The patient has **hepatitis C**, **weight loss**, **right upper quadrant pain**, and an **elevated alpha-fetoprotein (AFP)** level. These findings are most suggestive of **hepatocellular carcinoma (HCC)**, a common complication of chronic liver disease, particularly in patients with cirrhosis or hepatitis C. **Hamartomas** are benign and not typically associated with elevated AFP. **Cholangiocarcinoma** may present with similar symptoms but does not typically have elevated AFP. **Hepatocellular adenoma** is usually asymptomatic and does not present with such a high AFP level.

Hepatocellular Carcinoma (HCC)

Definition:
Primary malignant tumor of the liver, commonly associated with cirrhosis and chronic liver disease (e.g., hepatitis B, C).

Key Clinical Presentation:

Right upper quadrant pain.

Weight loss.

Jaundice.

Elevated AFP.

Investigations:

Best initial: Ultrasound of the abdomen.

Confirmatory: CT scan or MRI of the liver.

Diagnostic: Biopsy (if uncertain).

Treatment Outline:

First-line: Surgical resection or liver transplant if feasible.

Second-line: Ablation therapies (radiofrequency, transarterial chemoembolization).

Definitive: Transplantation for patients with cirrhosis and HCC.

Correct Answer:
D. Hepatocellular carcinoma ✅

MCQ 734:

A 37-year-old man presented to the Emergency Department with a 10-day history of frequent vomiting. He was dehydrated, and his electrocardiogram showed flattening of the T-wave. Which of the following is the most likely acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
The patient's prolonged vomiting is most likely to result in **metabolic alkalosis**, which is characterized by **loss of gastric acid**. Vomiting leads to the loss of hydrogen ions, increasing the pH of the body. The **flattening of the T-wave** on ECG is indicative of **hypokalemia**, which is a common finding in metabolic alkalosis due to potassium shift into cells in response to the alkalosis.
Metabolic alkalosis is the most likely acid-base disturbance in this patient.

Acid-Base Disorders

Definition:
Metabolic alkalosis is an increase in blood pH due to excessive loss of hydrogen ions or excess of bicarbonate.

Key Clinical Presentation:

Vomiting

Dehydration

Hypokalemia

Flattening of T-waves on ECG

Investigations:

Best initial: Serum electrolyte levels (especially potassium and bicarbonate).

Confirmatory: Arterial blood gas.

Treatment Outline:

First-line: Correction of underlying cause (e.g., fluid replacement with potassium).

Second-line: Acidifying agents in severe cases.

Correct Answer:
B. Metabolic alkalosis ✅

MCQ 735:

A 32-year-old man presented with a 2-week history of frequent attacks of watery diarrhea. On examination, he was severely dehydrated.
Blood pressure: 90/45 mmHg
Heart rate: 130 /min
Respiratory rate: 18 /min
Temperature: 37.2 °C
Which of the following is the most likely acid-base abnormality?

A. Metabolic acidosis
B. Metabolic alkalosis
C. Compensated metabolic acidosis
D. Compensated metabolic alkalosis

Explanation:
Severe dehydration and frequent watery diarrhea typically lead to **metabolic acidosis** due to the loss of bicarbonate in stool. The body compensates for this acidosis by increasing the respiratory rate to eliminate CO_2. **Hypovolemia** and **electrolyte imbalances** are also common findings. This patient's clinical presentation is consistent with **metabolic acidosis**.

Acid-Base Disorders

Definition:
Metabolic acidosis is a condition in which there is a decrease in pH due to excessive accumulation of acid or loss of bicarbonate.

Key Clinical Presentation:

Diarrhea

Dehydration

Tachycardia and hypotension

Investigations:

Best initial: Serum electrolytes and arterial blood gas.

Confirmatory: Urine analysis for anion gap.

Treatment Outline:

First-line: Fluid resuscitation and correction of electrolyte imbalances.

Second-line: Sodium bicarbonate in severe cases.

Correct Answer:
A. Metabolic acidosis ✅

MCQ 736:

A 43-year-old woman who recovered from acute pancreatitis 5 weeks ago, presented with epigastric pain and postprandial vomiting. Physical examination confirmed tenderness and fullness in the epigastric region.
Blood pressure: 127/70 mmHg
Heart rate: 70 /min
Respiratory rate: 16 /min
Temperature: 36.6 °C
Test Result Normal Values
WBC: 6.4 (4.5-10.5 x 10^9/L)
Amylase: 430 (24-151 IU/L)
CT scan of abdomen:
Large cystic collection abutting the posterior wall of the stomach.

Which of the following is the most likely pancreatic pathology?

A. Cancer
B. Abscess
C. Necrosis
D. Pseudocyst

Explanation:
The patient has a history of **acute pancreatitis** and now presents with a **cystic collection** seen on CT scan, which is suggestive of a **pseudocyst**. Pancreatic pseudocysts are common sequelae of pancreatitis, occurring weeks to months after the initial episode. They typically contain pancreatic fluid, and the absence of infection or necrosis suggests a benign pseudocyst rather than abscess or necrosis.

Pancreatic Disorders

Definition:
A pseudocyst is a collection of fluid rich in pancreatic enzymes, surrounded by a fibrous capsule, often resulting from pancreatitis.

Key Clinical Presentation:

Epigastric pain

Vomiting

Abdominal fullness

History of pancreatitis

Investigations:

Best initial: CT scan of the abdomen.

Confirmatory: Ultrasound or endoscopic ultrasound if needed.

Treatment Outline:

First-line: Observation for asymptomatic pseudocysts.

Second-line: Drainage (percutaneous or surgical) if symptomatic.

Definitive: Surgical drainage or resection if complications arise.

Correct Answer:
D. Pseudocyst ✅

MCQ 737:

A 50-year-old patient underwent a total thyroidectomy 5 hours ago. He has developed a large neck swelling associated with stridor and shortness of breath.
Blood pressure: 165/90 mmHg
Heart rate: 130 /min
Respiratory rate: 24 /min
Temperature: 37 °C

Which of the following is the most appropriate next step in management?

A. Tracheostomy
B. Close observation

C. Percutaneous drainage
D. Bedside wound exploration

Explanation:
The patient has developed a **neck swelling** associated with **stridor** and **shortness of breath** after thyroid surgery, which is concerning for **postoperative hematoma** compressing the airway. **Immediate exploration** of the surgical site is needed to evacuate the hematoma and relieve airway obstruction. **Tracheostomy** is not required unless there is failure to secure the airway. Close observation is insufficient given the symptoms of airway compromise.

Thyroid Surgery Complications

Definition:
Post-thyroidectomy hematoma can occur as a complication, causing compression of the airway and leading to respiratory distress.

Key Clinical Presentation:

Neck swelling

Stridor

Difficulty breathing after thyroid surgery

Investigations:

Best initial: Clinical evaluation and immediate examination of the surgical site.

Confirmatory: No imaging is needed if symptoms are suggestive.

Treatment Outline:

First-line: Immediate exploration and evacuation of the hematoma.

Second-line: Airway management (intubation, tracheostomy if needed).

Correct Answer:
D. Bedside wound exploration ✅

MCQ 738:

Which of the following is most commonly cured by having a splenectomy?

A. β-thalassemia
B. a-thalassemia
C. Sickle cell trait
D. Idiopathic thrombocytopenia purpura

Explanation:
Splenectomy is most commonly performed for **idiopathic thrombocytopenia purpura (ITP)**, as it helps improve platelet count by removing the spleen, which is responsible for platelet destruction in this condition. It is not typically used for thalassemia or sickle cell trait.

Hematologic Disorders

Definition:
Idiopathic thrombocytopenic purpura (ITP) is an autoimmune disorder characterized by low platelet count due to immune-mediated destruction in the spleen.

Key Clinical Presentation:

Bleeding tendency

Easy bruising

Petechiae

Investigations:

Best initial: Blood counts and peripheral smear.

Confirmatory: Clinical diagnosis after ruling out other causes of thrombocytopenia.

Treatment Outline:

First-line: Steroids, immunoglobulin.

Second-line: Splenectomy for refractory cases.

Correct Answer:
D. Idiopathic thrombocytopenia purpura ✅

MCQ 739:

A 25-year-old man is undergoing appendectomy for acute appendicitis. Upon entering the peritoneal cavity, the surgeon could not find the appendix.
Which of the following is the most effective way to localize the appendix?

A. Following the terminal ileum
B. Palpating the ileocaecal valve
C. Looking at the confluence of the taenia coli
D. Following the course of the right colic artery

Explanation:
The **most effective way to localize the appendix** is by **following the terminal ileum**. The appendix is typically located at the **junction of the ileum and cecum**, so following the terminal ileum allows for easier identification.

Abdominal Surgery

Definition:
Appendectomy is a surgical procedure to remove the appendix, typically performed in cases of appendicitis.

Key Clinical Presentation:

Right lower quadrant pain

Fever

Nausea and vomiting

Investigations:

Best initial: Clinical examination and ultrasound or CT scan.

Confirmatory: Exploration during surgery.

Treatment Outline:

First-line: Appendectomy (laparoscopic or open).

Second-line: Conservative management with antibiotics for uncomplicated cases.

Correct Answer:

A. Following the terminal ileum ✅

MCQ 739;

A 29-year-old woman underwent a left lower parathyroidectomy for primary hyperparathyroidism. Four months later, she presented with generalized fatigue, muscle and bone ache, and constipation (see lab results).
Which of the following is the most likely cause?

A. New adenoma
B. Missed adenoma
C. Parathyroid cancer
D. Parathyroid hyperplasia

Explanation:
The most likely cause in this patient is a **missed adenoma**. Parathyroidectomy is usually performed to remove an adenoma causing primary hyperparathyroidism, but sometimes the adenoma is missed during the initial surgery, leading to persistent symptoms. **Parathyroid hyperplasia** is a less common cause.

Endocrine Disorders

Definition:
Primary hyperparathyroidism is caused by excessive parathyroid hormone (PTH) secretion, often due to a parathyroid adenoma.

Key Clinical Presentation:

Hypercalcemia

Fatigue

Bone pain

Constipation

Investigations:

Best initial: Serum calcium and PTH levels.

Confirmatory: Imaging of the parathyroid glands (ultrasound, Sestamibi scan).

Treatment Outline:

First-line: Parathyroidectomy.

Second-line: Medical management with bisphosphonates or calcimimetics for refractory cases.

Correct Answer:
B. Missed adenoma ✅

MCQ 740

A 39-year-old woman presented with a 3-day history of central abdominal pain associated with nausea and vomiting for 1 day. She underwent a laparoscopic sleeve gastrectomy for obesity 6 years ago.
Examination confirmed a dehydrated patient with abdominal distention and exaggerated bowel sounds.
Which of the following is the most likely cause?

A. Adhesions
B. Internal hernia
C. Intussusception
D. Incisional hernia

Explanation:
The most likely cause in this patient is an **internal hernia**. After bariatric surgery such as sleeve gastrectomy, there is a risk of internal hernias developing, where part of the small intestine becomes trapped and causes bowel obstruction. Adhesions are also common but are more likely after a longer duration.

Bariatric Surgery Complications

Definition:
Internal hernia is a potential complication following bariatric surgery, where the bowel is trapped within mesenteric defects.

Key Clinical Presentation:

Abdominal pain

Nausea and vomiting

Bowel obstruction symptoms

Investigations:

Best initial: CT scan of the abdomen.

Confirmatory: Laparoscopy.

Treatment Outline:

First-line: Laparoscopic repair of the hernia.

Second-line: Open surgical approach for complicated cases.

Correct Answer:
B. Internal hernia ✅

MCQ 741:

Which of the following is the most effective way to minimize excessive scar formation?

A. Pre-operative steroid
B. Elimination of wound tension
C. Post-operative topical steroid
D. Closure with subcuticular suture

Explanation:
Eliminating wound tension is the most effective method for minimizing excessive scar formation. This can be achieved by proper surgical technique, tension-free closure, and sometimes the use of sutures that minimize tension on the wound edges.

Wound Healing

Definition:
Excessive scar formation, or hypertrophic scarring, can occur due to tension on the wound edges, infection, or poor healing conditions.

Key Clinical Presentation:

Raised, red scars

Poor wound healing

Keloid formation

Investigations:

Best initial: Clinical evaluation.

Confirmatory: Biopsy if excessive scarring or infection is suspected.

Treatment Outline:

First-line: Tension-free closure, possible use of steroids or silicone sheets for scar prevention.

Second-line: Steroid injections for existing scars.

Correct Answer:
B. Elimination of wound tension ✅

MCQ 742;

Which of the following is most important factor for small bowel adaptation?

A. Prolonged TPN
B. Enteral feeding
C. Parietal cell vagotomy
D. Reversal of jejunal limb

Explanation:
Enteral feeding is the most important factor in promoting small bowel adaptation after surgery. The presence of nutrients in the bowel stimulates the growth and function of the enterocytes, aiding in bowel adaptation and recovery. Prolonged TPN is not as effective in promoting adaptation.

Small Bowel Disorders

Definition:
Small bowel adaptation is the process by which the remaining small intestine compensates for resected portions to maintain nutrient absorption.

Key Clinical Presentation:

Diarrhea

Malabsorption

Weight loss

Investigations:

Best initial: Nutritional assessment and electrolyte balance.

Confirmatory: Imaging if anatomical changes are suspected.

Treatment Outline:

First-line: Enteral feeding, nutritional support.

Second-line: Surgical interventions if necessary.

Correct Answer:
B. Enteral feeding ✅

MCQ 743:

A 62-year-old man presented with sudden onset epigastric pain radiating to the back of 6 hours' duration. He has ischemic heart disease, on plavix and aspirin. Examination confirmed an ill looking and dehydrated patient, with generalized abdominal tenderness and absent bowel sounds (see report).
Erect chest X-ray: Air under the left hemidiaphragm.
Which of the following is the most likely diagnosis?

A. Gall stone disease
B. Acute pancreatitis
C. Perforated peptic ulcer
D. Gastric outlet obstruction

Explanation:
The **presence of air under the left hemidiaphragm** on an erect chest X-ray is a classic finding for **perforated peptic ulcer**, where the perforation allows air to escape into the peritoneal cavity. This condition presents with sudden epigastric pain, which often radiates to the back, and is associated with tenderness, guarding, and absent bowel sounds.

Gastrointestinal Emergencies

Definition:
Perforated peptic ulcer occurs when an ulcer in the stomach or duodenum breaches the wall, leading to leakage of gastric contents into the peritoneal cavity.

Key Clinical Presentation:

Sudden onset severe abdominal pain

Epigastric pain radiating to the back

Tenderness, guarding, and absent bowel sounds

Investigations:

Best initial: Abdominal X-ray (air under diaphragm)

Confirmatory: CT scan if needed

Treatment Outline:

First-line: Surgical repair (laparotomy and closure)

Second-line: Medical management and observation if the patient is stable (rare)

Correct Answer:
C. Perforated peptic ulcer ✅

MCQ 744:

A 23-year-old man sustained a closed right mid-shaft tibia-fibula fracture in a motor vehicle accident. He was alert, awake, complained of severe pain and numbness in his right lower extremity. Examination confirmed non-palpable dorsalis pedis and posterior tibial pulses.
Blood pressure 120/70 mmHg

Heart rate 85 /min
Respiratory rate 20 /min
Temperature 36.6 °C
Which of the following is the most appropriate management?

A. Fasciotomy
B. Angiography
C. Leg elevation and splint
D. Anticoagulation therapy

Explanation:
The patient presents with a **closed tibia-fibula fracture** along with signs of **vascular compromise**, as indicated by the non-palpable pulses in the lower extremity. This suggests possible **compartment syndrome** or arterial injury. The most appropriate management is **fasciotomy** to relieve the pressure in the compartment and restore blood flow, preventing long-term tissue damage.

Orthopedic Emergencies

Definition:
Compartment syndrome is a condition in which increased pressure within a muscle compartment impairs blood flow and tissue perfusion, potentially leading to irreversible damage.

Key Clinical Presentation:

Pain out of proportion to the injury

Numbness, loss of pulses, and swelling in the affected limb

Absence of pulses

Investigations:

Best initial: Clinical diagnosis

Confirmatory: Pressure measurement within the compartment

Treatment Outline:

First-line: Fasciotomy

Second-line: Surgical repair of arterial injury if present

Correct Answer:
A. Fasciotomy ✅

MCQ 745:

A 28-year-old woman was undergoing a laparoscopic cholecystectomy for gallstones. After creating pneumo-peritoneum and inserting the trocars, her heart rate was 40 beats/min. Which of the following is the most likely cause?

A. Overhydration
B. Cold gas insufflation
C. Increased venous return
D. Rapid stretching of parietal peritoneum

Explanation:
Cold gas insufflation during laparoscopic surgery is a well-known cause of **vagal response**, leading to bradycardia. The sudden stretching of the peritoneum and cooling effect of the insufflation can trigger the vagus nerve, causing a **decrease in heart rate** (known as vagal bradycardia).

Surgical Complications

Definition:
Vagal bradycardia can occur during laparoscopic procedures due to stimulation of the vagus nerve by cold insufflation or peritoneal stretching.

Key Clinical Presentation:

Bradycardia during laparoscopic procedures

Normal vital signs otherwise

Investigations:

Best initial: Clinical observation during surgery

Confirmatory: Electrocardiogram to confirm bradycardia

Treatment Outline:

First-line: Adjusting gas temperature, use of atropine if needed

Second-line: Monitoring and discontinuation of insufflation if needed

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Correct Answer:
B. Cold gas insufflation ✅

MCQ 746

A 25-year-old woman presented with a right breast lump, which has been increasing in size over the past year. Physical examination confirmed a 10x12 cm firm, mobile mass involving the upper half of the right breast with no palpable axillary lymph nodes (see report).
Core needle biopsy: Malignant phyllodes tumour.
Which of the following is the most appropriate next step?

A. Wide local excision
B. Whole body PET scan
C. Contrast CT scan of chest
D. Immunohistochemistry for hormone receptors

Explanation:
Phyllodes tumors are fibroepithelial tumors that are typically benign but can also present with malignant characteristics, as in this case. The treatment of choice for malignant phyllodes tumors is **wide local excision** to ensure clear margins and reduce the risk of recurrence. **Lymph node evaluation** may be considered if there is evidence of nodal involvement. Other imaging studies like PET or CT scans are not routinely indicated unless there is suspicion of metastatic disease.

Breast Cancer and Tumors

Definition:
Phyllodes tumors are rare fibroepithelial tumors of the breast that can be benign, borderline, or malignant. Malignant forms require wide excision to avoid recurrence.

Key Clinical Presentation:

A rapidly growing breast mass, typically larger than typical fibroadenomas

Absence of axillary lymphadenopathy in many cases

Investigations:

Best initial: Core needle biopsy

Confirmatory: Surgical excision with histopathological assessment

Treatment Outline:

First-line: Wide local excision

Second-line: Mastectomy if the tumor is large or poorly localized

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Correct Answer:
A. Wide local excision ✅

MCQ 747:

A 47-year-old single woman with a positive family history of breast cancer presented with a lesion found on routine mammography at the lower outer quadrant of the right breast (see report).
Image guided core needle biopsy: Atypical ductal hyperplasia.
Which of the following is the most appropriate next step in management?

A. Simple mastectomy
B. Wire localization excision
C. Chemoprevention therapy
D. 6-month follow-up mammography

Explanation:
Atypical ductal hyperplasia (ADH) is considered a premalignant lesion and is associated with an increased risk of breast cancer. The next step in managing ADH is **surgical excision** to ensure that there is no associated carcinoma. **Chemoprevention therapy** may be considered after excision, especially in high-risk individuals, but is not the next step in management for ADH. **Follow-up mammography** would not be sufficient without removal of the lesion.

Breast Cancer Risk Management

Definition:
Atypical ductal hyperplasia is characterized by abnormal cell growth in the milk ducts, which increases the risk of developing invasive breast cancer.

Key Clinical Presentation:

Detected as an incidental finding on biopsy or mammography

Typically asymptomatic

Investigations:

Best initial: Core needle biopsy or excisional biopsy for confirmation

Treatment Outline:

First-line: Surgical excision with margin evaluation

Second-line: Consideration of chemoprevention if excised tissue shows residual atypia

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Correct Answer:
B. Wire localization excision ✅

MCQ 747:

A 25-year-old patient sustained blunt trauma to the right chest wall. There is paradoxical movement of the right side of the chest. His vital signs are as follows:

Heart rate: 96 /min

Respiratory rate: 35 /min

Oxygen saturation: 84%

Chest X-ray shows fractures of the 2nd through 4th ribs at more than 1 site. What is the most appropriate initial management?

A. Fluid resuscitation
B. Assisted ventilation

C. Needle decompression
D. Thoracostomy chest tube

Explanation:

The most appropriate initial management for **paradoxical movement** and **rib fractures** indicating **flail chest** is **D. Thoracostomy chest tube**. Flail chest occurs when multiple rib fractures lead to a segment of the chest wall becoming detached from the rest of the thoracic cage, impairing breathing. The key management in this case is to manage the underlying pneumothorax or hemothorax with thoracostomy, to prevent respiratory compromise.

Topic Section:

Flail Chest Management

Definition:
Flail chest refers to the fracture of two or more ribs in two or more places, resulting in paradoxical chest wall movement.

Key Clinical Presentation:

Paradoxical chest wall movement

Respiratory distress

Tachypnea, tachycardia, hypotension

Investigations:

Best initial: **Chest X-ray**

Confirmatory: **CT scan** (if indicated for further detail)

Treatment Outline:

First-line: **Thoracostomy chest tube** (for managing pneumothorax/hemothorax)

Second-line: **Ventilatory support** (e.g., assisted ventilation if required)

Definitive: Surgical fixation in severe cases

Correct Answer:

D. Thoracostomy chest tube ✅

MCQ 748:

A 9-year-old child presented with blunt abdominal trauma. His vital signs are:

Blood pressure: 110/70 mmHg

Heart rate: 90 /min

Respiratory rate: 22 /min

Temperature: 36.6 °C

Test results:

Hb: 110 g/L (Normal range: 130-170 g/L)

CT scan abdomen reveals a 2 cm splenic laceration with perispheric fluid. What is the most appropriate management?

A. Non-operative
B. Laparoscopic splenectomy
C. Laparotomy and splenorrhaphy
D. Transfuse 2 units of packed RBCs

Explanation:

For a **splenic laceration** with **stable vital signs** and no signs of significant hemodynamic instability or large bleeding, **A. Non-operative management** is the most appropriate. Splenic injuries in children are often managed conservatively, with close monitoring and supportive care, including IV fluids and blood transfusions if needed.

Topic Section:

Management of Splenic Trauma

Definition:
Traumatic injury to the spleen, commonly from blunt trauma, leading to lacerations or rupture.

Key Clinical Presentation:

Abdominal pain (typically in the left upper quadrant)

Decreased hemoglobin in the case of bleeding

Investigations:

Best initial: **CT scan**

Confirmatory: **Ultrasound** for hemodynamic monitoring

Treatment Outline:

First-line: **Non-operative management** (close observation)

Second-line: **Surgical intervention** (splenorrhaphy or splenectomy if necessary)

Definitive: In severe cases, **splenectomy** (if not manageable conservatively)

Correct Answer:

A. Non-operative ✅

MCQ 749

Which of the following is the most appropriate biopsy for a limb sarcoma?

A. Incisional
B. Excisional
C. Core needle
D. Fine needle aspiration

Explanation:

A. Incisional biopsy is the most appropriate biopsy for a **limb sarcoma**. Excisional biopsy should be avoided for sarcomas as it can result in seeding of the tumor. Incisional biopsy, where only a portion of the tumor is removed, is preferred to obtain a definitive diagnosis without compromising surgical resection.

Topic Section:

Limb Sarcoma

Definition:
A limb sarcoma is a malignant tumor arising from connective tissues, often requiring biopsy to confirm the diagnosis.

Key Clinical Presentation:

Painless, firm mass

May present with rapid growth

Can be associated with pain if pressing on adjacent structures

Investigations:

Best initial: **Incisional biopsy**

Confirmatory: **Core needle biopsy** or **Fine needle aspiration** (if applicable)

Treatment Outline:

First-line: **Surgical excision** with wide margins

Second-line: **Chemotherapy** or **Radiotherapy** (in advanced or non-resectable cases)

Correct Answer:
A. Incisional biopsy ✅

MCQ 750

A 44-year-old man presented with a 6-month history of vague central abdominal pain. CT scan of abdomen: Presence of a 20x10 cm retroperitoneal mass. No enlarged mesenteric or para-aortic lymph nodes noted. Which of the following is the most likely diagnosis?

A. Lymphoma
B. Liposarcoma
C. Germ cell tumor
D. Metastatic testicular tumor

Explanation:

B. Liposarcoma is the most likely diagnosis. A large, well-defined retroperitoneal mass, without enlarged lymph nodes, suggests **liposarcoma**, which is a common soft tissue tumor in this area. Lymphoma typically presents with enlarged lymph nodes, and germ cell tumors and metastatic testicular tumors are less likely to present as large retroperitoneal masses without other clinical signs.

Topic Section:

Liposarcoma

Definition:
Liposarcoma is a malignant tumor of adipose tissue, most commonly arising in the retroperitoneum or extremities.

Key Clinical Presentation:

Painless, slowly enlarging mass

Often located in the retroperitoneum

May cause symptoms due to compression of surrounding structures

Investigations:

Best initial: **CT scan** or **MRI** (to assess the size and location)

Confirmatory: **Biopsy** (for histological confirmation)

Treatment Outline:

First-line: **Surgical resection**

Second-line: **Radiotherapy** or **Chemotherapy** (if inoperable or recurrent)

Definitive: **Surgical excision** with clear margins

Correct Answer:
B. Liposarcoma ✅

MCQ 751

A 30-year-old man was hit by a truck 4 hours ago. On examination, he has raised jugular venous pressure and diminished heart sounds.

Blood pressure: 90/60 mmHg
Heart rate: 140 /min
Respiratory rate: 32 /min
Temperature: 36.6 °C
Oxygen saturation: 85 %

What type of shock is the most likely diagnosis?

A. Septic
B. Cardiogenic
C. Haemorrhagic
D. Anaphylactic

Explanation:

B. Cardiogenic shock is the most likely diagnosis. In this patient with trauma and diminished heart sounds, **cardiac tamponade** is highly suggested, a complication of trauma leading to fluid accumulation in the pericardial sac, which impairs cardiac output. This leads to raised jugular venous pressure, hypotension, tachycardia, and diminished heart sounds (Beck's triad), characteristic of cardiogenic shock.

Topic Section:

Cardiogenic Shock

Definition:
Cardiogenic shock occurs when the heart is unable to pump sufficient blood to meet the body's needs, often due to severe cardiac dysfunction or mechanical obstruction.

Key Clinical Presentation:

Hypotension, tachycardia

Raised jugular venous pressure

Diminished heart sounds

Cold, clammy skin

Organ hypoperfusion (e.g., oliguria)

Investigations:

Best initial: **Echocardiography** or **CT scan** (if trauma-related)

Confirmatory: **Cardiac catheterization**

Treatment Outline:

First-line: **IV fluids**, **inotropic support** (e.g., dopamine, dobutamine)

Second-line: **Mechanical circulatory support** (e.g., intra-aortic balloon pump)

Definitive: **Treat underlying cause** (e.g., pericardiocentesis for tamponade, surgery for structural damage)

Correct Answer:
B. Cardiogenic ✅

MCQ 751

A 20-year-old patient complains of painless fresh bleeding post defecation over the last 6 months.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal fistula
D. Perianal abscess

Explanation:

B. Haemorrhoids are the most likely diagnosis in this patient. Painless, fresh rectal bleeding that occurs during or after defecation is characteristic of **haemorrhoids**, particularly internal hemorrhoids. The patient's complaint of long-term bleeding with no pain supports this diagnosis.

Topic Section:

Haemorrhoids

Definition:
Haemorrhoids are swollen and inflamed veins in the rectum and anus, causing symptoms such as bleeding, pain, and itching.

Key Clinical Presentation:

Painless rectal bleeding (bright red)

May prolapse or cause itching and discomfort

Blood typically seen on toilet paper or in stool

Investigations:

Best initial: **Clinical examination** (digital rectal exam, anoscopy)

Confirmatory: **Anoscopy** for internal hemorrhoids

Treatment Outline:

First-line: **Conservative measures** (high-fiber diet, stool softeners)

Second-line: **Rubber band ligation**, **sclerotherapy**

Definitive: **Surgical excision** for large or prolapsed hemorrhoids

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Correct Answer:
B. Haemorrhoids ✅

MCQ 752

A 40-year-old diabetic patient complains of a painful perianal swelling with fever for the last 3 days (see lab results).

Test Result Normal Values
Hb: 135 (130-170 g/L)
WBC: 16.4 (4.5-10.5 x 10^9/L)
Neutrophils: 80% (40-60%)
Random Glucose: 6.5 (3.9-5.5 mmol/L)

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:

C. Perianal abscess is the most likely diagnosis. The combination of **painful perianal swelling**, **fever**, and **leukocytosis** is typical of a **perianal abscess**. This condition often occurs in patients with diabetes and presents with localized tenderness and swelling around the anus.

Topic Section:

Perianal Abscess

Definition:
A perianal abscess is a collection of pus in the perianal or anal area due to infection of the anal glands.

Key Clinical Presentation:

Painful, fluctuating mass in the perianal region

Fever, erythema, and warmth over the swelling

Possible drainage of pus

Investigations:

Best initial: **Clinical examination** (digital rectal exam)

Confirmatory: **Ultrasound** or **CT scan** if needed for abscess localization

Treatment Outline:

First-line: **Incision and drainage**

Second-line: **Antibiotics** if cellulitis or systemic infection is present

Definitive: **Excision or marsupialization** in chronic cases or for recurrent abscesses

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Correct Answer:

C. Perianal abscess ✅

MCQ 753

A 20-year-old primi-gravida complains of severe perianal pain post defecation along with fresh bleeding per rectum for the last 4 days. She describes the pain to be sharp and persists long after the passage of stool. She denies any history of associated swelling or fever.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Haemorrhoids
C. Perianal abscess
D. Thrombosed piles

Explanation:

A. Anal fissure is the most likely diagnosis. The patient's description of **sharp pain** post defecation that **persists long after the stool** suggests an **anal fissure**. Fissures are commonly associated with bright red blood, particularly after passing hard stools.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a small tear in the skin of the anal canal, typically caused by the passage of hard stools or trauma to the anal region.

Key Clinical Presentation:

Severe, sharp pain during or after defecation

Bright red blood on the stool or toilet paper

May be associated with itching or spasms in the anal sphincter

Investigations:

Best initial: **Clinical examination**

Confirmatory: **Anoscopy** if needed

Treatment Outline:

First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**

Second-line: **Topical nitroglycerin** or **calcium channel blockers** for sphincter relaxation

Definitive: **Surgical lateral internal sphincterotomy** in chronic cases

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Correct Answer:
A. Anal fissure ✅

MCQ 754

A 50-year-old hypertensive diabetic complains of a 2-month history of perianal swelling with discharge. 2 months ago, he noticed a small swelling, which ruptured.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Perianal abscess
D. Thrombosed piles

Explanation:

B. Perianal fistula is the most likely diagnosis. A perianal fistula is a chronic condition that often follows the rupture of a perianal abscess, leading to the formation of an abnormal passage that discharges pus. The patient's history of swelling, rupture, and ongoing discharge strongly supports this diagnosis.

Topic Section:

Perianal Fistula

Definition:
A perianal fistula is a small tunnel that forms between the skin and the anus, often following an abscess.

Key Clinical Presentation:

Painless or mildly painful perianal discharge

History of a prior perianal abscess

Recurrent drainage of pus or serosanguinous fluid from the anal area

Investigations:

Best initial: **Clinical examination** and **digital rectal exam**

Confirmatory: **Anoscopy** or **MRI** for complex fistulas

Treatment Outline:

First-line: **Fistulotomy** or **Seton placement** for draining fistulas

Second-line: **Antibiotics** for infection control if necessary

Definitive: **Surgical fistulotomy** or **fistula repair** for recurrent or complex cases

---

Correct Answer:
B. Perianal fistula ✅

MCQ 755

A 30-year-old multigravida mother complains of severe perianal pain, swelling, and absolute constipation for the last 2 days. On examination, swollen anal cushions, which are irreducible and tender to touch, are noted.

Blood pressure: 120/90 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 37 °C

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Anal carcinoma
C. Perianal abscess
D. Thrombosed piles

Explanation:

D. Thrombosed piles is the most likely diagnosis. The patient presents with **irreducible**, **tender anal cushions** and symptoms of **perianal pain**, **swelling**, and **constipation**, which are typical signs of **thrombosed hemorrhoids** (thrombosed piles). The condition occurs when a clot forms inside the hemorrhoid, causing swelling and pain.

Topic Section:

Thrombosed Hemorrhoids

Definition:
Thrombosed hemorrhoids occur when blood clots form inside the veins of the hemorrhoidal plexus, causing painful swelling.

Key Clinical Presentation:

Sudden onset of **severe perianal pain**

Irreducible, tender anal mass

Associated with **constipation** and **straining**

Investigations:

Best initial: **Clinical examination** (digital rectal exam)

Confirmatory: **Anoscopy** for internal hemorrhoids

Treatment Outline:

First-line: **Conservative treatment** (fiber, stool softeners, pain relief)

Second-line: **Surgical excision** of the thrombosed hemorrhoid if severe

Definitive: **Excision** for persistent or recurrent cases

Correct Answer:
D. Thrombosed piles ✅

MCQ 756

A 36-year-old man presents with a 2-day history of constipation. On examination, he looks nervous and severe pain was experienced during parting of the perianal region. Linear cracks were noted at both 6 and 12 O'clock positions.

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Perianal fistula
C. Solitary rectal ulcer
D. Intersphincteric abscess

Explanation:

A. Anal fissure is the most likely diagnosis. The presence of **linear cracks** at the **6 o'clock** and **12 o'clock** positions is characteristic of an **anal fissure**. Fissures are typically caused by trauma (e.g., hard stools) and cause sharp pain during and after defecation.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a tear in the skin of the anal canal, often caused by the passage of hard stools or trauma to the anal area.

Key Clinical Presentation:

Sharp, severe pain during and after defecation

Linear crack at the **6 o'clock** or **12 o'clock** position

Bright red blood on the stool or toilet paper

Investigations:

Best initial: **Clinical examination**

Confirmatory: **Anoscopy**

Treatment Outline:

First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**

Second-line: **Topical nitroglycerin** or **calcium channel blockers**

Definitive: **Lateral internal sphincterotomy** in chronic cases

---

Correct Answer:
A. Anal fissure ✅

MCQ 757

A 65-year-old diabetic presents with a 4-day history of severe perianal pain with high-grade fever. On examination, there is no external swelling. On digital rectal examination, there is posterior bogginess and tenderness with slight purulent discharge coming from the anal canal.

Temperature: 38 °C

Which of the following is the most likely diagnosis?

A. Perianal fistula
B. Ulcerative colitis
C. Intersphincteric abscess
D. Condylomata acuminate

Explanation:

C. Intersphincteric abscess is the most likely diagnosis. The patient's history of **severe perianal pain** and **fever** with **bogginess** and **purulent discharge** on digital rectal examination suggests an **intersphincteric abscess**, a common cause of perianal abscesses.

Topic Section:

Intersphincteric Abscess

Definition:
An intersphincteric abscess is a collection of pus between the internal and external anal sphincters, often presenting with pain, swelling, and fever.

Key Clinical Presentation:

Severe perianal pain

Fever, often high-grade

Boggy, tender mass on digital rectal exam

Purulent discharge from the anal canal

Investigations:

Best initial: **Digital rectal exam** and **clinical examination**

Confirmatory: **Anoscopy** or **MRI** for complex cases

Treatment Outline:

First-line: **Incision and drainage**

Second-line: **Antibiotics** if cellulitis or systemic infection is present

Definitive: **Excision** or **marsupialization** in recurrent or chronic cases

---

Correct Answer:
C. Intersphincteric abscess ✅

MCQ 758

A 30-year-old patient complains of a 2-year history of perianal itching and discomfort. On perianal examination, there are cauliflower-like multiple lesions clustered together. They are pink to grey in color and tender to touch.

Which of the following is the most likely diagnosis?

A. Anal carcinoma
B. Rectal carcinoma
C. Fibroepithelial polyps
D. Condylomata acuminate

Explanation:

D. Condylomata acuminate is the most likely diagnosis. The presence of **cauliflower-like lesions** that are **tender to touch** and **pink to grey in color** suggests **condylomata acuminate** (genital warts), commonly caused by **human papillomavirus (HPV)** infection.

Topic Section:

Condylomata Acuminate (Genital Warts)

Definition:
Condylomata acuminate are benign, proliferative warts caused by the human papillomavirus (HPV), commonly affecting the genital or anal area.

Key Clinical Presentation:

Cauliflower-like lesions in the perianal or genital region

Pink to grey in color

Often tender to touch

May be associated with **itching** and **discomfort**

Investigations:

Best initial: **Clinical examination**

Confirmatory: **HPV DNA testing** or **biopsy** in cases of uncertainty

Treatment Outline:

First-line: **Topical treatments** (e.g., imiquimod, podophyllin)

Second-line: **Cryotherapy** or **laser ablation**

Definitive: **Excision** in persistent or large lesions

Correct Answer:
D. Condylomata acuminate ✅

MCQ 759

A 65-year-old complains of a 6-month change in bowel habits. On examination, he is pale, thin, and emaciated. His abdomen is slightly distended. On digital rectal examination, there is a lesion 2 cm from the anal verge. On proctoscopy, the lesion appears to be a cauliflower-like mass, which is friable to touch.

Which of the following is the most likely diagnosis?

A. Anal carcinoma
B. Rectal carcinoma
C. Fibroepithelial polyps
D. Condylomata acuminate

Explanation:

A. Anal carcinoma is the most likely diagnosis. The patient's presentation with **weight loss**, **anorexia**, and a **friable cauliflower-like mass** near the anal verge strongly suggests **anal carcinoma**.

Topic Section:

Anal Carcinoma

Definition:
Anal carcinoma is a malignant tumor arising from the anal canal or perianal skin, often associated with HPV infection.

Key Clinical Presentation:

Change in bowel habits, often with **rectal bleeding**

Weight loss and **emaciation**

Friable, cauliflower-like mass on proctoscopy

Palpable mass on digital rectal exam

Investigations:

Best initial: **Proctoscopy** and **biopsy**

Confirmatory: **CT scan** or **MRI** for staging

Treatment Outline:

First-line: **Chemoradiation** for early-stage disease

Second-line: **Surgical resection** for locally advanced disease

Definitive: **Chemoradiation** followed by **surgical resection** for advanced cases

---

Correct Answer:

A. Anal carcinoma ✅

MCQ 760

A 25-year-old presents with a 2-week history of severe perianal pain and an inability to pass stool for the last 2 days. History confirmed a haemorrhoidectomy 12 months ago. On examination, he is uneasy and in distress. On rectal examination, the finger could not be passed beyond the anal verge due to severe pain.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Which of the following is the most likely diagnosis?

A. Anal fissure
B. Anal stenosis
C. Anal carcinoma
D. Intersphincteric abscess

Explanation:

A. Anal fissure is the most likely diagnosis. The patient's history of **severe perianal pain** and **inability to pass stool**, with **severe pain on rectal examination**, suggests **anal fissure**, especially after **haemorrhoidectomy**. The acute onset of pain and difficulty with bowel movements is characteristic of this condition.

Topic Section:

Anal Fissure

Definition:
An anal fissure is a tear or crack in the skin around the anus, often caused by trauma, constipation, or straining during bowel movements.

Key Clinical Presentation:

Severe perianal pain during and after defecation

Bleeding, typically bright red, on the stool or toilet paper

Painful **spasm** during rectal examination

Investigations:

Best initial: **Clinical examination** and **digital rectal examination**

Confirmatory: **Anoscopy** for internal fissures

Treatment Outline:

First-line: **Topical analgesics** (e.g., lidocaine) and **fiber supplements**

Second-line: **Topical nitroglycerin** or **calcium channel blockers**

Definitive: **Lateral internal sphincterotomy** for chronic cases

---

Correct Answer:
A. Anal fissure ✅

MCQ 761

A 20-year-old weight-lifter complains of a 2-day history of bilious vomiting and abdominal distention. Examination revealed swelling in the left inguinal region, which is firm and tender.

Blood pressure: 100/70 mmHg
Heart rate: 116 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C

X-ray abdomen:
Multiple air-fluid levels with no air in the large bowel.

What type of hernia is the most likely diagnosis?

A. Obstructed
B. Irreducible
C. Incarcerated
D. Strangulated

Explanation:

D. Strangulated hernia is the most likely diagnosis. The **bilious vomiting** and **abdominal distention**, along with **tender swelling** in the inguinal region, suggest a **strangulated hernia**, where the blood supply to the hernia contents is compromised, potentially leading to bowel necrosis. The **X-ray showing air-fluid levels** and absence of air in the large bowel also points to bowel obstruction due to a hernia.

Topic Section:

Strangulated Hernia

Definition:
A strangulated hernia occurs when the blood supply to the hernia contents (typically bowel) is compromised, leading to ischemia and possible necrosis.

Key Clinical Presentation:

Severe abdominal pain with **vomiting** (bilious)

Tender swelling at the hernia site

Absence of bowel sounds or **abdominal distention**

Investigations:

Best initial: **Abdominal X-ray** showing air-fluid levels and bowel obstruction

Confirmatory: **CT scan** or **ultrasound** for hernia and vascular compromise

Treatment Outline:

First-line: **Emergency surgical exploration** and reduction of hernia

Second-line: **Resection of necrotic bowel** if needed

Definitive: **Herniorrhaphy** to repair the defect

Correct Answer:
D. Strangulated ✅

MCQ 762

A 25-year-old man was brought to the Emergency Department with a penetrating chest injury 45 minutes ago. He was initially fine in the ambulance but then collapsed. Emergency response personnel started resuscitation. On arrival, he had a weak thread pulse, raised JVP, and bilateral equal breath sounds.

Blood pressure: 90/60 mmHg
Heart rate: 140 /min
Respiratory rate: 30 /min
Temperature: 36.6 °C
Oxygen saturation: 85 %

Which of the following is the most likely diagnosis?

A. Flail segment
B. Pneumothorax
C. Cardiac tamponade
D. Pulmonary contusions

Explanation:

C. Cardiac tamponade is the most likely diagnosis. The patient's **raised JVP**, **weak thread pulse**, and **collapse** suggest **cardiac tamponade**, where blood accumulates in the pericardial sac, compressing the heart. This is a common consequence of blunt or penetrating chest trauma.

Topic Section:

Cardiac Tamponade

Definition:
Cardiac tamponade is the compression of the heart due to the accumulation of fluid (e.g., blood) in the pericardial sac, impairing heart function.

Key Clinical Presentation:

Hypotension with **weak pulse** (pulsus paradoxus)

Elevated jugular venous pressure (JVP)

Tachycardia

Muffled heart sounds

Dyspnea

Investigations:

Best initial: **Clinical examination** with **echocardiography** or **focused assessment with sonography for trauma (FAST)**

Confirmatory: **Chest X-ray** and **echocardiogram**

Treatment Outline:

First-line: **Pericardiocentesis** to relieve pressure

Second-line: **Pericardial window** if persistent tamponade

Definitive: **Surgical repair** if trauma-related

Correct Answer:
C. Cardiac tamponade ✅

MCQ 763

A young man was brought to the Emergency Department after he was hit in the chest while playing football. On examination, he was dyspnoeic and drowsy with GCS 11/15. Tracheal shifting to the left and raised JVP were noted. He has barely audible breath sounds on the right side.

Blood pressure: 80/50 mmHg
Heart rate: 130 /min
Respiratory rate: 32 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most likely diagnosis?

A. Flail segment
B. Cardiac tamponade
C. Pulmonary contusions
D. Tension pneumothorax

Explanation:

D. Tension pneumothorax is the most likely diagnosis. The combination of **tracheal shift**, **raised JVP**, **hypotension**, and **diminished breath sounds on the right** suggests a **tension pneumothorax**, where air is trapped in the pleural space, causing lung collapse and mediastinal shift.

Topic Section:

Tension Pneumothorax

Definition:
Tension pneumothorax occurs when air enters the pleural space and is unable to escape, leading to increased pressure on the lung and mediastinal structures, impairing respiratory and circulatory function.

Key Clinical Presentation:

Dyspnea and **hypoxia**

Tachycardia and **hypotension**

Tracheal deviation away from the affected side

Diminished breath sounds on the affected side

Investigations:

Best initial: **Clinical examination** with **needle decompression**

Confirmatory: **Chest X-ray** or **ultrasound**

Treatment Outline:

First-line: **Needle decompression** followed by **chest tube insertion**

Second-line: **Chest tube** for ongoing air leak

Definitive: **Surgical intervention** if persistent

Correct Answer:
D. Tension pneumothorax ✅

MCQ 764

A 70-year-old alcoholic complains of a 6-month history of gradual progressive dysphagia and weight loss. On examination, he is cachectic and pale. His abdomen is soft and non-tender. Digital rectal examination confirms altered coloured stool in the rectum.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Values:
WBC: 9.5 (4.5-10.5 x 109/L)
Hb: 90 (130-170 g/L)
Urea: 8 (2.75-7.4 mmol/L)
Creatinine: 120 (44-115 µmol/L)

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Oesophageal cancer
C. Acid peptic disease
D. Duodenal perforation

Explanation:

B. Oesophageal cancer is the most likely diagnosis given the patient's **gradual progressive dysphagia** and **weight loss**, which are hallmark symptoms of esophageal cancer. The patient's history of **alcoholism** is a significant risk factor for this condition.

Topic Section:

Esophageal Cancer

Definition:
Esophageal cancer is a malignant tumor that originates in the esophagus. It typically presents with symptoms related to obstruction or invasion of nearby structures.

Key Clinical Presentation:

Progressive dysphagia

Weight loss

Anorexia

Altered colored stool (could indicate bleeding)

Investigations:

Best initial: **Endoscopy** with **biopsy**

Confirmatory: **CT scan** of the chest and abdomen

Treatment Outline:

First-line: **Surgical resection** if feasible

Second-line: **Chemotherapy** and/or **radiotherapy**

Definitive: **Multidisciplinary approach** with surgery, chemotherapy, and radiotherapy

---

Correct Answer:
B. Oesophageal cancer ✅

MCQ 765

A 40-year-old man, alcoholic and smoker, is brought to the Emergency Department with a 4-hour history of severe upper abdominal pain radiating to the back. On examination, his abdomen is distended, tense, and tender. Digital Rectal examination confirms a collapsed empty rectum.

Blood pressure: 100/70 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 37 °C

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Pneumothorax
C. Duodenal perforation
D. Acalculous cholecystitis

Explanation:

A. Pancreatitis is the most likely diagnosis. The patient's **severe upper abdominal pain**, radiating to the back, along with **distension** and **tenderness**, points to **acute pancreatitis**, especially given the history of **alcohol consumption**. The collapsed rectum suggests no distal bowel obstruction, which is typical for pancreatitis.

Topic Section:

Acute Pancreatitis

Definition:
Acute pancreatitis is inflammation of the pancreas, often caused by gallstones or alcohol use, leading to digestive enzyme activation and pancreatic injury.

Key Clinical Presentation:

Severe upper abdominal pain radiating to the back

Nausea and **vomiting**

Abdominal distension

Tenderness on palpation

Raised amylase levels

Investigations:

Best initial: **Serum amylase** and **lipase**

Confirmatory: **CT scan** of the abdomen

Treatment Outline:

First-line: **Supportive care** (hydration, analgesia, fasting)

Second-line: **ERCP** if gallstones are the cause

Definitive: **Surgical management** for complications or if the cause is due to pancreatitis-related complications

---

Correct Answer:
A. Pancreatitis ✅

MCQ 766

A 55-year-old man was admitted with a 2-year history of recurrent upper abdominal pain attacks. History confirmed a recurrent colicky pain aggravated by morphine or its analogues.

Blood pressure: 110/70 mmHg
Heart rate: 96 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Values:
Indirect bilirubin: 5 (3.2-12.1 µmol/L)
Direct bilirubin: 10 (1.5-6.5 µmol/L)
Total bilirubin: 15 (3.5-16.5 µmol/L)
Alkaline phosphatase: 250 (39-117 IU/L)
Amylase: 150 (24-151 IU/L)

Ultrasound abdomen:
Prominent common bile duct along with minimal dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Pancreatitis
B. Biliary colic
C. Cholecystitis
D. Sphincter of Oddi dysfunction

Explanation:

D. Sphincter of Oddi dysfunction is the most likely diagnosis. The patient's **recurrent colicky pain**, **increased bilirubin**, and the history of morphine triggering pain point to **Sphincter of Oddi dysfunction**. This condition is characterized by impaired function of the sphincter of Oddi, which can cause intermittent biliary obstruction.

Topic Section:

Sphincter of Oddi Dysfunction

Definition:
Sphincter of Oddi dysfunction is a disorder of the biliary system caused by impaired relaxation or contraction of the sphincter of Oddi, leading to bile flow abnormalities.

Key Clinical Presentation:

Colicky abdominal pain, often postprandial

Pain exacerbated by morphine

Jaundice (if biliary obstruction occurs)

Investigations:

Best initial: **Ultrasound** or **MRCP**

Confirmatory: **Manometry of the sphincter of Oddi**

Treatment Outline:

First-line: **Endoscopic sphincterotomy**

Second-line: **Medication for pain relief**

Definitive: **Surgical intervention** if other treatments fail

---

Correct Answer:
D. Sphincter of Oddi dysfunction ✅

MCQ 767

A 20-year-old woman recently had an infant complains that she has been having recurrent episodes of upper abdominal pain since the third trimester. The pain radiates to her back and is associated with nausea. On examination, her abdomen is soft and non-tender. On deep palpation, there is a mass measuring 8 x 8 cm in the epigastric region, which appears to be non-pulsatile and non-tender.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Test Result Normal Value:
Lipase: 250 (0-160 IU/L)

Which of the following is the most likely diagnosis?

A. Hydatid cyst
B. Gall bladder mucocele
C. Pancreatic pseudocyst
D. Gastrointestinal stromal tumour

Explanation:

C. Pancreatic pseudocyst is the most likely diagnosis. The patient has **upper abdominal pain**, **nausea**, and a **non-pulsatile mass** in the epigastric region. The elevated **lipase** supports a pancreatic origin, making **pancreatic pseudocyst** the most likely diagnosis.

Topic Section:

Pancreatic Pseudocyst

Definition:
A pancreatic pseudocyst is a fluid-filled collection in or around the pancreas, usually following acute pancreatitis, that does not have an epithelial lining.

Key Clinical Presentation:

Epigastric pain, nausea

Mass in the epigastric region

Elevated lipase/amylase

Non-tender and non-pulsatile mass

Investigations:

Best initial: **Abdominal ultrasound** or **CT scan**

Confirmatory: **Endoscopic ultrasound** or **MRI**

Treatment Outline:

First-line: **Conservative management** with observation

Second-line: **Percutaneous drainage** for symptomatic pseudocysts

Definitive: **Surgical resection** for large or complicated pseudocysts

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Correct Answer:

C. Pancreatic pseudocyst ✅

MCQ 768

A 30-year-old patient presents to the Emergency Department with an 8-hour history of severe upper abdominal pain that is not relieved by oral medications. History confirmed recurrent similar events over the last year that were relieved by anti-spasmodic and anti-inflammatory drugs. Examination revealed jaundice, with a soft abdomen and marked tenderness in the right hypochondriac region.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 37.9 °C

Test Result Normal Values:
WBC: 16 (4.5-10.5 x 109/L)
Indirect bilirubin: 5 (3.2-12.1 µmol/L)
Direct bilirubin: 15 (1.5-6.5 µmol/L)
Total bilirubin: 20 (3.5-16.5 µmol/L)
Alkaline phosphatase: 400 (39-117 IU/L)
Amylase: 115 (24-151 IU/L)

Ultrasound:
Minimal CBD of 1.1 cm with dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Hepatitis
B. Cholangitis
C. Pancreatitis
D. Cholecystitis

Explanation:

B. Cholangitis is the most likely diagnosis given the patient's **jaundice**, **right upper quadrant tenderness**, and **dilated intrahepatic ducts** seen on ultrasound. The elevated **bilirubin** and **alkaline phosphatase** suggest obstruction of the biliary tree, which is characteristic of **cholangitis**.

Topic Section:

Cholangitis

Definition:
Cholangitis is an infection of the biliary system, often due to bile duct obstruction, typically caused by gallstones, strictures, or malignancy.

Key Clinical Presentation:

Fever

Jaundice

Right upper quadrant pain

Tachycardia

Tachypnea

Investigations:

Best initial: **Ultrasound**

Confirmatory: **ERCP** or **MRCP**

Treatment Outline:

First-line: **IV antibiotics** (e.g., piperacillin-tazobactam)

Second-line: **Endoscopic retrograde cholangiopancreatography (ERCP)** for biliary drainage

Definitive: **Surgical intervention** if necessary

Correct Answer:
B. Cholangitis ✅

MCQ 769

A 45-year-old patient complains of recurrent episodes of upper abdominal pain for the past 2 weeks with fever. On examination, the patient is jaundiced, abdomen is soft with marked tenderness all over the abdomen.

Blood pressure: 90/60 mmHg
Heart rate: 130 /min

Respiratory rate: 18 /min
Temperature: 38 °C

Test Result Normal Values:
Indirect bilirubin: 5.3 (3.2-12.1 µmol/L)
Direct bilirubin: 20.1 (1.5-6.5 µmol/L)
Total bilirubin: 25.3 (3.5-16.5 µmol/L)
Alkaline phosphatase: 450 (39-117 IU/L)
Amylase: 1400 (24-151 IU/L)

Ultrasound:
CBD of 1.4 cm with dilatation of intrahepatic ducts.

Which of the following is the most likely diagnosis?

A. Hepatitis
B. Cholangitis
C. Pancreatitis
D. Acute Cholecystitis

Explanation:

B. Cholangitis is the most likely diagnosis based on the patient's **jaundice**, **fever**, and **elevated bilirubin**. The **dilated common bile duct** and elevated **amylase** point toward a **biliary obstruction**, which could be complicated by infection.

Topic Section:

Cholangitis

Definition:
Cholangitis is an infection of the bile ducts, often caused by bacterial infections secondary to biliary obstruction.

Key Clinical Presentation:

Fever

Jaundice

Right upper quadrant pain

Tachycardia

Tachypnea

Investigations:

Best initial: **Ultrasound**

Confirmatory: **ERCP** or **MRCP**

Treatment Outline:

First-line: **IV antibiotics** (e.g., piperacillin-tazobactam)

Second-line: **Endoscopic retrograde cholangiopancreatography (ERCP)** for biliary drainage

Definitive: **Surgical intervention** if necessary

Correct Answer:
B. Cholangitis ✅

MCQ 770:

A 20-year-old woman presents with a lump in the right lower quadrant of her breast that has been present for the last 2 years. The lump was initially small but over the past 6 months has constantly been growing in size. On examination, the mass measures 8 x 10 cm, is mobile and not attached to underlying muscles. Overlying skin is thinned out due to pressure effect of the mass. There are no signs of surrounding inflammation or infection.

Which of the following is the most likely diagnosis?

A. Mastitis
B. Phyllodes
C. Galactocele
D. Fibroadenoma

Explanation:

B. Phyllodes is the most likely diagnosis given the **rapid growth**, **large size**, and **mobile mass** in a young woman. **Phyllodes tumors** are fibro-epithelial lesions that can grow rapidly and are often benign but may have malignant potential.

Topic Section:

Phyllodes Tumor

Definition:
Phyllodes tumors are rare fibroepithelial breast tumors that can be benign or malignant, characterized by rapid growth and a leaf-like architecture on histology.

Key Clinical Presentation:

Painless mass

Rapid growth

Non-tender, mobile

May have overlying skin changes

No inflammatory signs

Investigations:

Best initial: **Ultrasound**

Confirmatory: **Core needle biopsy** or **excision biopsy**

Treatment Outline:

First-line: **Wide excision**

Second-line: **Lumpectomy** or **mastectomy** for larger tumors or malignant variants

---

Correct Answer:
B. Phyllodes ✅

MCQ 771:

A 30-year-old patient was brought to the Emergency Department after being shot in the chest 1 hour ago. A tube thoracostomy on the left side was done for a pneumothorax. After 15 minutes, the underwater seal bottle is filled with 2 liters of blood.

Blood pressure: 90/60 mmHg
Heart rate: 120 /min

Respiratory rate: 22 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. Thoracotomy
B. CT scan chest
C. Tube thoracostomy
D. Angiographic embolization

Explanation:

A. Thoracotomy is the most appropriate management in this case of **massive hemothorax** with **persistent blood drainage** from the chest tube. A **thoracotomy** allows for control of bleeding and evacuation of the hemothorax.

Topic Section:

Massive Hemothorax

Definition:
Massive hemothorax is the accumulation of a large amount of blood in the pleural cavity, typically due to trauma.

Key Clinical Presentation:

Hypotension

Tachycardia

Breathlessness

Cyanosis

Shock

Investigations:

Best initial: **Chest X-ray**

Confirmatory: **CT scan** of the chest

Treatment Outline:

First-line: **Tube thoracostomy** for blood drainage

Second-line: **Thoracotomy** for ongoing bleeding

Definitive: **Surgical intervention** (e.g., embolization, thoracotomy)

Correct Answer:
A. Thoracotomy ✅

MCQ 772:

A 5-year-old child is brought to the Emergency Department after being stabbed in the right lower chest. His abdomen is distended. A focused abdominal sonogram for trauma is suggestive of free fluid in the abdomen.

Blood pressure: 100/80 mmHg
Heart rate: 110 /min
Respiratory rate: 20 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. Thoracotomy
B. Tube thoracostomy
C. Exploratory laparotomy
D. Angiographic embolization

Explanation:

C. Exploratory laparotomy is the most appropriate management for a child with **blunt abdominal trauma** and **free fluid** seen on sonogram, as it indicates possible **intra-abdominal injury**, requiring surgical exploration.

Topic Section:

Trauma Management in Children

Definition:
Trauma management in children involves rapid assessment and stabilization, followed by targeted diagnostic imaging and appropriate surgical intervention if needed.

Key Clinical Presentation:

Trauma-related symptoms

Abdominal distension

Shock

Tachycardia, tachypnea, hypotension

Investigations:

Best initial: **Focused abdominal sonography** or **CT scan**

Confirmatory: **Exploratory laparotomy** if free fluid or solid organ injury is suspected

Treatment Outline:

First-line: **Initial resuscitation** (IV fluids, oxygen)

Second-line: **Exploratory laparotomy** or **conservative management**

Definitive: **Surgical repair** or **interventional radiology**

Correct Answer:
C. Exploratory laparotomy ✅

MCQ 774:

A 66-year-old man recently underwent an anterior resection of the rectum and is making a slow recovery. He complains that in the middle of the night he has pain in the left calf. On examination, he has a tender left calf, which is slightly swollen and red.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Ultrasound Doppler:
Deep vein thrombosis in the left popliteal region extending up to the common femoral junction.

Which of the following is the most appropriate management?

A. Heparin
B. Warfarin
C. IVC filter
D. Enoxaparin

Explanation:

D. Enoxaparin is the most appropriate initial treatment for **deep vein thrombosis (DVT)** in this patient. **Enoxaparin** is a low molecular weight heparin (LMWH), commonly used for the initial treatment of DVT to prevent further clot formation. This is the first-line treatment until transition to oral anticoagulants like warfarin, if necessary.

Topic Section:

Deep Vein Thrombosis (DVT)

Definition:
DVT refers to the formation of a thrombus within the deep veins, most commonly in the legs, which can cause complications such as pulmonary embolism.

Key Clinical Presentation:

Pain, swelling, redness, and warmth in the affected leg

Tenderness over the affected vein

Positive Homan's sign (not diagnostic)

Investigations:

Best initial: **Ultrasound Doppler**

Confirmatory: **CT pulmonary angiography** (if pulmonary embolism is suspected)

Treatment Outline:

First-line: **Enoxaparin** or **unfractionated heparin** for anticoagulation

Second-line: **Warfarin** for long-term anticoagulation

Definitive: **IVC filter** (in cases of contraindications to anticoagulation)

Correct Answer:
D. Enoxaparin ✅

MCQ 774

A 12-year-old boy presents to the Emergency Department after a fall that led to a supra-condylar fracture. Presently his distal pulses are not palpable.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min

Which of the following is the most appropriate management?

A. Early K-wire
B. Hand elevation
C. Watchful waiting
D. Surgical exploration

Explanation:

D. Surgical exploration is the most appropriate management for a **supra-condylar fracture** with **absent distal pulses**. In this case, the absence of pulses suggests potential vascular injury, and surgical exploration is required to evaluate and repair any damage to the arteries or nerves.

Topic Section:

Supracondylar Fracture of the Humerus

Definition:
A supracondylar fracture is a common elbow fracture in children, typically occurring due to a fall on an outstretched hand.

Key Clinical Presentation:

Swelling and deformity around the elbow

Absent or diminished distal pulses

Pain and limited movement of the elbow joint

Investigations:

Best initial: **X-ray** (AP and lateral views of the elbow)

Confirmatory: **CT scan** (if displacement or complex fractures are suspected)

Treatment Outline:

First-line: **Surgical exploration** for vascular or nerve injury

Second-line: **Closed reduction and percutaneous pinning** (if stable fracture)

Definitive: **Open reduction and internal fixation** for severe fractures or complications

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Correct Answer:
D. Surgical exploration ✅

MCQ 775

A 5-month-old baby was admitted with an inability to tolerate food and non-billous vomiting over the past 2-days. On examination, he appears to be dehydrated. There is an olive shaped mass palpable in the epigastric region measuring 2 x 2 cm.

Which of the following is the most appropriate management?

A. Pyloromyotomy
B. Balloon dilatation
C. Gastrojejunostomy

Explanation:

A. Pyloromyotomy is the treatment of choice for **pyloric stenosis** in infants. The presence of an **olive-shaped mass** in the epigastric region along with **non-bilious vomiting** is characteristic of pyloric stenosis, and **pyloromyotomy** is the definitive surgical treatment to relieve the obstruction.

Topic Section:

Pyloric Stenosis

Definition:
Pyloric stenosis is a condition in infants where the **pyloric sphincter** becomes thickened, leading to **gastric outlet obstruction**.

Key Clinical Presentation:

Non-bilious vomiting (often after feeding)

Olive-shaped mass in the epigastric region

Dehydration and weight loss

Investigations:

Best initial: **Ultrasound** (shows thickened pylorus)

Confirmatory: **Upper GI contrast study** (shows delayed gastric emptying)

Treatment Outline:

First-line: **Pyloromyotomy** (surgical division of the hypertrophied pyloric muscle)

Second-line: **Balloon dilatation** (if surgery is not possible)

Correct Answer:
A. Pyloromyotomy ✅

MCQ 776

A 45-year-old patient complains of pain and bilateral groin swelling over the last 6-months. On examination, there are small bilateral inguino-scrotal swellings. Cough impulse is positive, and his abdomen is soft and non-tender.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C
Oxygen saturation: 95 %
BMI: 30 kg/m²

Which of the following is the most appropriate management?

A. Herniotomy
B. Lichtenstein repair
C. Laparoscopic mesh repair
D. Biological mesh placement

Explanation:

B. Lichtenstein repair is the most appropriate management for an **inguinal hernia**. This **open surgical repair** involves the placement of a **mesh** to reinforce the abdominal wall and prevent recurrence.

Topic Section:

Inguinal Hernia

Definition:
An inguinal hernia occurs when a portion of the intestine or fat protrudes through the abdominal wall or into the inguinal canal.

Key Clinical Presentation:

Groin swelling (that becomes more prominent with coughing or straining)

Cough impulse positive

Painful or painless mass

Investigations:

Best initial: **Physical examination**

Confirmatory: **Ultrasound** or **CT scan** (if complications or doubts)

Treatment Outline:

First-line: **Hernia repair** (Lichtenstein or laparoscopic mesh repair)

Second-line: **Watchful waiting** in asymptomatic patients

Definitive: **Surgical intervention** to prevent complications

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Correct Answer:

B. Lichtenstein repair ✅

MCQ 777

A 5-month-old baby is presented with bilateral groin swelling that has been present since birth. On examination, there are small bilateral inguino-scrotal swellings, which become prominent when the child cries. His abdomen is soft and non-tender.

Which of the following is the most appropriate management?

A. Herniotomy
B. Lichtenstein repair
C. Laparoscopic mesh repair
D. Biological mesh placement

Explanation:

A. Herniotomy is the most appropriate management for **congenital inguinal hernia** in an infant. The hernia needs to be repaired surgically, and **herniotomy** involves the removal of the hernial sac.

Topic Section:

Inguinal Hernia in Children

Definition:
An inguinal hernia in children is a common congenital condition where a portion of the intestine or fat protrudes through the inguinal canal.

Key Clinical Presentation:

Bilateral groin swelling that becomes more prominent with crying

Soft, non-tender abdomen

Visible mass in the scrotum or groin area

Investigations:

Best initial: **Physical examination**

Confirmatory: **Ultrasound** (to confirm contents of the hernia)

Treatment Outline:

First-line: **Herniotomy** (surgical removal of the hernial sac)

Second-line: **Laparoscopic repair** (in specific cases, e.g., recurrent hernias)

Correct Answer:
A. Herniotomy ✅

MCQ 778:

A 42-year-old woman complains of recurrent episodes of upper abdominal pain over the past 2 weeks with associated high-grade fever. On examination, she looks icteric. Her abdomen is soft with marked tenderness in the right hypochondriac region.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Test Result Normal Values:
Indirect bilirubin 10 3.2-12.1 µmol/L
Direct bilirubin 20 1.5-6.5 µmol/L
Total bilirubin 30 3.5-16.5 µmol/L
Alkaline phosphatase 250 39-117 IU/L
Amylase 100 24-151 IU/L
WBC 20 4.5-10.5 x 109/L

Which of the following is the most appropriate management?

A. CBD exploration
B. Laparoscopic cholecystectomy
C. Percutaneous cholecystostomy
D. Endoscopic retrograde cholangiopancreatography

Explanation:

D. Endoscopic retrograde cholangiopancreatography (ERCP) is the most appropriate management in this patient with **cholangitis**, as indicated by the presence of jaundice, fever, elevated bilirubin, and white blood cells. **ERCP** is both diagnostic and therapeutic, allowing for the removal of bile duct stones and drainage.

Topic Section:

Cholangitis

Definition:
Cholangitis is the inflammation of the bile ducts, commonly caused by bacterial infection due to a biliary obstruction.

Key Clinical Presentation:

Right upper quadrant pain, fever, and jaundice (Charcot's triad)

Elevated bilirubin and alkaline phosphatase

Investigations:

Best initial: **Ultrasound abdomen**

Confirmatory: **ERCP** for bile duct stone extraction and drainage

Treatment Outline:

First-line: **Antibiotics** (IV broad-spectrum) and **ERCP**

Second-line: **Cholecystectomy** (after infection resolution)

Definitive: **Cholecystectomy** + **CBD exploration** (if the source of obstruction is identified)

Correct Answer:
D. Endoscopic retrograde cholangiopancreatography ✅

MCQ 779:

A 45-year-old patient complains of perianal swelling, fresh bleeding per-rectum, and pain over the last 3 months. On examination, there is a mass 1 cm from the anal verge. She also has obstructive symptoms.

Blood pressure: 110/70 mmHg
Heart rate: 96 /min

Respiratory rate: 18 /min
Temperature: 36.6 °C

Biopsy:
Adenocarcinoma

MRI abdomen:
Localized lesion with cranio-caudal extension of 3 cm with associated lymphadenopathy

CT scan chest:
No evidence of metastasis

Which of the following is the most appropriate treatment?

A. Diversion colostomy
B. Low anterior resection
C. Concurrent chemoradiation
D. Abdominoperineal resection

Explanation:

C. Concurrent chemoradiation is the most appropriate treatment for locally advanced **anal adenocarcinoma**. In cases where surgical resection is not immediately appropriate due to the extent of disease or lymph node involvement, chemoradiation is often used first to downstage the tumor before surgery.

Topic Section:

Anal Cancer (Adenocarcinoma)

Definition:
Anal carcinoma is a malignant tumor arising from the anal canal, most commonly **squamous cell carcinoma**, though **adenocarcinomas** can also occur.

Key Clinical Presentation:

Perianal mass, bleeding, and pain

Obstructive symptoms

Lymphadenopathy

Investigations:

Best initial: **Proctoscopy and biopsy**

Confirmatory: **MRI abdomen** (for local extension and lymph node involvement)

Treatment Outline:

First-line: **Concurrent chemoradiation** for anal adenocarcinoma

Second-line: **Low anterior resection** (if suitable for surgery post-chemoradiation)

Definitive: **Abdominoperineal resection** (in cases of extensive local disease)

Correct Answer:
C. Concurrent chemoradiation ✅

MCQ 780:

A 30-year-old woman presents with a mass in the right upper quadrant of her breast for the last 4 years. According to the patient, the mass was initially small but over the past 2 years, it has been constantly growing in size. On examination, the mass measures 15 x 15 cm, is mobile and not attached to underlying muscle. Overlying skin is thinned out due to the mass effect.

Ultrasound breast:
Solid mass containing single or multiple, round or cleft-like cystic spaces and demonstrating posterior acoustic enhancement.

Which of the following is the most appropriate management?

A. Radiation therapy
B. Simple mastectomy
C. Neoadjuvant chemotherapy
D. Modified radical mastectomy

Explanation:

B. Simple mastectomy is the most appropriate management for a **breast mass** identified as likely **fibroadenoma**. The described **solid mass** with cystic components and acoustic enhancement on ultrasound points to this benign condition, and excision is typically curative.

Topic Section:

Fibroadenoma of the Breast

Definition:
Fibroadenomas are the most common benign tumors of the breast, usually affecting women in their 20s and 30s.

Key Clinical Presentation:

Painless, firm, well-defined, mobile mass

Overlying skin not affected

No lymphadenopathy

Investigations:

Best initial: **Ultrasound**

Confirmatory: **Core needle biopsy** (to differentiate from other lesions)

Treatment Outline:

First-line: **Excisional biopsy or simple mastectomy** for large or symptomatic fibroadenomas

Second-line: **Observation** if small and asymptomatic

Correct Answer:
B. Simple mastectomy ✅

MCQ 781:

A young woman underwent a laparoscopic cholecystectomy for symptomatic gallstones 1 day ago. Post-operatively, there are decreased breath sounds in the lower chest on auscultation.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C
Oxygen saturation: 90 %

Which of the following is the most appropriate management?

A. IV antibiotics
B. Bronchodilators
C. Ultrasound chest
D. Incentive spirometry

Explanation:

D. Incentive spirometry is the most appropriate management for **atelectasis** following surgery. It is a common post-operative complication, particularly after laparoscopic procedures, where shallow breathing can lead to **lung collapse** in the lower lung zones. **Incentive spirometry** helps improve lung expansion and oxygenation.

Topic Section:

Post-operative Atelectasis

Definition:
Atelectasis refers to the collapse of part or all of a lung due to alveolar deflation, which can occur after surgery, especially in patients with shallow breathing.

Key Clinical Presentation:

Decreased breath sounds on auscultation, particularly in the lower lobes

Shallow breathing and **tachypnea**

Hypoxia

Investigations:

Best initial: **Chest X-ray**

Confirmatory: **CT chest** (if there is suspicion of a more severe process)

Treatment Outline:

First-line: **Incentive spirometry**

Second-line: **Postural drainage** and **deep breathing exercises**

Definitive: **Chest physiotherapy** and **oxygen therapy** if required

---

Correct Answer:

D. Incentive spirometry ✅

MCQ 782::

A 25-year-old man underwent an open appendectomy for a perforated appendix. He was discharged on the 2nd post-operative day uneventfully. He returned on the 8th post-operative day complaining of pain and redness around the wound site. On examination, there is purulent discharge from the wound site. His abdomen is soft with tenderness over the wound site. Digital rectal examination was unremarkable.

Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate management?

A. IV antibiotics
B. Open drainage
C. CT scan abdomen
D. Exploratory laparotomy

Explanation:

A. IV antibiotics are the most appropriate management for **post-surgical wound infection** in this case. The patient presents with signs of a wound infection (pain, redness, purulent discharge) without signs of intra-abdominal pathology. **IV antibiotics** should be started while considering wound care and monitoring for further signs of complications.

Topic Section:

Post-operative Infections

Definition:
Infections that occur after surgery, typically due to bacterial colonization of the wound or internal sites.

Key Clinical Presentation:

Redness, pain, and purulent discharge at the wound site

Fever

Tachycardia

Investigations:

Best initial: **Wound culture**

Confirmatory: **CT scan abdomen** (if concerned about abscess formation or intra-abdominal complications)

Treatment Outline:

First-line: **IV antibiotics**

Second-line: **Wound care** and **drainage** if indicated

Definitive: **Exploratory surgery** if there are signs of abscess formation or further complications

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Correct Answer:
A. IV antibiotics ✅

MCQ 783:

A 55-year-old man underwent an open appendectomy for a perforated appendix. He presents to the Emergency Department on the 6th post-operative day complaining of severe abdominal pain, distension, and an inability to pass stool. On examination, there is feculent discharge from the wound site with signs of peritonitis.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Oxygen saturation: 95 %

Test Result Normal Values:
ABG HCO3- 10 22-28 mmol/L
ABG PCO2 4.8 4.7-6.0 kPa
pH 7.25 7.36-7.45
ABG PO2 11 10.6-14.2 kPa
Base access -12 -2 to 2 mmol/L

Ultrasound abdomen:
Moderate ascites with echoes and septations.

Which of the following is the most appropriate management?

A. Wound care
B. IV antibiotics
C. CT scan abdomen
D. Exploratory laparotomy

Explanation:

D. Exploratory laparotomy is the most appropriate management. The patient presents with signs of **peritonitis** (feculent discharge, abdominal distension, and pain) and a significant metabolic disturbance (low bicarbonate, acidosis). These findings are consistent with a **severe intra-abdominal infection** or **peritonitis**, necessitating an urgent **exploratory laparotomy** to identify and treat the source of the infection, such as a leaking anastomosis or abscess formation.

Topic Section:

Post-operative Peritonitis

Definition:
Infection and inflammation of the peritoneal cavity following surgery, often due to leakage or bacterial contamination.

Key Clinical Presentation:

Severe abdominal pain, distension, and feculent discharge from a surgical wound

Signs of systemic infection (tachycardia, fever)

Metabolic acidosis (low bicarbonate, base deficit)

Peritonitis on clinical examination

Investigations:

Best initial: **Ultrasound abdomen**

Confirmatory: **CT scan abdomen** or **exploratory laparotomy**

Treatment Outline:

First-line: **IV antibiotics** and **supportive care**

Second-line: **Exploratory laparotomy** to identify and manage the source of infection

Definitive: **Surgical intervention** (e.g., reoperation to repair leaks or drain abscesses)

Correct Answer:
D. Exploratory laparotomy ✅

MCQ 784:

A 45-year-old patient complains of a high-grade fever with chills and multiple spikes on the 4th post-operative day after a diverticular perforation surgery. On examination, his wound is healthy and his chest is clear to auscultation. On Digital rectal examination, there is bogginess anteriorly.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate treatment?

A. IV antibiotics
B. IV paracetamol
C. Abdominal cavity washout
D. Ultrasound guided drainage

Explanation:

A. IV antibiotics is the most appropriate treatment for **post-surgical infection** suspected to be intra-abdominal. The patient has a post-surgical fever, and the presence of **bogginess** on digital rectal examination suggests possible **intra-abdominal infection** or **abscess formation**. IV antibiotics should be started immediately, and further imaging or drainage may be required based on clinical progression.

Topic Section:

Post-surgical Infections (Intra-abdominal)

Definition:
Infections that arise after surgery, most commonly from leakage or contamination of the abdominal cavity.

Key Clinical Presentation:

Fever and chills after surgery

Boggy areas on digital examination or tenderness

Tachycardia and systemic signs of infection

Investigations:

Best initial: **Ultrasound abdomen** or **CT scan abdomen**

Confirmatory: **Drainage and culture** of peritoneal fluid if needed

Treatment Outline:

First-line: **IV antibiotics** and **supportive care**

Second-line: **Intra-abdominal drainage** (either percutaneous or surgical)

Definitive: **Exploratory laparotomy** or **washout** if needed to manage the infection source

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Correct Answer:
A. IV antibiotics ✅

MCQ 785

A 55-year-old man with no known co-morbidities underwent a Graham's patch repair for a perforated duodenal ulcer 10 days ago. On examination, his abdomen is soft and non-tender. On auscultation, there is decreased breath sounds on the right lower chest. Blood cultures obtained from the central and peripheral line of the patient are growing gram-positive cocci.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 18 /min
Temperature: 38 °C

Which of the following is the most appropriate treatment?

A. Chest physiotherapy
B. Removal of central line
C. Abdominal cavity washout
D. Ultrasound guided drainage

Explanation:

B. Removal of central line is the most appropriate treatment for suspected **central venous catheter-related infection** with **gram-positive cocci**. The patient has a **central venous line infection**, which is suggested by the growth of bacteria from both the central and peripheral blood cultures. The first step in management is to **remove the central line** and initiate appropriate **antibiotic therapy**.

Topic Section:

Central Line Infections

Definition:
Infection associated with the use of a central venous catheter, often caused by bacterial colonization at the insertion site or within the catheter.

Key Clinical Presentation:

Fever and chills after central line insertion

Positive blood cultures from **both central and peripheral lines**

Tachycardia and systemic signs of infection

Investigations:

Best initial: **Blood cultures** from both central and peripheral lines

Confirmatory: **Catheter tip culture** and imaging for potential complications

Treatment Outline:

First-line: **Removal of the central line** and **IV antibiotics**

Second-line: **Ultrasound or CT** if abscess formation is suspected

Definitive: **Surgical removal** if infection persists or complications arise

Correct Answer:
B. Removal of central line ✅

MCQ 786:

A patient that underwent a cholecystectomy earlier has pyrexia with temperature spikes. On examination, the wound is healthy. His abdomen is soft with mild tenderness in the right hypochondriac region. There is decreased air entry on the right side of the chest on auscultation.

Blood pressure: 110/70 mmHg
Heart rate: 120 /min
Respiratory rate: 20 /min
Temperature: 38.5 °C

Ultrasound: Collection of 10 x 15 cm in the gall bladder fossa.

Which of the following is the most appropriate management?

A. IV antibiotics
B. Chest physiotherapy
C. Ultrasound guided drainage
D. Endoscopic retrograde cholangiopancreatography

Explanation:

C. Ultrasound guided drainage is the most appropriate management. The patient has a **post-surgical collection** in the gall bladder fossa, most likely a **bile collection** or abscess. The management of this is **ultrasound-guided drainage**, which helps relieve the collection and prevent further complications. **IV antibiotics** may be started if there is suspicion of infection but drainage is key for proper resolution.

Topic Section:

Post-operative Abscess Management

Definition:
Abscess formation in the abdominal cavity following surgery, commonly after procedures like cholecystectomy or appendectomy.

Key Clinical Presentation:

Fever and **temperature spikes**

Localized tenderness and **decreased breath sounds** (if associated with pleural involvement)

Post-operative history of surgery (e.g., cholecystectomy)

Investigations:

Best initial: **Ultrasound abdomen**

Confirmatory: **CT scan abdomen** if further information is required

Treatment Outline:

First-line: **Ultrasound-guided drainage**

Second-line: **IV antibiotics** if infection is suspected

Definitive: **Surgical drainage** if percutaneous drainage fails or complications arise

---

Correct Answer:
C. Ultrasound guided drainage ✅

MCQ 787:

A 25-year-old underwent an open appendectomy for a perforated appendix 8 days ago and returns for follow-up. On examination, there is a seroma formation at the wound site, which is draining freely through a gaping wound. There are no signs of surrounding inflammation.

Blood pressure: 110/70 mmHg
Heart rate: 76 /min
Respiratory rate: 18 /min
Temperature: 36.6 °C

Which of the following is the most appropriate treatment?

A. Antibiotics
B. CT scan abdomen
C. Regular wound dressings
D. Ultrasound guided drainage

Explanation:

C. Regular wound dressings is the most appropriate treatment. This patient has **seroma formation** following appendectomy, which is a common post-operative complication. **Regular wound dressings** with proper hygiene will allow the seroma to resolve. There are no signs of infection or further complications, so antibiotics or drainage are unnecessary.

Topic Section:

Post-operative Seroma Management

Definition:
A **seroma** is a collection of **serous fluid** in a tissue cavity, usually occurring after surgery, particularly in areas with potential space or low tissue adherence.

Key Clinical Presentation:

Swelling at the surgical site

Drainage of clear fluid (serous) from the wound

No signs of infection (e.g., redness, fever)

Investigations:

Best initial: **Clinical examination**

Confirmatory: **Ultrasound** if there is concern about abscess or other collections

Treatment Outline:

First-line: **Regular wound dressings**

Second-line: **Compression bandage** or **aspiration** if the seroma persists

Definitive: **Surgical intervention** if seroma recurs or persists

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Correct Answer:
C. Regular wound dressings ✅

MCQ 788:

Which of the following is most likely involved in neuro-hormonal stress response?

A. Androgen
B. Parathyroid
C. Thyroid stimulating
D. Corticotrophin releasing

Explanation:

D. Corticotrophin releasing is the correct answer. The **neuro-hormonal stress response** involves the **hypothalamic-pituitary-adrenal (HPA) axis**, where the release of **corticotrophin-releasing hormone (CRH)** from the hypothalamus triggers the release of **ACTH** from the pituitary, leading to the secretion of **cortisol** from the adrenal glands. Cortisol is crucial in managing the body's stress response.

Topic Section:

Neuro-hormonal Stress Response

Definition:
The body's physiological response to stress involves the activation of the **sympathoadrenal system** and the **hypothalamic-pituitary-adrenal (HPA) axis**, leading to increased production of cortisol, catecholamines, and other hormones.

Key Clinical Presentation:

Increased heart rate, **blood pressure**

Elevated blood glucose

Suppression of non-essential functions (e.g., reproduction, digestion)

Investigations:

Best initial: **Serum cortisol level**

Confirmatory: **ACTH stimulation test** for adrenal insufficiency

Treatment Outline:

First-line: **Supportive care** in acute stress

Second-line: **Steroid therapy** if chronic adrenal insufficiency (Addison's disease) is suspected

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Correct Answer:

D. Corticotrophin releasing ✅

MCQ 789:

A 30-year-old woman underwent an exploratory laparotomy for a perforated duodenal ulcer. Post-operatively she developed septic shock syndrome.

Which of the following is the most common physiological response?

A. Hypokalaemia
B. Respiratory acidosis
C. Cellular anaerobic respiration
D. Increase glomerular filtration rate

Explanation:

C. Cellular anaerobic respiration is the correct answer. In **septic shock**, tissue hypoperfusion leads to **anaerobic metabolism**, resulting in the production of **lactic acid** and metabolic **acidosis**. This process reflects inadequate oxygen delivery to tissues, and the body shifts from aerobic to anaerobic pathways.

Topic Section:

Septic Shock

Definition:
A life-threatening condition caused by widespread infection, resulting in systemic inflammation and **impaired perfusion** to vital organs.

Key Clinical Presentation:

Hypotension, **tachycardia**, and **oliguria**

Fever, **tachypnea**, and **altered mental status**

Investigations:

Best initial: **Blood cultures**

Confirmatory: **Lactate levels** to assess severity

Treatment Outline:

First-line: **Fluid resuscitation** and **broad-spectrum antibiotics**

Second-line: **Vasopressors** if blood pressure does not respond to fluids

Definitive: **Source control** (e.g., drainage of abscess)

Correct Answer:
C. Cellular anaerobic respiration ✅

MCQ 790:

A 65-year-old patient is intubated in the Intensive Care Unit because of septic shock.

Which of the following will best indicate adequate systemic perfusion?

A. Cardiac index
B. Central venous pressure
C. Mixed venous oxygen saturation
D. Pulmonary capillary wedge pressure

Explanation:

C. Mixed venous oxygen saturation is the best indicator of **adequate systemic perfusion** in critically ill patients, especially those in shock. **Mixed venous oxygen saturation** (SvO2) reflects the balance between oxygen delivery and oxygen consumption. A low SvO2 indicates insufficient oxygen delivery to tissues, which is a hallmark of inadequate perfusion.

Topic Section:

Monitoring Shock and Perfusion

Definition:
In critically ill patients, monitoring **perfusion status** is crucial for managing **shock** and preventing organ failure. SvO2, along with other clinical indicators, helps guide therapy.

Key Clinical Presentation:

Hypotension, **tachycardia**, and **altered mental status**

Decreased urine output and **cool extremities**

Investigations:

Best initial: **Central venous oxygen saturation** (ScvO2 or SvO2)

Confirmatory: **Lactate levels**, **cardiac output measurement**

Treatment Outline:

First-line: **Fluid resuscitation** and **vasopressors**

Second-line: **Inotropes** if cardiac output is low

Definitive: **Address the underlying cause** (e.g., source control in sepsis)

Correct Answer:

C. Mixed venous oxygen saturation ✅

MCQ 791

A 55-year-old is scheduled to undergo laparoscopic cholecystectomy.

Which of the following is the most appropriate prophylactic antibiotic regimen?

A. Regular dose for 5 days
B. 3 doses postoperatively
C. Single dose with anaesthesia induction
D. Single dose 1 hour before skin incision

Explanation:

D. Single dose 1 hour before skin incision is the most appropriate prophylactic antibiotic regimen. For **laparoscopic cholecystectomy**, a **single dose of antibiotics** administered **1 hour before the skin incision** is the standard practice to reduce the risk of infection, especially **biliary infections**. Prolonged use or multiple doses are typically not necessary unless there are specific risk factors.

Topic Section:

Surgical Prophylaxis in Cholecystectomy

Definition:
Prophylactic antibiotics are administered before surgery to prevent postoperative infections.

Key Clinical Presentation:

Elective surgery (e.g., cholecystectomy)

High-risk patients (e.g., immunocompromised)

Investigations:

Best initial: **Clinical evaluation**

Confirmatory: Not needed for routine procedures

Treatment Outline:

First-line: **Single dose of antibiotics** (1 hour before incision)

Second-line: **Additional doses** if high risk for infection or extended surgeries

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Correct Answer:
D. Single dose 1 hour before skin incision ✅

MCQ 792:

A 33-year-old man was brought to the Emergency Department after a high velocity trauma. He had distended neck veins and markedly decreased breath sounds on the right side.

Heart rate: 105 /min
Blood Pressure: 90/60 mmHg
Oxygen saturation: 85%

Which of the following is the most likely cause?

A. Flail chest
B. Right hemothorax
C. Pericardial tamponade
D. Right tension pneumothorax

Explanation:

D. Right tension pneumothorax is the most likely cause. The clinical features, such as **distended neck veins** and **decreased breath sounds** on one side, suggest **tension pneumothorax**. This condition occurs when air enters the pleural space and cannot escape, causing **increased intrathoracic pressure**, leading to **hypoxia**, **hemodynamic instability**, and **distended neck veins**.

Topic Section:

Tension Pneumothorax

Definition:
A life-threatening condition where air accumulates in the pleural cavity and cannot escape, leading to **increased intrathoracic pressure** and **cardiovascular compromise**.

Key Clinical Presentation:

Hypotension

Distended neck veins

Decreased breath sounds on the affected side

Tracheal deviation (late finding)

Investigations:

Best initial: **Clinical diagnosis** (e.g., physical examination, imaging if available)

Confirmatory: **Chest X-ray** or **ultrasound**

Treatment Outline:

First-line: **Needle decompression** followed by **chest tube insertion**

Second-line: **Surgical intervention** if necessary

Correct Answer:
D. Right tension pneumothorax ✅

MCQ 793:

Which of the following is the most likely cerebral perfusion pressure in mmHg?

A. 40
B. 50
C. 60
D. 70

Explanation:

C. 60 is the most likely cerebral perfusion pressure (CPP) in mmHg. **Cerebral perfusion pressure (CPP)** is typically between **60-80 mmHg** in healthy adults. A **CPP of 60 mmHg** is generally considered adequate for maintaining **cerebral blood flow**.

Topic Section:

Cerebral Perfusion Pressure

Definition:
Cerebral perfusion pressure (CPP) is the pressure gradient driving blood to the brain, defined as the **mean arterial pressure (MAP)** minus the **intracranial pressure (ICP)**.

Key Clinical Presentation:

Decreased consciousness

Signs of brain injury (e.g., post-traumatic brain injury)

Hypotension

Investigations:

Best initial: **Intracranial pressure (ICP) monitoring**

Confirmatory: **Head CT scan** to assess structural brain damage

Treatment Outline:

First-line: **Optimization of blood pressure**

Second-line: **Surgical decompression** if necessary

Correct Answer:
C. 60 ✅

MCQ 794

A 35-year-old woman presented with partial thickness burns on her arms and leg due to hot water spillage 1 day ago, with wound edema and congestion.

Which of the following is the most likely responsible mediator?

A. Serotonin
B. Bradykinin
C. Prostaglandin I2
D. Thromboxane A2

Explanation:

B. Bradykinin is the most likely mediator responsible for the edema and congestion seen in **partial thickness burns**. **Bradykinin** plays a major role in the **inflammatory response**, contributing to **vasodilation**, **increased vascular permeability**, and **edema**.

Topic Section:

Burns and Inflammation

Definition:
Partial thickness burns involve the **epidermis and part of the dermis**, leading to **edema**, **pain**, and **congestion** due to **inflammatory mediators** like bradykinin.

Key Clinical Presentation:

Painful, **red, blistered skin**

Edema and **congestion** around the burn site

Investigations:

Best initial: **Clinical assessment**

Confirmatory: **Burn depth and severity assessment** (e.g., **Lund-Browder chart**)

Treatment Outline:

First-line: **Fluid resuscitation** and **wound care**

Second-line: **Topical antimicrobial agents**

Definitive: **Surgical debridement** if necessary

Correct Answer:
B. Bradykinin ✅

MCQ 795

A 55-year-old diabetic woman with steroid-dependent bronchial asthma underwent a right hemicolectomy for carcinoma of the caecum. Postoperatively, she was kept in the Intensive Care Unit (see lab results).

Blood pressure: 80/40 mmHg
Heart rate: 140 /min
Respiratory rate: 25 /min
Temperature: 35.1 °C
Oxygen saturation: 95 %

Test Result Normal Values
Sodium: 131 134-146 mmol/L
Potassium: 6.0 3.5-5.1 mmol/L
Bicarbonate: 18 21-28 mmol/L
Random Glucose: 2.3 3.9-5.5 mmol/L

Which of the following is the most likely clinical condition?

A. Thyroid storm
B. Adrenal insufficiency
C. Diabetic ketoacidosis
D. Septic shock syndrome

Explanation:

B. Adrenal insufficiency is the most likely diagnosis. The patient is showing signs of **hypotension**, **electrolyte imbalances (elevated potassium, low sodium)**, **hypoglycemia**, and **fever**. These are suggestive of **adrenal insufficiency**, especially in a post-operative setting with steroid dependence, which can precipitate **acute adrenal crisis**.

Topic Section:

Adrenal Insufficiency (Adrenal Crisis)

Definition:
A potentially life-threatening condition caused by **insufficient cortisol production**, often due to **acute stress** or **steroid withdrawal**.

Key Clinical Presentation:

Hypotension, **shock**

Electrolyte abnormalities (e.g., **hyperkalemia**, **hyponatremia**)

Hypoglycemia, **fever**

Investigations:

Best initial: **Serum cortisol level**

Confirmatory: **ACTH stimulation test** (low cortisol response)

Treatment Outline:

First-line: **IV hydrocortisone** and **fluids**

Second-line: **Electrolyte correction** (e.g., potassium, sodium)

Definitive: **Chronic steroid replacement** (e.g., hydrocortisone)

Correct Answer:
B. Adrenal insufficiency ✅

MCQ 795

A 13-year-old boy presents for elective surgery. He has a significant past history of bleeding tendency and has contracture of right knee and hemarthrosis.

Which of the following is the most likely underlying clinical disorder?

A. Haemophilia
B. Aplastic anaemia
C. Wilson's disease
D. Henoch-Schönlein purpura

Explanation:

A. Haemophilia is the most likely diagnosis. **Haemophilia** is an inherited bleeding disorder due to deficiency of clotting factors (factor VIII or IX). It commonly presents with **hemarthrosis** (bleeding into joints) and spontaneous bleeding, especially in **children**. The history of **bleeding tendency** and **contracture** (due to repeated joint bleeds) points towards this diagnosis.

Topic Section:

Haemophilia

Definition:
An X-linked recessive bleeding disorder due to deficiency of clotting factors (Hemophilia A - factor VIII deficiency; Hemophilia B - factor IX deficiency).

Key Clinical Presentation:

Spontaneous bleeding or **after minor trauma**

Hemarthrosis (bleeding into joints), especially knees, elbows, and ankles

Contractures and deformities due to repeated joint bleeds

Investigations:

Best initial: **Coagulation profile** (prolonged aPTT)

Confirmatory: **Factor VIII or IX assay**

Treatment Outline:

First-line: **Factor replacement therapy** (IV infusion of missing clotting factor)

Second-line: **Desmopressin** (for mild Hemophilia A)

Definitive: **Gene therapy** (emerging treatment)

Correct Answer:
A. Haemophilia ✅

MCQ 796

A 15-year-old girl underwent an exploratory laparotomy for a perforated appendix. On the 7th postoperative day, she remained febrile and developed respiratory distress requiring mechanical ventilation. On examination, her abdomen was distended, tense, and tender. A contrast-enhanced computed tomography (CT) was ordered. Post-CT, she developed bleeding from her mouth, trachea, and I/V puncture site.

Which of the following is the most likely diagnosis?

A. Haemophilia
B. Anaphylactic contrast reaction
C. Idiopathic thrombocytopenic purpura
D. Disseminated intravascular coagulation

Explanation:

D. Disseminated intravascular coagulation (DIC) is the most likely diagnosis. **DIC** is a condition characterized by **widespread clotting** and **bleeding**, often seen in critically ill patients, including those with sepsis or infection. The **fever, respiratory distress**, and **post-contrast bleeding** are typical of **DIC**, which is a **complication** that can occur after **severe infections** or **trauma**. The **CT scan findings** of abdominal tenderness and distension are consistent with a **complicated postoperative infection**.

Topic Section:

Disseminated Intravascular Coagulation (DIC)

Definition:
A life-threatening condition involving systemic activation of the coagulation cascade, leading to both **thrombosis** and **bleeding**.

Key Clinical Presentation:

Bleeding from multiple sites (e.g., mouth, puncture sites, mucosal membranes)

Fever, **respiratory distress**, and **organ dysfunction**

Abdominal distension and **tenderness** (in cases of sepsis)

Investigations:

Best initial: **D-dimer** and **PT/aPTT** (prolonged clotting times)

Confirmatory: **Fibrinogen levels** (low) and **platelet count** (low)

Treatment Outline:

First-line: **Supportive care**, including **anticoagulation** and **platelet transfusion**

Second-line: **Treatment of underlying cause** (e.g., infection, sepsis)

Definitive: **Addressing the underlying disorder** (e.g., sepsis, trauma)

Correct Answer:
D. Disseminated intravascular coagulation ✅

MCQ 797:

A 55-year-old woman complains of abdominal pain, distention, and bilious vomiting. On examination, she is pale, dehydrated, and her abdomen is soft, distended, with hyperdynamic gut sounds. She was diagnosed with gallstone disease 10 years ago (see report).

Which of the following is the most likely diagnosis?

A. Cholangitis
B. Pancreatitis
C. Cholecystitis
D. Gallstone ileus

Explanation:

D. Gallstone ileus is the most likely diagnosis. **Gallstone ileus** occurs when a gallstone passes through a fistula between the gallbladder and the intestine, leading to bowel obstruction. The **history of gallstone disease**, **bilious vomiting**, and **abdominal distension** with **hyperdynamic gut sounds** suggest an obstruction. The **CT scan findings** of air in the biliary system further confirm this diagnosis.

Topic Section:

Gallstone Ileus

Definition:
A rare complication of **cholelithiasis** where a gallstone passes into the intestinal tract, causing **bowel obstruction**.

Key Clinical Presentation:

Abdominal pain, **distention**, and **bilious vomiting**

History of gallstone disease

Hyperdynamic gut sounds and signs of bowel obstruction

Investigations:

Best initial: **Abdominal X-ray** (showing air in the biliary system)

Confirmatory: **CT scan abdomen**

Treatment Outline:

First-line: **Surgical removal of the stone**

Second-line: **Endoscopic or percutaneous intervention** if surgery is not feasible

Definitive: **Cholecystectomy** to prevent recurrence

Correct Answer:
D. Gallstone ileus ✅

MCQ 798

A 55-year-old hypertensive diabetic with ischemic heart disease woman presented with sudden onset of left-sided lower abdominal pain associated with vomiting for 6 hours. On examination, she is pale, dehydrated, and flushed. Peripheral pulses are regularly irregular. Her abdomen is distended, tender all over (see lab results).

Which of the following is the most likely diagnosis?

A. Diverticulitis
B. Ulcerative colitis
C. Perforated viscus
D. Mesenteric ischemia

Explanation:

D. Mesenteric ischemia is the most likely diagnosis. **Mesenteric ischemia** presents with **acute abdominal pain**, particularly in patients with **cardiac risk factors** such as **ischemic heart disease**, **hypertension**, and **diabetes**. The **sudden onset of pain**, **abdominal distention**, and **dehydration** with **irregular peripheral pulses** are indicative of **poor perfusion to the intestines**.

Topic Section:

Mesenteric Ischemia

Definition:
A condition caused by reduced blood flow to the intestines, leading to **ischemic damage** and **potential bowel infarction**.

Key Clinical Presentation:

Acute abdominal pain, often **severe and sudden onset**

Dehydration and **abdominal distention**

Risk factors: **Cardiac disease**, **hypertension**, and **diabetes**

Investigations:

Best initial: **CT angiography** of the mesenteric vessels

Confirmatory: **Mesenteric angiography**

Treatment Outline:

First-line: **Surgical exploration** to assess bowel viability

Second-line: **Embolectomy or bypass** for mesenteric artery blockage

Definitive: **Revascularization** and addressing the underlying cause (e.g., atrial fibrillation)

Correct Answer:
D. Mesenteric ischemia ✅

MCQ 799:

A 33-year-old woman underwent surgery for her right leg varicose veins. On the 2nd postoperative day, she complained of pain and paraesthesia on the medial aspect of her leg and foot. On examination, there is sensory loss of fine touch and pain in her right leg and foot, but no motor loss. Wounds are healthy.

Which of the following nerves is most likely injured?

A. Sciatic
B. Femoral
C. Obturator
D. Saphenous

Explanation:

D. Saphenous nerve is most likely injured. The **saphenous nerve**, which is a branch of the femoral nerve, supplies sensation to the medial aspect of the leg and foot. The description of **sensory loss** on the **medial side** with **no motor loss** suggests **saphenous nerve injury**, a common complication after varicose vein surgery.

Topic Section:

Saphenous Nerve Injury

Definition:
A sensory nerve injury often associated with **surgical procedures** around the **femoral artery** or **saphenous vein**, leading to loss of sensation along the medial leg and foot.

Key Clinical Presentation:

Sensory loss along the **medial leg** and **foot**

No motor involvement

Investigations:

Best initial: **Clinical examination** and **nerve conduction study** if needed

Treatment Outline:

First-line: **Conservative management** (rest, analgesia, compression)

Second-line: **Nerve block** or **surgical repair** if symptoms persist

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Correct Answer:
D. Saphenous ✅

MCQ 800

A 25-year-old complains of an inability to extend her fingers and numbness over her left wrist. On examination, there is paraesthesia of the skin over the left wrist and anatomical snuffbox, and an inability to extend her metacarpophalangeal joint. She also demonstrates wrist drop. Clinical injury to radial nerve is suspected.

Which of the following is the most likely site of injury?

A. In the axilla
B. Carpal tunnel
C. Olecranon process
D. Spiral groove of humerus

Explanation:

D. Spiral groove of humerus is the most likely site of injury. The **radial nerve** is most commonly injured in the **spiral groove** of the **humerus**, leading to **wrist drop** and **sensory deficits** in the **anatomical snuffbox**. The other options are less likely to cause this specific clinical presentation.

Topic Section:

Radial Nerve Injury

Definition:
Radial nerve injury can lead to **wrist drop** and **sensory loss** in the **dorsal hand** and **forearm**.

Key Clinical Presentation:

Wrist drop (inability to extend wrist and fingers)

Paraesthesia over the **anatomical snuffbox**

Injury in the **spiral groove** of the **humerus**

Investigations:

Best initial: **Clinical examination**

Confirmatory: **Nerve conduction studies**

Treatment Outline:

First-line: **Conservative management** (splinting, physical therapy)

Second-line: **Surgical exploration** if conservative management fails

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Correct Answer:
D. Spiral groove of humerus ✅

MCQ 801

A 55-year-old man complains of dark stool and weakness. He has peptic ulcer disease and endoscopy investigations have confirmed multiple ulcers in the antrum. He has been kept on medical therapy. On examination, his abdomen is soft and non-tender. Repeat endoscopy confirms multiple ulcers in the antrum (see lab results).
Test Result Normal Values
Hb 58 130-170 g/L
HCT 0.18 0.42-0.52
Which of the following is the most appropriate management?

A. Total gastrectomy
B. Partial distal gastrectomy
C. Endoscopic laser photocoagulation
D. Truncal vagotomy and pyloroplasty

Explanation:
This patient is presenting with **gastric bleeding** due to peptic ulcer disease and has significant anemia. The most appropriate management in this case would be **truncal vagotomy and pyloroplasty**, which addresses the underlying cause (gastric acid secretion) and prevents further ulcer formation and bleeding.

Correct Answer: D. Truncal vagotomy and pyloroplasty ✅

Topic Section:

Peptic Ulcer Disease (PUD)

Definition:
A condition characterized by the formation of ulcers in the stomach or duodenum due to acid or pepsin digestion of the mucosal lining.

Key Clinical Presentation:

Epigastric pain, typically relieved by food or antacids

Dark stools (melena) due to bleeding

Fatigue and weakness from anemia (as seen in this patient)

Investigations:

Best initial: Endoscopy

Confirmatory: Histology and biopsies if malignancy is suspected

Treatment Outline:

First-line: Proton pump inhibitors (PPIs), H. pylori eradication

Second-line: Surgery (if complications such as bleeding or perforation occur)

Definitive: Truncal vagotomy and pyloroplasty in refractory or complicated cases

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MCQ 802:

A 22-year-old complains of a 2-day history of right-sided chest pain and dyspnea. It started during landing from a long flight. On examination, he is tall, breath sounds are absent on the right side and percussion note is resonant (see report).
CXR: Right pneumothorax.
Which of the following is the most appropriate initial management?

A. Thoracentesis
B. Tube thoracostomy
C. Thoracoscopic blebectomy
D. Conservative management

Explanation:
The initial management of **simple pneumothorax** in a patient with stable vital signs is **conservative management**, including observation and oxygen therapy. In cases of large pneumothorax or symptoms worsening, **tube thoracostomy** may be required.

Correct Answer: D. Conservative management ✅

Topic Section:

Pneumothorax

Definition:
The presence of air in the pleural space causing lung collapse. Can be spontaneous, traumatic, or secondary to underlying lung disease.

Key Clinical Presentation:

Sudden-onset pleuritic chest pain

Dyspnea

Reduced breath sounds on the affected side

Investigations:

Best initial: Chest X-ray

Confirmatory: CT chest (in uncertain or complicated cases)

Treatment Outline:

First-line: Observation for small pneumothoraces

Second-line: Chest tube placement (tube thoracostomy) for larger pneumothoraces

Definitive: Surgery or thoracoscopic blebectomy for recurrent or persistent pneumothorax

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MCQ 803:

A 70-year-old woman is bed bound due to basal ganglia bleed and is having difficulty swallowing and has experienced significant weight loss. On examination, she is emaciated, and lethargic. Weak muscles of mastication and an absent gag reflex.
Which of the following is the most appropriate option to establish feeding?

A. Gastrostomy tube
B. Jejunostomy tube
C. Parenteral nutrition
D. Nasogastric tube feeding

Explanation:
In this case, **gastrostomy tube** is the most appropriate option for long-term feeding in a patient with difficulty swallowing due to neurological impairment. It provides a safe and reliable route for nutrition in patients with chronic swallowing dysfunction.

Correct Answer: A. Gastrostomy tube ✅

Topic Section:

Dysphagia and Feeding Tubes

Definition:
Dysphagia refers to difficulty swallowing, which can be due to a variety of causes, including neurological impairment, esophageal motility disorders, or structural abnormalities.

Key Clinical Presentation:

Difficulty swallowing solid or liquid foods

Weight loss due to poor intake

Aspiration risk, particularly in neurologically impaired patients

Investigations:

Best initial: Clinical assessment, swallowing assessment, and endoscopy if required

Confirmatory: Video fluoroscopy or manometry

Treatment Outline:

First-line: Nutritional support via feeding tubes (e.g., gastrostomy tube for long-term use)

Second-line: Parenteral nutrition (for short-term support in the hospital setting)

Definitive: Surgical intervention if necessary, such as correction of anatomical defects or therapy for underlying conditions

MCQ 804:

A 37-year-old woman complains of abdominal fullness, epigastric discomfort, and loss of appetite. She is recovering from an episode of acute pancreatitis 5 weeks ago. On examination, her abdomen is distended, and there is vague fullness in the epigastrium (see report).
CT scan:

Cyst about 24 x 18 cm in the lesser sac.
Which of the following is the most appropriate management?

A. Excision of cyst
B. Surgical drainage
C. Image-guided aspiration
D. Endoscopic Internal drainage

Explanation:
The presence of a large pancreatic cyst (likely a pseudocyst) in a patient recovering from acute pancreatitis suggests that it needs to be managed with **surgical drainage** if symptomatic. **Image-guided aspiration** is a temporary solution and can be used if there are concerns about the cyst's nature or complications. Surgical intervention, such as drainage, is the most appropriate in this setting.

Correct Answer: B. Surgical drainage ✅

Topic Section:

Pancreatic Pseudocyst (Pancreatic Pseudocyst)

Definition:
A fluid collection encapsulated by a wall of fibrous tissue, often a sequelae of acute pancreatitis, characterized by a lack of epithelial lining.

Key Clinical Presentation:

Abdominal pain

Nausea, vomiting

Abdominal fullness and distension

Investigations:

Best initial: CT scan or ultrasound for cyst size and location

Confirmatory: Endoscopic ultrasound or MRI for further characterization

Treatment Outline:

First-line: Surgical drainage, especially in symptomatic cases or large cysts

Second-line: Percutaneous drainage or endoscopic drainage in select cases

Definitive: Surgical resection if recurrent or complicated

MCQ 805:

A 65-year-old smoker with diabetes mellitus presents with a long history of intermittent buttock pain and claudication that has worsened over the past few weeks. On examination, there is atrophy of buttock muscles with decreased power and painful movements. The femoral pulses are weak (see report).
Angiogram:
Aortolilac insufficiency.
Which of the following is the most appropriate management?

A. Vasodilator therapy
B. Aortofemoral bypass
C. Axillobifemoral bypass
D. Lumbar sympathectomy

Explanation:
In a patient with **aortoiliac insufficiency**, the most appropriate management is **aortofemoral bypass** to restore blood flow to the lower extremities. This procedure is typically indicated for patients with significant symptoms or non-healing wounds due to peripheral arterial disease.

Correct Answer: B. Aortofemoral bypass ✅

Topic Section:

Aortoiliac Disease (Aortoiliac Disease)

Definition:
Atherosclerotic narrowing or blockage of the aorta and iliac arteries, leading to impaired blood flow to the lower limbs.

Key Clinical Presentation:

Intermittent claudication, particularly in the buttocks and thighs

Weak or absent femoral pulses

Muscle atrophy in the affected leg

Investigations:

Best initial: Doppler ultrasound, angiogram

Confirmatory: CT angiography or MRI angiography

Treatment Outline:

First-line: Surgical intervention (bypass surgery, angioplasty)

Second-line: Medical therapy with vasodilators and antiplatelet agents

Definitive: Bypass surgery for critical cases

MCQ 806:

A 45-year-old presents after a 50 feet fall. On examination, he looks pale, peripheral pulses are feeble, and his chest has decreased breath sounds on the left side (see image and report).
Blood pressure 100/60 mmHg
Heart rate 100 /min
Respiratory rate 20 /min
Temperature 36.6 °C
X-ray trauma series:
Left dislocated hip and calcaneal fracture.
Which of the following is the most appropriate management?

A. Left tube thoracostomy
B. Emergency thoracotomy
C. Aortography and stenting
D. Intubation and ventilation

Explanation:
Given the mechanism of trauma and presentation with **decreased breath sounds**, the most likely diagnosis is **hemothorax**. The **initial management** involves **tube thoracostomy** to evacuate blood and air from the pleural space, thereby restoring normal lung expansion and preventing respiratory distress.

Correct Answer: A. Left tube thoracostomy ✅

Topic Section:

Hemothorax (Hemothorax)

Definition:
Collection of blood in the pleural space, commonly caused by trauma, leading to impaired lung expansion.

Key Clinical Presentation:

Dyspnea

Decreased breath sounds on the affected side

Tachypnea, hypotension if severe

Investigations:

Best initial: Chest X-ray

Confirmatory: CT scan if needed for further evaluation

Treatment Outline:

First-line: Tube thoracostomy for drainage

Second-line: Surgical intervention if bleeding persists or is massive

Definitive: Control of the underlying cause, including surgery if needed

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MCQ 807:

A 25-year-old woman that is taking steroids for inflammatory bowel disease complains of abdominal pain and bilious vomiting. On examination, her abdomen is distended and tender in the right iliac fossa. She had a colonoscopy 2 weeks ago, which was normal. Contrast barium study was ordered (see report).
Blood pressure 110/70 mmHg
Heart rate 100 /min
Barium study:
Single stricture at terminal ileum, 1 cm from ileocecal valve.
Which of the following is the most appropriate management?

A. Stricturoplasty
B. Right hemicolectomy
C. Conservative medical management
D. Segmental resection with ileostomy

Explanation:
The **stricture** seen at the **terminal ileum** is most likely a **result of Crohn's disease**, especially in a patient with **inflammatory bowel disease**. **Stricturoplasty** is the treatment of choice for non-complicated strictures, as it preserves bowel length and function.

Correct Answer: A. Stricturoplasty ✅

Topic Section:

Crohn's Disease (Crohn's Disease)

Definition:
Chronic inflammatory bowel disease that can affect any part of the gastrointestinal tract, with a predilection for the ileum.

Key Clinical Presentation:

Abdominal pain and cramping

Diarrhea, weight loss, malnutrition

Possible fistulas or strictures

Investigations:

Best initial: Colonoscopy with biopsy

Confirmatory: Imaging (CT, MRI) for complications

Treatment Outline:

First-line: Medical management with corticosteroids and immunosuppressants

Second-line: Surgical intervention (stricturoplasty or resection) for complications or refractory disease

Definitive: Surgery for complications such as obstruction or perforation

MCQ 808:

A 24-year-old woman complains of abdominal pain, sweating, and palpitations. She was recently diagnosed with hypertension and has had a low response to medical therapy. On examination, her abdomen is soft and non-tender (see lab result and report).
Blood pressure: 215/110 mmHg
Heart rate: 124/min

Respiratory rate: 21/min
Temperature: 38.6 °C
Test Result: Catecholamines 2100 (<591 nmol/day)
Ultrasound: 4 cm mass in the right adrenal cortex.
Which of the following is the most appropriate management for the hypertension?

A. Beta blockers
B. ACE inhibitors
C. Alpha blockers
D. Calcium channel blocker

Explanation:
The presence of a mass in the right adrenal cortex and markedly elevated catecholamine levels suggests **pheochromocytoma**, a catecholamine-secreting tumor. **Alpha blockers** (e.g., phenoxybenzamine) are the most appropriate initial management in pheochromocytoma to control hypertension before surgery.

Correct Answer: C. Alpha blockers ✅

Topic Section:

Pheochromocytoma (Pheochromocytoma)

Definition:
A rare catecholamine-secreting tumor usually arising from the adrenal medulla, causing episodic or sustained hypertension.

Key Clinical Presentation:

Severe hypertension (episodic or sustained)

Abdominal pain, sweating, palpitations

Headache, tremor

Investigations:

Best initial: Measurement of plasma metanephrines or 24-hour urinary catecholamines

Confirmatory: CT or MRI of the abdomen for tumor localization

Treatment Outline:

First-line: Alpha blockers (e.g., phenoxybenzamine) to control blood pressure

Second-line: Beta blockers after alpha blockade for tachycardia control

Definitive: Surgical resection of the tumor

MCQ 809:

A 5-year-old child complains of a painful left testicular swelling. On examination, there is a firm swelling in his left hemi-scrotum, it is irreducible, with a negative transillumination test.
Which of the following is the most appropriate management?

A. Herniotomy
B. Orchidopexy
C. Herniorrhaphy
D. Tension-free repair

Explanation:
The child most likely has an **inguinal hernia** with incarcerated bowel or fat. **Herniotomy** (surgical repair of the hernia) is the most appropriate management for an irreducible inguinal hernia, as it involves removing the hernial sac and repairing the defect.

Correct Answer: A. Herniotomy ✅

Topic Section:

Inguinal Hernia (Inguinal Hernia)

Definition:
Protrusion of abdominal contents through the inguinal canal, often presenting as a swelling in the groin or scrotum.

Key Clinical Presentation:

Swelling or bulge in the groin or scrotum

Pain, especially with coughing or straining

Irreducible swelling, sometimes associated with bowel obstruction

Investigations:

Best initial: Physical examination and hernia reduction test

Confirmatory: Ultrasound or CT scan if the diagnosis is uncertain

Treatment Outline:

First-line: Herniotomy (surgical repair)

Second-line: Watchful waiting in asymptomatic cases

Definitive: Surgical repair with mesh or suture techniques

MCQ 810:

A 27-year-old woman complains of a 3-month history of right upper quadrant pain, fever, and malaise. On examination, she is febrile, her abdomen is soft and tender in the right upper quadrant. Indirect hemagglutination titer for *Echinococcus granulosus* is positive (see image).
Blood pressure: 110/70 mmHg
Heart rate: 95 /min
Temperature: 37.4 °C
Which of the following is the most appropriate management?

A. Albendazole therapy
B. Percutaneous drainage
C. Open cyst deroofing
D. Laparoscopic pericystectomy

Explanation:
The patient most likely has a **hydatid cyst** caused by *Echinococcus granulosus*. The most appropriate treatment for hydatid cysts involves **albendazole therapy** to reduce the risk of cyst rupture, along with surgical treatment in severe cases.

Correct Answer: A. Albendazole therapy ✅

Topic Section:

Hydatid Disease (Hydatid Disease)

Definition:
A parasitic infection caused by *Echinococcus granulosus*, commonly affecting the liver, lungs, or other organs.

Key Clinical Presentation:

Abdominal pain, nausea, vomiting

Jaundice in cases involving the liver

Fever and malaise in acute infections

Investigations:

Best initial: Serologic testing for *Echinococcus granulosus* antibodies

Confirmatory: Ultrasound or CT scan to visualize cysts

Treatment Outline:

First-line: Albendazole therapy for cyst management

Second-line: Percutaneous drainage or surgery for cyst removal

Definitive: Surgery (cyst deroofing or resection) in complicated cases

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MCQ 811:

A 55-year-old patient is diagnosed with a gastrointestinal stromal tumor of the stomach (4 x 3 cm), confined to the body of the stomach, no surrounding infiltration, no distant metastases, or lymphadenopathy.
Which of the following is the most appropriate management?

A. Gastrectomy
B. Endoscopic ablation
C. Systemic chemo-radiation
D. Wedge resection with clear margins

Explanation:
For **gastrointestinal stromal tumors (GISTs)** that are localized, **wedge resection** is the most appropriate management, as it allows for removal of the tumor with clear margins. Systemic therapy is reserved for high-risk or metastatic disease.

Correct Answer: D. Wedge resection with clear margins ✅

Topic Section:

Gastrointestinal Stromal Tumor (GIST)

Definition:
A type of tumor arising from the interstitial cells of Cajal in the gastrointestinal tract, most commonly in the stomach.

Key Clinical Presentation:

Abdominal pain, bloating

Vomiting or weight loss in larger tumors

Bleeding in some cases

Investigations:

Best initial: Endoscopy, CT scan

Confirmatory: Biopsy and immunohistochemistry for CD117 (KIT)

Treatment Outline:

First-line: Surgical resection (wedge resection)

Second-line: Imatinib (tyrosine kinase inhibitor) for advanced or metastatic disease

Definitive: Complete surgical resection with clear margins

MCQ 812:

A 35-year-old man with gastroesophageal reflux disease presents to discuss his family history of oesophageal and gastric carcinoma.
Which of the following is the main risk factor for oesophageal carcinoma?

A. Smoking
B. Idiopathic
C. Barrett's oesophagus
D. Long-standing achalasia

Explanation:
Barrett's oesophagus, a condition in which the squamous epithelium of the esophagus is replaced by columnar epithelium, is a major risk factor for **oesophageal adenocarcinoma**. It is caused by chronic acid reflux and is strongly linked to an increased risk of esophageal cancer.

Correct Answer: C. Barrett's oesophagus ✅

Topic Section:

Esophageal Carcinoma (Esophageal Carcinoma)

Definition:
Malignancy of the esophagus, which can be either squamous cell carcinoma or adenocarcinoma.

Key Clinical Presentation:

Dysphagia (progressive)

Weight loss

Odynophagia (painful swallowing)

Investigations:

Best initial: Endoscopy with biopsy

Confirmatory: CT scan or endoscopic ultrasound for staging

Treatment Outline:

First-line: Surgical resection (esophagectomy) for early-stage disease

Second-line: Chemotherapy and radiation for advanced disease

Definitive: Chemoradiation or surgery depending on the stage

MCQ 813:

A 26-year-old woman complains of right upper quadrant pain and nausea. She is 20 weeks pregnant. On examination, her abdomen is soft and non-tender.
Ultrasound: Cholelithiasis.
Which of the following is the most appropriate management?

A. Cholecystostomy tube
B. Open cholecystectomy
C. Conservative treatment
D. Laparoscopic cholecystectomy

Explanation:
In a pregnant woman with symptomatic cholelithiasis, the first-line treatment is usually **conservative management** with monitoring and medical therapy. If symptoms persist or there are complications, **laparoscopic cholecystectomy** may be considered in the second trimester, as it is the safest during this time.

Correct Answer: C. Conservative treatment ✅

Topic Section:

Cholelithiasis in Pregnancy (Cholelithiasis)

Definition:
Presence of gallstones in the gallbladder, which may or may not cause symptoms.

Key Clinical Presentation:

Right upper quadrant pain, nausea, vomiting

Jaundice (in severe cases)

Often presents during pregnancy or in patients with obesity, diabetes, or certain medications.

Investigations:

Best initial: Ultrasound of the abdomen

Confirmatory: Cholescintigraphy (if diagnosis is uncertain)

Treatment Outline:

First-line: Conservative management (pain control, observation, diet modifications)

Second-line: Laparoscopic cholecystectomy (safe during the second trimester)

Definitive: Surgery in symptomatic cases or if complications like cholecystitis or pancreatitis occur

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MCQ 814:

A 20-year-old woman presents with fever and pain in her right leg. On examination, her right leg is swollen, tender with multiple red streaks, and palpable inguinal nodes.
Which of the following is the most likely causative organism?

A. Bacteroides
B. Enterococcus
C. Klebsiella pneumoniae
D. Streptococcus pyogenes

Explanation:
The clinical presentation of fever, leg pain, redness, and streaks with inguinal lymphadenopathy is suggestive of **cellulitis**, which is most commonly caused by **Streptococcus pyogenes** (Group A Streptococcus).

Correct Answer: D. Streptococcus pyogenes ✅

Topic Section:

Cellulitis (Cellulitis)

Definition:
A common, often superficial skin infection, typically caused by **Streptococcus pyogenes** or **Staphylococcus aureus**.

Key Clinical Presentation:

Red, swollen, warm, and painful area of skin

Fever and malaise in severe cases

Lymphangitis or lymphadenopathy (if the infection spreads)

Investigations:

Best initial: Clinical diagnosis based on physical examination

Confirmatory: Blood cultures (if systemic infection is suspected)

Treatment Outline:

First-line: Antibiotics targeting Gram-positive organisms (e.g., penicillin or cephalexin)

Second-line: Vancomycin or clindamycin for MRSA

Definitive: Drainage of abscesses, if present

MCQ 815:

Which of the following is most commonly associated with retroperitoneal sarcoma?

A. Retroperitoneal bleeding
B. Systemic lymphadenopathy

C. Invasion into surrounding tissues
D. Compression of surrounding tissues

Explanation:
Retroperitoneal sarcomas typically **compress surrounding tissues**, as they grow in the confined space of the retroperitoneum. These tumors are often asymptomatic in the early stages and can present with nonspecific symptoms such as abdominal pain or a palpable mass.

Correct Answer: D. Compression of surrounding tissues ✅

Topic Section:

Retroperitoneal Sarcoma (Retroperitoneal Sarcoma)

Definition:
A rare malignant tumor arising from the mesenchymal tissues of the retroperitoneum.

Key Clinical Presentation:

Often asymptomatic early on

Abdominal pain, fullness, or mass in the abdomen

Nonspecific symptoms (e.g., weight loss)

Investigations:

Best initial: CT or MRI of the abdomen

Confirmatory: Biopsy of the mass

Treatment Outline:

First-line: Surgical resection with clear margins

Second-line: Radiation therapy (if the tumor is unresectable)

Definitive: Complete surgical resection

MCQ 816:

A 45-year-old patient complains of chills and rigors. He underwent a common bile duct exploration a few hours ago (see lab results).
Blood pressure: 110/70 mmHg

Heart rate: 80 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Test Result:

Hb: 130 (Normal)

WBC: 9.4 (Normal)

Neutrophils: 50% (Normal)
Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
Post-procedure fever with chills and rigors can indicate **sepsis**, which is an infection that has spread to the bloodstream and is causing a systemic inflammatory response. This can be a complication after procedures like bile duct exploration.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
A systemic inflammatory response to infection, leading to organ dysfunction and potential organ failure.

Key Clinical Presentation:

Fever, chills, and rigors

Tachycardia, hypotension

Organ dysfunction (e.g., respiratory distress, renal failure)

Investigations:

Best initial: Blood cultures, urinalysis, chest X-ray

Confirmatory: Cultures of the suspected infection source

Treatment Outline:

First-line: Broad-spectrum antibiotics (based on the suspected source)

Second-line: Source control (e.g., drainage of abscesses, removal of infected devices)

Definitive: Supportive care in an ICU setting if organ failure occurs

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MCQ 817:

A 45-year-old man is complaining of chills and rigors. He underwent an elective subtotal colectomy for ulcerative colitis 12 hours ago (see lab results).
Blood pressure: 110/70 mmHg
Heart rate: 110 /min
Respiratory rate: 18 /min
Temperature: 38 °C
Test Result:

Hb: 110

HCT: 0.40

WBC: 15

Neutrophils: 60%
Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
This patient is presenting with **SIRS (Systemic Inflammatory Response Syndrome)** following major surgery. The elevated white blood cell count, fever, and tachycardia are indicative of SIRS, a precursor to sepsis. While this patient is not in septic shock yet, these signs suggest he is at risk for sepsis.

Correct Answer: D. Systemic inflammatory response syndrome ✅

Topic Section:

Systemic Inflammatory Response Syndrome (SIRS)

Definition:
A systemic response to a variety of insults (e.g., infection, trauma, surgery) characterized by inflammation, tachycardia, fever, and leukocytosis.

Key Clinical Presentation:

Fever or hypothermia

Tachycardia, tachypnea

Leukocytosis or leukopenia

Investigations:

Best initial: Clinical diagnosis based on physical signs (e.g., fever, tachycardia)

Confirmatory: None (diagnosis is based on clinical criteria)

Treatment Outline:

First-line: Treat the underlying cause (e.g., infection, surgical complications)

Second-line: Supportive care (e.g., fluids, oxygen therapy)

Definitive: Management of the underlying cause (e.g., antibiotics for infection, surgical intervention for complications)

MCQ 818

A 65-year-old patient complains of chills and rigors 12 hours after undergoing removal of an infected ureteric stent. Prior to the cystoscopy, her urine culture was positive for E. Coli (see lab result).
Blood pressure: 110/70 mmHg
Heart rate: 96/min
Respiratory rate: 18/min
Temperature: 38.6 °C
Test Result:
WBC: 15.4 (Normal range: 4.5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia

C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
This patient has chills and rigors following the removal of an infected ureteric stent. The positive urine culture for **E. coli**, along with fever and elevated WBC, suggest **sepsis** (a systemic infection with potential organ dysfunction). While systemic inflammatory response syndrome (SIRS) can be present in the early stages of infection, the confirmed infection with E. coli makes **sepsis** the most likely diagnosis.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key Clinical Presentation:

Fever, chills, and rigors

Tachycardia, hypotension

Organ dysfunction (e.g., respiratory failure, renal failure)

Positive blood cultures

Investigations:

Best initial: Blood cultures, urinalysis, chest X-ray

Confirmatory: Cultures of the suspected infection source

Treatment Outline:

First-line: Broad-spectrum antibiotics

Second-line: Source control (e.g., drainage of abscesses, removal of infected devices)

Definitive: ICU support if needed for organ failure

MCQ 819:

A 65-year-old required intubation and transfer to the Intensive Care Unit after collapsing in the ward. 2 days previously, she underwent an operation for empyema of the gallbladder (see lab result).
Blood pressure: 110/70 mmHg
Heart rate: 110/min
Respiratory rate: 24/min
Temperature: 38.7 °C
Test Result:
WBC: 19.4 (Normal range: 4.5-10.5 x 10^9/L)

Which of the following is the most likely diagnosis?

A. Sepsis
B. Bacteraemia
C. Severe sepsis
D. Systemic inflammatory response syndrome

Explanation:
The patient, who has recently undergone surgery for empyema, is presenting with fever, tachycardia, and leukocytosis (elevated WBC). These findings, combined with a history of recent surgery, suggest **sepsis**. Since the patient has collapsed and required ICU transfer, this may indicate severe sepsis, but the most likely diagnosis given the available information is **sepsis** at this point.

Correct Answer: A. Sepsis ✅

Topic Section:

Sepsis (Sepsis)

Definition:
Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key Clinical Presentation:

Fever, chills, and rigors

Tachycardia, hypotension

Organ dysfunction (e.g., respiratory failure, renal failure)

Positive blood cultures

Investigations:

Best initial: Blood cultures, urinalysis, chest X-ray

Confirmatory: Cultures of the suspected infection source

Treatment Outline:

First-line: Broad-spectrum antibiotics

Second-line: Source control (e.g., drainage of abscesses, removal of infected devices)

Definitive: ICU support if needed for organ failure

MCQ 819:

A 24-month-old baby is brought in with a right-sided red inflamed hemiscrotum. On exploration, the right testis is viable, however, the cord is thick and oedematous.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orthitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a red, inflamed hemiscrotum with a thick and oedematous cord suggests **testicular torsion**, a surgical emergency that typically presents with severe pain and swelling of the scrotum. The viability of the testis and the oedema of the cord are characteristic of this condition, which requires immediate surgical intervention.

Correct Answer: A. Testicular torsion ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency characterized by the twisting of the spermatic cord, cutting off the blood supply to the testis.

Key Clinical Presentation:

Sudden onset of severe scrotal pain and swelling

Nausea and vomiting

Red, swollen scrotum

Absent cremasteric reflex

Investigations:

Best initial: Clinical diagnosis based on physical examination

Confirmatory: Doppler ultrasound (to assess blood flow)

Treatment Outline:

First-line: Immediate surgical exploration and detorsion

Second-line: Orchidopexy to prevent recurrence

Definitive: Surgery to salvage the testis and prevent future torsion

MCQ 820

A 7-month-old baby is brought in with a red and inflamed left hemiscrotum. Examination confirms a firm, tender, irreducible lump in the scrotum extending to the groin. The left testis cannot be palpated.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orthitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a **firm, tender, irreducible lump** in the scrotum that extends into the groin, with the testis not palpable, is most likely an **incarcerated inguinal hernia**. This occurs when a portion of the intestine or other abdominal contents protrude through the inguinal canal and becomes trapped, leading to swelling and tenderness.

Correct Answer: C. Incarcerated inguinal hernia ✅

Topic Section:

Incarcerated Inguinal Hernia (Inguinal Hernia)

Definition:
A condition in which part of the intestine or other abdominal contents becomes trapped in the inguinal canal, causing swelling, pain, and risk of strangulation.

Key Clinical Presentation:

Swelling or lump in the groin or scrotum

Pain and tenderness, especially when coughing or straining

Inability to reduce the lump manually

Vomiting and signs of bowel obstruction in severe cases

Investigations:

Best initial: Clinical examination

Confirmatory: Ultrasound or CT scan (if needed for complex cases)

Treatment Outline:

First-line: Immediate surgical intervention (herniorrhaphy)

Second-line: Conservative management if reducible (but not for incarcerated cases)

Definitive: Herniorrhaphy (surgical repair)

MCQ 821:

A 12-year-old presents with left-sided lower abdominal pain. Examination confirms a slightly elevated, tender left testis with redness and oedema of the scrotum.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
The presentation of a tender, swollen, and red scrotum with an elevated testis in a young patient is most consistent with **epididymo-orchitis**, an infection of the epididymis and testis,

commonly caused by a bacterial infection such as **E. coli** or **Chlamydia**. This condition is often associated with fever, dysuria, and urinary symptoms.

Correct Answer: B. Epididymo-orchitis ✅

Topic Section:

Epididymo-orchitis (Epididymo-orchitis)

Definition:
Infection of the epididymis and testis, often caused by bacterial pathogens such as **E. coli**, **Chlamydia**, and **Gonorrhea**.

Key Clinical Presentation:

Scrotal pain, swelling, and redness

Elevated testis with tenderness over the epididymis

Urinary symptoms such as dysuria or frequency

Fever

Investigations:

Best initial: Clinical examination and urinalysis

Confirmatory: Doppler ultrasound (to evaluate blood flow and exclude torsion)

Treatment Outline:

First-line: Antibiotics (e.g., ciprofloxacin, doxycycline)

Second-line: Pain management and scrotal elevation

Definitive: Surgical drainage (if abscess formation)

MCQ 822:

A 12-year-old presents with a 2-day history of right-sided scrotal pain. Examination confirms right testis with vertical orientation. There is an area of tenderness around the superior pole and the scrotum is red and oedematous.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Incarcerated inguinal hernia
D. Torsion of a testicular appendage

Explanation:
This patient's symptoms of scrotal pain, a vertically oriented testis, redness, and oedema are most indicative of **testicular torsion**, a surgical emergency. The testis rotates around its spermatic cord, leading to compromised blood flow and acute pain. Immediate intervention is necessary to salvage the testis.

Correct Answer: A. Testicular torsion ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency caused by the twisting of the spermatic cord, leading to impaired blood flow to the testis.

Key Clinical Presentation:

Sudden onset of severe scrotal pain

Vomiting and nausea

Swollen, tender scrotum

Horizontal or vertical orientation of the testis

Absence of the cremasteric reflex

Investigations:

Best initial: Clinical examination and Doppler ultrasound (to assess blood flow)

Confirmatory: Surgical exploration (if diagnosis is unclear)

Treatment Outline:

First-line: Immediate surgical exploration and detorsion

Second-line: Orchidopexy to prevent recurrence

Definitive: Surgery to preserve the testis

MCQ 823:

A 24-month-old is presented with redness of both hemi-scrotum. Examination confirms non-tender, redness and oedema of the scrotum extending to both groins.
Which of the following is the most likely diagnosis?

A. Testicular torsion
B. Epididymo-orchitis
C. Idiopathic scrotal oedema
D. Torsion of a testicular appendage

Explanation:
The **non-tender**, **non-painful**, and **bilateral** redness and oedema of the scrotum are characteristic of **idiopathic scrotal oedema**. This condition is benign and typically resolves on its own, unlike other causes of scrotal swelling, which are usually painful.

Correct Answer: C. Idiopathic scrotal oedema ✅

Topic Section:

Idiopathic Scrotal Oedema (Idiopathic scrotal oedema)

Definition:
A benign condition characterized by non-painful, bilateral swelling and redness of the scrotum, often resolving without intervention.

Key Clinical Presentation:

Non-tender, bilateral scrotal swelling

Redness and oedema without underlying testicular pathology

Often resolves spontaneously

Investigations:

Best initial: Clinical examination

Confirmatory: None needed, diagnosis based on clinical features

Treatment Outline:

First-line: Observation and reassurance

Second-line: Supportive care (e.g., scrotal elevation) if needed

Definitive: None required, condition is self-limiting

MCQ 824:

A 12-year-old patient presents with a 4-hour history of right-sided lower abdominal pain. Examination confirms a slightly elevated, tender right testis. The lie of the testis is horizontal with no redness or oedema of the scrotum.
Which of the following is the most appropriate management?

A. Oral analgesics
B. Oral antibiotics
C. Doppler ultrasound
D. Scrotal exploration

Explanation:
The **horizontal orientation** of the testis suggests **testicular torsion**, a surgical emergency. **Doppler ultrasound** is the most appropriate initial investigation to confirm the diagnosis, as it can assess blood flow to the testis. If torsion is confirmed, immediate surgical exploration is required.

Correct Answer: C. Doppler ultrasound ✅

Topic Section:

Testicular Torsion (Testicular Torsion)

Definition:
A surgical emergency caused by the twisting of the spermatic cord, leading to impaired blood flow to the testis.

Key Clinical Presentation:

Sudden onset of severe scrotal pain

Vomiting and nausea

Swollen, tender scrotum

Horizontal or vertical orientation of the testis

Absence of the cremasteric reflex

Investigations:

Best initial: Clinical examination and Doppler ultrasound (to assess blood flow)

Confirmatory: Surgical exploration (if diagnosis is unclear)

Treatment Outline:

First-line: Immediate surgical exploration and detorsion

Second-line: Orchidopexy to prevent recurrence

Definitive: Surgery to preserve the testis

MCQ 825:

A 45-year-old sustained a knife injury to his left wrist. Examination confirms a clean wound with division of the underlying tendons and median nerve.
Which of the following is the most appropriate management?

A. Debridement and primary closure
B. Primary repair of injured structures

Explanation:
The appropriate management for a clean knife injury with division of tendons and a nerve is **primary repair of the injured structures**. This includes repairing both the tendons and median nerve as early as possible to minimize the risk of complications like functional loss and nerve damage.

Correct Answer: B. Primary repair of injured structures ✅

Topic Section:

Tendon and Nerve Injury (Tendon and Nerve Injury)

Definition:
Injury to tendons and nerves caused by trauma, resulting in loss of function or sensation.

Key Clinical Presentation:

Trauma with visible lacerations or wounds

Tenderness and loss of function in the affected area

Decreased sensation or motor loss (in case of nerve injury)

Investigations:

Best initial: Clinical examination

Confirmatory: Imaging or exploration during surgery

Treatment Outline:

First-line: Primary repair of tendons and nerves as soon as possible

Second-line: Rehabilitation and therapy post-repair

MCQ 826:

A 72-year-old underwent a subtotal colectomy 5 days ago. His urine output over the past 8 hours has been 7 ml/hr.
Blood pressure 110/70 mmHg
Heart rate 90 /min
Temperature 36.6 °C
Which of the following is most appropriate to administer intravenously?

A. Diuretics
B. Inotropes
C. Antibiotics
D. 500 NS fluid challenge

Explanation:
The most appropriate management for a patient with low urine output post-surgery is to administer a **500 ml normal saline (NS) fluid challenge** to assess for possible hypovolemia. The low urine output in this case suggests possible fluid depletion rather than an intrinsic renal issue.

Correct Answer: D. 500 NS fluid challenge ✅

Topic Section:

Postoperative Acute Kidney Injury (Postoperative Acute Kidney Injury)

Definition:
Acute kidney injury (AKI) occurring after surgery, often due to hypovolemia, sepsis, or medication effects.

Key Clinical Presentation:

Decreased urine output post-surgery

Elevated serum creatinine or urea levels (if available)

Possible signs of fluid imbalance (e.g., hypotension)

Investigations:

Best initial: Urine output monitoring

Confirmatory: Serum creatinine and BUN levels

Treatment Outline:

First-line: Fluid resuscitation with normal saline or Ringer's lactate

Second-line: Diuretics if fluid overload is confirmed

Definitive: Dialysis if necessary

MCQ 827:

A 75-year-old man complains of severe diarrhoea for the past 2 days. He has been taking antibiotics for the past 2 weeks for a lower respiratory tract infection. Examination confirms generalized distension and tenderness.
Blood pressure 100/60 mmHg
Heart rate 110 /min
Respiratory rate 22 /min
Temperature 38 °C
Which of the following is the most appropriate management?

A. CT scan of chest
B. Diagnostic laparoscopy
C. Exploratory laparotomy
D. Stool for clostridium difficile toxins

Explanation:
Given the patient's recent antibiotic use and current symptoms of severe diarrhea, **Clostridium**

difficile infection (C. difficile) is the most likely diagnosis. The most appropriate test to confirm this diagnosis is stool testing for C. difficile toxins.

Correct Answer: D. Stool for clostridium difficile toxins ✅

Topic Section:

Clostridium Difficile Infection (Clostridium Difficile Infection)

Definition:
A bacterial infection caused by Clostridium difficile, typically following antibiotic use, leading to diarrhea, fever, and abdominal pain.

Key Clinical Presentation:

Watery diarrhea, often with blood

Abdominal cramps and tenderness

Recent history of antibiotic use

Investigations:

Best initial: Stool culture or toxin test for C. difficile

Confirmatory: PCR for C. difficile toxins

Treatment Outline:

First-line: Oral vancomycin or fidaxomicin

Second-line: Metronidazole

Definitive: Surgical intervention if severe complications (e.g., megacolon)

MCQ 828:

A 60-year-old woman presents with a 12-hour history of severe pain in the epigastrium radiating to her back. It is associated with nausea and 2 episodes of vomiting. She has a history of cholelithiasis.
Abdominal examination confirms generalized tenderness in abdomen.
Which of the following is the most likely diagnosis?

A. Biliary colic
B. Acute pancreatitis
C. Acute cholecystitis
D. Empyema gallbladder

Explanation:
The patient's presentation of severe epigastric pain radiating to the back, nausea, vomiting, and a history of cholelithiasis is most consistent with **acute pancreatitis**. The pain is typically more severe and radiates to the back, distinguishing it from biliary colic or acute cholecystitis.

Correct Answer: B. Acute pancreatitis ✅

Topic Section:

Acute Pancreatitis (Acute Pancreatitis)

Definition:
Inflammation of the pancreas, often caused by alcohol use, gallstones, or other factors, leading to abdominal pain, nausea, vomiting, and elevated pancreatic enzymes.

Key Clinical Presentation:

Severe epigastric pain radiating to the back

Nausea and vomiting

History of risk factors (e.g., alcohol, gallstones)

Investigations:

Best initial: Serum amylase and lipase

Confirmatory: CT scan if diagnosis is uncertain or complications suspected

Treatment Outline:

First-line: Supportive care, IV fluids, and pain management

Second-line: Nutritional support (oral or enteral feeding)

Definitive: Surgical management if complications (e.g., abscess, necrosis)

MCQ 829:

A 7-day-old is presented for circumcision. Examination confirms urethral opening on the dorsal surface.
Which of the following is the most likely diagnosis?

A. Epispadias
B. Coronal hypospadias
C. Glandular hypospadias
D. Mid-penile hypospadias

Explanation:
The urethral opening on the dorsal surface of the penis is characteristic of **epispadias**. This congenital condition involves the malformation of the urethra, with the opening located on the dorsal aspect of the penis, as opposed to the ventral location in hypospadias.

Correct Answer: A. Epispadias ✅

Topic Section:

Epispadias (Epispadias)

Definition:
A congenital defect where the urethral opening is located on the dorsal surface of the penis.

Key Clinical Presentation:

Urethral opening on the dorsal side of the penis

May be associated with other genital anomalies

Investigations:

Best initial: Physical examination

Confirmatory: Surgical evaluation

Treatment Outline:

First-line: Surgical repair, often in infancy

Second-line: Follow-up care for any associated abnormalities

MCQ 830:

A 3-year-old uncircumcised boy has a history of foreskin ballooning during micturition. Examination confirms a ring of white scarred tissue and an inability to retract foreskin. Which of the following is the most likely diagnosis?

A. Chordee
B. Phimosis
C. Hypospadias
D. Paraphimosis

Explanation:
The inability to retract the foreskin with a history of ballooning during micturition suggests **phimosis**, a condition where the foreskin is too tight to be retracted over the glans, often due to scarring or congenital tightness.

Correct Answer: B. Phimosis ✅

Topic Section:

Phimosis (Phimosis)

Definition:
A condition where the foreskin of the penis cannot be retracted over the glans due to tightness or scarring.

Key Clinical Presentation:

Difficulty retracting the foreskin

Ballooning of the foreskin during urination

Scarring or fibrotic ring at the tip of the foreskin

Investigations:

Best initial: Physical examination

Confirmatory: If necessary, referral for urology evaluation

Treatment Outline:

First-line: Topical steroid application to loosen the foreskin

Second-line: Surgical intervention (circumcision) if conservative methods fail

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MCQ 831:

A 3-day-old is presented for circumcision. Examination confirms chordee and hooded foreskin. Urethral opening is at mid-penile level.
Which of the following is the most appropriate management?

A. Open circumcision
B. Plastibell circumcision
C. Circumcision using gumco
D. Referral to pediatric surgeon

Explanation:
Given the presence of **chordee** (a curvature of the penis) and **hooded foreskin**, along with a mid-penile urethral opening, **referral to a pediatric surgeon** is the most appropriate course of action. Chordee often requires correction alongside any other associated anomalies, and this type of repair is best done by a specialist.

Correct Answer: D. Referral to pediatric surgeon ✅

Topic Section:

Chordee (Chordee)

Definition:
A congenital condition characterized by a downward curvature of the penis, often associated with other anomalies such as hypospadias.

Key Clinical Presentation:

Curvature of the penis during erection

Possible associated hypospadias or other genital anomalies

Investigations:

Best initial: Physical examination

Confirmatory: Surgical evaluation and correction

Treatment Outline:

First-line: Surgical correction, often in infancy or early childhood

Second-line: Follow-up care for any other associated abnormalities

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MCQ 832:

What color of nipple discharge is most commonly associated with intraductal papilloma of the breast?

A. Red
B. Black
C. White
D. Green

Explanation:
Intraductal papillomas often present with **serous or blood-tinged nipple discharge**, and **red** is the most common color of discharge associated with this benign condition.

Correct Answer: A. Red ✅

Topic Section:

Intraductal Papilloma (Intraductal Papilloma)

Definition:
A benign tumor that forms within the milk ducts of the breast, commonly presenting with nipple discharge.

Key Clinical Presentation:

Blood-tinged or serous nipple discharge

A palpable mass or no palpable mass

Investigations:

Best initial: Mammography or ultrasound

Confirmatory: Core biopsy for diagnosis

Treatment Outline:

First-line: Surgical excision

Second-line: Monitoring or ductal excision if necessary

MCQ 843:

A 60-year-old woman presents with a mass in her right breast. Clinical examination confirms a hard, non-tender irregular lump with skin tethering.
Which of the following is the most likely diagnosis?

A. Breast cyst
B. Duct ectasia
C. Fibroadenoma
D. Carcinoma of breast

Explanation:
A hard, irregular mass with skin tethering in an older woman is highly suggestive of **breast carcinoma**, particularly invasive ductal carcinoma.

Correct Answer: D. Carcinoma of breast ✅

Topic Section:

Breast Cancer (Breast Cancer)

Definition:
Malignant tumor arising from the cells of the breast tissue, commonly presenting as a lump.

Key Clinical Presentation:

Hard, irregular mass in the breast

Skin changes such as dimpling or tethering

Possible nipple discharge or retraction

Investigations:

Best initial: Mammography or ultrasound

Confirmatory: Biopsy for histopathological diagnosis

Treatment Outline:

First-line: Surgical excision (lumpectomy or mastectomy)

Second-line: Chemotherapy, radiation, and hormone therapy

Definitive: Based on tumor stage and hormone receptor status

MCQ 835:

A 35-year-old woman presents with difficulty in breastfeeding her baby. Examination confirms unilateral nipple inversion. Gentle pressure exerts the nipple and it is transversely slitted. Which of the following is the most likely diagnosis?

A. Breast cyst
B. Duct ectasia
C. Fibroadenoma
D. Carcinoma of breast

Explanation:
Nipple inversion with a transverse slit appearance on pressure suggests **duct ectasia**, a benign condition involving the dilatation of mammary ducts. It is commonly associated with nipple inversion and occurs in women in their 30s and 40s.

Correct Answer: B. Duct ectasia ✅

Topic Section:

Duct Ectasia (Duct Ectasia)

Definition:
Dilatation of the breast ducts, commonly affecting the subareolar ducts, often presenting with nipple inversion, discharge, and pain.

Key Clinical Presentation:

Nipple inversion or retraction

Nipple discharge (often green or brown)

Pain or tenderness in the breast

Investigations:

Best initial: Physical examination

Confirmatory: Ultrasound or mammography

Treatment Outline:

First-line: Conservative management, including warm compresses and anti-inflammatory drugs

Second-line: Surgical intervention (excision of affected ducts) if symptoms persist

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MCQ 836

A 28-year-old lactating mother presents with pain in her right breast. Examination confirms a 3 cm tender swelling with red and hot overlying skin. Temperature 37.9 °C.
Which of the following is the most likely diagnosis?

A. Duct ectasia
B. Fibroadenosis
C. Fibroadenoma
D. Breast Abscess

Explanation:
The combination of tenderness, redness, and warmth over the affected area, along with fever, points towards a **breast abscess**, a common complication of breastfeeding due to blocked milk ducts or mastitis.

Correct Answer: D. Breast Abscess ✅

Topic Section:

Breast Abscess (Breast Abscess)

Definition:
A localized collection of pus in the breast tissue, commonly occurring in lactating women due to infection of the milk ducts.

Key Clinical Presentation:

Painful, swollen, and red area on the breast

Fever and malaise

Tenderness with fluctuation

Investigations:

Best initial: Physical examination

Confirmatory: Ultrasound to confirm the presence of an abscess

Treatment Outline:

First-line: Antibiotics and drainage (needle aspiration or incision and drainage)

Second-line: Surgery for recurrent or complicated cases

MCQ 837

A 36-year-old woman presents with pain in both breasts associated with multiple lumps that become prominent during her premenstrual phase. Examination confirms multiple nodules of variable size and consistency. Examination of the axillae is normal. Ultrasound: Multiple bilateral cysts of variable sizes, and 2 solid oval shape lesion in the left breast.
Which of the following is the most likely diagnosis?

A. Fibroadenomas
B. Chronic mastitis
C. Fibrocystic disease
D. Carcinoma of breast

Explanation:
The most likely diagnosis is **fibrocystic disease**, which is characterized by multiple cysts and fibrous tissue changes in the breast, often exacerbated by hormonal fluctuations (e.g., premenstrual phase). This is a benign condition.

Correct Answer: C. Fibrocystic disease ✅

Topic Section:

Fibrocystic Disease (Fibrocystic Breast Disease)

Definition:
A benign condition involving the development of multiple cysts and fibrous tissue in the breast, often related to hormonal changes.

Key Clinical Presentation:

Multiple lumps in the breast that vary in size and consistency

Pain or tenderness, especially around the menstrual cycle

No axillary lymph node involvement

Investigations:

Best initial: Physical examination and ultrasound

Confirmatory: Fine needle aspiration (FNA) if necessary

Treatment Outline:

First-line: Conservative management with pain relief (NSAIDs, warm compresses)

Second-line: Hormonal treatments if symptoms are severe

MCQ 838:

A 24-year-old lactating mother presents with pain in her right breast. Examination confirms a 5 cm tender swelling. The overlying skin is red, inflamed, and thinned out. Temperature 38.7 °C. What is the most appropriate management?

A. Lump excision
B. Needle aspiration
C. Incision and drainage
D. Conservative management with antibiotics

Explanation:
This presentation, along with fever, suggests a **breast abscess**, which is typically treated with incision and drainage (I&D) if an abscess has formed. Antibiotics are also necessary.

Correct Answer: C. Incision and drainage ✅

Topic Section:

Breast Abscess (Breast Abscess)

Definition:
A localized infection in the breast that leads to the formation of a pus-filled cavity, often seen in lactating women due to blocked ducts.

Key Clinical Presentation:

Painful, swollen, warm, and erythematous breast

Fever and chills

Fluctuant mass

Investigations:

Best initial: Physical examination

Confirmatory: Ultrasound to confirm the presence of an abscess

Treatment Outline:

First-line: Antibiotics and incision & drainage (I&D)

Second-line: Surgical drainage for recurrent or severe cases

MCQ 839:

A 25-year-old man underwent surgery for inflammatory bowel disease. He requires regular replacement of fat-soluble vitamins parenterally. Which part of the bowel was most likely removed?

A. Distal ileum
B. Mid jejunum
C. Transverse colon
D. Descending colon

Explanation:
The **distal ileum** is responsible for absorbing fat-soluble vitamins (A, D, E, K). Removal of this section, commonly in surgeries for inflammatory bowel disease (e.g., Crohn's disease), can lead to malabsorption of these vitamins.

Correct Answer: A. Distal ileum ✅

Topic Section:

Ileal Resection (Ileal Resection)

Definition:
Surgical removal of the ileum, which may occur in conditions like Crohn's disease or other inflammatory bowel diseases.

Key Clinical Presentation:

Malabsorption of nutrients, particularly fat-soluble vitamins

Diarrhea or weight loss in some cases

Fatigue or other signs of vitamin deficiency

Investigations:

Best initial: Nutritional assessment

Confirmatory: Lab tests for vitamin deficiency (e.g., serum vitamin levels)

Treatment Outline:

First-line: Vitamin supplementation (fat-soluble vitamins)

Second-line: Management of malabsorption with dietary modifications or medications

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MCQ 840:

A 75-year-old woman presents with a 7-day history of severe pain in the epigastrium. The pain is relieved by vomiting and becomes worse when eating. History confirms that she has been taking medication for joint pain for the last 2 months. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Boerhaave's syndrome
D. Perforated peptic ulcer

Explanation:
This patient's history of recent medication use (non-steroidal anti-inflammatory drugs) and epigastric pain relieved by vomiting points towards **acute gastritis**. NSAIDs are commonly associated with causing gastritis, which can present with the described symptoms.

Correct Answer: B. Acute gastritis ✅

Topic Section:

Acute Gastritis (Acute Gastritis)

Definition:
Inflammation of the gastric mucosa, often triggered by factors like NSAIDs, alcohol, infections, or stress.

Key Clinical Presentation:

Epigastric pain, often relieved by vomiting

Nausea and vomiting

Use of NSAIDs or alcohol

Investigations:

Best initial: Clinical evaluation and history

Confirmatory: Endoscopy for mucosal appearance

Treatment Outline:

First-line: Proton pump inhibitors (PPIs), H2 blockers, antacids

Second-line: Discontinuation of the offending agent (e.g., NSAIDs)

Definitive: Endoscopic intervention if complications arise (e.g., bleeding)

MCQ 841:

A 60-year-old patient complains of sudden onset of upper abdominal pain after a few bouts of vomiting. Examination confirms a sick patient with tenderness in the epigastrium and supraclavicular subcutaneous emphysema. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Perforated peptic ulcer
D. Boerhaave’s syndrome

Explanation:
The presence of **supraclavicular subcutaneous emphysema** following vomiting, with severe upper abdominal pain, strongly suggests **Boerhaave’s syndrome**, a spontaneous rupture of the esophagus, often due to forceful vomiting.

Correct Answer: D. Boerhaave’s syndrome ✅

Topic Section:

Boerhaave's Syndrome (Boerhaave's Syndrome)

Definition:
Spontaneous rupture of the esophagus, typically following forceful vomiting, leading to mediastinal air leakage and subcutaneous emphysema.

Key Clinical Presentation:

Severe chest or epigastric pain following vomiting

Subcutaneous emphysema in the neck or chest

Shock or sepsis signs

Investigations:

Best initial: Chest X-ray (may show pneumomediastinum or subcutaneous emphysema)

Confirmatory: Contrast esophagram or CT scan

Treatment Outline:

First-line: Emergency surgery or drainage

Second-line: Broad-spectrum antibiotics

Definitive: Surgical repair of the esophageal tear

MCQ 842:

A 25-year-old man presents with a 6-month history of heartburn. The pain becomes worse at night after lying down. On lifting weight, he experiences a bitter taste in mouth. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Boerhaave's syndrome
D. Perforated peptic ulcer

Explanation:
This patient's history of heartburn and the symptom of bitter taste on lifting weight points toward **gastroesophageal reflux disease (GERD)**, which causes acid reflux, leading to symptoms such as heartburn, regurgitation, and a bitter taste in the mouth.

Correct Answer: A. Esophagitis ✅

Topic Section:

Gastroesophageal Reflux Disease (GERD)

Definition:
A condition where stomach acid or bile irritates the esophagus, causing symptoms like heartburn and regurgitation.

Key Clinical Presentation:

Heartburn, especially after eating or lying down

Regurgitation and bitter taste

Sore throat or cough in some cases

Investigations:

Best initial: Clinical evaluation

Confirmatory: Endoscopy if needed

Treatment Outline:

First-line: Proton pump inhibitors (PPIs)

Second-line: H2 receptor antagonists or antacids

Definitive: Lifestyle modifications (e.g., weight loss, avoiding triggers)

MCQ 843:

A 55-year-old man presents with sudden, severe and constant pain in his epigastrium for the past 2 hours. History confirms that he has been on medication for joint pain for the last 2 years. Examination confirms a very sick patient with sunken eyes. Abdomen is tender and rigid to palpation. Which of the following is the most likely diagnosis?

A. Esophagitis
B. Acute gastritis
C. Perforated peptic ulcer
D. Boerhaave's syndrome

Explanation:
This patient's severe, sudden epigastric pain, along with rigidity on abdominal examination, suggests **perforated peptic ulcer**, which occurs due to erosion of the gastric or duodenal lining leading to free perforation into the peritoneal cavity.

Correct Answer: C. Perforated peptic ulcer ✅

Topic Section:

Perforated Peptic Ulcer (Perforated Peptic Ulcer)

Definition:
A complication of peptic ulcer disease where an ulcer erodes through the stomach or duodenal wall, leading to leakage of gastric contents into the peritoneum.

Key Clinical Presentation:

Sudden, severe epigastric pain

Abdominal rigidity (signs of peritonitis)

Recent history of NSAID use or gastritis

Investigations:

Best initial: X-ray (free air under diaphragm)

Confirmatory: CT scan or laparoscopy

Treatment Outline:

First-line: Surgery (laparotomy with ulcer repair)

Second-line: Broad-spectrum antibiotics for infection

Definitive: Surgical closure with omental patch and acid suppression

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MCQ 844:

A 50-year-old man presents with a long-standing history of indigestion and epigastric discomfort. He has poor appetite and has lost 10 kg recently. Examination confirms wasting of facial muscles, palpable supraclavicular lymph nodes and dull epigastric mass. Which of the following is the most likely diagnosis?

A. Liver metastasis
B. Chronic peptic ulcer
C. Stomach carcinoma
D. Chronic pancreatitis

Explanation:
This patient’s symptoms, weight loss, and epigastric mass, combined with palpable

supraclavicular lymph nodes, suggest **stomach carcinoma**, a malignancy commonly associated with weight loss and systemic signs of cancer.

Correct Answer: C. Stomach carcinoma ✅

Topic Section:

Stomach Carcinoma (Gastric Cancer)

Definition:
Malignancy of the stomach, often presenting with vague upper abdominal symptoms, weight loss, and sometimes lymphadenopathy.

Key Clinical Presentation:

Epigastric pain or discomfort

Weight loss and anorexia

Palpable supraclavicular lymph nodes (Virchow's node)

Investigations:

Best initial: Endoscopy with biopsy

Confirmatory: CT scan for staging

Treatment Outline:

First-line: Surgical resection (gastrectomy)

Second-line: Chemotherapy (for metastases)

Definitive: Chemoradiation therapy if applicable

MCQ 856:

A 65-year-old man presents with a 72-hour history of severe pain in the epigastrium radiating to his back with associated nausea and 2 episodes of vomiting. History confirms that he is a known case of cholelithiasis. Abdominal examination confirms generalized tenderness in the abdomen. US-guided aspiration of the intra-abdominal fluid grows E.coli. Blood pressure 100/70 mmHg, Heart rate 110 /min, Respiratory rate 22 /min, Temperature 38.6 °C. Test Result:
Amylase: 576 (24-151 IU/L)
Lipase: 300 (0-160 IU/L)

C-reactive peptide: 135 (<8.2 mg/L).
What is the most likely explanation for the bacterial growth?

A. Bowel perforation
B. Bacterial translocation
C. Haematogenous spread
D. Exogenous contamination

Explanation:
This patient's history and presentation of pain with nausea, vomiting, and raised inflammatory markers, along with the bacterial growth of E. coli from intra-abdominal fluid, are consistent with **bacterial translocation**. This occurs when bacteria from the gastrointestinal tract, often due to intestinal ischemia or perforation, move to the peritoneal cavity and cause infection.

Correct Answer: B. Bacterial translocation ✅

Topic Section:

Bacterial Translocation (Bacterial Translocation)

Definition:
The movement of bacteria from the gastrointestinal tract to sterile areas like the peritoneal cavity, bloodstream, or lymphatic system.

Key Clinical Presentation:

Abdominal pain and tenderness

Nausea and vomiting

Fever and signs of sepsis

Investigations:

Best initial: Blood and fluid cultures

Confirmatory: CT scan if perforation or abscess is suspected

Treatment Outline:

First-line: Antibiotics targeting E. coli and other common peritoneal pathogens

Second-line: Surgical management if there is a source of perforation or abscess

Definitive: Surgical intervention for any underlying pathology (e.g., bowel perforation)

MCQ 857:

A 25-year-old woman presents with a longstanding history of lower abdominal and right hypochondrial pain. Diagnostic laparoscopy confirms pelvic and perihepatic adhesions. Which of the following is most likely responsible?

A. E.coli
B. Chlamydia
C. Bacteroides
D. Streptococci

Explanation:
Chronic pelvic pain with adhesions, particularly in the setting of previous infections, is most commonly caused by **Chlamydia**, a common sexually transmitted infection that can lead to pelvic inflammatory disease (PID) and subsequent adhesions in the pelvic and perihepatic regions.

Correct Answer: B. Chlamydia ✅

Topic Section:

Pelvic Inflammatory Disease (PID)

Definition:
An infection of the female reproductive organs, commonly caused by sexually transmitted infections, leading to inflammation and the formation of adhesions.

Key Clinical Presentation:

Chronic pelvic pain

Abnormal vaginal discharge

History of sexually transmitted infections

Investigations:

Best initial: Pelvic ultrasound or laparoscopy

Confirmatory: Culture for Chlamydia or Gonorrhea

Treatment Outline:

First-line: Antibiotics targeting Chlamydia and Gonorrhea

Second-line: Pain management and surgical removal of adhesions if needed

Definitive: Prevention and treatment of STIs

MCQ 858:

A 25-year-old woman presents with a 6-hour history of lower abdominal pain associated with nausea and 1 episode of vomiting. Examination confirms minimal tenderness in the suprapubic and left iliac fossa region. Urine dipstick and pregnancy tests are normal. Blood pressure 110/70 mmHg, Heart rate 76 /min, Temperature 36.6 °C. Test Result:
Hb 110 (130-170 g/L Male, 120-160 Female),
WBC 9 (4.5-10.5 x 10^9/L),
Neutrophils 50 (40-60%).
Which of the following is the most appropriate management?

A. US abdomen
B. Abdominal X-ray
C. Diagnostic laparoscopy
D. Discharge on oral analgesics

Explanation:
This patient's mild symptoms and lack of alarming signs suggest a likely benign condition, such as a minor gastrointestinal issue. **Ultrasound abdomen** is the most appropriate initial investigation, especially to rule out gynecological or urinary tract issues like ovarian cysts or UTI.

Correct Answer: A. US abdomen ✅

Topic Section:

Abdominal Pain (Abdominal Pain)

Definition:
Pain arising from the abdominal organs, often requiring clinical evaluation and imaging to identify the underlying cause.

Key Clinical Presentation:

Abdominal pain with associated symptoms such as vomiting, diarrhea, or tenderness

Clinical history is essential in differentiating between gastrointestinal, gynecological, or other causes

Investigations:

Best initial: Abdominal ultrasound for gynecological and gastrointestinal pathology

Confirmatory: CT scan if needed for further evaluation

Treatment Outline:

First-line: Symptomatic treatment for benign conditions

Second-line: Surgical intervention if indicated

Definitive: Address the underlying cause based on diagnosis

MCQ 859:

A 55-year-old is admitted for an elective laparoscopic cholecystectomy. He is diabetic and hypertensive for 5 years. History confirms a recent admission in Intensive Care Unit for anterior wall MI 6 weeks ago. What is the most appropriate management?

A. Open cholecystectomy during this admission
B. Laparoscopic cholecystectomy after 6 months
C. Laparoscopic Cholecystectomy during this admission
D. Per-cutaneous cholecystostomy during this admission

Explanation:
For a patient who recently had a myocardial infarction (MI), it is essential to **delay elective surgery** for at least 6 weeks after MI to allow the heart to heal. **Laparoscopic cholecystectomy after 6 months** is the most appropriate approach for this patient.

Correct Answer: B. Laparoscopic cholecystectomy after 6 months ✅

Topic Section:

Laparoscopic Cholecystectomy (Laparoscopic Cholecystectomy)

Definition:
A minimally invasive procedure to remove the gallbladder, typically performed for symptomatic gallstones or cholecystitis.

Key Clinical Presentation:

Right upper quadrant pain

History of cholelithiasis or cholecystitis

Investigations:

Best initial: Abdominal ultrasound

Confirmatory: Liver function tests or CT scan if needed

Treatment Outline:

First-line: Laparoscopic cholecystectomy for symptomatic gallstones

Second-line: Percutaneous cholecystostomy for high-risk patients

Definitive: Surgical resection if indicated

MCQ 860:

A 65-year-old patient is admitted for elective ventral hernia repair. He is diabetic and hypertensive for 10 years. Examination confirms raised JVP, bilateral basal crepitation on chest auscultation and pedal edema. What is the most appropriate next step in management?

A. Open repair as scheduled
B. Laparoscopic repair as scheduled
C. Surgery after optimization at later date
D. No surgery unless presents with obstruction

Explanation:
This patient shows signs of **congestive heart failure** (elevated JVP, basal crepitations, pedal edema). **Surgery should be delayed** until the patient is optimized to avoid perioperative complications related to heart failure.

Correct Answer: C. Surgery after optimization at later date ✅

Topic Section:

Congestive Heart Failure (CHF)

Definition:
A condition in which the heart is unable to pump blood effectively, leading to fluid accumulation in the lungs and peripheral tissues.

Key Clinical Presentation:

Shortness of breath, edema, orthopnea, elevated JVP

History of hypertension, diabetes, or coronary artery disease

Investigations:

Best initial: Chest X-ray and echocardiogram

Confirmatory: BNP levels and cardiac MRI

Treatment Outline:

First-line: Diuretics, ACE inhibitors, beta-blockers

Second-line: Digoxin, aldosterone antagonists

Definitive: Lifestyle modifications, heart transplantation if needed

MCQ 861:

A 45-year-old woman complains of dysphagia to liquids, retrosternal pain on swallowing, and regurgitation of food. Which of the following has the highest diagnostic value?

A. Barium swallow
B. CT with contrast
C. Upper gastrointestinal endoscopy
D. Lower oesophageal sphincter manometry

Explanation:
The patient's symptoms of dysphagia, retrosternal pain, and regurgitation suggest a possible esophageal motility disorder or mechanical obstruction. **Upper gastrointestinal endoscopy** is the most appropriate diagnostic tool, allowing direct visualization of the esophagus, and can help diagnose conditions like esophagitis, strictures, or malignancy.

Correct Answer: C. Upper gastrointestinal endoscopy ✅

Topic Section:

Esophageal Disorders (Esophageal Disorders)

Definition:
Disorders affecting the esophagus that can cause dysphagia, regurgitation, and retrosternal pain, including motility disorders and mechanical obstructions.

Key Clinical Presentation:

Dysphagia (difficulty swallowing)

Regurgitation of food

Retrosternal pain

Investigations:

Best initial: Upper gastrointestinal endoscopy

Confirmatory: Manometry for motility disorders, Barium swallow

Treatment Outline:

First-line: Medical therapy (e.g., proton pump inhibitors for reflux) or surgical intervention for mechanical obstruction

Second-line: Endoscopic procedures for strictures or dilation

Definitive: Surgery or long-term management depending on underlying cause

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MCQ 862:

Which of the following is the most objective method for identifying high-risk patients for peri-operative morbidity and mortality?

A. Metabolic equivalent tasks
B. Revised cardiac risk index of Lee
C. Cardiopulmonary exercise testing
D. American society of anaesthesiologist scoring

Explanation:
The **Revised cardiac risk index of Lee** is the most objective and widely used tool for identifying high-risk patients for perioperative complications, particularly in relation to cardiovascular risk. It takes into account factors such as age, heart disease history, and type of surgery.

Correct Answer: B. Revised cardiac risk index of Lee ✅

Topic Section:

Perioperative Risk Assessment (Perioperative Risk Assessment)

Definition:
Assessment of a patient’s risk of morbidity and mortality during or after surgery, taking into account comorbidities and the type of surgical procedure.

Key Clinical Presentation:

History of cardiovascular disease

Age and overall physical condition

Type of surgery

Investigations:

Best initial: Revised cardiac risk index of Lee, American Society of Anesthesiologists (ASA) score

Confirmatory: Cardiopulmonary exercise testing if needed

Treatment Outline:

First-line: Preoperative optimization (e.g., managing diabetes, hypertension, or smoking cessation)

Second-line: Close monitoring during and after surgery

Definitive: Surgical approach tailored to the patient's risk profile

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MCQ 863:

A 45-year-old is brought to the Emergency Department following a high-speed motor vehicle accident. Examination confirms seat belt sign. CT: Chance fracture of lumbar vertebrae. Which of the following is the most likely finding on an abdominal CT?

A. Injury to vena cava
B. Stomach perforation
C. Duodenal perforation
D. Isolated jejunal blowout

Explanation:
In trauma with a seat belt sign and Chance fracture (a type of vertebral fracture), the most likely

abdominal injury is a **duodenal perforation** due to the compression and shearing forces during impact. The duodenum is particularly vulnerable in these types of trauma.

Correct Answer: C. Duodenal perforation ✅

Topic Section:

Abdominal Trauma (Abdominal Trauma)

Definition:
Injury to the abdomen, often resulting from blunt trauma, that may involve solid organs, hollow viscera, or vascular structures.

Key Clinical Presentation:

Abdominal pain or tenderness

Positive signs of trauma (e.g., seatbelt sign)

Abdominal distension and guarding

Investigations:

Best initial: CT scan of the abdomen, Focused Assessment with Sonography for Trauma (FAST)

Confirmatory: Diagnostic peritoneal lavage or exploratory laparotomy if needed

Treatment Outline:

First-line: Conservative management for minor injuries

Second-line: Surgical intervention for major injuries (e.g., bowel perforation)

Definitive: Surgical repair for injuries involving hollow organs or vascular structures

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MCQ 864:

A 25-year-old is brought to the Emergency Department after being rescued from a burning building. Examination confirms a slightly confused patient with singed facial and nasal hair. Which of the following is the most appropriate next step in management?

A. Elective intubation
B. Observation in ICU

C. Avoid medico-legal issues
D. Advice on pain management

Explanation:
For a patient who has been exposed to smoke inhalation and burns, **elective intubation** is critical to secure the airway, especially with signs of inhalation injury (e.g., singed facial hair). Intubation should be performed before any further deterioration occurs.

Correct Answer: A. Elective intubation ✅

Topic Section:

Burns and Smoke Inhalation Injury (Burns and Smoke Inhalation Injury)

Definition:
Injury from exposure to heat or chemicals causing damage to the skin, airways, and other tissues, often with a risk of inhalation injury from smoke.

Key Clinical Presentation:

Singed facial or nasal hair

Respiratory distress, confusion

Burns over a large body surface area

Investigations:

Best initial: Clinical evaluation, arterial blood gases (ABG), chest X-ray

Confirmatory: Bronchoscopy for assessing the extent of airway injury

Treatment Outline:

First-line: Airway management, oxygen supplementation, and intubation if needed

Second-line: Fluid resuscitation (e.g., Parkland formula)

Definitive: Burn care and long-term management of inhalation injuries

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MCQ 865:

A 45-year-old is presented following poly-trauma. He requires intubation to protect airway obstruction. After resuscitation and management, he is transferred to ICU. Which of the following is the most appropriate method to clear the cervical spine?

A. CT scan
B. Clinical judgement
C. MRI of cervical spine
D. AP and Lateral view cervical spine x-ray

Explanation:
In a poly-trauma patient, especially when there is concern for cervical spine injury, **CT scan** is the most appropriate method for clearing the cervical spine. It is highly sensitive for detecting fractures and dislocations of the cervical spine, and it can be performed rapidly, making it the best initial imaging choice in trauma cases.

Correct Answer: A. CT scan ✅

Topic Section:

Cervical Spine Injury Management (Cervical Spine Injury Management)

Definition:
Injuries to the cervical spine can lead to severe neurological compromise, and their management is crucial in poly-trauma patients.

Key Clinical Presentation:

Trauma to the neck or head

Mechanism of injury suggesting potential cervical spine involvement

Neurological deficits such as weakness, sensory loss, or paralysis

Investigations:

Best initial: CT scan for rapid assessment of fractures or dislocations

Confirmatory: MRI for soft tissue damage or ligamentous injuries

Treatment Outline:

First-line: Immobilization of the neck, and if injury is suspected, CT scan of the cervical spine

Second-line: MRI if soft tissue injury or spinal cord involvement is suspected

Definitive: Surgical intervention for fractures or instability

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MCQ 866:

Which of the following is the most appropriate strategy to avoid air bag injury in children under the age of 12 years?

A. Restrain in back seats
B. Restrain in front seats
C. Rear facing in front seats
D. Booster seats in front seats

Explanation:
The most appropriate strategy for children under the age of 12 years to avoid airbag injury is to **restrain in back seats**. The back seat is the safest place for children, especially when they are under 12 years old, as it reduces the risk of injury from airbags and is generally safer in the event of a collision.

Correct Answer: A. Restrain in back seats ✅

Topic Section:

Child Passenger Safety (Child Passenger Safety)

Definition:
Ensuring the safety of children during vehicular transport through appropriate restraint systems.

Key Clinical Presentation:

Child safety concerns during travel

Risk of injury due to improper seat placement or restraint system

Investigations:

Best initial: Evaluate the child's current restraint system and seating position

Confirmatory: Review of safety guidelines from organizations like NHTSA

Treatment Outline:

First-line: Proper restraint in back seat for children under 12 years

Second-line: Proper booster seat for children who outgrow car seats but are too small for standard seat belts

Definitive: Use of age-appropriate restraints for all children

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MCQ 867:

A 50-year-old woman comes to the clinic with numbness in the lower part of the ear, lower face, and upper neck. History reveals, she underwent a lateral neck dissection. Examination shows, loss of sensation over both surfaces of the lower part of the pinna and skin overlying the angle of the mandible on the same side of the operation. Which of the following nerves is most likely injured?

A. Third occipital
B. Great auricular
C. Lesser occipital
D. Greater occipital

Explanation:
The **Great auricular nerve** is responsible for sensory innervation to the lower part of the ear, the lower face, and the upper neck, which matches the clinical presentation of this patient after lateral neck dissection. This nerve is commonly injured during such procedures.

Correct Answer: B. Great auricular ✅

Topic Section:

Nerve Injuries in Neck Surgeries (Nerve Injuries in Neck Surgeries)

Definition:
Nerve injuries that occur as a complication of surgical procedures in the neck, such as lateral neck dissection.

Key Clinical Presentation:

Numbness or sensory loss in the ear, face, or neck after surgery

Specific patterns of sensory loss based on the affected nerve

Investigations:

Best initial: Clinical examination and history of surgery

Confirmatory: Nerve conduction studies if necessary

Treatment Outline:

First-line: Conservative management with observation and physiotherapy

Second-line: Surgical repair if function does not improve or if significant symptoms persist

Definitive: Nerve grafting or other surgical interventions in severe cases

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MCQ 868:

A 25-year-old man is brought to the Emergency Room after a motor vehicle accident. Examination shows, he has a severed nerve in the right upper limb, resulting in a loss of adduction of all the digits of the right hand. Which of the following is the most likely affected nerve?

A. Ulnar
B. Median
C. Musculocutaneous
D. Upper subscapular

Explanation:
The **ulnar nerve** innervates the muscles responsible for adduction of the fingers (including the interossei muscles). A lesion to this nerve would result in an inability to adduct the fingers, as described in this patient.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Nerve Injuries (Ulnar Nerve Injuries)

Definition:
Damage to the ulnar nerve, leading to loss of function in the hand, particularly affecting the small and ring fingers and the ability to adduct the digits.

Key Clinical Presentation:

Loss of finger adduction

Weakness or atrophy of the intrinsic hand muscles

Sensory loss along the ulnar aspect of the hand

Investigations:

Best initial: Clinical evaluation

Confirmatory: Nerve conduction studies or electromyography

Treatment Outline:

First-line: Conservative management or splinting

Second-line: Surgical exploration and repair if necessary

Definitive: Nerve repair or grafting in cases of severe injury

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MCQ 869:

A 26-year-old man is brought to the Emergency Room after sustaining a bullet injury to his arm. Examination shows his median nerve was damaged. Which of the following is the most likely clinical finding?

A. Ape hand
B. Claw hand
C. Wrist drop
D. Waiter's tip hand

Explanation:
Damage to the **median nerve** results in **Ape hand** deformity, characterized by the inability to oppose the thumb, leading to a loss of thumb opposition and flattening of the thenar eminence.

Correct Answer: A. Ape hand ✅

Topic Section:

Median Nerve Injuries (Median Nerve Injuries)

Definition:
Injury to the median nerve, typically resulting in loss of function in the thumb and the hand, affecting the ability to oppose the thumb and flex the wrist.

Key Clinical Presentation:

Weakness in thumb opposition

Flattening of the thenar eminence (Ape hand)

Sensory loss along the radial side of the palm and fingers

Investigations:

Best initial: Clinical evaluation and history of injury

Confirmatory: Nerve conduction studies or electromyography

Treatment Outline:

First-line: Conservative management with splinting

Second-line: Surgical exploration and repair if necessary

Definitive: Nerve repair or grafting for severe injuries

MCQ 870:

A 19-year-old patient comes to the clinic because he is unable to bring the fork to his mouth to feed himself. History reveals the patient had a closed head injury in a motor vehicle accident 10 days ago. Which of the following is the most likely involved area?

A. Cerebellum
B. Parietal lobe
C. Occipital lobe
D. Temporal lobe

Explanation:
The **parietal lobe** is primarily responsible for the integration of sensory information and motor control, particularly for tasks such as feeding oneself. Damage to this area can lead to difficulty with motor coordination (i.e., difficulty bringing a fork to the mouth).

Correct Answer: B. Parietal lobe ✅

Topic Section:

Parietal Lobe Injuries (Parietal Lobe Injuries)

Definition:
Damage to the parietal lobe, which is involved in sensory processing, spatial orientation, and motor control.

Key Clinical Presentation:

Difficulty with motor tasks (e.g., feeding oneself, dressing)

Impaired sensory processing

Hemispatial neglect

Investigations:

Best initial: CT scan or MRI of the brain

Confirmatory: Neuropsychological testing

Treatment Outline:

First-line: Physical therapy and occupational therapy to improve motor coordination

Second-line: Cognitive rehabilitation if necessary

MCQ 871:

A 24-year-old woman comes to the clinic after noticing a lump in her breast. Examination shows a firm, painless, and freely movable 3 cm mass in the left breast. The mass has grown slowly over the past year and does not change during the menstrual cycle. Which of the following is the most likely diagnosis?

A. Fat cyst
B. Fibroadenoma
C. Fibro-cystic changes
D. Intraductal papilloma

Explanation:
The **fibroadenoma** is the most likely diagnosis. It is a benign, firm, painless, and freely movable mass that grows slowly and is typically non-cyclical in nature. Fibroadenomas are common in young women and are not influenced by hormonal cycles.

Correct Answer: B. Fibroadenoma ✅

Topic Section:

Fibroadenoma of the Breast (Fibroadenoma of the Breast)

Definition:
A benign tumor of the breast made up of both glandular and stromal tissue.

Key Clinical Presentation:

Firm, painless, mobile mass

Often seen in younger women (typically 20-30 years old)

Does not change during the menstrual cycle

Investigations:

Best initial: Clinical examination and mammography or ultrasound

Confirmatory: Core needle biopsy if needed

Treatment Outline:

First-line: Observation if asymptomatic

Second-line: Surgical excision if symptomatic or increasing in size

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MCQ 872:

A 21-year-old patient presents with an abnormal left testicle. On examination, the area adjacent to the left testicle feels like a bag of worms, which gets larger with a valsalva maneuver. Which of the following is the most likely diagnosis?

A. Hydrocele
B. Varicocele
C. Spermatocele
D. Left inguinal hernia

Explanation:
A **varicocele** is the most likely diagnosis. It is characterized by a "bag of worms" appearance on examination and is caused by dilated veins within the pampiniform plexus of the scrotum, often becoming more prominent with a valsalva maneuver.

Correct Answer: B. Varicocele ✅

Topic Section:

Varicocele (Varicocele)

Definition:
Enlargement of the veins within the scrotum, typically affecting the left side.

Key Clinical Presentation:

"Bag of worms" feeling on palpation

Symptoms may include dull aching or heaviness in the scrotum

Can be exacerbated by standing or valsalva maneuver

Investigations:

Best initial: Clinical examination

Confirmatory: Doppler ultrasound

Treatment Outline:

First-line: Observation if asymptomatic

Second-line: Surgical intervention or embolization if causing symptoms or infertility

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MCQ 873:

A 35-year-old man patient comes to the clinic with a painless ulcer on his penis. History reveals, he had unprotected sexual encounter 3 weeks ago. Examination shows that the lesion has a clean base with sharp margins, and there are non-tender palpable regional lymph nodes. Which of the following is the most likely diagnosis?

A. Syphilis
B. Chancroid
C. Gonorrhoea
D. Granuloma inguinale

Explanation:
The presence of a painless ulcer with clean base and sharp margins, along with non-tender regional lymphadenopathy, suggests **syphilis**. This is the characteristic presentation of primary syphilis, caused by *Treponema pallidum*.

Correct Answer: A. Syphilis ✅

Topic Section:

Syphilis (Syphilis)

Definition:
A sexually transmitted infection caused by *Treponema pallidum* that presents in stages, with primary syphilis characterized by a painless chancre (ulcer).

Key Clinical Presentation:

Painless ulcer (chancre) with clean base and sharp margins

Non-tender regional lymphadenopathy

Resolves on its own but requires treatment to prevent progression

Investigations:

Best initial: Serologic testing (VDRL, RPR)

Confirmatory: Dark field microscopy or PCR for *Treponema pallidum*

Treatment Outline:

First-line: Penicillin G (IM)

Second-line: Doxycycline for penicillin-allergic patients

MCQ 874:

A 53-year-old man presents to the hospital with fever and pain in the right knee for 2 days. On physical examination, the right knee was swollen and red with a limited range of motion. A knee joint aspiration was performed. Then, the patient started empirically on Cloxacillin. On day 3 after admission, a further report was received: Gram stain showed numerous white blood cells and gram-positive cocci in clusters. Bacteriology (3 days later): *Staphylococcus aureus* that was resistant to cefoxitin. Which of the following is the most appropriate action?

A. Add gentamicin
B. Start vancomycin
C. Continue cloxacillin
D. Arthroscopic washout

Explanation:
Since *Staphylococcus aureus* resistant to cefoxitin (suggesting methicillin resistance), **vancomycin** is the most appropriate choice for resistant infections, especially for osteoarticular infections.

Correct Answer: B. Start vancomycin ✅

Topic Section:

Methicillin-resistant Staphylococcus aureus (MRSA) (Methicillin-resistant Staphylococcus aureus)

Definition:
A strain of *Staphylococcus aureus* that is resistant to beta-lactam antibiotics, including methicillin, and requires alternative antibiotics for treatment.

Key Clinical Presentation:

Localized infection (e.g., skin, bone, joint)

Can cause septic arthritis, osteomyelitis, or soft tissue infections

Fever, redness, and swelling at the site of infection

Investigations:

Best initial: Gram stain and culture of aspirated fluid

Confirmatory: PCR or antimicrobial susceptibility testing

Treatment Outline:

First-line: Vancomycin or clindamycin for MRSA

Second-line: Linezolid or daptomycin for severe cases

Definitive: Drainage of infected joints or tissue if required

MCQ 875:

A 75-year-old man presents to the Outpatient Department, complaining of back pain and increasing difficulty with passing urine. Test Result: Prostate-specific antigen 84 (0-4 µg/L), Alkaline phosphatase 410 (39-117 IU/L), Albumin 41 (34-56 g/L), Gamma glutamyltransferase 19 (6 to 37 IU/L). Which of the following is the most likely diagnosis?

A. Prostatitis
B. Prostatic cancer
C. Urinary bladder cancer
D. Benign prostate hyperplasia

Explanation:
The elevated prostate-specific antigen (PSA) levels and alkaline phosphatase are indicative of **prostatic cancer**, particularly when accompanied by symptoms of back pain and urinary difficulty. Elevated PSA in combination with bone pain can be a sign of metastatic prostate cancer, making it the most likely diagnosis.

Correct Answer: B. Prostatic cancer ✅

Topic Section:

Prostatic Cancer (Prostate Cancer)

Definition:
A malignant tumor originating from the prostate gland, often characterized by elevated PSA levels and symptoms of urinary obstruction.

Key Clinical Presentation:

Back pain (especially if metastasis to bone occurs)

Urinary symptoms (difficulty passing urine, frequency, urgency)

Elevated PSA levels

Weight loss in advanced stages

Investigations:

Best initial: Prostate-specific antigen (PSA)

Confirmatory: Transrectal ultrasound-guided biopsy of the prostate

Treatment Outline:

First-line: Radical prostatectomy, radiation therapy, or androgen deprivation therapy (ADT)

Second-line: Chemotherapy for metastatic disease

Definitive: Surgery or radiation based on disease staging

MCQ 876:

A 60-year-old man presents to the clinic with pain in the left flank, weight loss, and haematuria. Examination confirms a firm mass in the lumbar region. Blood pressure 158/98 mmHg, Heart rate 76 /min, Respiratory rate 18 /min, Temperature 36.6°C. Which of the following imaging is the most diagnostic?

A. MRI
B. CT scan
C. Ultrasound
D. Radionuclide scan

Explanation:
A **CT scan** is the most diagnostic for kidney masses, particularly when evaluating for potential renal cell carcinoma (RCC), which can present with flank pain, weight loss, and haematuria. CT scan offers detailed imaging for detecting tumors and metastases, as well as for staging.

Correct Answer: B. CT scan ✅

Topic Section:

Renal Cell Carcinoma (RCC) (Renal Cell Carcinoma)

Definition:
A malignant tumor originating from the renal epithelium, often presenting with flank pain, hematuria, and a palpable abdominal mass.

Key Clinical Presentation:

Flank pain

Hematuria

Weight loss

Palpable abdominal mass in some cases

Investigations:

Best initial: Ultrasound for initial evaluation

Confirmatory: Contrast-enhanced CT or MRI of the abdomen

Treatment Outline:

First-line: Nephrectomy (partial or radical) for localized disease

Second-line: Targeted therapy (e.g., tyrosine kinase inhibitors) for metastatic disease

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MCQ 877:

A 30-year-old patient presents to the clinic with a pelvic fracture, due to blunt trauma. A retrograde urethrography demonstrates disruption of the membranous urethra. Which of the following is the most appropriate first step in management?

A. Perineal repair
B. Retropubic repair
C. Suprapubic catheter
D. Transurethral catheter

Explanation:
The most appropriate initial step is to place a **suprapubic catheter** to decompress the bladder and avoid further injury to the urethra. This is particularly important when there is urethral injury, as inserting a transurethral catheter may worsen the injury.

Correct Answer: C. Suprapubic catheter ✅

Topic Section:

Urethral Injury (Urethral Injury)

Definition:
Injury to the urethra, often resulting from blunt trauma or pelvic fractures. This can lead to urinary retention and possible infection if not managed appropriately.

Key Clinical Presentation:

Blood at the urethral meatus

Difficulty passing urine

Pelvic fracture history

Tenderness over the pelvic region

Investigations:

Best initial: Retrograde urethrography

Confirmatory: Cystoscopy or further imaging as needed

Treatment Outline:

First-line: Suprapubic catheter insertion to avoid worsening urethral injury

Second-line: Surgical repair based on the type of injury

Definitive: Urethral repair (perineal or retropubic approach depending on injury site)

MCQ 878:

A 2-year-old boy is brought to the clinic with a large left-sided abdominal mass. IV pyelogram shows, the mass appears to arise from the left kidney and distort and displace the collecting system. Chest X-ray: Multiple pulmonary nodules. Which of the following is the most likely diagnosis?

A. Wilm’s tumour
B. Neuroblastoma
C. Rhabdomyosarcoma
D. Non-Hodgkin lymphoma

Explanation:
Wilm’s tumor is the most likely diagnosis. It is the most common renal malignancy in children and typically presents as an abdominal mass with or without symptoms. The presence of pulmonary nodules suggests metastatic spread, which is common in advanced stages of Wilm’s tumor.

Correct Answer: A. Wilm’s tumour ✅

Topic Section:

Wilm's Tumor (Wilms' Tumor)

Definition:
A malignant kidney tumor commonly seen in children, typically presenting as a painless abdominal mass.

Key Clinical Presentation:

Abdominal mass

Hematuria (less common)

Hypertension in some cases

Pulmonary metastasis in advanced stages

Investigations:

Best initial: Abdominal ultrasound

Confirmatory: CT scan or MRI for staging, biopsy if necessary

Treatment Outline:

First-line: Nephrectomy and chemotherapy

Second-line: Radiation for advanced or recurrent disease

Definitive: Surgery (nephrectomy) followed by chemotherapy based on staging

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MCQ 879:

A 30-year-old patient presents with recurrent urinary tract infections, polyuria, and electrolyte imbalance. Ultrasound: Bilaterally enlarged kidneys with multiple, variably-sized and thin-walled cysts distributed throughout the renal parenchyma. Which of the following is the most likely diagnosis?

A. Carcinoma bladder
B. Renal calculi disease
C. Medullary sponge kidney
D. Polycystic kidney disease

Explanation:
Polycystic kidney disease (PKD) is the most likely diagnosis. It is characterized by multiple cysts in the kidneys, leading to renal enlargement and dysfunction. It is often associated with recurrent urinary tract infections and electrolyte imbalances as the disease progresses.

Correct Answer: D. Polycystic kidney disease ✅

Topic Section:

Polycystic Kidney Disease (Polycystic Kidney Disease)

Definition:
A genetic disorder characterized by the growth of numerous cysts in the kidneys, leading to renal enlargement and progressive renal failure.

Key Clinical Presentation:

Enlarged kidneys (palpable in some cases)

Recurrent urinary tract infections

Hypertension

Electrolyte imbalances

Investigations:

Best initial: Ultrasound to detect renal cysts

Confirmatory: Genetic testing (ADPKD1 and ADPKD2 mutations)

Treatment Outline:

First-line: Blood pressure control (ACE inhibitors or ARBs), pain management

Second-line: Dialysis and kidney transplant for advanced renal failure

Definitive: Kidney transplant in end-stage disease

MCQ 880:

A 80-year-old man presents with dull aching pain in the loins. Test Result: Creatinine 280 (44-115 µmol/L), Urea 12.5 (2.75-7.4 mmol/L). Ultrasound abdomen: Bilateral hydronephrosis. Which of the following is the most likely diagnosis?

A. Stricture of urethral meatus
B. Prostatic enlargement
C. Neoplasm of bladder
D. Pelvic cancer

Explanation:
Bilateral hydronephrosis with elevated creatinine and urea is suggestive of obstructive uropathy. The most likely cause in this elderly male is **prostatic enlargement**, which can lead to urinary retention and bilateral hydronephrosis by obstructing the flow of urine.

Correct Answer: B. Prostatic enlargement ✅

Topic Section:

Prostatic Enlargement (Benign Prostatic Hyperplasia)

Definition:
Benign enlargement of the prostate gland commonly affecting older men, leading to urinary obstruction and other lower urinary tract symptoms.

Key Clinical Presentation:

Urinary frequency, especially at night (nocturia)

Difficulty initiating urination

Weak urinary stream

Sensation of incomplete bladder emptying

Pain in the loins or lower abdomen

Investigations:

Best initial: Digital rectal exam (DRE) and PSA measurement

Confirmatory: Ultrasound of the prostate, cystoscopy if needed

Treatment Outline:

First-line: Alpha-blockers (e.g., tamsulosin)

Second-line: 5-alpha-reductase inhibitors (e.g., finasteride)

Definitive: Surgical options like transurethral resection of the prostate (TURP) for refractory cases

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MCQ 881:

A 65-year-old male comes to the clinic with a mild intermittent urinary flow reduction. Rectal examination, urinalysis and prostate specific antigen studies are normal. Ultrasound prostate: Enlarged median lobe. Which of the following is the best way to investigate?

A. Annual renal function monitoring
B. Periodic PSA measurement
C. Beta-blocker therapy
D. Cystoscopy

Explanation:
The normal PSA, rectal exam, and urinalysis suggest benign prostatic hypertrophy (BPH) rather than prostate cancer. The **best next step** for further investigation would be **periodic PSA**

measurement to monitor the condition, as well as symptomatic treatment with alpha-blockers if required.

Correct Answer: B. Periodic PSA measurement ✅

Topic Section:

Benign Prostatic Hyperplasia (BPH)

Definition:
Non-cancerous enlargement of the prostate gland, leading to lower urinary tract symptoms (LUTS).

Key Clinical Presentation:

Increased frequency of urination, particularly nocturia

Weak stream, difficulty initiating urination

Sensation of incomplete emptying

No fever, pain, or hematuria

Investigations:

Best initial: Digital rectal exam (DRE) and PSA

Confirmatory: Ultrasound of prostate, cystoscopy if necessary

Treatment Outline:

First-line: Alpha-blockers (e.g., tamsulosin)

Second-line: 5-alpha-reductase inhibitors (e.g., finasteride)

Definitive: Surgical options like TURP

MCQ 882:

A 82-year-old man presents to the clinic with inability to urinate. The patient complains of severe pain and bloating in the lower abdomen, painful and urgent need to urinate. Which of the following is the most appropriate management?

A. Semi-urgent prostatectomy
B. Antibiotic for a urinary infection

C. Foley catheterisation and culture urine
D. Cystoscopy and Transurethral Resection Prostatectomy

Explanation:
The patient is presenting with acute urinary retention, likely due to benign prostatic hyperplasia (BPH) or other obstruction. The most immediate and appropriate intervention is **Foley catheterization** to relieve urinary retention, followed by a urine culture to rule out infection.

Correct Answer: C. Foley catheterisation and culture urine ✅

Topic Section:

Acute Urinary Retention (Acute Urinary Retention)

Definition:
Sudden inability to urinate, often due to obstruction such as benign prostatic hyperplasia, bladder stones, or other conditions.

Key Clinical Presentation:

Severe pain and bloating in the lower abdomen

Inability to void, with a strong urge to urinate

Firm, distended bladder on palpation

Investigations:

Best initial: Clinical examination and bladder ultrasound

Confirmatory: Urine culture, cystoscopy, and prostate examination

Treatment Outline:

First-line: Foley catheterization to relieve the retention

Second-line: Treat underlying cause (e.g., alpha-blockers for BPH)

Definitive: Surgery for persistent obstruction or cause

MCQ 883:

A 70-year-old, non-smoking man presents with increased urinary frequency, especially at night. The patient feels that the bladder is not fully empty after voiding, and the urine stream

has slightly diminished. The patient's high blood pressure is well controlled with a calcium channel blocker. There is no associated fever, weight loss, haematuria, or pain with voiding. The digital rectal examination was significant for a moderate enlargement of the prostate, but is otherwise normal. Results of urinalysis and kidney function tests were normal. PSA antigen 1 (< 4 ng/ml). Which of the following is the most appropriate initial management?

A. Cystoscopy
B. Open prostatectomy
C. Alpha-blocker therapy
D. Transurethral Resection Prostatectomy

Explanation:
The patient likely has **benign prostatic hyperplasia (BPH)**, as indicated by urinary frequency, nocturia, and weak stream with normal prostate exam findings. **Alpha-blocker therapy** (e.g., tamsulosin) is the first-line treatment for symptomatic BPH, improving urine flow and relieving symptoms.

Correct Answer: C. Alpha-blocker therapy ✅

Topic Section:

Benign Prostatic Hyperplasia (BPH)

Definition:
Enlargement of the prostate gland, often leading to urinary obstruction and lower urinary tract symptoms.

Key Clinical Presentation:

Increased frequency of urination

Nocturia

Weak stream, difficulty starting urination

Sensation of incomplete emptying of the bladder

Investigations:

Best initial: Digital rectal exam (DRE) and PSA level

Confirmatory: Ultrasound of prostate and cystoscopy if needed

Treatment Outline:

First-line: Alpha-blockers (e.g., tamsulosin)

Second-line: 5-alpha-reductase inhibitors (e.g., finasteride)

Definitive: Surgical options like TURP for refractory cases

MCQ 884:

A 56-year-old multiparous patient on oral oestrogen, complains of painless urine incontinence when coughing, sneezing, or laughing for 1 year. On examination, there was mild vaginal atrophy and constrictor muscle laxity. Urinary incontinence is demonstrated in the lithotomy position with Valsalva manoeuvre. Urinalysis: Normal. Which of the following would be the recommended management?

A. Beta blocker
B. Kegel exercises
C. Periurethral bulking
D. Post-coital antibiotics

Explanation:
The patient's symptoms are indicative of **stress urinary incontinence** (SUI), which is commonly seen in multiparous women, particularly with vaginal atrophy and laxity of the pelvic floor muscles. The most effective initial treatment is **Kegel exercises**, which strengthen the pelvic floor muscles and help manage stress incontinence.

Correct Answer: B. Kegel exercises ✅

Topic Section:

Stress Urinary Incontinence (SUI)

Definition:
Involuntary leakage of urine during activities that increase intra-abdominal pressure (e.g., coughing, sneezing, laughing, or exercising), due to weakening of the pelvic floor muscles or damage to the urethral sphincter.

Key Clinical Presentation:

Urinary incontinence associated with coughing, sneezing, or physical activity

Palpably distended bladder, especially after voiding

Vaginal atrophy or pelvic floor laxity may be present in older women

Investigations:

Best initial: Clinical examination and history

Confirmatory: Urodynamic testing if needed

Treatment Outline:

First-line: Kegel exercises, lifestyle changes (weight loss, fluid management)

Second-line: Periurethral bulking agents, vaginal pessaries

Definitive: Surgery, such as sling procedures, in refractory cases

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MCQ 885:

A 37-year-old man presents to the clinic with flank pain, haematuria and burning sensation while urinating. The pain became more severe and constant in the last several days. Which of the following is the most appropriate investigation?

A. Nuclear renal scan
B. Ultrasound of Kidneys
C. CT of Abdomen without contrast
D. CT of Abdomen with oral and IV contrast

Explanation:
Given the patient's symptoms of flank pain, hematuria, and dysuria, the most likely diagnosis is a **urinary stone**. A **CT of the abdomen with contrast** is the most appropriate investigation as it provides detailed imaging to detect stones in the kidneys, ureters, and bladder, including those that may not be visible on ultrasound.

Correct Answer: D. CT of Abdomen with oral and IV contrast ✅

Topic Section:

Urinary Tract Stones (Urolithiasis)

Definition:
Formation of stones in the urinary tract, commonly in the kidneys, ureters, or bladder, resulting in pain, hematuria, and sometimes obstruction.

Key Clinical Presentation:

Severe flank pain (renal colic), hematuria, and burning sensation on urination

Pain often radiates to the groin or lower abdomen

Symptoms may worsen with movement or voiding

Investigations:

Best initial: Non-contrast CT scan or ultrasound

Confirmatory: CT of abdomen with contrast for detailed stone imaging

Treatment Outline:

First-line: Conservative management with hydration and analgesia for small stones

Second-line: Extracorporeal shock wave lithotripsy (ESWL) for larger stones

Definitive: Surgical intervention (ureteroscopy or nephrolithotomy) for large or impacted stones

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MCQ 886:

A 70-year-old man presents with urinary incontinence. The patient’s bladder is palpably distended after voicing. The patient stated that he voids frequently, although he has difficulty initiating the urine stream. Which of the following is the most likely type of incontinence?

A. Urge
B. Stress
C. Reflex
D. Overflow

Explanation:
The patient’s presentation of urinary retention, difficulty initiating urination, and a palpable distended bladder suggests **overflow incontinence**, which occurs when the bladder cannot empty fully, leading to frequent, incomplete voiding and leakage due to overflow.

Correct Answer: D. Overflow ✅

Topic Section:

Overflow Urinary Incontinence

Definition:
Involuntary leakage of urine due to an overfull bladder, often caused by incomplete bladder emptying, leading to urine retention and leakage.

Key Clinical Presentation:

Palpably distended bladder, frequent small voids with difficulty initiating urination

Leakage of urine, especially at night

May be associated with neurological conditions or bladder outlet obstruction

Investigations:

Best initial: Post-void residual volume measurement (bladder scan)

Confirmatory: Urodynamic studies to assess bladder function

Treatment Outline:

First-line: Bladder drainage via intermittent catheterization or indwelling catheter

Second-line: Alpha-blockers or surgical intervention for bladder outlet obstruction

Definitive: Surgery for bladder outlet obstruction or neurogenic bladder management

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MCQ 887:

A 40-year-old man comes to the clinic with pain in the lower back, and blood in the urine. IVP shows a non-opaque filling defects in the renal pelvis. Ultrasound revealed dense echoes and acoustic shadowing. Which of the following is the most likely diagnosis?

A. Tumour
B. Blood clot
C. Uric acid stone
D. Sloughed renal papilla

Explanation:
The findings of non-opaque filling defects on IVP, along with dense echoes and acoustic shadowing on ultrasound, are highly suggestive of a **urinary stone**, particularly a **uric acid stone**, which is radiolucent and may present with these features.

Correct Answer: C. Uric acid stone ✅

Topic Section:

Uric Acid Stones

Definition:
A type of kidney stone that forms in acidic urine, often seen in conditions like gout, dehydration, or metabolic syndrome.

Key Clinical Presentation:

Severe flank pain (renal colic), hematuria, dysuria

Pain often radiates to the groin or lower abdomen

Investigations:

Best initial: Non-contrast CT or ultrasound for stone detection

Confirmatory: Urinary pH measurement and stone analysis (if available)

Treatment Outline:

First-line: Hydration, alkalinization of urine (e.g., potassium citrate)

Second-line: Lithotripsy or ureteroscopy for larger stones

Definitive: Surgical removal for large or obstructing stones

MCQ 889:

A 74-year-old man is referred to the thoracic surgeon with dysphagia of solid food for six weeks and marked weight loss for 2 months. Oesophageal endoscopy shows a stenotic mass in the mid oesophagus. Chest X-ray: Normal. Which of the following is the most likely diagnosis?

A. Lymphoma
B. Adenocarcinoma
C. Squamous cell carcinoma
D. Extension from neuroendocrine cancer

Explanation:
This patient presents with dysphagia, weight loss, and a stenotic mass seen on oesophageal endoscopy. The most likely diagnosis is **squamous cell carcinoma** of the esophagus, which is more commonly located in the mid to upper oesophagus, particularly in patients with risk factors such as smoking and alcohol use.

Correct Answer: C. Squamous cell carcinoma ✅

Topic Section:

Oesophageal Squamous Cell Carcinoma

Definition:
A malignant tumor originating in the squamous epithelial cells of the esophagus, commonly associated with smoking, alcohol, and other environmental factors.

Key Clinical Presentation:

Dysphagia, initially to solids, progressing to liquids

Unintentional weight loss

Painful swallowing (odynophagia)

Regurgitation of food

Investigations:

Best initial: Oesophagoscopy with biopsy for histopathology

Confirmatory: Endoscopic ultrasound for staging

Treatment Outline:

First-line: Chemoradiotherapy for localized disease

Second-line: Esophagectomy with lymph node dissection

Definitive: Combination of surgery, radiation, and chemotherapy for advanced cases

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MCQ 890:

An army soldier comes to the Outpatient Clinic complaining of painful flat feet after several years of service, which included thousands of kilometers of marching in the desert. The pain was acute in the medial aspect of his sole. Which of the following structures is the most likely strained?

A. Tendon achilles
B. Spring ligament

C. Flexor retinaculum
D. Extensor retinaculum

Explanation:
The pain on the medial aspect of the sole with prolonged marching in a soldier suggests strain on the **spring ligament**, which supports the medial arch of the foot. The spring ligament is crucial in maintaining the arch, and its injury can lead to pain in the medial foot.

Correct Answer: B. Spring ligament ✅

Topic Section:

Spring Ligament (Plantar Calcaneonavicular Ligament)

Definition:
A ligament in the foot that supports the medial longitudinal arch, attaching the calcaneus to the navicular bone.

Key Clinical Presentation:

Pain in the medial arch, especially with prolonged standing or activity

Tenderness over the spring ligament

Investigations:

Best initial: Clinical examination and history

Confirmatory: MRI if conservative management fails

Treatment Outline:

First-line: Rest, ice, and orthotic support

Second-line: Physical therapy, NSAIDs

Definitive: Surgical repair if conservative measures fail

MCQ 891:

A 65-year-old woman comes to the Outpatient Clinic complaining of pain in her left hand and fingers from typing on the keyboard for a while. Investigations reveal insufficient blood flow into the superficial palmar arch. Occlusion of which of the following arteries is the most likely cause?

A. Ulnar
B. Radial
C. Anterior interosseous
D. Posterior interosseous

Explanation:
The **ulnar artery** is the primary source of blood supply to the superficial palmar arch, and occlusion here can result in insufficient blood flow to the hand and fingers, particularly in the context of repetitive activities like typing.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Artery Occlusion

Definition:
Occlusion of the ulnar artery, which can lead to decreased blood flow to the hand, especially the medial side.

Key Clinical Presentation:

Pain, numbness, or tingling in the hand and fingers

Decreased blood flow in the affected hand

May be exacerbated by repetitive activities (e.g., typing)

Investigations:

Best initial: Doppler ultrasound to assess blood flow

Confirmatory: Angiography in cases of persistent symptoms

Treatment Outline:

First-line: Conservative management with rest and anti-inflammatory medications

Second-line: Surgical intervention in cases of persistent ischemia or claudication

Definitive: Arterial bypass or thrombectomy for severe cases

MCQ 892:

As part of physical examination to evaluate the lower limb function of a football player, the physician asked him to stand on his tiptoes. Which of the following nerves was the physician testing?

A. Tibial
B. Femoral
C. Deep fibular
D. Superficial fibular

Explanation:
The ability to stand on tiptoes tests the function of the **tibial nerve**, which innervates the gastrocnemius and soleus muscles, responsible for plantar flexion at the ankle.

Correct Answer: A. Tibial ✅

Topic Section:

Tibial Nerve Function

Definition:
The tibial nerve is responsible for motor innervation to muscles of the posterior compartment of the leg, including the gastrocnemius and soleus, which control plantar flexion of the foot.

Key Clinical Presentation:

Inability to plantar flex the foot or stand on tiptoes when the tibial nerve is damaged

Weakness in foot inversion and toe flexion

Investigations:

Best initial: Clinical examination of foot movements

Confirmatory: Electromyography (EMG) if necessary

Treatment Outline:

First-line: Physical therapy and conservative management

Definitive: Surgical intervention for nerve decompression or repair in severe cases

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MCQ 893:

A 25-year-old man is brought to the Emergency Room after a motor vehicle accident. Examination shows, he has a severed nerve in the right upper limb, resulting in a loss of adduction of all the digits of the right hand. Which of the following is the most likely affected nerve?

A. Ulnar
B. Median
C. Musculocutaneous
D. Upper subscapular

Explanation:
The **ulnar nerve** innervates the intrinsic muscles of the hand responsible for adducting the fingers, including the palmar interossei and the adductor pollicis. Damage to the ulnar nerve results in loss of adduction of the digits.

Correct Answer: A. Ulnar ✅

Topic Section:

Ulnar Nerve Injury

Definition:
Damage to the ulnar nerve can result in impaired hand function, particularly the loss of fine motor control and grip strength, including the ability to adduct the fingers.

Key Clinical Presentation:

Weakness in hand grip and finger adduction

Claw hand deformity (if prolonged damage)

Numbness along the ulnar side of the hand and forearm

Investigations:

Best initial: Clinical examination for hand movements and grip strength

Confirmatory: Electromyography (EMG) to assess nerve function

Treatment Outline:

First-line: Conservative management with hand therapy

Second-line: Splinting or bracing for claw hand deformity

Definitive: Nerve repair or decompression if necessary

MCQ 894:

A 20-year-old man comes to the clinic complaining of knee pain. History reveals he received a severe blow on the inferolateral side of the left knee joint while playing football. Radiographic examination revealed a fracture of the head and neck of the fibula. Which of the following nerves is the most likely to be damaged?

A. Tibial
B. Deep peroneal
C. Common peroneal
D. Superficial peroneal

Explanation:
The **common peroneal nerve** is most likely to be damaged in fractures of the fibula, particularly at the head and neck. This nerve passes around the fibular head and supplies the muscles responsible for dorsiflexion and eversion of the foot. Damage to this nerve causes foot drop and sensory loss in the dorsum of the foot.

Correct Answer: C. Common peroneal ✅

Topic Section:

Common Peroneal Nerve Injury

Definition:
Injury to the common peroneal nerve, often resulting from trauma to the fibular head or neck, leading to motor and sensory deficits.

Key Clinical Presentation:

Foot drop (inability to dorsiflex the foot)

Loss of sensation on the dorsum of the foot and lateral aspect of the lower leg

Weakness in eversion of the foot

Investigations:

Best initial: Clinical examination to assess motor and sensory function

Confirmatory: Electromyography (EMG) or nerve conduction studies

Treatment Outline:

First-line: Conservative management with ankle-foot orthotics

Second-line: Physical therapy to improve foot function

Definitive: Nerve repair or decompression if necessary

MCQ 895:

A 25-year-old man is brought to the Emergency Room after a severe road traffic accident. Examination shows he has an unstable left knee joint and it was possible to pull the tibia abnormally forwards on the femur. Which of the following ligaments is most likely injured?

A. Fibular collateral
B. Tibial collateral
C. Anterior cruciate
D. Posterior cruciate

Explanation:
The **anterior cruciate ligament (ACL)** is most likely injured in cases of instability and abnormal forward movement (anterior drawer sign) of the tibia relative to the femur. This is a common injury in trauma, especially in road traffic accidents and sports.

Correct Answer: C. Anterior cruciate ✅

Topic Section:

Anterior Cruciate Ligament Injury

Definition:
A tear or sprain of the anterior cruciate ligament (ACL), which stabilizes the knee by preventing anterior displacement of the tibia relative to the femur.

Key Clinical Presentation:

Instability or "giving way" of the knee

Pain and swelling shortly after injury

Positive anterior drawer test or Lachman test

Investigations:

Best initial: Physical examination with Lachman and anterior drawer tests

Confirmatory: MRI to confirm ligament rupture

Treatment Outline:

First-line: Conservative management with physical therapy (for minor sprains)

Second-line: ACL reconstruction surgery for significant tears

Definitive: Surgical intervention for severe ACL injuries with persistent instability

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MCQ 896:

A 35-year-old man is brought to the Emergency Room after a road traffic accident. X-ray shows a displaced fracture of the midshaft of the humerus. Which of the following is the most likely clinical complication?

A. Wrist drop
B. Claw hand
C. Waiter's tip hand
D. Klumpke's paralysis

Explanation:
The **radial nerve** is at risk in fractures of the midshaft of the humerus. Damage to the radial nerve can result in **wrist drop**, characterized by the inability to extend the wrist and fingers.

Correct Answer: A. Wrist drop ✅

Topic Section:

Radial Nerve Injury

Definition:
Injury to the radial nerve, typically caused by midshaft humeral fractures, leading to motor and sensory deficits.

Key Clinical Presentation:

Inability to extend the wrist and fingers (wrist drop)

Sensory loss on the dorsum of the hand and forearm

Investigations:

Best initial: Clinical examination for motor and sensory loss

Confirmatory: Nerve conduction studies or EMG

Treatment Outline:

First-line: Conservative management, including wrist splints and physical therapy

Second-line: Surgical intervention for severe or persistent cases

Definitive: Surgical repair or decompression in cases of nerve compression or severe injury

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MCQ 897:

A 24-year-old man came to the hospital with pain, swelling, and difficulty in moving the arm. Examination shows he has a humeral fracture. X-ray: The fracture was at the level of the surgical neck. Which of the following nerves was most likely injured?

A. Ulnar
B. Radial
C. Axillary
D. Median

Explanation:
The **axillary nerve** is most likely injured in fractures of the surgical neck of the humerus. The axillary nerve innervates the deltoid and teres minor muscles and is responsible for shoulder abduction. Injury to this nerve results in shoulder weakness and sensory loss over the deltoid area.

Correct Answer: C. Axillary ✅

Topic Section:

Axillary Nerve Injury

Definition:
Damage to the axillary nerve, often caused by fractures of the surgical neck of the humerus, leading to motor and sensory deficits.

Key Clinical Presentation:

Weakness in shoulder abduction

Sensory loss over the deltoid region

Muscle atrophy of the deltoid muscle

Investigations:

Best initial: Physical examination for shoulder movement and sensory loss

Confirmatory: Electromyography (EMG) and nerve conduction studies

Treatment Outline:

First-line: Conservative management, including physical therapy to strengthen the shoulder

Second-line: Surgical intervention for persistent weakness

Definitive: Surgical repair if nerve is severed or severely compressed

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MCQ 898:

A 41-year-old woman developed atrophy of the thenar eminence, but the sensation over it is intact. Which of the following nerves is most likely damaged?

A. Ulnar
B. Radial
C. Axillary
D. Median

Explanation:
The **median nerve** innervates the thenar eminence muscles (responsible for thumb movements). Atrophy of the thenar eminence with preserved sensation suggests median nerve injury, which commonly occurs in conditions like carpal tunnel syndrome.

Correct Answer: D. Median ✅

Topic Section:

Median Nerve Injury

Definition:
Injury to the median nerve, which provides motor innervation to the thenar muscles and sensation to the lateral palm and fingers.

Key Clinical Presentation:

Atrophy of the thenar eminence (e.g., "ape hand" deformity)

Weakness in thumb opposition and flexion

Sensory loss in the lateral palm and digits

Investigations:

Best initial: Clinical examination for motor function and sensory loss

Confirmatory: Nerve conduction studies or electromyography (EMG)

Treatment Outline:

First-line: Conservative management (splints, physical therapy) for mild cases

Second-line: Surgical decompression (e.g., carpal tunnel release)

Definitive: Surgical repair if the nerve is severely compressed or severed

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MCQ 899:

A 27-year-old pianist with known carpal tunnel syndrome experiences difficulty in finger movements. Which of the following muscles is most likely paralyzed?

A. Thenar muscles
B. Dorsal interossei
C. Palmar interossei
D. Medial two lumbricals

Explanation:
The **thenar muscles** are most likely to be paralyzed in carpal tunnel syndrome, as the median nerve innervates the muscles responsible for thumb opposition and flexion. This condition leads to weakness and atrophy of the thenar eminence.

Correct Answer: A. Thenar muscles ✅

Topic Section:

Carpal Tunnel Syndrome

Definition:
Compression of the median nerve at the wrist, leading to motor and sensory deficits in the hand, particularly affecting the thenar muscles.

Key Clinical Presentation:

Numbness, tingling, and weakness in the thumb and first two fingers

Atrophy of the thenar eminence (e.g., "ape hand" deformity)

Positive Tinel's sign or Phalen's maneuver

Investigations:

Best initial: Physical examination for weakness and sensory loss

Confirmatory: Nerve conduction studies

Treatment Outline:

First-line: Wrist splints, rest, and anti-inflammatory medications

Second-line: Corticosteroid injections or physical therapy

Definitive: Surgical decompression (carpal tunnel release) for severe cases

MCQ 900

A 52-year-old woman slipped and fell, and she complains of being unable to extend her leg at the knee joint. Which of the following muscles is most likely paralyzed?

A. Sartorius
B. Biceps femoris
C. Semitendinosus
D. Quadriceps femoris

Explanation:
The **quadriceps femoris** is responsible for extending the knee. If this muscle is paralyzed, the patient will be unable to extend her leg at the knee joint. The quadriceps is innervated by the femoral nerve.

Correct Answer: D. Quadriceps femoris ✅

Topic Section:

Quadriceps Femoris Muscle

Definition:
The quadriceps femoris is a large muscle group that includes four muscles: rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius. It is responsible for knee extension.

Key Clinical Presentation:

Inability to extend the knee joint

Weakness or atrophy in the anterior thigh

Inability to perform squatting or rising from a seated position

Investigations:

Best initial: Clinical examination and neurological assessment

Confirmatory: MRI or EMG to assess nerve or muscle damage

Treatment Outline:

First-line: Physical therapy and rehabilitation

Definitive: Surgery if there is nerve damage or significant muscle injury

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MCQ 901:

A 45-year-old man was brought to the Emergency Department after a motor vehicle accident with severe pain in his left arm. X-rays confirmed left humeral and ulnar fractures. After an open reduction and internal fixation, the patient was unable to extend his forearm, wrist, and fingers. Which of the following is the most likely damaged?

A. Ulnar nerve at medial epicondyle
B. Median nerve at the cubital fossa
C. Radial nerve above the spiral groove
D. Median nerve at the lateral epicondyle

Explanation:
The **radial nerve** is most commonly damaged in humeral fractures, particularly at the spiral groove. Injury to the radial nerve results in an inability to extend the forearm, wrist, and fingers, commonly referred to as "wrist drop."

Correct Answer: C. Radial nerve above the spiral groove ✅

Topic Section:

Radial Nerve Injury

Definition:
The radial nerve innervates the muscles responsible for extension of the arm, wrist, and fingers. Damage to the nerve, especially at the humeral level, leads to weakness in wrist and finger extension.

Key Clinical Presentation:

Inability to extend the wrist and fingers (wrist drop)

Sensory loss on the dorsal aspect of the hand and forearm

Investigations:

Best initial: Clinical examination to assess motor and sensory function

Confirmatory: Nerve conduction studies or EMG

Treatment Outline:

First-line: Conservative management with wrist splints and physical therapy

Second-line: Surgical decompression or nerve repair if necessary

Definitive: Surgical repair for severe cases

MCQ 902:

A 25-year-old man presents to the hospital with an open fracture of his right leg, after a motorbike accident on a farm. Later, he develops severe pain, necrosis, and blue discoloration of the skin, and there is gas in the deep tissues of the leg. Which of the following is the most likely causing agent?

A. Actinomyces israelii
B. Staphylococcus aureus
C. Clostridium perfringens
D. Mycobacterium ulcerans

Explanation:
Clostridium perfringens is the most likely causing agent in this case, as it is associated with gas gangrene, a severe soft tissue infection that causes tissue necrosis and gas production. This is typically seen in trauma involving open fractures.

Correct Answer: C. Clostridium perfringens ✅

Topic Section:

Gas Gangrene

Definition:
Gas gangrene is a life-threatening soft tissue infection caused primarily by **Clostridium perfringens**. It leads to tissue necrosis, sepsis, and gas formation in tissues.

Key Clinical Presentation:

Severe pain and swelling in the affected area

Skin discoloration (blue, black)

Gas production in tissues (crepitus on palpation)

Rapid progression of necrosis

Investigations:

Best initial: Clinical examination and X-rays to detect gas in the tissues

Confirmatory: Cultures from the infected tissue for **Clostridium perfringens**

Treatment Outline:

First-line: Surgical debridement and high-dose IV antibiotics (penicillin + clindamycin)

Second-line: Hyperbaric oxygen therapy

Definitive: Surgical removal of necrotic tissue and supportive care

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MCQ 903:

A 26-year-old man complained of pain in his buttocks, lower back, and stiffness which improved after moving around and doing exercises. The patient had similar complaints a year ago. Which of the following is the most likely diagnosis?

A. Hyperuricemia
B. Psoriatic arthritis
C. Reactive arthritis
D. Ankylosing spondylitis

Explanation:
Ankylosing spondylitis is the most likely diagnosis, as it typically presents with chronic low back

pain and stiffness that improves with movement. It commonly affects young men and is associated with sacroiliac joint involvement.

Correct Answer: D. Ankylosing spondylitis ✅

Topic Section:

Ankylosing Spondylitis

Definition:
Ankylosing spondylitis is a chronic inflammatory arthritis primarily affecting the spine and sacroiliac joints, often leading to fusion of the joints.

Key Clinical Presentation:

Chronic low back pain and stiffness (worse in the morning or after rest)

Improvement with physical activity

Sacroiliac joint tenderness

Reduced spinal mobility

Investigations:

Best initial: Clinical examination and history

Confirmatory: MRI of the sacroiliac joints or HLA-B27 testing

Treatment Outline:

First-line: Nonsteroidal anti-inflammatory drugs (NSAIDs)

Second-line: Disease-modifying antirheumatic drugs (DMARDs) like TNF inhibitors

Definitive: Long-term physical therapy and disease management

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MCQ 904:

A 36-year-old man has unilateral mild pain in the neck, which radiates to the right interscapular and shoulder regions. The pain is followed by sharp, electric shock-like pain, which goes down the arm with numbness, weakness, and loss of tendon reflexes. Which of the following is the most likely diagnosis?

A. Whiplash injury
B. Cervical disc prolapse
C. Rotator cuff tendonitis
D. Polymyalgia rheumatica

Explanation:
Cervical disc prolapse is the most likely diagnosis, as it can cause neck pain radiating to the shoulder and arm with associated neurological symptoms (numbness, weakness, loss of tendon reflexes).

Correct Answer: B. Cervical disc prolapse ✅

Topic Section:

Cervical Disc Prolapse

Definition:
Cervical disc prolapse occurs when a disc in the cervical spine herniates, pressing on adjacent nerve roots, causing pain, numbness, weakness, and loss of reflexes.

Key Clinical Presentation:

Neck pain radiating to the shoulder or arm

Numbness and tingling in the arm or hand

Weakness in the upper limbs

Loss of tendon reflexes

Investigations:

Best initial: Clinical examination and neurological assessment

Confirmatory: MRI of the cervical spine

Treatment Outline:

First-line: Conservative management with pain control, physical therapy, and rest

Second-line: Epidural steroid injections for refractory cases

Definitive: Surgery (discectomy or fusion) for severe or persistent symptoms

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MCQ 905:

A 58-year-old healthy patient has a six-month history of central lower back pain. The pain is present on awakening and, without treatment, resolves after about 30 minutes. Acetaminophen seemed to help it resolve sooner. Morning examination showed mild parasplinal muscle spasm with no neurological changes. MRI Lumbar: Mild lumbar spinal stenosis. Which of the following is the most appropriate management?

A. Biofeedback
B. Physical therapy
C. Lumbar laminectomy
D. Epidural steroid injection

Explanation:
The most appropriate management for mild lumbar spinal stenosis is **physical therapy**. Conservative measures like physical therapy and lifestyle modifications are effective in managing symptoms in mild cases.

Correct Answer: B. Physical therapy ✅

Topic Section:

Lumbar Spinal Stenosis

Definition:
Lumbar spinal stenosis is the narrowing of the spinal canal in the lumbar region, often leading to compression of the spinal nerves.

Key Clinical Presentation:

Central lower back pain, often worse on waking

Pain relieved by standing or walking

Numbness or weakness in the legs in severe cases

Mild parasplinal muscle spasm on examination

Investigations:

Best initial: Clinical examination

Confirmatory: MRI of the lumbar spine

Treatment Outline:

First-line: Physical therapy, weight management, and nonsteroidal anti-inflammatory drugs (NSAIDs)

Second-line: Epidural steroid injections for refractory cases

Definitive: Surgical intervention (laminectomy) for severe symptoms or nerve compression

A 24-year-old was running and suddenly experienced severe left leg pain. The patient was limping because of the pain, which is relieved by passive stretching. Which of the following is the most commonly involved muscle?

A. Soleus
B. Gastrocnemius
C. Tibialis posterior
D. Peroneus longus

Explanation:
The most commonly involved muscle in this scenario is the **gastrocnemius**, which is a calf muscle that is prone to strain during sudden, high-intensity activities like running. The pain relieved by passive stretching is characteristic of a muscle strain, commonly affecting the gastrocnemius.

Correct Answer: B. Gastrocnemius ✅

Topic Section:

Gastrocnemius Muscle Strain

Definition:
A gastrocnemius muscle strain is an injury that occurs when the muscle fibers are stretched beyond their limits, causing pain, swelling, and weakness.

Key Clinical Presentation:

- Sudden onset of calf pain during running or sudden movements
- Limping due to pain
- Pain relieved by passive stretching
- Tenderness over the muscle

Investigations:

- **Best initial:** Clinical examination to assess muscle tenderness and range of motion

- **Confirmatory:** Ultrasound or MRI if necessary for severe cases

Treatment Outline:

- **First-line:** Rest, ice, compression, and elevation (R.I.C.E)
- **Second-line:** Physical therapy for strengthening and flexibility
- **Definitive:** Rarely required; surgery for severe tears

MCQ 906:

A 23-year-old athlete presented with foot pain while standing and walking. On physical exam, there was significant tenderness on the midline plantar surface of the foot. The patient denied any history of trauma. Which of the following is the most likely diagnosis?

A. Hallux valgus
B. Hallux rigidus
C. Plantar fasciitis
D. Tarsal tunnel syndrome

Explanation:
Plantar fasciitis is the most likely diagnosis in this case. It is characterized by pain and tenderness in the plantar surface of the foot, particularly near the heel or midfoot, and is often aggravated by standing or walking. It is the most common cause of heel pain in athletes and non-athletes alike.

Correct Answer: C. Plantar fasciitis ✅

Topic Section:

Plantar Fasciitis

Definition:
Plantar fasciitis is the inflammation of the plantar fascia, a thick band of tissue running along the bottom of the foot, typically leading to heel pain.

Key Clinical Presentation:

- Pain in the heel or midfoot, often worse in the morning or after prolonged standing or walking
- Tenderness over the midline of the plantar surface of the foot

- Absence of history of trauma

Investigations:

- **Best initial:** Clinical examination with palpation of the plantar fascia
- **Confirmatory:** X-rays to rule out other causes, MRI if needed in chronic or severe cases

Treatment Outline:

- **First-line:** Rest, stretching exercises, and NSAIDs
- **Second-line:** Orthotic insoles, physical therapy, and corticosteroid injections
- **Definitive:** Surgery (rare) for refractory cases

MCQ 907:

A 24-year-old patient with a history of epilepsy is brought to the Emergency Room with complaints of left shoulder pain. The shoulder contour is flat and the arm is fixed in adduction and internal rotation. Which of the following is the most likely type of shoulder dislocation?

A. Inferior
B. Subglenoid anterior
C. Subcoracoid anterior
D. Subacromial posterior

Explanation:
A **subcoracoid anterior** shoulder dislocation is the most likely diagnosis in this case. The arm is held in adduction and internal rotation, which is characteristic of an anterior dislocation, specifically subcoracoid, which is the most common type of anterior dislocation.

Correct Answer: C. Subcoracoid anterior ✅

Topic Section:

Anterior Shoulder Dislocation

Definition:
An anterior shoulder dislocation occurs when the humeral head is displaced out of the glenoid fossa in an anterior direction. It is the most common type of shoulder dislocation.

Key Clinical Presentation:

- Severe shoulder pain and inability to move the arm
- The arm is held in adduction and internal rotation
- Flattening of the shoulder contour

Investigations:

- **Best initial:** Clinical examination and assessment of shoulder position
- **Confirmatory:** X-ray to confirm the type and severity of dislocation

Treatment Outline:

- **First-line:** Closed reduction under sedation
- **Second-line:** Analgesics and post-reduction immobilization
- **Definitive:** Surgery if recurrent or associated with fractures or nerve damage

MCQ 909:

A 2-year-old child who refuses to walk since the day before, presents to the clinic. The parent states that the child was playing and tripped over a toy. While falling, the right leg was caught in the toy and twisted. Since then, the child has wanted to be carried. The past medical history is normal and immunizations are current. Which of the following is the most likely diagnosis?

A. Spiral fracture of the right tibia
B. Spiral fracture of the right femur
C. Soft tissue swelling of the right ankle
D. Chip fracture of the proximal right tibia

Explanation:
The most likely diagnosis is **spiral fracture of the right tibia**. A common injury in toddlers is a spiral fracture due to twisting or torsion forces during a fall or tripping. These fractures are typically seen in non-ambulatory children and occur in the long bones, especially the tibia, when the leg is caught and twisted.

Correct Answer: A. Spiral fracture of the right tibia ✅

Topic Section:

Spiral Fracture of Tibia in Children

Definition:
A spiral fracture occurs when a bone is subjected to a rotational or twisting force, leading to a fracture that spirals around the bone.

Key Clinical Presentation:

- Refusal to walk, pain, and tenderness in the lower leg
- A twisting injury or fall that causes the leg to be caught or rotated
- Visible deformity or swelling over the affected leg

Investigations:

- **Best initial:** X-ray of the leg to confirm the fracture
- **Confirmatory:** CT scan if further detail is needed

Treatment Outline:

- **First-line:** Immobilization with a cast for minor fractures
- **Second-line:** Surgical intervention for displaced or severe fractures
- **Definitive:** Surgical repair for complex fractures or if the child is non-ambulatory

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MCQ 910:

A 23-year-old patient presented to the Emergency Department with pain in his finger that was caused by a hyperextension injury. An acute pain in the right second finger developed instantly. On examination, the patient cannot flex the distal phalanx of the right second finger, and has tenderness and swelling over the volar aspect of the distal phalanx of that finger. The patient also complained of pain and tenderness at the palm. Which of the following is the most likely resulting injury?

A. Rupture of the flexor profundus tendon
B. Rupture of the flexor superficialis tendon
C. Extra-articular fracture of the distal phalanx
D. Intra-articular fracture of the middle phalanx

Explanation:
The most likely injury is **rupture of the flexor profundus tendon**, as this injury is commonly associated with an inability to flex the distal phalanx. This condition, known as "jersey finger,"

occurs when the flexor tendon is ruptured during hyperextension injuries, leading to an inability to flex the finger at the distal joint.

Correct Answer: A. Rupture of the flexor profundus tendon ✅

Topic Section:

Jersey Finger (Rupture of Flexor Profundus Tendon)

Definition:
Jersey finger is an injury where the flexor profundus tendon is ruptured, often due to a forceful hyperextension of the finger.

Key Clinical Presentation:

- Inability to flex the distal phalanx of the affected finger
- Pain, tenderness, and swelling in the volar aspect of the distal phalanx
- Occurs after a hyperextension injury

Investigations:

- **Best initial:** Clinical examination and observation of finger movements
- **Confirmatory:** MRI or ultrasound to confirm tendon rupture

Treatment Outline:

- **First-line:** Surgical repair of the ruptured tendon
- **Second-line:** Post-operative rehabilitation and therapy
- **Definitive:** Tendon repair and finger splinting

MCQ 911:

A 26-year-old professional football player is brought to the Emergency Room after being hit on the lateral side of the left knee. The knee is buckled, and the patient is in severe pain. On examination, there was a swelling over the medial aspect of the left knee. There was laxity when the valgus stress test was performed on the knee. Both the Lachman test and McMurray test were negative. Which of the following is the most likely type of injury?

A. Lateral meniscus tear
B. Medial meniscus tear
C. Lateral collateral ligament sprain
D. Medial collateral ligament sprain

Explanation:
The most likely injury is a **medial collateral ligament (MCL) sprain**. The swelling on the medial side of the knee and laxity during the valgus stress test suggests damage to the MCL, which is stressed during valgus forces (such as when hit on the lateral side of the knee).

Correct Answer: D. Medial collateral ligament sprain ✅

Topic Section:

Medial Collateral Ligament (MCL) Injury

Definition:
The MCL is a ligament that stabilizes the inner aspect of the knee. It can be injured when excessive force is applied to the outer side of the knee.

Key Clinical Presentation:

- Pain and swelling on the medial aspect of the knee
- Positive valgus stress test indicating laxity of the MCL
- No significant findings on tests for ACL or meniscal tears

Investigations:

- **Best initial:** Clinical examination with valgus stress test
- **Confirmatory:** MRI if further evaluation of soft tissues is needed

Treatment Outline:

- **First-line:** Rest, ice, compression, elevation (R.I.C.E) and NSAIDs for pain relief
- **Second-line:** Bracing and physical therapy for rehabilitation
- **Definitive:** Surgery in case of severe tears or instability

912:

A 12-year-old overweight boy complains of severe left hip pain after minor trauma. He cannot bear his weight on the left leg. The boy denied any history of fever. Physical examination

confirmed that the left lower leg was externally rotated and the range of motion of the left hip joint is restricted and painful. Laboratory results were normal. Which of the following is the most likely diagnosis?

A. Septic arthritis
B. Femoral neck fracture
C. Slipped capital femoral epiphysis
D. Developmental dysplasia of the hip

Explanation:
The most likely diagnosis is **slipped capital femoral epiphysis (SCFE)**. This condition is commonly seen in overweight children and adolescents, particularly those who are 10-16 years old. It involves the displacement of the femoral head at the growth plate, and typically presents with hip pain, restricted motion, and the affected leg in external rotation.

Correct Answer: C. Slipped capital femoral epiphysis ✅

Topic Section:

Slipped Capital Femoral Epiphysis (SCFE)

Definition:
SCFE is a condition where the femoral head displaces posteriorly and inferiorly relative to the femoral neck, typically affecting adolescents during growth spurts.

Key Clinical Presentation:

- Hip pain, which may radiate to the groin or thigh
- Leg in external rotation and limited range of motion of the hip joint
- Typically affects overweight adolescents

Investigations:

- **Best initial:** X-ray of the hip, showing slippage of the femoral head
- **Confirmatory:** MRI if X-ray findings are inconclusive

Treatment Outline:

- **First-line:** Surgical pinning of the femoral head to prevent further displacement
- **Second-line:** Avoid weight-bearing and use crutches until surgery

- **Definitive:** Surgical fixation with screws or pins

MCQ 913:

An 80-year-old woman presents with a history of bilateral leg pain and numbness, preventing her from walking more than 50 meters for over 2 years. Her symptoms are worse when going downhill. On the other hand, she can walk further if she bent forward on a walker or if she is going uphill. When she reaches the limit, her leg pain stops her from going any further. Physical examination showed a forward stooped gait. Spine and neurological examination were normal. Both pedal pulses are present. Which of the following is the most likely diagnosis?

A. Lumbar spinal stenosis
B. Lumbar discogenic pain
C. Peripheral vascular disease
D. Lumbosacral spondylolisthesis

Explanation:
The most likely diagnosis is **lumbar spinal stenosis**. The patient's symptoms of leg pain, numbness, and the relief she experiences with flexion (bending forward) are characteristic of spinal stenosis, which is caused by narrowing of the spinal canal, leading to compression of nerves. The fact that the pain is worse with downhill walking and relieved by leaning forward is typical.

Correct Answer: A. Lumbar spinal stenosis ✅

Topic Section:

Lumbar Spinal Stenosis

Definition:
Lumbar spinal stenosis is the narrowing of the lumbar spinal canal, leading to compression of the nerve roots, causing pain, numbness, and weakness in the legs.

Key Clinical Presentation:

- Leg pain, numbness, and weakness that worsen with walking, especially downhill
- Relief with leaning forward (e.g., using a walker)
- Forward stooped gait

Investigations:

- **Best initial:** MRI of the lumbar spine to evaluate for canal narrowing and nerve compression
- **Confirmatory:** CT scan if MRI is inconclusive

Treatment Outline:

- **First-line:** Conservative management with physical therapy, NSAIDs, and activity modification
- **Second-line:** Epidural steroid injections for symptomatic relief
- **Definitive:** Surgical decompression (laminectomy) for severe cases

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MCQ 914:

A 8-year-old boy presents to the Emergency Department after a fall on his left forearm. On examination, there was swelling and a deformity with a 1 cm wound on the volar aspect of his forearm. Which of the following is the first step in management?

A. Closed reduction and below elbow cast
B. Closed reduction and above elbow cast
C. Discharge the patient with oral antibiotic
D. Surgical debridement, irrigation, and fixation

Explanation:
The most appropriate first step is **surgical debridement, irrigation, and fixation**. The presence of a wound and the deformity in the forearm suggest an open fracture, which requires urgent debridement, cleaning, and stabilization of the fracture. Oral antibiotics alone are not sufficient for open fractures.

Correct Answer: D. Surgical debridement, irrigation, and fixation ✅

Topic Section:

Open Fractures

Definition:
An open fracture is when the bone breaks through the skin, creating a communication between the external environment and the bone.

Key Clinical Presentation:

- Pain, swelling, and deformity at the fracture site
- Open wound with visible bone or bone protruding
- Risk of infection

Investigations:

- **Best initial:** X-ray to assess the fracture
- **Confirmatory:** CT scan if more detailed visualization of the fracture is needed

Treatment Outline:

- **First-line:** Surgical debridement, irrigation, and fixation with either external fixation or internal fixation
- **Second-line:** Broad-spectrum IV antibiotics for infection prevention
- **Definitive:** Bone healing and rehabilitation with follow-up care

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MCQ 915:

A 30-year-old man was hit by a car. He was brought to the Emergency Department within 1 hour, and cleared by the trauma team. He was found to have an isolated injury to his left leg. An examination revealed tenderness over the shin and abrasions with a small oozing wound. IV antibiotics and tetanus prophylaxis were started. Which of the following is the most appropriate next step?

A. Discharge from emergency department with above knee cast
B. Urgent surgical debridement and intramedullary nailing
C. Closed reduction followed by external fixation
D. Discharge on oral antibiotic for 3 days

Explanation:
The most appropriate next step is **urgent surgical debridement and intramedullary nailing**. The presence of an open fracture with oozing wounds necessitates debridement to prevent infection and to stabilize the bone, followed by fixation with intramedullary nails to ensure alignment and healing.

Correct Answer: B. Urgent surgical debridement and intramedullary nailing ✅

Topic Section:

Management of Open Fractures

Definition:
Open fractures involve the bone being exposed to the external environment, putting the patient at high risk for infection.

Key Clinical Presentation:

- Open wound with exposed bone or tissue
- Pain, swelling, and deformity at the fracture site
- History of trauma, often with a wound, abrasions, or skin penetration

Investigations:

- **Best initial:** X-ray to assess fracture alignment and bone injury
- **Confirmatory:** CT scan for complex fractures or to assess bone involvement

Treatment Outline:

- **First-line:** Surgical debridement and fixation (internal or external)
- **Second-line:** IV antibiotics and tetanus prophylaxis
- **Definitive:** Bone healing and rehabilitation with follow-up care

MCQ 916:

A 25-year-old obese patient complained of numbness and tingling of the little and ring fingers on the left hand. The patient said that the symptoms are worse when the hands are raised. The elevated arm stress test is positive on the left. Which of the following is the most likely diagnosis?

A. Carpal tunnel syndrome
B. Thoracic outlet syndrome
C. Chronic shoulder subluxation
D. Ulnar artery thrombophlebitis

Explanation:
The most likely diagnosis is **thoracic outlet syndrome** (TOS). The patient's symptoms of numbness and tingling in the little and ring fingers, worsened with arm elevation, are classic signs of TOS. The positive elevated arm stress test further supports this diagnosis, as it indicates

compression of the neurovascular bundle at the thoracic outlet, leading to impaired circulation and nerve compression in the upper extremity.

Correct Answer: B. Thoracic outlet syndrome ✅

Topic Section:

Thoracic Outlet Syndrome (TOS)

Definition:
Thoracic outlet syndrome is a condition caused by compression of the neurovascular structures (brachial plexus, subclavian artery, or vein) in the thoracic outlet, leading to symptoms in the arm and hand.

Key Clinical Presentation:

- Numbness, tingling, and weakness in the arm and hand, especially the ulnar nerve distribution (little and ring fingers)
- Worsening of symptoms with arm elevation
- Pain and swelling in the arm
- Positive elevated arm stress test

Investigations:

- **Best initial:** Clinical diagnosis with history and physical exam
- **Confirmatory:** MRI or neurophysiological tests to assess nerve compression

Treatment Outline:

- **First-line:** Physical therapy and postural adjustments to relieve compression
- **Second-line:** Surgical decompression if symptoms persist or worsen

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MCQ 917:

A 13-year-old boy with sickle cell disease, presents with severe unilateral hip pain, which has developed over the past few weeks. Which of the following is the most likely diagnosis?

A. Avascular necrosis
B. Rheumatic fever

C. Psoriatic arthritis
D. Still’s disease

Explanation:
The most likely diagnosis is **avascular necrosis** (AVN). Children with sickle cell disease are at high risk of developing AVN, particularly in the hip joint, due to impaired blood flow and vaso-occlusion. The gradual onset of hip pain, along with a history of sickle cell disease, makes AVN the most likely diagnosis.

Correct Answer: A. Avascular necrosis ✅

Topic Section:

Avascular Necrosis (AVN) in Sickle Cell Disease

Definition:
Avascular necrosis is the death of bone tissue due to a lack of blood supply, often affecting the femoral head in sickle cell disease due to microvascular occlusions.

Key Clinical Presentation:

- Gradual onset of joint pain, often in the hip or shoulder
- Pain exacerbated by weight-bearing or movement
- History of sickle cell disease or other conditions causing impaired blood flow

Investigations:

- **Best initial:** X-ray of the affected joint
- **Confirmatory:** MRI is the gold standard for detecting early AVN changes

Treatment Outline:

- **First-line:** Conservative management with rest and analgesics
- **Second-line:** Core decompression surgery for early cases
- **Definitive:** Hip replacement for advanced cases

MCQ 918:

A 30-year-old patient presents with a history of pain and swelling of the right knee. Aspirate of knee: Colour Yellow (Clear), Appearance Opaque (Transparent), Viscosity Variable (High), WBC 15.2 (< 200 /ml), PMNs 80% (< 25%), Mucin clot Poor (Good). Which of the following is the most likely diagnosis?

A. Gouty arthritis
B. Septic arthritis
C. Rheumatic arthritis
D. Pseudogout arthritis

Explanation:
The most likely diagnosis is **septic arthritis**. The high white blood cell count, predominance of polymorphonuclear leukocytes (PMNs), and poor mucin clot formation all suggest a bacterial infection in the joint, which is consistent with septic arthritis. This condition requires urgent treatment to prevent joint destruction.

Correct Answer: B. Septic arthritis ✅

Topic Section:

Septic Arthritis

Definition:
Septic arthritis is an infection of the joint, usually caused by bacteria, leading to inflammation, pain, and swelling.

Key Clinical Presentation:

- Sudden onset of joint pain, swelling, and erythema
- Fever and systemic signs of infection
- Joint aspiration showing high WBC count with predominance of neutrophils

Investigations:

- **Best initial:** Joint aspiration for analysis of synovial fluid (WBC count, Gram stain, and culture)
- **Confirmatory:** Blood cultures to identify the causative organism

Treatment Outline:

- **First-line:** IV antibiotics based on culture results

- **Second-line:** Surgical drainage if necessary
- **Definitive:** Joint lavage and debridement if the infection is not controlled with antibiotics

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MCQ 919:

An 8-year-old boy complained of pain in his right hip and was unable to walk, associated with high fever for the last five days. Physical examination confirmed that the active and passive range of motion of the hip was painful and restricted. Pelvic and hip X-rays were normal. Ultrasound of the hip showed a right hip fluid collection. Blood pressure 120/65 mmHg, Heart rate 110 /min, Respiratory rate 22 /min, Temperature 39.2° C. Test Result: Hb 125 (112-165 g/L), Platelets count 210 (150-400 x 10^9/L), WBC 9.8 (4.5-10.5 x 10^9/L), ESR 78 (2-10 mm/h), C-reactive peptide 2.0 (0-0.5 mg/dl). Which of the following is the most appropriate next step in management?

A. Hip MRI
B. Bone scan
C. Hip CT scan
D. Hip aspiration

Explanation:
The most appropriate next step is **hip aspiration**. The child likely has septic arthritis or osteomyelitis, and joint aspiration is essential to obtain a definitive diagnosis and guide treatment. The elevated ESR, CRP, and fever support an infectious process, and ultrasound showing fluid collection in the hip suggests a septic joint.

Correct Answer: D. Hip aspiration ✅

Topic Section:

Septic Hip in Children

Definition:
Septic arthritis of the hip is a bacterial infection of the hip joint, often caused by hematogenous spread, leading to inflammation, pain, and potential joint destruction if untreated.

Key Clinical Presentation:

- Sudden onset of hip pain, fever, and limited range of motion
- Tenderness on palpation of the hip joint

- Elevated inflammatory markers (ESR, CRP)

Investigations:

- **Best initial:** Hip aspiration to identify causative organism
- **Confirmatory:** MRI for detailed assessment of joint and surrounding structures

Treatment Outline:

- **First-line:** IV antibiotics based on culture results
- **Second-line:** Surgical drainage if the infection is not controlled with antibiotics
- **Definitive:** Joint lavage and debridement if necessary to prevent joint destruction

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MCQ 920:

An 8-year-old boy complained of pain in his right hip and was unable to walk, associated with high fever for the last five days. Physical examination confirmed that the active and passive range of motion of the hip was painful and restricted. Pelvic and hip X-rays were normal. Ultrasound of the hip showed a right hip fluid collection. Blood pressure 120/65 mmHg, Heart rate 110 /min, Respiratory rate 22 /min, Temperature 39.2° C. Test Result: Hb 125 (112-165 g/L), Platelets count 210 (150-400 x 10^9/L), WBC 9.8 (4.5-10.5 x 10^9/L), ESR 78 (2-10 mm/h), C-reactive peptide 2.0 (0-0.5 mg/dl). Which of the following is the most appropriate next step in management?

A. Hip MRI
B. Bone scan
C. Hip CT scan
D. Hip aspiration

Explanation:
The most appropriate next step is **hip aspiration**. The child likely has septic arthritis, which requires joint aspiration to identify the causative organism and guide treatment. The elevated ESR, CRP, and fever, along with ultrasound showing fluid collection, all point to an infectious process in the hip joint. Hip aspiration is necessary for both diagnostic and therapeutic purposes.

Correct Answer: D. Hip aspiration ✅

Topic Section:

Septic Hip in Children

Definition:
Septic arthritis of the hip is a bacterial infection of the hip joint, often caused by hematogenous spread, leading to inflammation, pain, and potential joint destruction if untreated.

Key Clinical Presentation:

- Sudden onset of hip pain, fever, and limited range of motion
- Tenderness on palpation of the hip joint
- Elevated inflammatory markers (ESR, CRP)

Investigations:

- **Best initial:** Hip aspiration to identify causative organism
- **Confirmatory:** MRI for detailed assessment of joint and surrounding structures

Treatment Outline:

- **First-line:** IV antibiotics based on culture results
- **Second-line:** Surgical drainage if the infection is not controlled with antibiotics
- **Definitive:** Joint lavage and debridement if necessary to prevent joint destruction

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MCQ 921:

A 42-year-old woman gives a history of recurrent fractures and the passage of stones in the urine preceded by renal colic. Family history and a clinical examination were normal. Test Result: Calcium 2.60 (2.15-2.62 mmol/L), Albumin 41 (34-56 g/L), Phosphate 1.1 (0.82-1.51 mmol/L), Parathyroid hormone 402 (1.1-5.3 pmol/L). Plain X-ray abdomen: Multiple radio-opaque shadows in the renal area. Which of the following is the most appropriate investigation?

A. Sestamibi scan
B. Alkaline phosphatase
C. 24-hour urine calcium
D. Intravenous pyelogram

Explanation:
The most appropriate investigation is **24-hour urine calcium**. The patient presents with hypercalcemia and recurrent kidney stones, which suggests possible primary

hyperparathyroidism. Measuring calcium levels in a 24-hour urine collection will help determine the calcium excretion and further confirm the diagnosis. Elevated urinary calcium excretion is often seen in hyperparathyroidism.

Correct Answer: C. 24-hour urine calcium ✅

Topic Section:

Primary Hyperparathyroidism

Definition:
Primary hyperparathyroidism is characterized by excessive secretion of parathyroid hormone (PTH), leading to increased serum calcium levels and often causing kidney stones, fractures, and other complications.

Key Clinical Presentation:

- Hypercalcemia with symptoms such as fatigue, bone pain, and kidney stones
- History of recurrent fractures and renal colic
- Elevated parathyroid hormone (PTH) levels

Investigations:

- **Best initial:** Serum calcium and PTH levels
- **Confirmatory:** 24-hour urine calcium to assess calcium excretion
- **Imaging:** Neck ultrasound or sestamibi scan for parathyroid adenoma

Treatment Outline:

- **First-line:** Parathyroidectomy if an adenoma is confirmed
- **Second-line:** Medical management with bisphosphonates or calcimimetics in non-surgical candidates

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MCQ 922:

A 48-year-old man who is a smoker for the last 20 years, had a routine exam. Test Result: Calcium 3.1 (2.15-2.62 mmol/L). Chest X-ray: 4 cm diameter left hilar mass. Which of the following is the most likely diagnosis?

A. Metastatic carcinoma
B. Adenocarcinoma of lung
C. Advanced bronchiectasis
D. Squamous cell carcinoma of lung

Explanation:
The most likely diagnosis is **squamous cell carcinoma of the lung**. The patient's smoking history, along with the presence of a left hilar mass on chest X-ray, points towards lung cancer, particularly squamous cell carcinoma, which is known to arise in the central part of the lung (hilar region). Elevated calcium levels (hypercalcemia) can be seen in squamous cell carcinoma due to parathyroid hormone-related peptide (PTHrP) production.

Correct Answer: D. Squamous cell carcinoma of lung ✅

Topic Section:

Squamous Cell Carcinoma of the Lung

Definition:
Squamous cell carcinoma of the lung is a type of non-small cell lung cancer (NSCLC) that typically arises in the central airways and is strongly associated with smoking.

Key Clinical Presentation:

- Cough, hemoptysis, and weight loss
- Hypercalcemia (due to PTHrP secretion)
- Chest pain or dyspnea in advanced cases

Investigations:

- **Best initial:** Chest X-ray or CT scan for mass detection
- **Confirmatory:** Bronchoscopy and biopsy for histological diagnosis

Treatment Outline:

- **First-line:** Surgery (lobectomy) if the tumor is resectable
- **Second-line:** Chemotherapy and radiation for advanced stages

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MCQ 923:

A 19-year-old sustained head trauma following road traffic accident. Skull X-ray: A skull base fracture passing through the jugular foramen. Which of the following is most likely symptom?

A. Loss of abduction of the eye
B. Ipsilateral vocal cord paralysis
C. Sensory loss over the zygoma
D. Paralysis of muscles of mastication

Explanation:
The most likely symptom is **ipsilateral vocal cord paralysis**. A fracture involving the jugular foramen can damage the glossopharyngeal (CN IX) and vagus (CN X) nerves, leading to vocal cord paralysis on the same side. This occurs because the vagus nerve controls the muscles of the larynx, including those responsible for vocal cord movement.

Correct Answer: B. Ipsilateral vocal cord paralysis ✅

Topic Section:

Jugular Foramen Syndrome

Definition:
Jugular foramen syndrome is caused by the compression or injury of the cranial nerves (IX, X, XI) passing through the jugular foramen.

Key Clinical Presentation:

- Ipsilateral vocal cord paralysis (due to vagus nerve involvement)
- Loss of sensation in the posterior tongue (due to glossopharyngeal nerve involvement)
- Shoulder weakness (due to accessory nerve involvement)

Investigations:

- **Best initial:** CT or MRI of the skull to identify the fracture
- **Confirmatory:** Neurological examination and laryngoscopy for vocal cord assessment

Treatment Outline:

- **First-line:** Observation and supportive care
- **Second-line:** Surgical intervention if nerve damage is extensive